

MAGNETOM Symphony Tim

MR

Online-Help

System

VB19A

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Print No.: MR06-015.880.04.02.02
Replaced: MR06-015.880.04.01.02

English
Doc. Gen. Date: 03.14

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1 General Information

HP-Z0000YPR-210206-02-001

n.a.

1.1 General

HP-Z0000YPR-061005-02-001

n.a.

1.1.1 Disclaimer

HP-Z0000YPR-061005-02-003

System Type:Avanto
Espreo
TrioTim
SymphonyTim
ClinScan

CAUTION
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The installation and service of the equipment described herein are to be performed by qualified personnel who are employed by Siemens or one of its affiliates or who are otherwise authorized by Siemens or one of its affiliates to provide such services.

- ⇒ Assemblers and other persons who are not employed by or otherwise directly affiliated with or authorized by Siemens or one of its affiliates are directed to contact one of the local offices of Siemens or one of its affiliates before attempting installation or service procedures.

1.1.2 Introduction

HP-Z0000YPR-061005-02-004

System Type: Avanto
 Espree
 TrioTim
 SymphonyTim
 ClinScan

This document provides an overview of the available service software features, a detailed description of local and remote access modes, and information regarding the Service Key used with the variable password protection mechanism.

1.1.3 Features

 HP-Z0000YPR-061005-02-005

System Type:Avanto
 Espreo
 TrioTim
 SymphonyTim
 ClinScan

The Service Software package consists of the following parts:

- An HTML-based **Service Software User Interface** (General) for local and remote use.
- **Local Options** (General) for the customer to grant remote access to the system, and for viewing the service session (Remote Service) and the maintenance bar.

All Service Software functions are TCP/IP-based and can be accessed simultaneously, allowing, e.g., EventLog evaluation while testing is in progress.

All Service Software functions are also protected against unauthorized use by the Service Key mechanism.

- Language-dependent buttons or windows: Some buttons and pop-up windows are language-dependent in the installed Windows version (English, German, etc.).
- Scrolling/paging backward/forward: In some menus, scrolling “backward/forward” is not performed via right mouse-click or the buttons. In these cases, the navigation frame (left side of screen) is used.
- Hyperlink: A blue underlined text string indicates that there is a link to this section.
- Printing help pages: To print the online help page currently displayed, press **Ctrl + p**.

1.1.4 Abbreviations

HP-Z0000YPR-061005-02-006

System Type: Avanto
 Espree
 TrioTim
 SymphonyTim
 ClinScan

Item	Explanation
^	Control key (<CTRL>)
AET	Application Entity Title - Name of DICOM service participating in the communication, either as a user (SCU) or the provider (SCP).
API	Application Programmers Interface
bps	Bit per second
ASCII	American Standard Code for Information Interchange
CAP	Common Application Program
CSA	Common Software Architecture
C.S.E.	Customer Service Engineer
cmd	Windows NT command console
CompMgr	Component Manager
DHCP	Dynamic Host Configuration Protocol - assignment procedure of the numeric IP address to nodes, in which blocks of addresses are leased to clients on a first-come, first-served basis. Typically, DHCP is combined with WINS, so an IP address is both dynamically assigned and automatically associated with a domain name. This can be combined with DNS, resulting in a dynamically assigned numeric IP address that is automatically associated with both NetBIOS and Internet names.
DICOM	Digital Imaging & Communication Network Protocol
diruse	directory usage
DNS	Domain Name Service - address resolution procedure which associates numeric IP addresses with Internet-style names.
FTP	File Transfer Protocol - a client-server application that transfers files from one computer to another.
Gateway	A display that controls traffic between two networks using factors above the network layer.
GSM	Global Session Manager
GUI	Graphical User Interface

HIS	Hospital Information System
HQ	Headquarters
HTML	Hyper-Text Mark-up Language
HTTP	Hyper-Text Transfer Protocol
IP	Internet Protocol (See also TCP/IP)
ISDN	Integrated Services Digital Network
LAN	Local Area Network
LMHOSTS	Address resolution procedure which statically associates numeric IP addresses with NetBIOS names. The LMHOSTS file is checked to see whether the name is being addressed in this file.
MED	Siemens Medical Solutions Group
MOD	Magnetic Optical Disk
NetBIOS	Network Basic Input-Output System - a collection of network services and programming interfaces. NetBIOS services run on network protocols such as TCP/IP.
NIC	Network Interface Card - an expansion card that provides a physical connection to the network.
OS	Operating system
PPP	Point-to-Point Protocol; establishes a transparent TCP/IP connection via a modem or router connection.
QA	Quality Assurance
RAS	Remote Access Services - a Windows NT service that supports dial-in and dial-out connections using modems and ISDN.
RIP	Routing Information Protocol. Common protocol for managing router information within a local area network.
Router	A multiport computer or piece of dedicated hardware that examines network-layer addresses of packets on a network and, if necessary, retransmits them on a different network.
rti	runtime interface tool
SAM	Service Access Manager
SCP	Service Class Provider - Server waiting for a DICOM communication.
SCU	Service Class User - Initiator of a DICOM communication.
SPI	SCSI-3 Parallel Interface
SRS	Siemens Remote Service
TCP/IP	Transport Control Program / Internet Protocol - the most widely used standard for networking.

TCP/IP address	Four-digit period-separated code to identify nodes on a network.
UI	User Interface
USC	UPTIME Service Center
W2K	Windows 2000
WAN	Wide Area Network
WINS	Windows Internet Name Service - address resolution procedure which dynamically associates numeric IP addresses with NetBIOS names. It is possible to combine DNS and WINS.
WWW	World Wide Web

1.1.5 Definitions

HP-Z0000YPR-061005-02-007

System Type:Avanto
 Espree
 TrioTim
 SymphonyTim
 ClinScan

Short form	Description
Batch Mode	A transfer mode that is executed without interactive user action; in other words, a pre-scheduled, unattended action.
RDIAG	Describes the Remote Diagnostics capabilities
RDIAG Server	This definition is no longer used. It has been replaced by RemoteServer .
Remote Server	Central Server equipped with modems, routers, and a communication package to enable dial-up connections from the Service Center to the remote systems.
Remote Service	Ability to log into a system from a remote location to employ various service software functions.
Remote Terminal	An office PC or Service PC located remotely from the system.
Service software	All functions provided for service personnel to install, tune, maintain, and service the system.
Service Key	A time-dependent, variable service password, unique to a customer site.
System Management	Enhanced remote service features, such as proactive event monitoring and software distribution which is enabled by the installed Managed Note Package containing several agents.

1.1.6 Service Software UI

HP-Z0000YPR-061005-02-008

System Type: Avanto
Espreo
TrioTim
SymphonyTim
ClinScan

The service software UI is a set of automated functions for installing, tuning, maintaining, and servicing the system.

Description

The user interface (UI) is based on the WWW technologies **HTML** and **JavaScript** and runs on the Microsoft internet browser (**Internet Explorer 6.0**). Therefore, the service software UI is independent of hardware and software: it can be run locally on the console or remotely on a Service PC.

The following menus are available:

- Authentication menu
- Home menu
- Viewing menu
- Bars

NOTE

The menus have been optimized for a resolution of 800x600 pixels so that they can also be displayed on the Service laptop.

Authentication menu

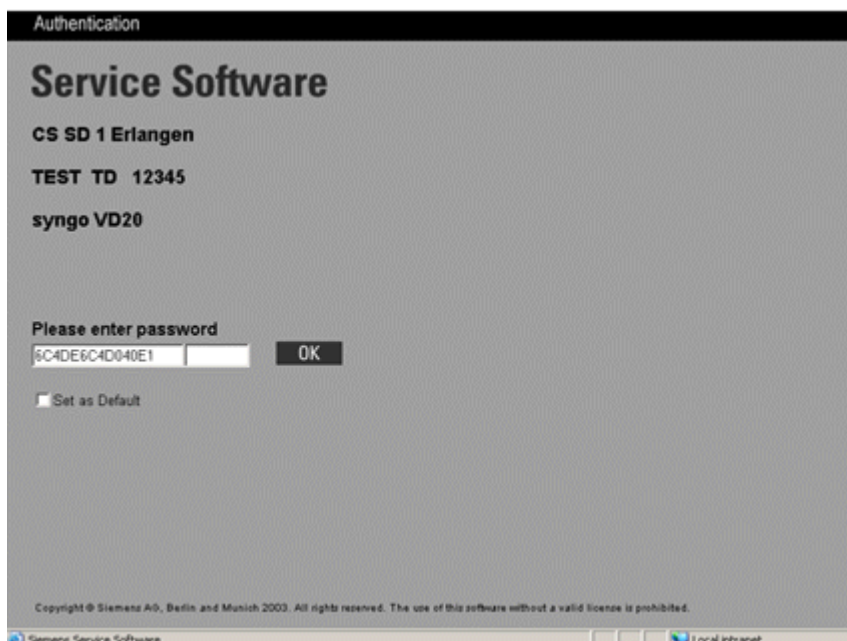


Fig. 1: Authentication Page

The Service Software is protected by a **Service Key** to prevent unauthorized access. Service Keys are valid for one particular system only for a limited time period (e.g. one year).

As of today, 7 service levels have been defined. The levels are grouped into Free, Basic, Advanced, Expert, and Siemens (Level 7). The higher the level, the more rights the user of the Service Software has to access certain functions.

Service level 0 (without Service Key) is the so-called Customer Level. Only the Site Info configuration and Licensing configuration can be accessed.

The last 6 digits of the Service Key have to be entered into the right field every time a login is performed. The left part (14 digits) is memorized by the system if the **Select as Default** checkbox is selected.

When the validity of the Service Key has expired, or when an incorrect Service Key has been entered, a pop-up window is displayed with the appropriate Service Key identification.

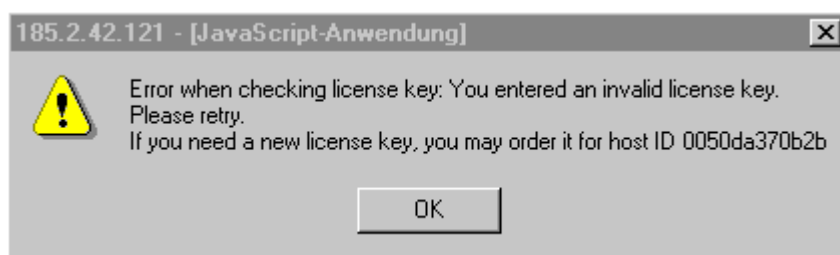


Fig. 2: Incorrect or invalid Service Key

Home menu

When a valid Service Key has been entered, the **Home Menu** is displayed. The menu contains the various service functions.

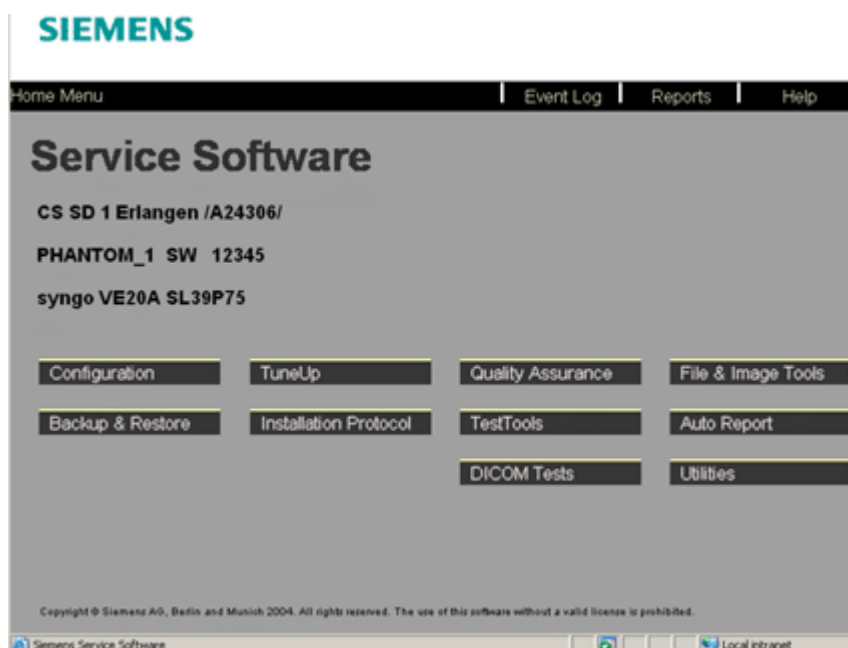


Fig. 3: Home menu

The number of service functions varies from modality to modality. Please refer to the various service functions in the Overview menu for detailed information on the service functions supplied.

Viewing menu

As an example, the **Online Help** window is displayed.

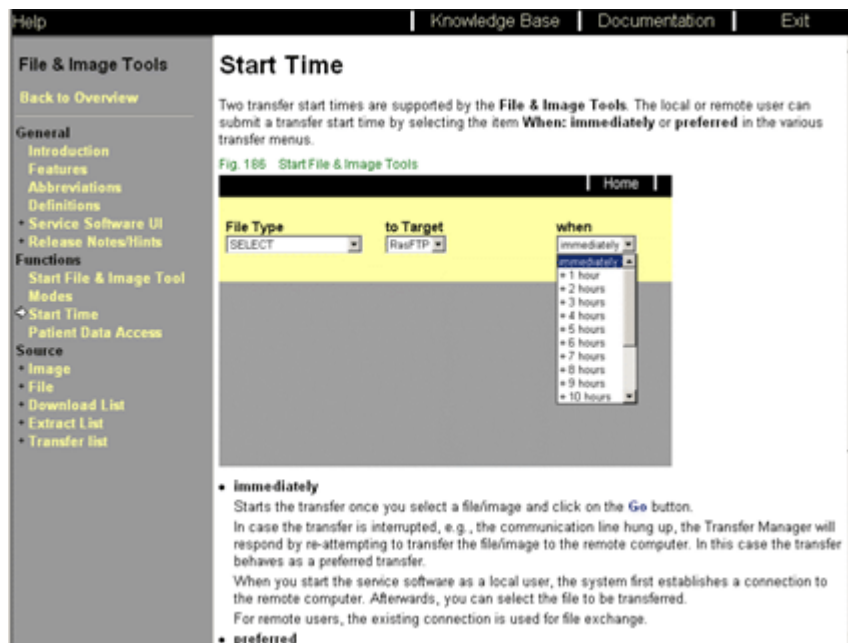


Fig. 4: Viewing Menu

The Viewing menu is divided into two frames, the **Navigation Frame** and the **Function Frame**.

A structured index is displayed in the **Navigation Frame**. A '+' in front of the hyperlink indicates that another sub-chapter is available. Selecting the sub-chapter will change the '+' to a '->' and opens the sub-chapter. Selecting another hyperlink will change the '->' back to a '+' and the sub-chapter will be closed again.

As soon as the lowest level has been reached, an arrow will be displayed in front of the hyperlink and the corresponding information will be displayed in the **Function Frame**.

Bars

Two different bars are supported by the syngo service software, the **Navigation Bar** and the **Action Bar**.



Fig. 5:

The **Navigation Bar** has the following functions:

- Shows the name of the function, here **Home Menu**.
- Navigates the user through the service software. Depending on the displayed menu, you will find different buttons (here, for example, Event Log, Reports, etc.).
- The navigation bar enables frequently used applications to be accessed quickly. Functions that are started from the **Navigation Bar** include, e.g., Event Log, Reports, Documentation, and Help. These functions will always open an additional window that can be used in parallel to the main function, e.g., **File & Image, Backup & Restore** etc.



Fig. 6: Action Bar

The **Action Bar** has the following function:

- The Action Bar contains buttons that affect the behavior of the function, e.g., Save, Go, Abort, etc.
- The current status is displayed in the left area of the Action Bar, e.g., "Status: 84 messages found".

1.1.7 Release Notes/Hints

HP-Z0000YPR-061005-02-009

System Type: Avanto
Espreo
TrioTim
SymphonyTim
ClinScan

Configuration settings

After configuration changes have been made; we recommend that you use **Reports > Configuration > Show as PDF** then use the save function in “Acrobat Reader” to save a copy of the file in the location C:\MedCom\service\html\seso\sitedb\log e.g. with the syntax: (YYYY-MM-DD) **2014-12-18_Configuration_Report.pdf** so that you can compare the current configuration with a previous configuration.

2 Configuration

HP-Z0000YPR-101005-02-001	
System Type:	.Avanto
	Espre
	TrioTim
	SymphonyTim
	ClinScan
n.a.	

2.1 General remarks

HP-Z0000YPR-101005-02-002

System Type: Avanto
Espreo
TrioTim
SymphonyTim
ClinScan

n.a.

2.1.1 Introduction

 HP-Z0000YPR-101005-02-003

System Type:Avanto
 Espreo
 TrioTim
 SymphonyTim
 ClinScan

NOTE

Most of the pictures are screen dumps from syngo only consoles, the syngo MR specific issues are only described in text form.

Performing the configuration from a remote location assumes that **Full Access mode** and a **level 7** license are available; i.e., a corresponding service key is required. With **limited access**, you can display the configuration data, but you are not able to modify any settings.

To configure your system, select the options required from the **list of system options**. Options that are not selected will not be provided in the following configuration menus.

To **store any modifications or new entries**, it is necessary to click the 'Save' button. In the Navigation Bar, you will get the message `shutdown pending`, indicating that a restart is required to activate the modifications. This restart is performed automatically when you click the **Home** button. Depending on the modifications, either a restart of the application or a reboot of the system is performed. It may take several minutes until the system is up again; do not interrupt the restart procedure.

NOTE

To exit Configuration, always select the Home button in the Action Bar. Never exit Configuration by closing the web browser with the X button.

Once you click the **Home** button, a new Configuration Report is created which can be seen under **Reports**.

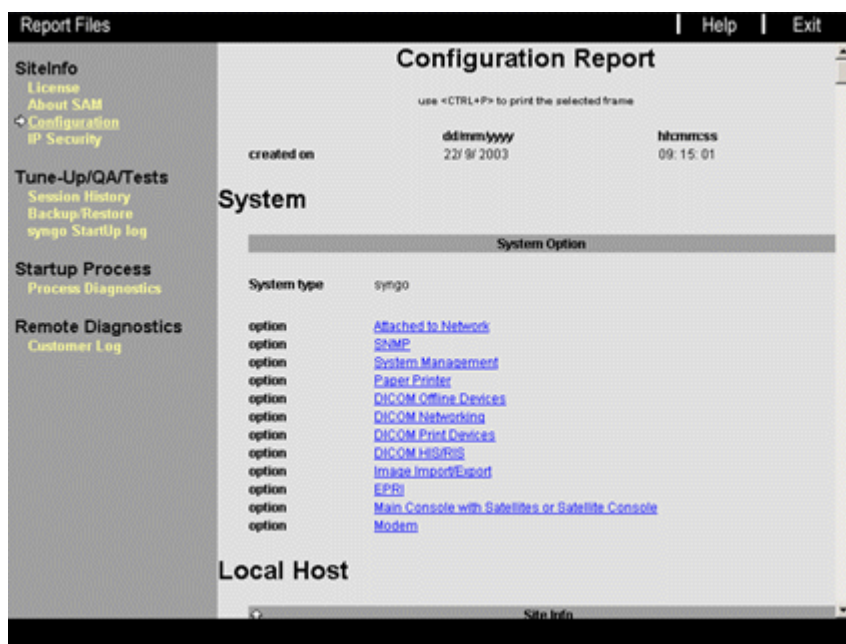


Fig. 7: Report - Site Info - Configuration

2.1.2 Features

HP-Z0000YPR-101005-02-004

System Type:Avanto
 Espreo
 TrioTim
 SymphonyTim
 ClinScan

Configuration allows for easy setup and customizing of your system, including networking, DICOM, customer user interface, image database, Remote Diagnostics, and other service functions.

- **General Site and System Information**
- **Properties of the local area network**
- **Properties of the wide area network**
- **Image database location and usage**
- **A system that works as a Satellite Console**
- **Remote Diagnostics functions (mail and ftp addresses for Auto Report, File & Image Tools)**
- **Other service functions (Event Log, Backup & Restore)**
- **DICOM functions**
- **System Management settings**
- **Customer user interface**
- **Measurement (*MRC* only e.g.: *System Type, Meas. Settings, Avail. Coils*)**
- **System Utilization (*MrStatistic*: To meet legal aspects, the customer has to sign a contract in advance before activating the *System Utilization* option)**

2.2 Functions

HP-Z0000YPR-101005-02-005	
System Type:	Avanto
	Espre
	TrioTim
	SymphonyTim
	ClinScan
n.a.	

2.2.1 List of System Options

HP-Z0000YPR-101005-02-006

System Type:Avanto
 Espree
 TrioTim
 SymphonyTim
 ClinScan

First, specify the options you want to configure by selecting the corresponding checkboxes. Only the selected options will be available in the subsequent configuration pages.



Fig. 8: System Options

The following settings are available after the system option is selected:

Tab. 1 List of System Options

Enabling Help Pages

Depending on modality type information will be displayed between the configuration masks.

System Type

Select the appropriate syngo MR system type to preset the site info mask with the values for the Material No., Rdiag Prefix and Modality.

Attached to Network

TCP/IP LAN (Local Network) for the local system

Modem

Not supported by syngo MR systems

Main Console with Satellites or Satellite Console

When the local system has a main console and serves satellites (MRSC) consoles.

DICOM Print Devices

DICOM Basic Print service for laser cameras

DICOM Offline Devices

DICOM Media Storage on off-line media such as CD or DVD

DICOM Networking

DICOM Storage and Query/Retrieve to remote network nodes

DICOM HIS/RIS	DICOM Basic Worklist for Radiology (Hospital) Info Systems
Image Import/Export	Import/export of DICOM images from/to the file system
Paper printer	Paper formats for postscript printers
EPRI	Electronic Patient Record Interface
System Management	System Management Agent

2.2.2 Initial Setup

HP-Z0000YPR-101005-02-007

System Type:Avanto
Espreo
TrioTim
SymphonyTim
ClinScan

When you configure the syngo-based system for the first time (usually during the System Test), or when you add a function in the “list of system options“, you will be guided through the configuration procedure. Click on the > button to perform the configuration step-by-step.

2.2.3 Local Host

HP-Z0000YPR-101005-02-008

System Type: Avanto
 Espree
 TrioTim
 SymphonyTim
 ClinScan

The Local Host section provides a set of functions that allow you to enter general site information and to configure the local networking properties of your system.

Site Info

This item allows you to enter general system and site-specific information for easy identification of your system.

Step 1: Site-specific Data

The headline of this menu shows the modality type (CT, MR, ...), the product name, and the software version. The values are set automatically during software installation and by selecting the "System Type" in the previous mask (System Options).

System Options		Reports		Home		Help	
CS syngo syngo VE25A SL07P08		[mm/dd/yyyy]		[hh:mm]			
Software Installation		03/30/2005		12:06			
handed over		01 / 07 / 2003					
System		Customer and Address					
Material No		4523678		Name		tdseso	
Rdiag Prefix		SW		* Id		232	
* Serial No		12345		Hospital		CS SD 1	
Department		Department		Street		Street	
Station Name		TDSESO		Street No		StreetNo	
Service		Remote Service Center		Zip Code		ZIP	
		USA		Phone No		Phone	
Service Center		ServiceCenter		City		Erlangen/A24306/	
Site Identification No		0123456789		District		District	
Service Center Phone		99999		* Country		Germany	
				Administrator		Administrator	

Fig. 9: Site Info menu

Correct entries into the **marked (*) fields** are important for sufficient node registration in case **System Management** is enabled.

- **Software installation**

This date is set automatically during software installation; it cannot be changed.

- **Handover**

Enter the date and time when the system was handed over to the customer. This date has no effect on the software. Like many other data in this menu, it serves only as information for the service engineer.

System

- **Material No.**

This number is filled in automatically after you select the “System Type” in the previous mask (System Options).

- **Rdiag Prefix**

This number is filled in automatically after you select the “System Type” in the previous mask (System Options). It is used to create unique file names for remote transfer (file&image tools or auto reporting).

- **Serial no.**

The serial number is used for the DICOM Unique Identifiers within the image header attributes, as well as for the file name and images. Therefore, this field is mandatory. The contents of the serial number are checked according to the selected “System Type” in the previous mask (System Options).

NOTE

The serial number is mandatory. It may not include any characters other than numbers from 0 to 9.

- **Department**

Enter the department where the system is located

- **Station Name**

The Station Name is entered into the DICOM header of each image (non-mandatory attributes 0008, 1010). It is used by certain products (Agfa Impax) for routing purposes. You can enter up to 16 characters whereby ‘\’ should not be used. As a default, enter the Computer Name of your product.

NOTE

If the Computer Name needs to be changed, it has to be clarified whether the Station Name has to be changed too.

Service

The Site Identification No. and Service Center Phone are displayed in the customer user interface under “Options - Maintenance”. The customer sees the Service phone number and can identify his product by the Site Identification number.

- **Remote Service Center**

Select the location of the Siemens Remote Service Server, to which the system shall be connected for remote service.

By this selection the **SRS routing** (SRS IP-address), **FTP** target and **Mail** target are pre-configured according to the modality depending pre-setting within the “preconfig_remotehost.xml” file.

NOTE

Whenever the “Site Info” is saved the FTP and Mail target with the pre-configured default logical name are set back to default values.

Worldwide exist three SRS Server for three different time zones:

Europe (used for Europe, Africa and Middle East)

USA (used for North- and South America)

Asia (used for Asia and Australia)

- **Service Center**

Enter the location of the Service Center responsible for this system in this box.

- **Site identification no.**

The syntax of the ID is different in each country, and can be found in your Service Management System. The ID is used to define the local system in the Service Center.

- **Service center phone**

Enter the phone number of the appropriate Service Center.

Customer and Address

- **Name**

Enter the name of the customer. In certain cases, it may be the same as the hospital name (see the adjacent box).

- **Id**

Enter the identification number of the customer. The syntax of the ID is different in each country, and can be found in your Service Management System.

- **Hospital, Street, Street No., ZIP Code, Phone No., City, State/Province, Country**

Enter the appropriate data into these boxes. This information may be helpful during a remote service session and is shown in the report files (not included in this version).

- **Country**

Select the country of installation from the drop down list.

- **Administrator**

Enter the name of the customer’s administrator who is responsible for this system in this box. This information is helpful during remote and local service sessions.

Step 2: Time Zone Settings

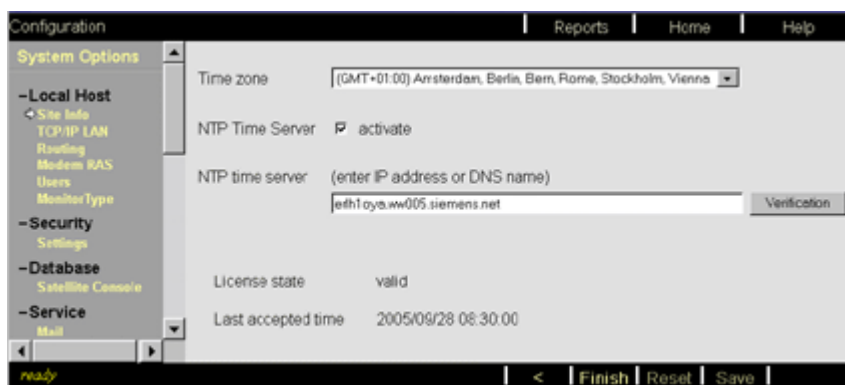


Fig. 10: Site Info menu 2 - time zone

- **NTP Time Server** (Network Time Protocol)

If you select this box, you may enter the IP address of the NTP server for the customer's site. In this case, you will not be able to enter the date and time manually.

With the **Verification** button, beside the server address input field, the connection to the entered server address can be tested

NOTE

A check is performed to determine if the time provided by the NTP server is within the date and time range of the application licenses. If not, the NTP Server will not be accepted.

- If no NTP server (Network Time Protocol) is available, manually enter the **Time zone, date and time**. In addition, **Daylight Time** should be set to automatic.

Console Type

This item allows you to select Console Type of your syngo MR system (MRC or MRSC).

Console Options

Define if the Main Console has an satellite console connected.

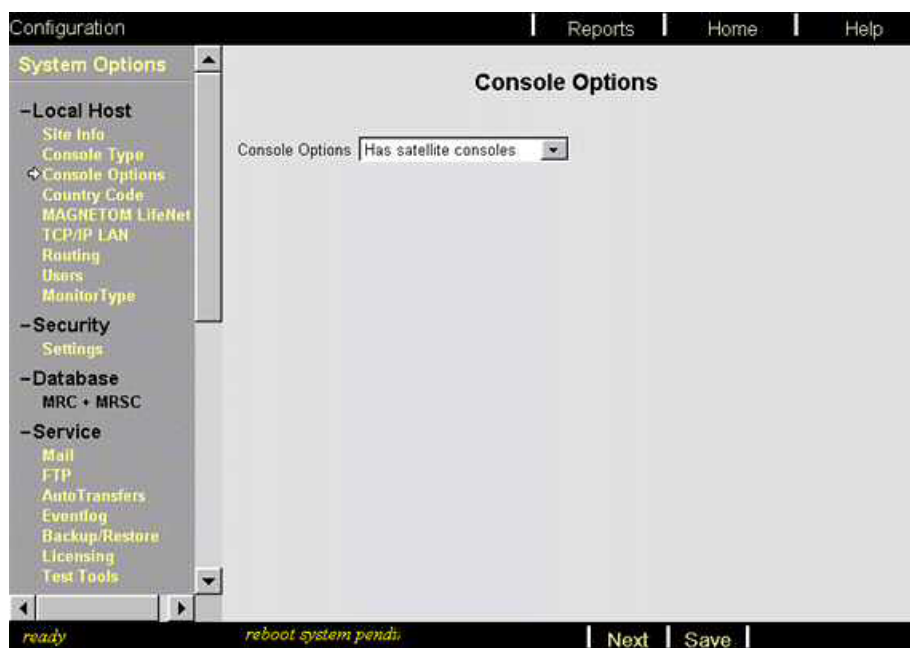


Fig. 11: Console Options

Country Code

Select Country where the system is located.

MAGNETOM LifeNet

LifeNet is an integrated customer care solution - LifeNet provides following functionalities:

- Newsticker - Cutting-edge news around the MAGNETOM system
- Automatic download of Phoenix images under the patient name Phoenix

Set the MAGNETOM **LifeNet** option to **Enabled**

NOTE

Only Disable the **LifeNet** option on customer wish or if the system has no remote connectivity.

TCP/IP LAN

The TCP/IP LAN menu allows you to configure the local networking properties of your system. This menu will not be available if “attached to network” is not selected in the “List of System Options menu“. The default TCP/IP address 192.168.0.0 is set automatically.

NOTE

This system supports only the TCP/IP protocol.

Prerequisites

When you configure the TCP/IP LAN properties, you need to decide what sort of TCP/IP LAN support your system requires. **It is the responsibility of the Project Manager to obtain the necessary data from the hospital's network administrator or from the Siemens Uptime Service Center.**

NOTE

If no IP address is provided by the customer, you need to request a unique IP address from your Uptime Service Center. They can create unique IP addresses by using the "IP Reservation Tool" in the TD Intranet (Product Information - Applications and Utilities). In this case, no DHCP, DNS, or WINS needs to be configured.

NOTE

It is recommended not to configure WINS or DNS for syngo products. (Netstat -a returns putpunt ~1h when WINS and DNS are configured)

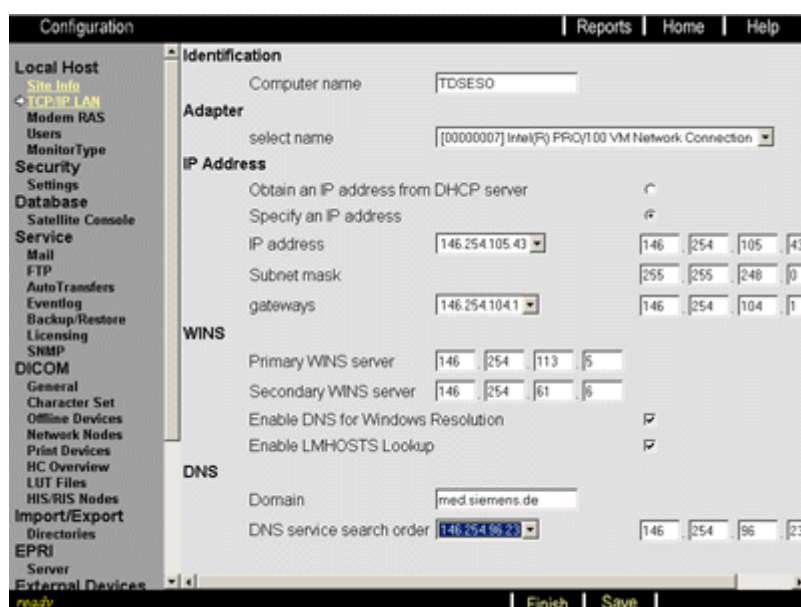


Fig. 12: TCP/IP LAN menu

Identification

- **Computer Name**

The computer name (host name, node name) identifies this system on the network. It is used internally (hosts file) or by the Domain name server to map your system to your corresponding Internet address.

NOTE**Restrictions for the computer name:**

The computer name (= host name) may only be entered in capital letters. It may include up to 14 characters. The characters “_” and “-” should be avoided, as they could cause problems (even though it is possible to enter them).

The computer name may not consist of only numeric characters or start with a numeric character.

NOTE

When a Domain name server is used, the computer name should be the same as the name entered at the DNS.

NOTE**Changing the computer name:**

- When changing this entry, remember that the computer name is used internally by the Versant database functions. Therefore, it is recommended to save the images before changing the computer name.

- The local AET (Application Entity Title) is generated automatically from the computer name. Changing the computer name generates a pop-up window, allowing you to select whether the AET should be changed as well. As soon as DICOM connections are configured, the original AET should be kept.

- Changing the local computer name (host name) requires a reinstallation of the system management agents (if they have already been installed).

- If a DNS (Domain Name Server) is used, make sure that the DNS is updated.

NOTE

If the Computer Name needs to be changed, it has to be clarified whether the Station Name has to be changed too.

Adapter

- **Select Name**

This drop-down box lists all available network adapters in your system.

NOTE

The settings entered in the menu below refer to the adapter selected.

IP Address

- **Obtain an IP address from DHCP server**

DHCP is an acronym for Dynamic Host Configuration Protocol, which offers dynamic configuration of IP addresses and related information. DHCP provides safe, reliable, and simple TCP/IP network configuration, prevents address conflicts, and helps conserve the use of IP addresses through centralized management of address allocation. Consequently, an IP address does not have to be specified.

Ask the network administrator or the Project Manager whether DHCP is available, and if so, whether it should be used.

NOTE

IP Address configuration via DHCP has not been definitively tested with this software version!!

NOTE

For DICOM transfer (HIS/RIS network), it is necessary to enter an IP address for each connected system. Therefore the IP addresses of these systems should not be changed by the DHCP server.

DHCP provides the possibility of assigning a specific IP address to a specific system. Make sure that the DHCP server always uses the same IP address for each connected system and also for the system itself.

- **Specify an IP address**

Select this option if DHCP is not available in your LAN or is not used for this system.

- **IP Address**

Enter a unique four-digit dotted code for the selected adapter to identify your system on the LAN. (IP address, Subnet mask, and Gateway completely identify the logical location of your system). Each adapter receives its own unique address. Duplicate IP addresses on a network can cause all sorts of problems, including lockups.

NOTE

Changing the local IP address requires a reinstallation of the system management agents (if they have already been installed).

NOTE

If no IP address is provided by the customer, you need to request a unique IP address from your Uptime Service Center. They can create unique IP addresses by using the "IP Reservation Tool" in the TD Intranet (Product Information - Applications and Utilities). In this case, no DHCP, DNS, or WINS needs to be configured.

NOTE

Each network adapter may have up to five IP addresses assigned to it. Multiple addresses are commonly used in networking environments with multiple domains and subnets sharing the same wire.

- **Subnet Mask**

This number is used to mathematically “mask” IP addresses on the network. It is absolutely essential that this entry is correct. If it is entered incorrectly, you will not be able to “see” the other nodes on the network. All systems using the same addressing schemes have to use the same subnet mask in order to see each other.

- **Gateway**

This entry indicates where TCP/IP traffic should be sent when its address indicates that it needs to be forwarded to another logical network. You need to provide a gateway address for every network adapter in your system. This address points to a router - either another computer or a piece of dedicated hardware - responsible for forwarding packets to hosts that are not on the currently selected adapter’s subnet.

NOTE

You may enter other gateway addresses in addition to the default address. You may need to do this if your system is connected to multiple network segments.

NOTE

If your system is connected via an ISDN router to your headquarters Siemens Remote Server, enter the TCP/IP address of that ISDN router here as well.

TCP/IP Naming Services

TCP/IP requires a mapping table between the IP address and the host name (computer name, node name). This mapping can be done locally on the system or on centralized name servers. The local mapping table is the hosts file or, for WINS, the lmhosts file. Centralized name servers are DNS or WINS.

If both name servers are available and configured, the local system looks for the IP address of a given computer name in the following order:

- Netbios cache on local system (see command nbtstat -c)
- WINS (Windows Internet Naming Sever)
- DNS (Domain Name Server)
- lmhosts (local hosts file administered by the WINS)
- hosts (filled in by the Siemens configuration tool)
- DNS (even if DNS is not configured, the system will look for a DNS if the mapping tables listed above are not able to find an IP address for the computer name)

WINS (Windows Internet Naming Service)

NOTE

Do not configure a WINS server if it is not available. Otherwise, you may run into problems during Windows XP startup.

- **Primary WINS Server**

This entry applies if you use TCP/IP to connect to other Microsoft networking systems. To use the TCP/IP transport protocol with NT’s built-in networking, NetBios machine names (computer name) have to be mapped to the corresponding IP addresses for

each machine. This mapping can be managed through the use of a text database called LMHOSTS, with DNS, or through Windows Internet Naming Service Server (WINS), or a combination of all three.

WINS is the easiest method, and the only method that supports dynamic IP address assignment through DHCP.

You can specify a primary and secondary WINS Server address to be used by each adapter.

- **Secondary WINS Server**

Refer to the section above.

- **Enable DNS for Windows Resolution**

Windows NT 4.0 allows you to convert NetBIOS names to IP addresses using a text database called lmhosts, with DNS, with WINS servers, or a combination of the three.

If you want to use DNS for name-to-address mapping, enable this checkbox. Unlike for WINS, it is necessary for the network administrator to enter the node name and the appropriate IP address into the Domain Name Server.

- **Enable LMHOSTS Lookup**

If you want to use DNS for name-to-address mapping, enable this checkbox. The LMHOSTS file is administered by the WINS.

DNS (Domain Name Server)

NOTE

Do not configure an DNS if it is not available. Otherwise, you may experience problems during startup of Windows XP.

- **Domain**

If your network supports DNS addressing for name resolution, type in the domain name here. Please note: this is not necessarily the same as your Windows NT Domain name.

- **DNS Service Search Order**

Enter the IP address of the available domain name servers. To enter more than one DNS, select the add button. The local system will invoke the DNS in the order entered in this field.

Modem RAS

NOTE

***syngo MR* systems don't support RAS (modem) connections.**

Routing

The connection to SRS (Siemens Remote Server) requires the configuration of the route to the **SRS gateway** (SRS Router).

The SRS gateway can either be configured on the system via a **static route** or as **default gateway**.

The configuration of the SRS gateway as **static route** is recommended if the system and the SRS gateway (router) belonging to the same network (subnet).

Is the syngo system in another subnet located as the SRS gateway, then it's recommended to configure it as **default gateway** under section TCP/IP LAN.

Therefore the hospital network administrator has to be involved to configure the default gateway so, that the SRS gateway (SRS Router) can be reached by the system.

Routing Configuration



Fig. 13: Static Route Configuration Page

The **Configure Gateway** page provides the possibility to configure the **static route** to the SRS.

Next gateway to SRS:

enter the IP address of the SRS gateway (SRS Router) to create a static route. Skip this configuration if the SRS gateway is already configured as default gateway in section TCP/IP LAN.

The **Test** button verifies the connection to the gateway by a ping command.

Remote Service Center:

It shows the target SRS to whom the system belongs to according to the timezone of the system location. It is selected and configured in section **Site Info**.

The **Test Route** button verifies the connection to the SRS Access Server by a ping command.

NOTE

If the ping fails, it may not be bound to a problem in the network itself. It may also be caused by an automatic suppression of the ping, because of security reason.

Please contact the network administrator in this type of case, if the connection to the SRS cannot be established successfully.

Control RIP Listener

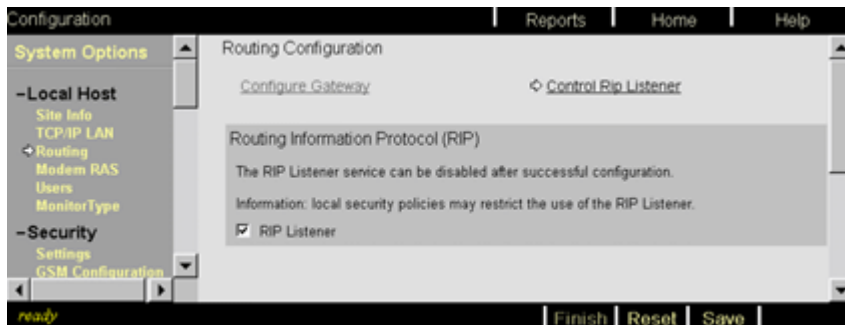


Fig. 14: Rip Listener Control Window

The **RIP Listener Control** window is accessible by the link within the **Routing Configuration** window. (RIP = Routing Information Protocol)

The **RIP Listener** is an implemented tool to listen on router information within the network. If the RIP is enabled on the SRS gateway (router) the Listener is able to collect the routing information and updates the system routing table by itself.

If **RIP Listener** is selected it listens to RIP information on the network only for a period of approx. 15min after the boot up of the system and gets than automatically inactive.

If not selected the RIP Listener is deactivated all the time.

Users

The “Users” menu lets you set the account name and password for the local system application and the password for the local PPP account.

NOTE

When modifying user properties, use only the “Users” mask provided by the Service Tools and the User Management dialogs offered by syngo. If you use the tools provided by the Windows operating system, the integrity and operation of the system cannot be ensured.

In addition, with **Autologin** you may select whether the application should start automatically during your system’s boot procedure. If **Autologin** is enabled (checked) not Windows-Login UI will appear.

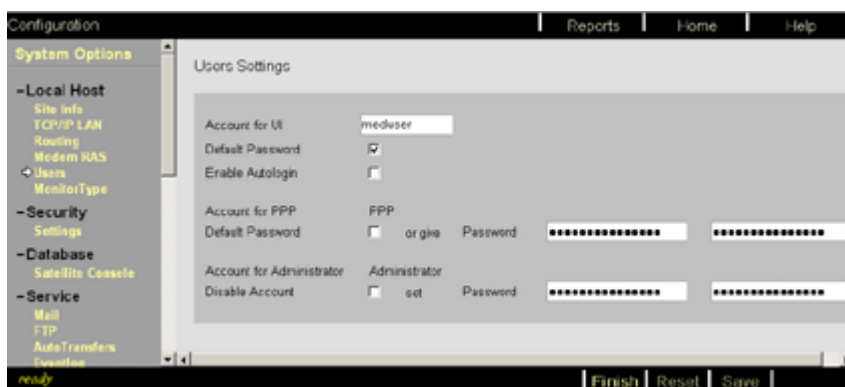


Fig. 15: Users menu

- **Account for UI**

The default account is set to “meduser“. Enter this account and the appropriate password to start the application. If Autologin is enabled, the request for this account will be skipped during startup.

- **Default Password**

When selecting **Default Password**, a password is created according to certain rules known to the service engineer. If you do not select this item, you need to enter a password and confirm it.

NOTE

Please be aware that if there is a main or satellite console, you should never select the Default Password, since this password is unique for each system (for this reason, it cannot be the same for the main and satellite console)!

- **Enabling Auto Login**

With **Autologin**, you can set whether the application should start automatically during your system’s boot procedure. If you need to enter the system as “Administrator“, you have to disable **Autologin** and restart the system. In this case, you will be asked for an account.

Account for PPP

The PPP (Point-to-Point Protocol) account is necessary for remote login to the system via RAS (modem line). PPP is not used in the case of a router connection (Ethernet line).

- **Default Password**

When selecting **Default Password**, a password is created according to certain rules known to the service engineer. If you do not select this item, you need to enter a password and confirm it.

Account for Administrator

According to the security regulations, it is not allowed to use a predefined administrator password. It has to be set for each system during startup (you will be forced to do this during the initial setup).

- **Random Password**

If **Random Password** is selected, a password will be created by the system, which is not visible and therefore not known. As a result, it will not be possible to login to the system again as user “administrator“, which may be necessary for security purposes.

If you need to login as administrator (e.g. for security and user settings or to install drivers) you need to enter a password and confirm it, instead of using the random password.

Monitor Type

To set the correct gray scale values or color for the display, the application needs to know the monitor type.

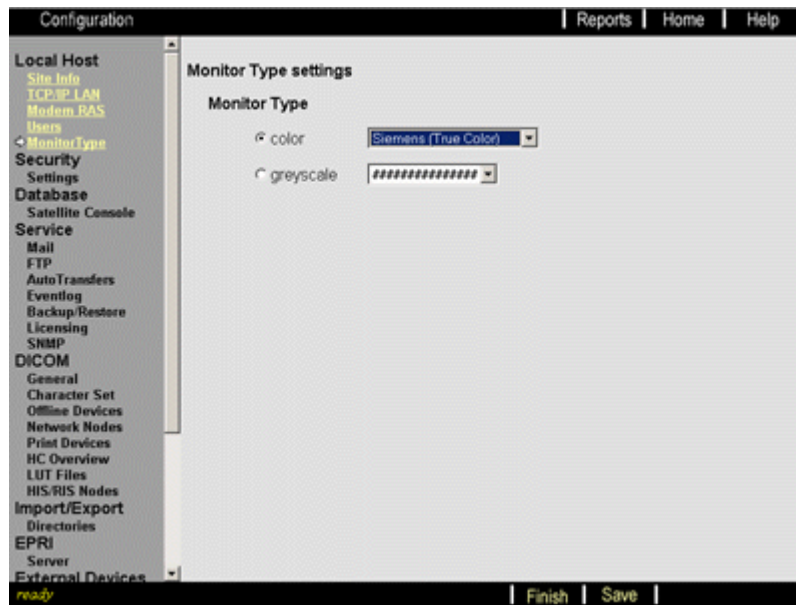


Fig. 16: Monitor Type menu

Four different settings are possible for Simomed gray scale monitors (Simomed I, II, III, 32). The corresponding setting for a certain device is preset at the factory.

2.2.4 Security

```
----- HP-Z0000YPR-101005-02-009 -----
System Type: . . . . . Avanto
                  Espree
                  TrioTim
                  SymphonyTim
                  ClinScan
```

Within the security configuration certain security level and functionalities, like e.g. User Management and Remote Date Encryption, can be configured.

Settings



Fig. 17: Safety Settings

Security Properties

NOTE

Setting the security properties is possible only one time!!! Deactivation is not possible due to security reasons (e.g., it is not possible to deactivate audit trailing after it has already been set).

- User Management**
User management is available from the Microsoft Windows platform. For more details, please refer to the system help pages, as this is not part of the service software.

NOTICE

Do only use the User Mangement dialogs offered by syngo and not those offered by the Windows operating system.

⇒ **Otherwise integrity and operating of the system can not be guaranteed.**

- **Data Access Check**

Adjustment of data access checks is available from the operating system (snap-in is started when you login under user “Administrator”). For more details, please refer to the system help pages, as this is not part of the service software.

- **Functional checks**

Adjustment of functional checks is available from the operating system (snap-in will be started when you login under user “Administrator”). For more details, please refer to the system help pages, as this is not part of the service software.

- **Auditing**

An audit trail according to the HIPAA regulations will be activated. For more details, please refer to the system help pages, as this is not part of the service software.

DICOM Access

- **Enable trusted host functionality**

By default, syngo permits only TCP/IP access from known hosts. This is the case as soon as the host name and IP address is configured, e.g. as network node or HIS/RIS node. Enable trusted hosts functionality checks in addition to the AET of DICOM nodes.

NOTE

Entering the AET of the sender in the configuration as a Storage AET causes this node to be entered in the list of the transfer destinations. Even if the sender cannot receive images (no Storage Provider), it will be shown in the User Interface as a transfer destination. To prevent this, you may enter the AET of the sender as “Storage Commitment” AET (workaround until a better solution is provided by syngo). In this case, the sender AET is accepted as a trusted host but does not appear in the transfer list.

Remote access

- **SSL Encryption**

The HTTP protocol will be used to transfer the service user interface from the web server (syngo) to the web browser (Internet Explorer). Selecting this button enables the secure HTTPS protocol instead of HTTP. HTTPS uses the SSL (Secure Socket Layer) protocol for authentication and data encryption.

SSL Encryption should be the default setting. Please be aware that HTTPS provides lower performance than HTTP.

NOTE

Selecting the SSL (HTTPS) protocol requires that the client (Siemens Remote Server) is set to HTTPS as well! In this case, syngo uses only HTTPS and not HTTP.

2.2.5 Database

HP-Z0000YPR-101005-02-010

This screen is only available for systems where the “**Local Host**” **Console Options** are configured for “**configured to Has satellite consoles**”.

Main Console (MRC + MRSC)

The screenshot shows the 'Configuration' window with a sidebar on the left containing a tree view of system options. The 'Database' section is expanded, showing 'MRC + MRSC'. The main panel is titled 'Satellite Consoles' and contains a form with the following fields and controls:

- 'Satellite Consoles': A pull-down menu currently showing 'define new'.
- 'Host Name': A text input field containing 'MRSC'.
- 'TCP/IP address': A text input field containing '172.16.6.52'.
- Buttons: 'Add', 'Del', 'Find', and 'Test'.

At the bottom of the window, a status bar displays 'ready', 'reboot system pending', and navigation buttons 'Finish', 'Reset', and 'Save'.

Fig. 18: Database MRC + MRSC

Satellite Consoles

Select the satellite consoles from the pull-down menu or enter a new one with “define new”. In this case, you need to enter the host name and TCP/IP address of a connected satellite console. Click on the **Add** button to save the data for the entered satellite console (host name and TCP/IP address). More than one satellite console may be entered and saved each time with the **Add** button.

If the selection box is set to a certain satellite console, you may remove this console by clicking the **Del** button.

- **Host Name**

Enter the host name of the satellite console.

NOTE**Restrictions for the host name:**

The host name may contain up to 15 characters. The characters “_” and “-” should be avoided, since they could cause problems (even though it is possible to enter them). The computer name may not consist of just numeric characters or start with a numeric character.

- **Find**

Select the **Find** button to check whether the host name is known at the WINS or Domain Name Server (DNS). If the name is found, the IP address is entered in the TCP/IP address box.

- **TCP/IP Address**

If a central name server (DNS, WINS) is not used, you need to enter the IP address of the MRSC manually.

- **Test**

Click on the **Test** button to send a PING command to the corresponding IP address.

A TCP/IP ping may fail for the following reasons:

- The remote host is turned off or not in the same network (or subnet), or the gateway is not configured.
- The TCP/IP address is configured incorrectly.
- Any networking device (router, switch, bridge,...) denies access to the other network.

Satellite Consoles (MRSC)

The screenshot shows the 'Configuration' window for 'Database MRSC'. The left sidebar lists various configuration categories. The main area contains several sections: 'Database Type' with radio buttons for 'Local Database' and 'Remote Database'; 'Database Properties' with a 'Database Offset of local DBs' field set to '105'; 'Main Console' with 'Host Name' (MRC7) and 'TCP/IP address' (172.16.6.51) fields, and 'Find' and 'Test' buttons; and 'User' with 'Account' (meduser) and 'Password' fields. A note at the bottom of the User section states: 'You must use identical account and password on satellite and main console.' At the bottom of the window are 'Finish', 'Reset', and 'Save' buttons.

Fig. 19: Database MRSC

Database Type

- **Local Database**

Only to be configured for standalone satellite consoles.

- **Remote Database**

If this radio button is selected, your system will be configured as a satellite console (MRSC) and will use the database of the main console (MRC).

Database Properties

- **Offset Database of local DBs [100...65500]**

Set the ID of your local Versant database. Each system in a network uses a unique offset database identifier value, e.g., 105 for the 1st satellite console and 205 for the second one.

If your system does not access other system's databases and other systems do not access the databases in your system, the database identifier offset value is irrelevant.

NOTE

Changing the database identifier offset value will subsequently delete all your databases on this system. During the next reboot, new databases will then be created.

Main Console

If the selection box is set to "define new", you may enter the host name and TCP/IP address of a connected main console. Click on the **Add** button to save the data for the main console entered (host name and TCP/IP address).

If the selection box is set to a specific main console, you may remove this console by clicking the **Del** button.

Main Console

Set the host name and IP address of the main console.

- **Host Name**

Enter the host name of the main console.

- **Find**

Select the **Find** button to check whether the host name is known in the WINS or Domain Name Server (DNS). If the name is found, the IP address is entered in the TCP/IP address box.

- **TCP/IP address**

If no central name server (DNS, WINS) is used, you need to enter the IP address manually.

- **Test**

Click on the **Test** button to send a PING command to the corresponding IP address.

A TCP/IP ping may fail for the following reasons:

- The remote host is turned off or not in the same network (or subnet), or the gateway is not configured.
- The TCP/IP address is configured incorrectly.
- Any networking device (router, switch, bridge,...) denies access to the other network.

User

To ensure that the account and password of the satellite console and the main console are identical, enter these values again in the appropriate boxes.

The values should be the same as the ones already entered for the UI user in the “Users” mask under Local Host.

- **Account**

Enter the UI user name as it is defined in the “Users” mask of the main console (usually, it is “meduser”).

- **Password**

Enter the password and confirm it as it is defined in the “Users” mask of the main console for the UI user (do not use the Default Password at Local Host - Users).

2.2.6 Service

HP-Z0000YPR-101005-02-011

The Service section provides a set of options that allow you to configure Remote Diagnostics functions (Auto Report, File&Image Tools) and Service Software functions (Event Log, Backup&Restore).

Mail

This menu item allows you to configure the **mail target for the Auto Report function and for the Service center provided in the Customer User Interface** at Option - Remote Service.

A target is defined by a mail server and a mail account on that server and may be reachable via LAN.

The procedure is performed in two steps. First you have to configure the host name and IP address of the mail targets. (RAS is not supported by **syngo MR** systems) Enter the target e-mail address. For the Siemens Remote Server (SRS), you need to enter an e-mail address known at the mail server connected to the Siemens Remote Server.

The host name and IP addresses for the SRS routers (LAN):

Tab. 2 SRS router addresses

Country	Rdiag server name	IP of Access Server
Europe, Africa	LUX09505	194.138.39.18
North / South America	SRSACC1	129.73.116.92
Asia	sgpt806x	194.138.243.178

Step 1: TCP/IP Settings for Mail Targets

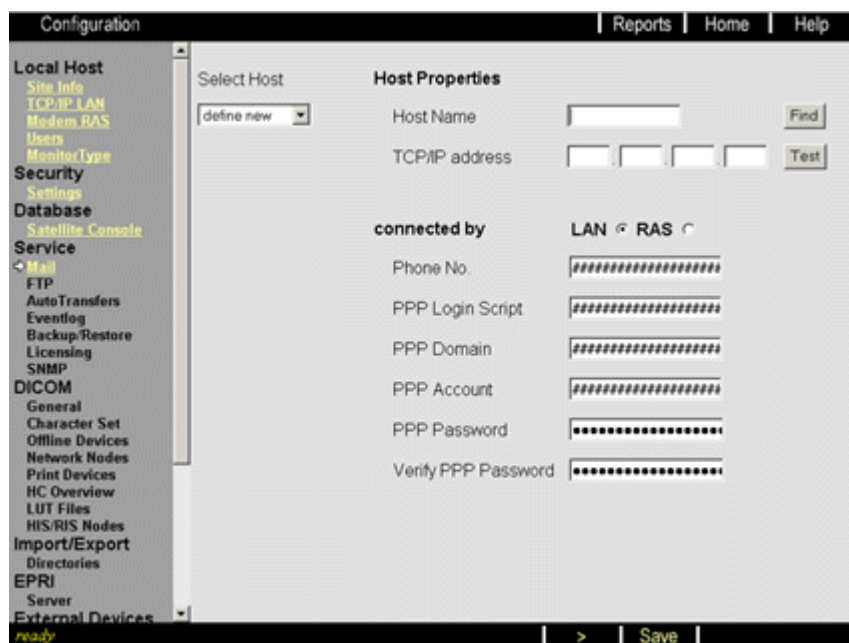


Fig. 20: Mail - step 1

Select Host

Click on the **Select Host** drop-down menu to see whether the mail targets are already listed.

If the mail target is not listed, you need to enter the host name and the IP address.

Host Properties

- **Host Name**
Enter the host name of the mail target.

NOTE

Restrictions for the Host Name:

The host name may contain up to 15 characters. The characters “_” and “-” should be avoided, since they could cause problems (even though it is possible to enter them). The computer name may not consist of just numeric characters or start with a numeric character.

- **Find**
Select the **Find** button to check whether the host name is known at the WINS or the Domain Name Server (DNS). If the name is found, the IP address is entered in the TCP/IP address box.
- **TCP/IP Address**
If no DNS is used, you need to manually enter the IP address into these boxes.
RAS is not supported by **syngo MR** systems.

- **Test**

Click on the **Test** button to send a PING command to the corresponding IP address.

A TCP/IP ping may fail for the following reasons:

- The remote host is turned off or does not answer on ping or not in the same network (or sub net), or the gateway is not configured.
- The route to the target TCP/IP address is not configured.
- An network device might deny access to the other network.

Connected by LAN / RAS

If the mail target is connected via a router, you need to select the **LAN** button. (**RAS** is not supported by **syngo MR** systems)

For syngo MR systems it's recommended to use one of the following e-mail addresses:

Tab. 3 SRS mail addresses

Country	e-mail address
Europe, Africa, and Asia	MRS MC.med@siemens.com
North / South America	MR UPTIME@sms.siemens.com

Step 2: Settings for Mail Target

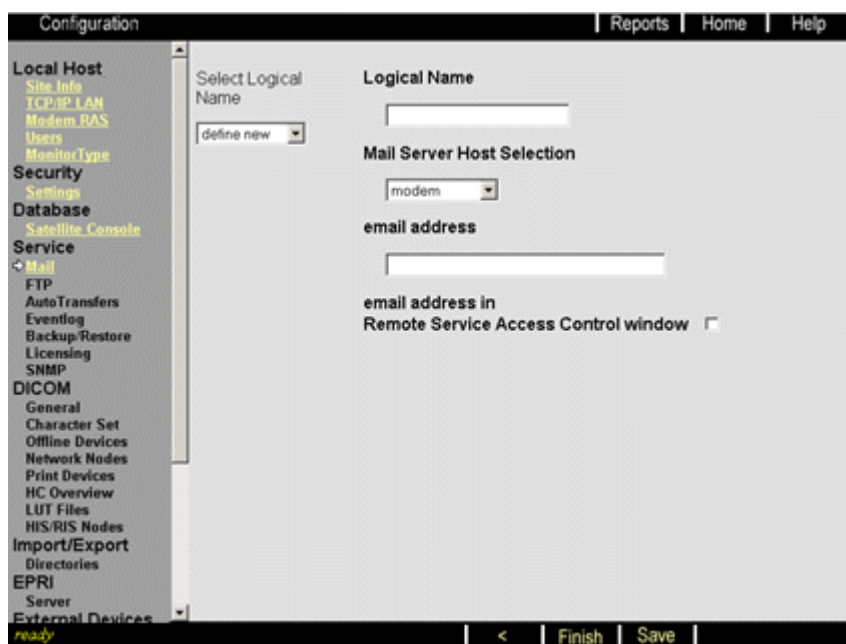


Fig. 21: Mail - step 2

- **Selecting the Logical Name**

Click on the “Select Logical Name” drop-down menu to see whether the mail target is already listed. Otherwise, you need to enter it in the “Logical Name” menu item.

- **Logical Name**

This name is used by the Auto Report tool for target selection. For **syngo MR** systems it's recommended to use the logical name: **SRS_mail** ↵

- **Mail Server Host Selection**

Select the corresponding host name for the system used as the mail target that was defined in the previous menu.

- **E-mail Address**

Enter the mail address of the mail server to which you want the auto report (for example) to be sent. For the Siemens Remote Server, you should enter an account name at the server that defines where the auto reports should be sent. For syngo MR system it's recommended to use one of the following addresses:

Tab. 4 SRS mail addresses

Country	e-mail address
Europe, Africa, and Asia	<i>MRSMC.med@siemens.com</i>
North / South America	<i>MRUPTIME@sms.siemens.com</i>

If you want to send the auto report (for example) to a particular mail address at headquarters, you can enter any mail address e.g. (name@siemens.com) instead of the account name.

- **E-mail Address in the Remote Service Access Control Window**

The User Interface under "Option - Remote Service Access Control" allows the customer to send an e-mail to (for example) the local Service department. If the mail target that was just configured is to be this customer-related mail target, you need to select this box.

FTP

This menu item allows you to **configure the file transfer target (FTP) for the Auto Report and for the File&Image Tool function.**

The procedure is performed in two steps. First you have to configure the host name and IP address of the FTP target(s). Then you have to enter the FTP account and password.

Step 1: Settings for FTP Targets.

Fig. 22: FTP menu - step 1

Select Host

Click on the “Select Host” drop-down menu to see whether the FTP targets are already listed. If so, and if the related data are correct (including RAS settings), you are done and may continue with the next menu.

If the FTP target is not listed, you need to enter the host name and the IP address.

Host Properties

- **Host Name**

Enter the host name of the FTP target.

NOTE**Restrictions for the Host Name:**

The host name may contain up to 15 characters. The characters “_” and “-” should be avoided, since they could cause problems (even though it is possible to enter them). The computer name may not consist of just numeric characters or start with a numeric character.

- **Find**

Select the **Find** button to check whether the host name is known in the WINS or Domain Name Server (DNS). If the name is found, the IP address is entered in the TCP/IP address box.

- **TCP/IP Address**

If no DNS is used, you need to manually enter the IP address into these boxes. RAS requires two IP addresses - one for the local system and the second for the remote system. For RAS, you need to enter the second IP address here.

- **Test**

Click on the **Test** button to send a PING command to the corresponding IP address.
RAS is not supported by **syngo MR** systems.

A TCP/IP ping may fail for the following reasons:

- The remote host is turned off or does not answer on ping or not in the same network (or sub net), or the gateway is not configured.
- The route to the target TCP/IP address is not configured.
- An network device might deny access to the other network.

Connected by LAN / RAS

If the mail target is connected via a router, you need to select the **LAN** button.

(RAS is not supported by **syngo MR** systems)

Step 2: Settings for FTP Targets

Fig. 23: FTP menu - step 2

- **Selecting the Logical Name**

Click on the “Select Logical Name” drop-down menu to see whether the FTP target is already listed. Otherwise, you need to enter it in the “Logical Name” menu item.

- **Logical Name**

This name is used by the Auto Report and File&Image Transfer tool for target selection. For **syngo MR** systems it's recommended to use the logical name: **SRS_ftp** ↵

- **FTP Host Selection**

Select the corresponding host name for the system used as the FTP target that was defined in the previous menu.

- **FTP User**

Enter the FTP account on the transfer target server to which you want the images file (for example) to be sent, e.g. the default SRS ftp account: **rdiagftp** (note case-sensitive).

- **Password**

Enter the password for the FTP user account on the transfer target server and confirm it in the next box. Ask your SRS responsible for the **rdiagftp** password.

Auto Transfer

If the customer agrees, automatic transfer (i.e., without user interaction) of auto reporting messages or System Management Event messages may be enabled here. Deactivate these options if the customer does not want automatic transfer of auto reporting messages, e.g. to the HSC.

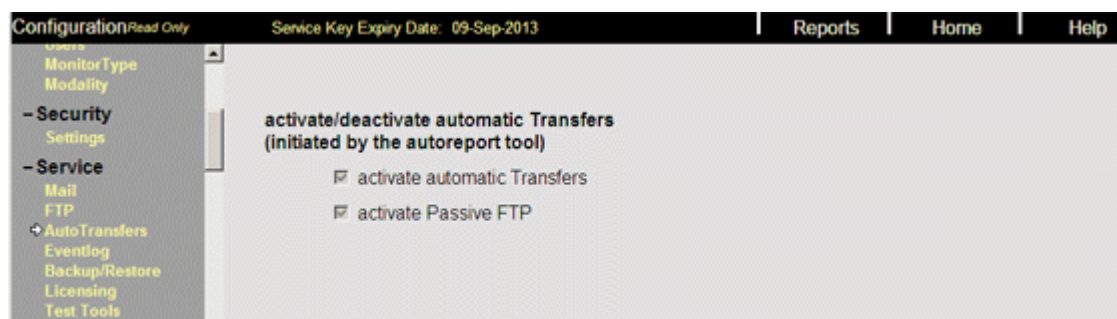


Fig. 24: Configuration - Service - AutoTransfers

Event log

The event log function allows you to customize the Event Log Viewer of the service software

The syngo Event Log Viewer provides a message filter for the three facilities Service, Developer and Customer (see Online Help Event log).



Fig. 25: Event Log menu

- **Log only Service Messages**

Only events which are classified for the facility Service (including Customer events) will be written into the event log file. In addition, all events with the sequence number 0.

- **Display Service messages**

Only service-relevant messages will be displayed in the event log window (while all errors are being logged). Events with sequence number 0 will not be displayed if they are classified as an engineering event.

- **Display Sequence Number**

It can be selected if the even Sequence Number shall be displayed in the Event Log Viewer. An events can be the cause of additional events. Events with the sequence number 0 are the first events logged for a certain problem.

Backup/Restore

Backup/Restore allows you to specify the data to be saved and restored when executing the Backup & Restore function of your service software. The time and destination for backup and restore is not set in the configuration platform, but rather at Backup/Restore.

Specify to which packages, groups, directories and files the Backup&Restore procedure should be applied, and which batch procedure is to be executed when performing the corresponding action.

The Restore procedure permits the selection of all packages and groups or of only particular groups or packages.

NOTE

Auto Backup is not supported for this software version.

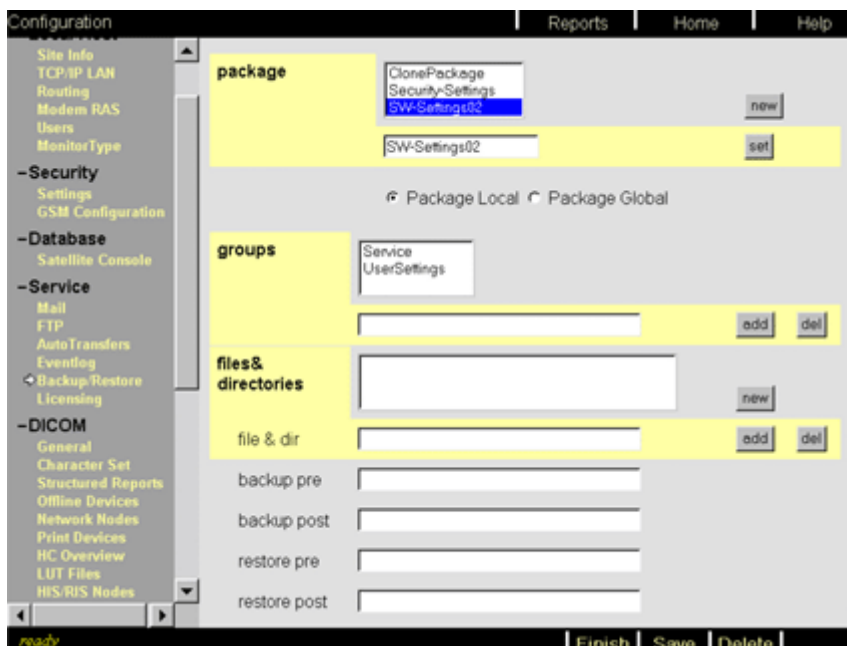


Fig. 26: Backup / Restore Local Menu - Package Local

Package

Define a new backup package by entering a new name in the text box. It is recommended that the existing packages should not be modified.

Local Package

Define the properties according to the file selected above.

- **Groups**

For the package selected above, groups can be defined as before by entering a new name in the text box and choosing **add** or selecting an existing group from the list box.

- **Files&Directories**

The file(s) and directory(ies) have to be entered for the selected group in the selected package:

- **file & dir**

Add a new file or directory to be backed up by entering the name in this text box and clicking the **Add** button.

Delete a file or directory by selecting it from the list box and then clicking the **Del** button.

- **backup pre**

Enter (add) or delete (del) a procedure command to be performed before backing up the particular file.

- **backup post**

Enter (add) or delete (del) a procedure command to be performed after backing up the particular file.

- **restore pre**

Enter (add) or delete (del) a procedure command to be performed before restoring the particular file.

- **restore post**

Enter (add) or delete (del) a procedure command to be performed after restoring the particular file.

Global Package

If you select this radio button, a different menu appears that allows you to set properties relevant to the package selected above.

Fig. 27: Backup / Restore Local Menu - Package Global

- **Command Before Backup Package**
Enter a procedure command to be performed prior to the backup.
- **Alert Before Backup Package**
Enter a text that will pop up in a window prior to backup.
- **Command After Backup Package**
Enter a procedure command to be performed after the backup.
- **Alert After Backup Package**
Enter a text that will pop up in a window after backup.
- **Command Before Restore Package**
Enter a procedure command to be performed prior to the restore.
- **Alert Before Restore Package**
Enter a text that will pop up in a window prior to restore.
- **Command After Restore Package**
Enter a procedure command to be performed after the restore.
- **Alert After Restore Package**
Enter a text that will pop up in a window after restore.
- **Count Archives**
Enter the number of backups to be saved. If 3 is entered, the last 3 backups will be stored and the fourth will be deleted.

Licensing

Several syngo applications are protected with an application license key. The licensing function in the Service Software allows you to display and modify application licenses.

It provides also the mechanism to add a new application license to the system. Import of new license files is possible locally and remotely.

NOTE

It is only possible to add a license file with all required licenses. It is not possible to edit or add a single license entry.

The Service license key is entered on the start page of the service function and not in this menu.

Application and Service license keys can be obtained from your UPTIME Service Center.

The license user interface is divided into 4 parts :

- General Settings
- Select New License File
- License Details
- License Server Settings

General Settings

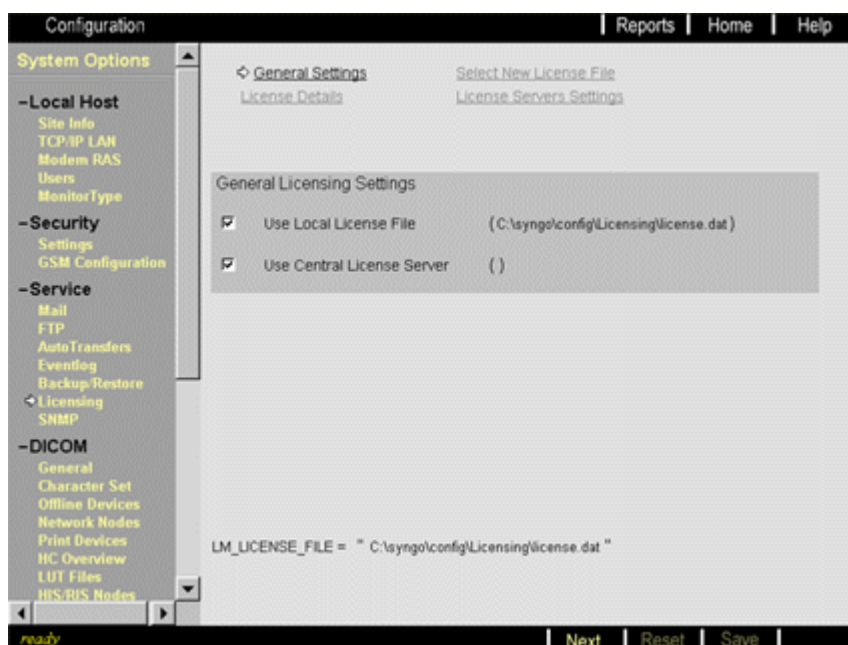


Fig. 28: Licensing - General Settings

Select the source of the license file.

- **Use Local License file**

The local license file **C:\MedCom\config\Licensing\license.dat** will be used.

- **Use Central License Server**

Select this menu item in case the Central License Server from **FLEXIm** shall be used.

Select New License File

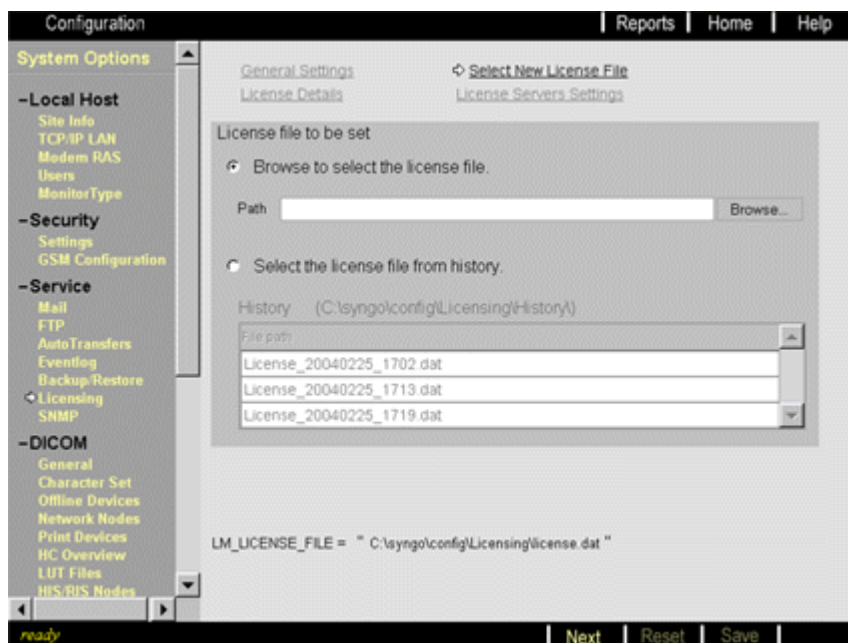


Fig. 29: Licensing - Select New License File

- **Browse to select the license file**

The 'Browse' button opens a window to browse through your LOCAL system.

Even when you logged in remotely, the local file system will be displayed. This allows to import a license file from a local PC via the remote line. As transfer protocol HTTP file transfer will be used.

To import the selected license file at the browser window, push the **Save** button.

NOTE

It is not possible to append licenses to the existing one using this function, the entire file will be overwritten.

NOTE

During the import of a licenses file a backup of the original file will be copied into the History directory. The date-time stamp will be added to the filename. If the new license file fails, you may re-import the previous license file from History .

- **Select the license file from history**

In case the import of the new license file fails, this menu item provides the possibility to import the backup of the original license file from the History directory.

License Details

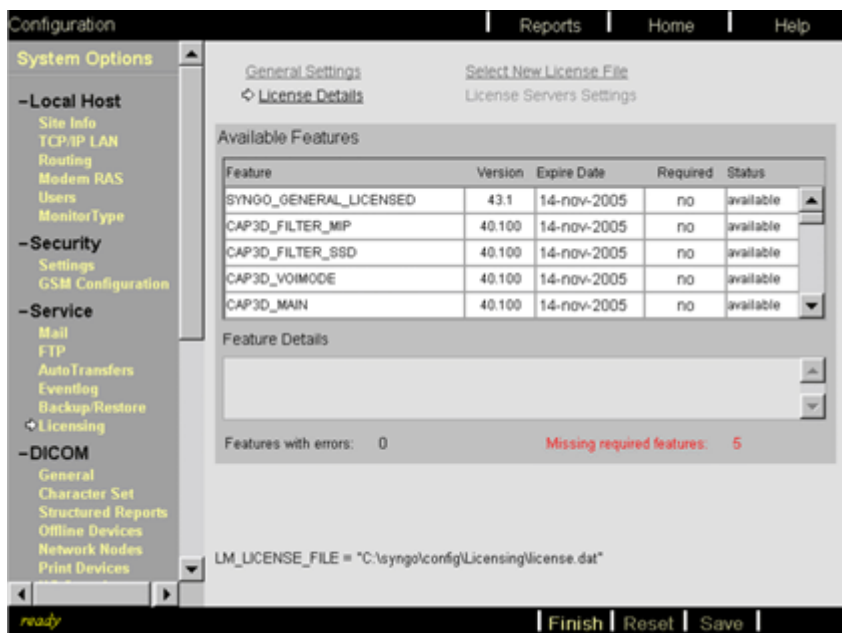


Fig. 30: License Details

- This menu item provides a list of all entered licenses and their status. It is not possible to add or edit any of these entries. If any license key has expired, the whole license.dat file needs to be exchanged.

License Network Server Settings

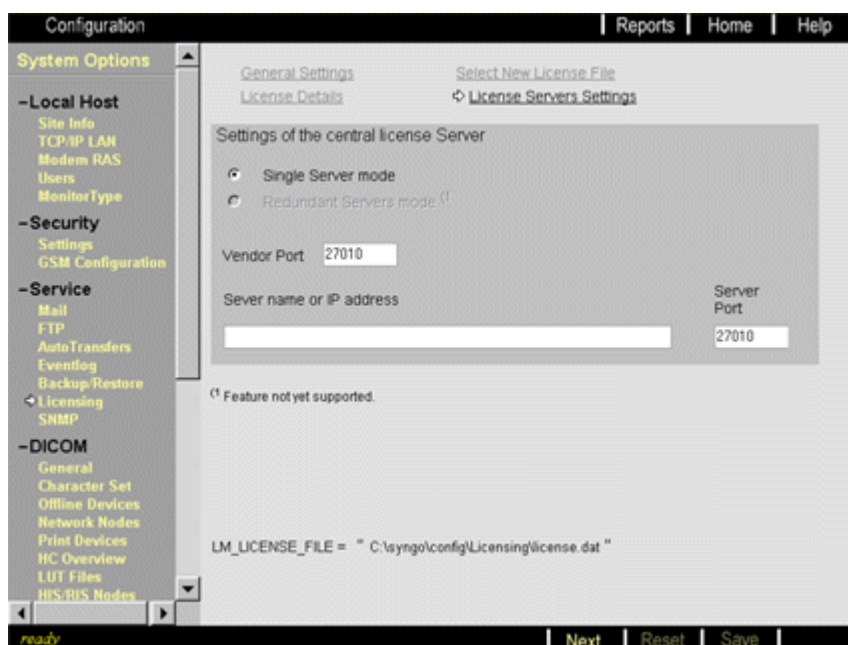


Fig. 31: Licensing - Server settings

This menu is only available if 'Use Central License Server' is selected under 'General Settings'.

- **Vendor Port**

It is not necessary to change the port number, as FlexLM uses by default the port number 27010.

- **Server Port**

The default port number is 27000. The port number shall be from 27000 to 27009, since these numbers are automatically scanned by the applications.

- **Server name or IP address**

Enter the host name or IP address of the FlexLM Server

Test Tools

This menu item allows you to configure the Auto Diagnoser for Test Tools.

The AutoDiagnoser automatically analyses multiple test results, makes a summary diagnosis of the test results and presents a diagnosis based on these results.

Selection	Recommended setting
Auto Diagnoser	Enabled
Auto Supervision	Enabled

2.2.7 DICOM

HP-Z0000YPR-101005-02-012

DICOM allows you to specify the parameters for the DICOM communication interfaces for your system and for the remote systems.

General remarks

In this section, you define the Application Entity Titles for the local system and general time-out settings for all DICOM processes.

Step 1: General - Local AETs

In this section, you define the Application Entity Title for the local system. Since 3 different Windows NT servers (processes) are running, 3 AEs are also available by default.

NOTE

The default AET name is the computer name. Changing the computer name could also change the AETs. A pop-up window lets you select whether or not the AET should also be changed.

Definition of the Application Entity Title

According to Part 8 of the DICOM Standard, the AET identifies a UNIQUE DICOM Service or application on a specific system in the NETWORK. The name consists of characters, excluding the control characters LF, FF, CR, and ESC. The AET is limited to a length of 16 bytes.

It is permitted to use the same AET for different DICOM services, e.g., Storage, Basic Print, Modality Worklist, etc. However, you cannot use the same AET twice for the same DICOM service.

Some configuration tools in Siemens products allow the same AET to be entered twice for the same service. In this case, the Merge tool kit will always select the destination of the first AET.

All Siemens products that support both DICOM and the PACSnet protocol use the PLA prefix (first 10 characters of the PACSnet Logical Address) and a 4-character suffix, e.g., 049SA1DC01DRSP.

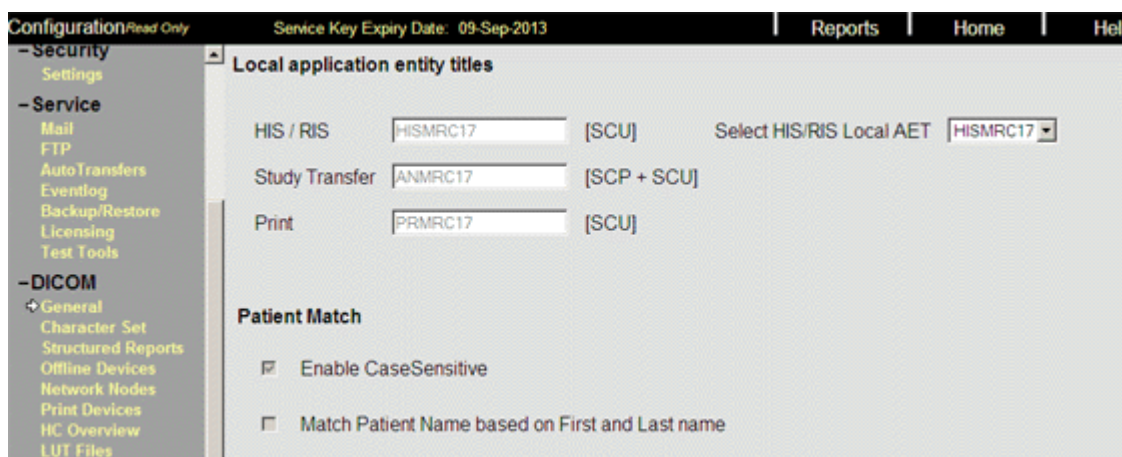


Fig. 32: Configuration - DICOM - General

- **HIS/RIS Server**

This server performs the DICOM Modality Worklist Services. Usually, this AET needs to be known at the RIS server as well as for authentication purposes. The default AET is the HOST NAME (upper case). Please be aware that changing the local host name may also change the AET.

- **Study Transfer Server**

This server performs the Storage and Query/Retrieve DICOM Services. Usually, this AET needs to be known at the remote nodes for authentication purposes. The default AET is the HOST NAME (upper case). Please be aware that changing the local host name may also change the AET.

- **Print server**

This server performs the Basic Print Service. Usually, this AET needs to be known at the camera as well as for authentication purposes. The default AET is the HOST NAME (upper case). Please be aware that changing the local host name may also change the AET.

NOTE

The port number of the Study Transfer Provider (SCP) is 104. The port numbers of the HIS/RIS and PRINT SCUs are not relevant because they change with each association and are not required for configuration.

Step 2: General DICOM Settings

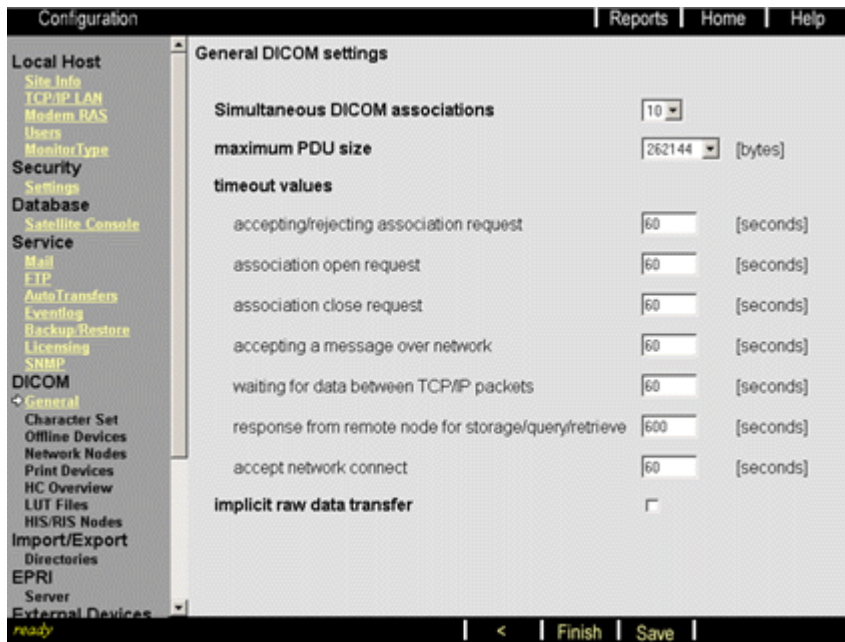


Fig. 33: General DICOM settings menu

- **Simultaneous DICOM Associations**

Set the number of DICOM associations to be handled simultaneously. Reduce the number if there are performance problems.

- **Maximum PDU Size**

The Data Unit Packet size defines how many bytes of data are collected at the interface between TCP/IP and DICOM. The higher the value of the PDU size, the better the performance. During the association negotiation at the beginning of each DICOM session, the maximum PDU size is negotiated between the two involved systems - the lower one is used.

Timeout Values

- **Accepting/Rejecting Association Request**

The ARTIM_TIMEOUT is set in the mergecom.pro file and is the time out to wait for an association request or to wait for a peer to shut down an association.

- **Association Open Request**

The ASSOC_REPLY_TIMEOUT is set in the mergecom.pro file and is the time out to wait for a reply to an association request.

- **Association Close Request**

The RELEASE_TIMEOUT is set in the mergecom.pro file and is the time out to wait for a reply to an association release.

- **Accepting a Message over Network**

The WRITE_TIMEOUT is set in the mergecom.pro file and is the time out to wait for a network write to be accepted.

- **Waiting for Data between TCP/IP Packets**

This INACTIVITY_TIMEOUT is active during the message transfer (e.g. image) and needs to be increased when performance problems in the network occur (switches, routers).

- **Response from Remote Node for Storages' Query/Retrieve**

This is the time out between an C_MOVE request (Search - Import) and the import (retrieve) of the selected images.

- **Accept Network Connect**

This CONNECT_TIMEOUT is set in the mergecom.pro file and is the time out to wait for a network connect to be accepted.

- **Implicit Raw Data Transfer**

Allows general transfer of raw data to a remote DICOM destination. In addition, it is necessary to select 'raw data' in the customer user interface when transferring images. The raw data is sent in private image header attributes, resulting in a very large image header.

Character set

In this menu, you define the character set which is used when a new image or object is created on the local system. As soon as images or objects are imported (Storage, Query/Retrieve, Worklist) to the system, the character set of the imported objects is used.



Fig. 34: Character Set menu

DICOM Default Character Sets

Depending on the country and the institutions environment, it offers a selection of different standard character sets:

- **Default Repertoire, ISO_IR6:** The most popular character set is the 7-bit ASCII Code. The registry entry of ASCII in ISO 2375 is ISO_IR 6, which is used as the defined term in the DICOM Standard for the 'Specific Character Set' attribute 0008,0005 (although DICOM defines just a subset of ISO_IR 6).
- **Latin Alphabet No.1, ISO_IR100:** The 8-bit character sets are defined in ISO 8859, which is a series of more than 10 standardized multilingual graphic character sets like:
 - Latin1 (West European) **ISO_IR 100**
 - Latin2 (East European) **ISO_IR 101**
 - Latin3 (South European) **ISO_IR 109**
 - Latin4 (North European) **ISO_IR 110**
 - Cyrillic **ISO_IR 144**
 - Arabic **ISO_IR 127**
 - Greek **ISO_IR 126**
 - Hebrew **ISO_IR 138**
 - Latin5 (Turkish) **ISO_IR 148**
- **Japanese, Katakana, ISO_IR13:** The Japanese character set description is composed of two character sets, ISO_IR 14 (Japanese version for Roman Characters) and ISO_IR 13 (Katakana character set). More detailed information can be found in the DICOM Standard, Section 5, Appendix H, and Section 3 C.12, General Modules.
- **Unicode in UTF-8, ISO_IR192**
Unicode is the common stand and supports all other DICOM supported character sets, except GB18030. The option character set extension is not possible.
- **Chinese, GB18030**
Used for chinese language. This character set is even larger and includes the Unicode character set. The option character set extension is not possible.

Allow character set extension

DICOM allows you to use more than one character set in an object. If you want to enter Swedish characters and Russian characters, for example, you need to switch from one code to the other in one string by adding a certain Escape Sequence. What is referred to as this 'code extension' technique is defined in ISO 2022. As a result, the defined term for the 'Specific Character Set' attribute 0008,0005 receives the prefix ISO 2022 to show the application that code extension is being used (e.g., ISO 2022 IR 100).

With this prefix, the application is prepared for decoding the Escape Sequences. Otherwise, it can fail when an Escape Sequence appears.

Structured Reports

A configuration interface is provided to offer the possibility to assign a certain viewing application to a certain Structured Report type of e.g. another vendor. The Report type is specified by the Template ID and Mapping Resource as part of the DICOM header of an image, whereas the assigned Siemens viewer for that system is already predefined.

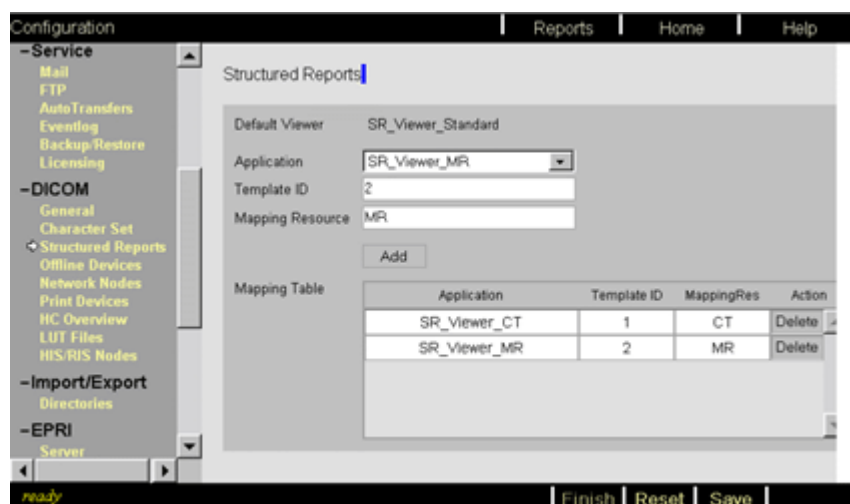


Fig. 35: Configuration Structure of Report View

Default Viewer

The default viewer is pre-configured and can not be changed.

Application

It provides a list of installed and known viewing applications on the system to select the proper viewer for a Structured Report type.

Template ID

DICOM header information to specify the Structured Report type together with the **Mapping Resource**.

Mapping Table

It shows the list of configured combinations of viewing application and concerned Template ID / Mapping Resource.

A new combination can be inserted into the table by selecting the viewing application out of the **Application** select box and entering the concerning **Template ID** and **Mapping Resource**.

The insert into the table is executed with the **Add** button and is permanently stored with **Save**.

A complete table row (stands for a viewing application and report type combination) can be deleted by the **Delete** button behind each row.

Offline devices

In this section, the settings for the DICOM offline devices (Media Storage) on the local host are defined.

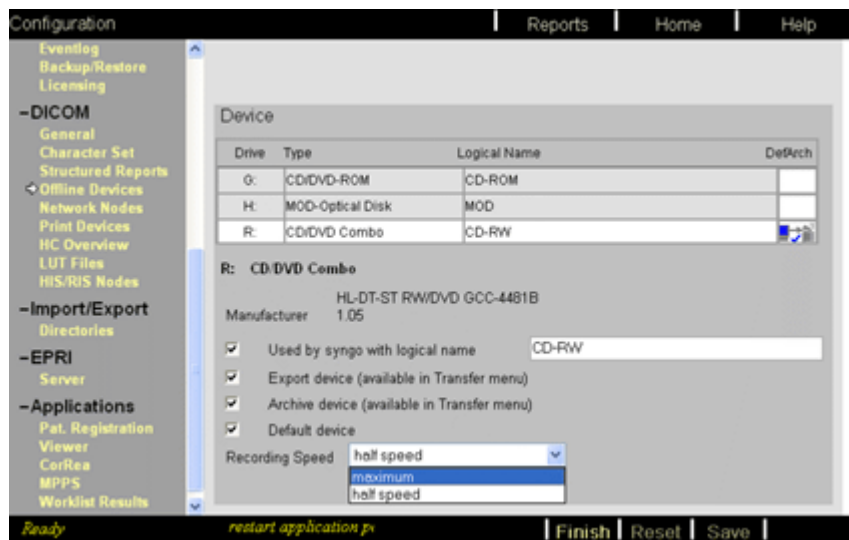


Fig. 36: DICOM - Offline Devices menu

The available Media Storage devices of the system are automatically detected by syngo and shown in the device table with the assigned drive letter.

The particular device is selected by mouse click.

Device Type



Fig. 37: Offline Devices Type

Due to the limitation of the operating system (Removable Media Abstraction Layer) the system is only able to detect devices of type CD/DVD Combo. Therefore those devices are recognized and no combo box are offered for changing the device type. Others have to be allocated to the proper device type.

- **CD/DVD Combo**

Select for combined CD and DVD Burner.

- **CD-R**

Select for CD Burner only.

- **CD/DVD-ROM**

Select for CD and DVD Reader only, without writing capability.

- **MOD-Optical Disk**

Select if the device is a MOD (Only supported by syngo MR upgrade systems from NUMARIS 3 / 3.5).

NOTE

Please be aware that syngo might not distinguish between CD-ROM and CD-RW.

Insert a CD to verify if you configure the correct drive letter for the corresponding drive!

Special syngo device settings

- **Used by syngo with logical name**

Enables the device to be used in syngo. If enabled an **unique** logical name has to be given (free text).

- **Export Device**

Defines if the device shall be used as export device within syngo.

NOTE

It's limited to the device types CD-R and CD/DVD Combo only, which can be selected for export.

- **Archive device**

This check box is available depending from the system type (modality). It defines if the device shall be used as archive device.

NOTE

Only one CD-R or CD/DVD Combo drive can be defined as archive device.

- **Default device**

Defines if the device shall be used as default device. Only one device per system can be defined. The defined device is marked as default within the device table by an icon in the last column (DefArch).

It's not applicable for CD-R drives.

Recording Speed

The recording speed of the selected device can be configured for half or maximum speed.

See following table for recommended logical names:

Drive	Type	Logical Name
R:	CD	CD-ROM or DVD-ROM
S:	CD-R or CD/DVD Combo	CD-RW or DVD-RW

Network nodes

Definition of DICOM nodes for Storage (send and receive) and Query/Retrieve. Please note that DICOM printers and the Radiology Information System (RIS) are not configured here.

The Image transfer or Query/Retrieve targets entered can be selected in the “Transfer - Send to...” menu of the viewer or patient browser.

The Query/Retrieve targets entered may be selected in the viewer or patient browser menus “Patient - Search”.

NOTE

DICOM nodes that send only to the local system need to be configured as well, since their IP addresses will be checked. One problem is that these nodes are also visible in the viewer or patient browsers “Transfer - Send to...” menu as a send destination. To prevent this, enter just the TCP/IP settings (Step 1) and skip the DICOM settings (Step 2).

Required Data

It is the responsibility of the Project Manager to obtain the following data from the customer.

- Host name and TCP/IP address of the remote host.
- Application Entity Title (AET) and port number of the corresponding DICOM processes on the remote host
- Transfer syntax (JPEG, Explicit Big Endian, ...)
- Which DICOM service and in which direction (SCU, SCP)
- Usually the remote systems need to be configured as well. It is recommended that the service engineer from the other vendor be on-site at the same time.

Step 1: TCP/IP Settings for DICOM Network Nodes

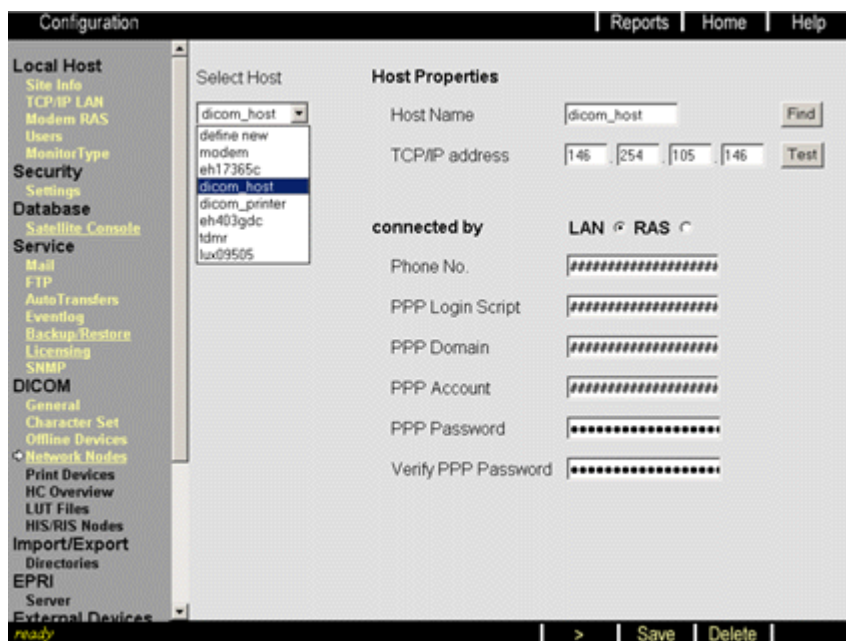


Fig. 38: DICOM Network Nodes menu - step 1

Select Host

Click on the "Select Host" drop-down menu to see whether the DICOM node is already listed.

If the DICOM node is not listed, you need to enter the host name and the IP address.

Host Properties

- Host Name**

Enter the host name of the DICOM node.

NOTE**Restrictions for the Host Name:**

The host name may contain up to 15 characters. The characters "_" and "-" should be avoided, since they could cause problems (even though it is possible to enter them). The computer name may not consist of just numeric characters or start with a numeric character.

- Find**

Select the **Find** button to check whether the host name is known at the WINS or Domain Name Server (DNS). If the name is found, the IP address is entered in the TCP/IP address box.

NOTE

Find works for RAS only if the RAS connection has been established previously.

- **TCP/IP Address**

If no DNS is used, you need to manually enter the IP address into these boxes.

RAS requires two IP addresses - one for the local system and the second for the remote system. For RAS, you need to enter the second IP address.

- **Test**

Click on the **Test** button to send a PING command to the corresponding IP address.

A TCP/IP ping may fail for the following reasons:

- The remote host is turned off, doesn't answer on ping or is not in the same network (or sub net), or the gateway is not configured in the routing table.
- Any networking device (router, switch, bridge,...) denies access to the target network.

Connected by LAN / RAS

If the DICOM node is connected via Ethernet (including a router), you need to select the **LAN** button.

NOTE

(RAS) Modems are not supported by syngo MR systems.

Step 2: DICOM Settings for DICOM Nodes

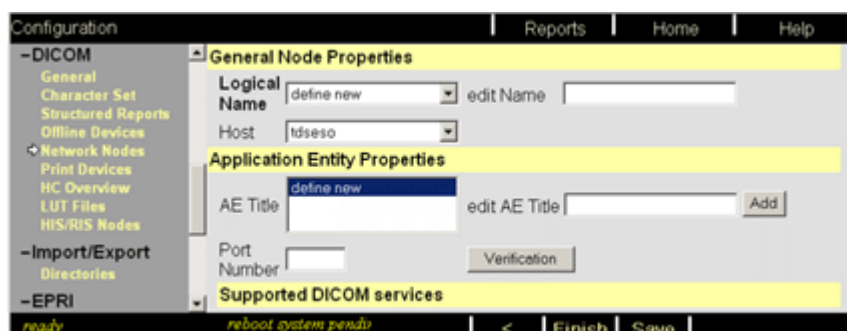


Fig. 39: DICOM Network Nodes Menu - step 2

General Node Properties

- **Logical Name**

Select the logical name for the DICOM device or define a new one in the 'Edit Name' box. The logical name is displayed in the customer user interface at "Transfer - Send to..." for Storage and at "Patient - Search" for Query/Retrieve.

- **Edit Name**

Create or update a logical name for the DICOM device.

- **Host**

Select the host name (computer name) of the DICOM device.

Application Entity Properties

- **AE Title**

Select the Application Entity Title for the DICOM device or define a new one in the “Edit AE Title” box. The AETs already configured are shown in this pull-down menu.

NOTE

Restrictions for the AET:

- **Maximum 16 characters are allowed**
 - **BLANKS and " (double quotes) are not allowed**
 - **You may use alphabetic and numeric characters as well as “_” and “-”.**
 - **It is recommended to write AETs in upper case.**
-

- **Edit AE Title**

Enter the Application Entity Title of the DICOM device.

- **Port Number**

Enter the Application Entity Title and port number of the DICOM device and click on the Add button. The port number is the TCP/IP port where the DICOM process (defined by the AET) at the DICOM device is listening.

- **Add / Del**

When selecting a previously configured AET, the Add button will change to the Del button to delete this AET.

- **Verification**

DICOM verification is a DICOM service for test purposes. What is called a C-ECHO command is transferred to the selected DICOM service, which is defined by the IP address, the port number, and the AET.

A C-ECHO may fail for the following reasons:

- TCP/IP address, AET, or port number is not configured correctly.
- The DICOM process at the remote host is not running.
- Certain products check the IP address or AET of the sending system. If the AET or IP address of the local system is not entered correctly there, DICOM verification will fail.
- The remote host does not support DICOM Verification as a Service Class Provider (see DICOM Conformance Statement).

Supported DICOM Services

Select which DICOM services are to be performed by the particular system.

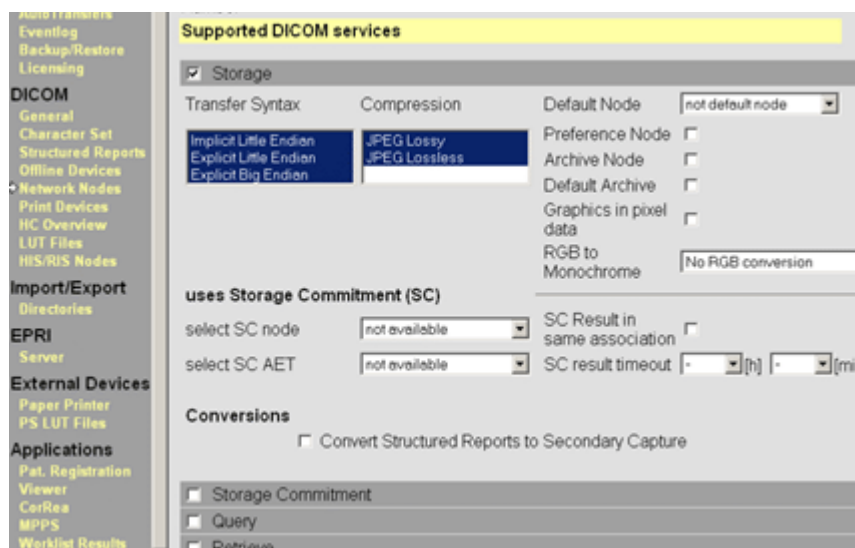


Fig. 40:

Storage

- **Transfer Syntax / Compression**

Select the transfer syntaxes proposed by the local system to the remote DICOM node during the association negotiation. When sending images to the DICOM node, this node selects one of the proposed transfer syntaxes as the result transfer syntax.

NOTICE

This setting has no effect when receiving images.

⇒ **Syngo will accept all transfer syntaxes, regardless of what is selected in this menu.**

NOTICE

How can you deselect both compression transfer syntaxes from the selection box?

⇒ **Just click on one of them to remove it. The second one can be removed by also selecting the CTRL button.**

- **Implicit Little Endian:** Implicit (without) value representation (string, integer, date, ...). The value representation for each attribute will not be sent in the image header and needs to be added at the destination using a data dictionary file.
Little Endian is a synonym for sending the least significant bit first (byte swap).
- **Explicit Little Endian:** Explicit (with) value representation (string, integer, date, ...). The value representation for each attribute is sent in the image header.
Little Endian is a synonym for sending the least significant bit first (byte swap).
- **Explicit Big Endian:** Explicit (with) value representation (string, integer, date, ...). The value representation for each attribute is sent in the image header.
Big Endian is a synonym for sending the most significant bit first (byte swap).

- **JPEG Lossy:** JPEG defines a set of compression algorithms. With Lossy JPEG, the transferred image will lose data. The quality factor can be selected in the customer user interface; this affects how much information will be lost. This transfer syntax should be selected only if required by the customer. It will usually be used for WAN (modem, ISDN) when the destination node does not need the full image quality.
- **JPEG Lossless:** This transfer syntax results in a much lower compression factor, but no data will be lost.
- **Default Node**
Define the order for display in the transfer menu by selecting the “first default” / “second default” or “no default” entry. Only the 1st and 2nd nodes will be displayed in the customer user interface under “Transfer - Send to ...”. For the other nodes, an additional pull-down menu needs to be opened in the customer user interface.
- **Preference Node**
One DICOM storage device may be selected as the preference node. This affects only the keyboard settings; when you click on the + button, the selected study is transferred to this preference node.
- **Archive Node**
If you check this box, the studies sent to the DICOM device will receive an “Archive flag” in the patient browser in the customer user interface.
- **Default Archive**
If you check this box, this DICOM device is set as the Archive device in the customer user interface.
- **Graphics in Pixel Data**
By default, syngo-based systems write the overlay data (annotations, distance, angle, ROI, ...) to the attribute 6000.3000 and also as private attributes (group 0029). The private attributes can be used only by other syngo-based systems. They make it possible to reanimate the overlays (change them on the destination).

Several applications from other vendors are not able to read the overlays in 6000.3000. In this case, you need to select this button. The overlay data is then shifted in the pixel data to bit 12. If there are images using more than 12 bits, this shift cannot be performed.
- **RGB to Monochrome**
It can be configured that an RGB color image will be converted and sent as a monochrome image.
 - **No RGB Conversion**
All images with photometric information (RGB) are sent without color conversion
 - **Convert RGB Color Images**
If certain Ultrasound specific DICOM attribute are recognized, **but no color data present**, the image is converted from 3x8 bit RGB into 8 bit gray scale value and certain attributes are changed before sending. In addition all private attributes are removed
 - **Convert Ultrasound Non-Color Data**
If certain Ultrasound specific DICOM attribute are recognized the image is converted from 3x8 bit RGB into 8 bit gray scale value and certain attributes are changed before sending. In addition all private attributes are removed

Conversions

The check box provides the ability to configure node-specific whether DICOM Structured Report or DICOM Secondary Capture is supported. In this way it's be possible to send DICOM Secondary Capture results and reports to nodes that don't support DICOM Structured Report. Per default convert to Secondary Capture is unchecked.

Uses Storage Commitment

Here you need to define the Storage Commitment provider to whom the commitment is to be sent after the image transfer (Storage). It is not necessarily the same device as the storage destination, e.g., you can send the images to a workstation which automatically forwards them to an archive. In this case, you may send the Storage Commitment request to the archive device and not to the workstation.

- **Select SC Node**

Select the host name of the Storage Commitment provider.

- **Select SC AET**

Select the Application Entity Title of the Storage Commitment provider.

- **SC Result in Same Association**

If this box is selected, the Storage Commitment association will stay open until the response from the provider is received or the "SC result time out" is exceeded.

- **SC Result Timeout**

The Time out for Storage Commitment response applies both to results in the same or in a new association.

Storage commitment

Select this box to configure the appropriate DICOM device as Storage Commitment Provider (and supports this DICOM Service). It is not necessarily the Storage Commitment provider used after the image transfer from the local device (see "Uses Storage Commitment").

Query

Selecting this box assumes that the AET shown is a Query provider. The logical name will be displayed in the customer user interface in "Patient - Search". The Query service results in a list of the selected patients / studies of the DICOM node just configured.

Retrieve

To be able to retrieve (get) the selected images from the DICOM node, it is also necessary to select the "Retrieve" box. Usually, Query and Retrieve need to be selected.

The retrieve command initiates a Storage service at the remote DICOM node to send the images to the local system.

Print Devices

This section defines the DICOM laser cameras connected to your system via DICOM Basic Print. A postscript paper printer is not a DICOM printer! This system does not support the SPCI/SPDI camera interface.

Required Data

It is the responsibility of the Project Manager to obtain the required data from the customer.

- Host name and TCP/IP address of the remote host.
- Application Entity Title (AET) and port number of the corresponding DICOM processes on the remote host
- Usually, it is necessary for the camera to be configured as well. It is recommended that the service engineer from the printer vendor be on-site at the same time.

NOTE

If you need to delete a print device, the correct procedure is to start with menu 3 to menu 1. Otherwise, problems may result with the filming tab card in the user interface, registry settings, etc.

Step 1: TCP/IP Settings for DICOM Camera

The screenshot shows the 'Configuration' window for 'DICOM Print Devices'. The left sidebar contains a tree view with categories: Local Host, Security, Database, Service, and DICOM. Under 'DICOM', 'Print Devices' is selected. The main panel is titled 'Host Properties' and includes a 'Select Host' dropdown menu currently showing 'dicom_printer'. Below this are fields for 'Host Name' (dicom_printer) and 'TCP/IP address' (146.254.105.146). There are also fields for 'connected by' (LAN), 'Phone No.', 'PPP Login Script', 'PPP Domain', 'PPP Account', 'PPP Password', and 'Verify PPP Password'. The status bar at the bottom indicates 'ready' and has buttons for '>', 'Save', and 'Delete'.

Fig. 41: DICOM Print Devices - step 1

Select Host

Click on the “Select Host” drop-down menu to see whether the DICOM camera is already listed. If so, and if the related data are correct (including RAS settings), you are done and may continue with the next menu.

If the DICOM camera is not listed, you need to enter the host name and the IP address.

Host Properties

- **Host Name**

Enter the host name of the DICOM camera.

NOTE**Restrictions for the Host Name:**

The host name may contain up to 15 characters. The characters “_” and “-” should be avoided, since they could cause problems (even though it is possible to enter them). The computer name may not consist of just numeric characters or start with a numeric character.

- **Find**

Select the **Find** button to check whether the host name is known at the WINS or Domain Name Server (DNS). If the name is found, the IP address is entered in the TCP/IP address box.

Find works for RAS only if the RAS connection has been established previously.

- **TCP/IP Address**

If no DNS is used, you need to manually enter the IP address into these boxes.

RAS requires two IP addresses - one for the local system and the second for the remote system. For RAS, you need to enter the second IP address here.

- **Test**

Click on the **Test** button to send a PING command to the corresponding IP address.

Test works for RAS only if the RAS connection has been established previously. The first ping may fail because of a time out, since it takes time to establish the line.

A TCP/IP ping may fail for the following reasons:

- The remote host is turned off or not in the same network (or sub net), or the gateway is not configured.
- The TCP/IP address is configured incorrectly.
- Any networking device (router, switch, bridge,...) denies access to the other network.

Connected by LAN / RAS

For performance reasons, a DICOM laser camera should not be connected via RAS.

Step 2: DICOM Settings for DICOM Camera

Fig. 42: DICOM Print Devices - step 2

General Node Properties

- **Logical Name**

Select the logical name for the DICOM camera or define a new one in the “edit Name” box. Unlike “network nodes” and “HIS/RIS nodes”, the logical name is not visible at the customer user interface. It is used only as a link to the next menu (step 3).

- **Edit Name**

Create or update a logical name for the DICOM camera.

- **Host**

Select the host name (computer name) of the DICOM camera.

Application Entity Properties

- **AE Title**

Select the Application Entity Title for the DICOM camera or define a new one in the field “edit AE Title”. The AETs already configured are shown in this pull-down menu.

NOTE**Restrictions for the AET:**

- Maximum 16 characters are allowed
- BLANKS and " (double quotes) are not allowed
- You may use alphabetic and numeric characters as well as “_” and “-”.
- It is recommended to write AETs in upper case.

- **Edit AE Title**

Enter the Application Entity Title of the DICOM camera.

- **Port Number**

Enter the port number of the DICOM camera and click on the Add button.

The port number is the TCP/IP port where the DICOM process (defined by the AET) at the DICOM camera is listening.

- **Add / Del**

If a previously configured AET is selected, the **Add** button changes to the **Del** button, allowing you to delete this AET.

- **Verification**

DICOM verification is a DICOM service for test purposes. What is called a C-ECHO command is transferred to the selected DICOM service, which is defined by the IP address, the port number, and the AET.

A C-ECHO may fail for the following reasons:

- TCP/IP address, AET, or port number is not configured correctly.
- The DICOM process at the remote host is not running.
- Certain products check the IP address or AET of the sending system. If the AET or IP address of the local system is not entered correctly there, DICOM verification will fail.
- The remote host does not support DICOM Verification as a Service Class Provider (see DICOM Conformance Statement).

Step 3: Camera Parameter Settings

The screenshot shows the 'Configuration' window for 'DICOM Print Devices'. The left sidebar contains a tree view with categories like Local Host, Site Info, TCP/IP LAN, Modem RAS, Users, MonitorType, Security, Settings, Database, Satellite Console, Service, Mail, ETP, AutoTransfers, Eventlog, Backup/Restore, Licensing, SNMP, DICOM, General, Character Set, Offline Devices, Network Nodes, Print Devices, HC Overview, LUT Files, HIS/RIS Nodes, Import/Export, Directories, EPRI, Server, and External Devices. The main area is titled 'Configuration' and has tabs for 'Reports', 'Home', and 'Help'. It contains several sections: 'Select HC device' with a 'define new' button; 'General Settings' with fields for Type (DICOM printer), Class (egfedrystar2000), DICOM Node, Location, and Comment (No Comment); 'Filing Properties' with fields for Hold printed filmjobs (10), Pixel Size (85 x 85), Film Sheet Formats Portrait (define new), Film Sheet Formats Landscape (define new), Medium type (blue film), and Film destination (processor). A 'Finish' button is at the bottom right.

Fig. 43: DICOM Print Devices - step 3

- **Select HC Device**

Previously configured DICOM cameras may be selected in this pull-down menu.

- **HC Device**

Enter the name of the camera that is displayed in the customer user interface on the Print tab card.

- **Default Camera**

If this box is selected, the camera will be set as the default camera. The other cameras need to be selected in a pull-down menu on the Print tab card in the customer user interface.

General Settings

- **Type**

You may select from the following camera types:

- **DICOM printer** : Select this item for all DICOM cameras and for SPDI/SPCI cameras connected via a DICOM interface box.
- **SPCI printer** : This item was originally reserved for a DICOM interface box from Siemens to SPDI/SPCI. This box is not released; as a result, this menu item is obsolete.

- **Class**

A pull-down menu with the released DICOM cameras is provided. Select your camera. Call Service if your camera is not available in the list.

- **DICOM Node**

The logical names defined in Step 2 of the camera configuration are displayed in this pull-down menu. Select the appropriate camera.

- **Location**

Enter the location of the camera. This information is displayed only in this configuration menu.

- **Comment**

Enter comments such as the help desk phone number of the camera vendor. This information is displayed only in this configuration menu.

Filming Properties

The screenshot shows the 'Configuration' window for 'DICOM Print Devices' at step 3. The left sidebar contains a tree view with categories: Local Host, Security, Service, and DICOM. Under 'DICOM', 'Print Devices' is selected. The main panel is titled 'Configuration' and has tabs for 'Reports', 'Home', and 'Help'. The 'Configuration' tab is active, showing a 'Select HC device' section with a 'define new' button. Below this is the 'Filming Properties' section, which includes the following fields: 'Hold printed filmjobs' (set to 10), 'min density' (set to 20, range [0.1]), 'Pixel Size [1/1000mm]' (set to 85 x 85), 'Film Sheet Formats Portrait' (set to 'define new'), 'Film Sheet Formats Landscape' (set to 'define new'), 'Medium type' (set to 'blue film'), 'Film destination' (set to 'processor'), 'color appearance' (set to 'Grayscale 8 bit'), 'background' (set to 'black'), and 'Transformation' (set to 'no magnification'). At the bottom of the window, there is a 'Finish' button.

Fig. 44: DICOM Print Devices - step 3

- **Hold printed Film Jobs**

Even if the film has been printed successfully, the most recent films remain in the print queue. The number of films in the queue may be defined here.

- **Min. Density**

Set the minimum density value for the camera. The maximum density value is modality-dependent and can be set in the “LUT files” menu.

- **Pixel Size [1/1000 mm]**

This value is entered automatically, depending on the selected camera (class). This value may be modified here.

- **Film Sheet Formats**

Portrait / Landscape The default film sheet formats are displayed, depending on the selected camera (class). They may be modified or additional formats may be defined.

- **Number of Pixels [rows, columns]**

The formats are automatically filled in, depending on the selected camera (class). In the boxes below, the number of pixels (row and column) may be modified depending on the medium sheet size.

- **Medium Type**

Depending on the film used, select between “blue film”, “clear film”, and “paper”.

- **Film Destination**

You may select between “magazine” and “processor”. If you select magazine, the exposed film sheet needs to be developed in a darkroom. A processor will automatically develop the exposed film sheet.

- **Color Appearance**

You can select between monochrome (black & white) or color. This setting is necessary if you want to print color images (Ultrasound) to a camera that supports color.

- **24 bit Color:**

In this case, the camera negotiates the “Basic Color Print Management” and transfers the pixel data in RGB photometric interpretation.

- **8 bit Gray Scale:**

In this case, the camera negotiates the “Basic Gray Scale Print Management” and transfers the pixel data in MONOCHROME2 photometric interpretation with 8-bit pixel depth.

- **12 bit Gray Scale:**

In this case, the camera negotiates the “Basic Gray Scale Print Management” and transfers the pixel data in MONOCHROME2 photometric interpretation with 12-bit pixel depth (16 bit allocated for each pixel).

- **Background**

The background of the film may be set to black or white.

- **Transformation**

This value is sent to the camera if pixel interpolation is necessary. Usually the entire film sheet is calculated at the local system and recalculation is therefore not necessary at the camera. As a result, this parameter has no effect on the camera.

But be aware that an incorrect setting may not be accepted by a camera.

Nevertheless there are more and more high-resolution cameras available, whereby syngo sends only e.g. the half pixel matrix (each second pixel value). In this case the camera has to interpolate.

The following interpolation algorithms are possible:

Replicate: The interpolated value is identical to the last pixel value

Bilinear: The interpolated value (linear) is between the last pixel value and the next pixel value.

Cubic: The interpolated value (curve) is in between the last pixel value and the next pixel value.

No Magnification: Leaves the image as is.

HC Overview

The “Hard Copy Overview” menu allows the customer to adjust and test certain camera settings. There is no need for the field service engineer to use this menu, since the settings and tests can also be performed in the “hardcopy device” menu.



Fig. 45: DICOM HC Overview menu

These menu items are already described in the “hardcopy devices” menu.

LUT Files

This menu makes it possible to import, remove, and select LUT files (Look Up Table) for DICOM Basic Print. Currently, no tool is available for modifying a lookup table (LUT editor).

Step 1: Import of additional LUT Files

NOTE

Normally it's not required to import additional LUT's, click > to select the syngo MR LUT for your console.

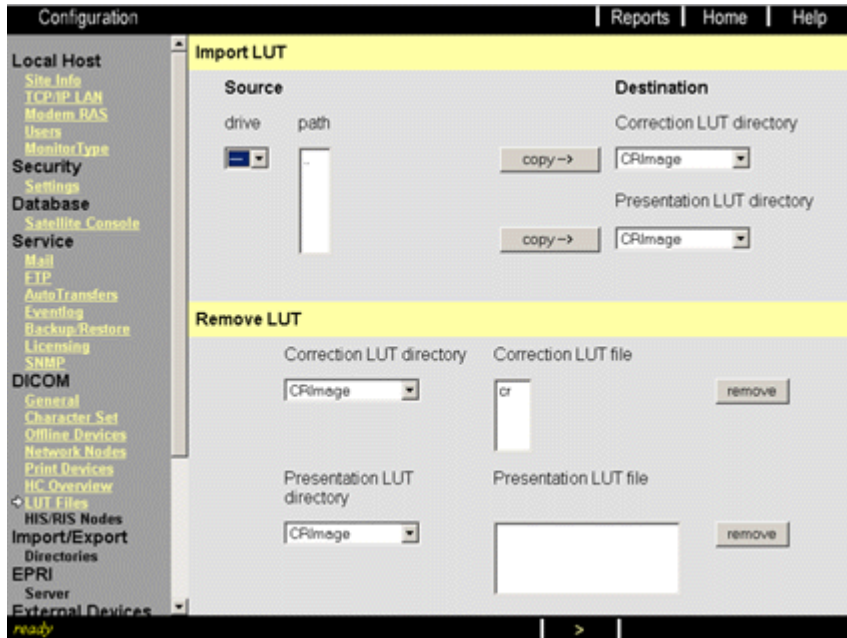


Fig. 46: DICOM LUT Files menu (import / remove)

Import LUT

If new LUT files for one or more modalities are available on a removable medium, e.g., floppy or CD, these files may be selected on the source drive and copied to the dependent LUT directories on the local system.

- **Source**

Select the drive and path where the appropriate LUT file is located.

- **Destination**

Select the destination LUT directory, depending on the image type. Click on the **Copy** button to transfer the selected LUT file.

Remove LUT

LUT files that are not needed may be removed from LUT directories on the local host by selecting the appropriate LUT file and clicking the **Remove** button.

Step 2a: Set LUT files - Correction LUT

This setting is used when the DICOM camera is set to a linear LUT and is not calibrated according to the "Gray Scale Standard Display Function" defined in the DICOM Standard (**Barten** curve).

The modality-specific LUTs (now called Correction LUTs) are used for diagnostics and for correcting the camera-specific optical density values. It has no effect on the monitor.

There may be 8-bit LUTs and 16-bit LUTs in the LUT selection box. This does not mean that one kind of LUT is made for 8-bit images and the other for 16-bit images. It means that the 8-bit LUTs have only 256 values to describe the LUT and the 16-bit LUTs have 65 kbits and are therefore much more precise.

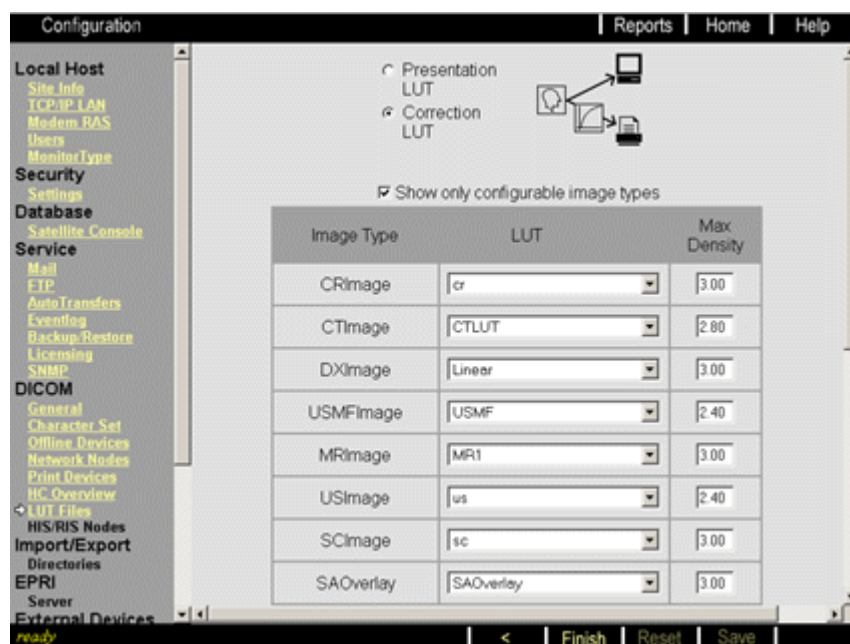


Fig. 47: LUT Files - Correction LUT enabled

Recommended settings for syngo MR systems: Select Correction LUT

	MRImage	SCImage	Your HC Device (Camera)
LUT	mr1Corr16b.lut	mr1Corr16b.lut	No LUT (Linear)
Max Density	3.0	3.0	

- **Presentation LUT**

The Presentation LUT is a camera-independent, modality-specific LUT which assumes that the camera is calibrated according to the **Barten** curve specified in the DICOM Standard, Section 14 - Gray Scale Standard Display Functions.

- **Correction LUT**

The modality-specific 'Correction LUT' needs to be selected if the camera is not calibrated according to the DICOM standard but according to a linear LUT.

- **Show only configurable image types**

Only image types in which Look up tables are loaded (imported) are shown.

- **Max. Density**

The Max. Density value needs to be set for each LUT. Usually it is not necessary to change these values, since they have already been verified by the lab and imported together with the LUT. The Min. Density value is set once in the Print Device menu.

- **Overall Maximum Density**

If images from several modalities are used in the virtual film sheet, this overall max. density value is used to recalculate the image presentation for each image.

- **Interpolation for Printing**

The entire film sheet is calculated in the local system and then sent in the page mode to the camera. To fit the images into the film sheet, pixel interpolation is necessary. The optimum value has already been evaluated at the laboratory. For this reason, modification of this parameter is not necessary. This setting has no effect on the monitor.

The following interpolation algorithms are possible:

Replicate: The interpolated value is identical to the last pixel value.

Bilinear: The interpolated value (linear) is between the last pixel value and the next pixel value

Cubic: The interpolated value (curve) is between the last pixel value and the next pixel value.

No Magnification: Leaves the image as is.

- **Mode**

Several modes may be selected to adjust the smoothing factor for the cubic interpolation algorithm.

Step 2b: Set LUT files - Presentation LUT enabled

In a Multi Vendor environment, it is often necessary to use the Presentation LUT defined by the DICOM Standard, Section 14 -> **Gray Scale Standard Display Function**. Since this LUT is new, it may not yet be available for certain modalities. This LUT is used for printing as well as for the monitor!

There may be 8-bit LUTs and 16-bit LUTs in the LUT selection box. This does not mean that one kind of LUT is made for 8-bit images and the other for 16-bit images. It means that the 8-bit LUTs have only 256 values to describe the LUT and the 16-bit LUTs have 65 kbits and are therefore much more precise.

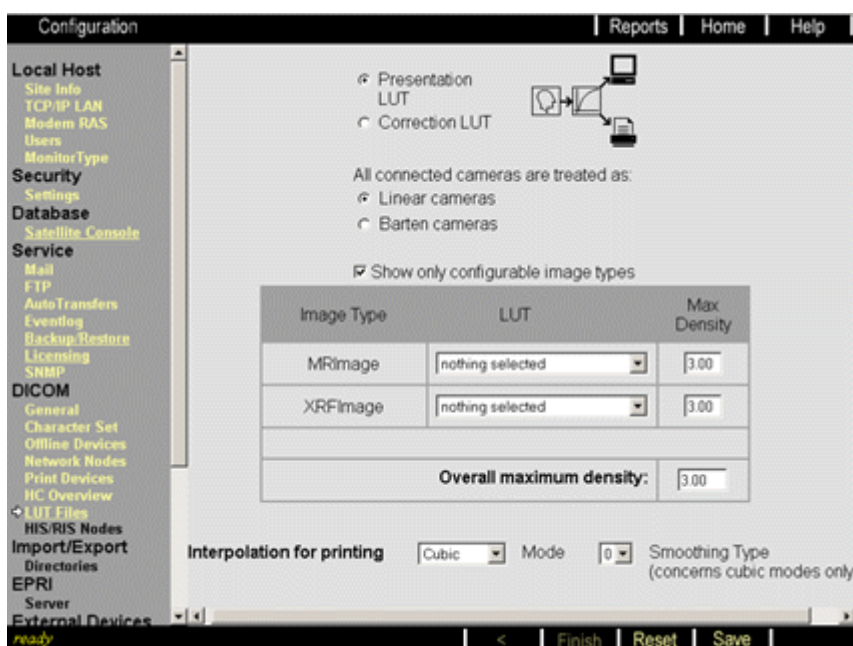


Fig. 48: LUT file - Presentation LUT enabled - Linear cameras

- **Presentation LUT**

The Presentation LUT is a camera-independent, modality-specific LUT which assumes that the camera is calibrated according to the Barten curve specified in the DICOM Standard, Section 14 - Gray Scale Standard Display Functions.

- **Correction LUT**

The modality-specific “Correction LUT” needs to be selected if the camera is not calibrated according to the DICOM standard but according to a linear LUT.

All connected cameras are treated as: (appears only if the Barten LUT is selected)

- **Linear Cameras**

The “Correction LUT” usually should be selected for cameras calibrated with a linear LUT. For certain new modalities, only a Presentation LUT exists. Therefore, the Presentation LUT needs to be selected. In addition, this button indicates that the camera does not support Presentation LUTs.

In this case, a Barten LUT is added to correct the difference between the Barten and the linear LUT.

It is not possible to mix cameras that support (and are set to) this Gray Scale Standard Display Function with cameras that do not support GSDF!

- **Barten Cameras**

In this case, all connected cameras need to support calibration according to the DICOM Standard with what is called the Barten curve, which defines luminance values as a function of the pixel value.

This calibration makes it possible to send a camera-independent Presentation LUT.

It is not possible to mix cameras that support (and are set to) this Gray Scale Standard Display Function with cameras that do not support GSDF!

- **Show only configurable image types**

Only image types in which look-up tables are loaded (imported) are shown.

- **Max. Density**

The Max. Density value needs to be set for each LUT. Usually it is not necessary to change these values, since they have already been verified and imported together with the LUT. The Min. Density value is set once in the Print Device menu.

- **Overall Maximum Density**

If images from several modalities are used in the virtual film sheet, this overall max. density value is used to recalculate the image presentation for each image.

- **Interpolation for Printing**

The entire film sheet is calculated in the local system and then sent in the page mode to the camera. To fit the images onto the film sheet, pixel interpolation is necessary. The optimum value has already been evaluated at the laboratory. For this reason, modification of this attribute is not necessary. This setting has no effect on the monitor.

The following interpolation algorithms are possible:

Replicate: The interpolated value is identical to the last pixel value.

Bilinear: The interpolated value (linear) is between the last pixel value and the next pixel value.

Cubic: The interpolated value (curve) is between the last pixel value and the next pixel value.

No Magnification: Leaves the image as is.

- **Mode**

Several modes may be selected to adjust the smoothing factor for the cubic interpolation algorithm.

HIS/RIS Nodes

Configuration of a Hospital Information System (HIS) or Radiology Information System (RIS) that supports DICOM Basic Worklist or the Detached Study Transfer Service. In many cases, the RIS is connected via an interface box such as the Mitra Broker (MagicLink HL7).

Required Data

It is the responsibility of the Project Manager to obtain the required data from the customer.

- Host name and TCP/IP address of the remote host.
- Application Entity Title (AET) and port number of the corresponding DICOM processes on the remote host
- Usually the camera has to be configured as well. It is recommended that the service engineer from the printer vendor be on-site at the same time.

Step 1: TCP/IP Settings for HIS/RIS Server

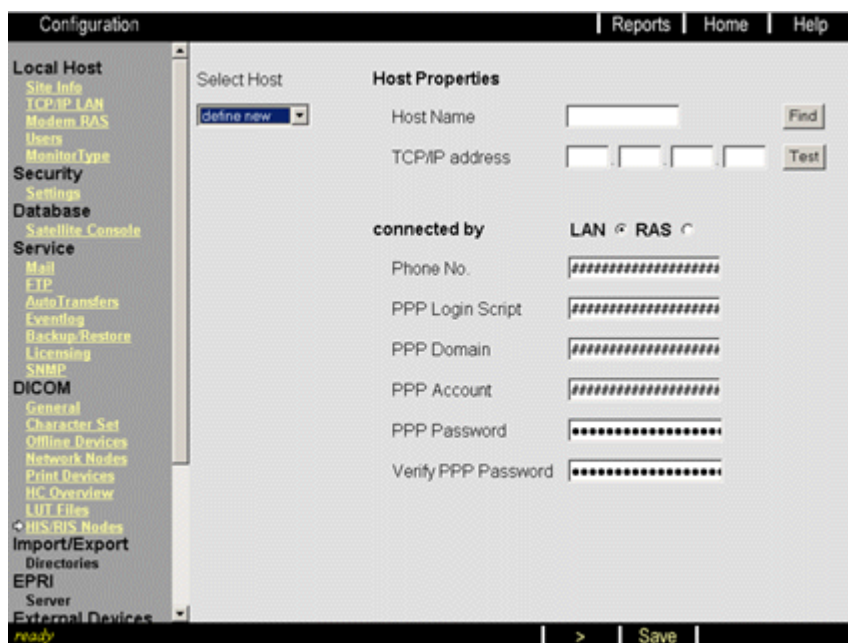


Fig. 49: DICOM HIS / RIS Nodes menu - step 1

Select Host

Click on the “Select Host” drop-down menu to see whether the HIS/RIS node is already listed. If so, and if the related data are correct, you are done and may continue with the next menu.

Host Properties

- **Host Name**

Enter the host name of the HIS/RIS server.

NOTE**Restrictions for the Host Name:**

The host name may contain up to 15 characters. The characters “_” and “-” should be avoided, since they could cause problems (even though it is possible to enter them). The computer name may not consist of just numeric characters or start with a numeric character.

- **Find**

Select the **Find** button to check whether the host name is known at the WINS or Domain Name Server (DNS). If the name is found, the IP address is entered in the TCP/IP address box.

- **TCP/IP Address**

If no DNS is used, you need to manually enter the IP address into these boxes.

- **Test**

Click on the **Test** button to send a PING command to the corresponding IP address. A TCP/IP ping may fail for the following reasons:

- The remote host is turned off or not in the same network (or subnet), or the gateway is not configured.
- The TCP/IP address is configured incorrectly.
- Any networking device (router, switch, bridge,...) denies access to the other network.

Connected by LAN / RAS

For performance reasons, an HIS/RIS should not be connected via RAS.

Step 2: DICOM Settings for HIS/RIS Server

Fig. 50: DICOM - HIS / RIS Nodes menu

General Node Properties

- **Logical Name**

Enter a logical name for the HIS/RIS node. Since the DICOM worklist requests are performed in the background and only one HIS/RIS is selectable, the logical name is not displayed in the customer user interface.

- **Edit Name**

Create or update a logical name for a HIS/RIS node.

- **Host**

Select the host name (computer name) of the HIS/RIS server.

Application Entity Properties

- **AE Title**

Select the Application Entity Title for the HIS/RIS or define a new one in the “Rdit AE Title” box. The AETs already configured are shown in this pull-down menu.

NOTE**Restrictions for the AET:**

- Maximum 16 characters are allowed
- BLANKS and " (double quotes) are not allowed
- You can use alphabetic and numeric characters as well as “ _ “ and “_“.
- It is recommended to write AETs in upper case.

- **Edit AE Title**

Enter the Application Entity Title of the HIS/RIS server and click on the **Add** button.

- **Port Number**

Enter the port number of the HIS/RIS server and click on the **Add** button.

The port number is the TCP/IP port where the DICOM process (defined by the AET) in the HIS/RIS server is listening.

- **Add / Del**

If a previously configured AET is selected, the **Add** button changes to the **Del** button, allowing you to delete this AET.

- **Verification**

DICOM verification is a DICOM service for test purposes. What is called a C-ECHO command is transferred to the selected DICOM service, which is defined by the IP address, the port number, and the AET.

A C-ECHO may fail for the following reasons:

- TCP/IP address, AET, or port number is not configured correctly.
- The DICOM process at the remote host is not running.
- Certain products check the IP address or AET of the sending system. If the AET or IP address of the local system is not entered correctly there, DICOM verification will fail.
- The remote host does not support DICOM Verification as a Service Class Provider (see DICOM Conformance Statement).

Supported DICOM Services

Basic Worklist

Basic Worklist is the DICOM Service **Modality Worklist** and is used to retrieve patient and study information from the Information System (e.g. RIS).

- **Query Waiting Time**

This is the timeout of the DICOM Worklist process waiting for a response from the HIS/RIS server.

- **Max. Query Match Number**

This parameter is used to limit the number of worklist entries to be transferred.

- **Query Interval**

Enter the time in minutes between DICOM Worklist requests (polling).

NOTE

To activate the automatic DICOM Worklist polling mechanism, it is also necessary to set 'HIS/RIS update Worklist in Background' in the customer user interface under 'Option - Configuration - Patient Registration'.

- **Automatic Removal of Cancelled\Rescheduled Requests**

If an examination was sent in a previous worklist and is therefore entered in the Scheduler Database, it is removed automatically if it is no longer sent in the worklist.

Performed Procedure Step

Prior to and after patient acquisition, information data is sent to the Information System (e.g. RIS) by the DICOM Service "**Modality Performed Procedure Step**". Enable this checkbox if the selected Information System is the destination for this DICOM service.

- **Default Catalog Name**

MPPS uses codes instead of readable values. Enter the name of the coding catalog used in the Information System.

- **Default Catalog Version**

MPPS uses codes instead of readable values. Enter the version of the coding catalog used in the Information System.

2.2.8 Import/Export

HP-Z0000YPR-101005-02-014

The system provides the function to import or export DICOM images from / to the file system. In this section, you need to specify the location of the import / export directories. These may be located in the system or in a shared directory.

In the case of shared directories, you first need to configure the host where the shared directory is located (Step 1).

To import / export DICOM images, select “Transfer - Import from Off-Line” or “Import / Export to Off-Line” in the customer user interface.

NOTE

Important In order to *Burn Offline files* the path to the “*Local system*” with the logical name e.g. “*CDR_OFFLINE*” must be set to *C:\MedCom\temp\CDR_OFFLINE* ↵

Step 1: TCP/IP Settings for Remote Import / Export Directory

The screenshot shows the 'Configuration' window with the 'Import/Export' menu path selected: **Local Host** > **Site Info** > **TCP/IP LAN** > **Modem RAS** > **Users** > **MonitorType** > **Security** > **Settings** > **Database** > **Satellite Console** > **Service** > **Mail** > **ETP** > **AutoTransfers** > **Eventlog** > **Backup/Restore** > **Licensing** > **SNMP** > **DICOM** > **General** > **Character Set** > **Offline Devices** > **Network Nodes** > **Print Devices** > **HC Overview** > **LUT Files** > **HIS/BIS Nodes** > **Import/Export** > **Directories** > **EPRI** > **Server** > **External Devices**. The 'Host Properties' section is active, showing fields for Host Name (tdmr), TCP/IP address (146.254.119.127), and various PPP settings. The 'connected by' section is set to LAN.

Fig. 51: Directories menu step 1

Select Host

Click the “Select Host” drop-down menu to see whether the host is already listed. If so, and if the related data are correct, you are done and may continue with the next menu.

Host Properties

- **Host Name**

Enter the host name of the host (shared directory).

NOTE**Restrictions for the Host Name:**

The host name may contain up to 15 characters. The characters “_” and “-” should be avoided, since they could cause problems (even though it is possible to enter them). The computer name may not consist of just numeric characters or start with a numeric character.

- **Find**

Select the **Find** button to check whether the host name is known at the WINS or Domain Name Server (DNS). If the name is found, the IP address is entered in the TCP/IP address box.

Find works for RAS only if the RAS connection has been established previously.

- **TCP/IP Address**

If no DNS is used, you need to manually enter the IP address into these boxes.

RAS requires two IP addresses - one for the local system and the second for the remote system. For RAS, you need to enter the second IP address.

- **Test**

Click on the **Test** button to send a PING command to the corresponding IP address.

Test works for RAS only if the RAS connection has been established previously. The first ping may fail because of a time out, since it takes time to establish the line.

A TCP/IP ping may fail for the following reasons:

- The remote host is turned off or not in the same network (or sub net), or the gateway is not configured.
- The TCP/IP address is configured incorrectly.
- Any networking device (router, switch, bridge,...) denies access to the other network.

Connected by LAN / RAS

If the host is connected via Ethernet (including a router), you need to select the **LAN** button. If the host.

NOTE

RAS (modems) are not supported by syngo MR systems.

Step 2: Definition of Import / Export Directories

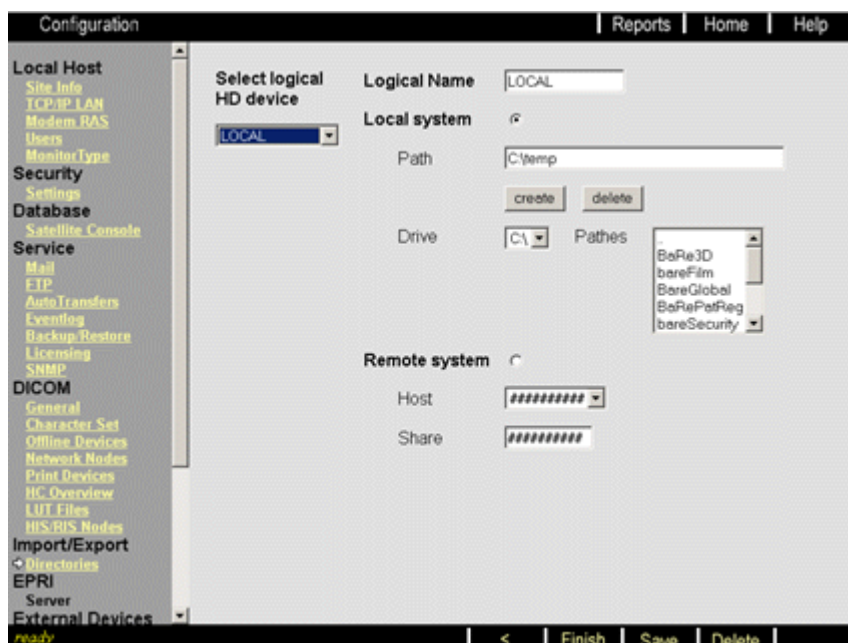


Fig. 52: Directories menu step 2

- **Select logical HD device**

Check if a logical name for the import / export directory is already defined.

By default, the C:\Temp directory is defined, **please change / create to:**

CDR_OFFLINE **C:\MedCom\temp\CDR_OFFLINE** ↵

It is also recommended that the following directories are created:

DICOM Tools	C:\MedCom\Service\dst\img	(Image directory for DICOM Test Tools)
Floppy	A:\	Floppy drive

- **Logical Name**

If the logical name for the import / export directory is not yet defined, you need to enter it here. This name is displayed in the customer user interface under "Transfer - Off-line - Export / Import". For the

- **Local System**

If the import / export directory is located on the local system, select this radio button.

- **Path / create / delete**

The import / export directory selected via the Drive and Paths boxes is displayed; if they have been newly entered, the directory may be deleted or created.

- **Drive**

Select the appropriate drive name from the pull-down menu.

- **Paths**

Depending on the selected drive, you will receive the defined directories. Select the directory to be used as the import / export directory.

- **Remote System**

If the import / export directory is located on a remote system, you need to specify the host and shared directory.

- **Host**

Enter the host name (computer name) defined in the previous menu.

- **Share**

The shared directories on the selected host are displayed. Select the one to be used as the import / export directory.

2.2.9 EPRI

HP-Z0000YPR-101005-02-015

The EPRI (Electronic Patient Record Interface) is a web interface (Internet Explorer browser) to a server, which provides additional data regarding the selected patient.

Server

The customer user interface includes a button for opening a web browser with a link to an EPR server. The URL and the type of EPR server need to be specified in this menu.

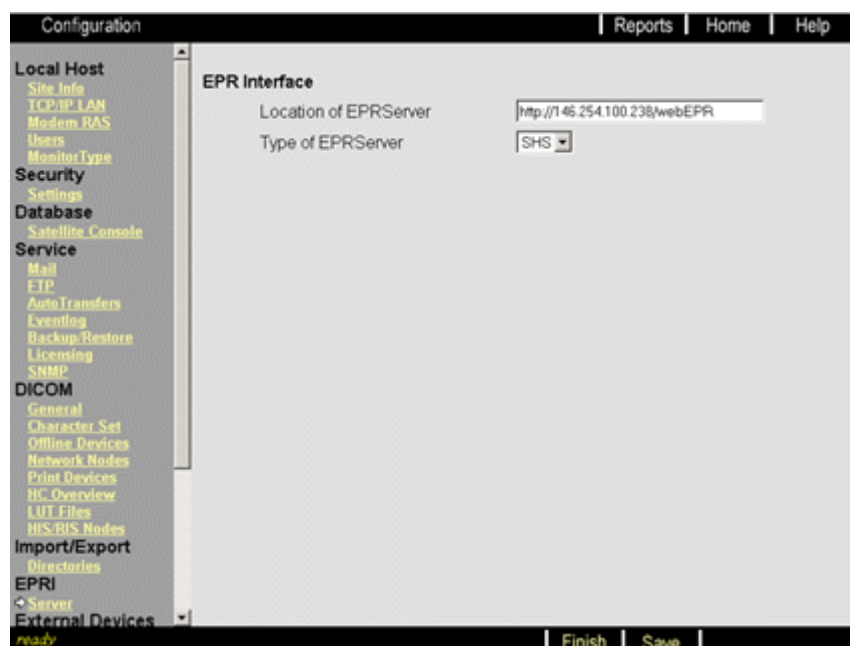


Fig. 53: Electronic Patient Record

- **Location**
URL of the Electronic Patient Record server
- **Type of EPR Server**
Currently only the WebEPR server from Siemens HS is released (as the default server).

2.2.10 External Devices

```
HP-Z0000YPR-101005-02-016
System Type: . . . . .Avanto
               Espree
               TrioTim
               SymphonyTim
```

For possible connected paper printer to the system some adaptations are possible like paper format and import of Postscript Printer Look Up Table.

Paper printer

The driver for the paper printer needs to be installed according to the printer instructions under the administration account for Window NT / 2000. This menu allows you to set the media format.



Fig. 54: Paper Printer

PS LUT Files

This menu provides the possibility of importing, removing, and selecting Postscript LUT files (Look Up Table) for Paper Print.

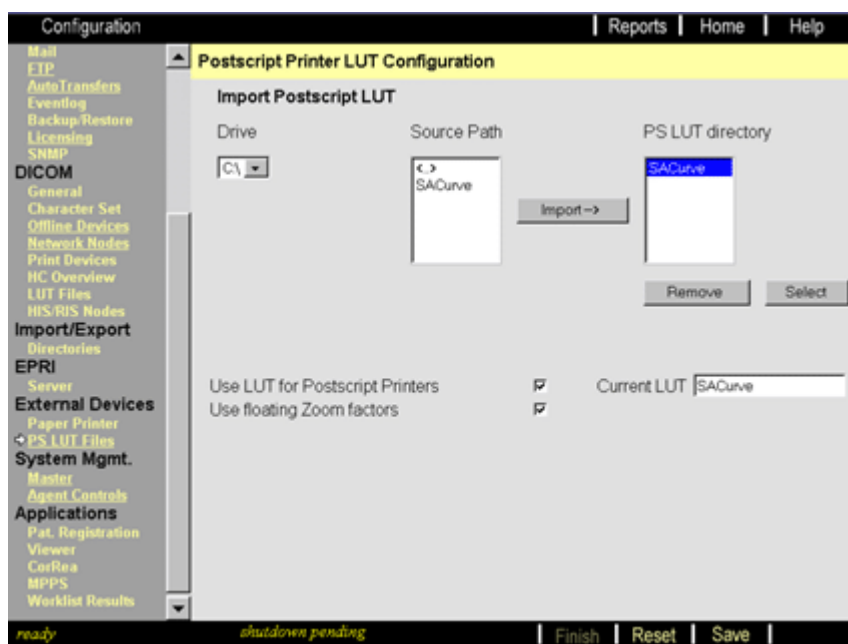


Fig. 55: Postscript LUT Files menu (import / remove)

Import Postscript LUT

If new Postscript LUT files are available on a removable medium, e.g. floppy or CD, these files may be selected on the source drive and copied to the dependent LUT directory on the local system.

- **Import**

Select the 'Drive' and 'Source Path' where the appropriate LUT file is located. By selecting the **Import** button the selected LUT will be copied to the 'PS LUT directory'.

- **Select**

By means of the **Select** button the appropriate LUT in the 'PS LUT directory' will be selected as the 'Current LUT'.

- **Remove**

This button allows to remove a LUT File in the 'PS LUT directory'.

- **Use LUT for Postscript Printers**

Selection of this box will activate the 'Current LUT'.

- **Use floating Zoom factors**

For syngo MR-systems it's recommended to activate **Use floating Zoom factors**.

Zoomed images with an not whole zoom factor (e.g. 2.3) may result in a bad image quality of the printer. In this case activate this box, which alternates the not whole zoom factor into the next smaller whole zoom factor (zoom factor 2.3 will be changed to 2.0).

2.2.11 System Mgmt

HP-Z0000YPR-111005-02-998

System Management enables additional service functionalities such as proactive event monitoring.

For this functionality, the so-called Managed Node Package (MNP), which delivers the required agents on the system, must be pre-installed.

The System Management configuration specifies the backend server and installs and activates all agents.

NOTE

Due to a bug in this software version, a system-specific certificate must be generated before *System Management* is configured.

Creation of System Management certificate

- In the **Service UI** select **Utilities**
- Select **Escape to OS** in the drop-down menu **Source**
- Enter the following command in the **Parameters:** field **ovcoreid -create -force** ↵

Master

In this mask, the automatically assigned System Management master server (server name and IP address), which is taken from the selected **Remote Service Center** in the **Site Info** mask, is shown.

SRS	SD Server	IP
Test (not_for_Customer_Site)	FTHW9EEA	194.138.36.84

Define Other ☐

Software Distribution Server IP Address

Server Network Address Translation ☐

Asset Management Server TCP/IP address

Event Management Server TCP/IP address

Software Distribution Server TCP/IP address

Fig. 56: System Management Master

Checkboxes and buttons:

- **Define Other**

If a different System Management master server needs to be configured for test purposes or if the server differs from the predefined one, this checkbox is marked. This allows you to fill in the differing Software Distribution Server IP Address in the IP entry field.

- **Server Network Address Translation**

The three (3) listed servers (Asset, Event, Software Distribution Server) are the backend servers for the System Management functionalities. In a standard network environment, the IP addresses for these servers are retrieved from the SD Server already configured during the registration procedure. In some cases, the network administration only allows their own address range, which would consequently suppress any traffic with the original IP address. Therefore, the **Server Network Address Translation** box must be checked and addresses within the allowed address range must be entered for each server.

- **Test**

The Test button initiates a PING command to the configured SD Server IP address.

A TCP/IP ping may fail for the following reasons:

- The remote host is turned off or not in the same network (or subnet), or the gateway is not configured.
- The TCP/IP address is configured incorrectly.
- Any networking device (router, switch, bridge, etc.) denies access to the other network.

NOTE

The Reset button is available only if a value has been changed. Use the Next button to switch to the Agent Control page, keeping the entered values. Use the Go button on the Agent Control page to save the System Management configurations.

Save is possible only if the System Management server is online!

Agent Controls

In this mask, you may start and stop the System Management agents. The Event Management agent and the Software Distribution agent are combined in what is called the Managed Node Package, which needs to be installed to enable the System Management functionality.

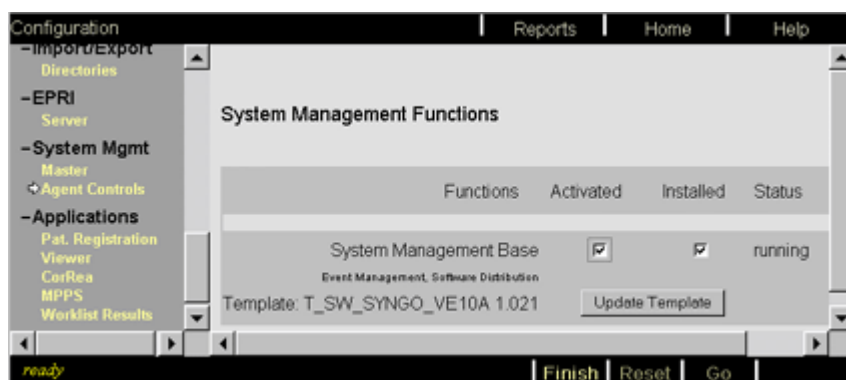


Fig. 57: System Management Agent Controls

- **Activated**

With this checkbox, the System Management agents can be started (and stopped).

- **Installed**

With this checkbox, the System Management agents are installed.

Prior to installing System Management agents via this window, the Manage Node Package must be pre-installed (via the installation CD). The progress of the installation is shown by a Progress bar on the Status bar (duration approx. 15min).

Unchecking the installed checkbox deinstalls the System Management agents. The agents are completely removed from the system, but the pre-installed Managed Node Package is still preserved.

- **Status**

You can see the status of the agents. A status value is underlined if further status information (error information) is available. Hovering the cursor over it or clicking it will display the message.

Status values are: **running**, **partial running**, **sleeping**, **error**.

- **Update Template**

A template is a configuration set telling the System Management agents what to do on the system and therefore differs between the various modalities. The templates are loaded from the System Management server during the registration procedure. They are maintained by Headquarters and are updated according to the monitoring needs at the system (proactive event monitoring).

Use this button to manually initiate any available template updates which are then downloaded from the System Management server. The current version of the template is shown next to the button.

2.2.12 Applications

HP-Z0000YPR-111005-02-001

Menus for setting application parameters are available under Applications. These settings are usually entered by the application specialist and not by the service engineer.

Pat. Registration

The attributes for the patient registration platform can be set in this menu.

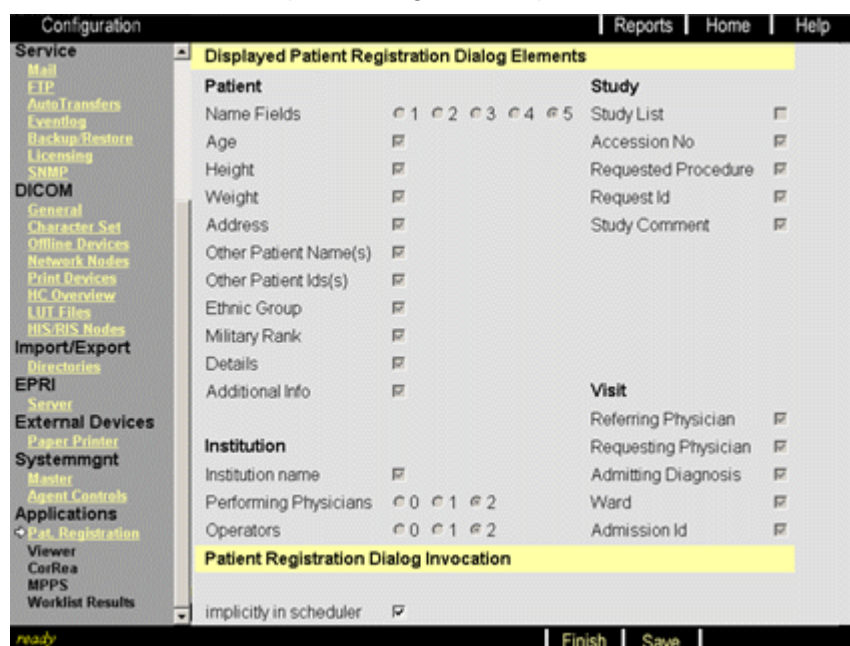


Fig. 58: Patient Registration

- **Displayed Patient Registration Dialog Elements**

Most of these attributes are predefined for the modality and cannot be changed here.

- **Patient Registration Dialog Invocation**

The Patient Registration platform may be opened from the Scheduler database by double-clicking it. If all required data are available for the patient examination, the Patient Registration platform is skipped and the Examination platform will already be opened.

Selecting the “implicitly in scheduler” box prevents the Patient Registration platform from being skipped before the Examination platform is opened.

Viewer

This menu lets you configure viewer settings for loading new studies or series. If the boxes are not selected, up to 3 studies will be kept open in the application and may be selected in the user interface (viewer) in the upper right corner (folder icons).



Fig. 59: Viewer

CorRea (Correct & Rearrange)

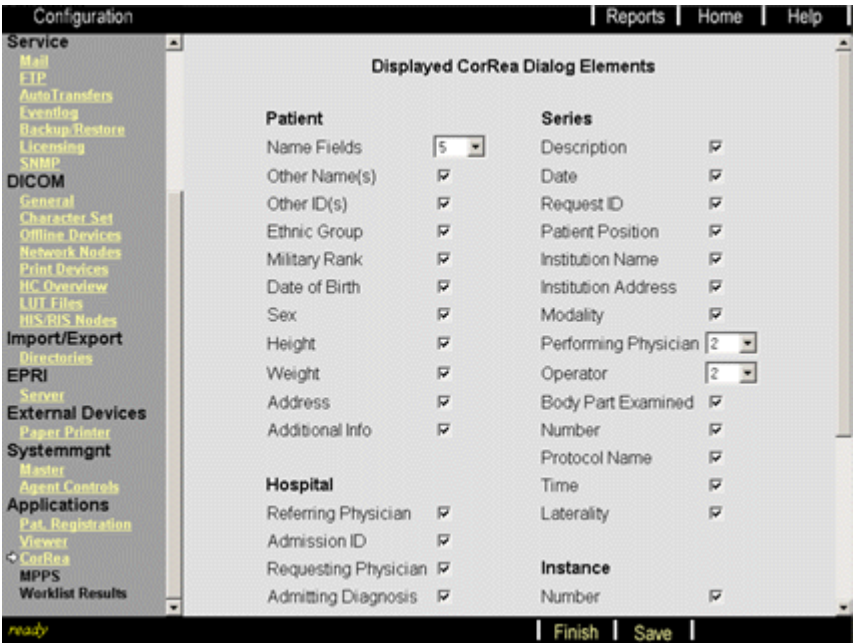


Fig. 60: Correct and rearrange (upper page)



Fig. 61: Correct and rearrange (lower page)

This menu allows you to set attributes which may be modified in the Correct&Rearrange platform.

MPPS (Modality Performed Procedure Steps)

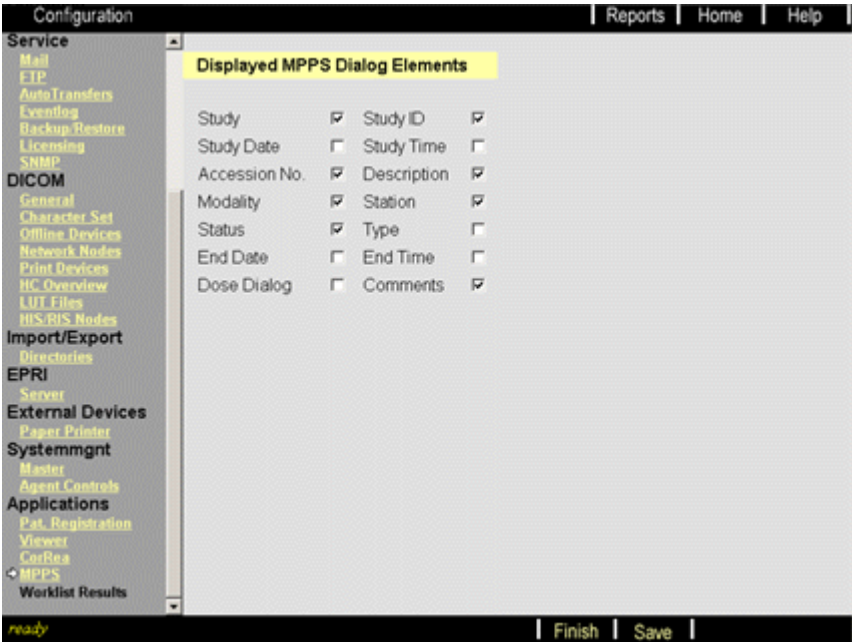


Fig. 62: Modality Performed Procedure Steps

These attributes are displayed in the customer user interface when a MPPS message is sent to the RIS and are set in this menu.

Worklist Results

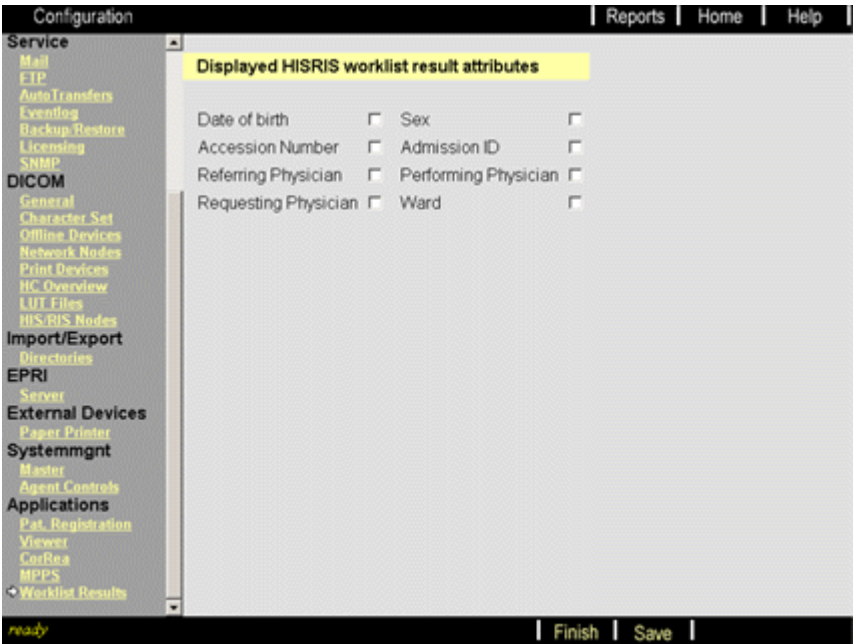


Fig. 63: Worklist Result

The attributes displayed after a DICOM worklist request in the Patient Registration form are set in this menu.

2.2.13 Measurement

HP-Z0000YPR-080206-02-999

The measurement section allows you to configure system-specific settings and measurements for the syngo MR-specific hardware and available options.

System Type

A document / CD-ROM is included with the system type of the syngo MR system delivery.

Meas. Settings

A document / CD-ROM is included with the delivery of the syngo MR system pre-configured measurement system configuration for your system type and options.

WARNING
--

The Specific Absorption Rate (SAR) is different for each country.

⇒ **For this reason, you must select the SAR guidelines that correspond to local guidelines and regulations.**

WARNING
--

The Specific Stimulation Monitor is different for each country.

⇒ **For this reason, you must select the Stimulation Monitor that corresponds to local guidelines and regulations.**

Avail. Coils

A document / CD-ROM is included with the delivery of the syngo MR system with site-specific data configured with all delivered coils.

Coils that are configured in the **Avail. Coils** configuration are automatically included in all automated workflows e.g. **QA** and **Maintenance**.

New coils are automatically detected and included in the coil list when plugged in.

NOTE

If a coil is removed from the system, deselect the checkbox for the respective coil.

2.2.14 SystemUtilization

HP-Z0000YPR-080206-02-998

System Utilization provides log and report data regarding the usage of syngo MR applications.

NOTE

To meet legal requirements, the customer must sign a contract in advance. Please contact your service organization before activating System Utilization. AutoReport for SYSUTIL must be configured before activating System Utilization.

NOTE

Only activate System Utilization for systems if the customer agrees to "Utilization Management".

3 Utilities

HP-Z0000YPR-111005-02-997

n.a.

3.1 General

3.1.1 Introduction

HP-Z0000YPR-111005-02-996

This section describes the **Utilities** functions.

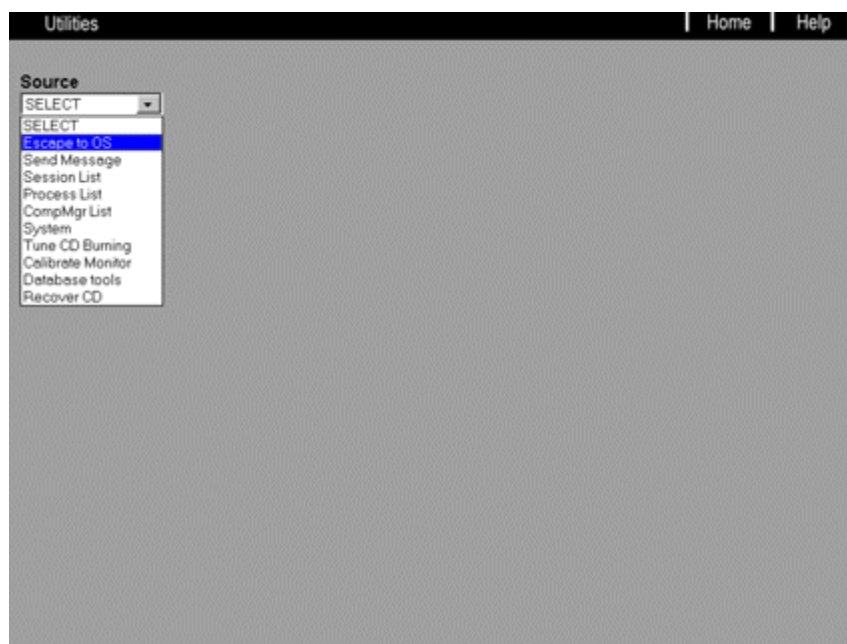


Fig. 64: Source

The **Utilities** functions include:

- Escape to OS
- Send Message
- Session List
- Process List
- CompMgr List
- System
- Tune CD Burning
- Calibrate Monitor
- Database Tools
- Recover CD

3.1.2 Features

HP-Z0000YPR-111005-02-995

With **Escape to OS**, you can enter any command at the OS level.

Send Message is useful for sending a message to the site.

The **Session List** provides information about the active sessions.

The **Process List** shows all active processes.

CompMgr List shows all active software components.

With **System**, you can shut down and restart the system or the application.

Tune CD Burning ensures that the customer can work properly in parallel to burning a CD.

The “**Calibrate Monitor**” tool adjusts the monitor for medical diagnostics.

The “**Database Tools**” help you check and repair the database.

The “**Recover CD**” tool will help to recover defective CDs.

3.2 Functions

3.2.1 Escape To OS

HP-Z0000YPR-111005-02-006

This function enables you to execute a command-driven program on the system. It provides the capabilities of a regular Command Prompt on the local system. Several predefined commands are available:

CAUTION

This function is mainly used with the support of headquarters.

⇒ Some commands may effect the syngo functionality.

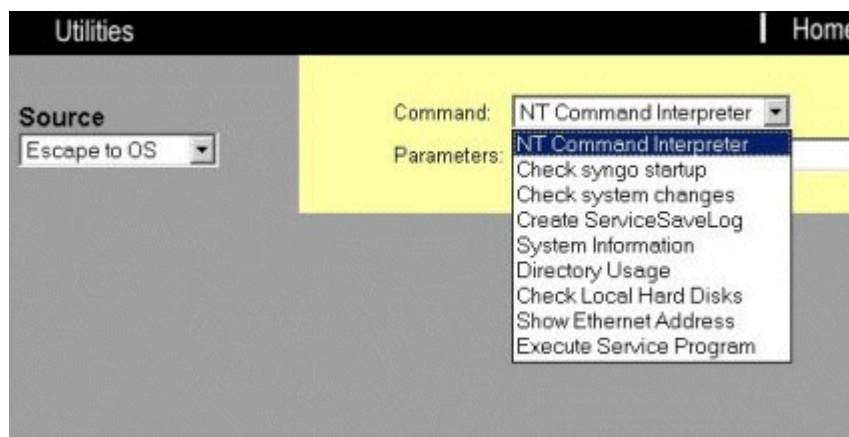


Fig. 65: Escape to OS - Commands

Most of these functions can be used without any parameters.

Use only parameters that are either supported by the development department or are displayed when you click **Go**.

- NT Command Interpreter
- Check syngo Startup
- Check System Changes
- Create ServiceSaveLog
- System Information
- Directory Usage
- Check Local Hard Disks
- Show Ethernet Address
- Execute Service Program

Remote - NT Command Interpreter

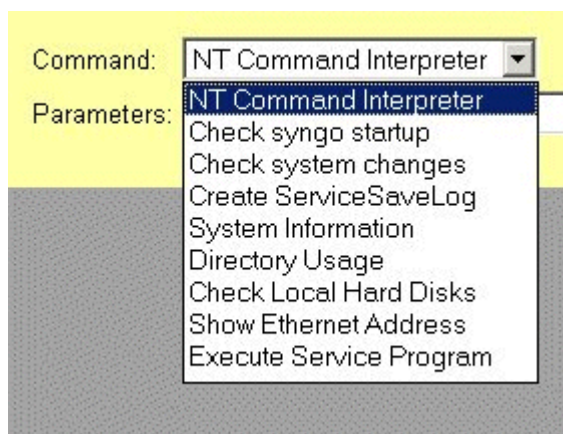


Fig. 66: *Escape to OS - NT Command Interpreter*

Patient Data Access

To start the **Escape to OS - NT Command Interpreter** remotely, you need the "Full access" service level.

When the **Patient Data Access** at the system is disabled, the following pop-up is displayed.



Fig. 67: *Patient Data Access is required*

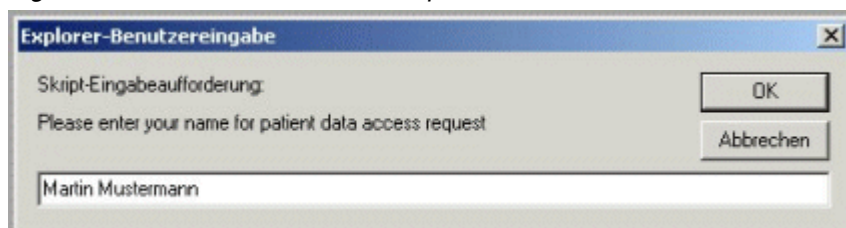


Fig. 68: *For Patient Data Access your name is required*

You need to enter your name. This name is used in the log file to indicate who accessed the patient data.

This is necessary to ensure patient data security.



Fig. 69: *SAM requests Patient Data Access at the remote system*

SAM requests patient data access at the system and the user at the system can enable the access.

Then you get this popup window:



Fig. 70: Patient Data Access changed to enabled

Local NT Command Interpreter

Escape to OS - NT Command Interpreter

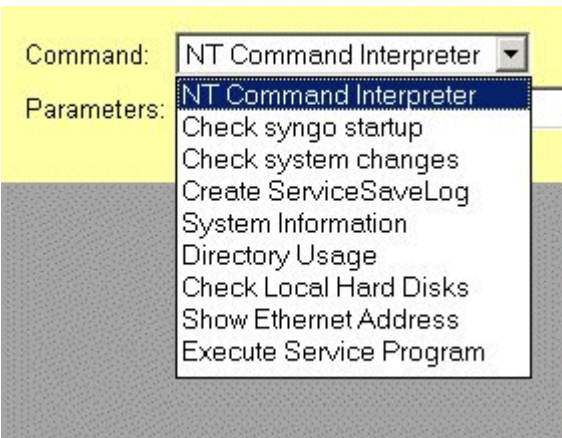


Fig. 71: Escape to OS - NT Command Interpreter

NT Command Interpreter

Starts a new instance of the Windows NT command interpreter CMD.EXE. This command interpreter accepts commands as they are typed in the Command Prompt window. Any program can be started, even programs with a graphical user interface. The GUI, of course, will be started on the local system only.

Helpful commands

For network tests:

Commands at OS level

IPCONFIG/ALL

PING magic1.hospital.org

Description

Check the local IP configuration

Check the low-level connectivity. Try using the name first. If that fails, try the IP address. If this does not work, even with the IP address, the problem is probably due to the NIC (Network Interface Card) drivers or routing. If ping works, but something high-level does not work, perform a test with NBTSTAT.

Commands at OS level	Description
NBTSTAT -c	See the IP address determined by the station to be associated with a name. If addressing is wrong, edit the LMHOSTS file (check HOSTS as well and/or WINS database); to reset the address, use the command NBTSTAT -R .
NETSTAT -rn	Determine whether the local and remote hosts have a default gateway specified.
NSLOOKUP magic1.hospital.org	If DNS is used, determine the corresponding Internet address. Reverse name lookup works as well.
Route	input of a static route (defines a route of a connection to a node)
TRACERT 155.2.42.117	Trace routing to a destination host (over a maximum of 30 hops).

Various commands:

Commands at OS level	Description
calc	start the Calculator
clipbrd.exe	start the ClpBook Viewer
compmgmt.msc	Computer Manager(Services)
control	start Windows control panel
control appwiz.cpl	Add/Remove Programs
control Desk.cpl	Display properties
control modem.cpl	display Modem settings
control srvmgr.cpl	User Manager
dblist	Database listing
eventvwr	start Windows Event Viewer
explorer.exe	start Windows File Explorer
ncpa.cpl	start Windows File Explorer
notepad	start the Notepad
opcagt	Status of System Management
regedit	start the Registry Editor
rti startPmgt appl	start the Application (syngo)
rti stopPmgt appl	stop the Application(syngo)
shutdown -l	[Administrator] Shutdown of the syngo Application
start	Start of batch files
startcm	Start Component Manager
stopcm	stop the syngo application and the Component Manager
taskmgr.exe	start Windows task manager
timedate	opens Date and Time Properties
type <file>	display of File content

Check syngo Startup

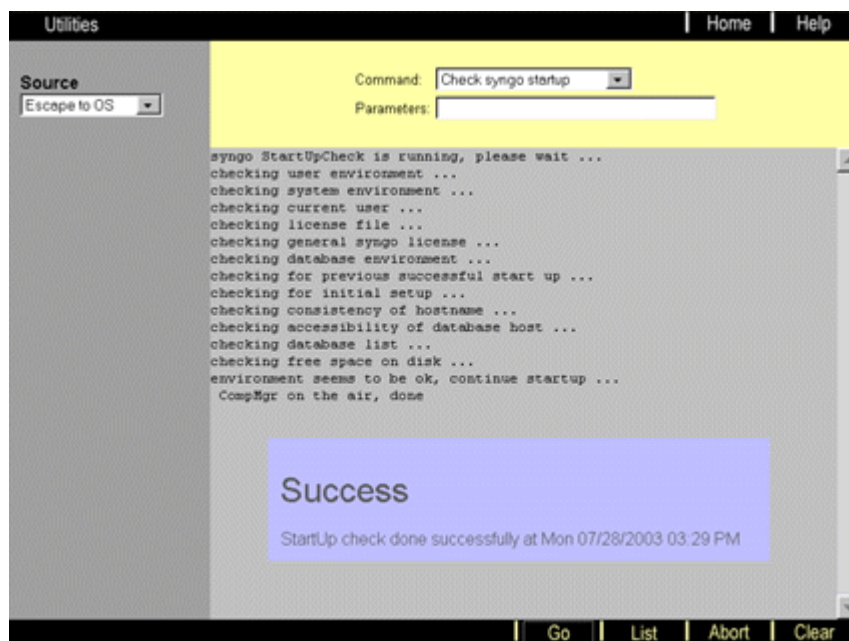


Fig. 72: Escape to OS - Check syngo Startup

The following items are checked and displayed: user environment, system environment, current user, database environment, previous successful startups, and free space on disk.

Check System Changes

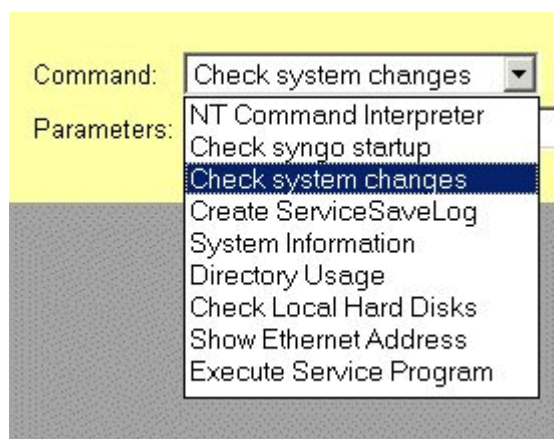


Fig. 73: Escape to OS - Check syngo Startup

After the first installation of the syngo software, the system will generate a checksum.

When you start this utility, the system will generate another checksum and compare it with the one created first.

This is a simple check to ensure that the system is not infected with viruses.

Create ServiceSaveLog



Fig. 74: Escape to OS - Create ServiceSaveLog

The description of the ServiceSaveLog is the same as for SaveLog customer.

Please go **back to overview** and then to **Customer -> SaveLog**.

System Information

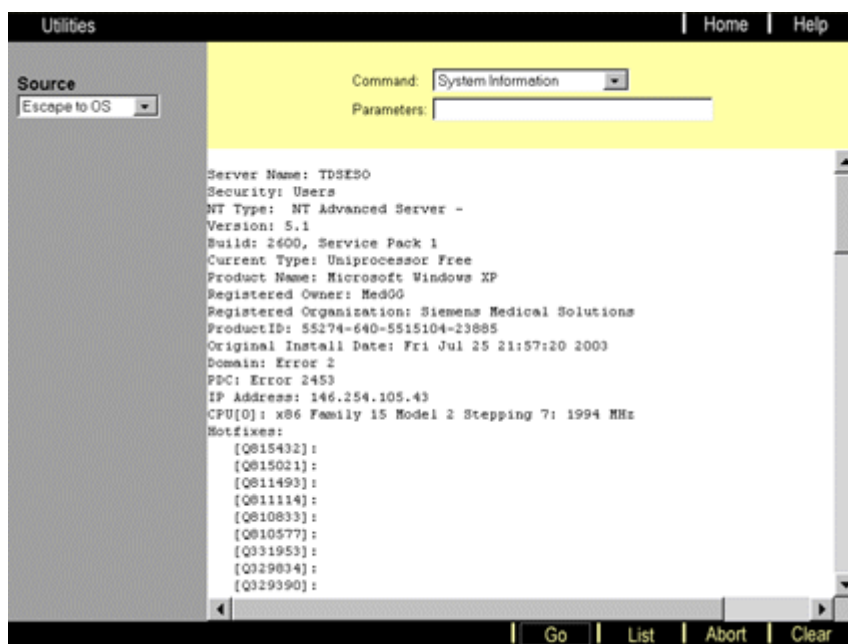


Fig. 75: Escape to OS - System Information

System Info shows the network IP address, host name, NT version and service pack, as well as the stopped and running processes.

Directory Usage

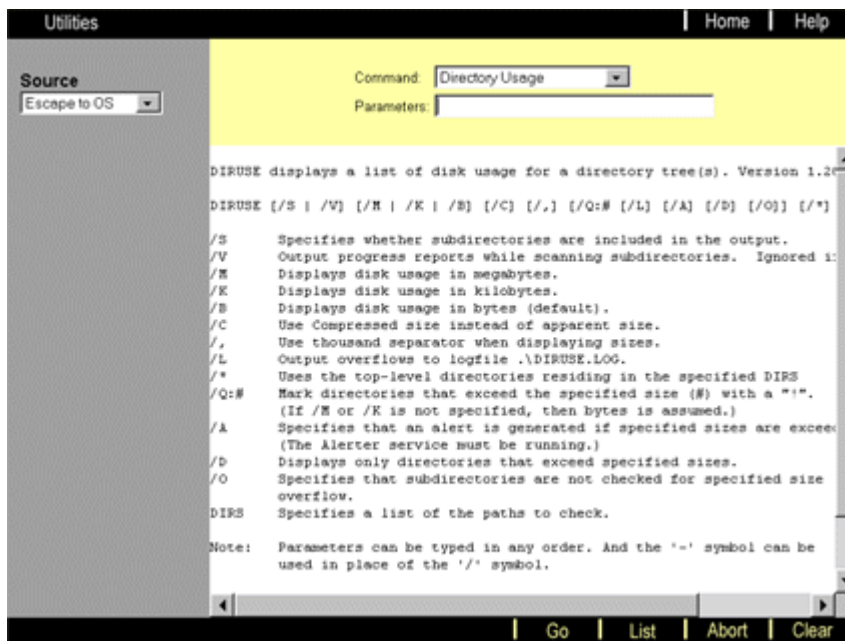


Fig. 76: Escape to OS - Directory Usage

DIRUSE displays a list of disk usage for a directory tree(s). Required parameters are listed. To obtain the corresponding list, click **Go** without a parameter.

Check Local Hard Disk

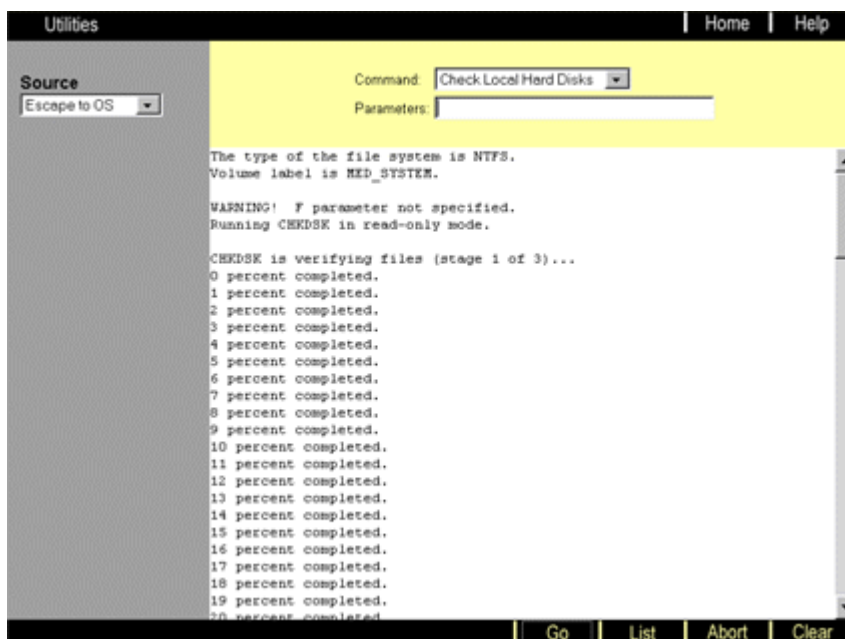


Fig. 77: Escape to OS - Check Local Hard Disk

All of the system hard disks can be checked using this command.

Show Ethernet Address

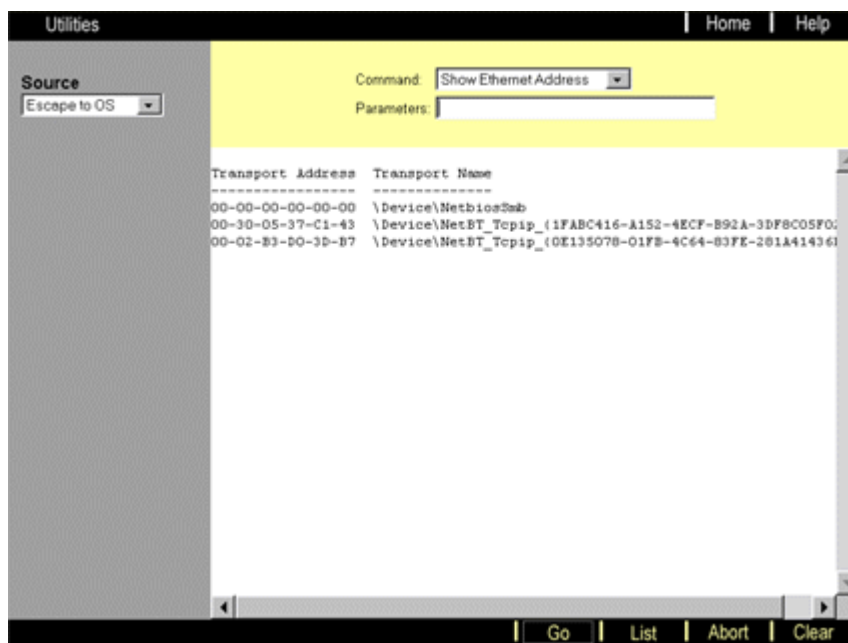


Fig. 78: Escape to OS - Show Internet Address

Show internet address will display the so-called MAC address from the ethernet card of the PC.

Execute Service Program

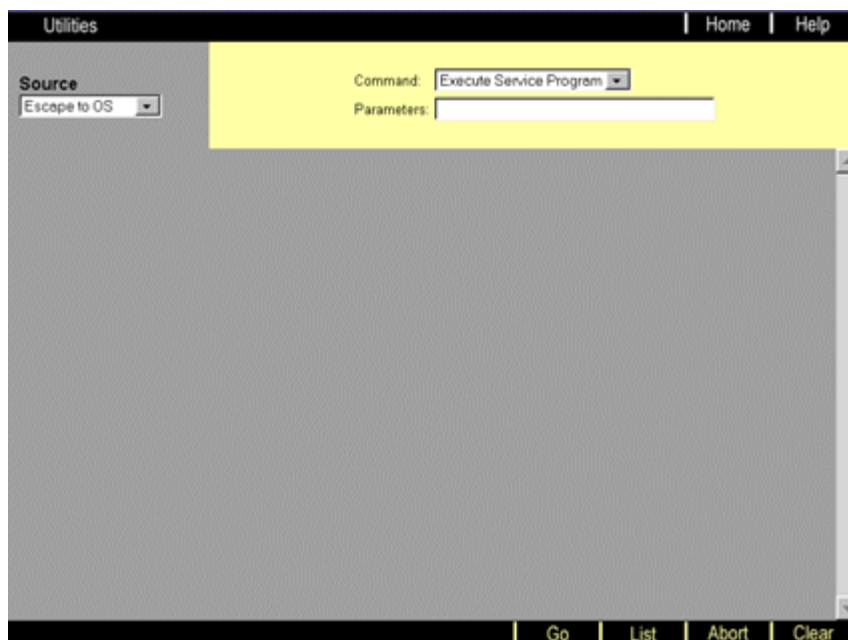


Fig. 79: Escape to OS - Execute Service Program

Execute Service Program

This command allows you to start service programs manually. This should be performed by experienced users (developers) only.

3.2.2 Send Message

HP-Z0000YPR-111005-02-007

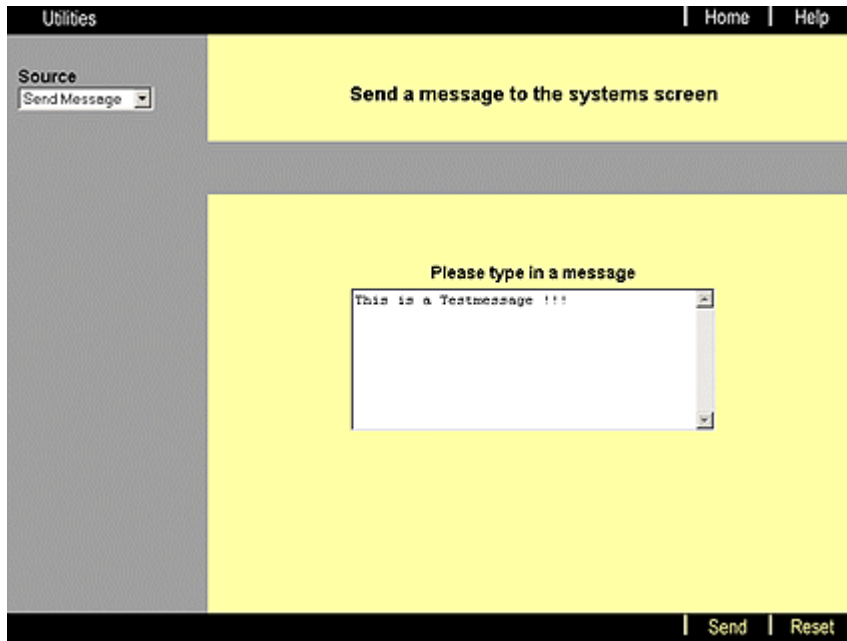


Fig. 80: Send Message - Send a message to the site.

It is possible to **send a message** to the customer during a remote session. The CSE simply needs to enter a text in the message text area and click the Send button to send the message.

As a result, the following message box will be displayed on the console.

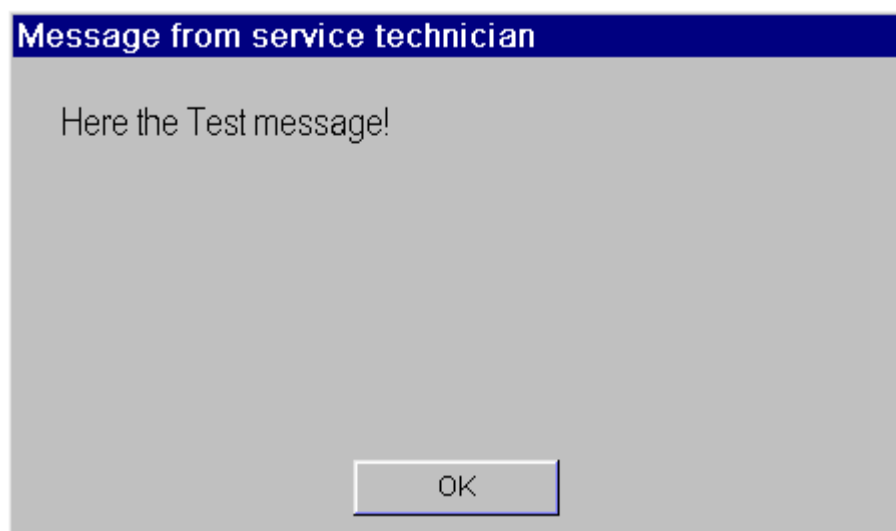


Fig. 81: Send message - Received message from the CSE

The message box must be confirmed by the customer by clicking the OK button. The message text will be logged in the Event log.

3.2.3 Session List

HP-Z0000YPR-111005-02-992

Session Key	Service Level	Started	Last Access	Status	Client
0000CEWV8IUDWVCDEFWFREGFDGE	SIEMENS	15:28:30, 28-Jul-03	15:38:35, 28-Jul-03	alive	127.0.0.1, local

Fig. 82: Session List

- **Session Key**
This is a session ID and does not contain the Service Key.
- **Service Level**
The service server of the remote access is displayed for each session.
- **Started**
The date and time at which the session is started will be displayed.
- **Last Access**
The date and time of the last access from each session is displayed.
- **Status**
Shows the current status of a session.
- **Client**
Shows the IP address of the remote client.

This displays all current sessions. Click **Refresh** to refresh the list.

To close all sessions, go to menu item **Options -> Remote access**, and switch to “no access”. (see also **Customer UI -> Remote Service**)

3.2.4 Process List

HP-Z0000YPR-111005-02-991

Pid	Process Name	Processor Time	Privileged	User	Mem Usage
0	Idle	000:35:22.671	100%	0%	20k
4	System	000:00:09.671	100%	0%	60k
544	smss	000:00:00.015	0%	100%	86k
592	csrss	000:00:08.515	80%	20%	1776k
620	winlogon	000:00:01.499	42%	58%	2308k
664	services	000:00:05.734	71%	29%	2924k
680	lsass	000:00:03.296	57%	43%	804k
852	svchost	000:00:00.218	43%	57%	2264k
932	svchost	000:00:17.984	27%	73%	11536k
1016	svchost	000:00:00.359	26%	74%	1176k
1076	spoolsv	000:00:00.156	10%	90%	2612k
1188	svchost	000:00:00.078	40%	60%	1924k
1248	inetinfo	000:00:02.312	65%	35%	6388k
1268	mgabg	000:00:00.812	4%	96%	320k
1296	NMSSvc	000:00:00.562	67%	33%	1656k
1320	serema	000:00:00.015	0%	100%	340k
1356	Sam	000:00:00.953	77%	23%	6776k
1424	snmp	000:00:00.078	40%	60%	1776k
1448	versantd	000:00:00.156	70%	30%	796k
1484	CsaCompMgrInit	000:00:00.031	50%	50%	444k
1924	CsaSecServer	000:00:04.890	40%	60%	19288k
2032	TransferMgr	000:00:00.031	50%	50%	1376k
144	AutoReport	000:00:00.031	50%	50%	360k

Fig. 83: Process List

- **Pid**

The ID of a process is displayed.

- **Process Name**

This is the name of the running process.

- **Processor Time**

The estimated processor time for a certain process.

- **Privileged**

This is the load in % of the Kernel mode.

- **User**

This is the load in % of the User mode.

- **Mem Usage**

Memory used for a certain process.

Information is provided about the processes currently running. Click **Refresh** to refresh the list.

3.2.5 CompMgr List

HP-Z0000YPR-111005-02-990

This function shows the software components currently running.

Process Name	Pid	State	Executable	Cre
BeginInfrastructureSection	0	NotExist		on_d
RemoveTmpFiles	0	NotExist	C:\WINNT\system32\cmd.exe	at_b
ResetErrorCondition	0	NotExist	CompMgrConfig.exe	at_st
ServiceBootHook	0	NotExist	ServiceBootHook.bat	at_st
StartupCheck	0	NotExist	Startup_Check.bat	at_st
EndInfrastructureSection	0	NotExist		on_d
BeginServerSection	0	NotExist		on_d
gdbstop	0	NotExist	C:\syngo\utils\stop_all_gdb.bat	at_st
WaitForRemote	0	NotExist	C:\syngo\utils\WaitForRemote.bat	at_st
checkDbid	0	NotExist	C:\syngo\utils\OscCmdWrapper.bat	at_st
gdbstop_startup	0	NotExist	C:\syngo\utils\OscCmdWrapper.bat	at_st
exchb_remove	0	NotExist	C:\syngo\utils\OscCmdWrapper.bat	on_d
exchb_stop	0	NotExist	C:\syngo\utils\ExchbStop.bat	at_lo
exchb_startup	0	NotExist	C:\syngo\utils\OscCmdWrapper.bat	at_st
PxCheckForDiskSpace	0	NotExist	CheckForDiskSpace.bat	at_st
patex_container	3812	Running		on_d
patex_server	3812	Running		at_st
IPCheckForDiskSpace	0	NotExist	CheckForDiskSpace.bat	at_st
IMemoryServer	3920	Running	IMemoryServer.exe	at_st
IPTServer	4068	Running	IPTServer.exe	at_st
gi_container	0	NotExist		on_d

Fig. 84: Component Manager List

Click **Refresh** to refresh the list.

- **Process Name**

A logical name for the process or component to be used in commands and other actions.

- **Pid**

The Process ID is 0 if the process is not running. Components obtain the Pid from the hosting container.

- **State**

Current state of the component or process.

- **Executable**

The executable file name.

- **Creation at**

Specifications relating to when a component or process is started. at_boot = after system startup, at_startup = if user logged in.

- **Lifetime**

Specifications relating to how long a component or process should live.

- **Lifetime value**

A number based on the "Lifetime" parameter

Source	Executable	Creation at	Lifetime	Lifetime value	Faults	Path type	On exit	Cwd
CompMgr List		on_demand	forever	30	0	application	ignore	
	ve	at_boot	forever	30	0	executable	ignore	
	system32\cmd.exe	at_boot	forever	30	1	executable	ignore	
	onfig.exe	at_startup	by_application	30	1	executable	ignore	
	ieck.bat	at_startup	by_application	30	0	executable	ignore	
		on_demand	forever	30	0	application	ignore	
		on_demand	forever	30	0	application	ignore	
		on_demand	by_components	1	0	backend	ignore	
		at_startup	by_application	30	0	component	restart	
	tlis\stop_all_gobs.bat	at_shutdown	forever	30	0	executable	ignore	
	tlis\WaitForRemote.bat	at_startup	by_application	30	1	executable	ignore	
	tlis\OscCmdWrapper.bat	at_startup	by_application	30	1	executable	ignore	
	tlis\OscCmdWrapper.bat	at_startup	forever	30	1	executable	ignore	
	tlis\OscCmdWrapper.bat	on_demand	forever	30	0	executable	ignore	
	tlis\EschbStop.bat	at_logoff	forever	30	0	executable	ignore	
	tlis\OscCmdWrapper.bat	at_startup	forever	30	1	executable	ignore	
	hiskSpace.bat	at_startup	by_application	30	1	executable	ignore	
		on_demand	by_components	1	0	backend	ignore	
		at_startup	by_application	30	0	component	restart	
	hiskSpace.bat	at_startup	by_application	30	1	executable	ignore	

Fig. 85: Component Manager List - Scrolled to the right

- **Faults**

A count of how often a component or process was terminated. Note: This is not an error if e.g. a batch file terminates a component or process.

- **Path type**

Type of executable. Can be either an external executable, backend or frontend.

- **On exit**

Specifications regarding whether the component or process should be restored in the case of an exception.

- **Cwd**

Current working directory

3.2.6 System

HP-Z0000YPR-111005-02-989

Here you can shut down or restart the system or an application.

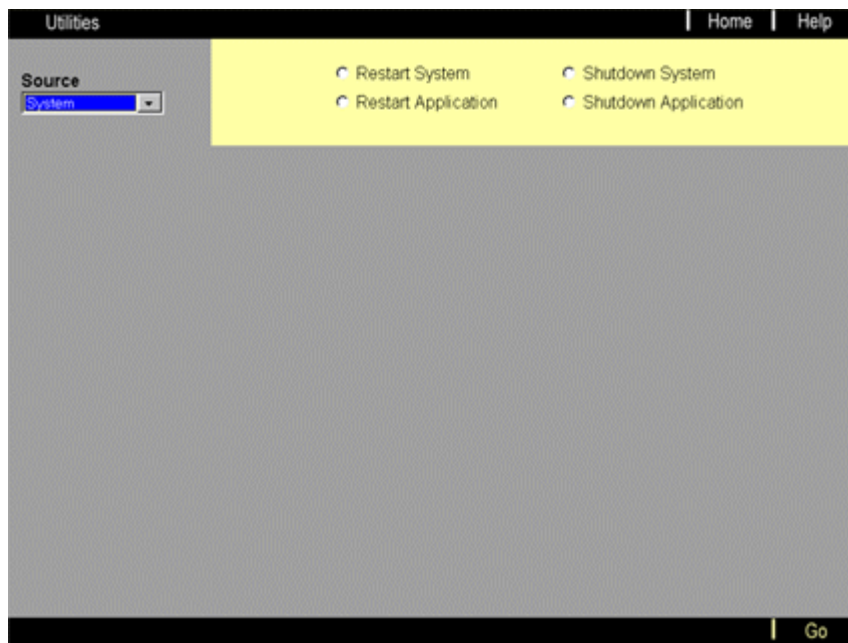


Fig. 86: System

3.2.7 Tune CD Burning

HP-Z0000YPR-111005-02-988

This tool is automatically started at the time of a new installation.

This tool must be started when a hardware part is modified or upgraded with hardware options.



Fig. 87: Tune CD Burning

“Tune CD Burning” checks the performance of a system to ensure the data stream for CD burning.

This information is used to ensure that the user can work properly in parallel to the burning process.

During the burning process, data is stored at a specific location on the hard disk.

“Tune CD Burning” requires approx. 5 minutes and runs in the background. After completion the system must be rebooted.



Fig. 88: Popup message after approx. 5 minutes.



If “Tune CD Burning” is running,

⇒ CDs may not be burned at the same time.

3.2.8 Calibrate Monitor

HP-Z0000YPR-111005-02-987

This tool calibrates the monitor for medical diagnostics
with the Minolta Luminance Meter LS100

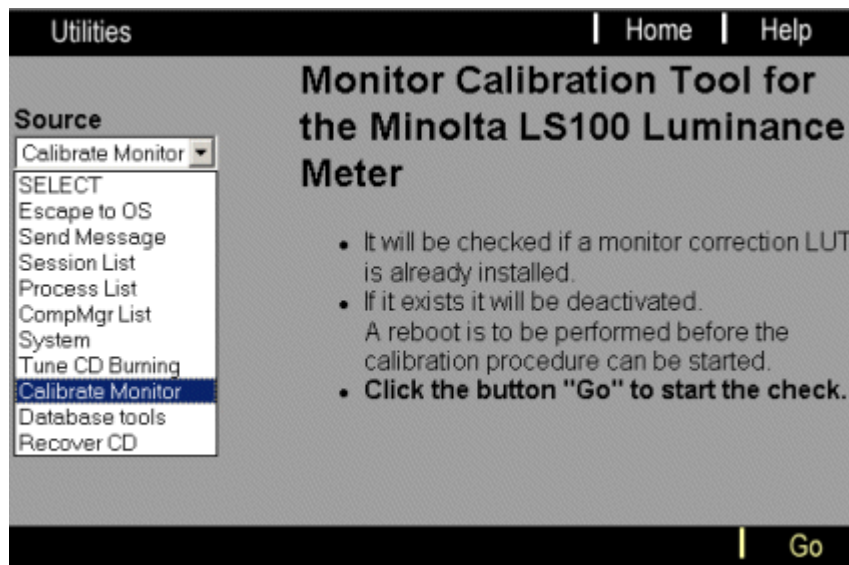


Fig. 89: Calibrate Monitor

Click **Go** to check whether a monitor correction LUT is already installed.

If yes, it will be deactivated. If the file name is "%medhome%\config\common\LUT\monitor.lut", it will be renamed "monitor.lut.bak".

A reboot will be performed.

If no LUT is installed, you may proceed with the calibration immediately.

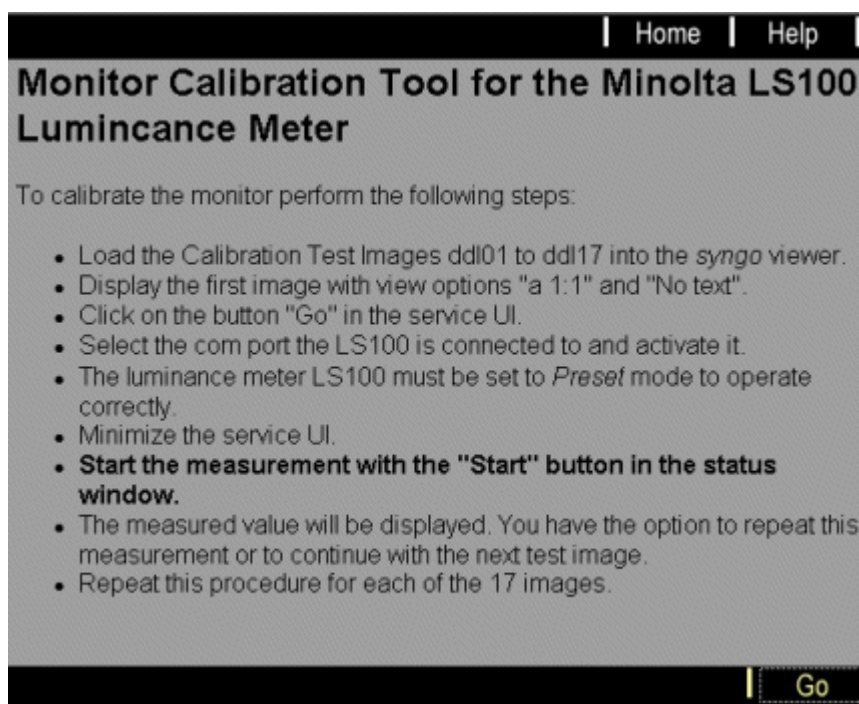


Fig. 90: Calibrate Monitor #1

After selecting **Go**, the ActiveX control for receiving data from the LS100 is displayed.

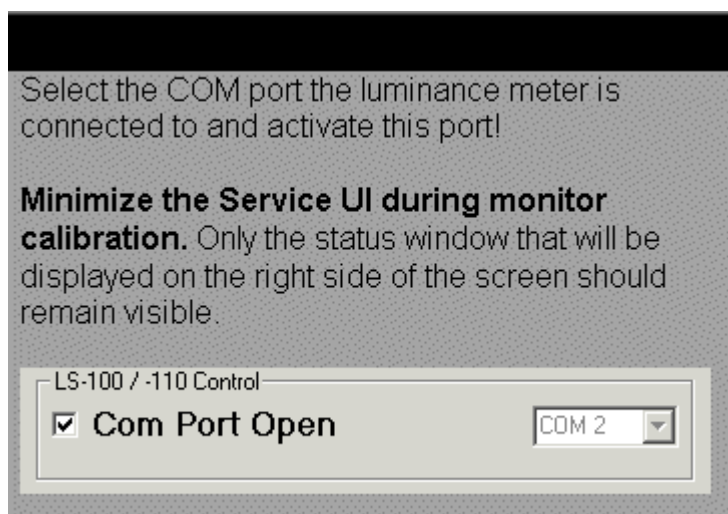


Fig. 91: Calibrate Monitor #2

Select the com port to which the luminance meter is connected and activate it with the **Com Port Open** button.

The calibration mode of the LS100 has to be set to "PRESET" in the measuring mode of "ABS". Otherwise, a warning will be displayed.

The 17 GSDF test patterns "ddl_01.dcm" to "ddl_17.dcm" have to be loaded into the syngo viewer.

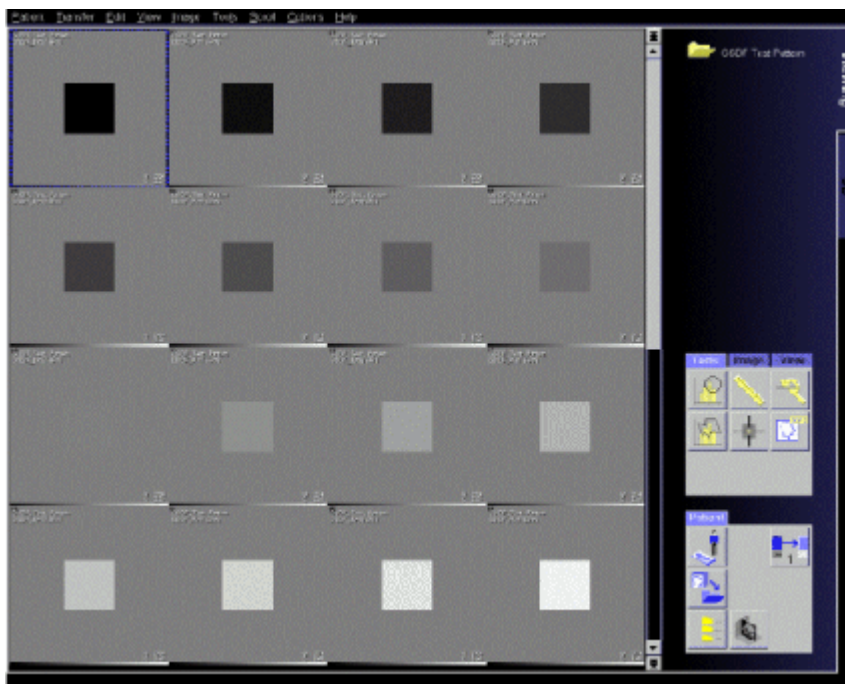


Fig. 92: Calibrate Monitor #3

To measure the luminance of the target (square region in the middle of the image), each image must be displayed with view options “a 1:1” and “No text”.

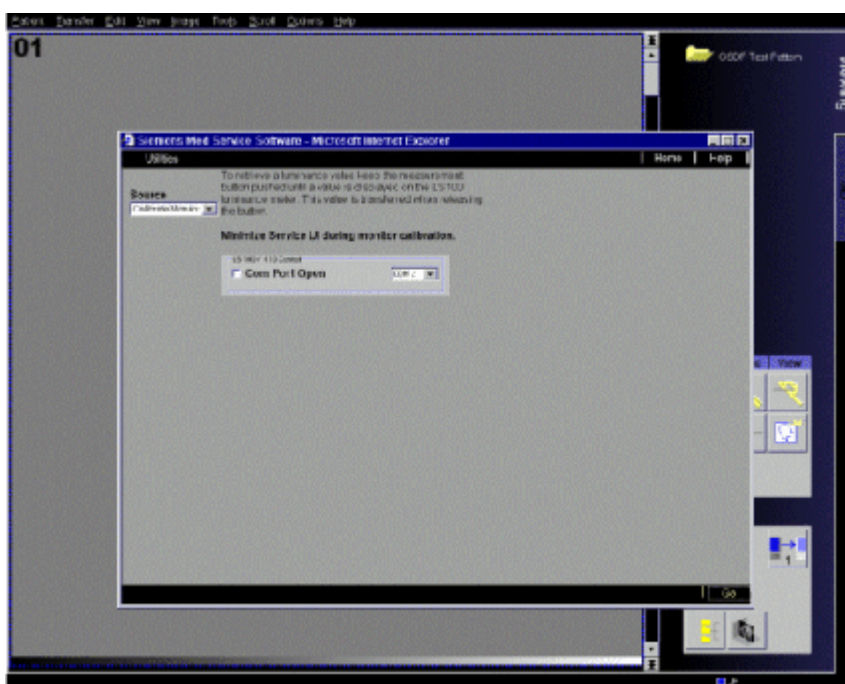


Fig. 93: Calibrate Monitor #4

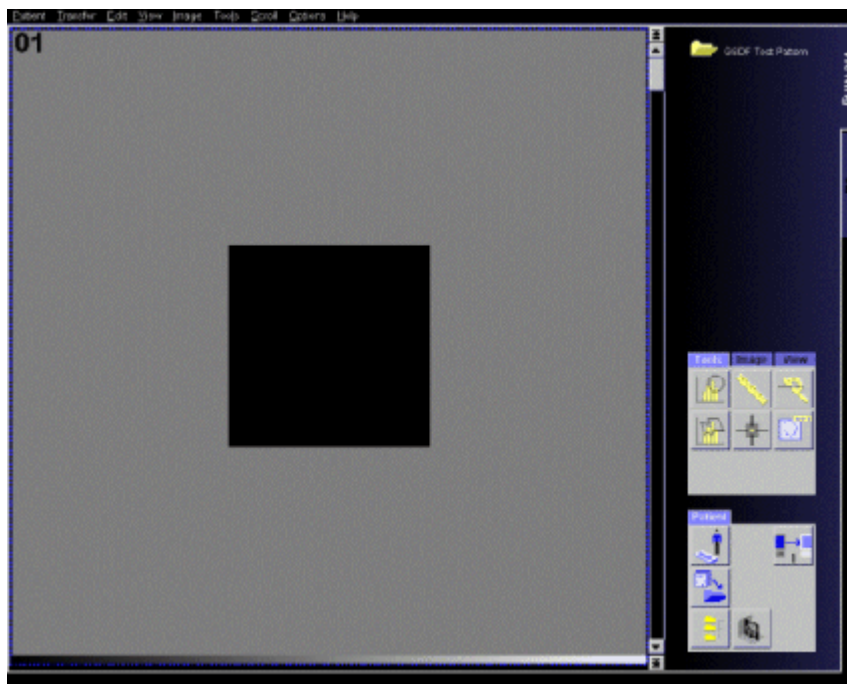


Fig. 94: Calibrate Monitor #5

Minimize the service UI so that only the calibration image is displayed on-screen. Only the narrow window on the right side should stay on-screen.

Click the Start button to measure the first value.

The value is displayed and may be accepted or rejected.

If the value is rejected, the last measurement will be repeated.

A list of all measured values will then be displayed.

If the values are accepted, they will be saved in a temporary file and a monitor correction LUT will be calculated.

The LUT file will be saved to “%medhome%\config\common\LUT\monitor.lut” and a reboot will be initiated to activate the monitor correction.

3.2.9 Database Tools

HP-Z0000YPR-111005-02-986

These tools are used to remove or repair the existing database.

Diagnostics

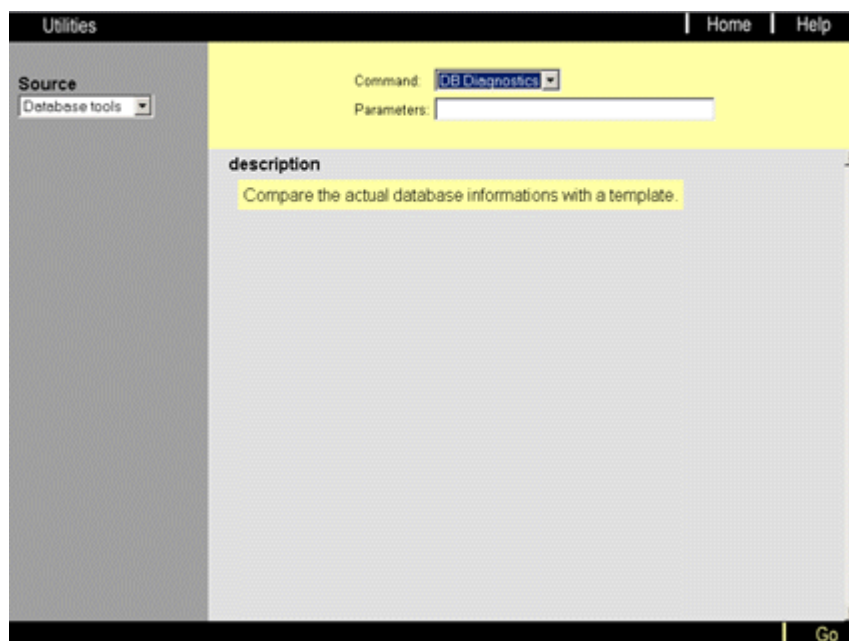


Fig. 95: Database Diagnostics

Click the **Go** button for the following results. This test takes a while.

Results

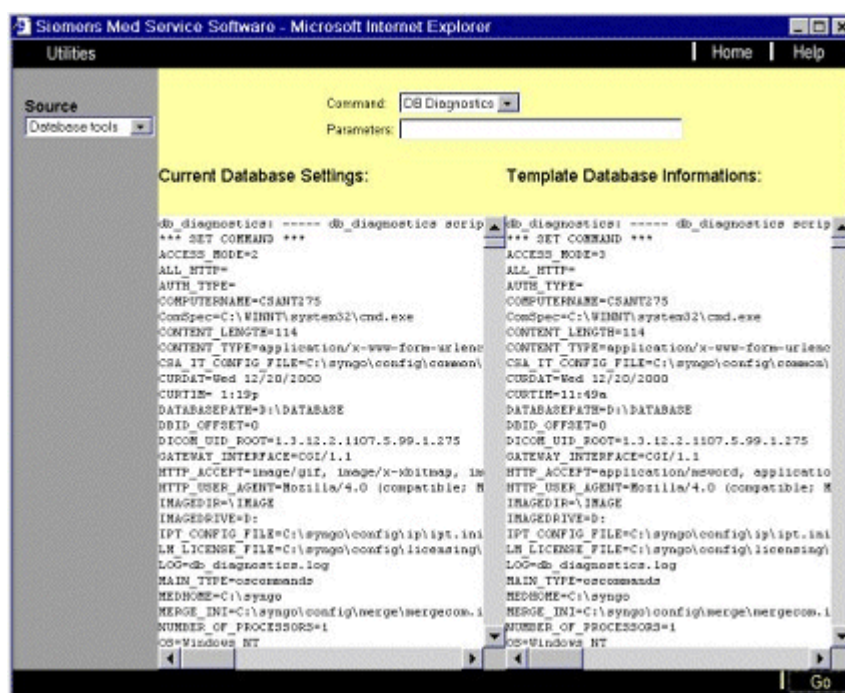


Fig. 96: Diagnostic Results

Current Database Settings: This is the current log file located under %MED-HOME%/log/db_diagnostics.log

Template Database Information: This is an example of a system containing a valid database. This information will never change.

If the CSE determines a significant difference between both sides, the file %MED-HOME%/log/db_diagnostics.log must be sent to headquarters for evaluation.

Remove

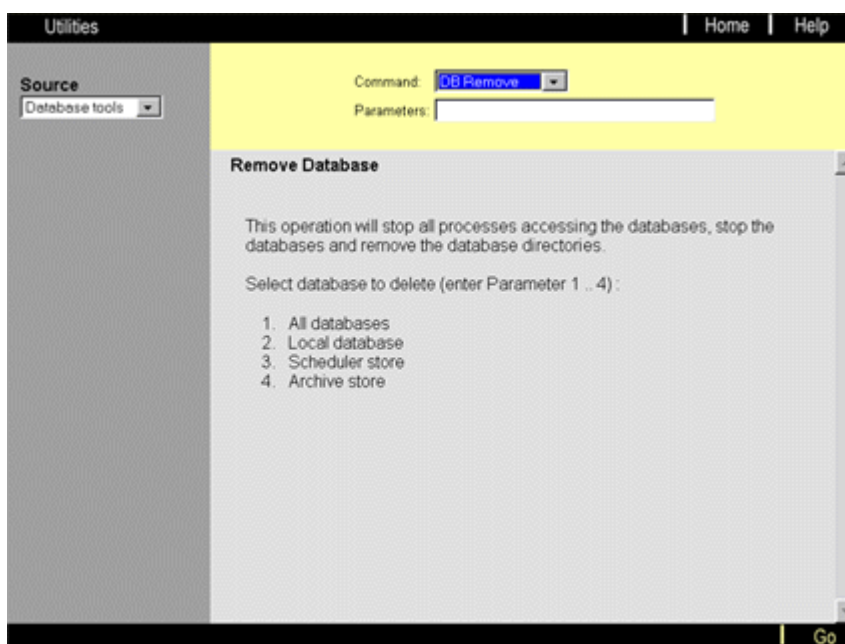


Fig. 97: Utilities - Remove Database

You can either select all databases or an individual database (store).

Enter one number (1-4) in the Parameters field and then click “Go”.

- 1 = All databases

Removes (2) Local Database, (3) Scheduler store and (4) Archive store.

- 2 = Local database

This is the database for patient images and patient data.

- 3 = Scheduler store

The HIS / RIS orders will be stored in this store. If it is removed, a new store will be created with the next HIS / RIS order.

- 4 = Archive store

When reading a CD, the data is stored to this archive store for a fast read.

This store will be automatically deleted when you start the system or by using this feature (remove database with parameter 4).

3.2.10 Recover CD

HP-Z0000YPR-111005-02-985

The customer is able to store several sessions on a single CD.

On rare occasions, it is possible that the newest session cannot be recorded correctly. In this case, the entire CD is unreadable.

The CSE can use the “Recover CD” tool to create a new session with pointers to the last “good” session and close it.

The last (“bad”) session will be lost, but all the others will be readable.

This procedure requires sufficient space on the CD and that the previous session is still readable.

Recovery may not always be possible.

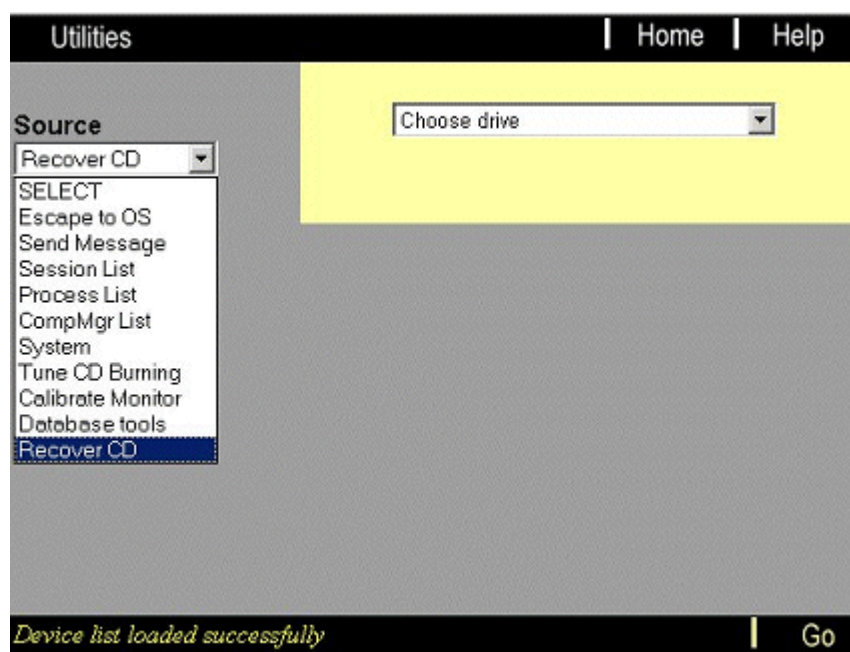


Fig. 98: Select Recover CD

Select **Recover CD** from the source drop-down menu.



Fig. 99: Select CD Drive to recover the CD

Select the CD burner drive to recover the CD.

4 Reports

4.1 General

HP-Z0000YPR-111005-02-016

n.a.

4.1.1 Introduction

HP-Z0000YPR-111005-02-017

The Report function is used to get an overall overview of the actual configuration. It provides additional access (by remote connection) to important system log files.

4.1.2 Features

HP-Z0000YPR-111005-02-982

The information and log files are clustered in groups as follows:

- SiteInfo
- Tune-Up/QA/Tests
- Startup Process
- Remote Diagnostics

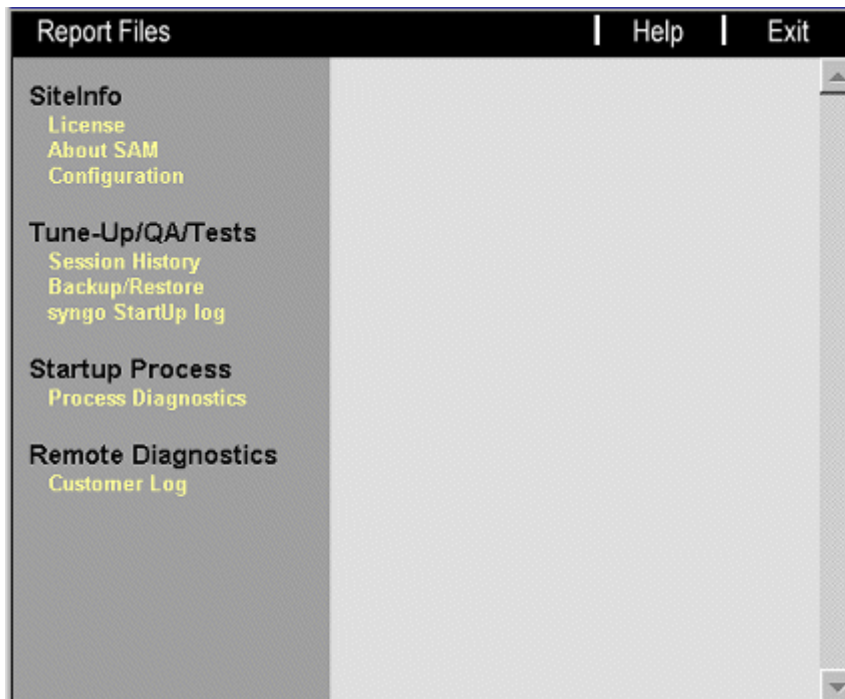


Fig. 100: Report files

4.2 Functions

HP-Z0000YPR-111005-02-981

n.a.

4.2.1 Handling

HP-Z0000YPR-111005-02-980

- Activate the **Reports** button in the Navigation bar of the Service Home menu to open a separate window. Available log files of service programs are shown in the left frame of the opened window. You can select an entry by clicking one of the specific reports as required.
- Activate the **Help** button in the Reports Navigation bar to obtain context-sensitive help information.
- Activate the **Exit** button in the Reports Navigation bar to close the Reports window.

4.2.2 SiteInfo

HP-Z0000YPR-111005-02-021

This menu entry provides information regarding the license, SAM, and the configuration for this specific site.

License

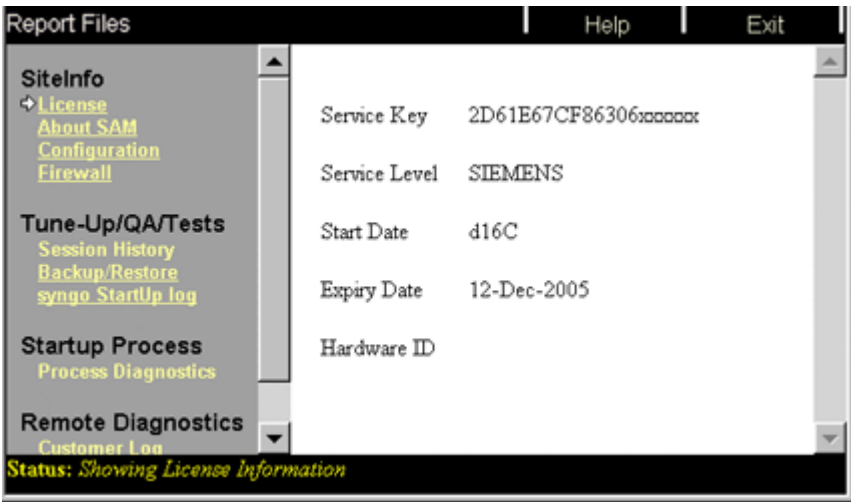


Fig. 101: License

Important license information is displayed, such as expiration date and hardware ID.

About SAM

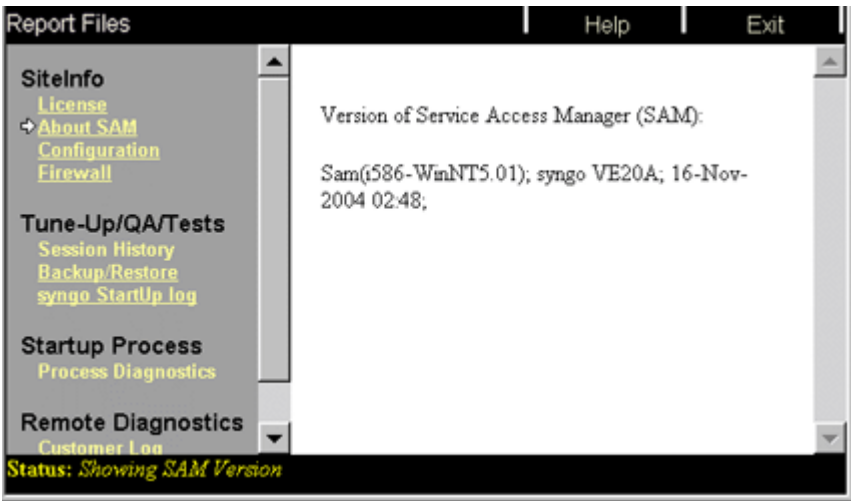


Fig. 102: About SAM

Displays the current version of the "Service Access Manager" (SAM). The SAM controls all the activities within the syngo Service Software and their modules.

Configuration

Configuration Report

use <CTRL+P> to print the selected frame

	dd/mm/yyyy	hh:mm:ss
created on	09/ 5/ 2003	14: 20: 20

System

System Option	
System type	syngo
option	Attached to Network
option	Modem
option	Main Console with Satellites or Satellite Console
option	DICOM Print Devices
option	DICOM Offline Devices
option	DICOM Networking
option	DICOM HIS/RIS
option	Image Import/Export
option	Paper Printer
option	EPRI
option	System Management

Local Host

↑	Site Info
---	-----------

Fig. 103: Configuration

A report of the system configuration is displayed. The report is divided into several options. Click an option to obtain the corresponding information. The white arrow within the Subtitle bar brings you back to the top of the page.

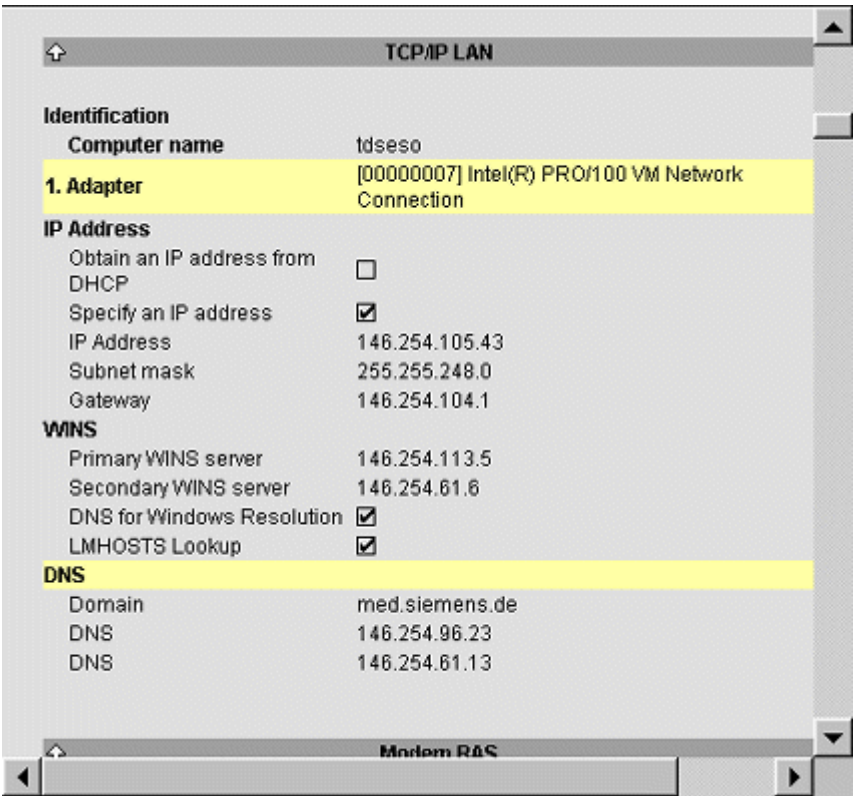


Fig. 104: Configuration - Network

Firewall

Displays the current Windows XP firewall (security) configuration. The firewall configuration displays which ports and programs are enabled on the system.

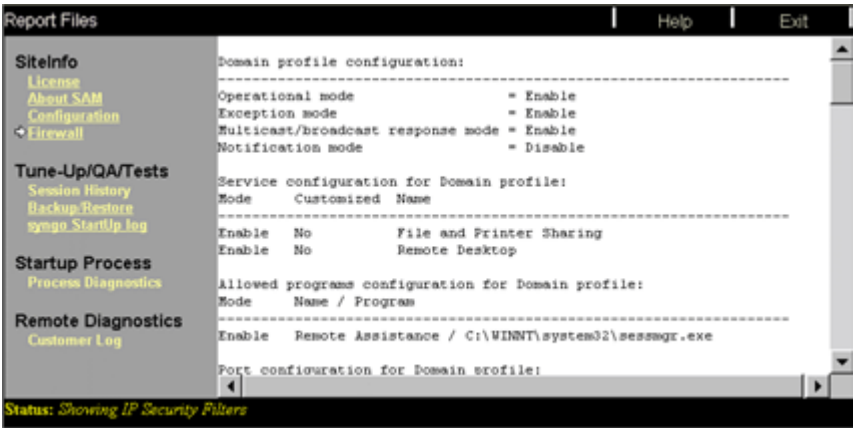


Fig. 105: Firewall Info Screen

4.2.3 Tune-Up/QA/Tests

HP-Z0000YPR-111005-02-978

This menu entry provides information on the status of the system with regard to:

- Tune-up
- Quality assurance
- Test functions (e.g. results of the latest measurements)
- Backup & Restore
- syngo Startup log

Session History

Session History	
on Wednesday, 14-May-2003, 10:16:55	
Backup SW-Settings02	14-May-2003 10:16:55
Restore D:TDSESO_SW-Settings02_20030428_110100_ok.ar	14-May-2003 09:14:52
Backup SW-Settings02	28-Apr-2003 12:42:36
Backup SW-Settings02	28-Apr-2003 11:01:00
Restore D:TDSESO_SW-Settings02_20030403_090704_ok.ar	23-Apr-2003 13:08:28

Fig. 106: Session History

Use this link to view the session log file. It displays the history of all measurements, quality assurance and tests performed.

Backup & Restore

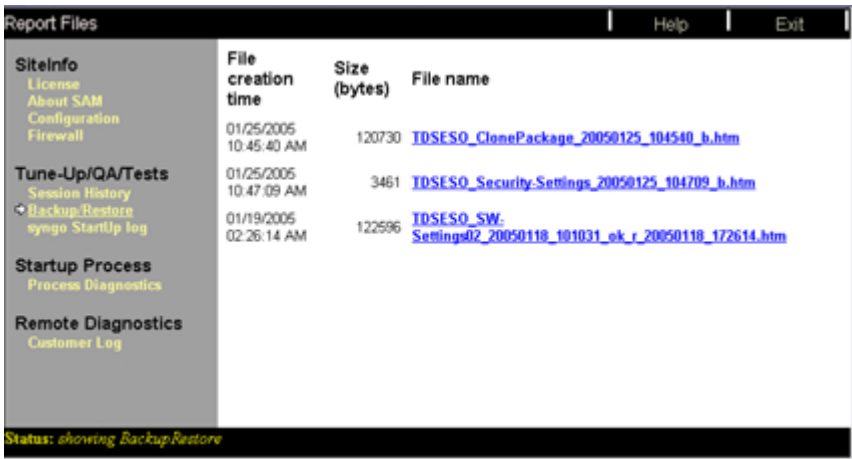


Fig. 107: Backup and Restore

Displays an overview of all backups performed on the system. It also provides a link to the log file that was written during the backup routine.

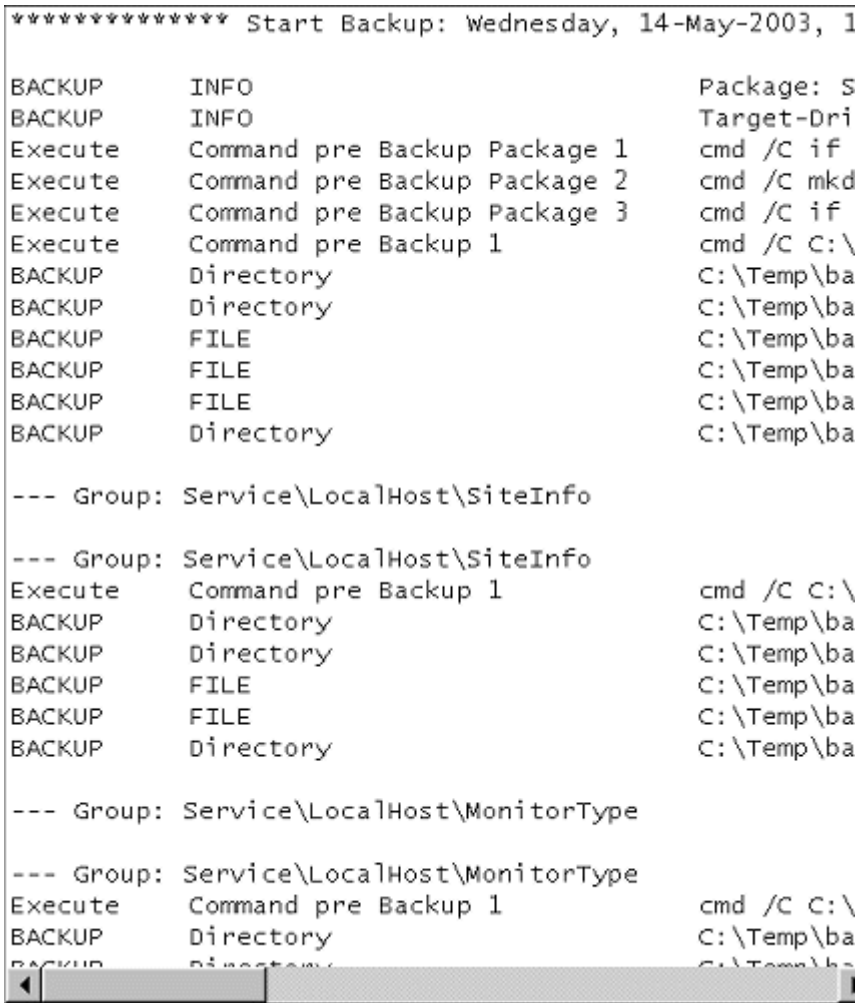


Fig. 108: detailed info for a saveset

This report shows which files / directories were saved and which programs were executed.

syngo Startup log

```

syngo_StartUpCheck[1].log - Notepad
File Edit Format View Help
Activate C:\syngo\utils\Startup_Check.bat at wed 04/23/2003 on hc
Startup check /install finished at wed 04/23/2003 10:33 AM
"C:\syngo\utils\Startup_Check.bat run at wed 04/23/2003 10:37 AM f
found different user TEMP = C:\DOCUME~1\meduser\LOCALS~1\Temp and s
setting user TEMP to C:\Temp
found different user TMP = C:\DOCUME~1\meduser\LOCALS~1\Temp and sy
set user TMP to C:\Temp
Checking for 'UserSettings.bat meduser'
Performing 'UserSettings.bat meduser'

UserSettings: ----- UserSettings script started at wed 04/23/2003
Colordepth = 32
FontSetting return-code: 0

UserSettings: ----- UserSettings script finished at wed 04/23/2003
details of C:\syngo\temp\syngo_RunOnce.cmd
regsvr32 /c /s C:\syngo\bin\CapAtComCollection.dll
regsvr32 /c /s C:\syngo\bin\SecClientComComp.dll
output of C:\syngo\temp\syngo_RunOnce.cmd
Please enter license
Startup check launching error page at wed 04/23/2003 10:37 AM ...
rti->getNumOfProcs() failed, 0 entry(s) found
"C:\syngo\utils\Startup_Check.bat run at wed 04/23/2003 12:06 PM f
Checking for 'UserSettings.bat meduser'
UserSettings.bat meduser
Not necessary to perform 'UserSettings.bat meduser'
Please enter license
Startup check launching error page at wed 04/23/2003 12:06 PM ...
rti->getNumOfProcs() failed, 0 entry(s) found
"C:\syngo\utils\Startup_Check.bat run at wed 04/23/2003 01:19 PM f
Checking for 'UserSettings.bat meduser'
UserSettings.bat meduser
Not necessary to perform 'UserSettings.bat meduser'
wait for CM finished after 0 counts
Startup check done successfully at wed 04/23/2003 01:19 PM
Success
Startup check finished at wed 04/23/2003 01:19 PM
Startup check /version /init finished at wed 04/23/2003 01:23 PM
Checking for 'UserSettings.bat meduser'
UserSettings.bat meduser
Not necessary to perform 'UserSettings.bat '
wait for CM finished after 0 counts
Startup check done successfully at Tue 05/13/2003 01:47 PM
Success
Startup check /service finished at Tue 05/13/2003 01:47 PM
Startup check /version finished at Tue 05/13/2003 01:53 PM
  
```

Fig. 109: syngo Startup log

This report displays detailed information on the startup of syngo.

4.2.4 Startup Process

HP-Z0000YPR-111005-02-977

This menu entry provides information on the system startup processes.

Process Diagnostics

The screenshot shows a web application window titled "Report Files" with a sidebar on the left and a main content area on the right. The sidebar contains the following links:

- SiteInfo
 - License
 - About SAM
 - Configuration
- Tune-Up/QA/Tests
 - Session History
 - Backup/Restore
 - syngo StartUp log
- Startup Process
 - Process Diagnostics** (selected)
- Remote Diagnostics
 - Customer Log

The main content area is titled "localhost - /log/" and displays a table of log files. The table has three columns: Date/Time, PID, and File Name. The first row is highlighted with a blue background.

[To Parent Directory]		
5/14/2003 10:18 AM	1133069	3dBackup.
5/14/2003 9:16 AM	1122176	3dRestore
5/9/2003 2:20 PM	34146	All Cfg.x
5/8/2003 12:48 PM	34371	All Cfg.o
5/20/2003 3:28 PM	<dir>	Auditing
5/14/2003 10:17 AM	29099	bareServi
5/9/2003 2:20 PM	26	CFC Cfg.l
5/20/2003 12:58 PM	167	checkDbId
5/20/2003 8:46 AM	167	checkDbId
5/21/2003 8:58 AM	167	checkDbId
5/20/2003 12:59 PM	20619	create al
5/20/2003 8:47 AM	20617	create al
5/21/2003 8:58 AM	20617	create al
5/9/2003 2:20 PM	11596	CreateRep
4/23/2003 1:21 PM	968	CreateSer
5/20/2003 8:46 AM	68	CsaCM Pro
5/14/2003 2:33 PM	68	CsaCM Pro
5/14/2003 2:01 PM	68	CsaCM Pro
5/20/2003 3:28 PM	68	CsaCM Pro
5/20/2003 8:46 AM	41	CsaCM Pro
5/14/2003 2:33 PM	41	CsaCM Pro
5/14/2003 2:01 PM	41	CsaCM Pro
5/20/2003 3:28 PM	41	CsaCM Pro
5/20/2003 1:11 PM	22143	CsaCompMg
5/16/2003 10:01 AM	1108	CsaCompMg
5/14/2003 2:31 PM	910	CsaCompMg
5/21/2003 9:01 AM	6103	CsaCompMg
5/21/2003 8:59 AM	16386	CsaCompMg
5/21/2003 8:58 AM	5836	CsaCompMg
5/20/2003 1:11 PM	615	CsaCompMg
5/20/2003 1:11 PM	6627	CsaCompMg
5/21/2003 8:58 AM	0	CsaCompMg

Fig. 110: Process Diagnostics

Use this link to view the available startup process log files. An overview of all the corresponding log files for Process Diagnostics is displayed in the right-hand frame of the window. Click one of the hyperlinks to view the desired log file. The text editor starts automatically.

```

3dBackup[1].log - Notepad
File Edit Format View Help

#####
3D Backup Logfile
.
      Date of Backup: wed 05/14/2003
#####
.
10:18:03.00 starting Backup.bat...
10:18:03.00 displaying version info:
10:18:03.00 retrieving HKLM\SOFTWARE\Siemens\MedCom\Config\Modality
HKEY_LOCAL_MACHINE\SOFTWARE\Siemens\MedCom\Config\Modality\Global\S
      type = string
      predicate = [a-zA-Z0-9/\_~]*
      v1 = VC20H
10:18:03.00 retrieving HKLM\SOFTWARE\Siemens\MedCom\Config\Modality
HKEY_LOCAL_MACHINE\SOFTWARE\Siemens\MedCom\Config\Modality\Global\S
      type = string
      predicate = [a-zA-Z ->/\_~]*
      v1 = VD20I SL88P217
10:18:03.02 retrieving HKLM\SOFTWARE\Siemens\syngo\COMPONENTS\sw_3c
HKEY_LOCAL_MACHINE\SOFTWARE\Siemens\syngo\COMPONENTS\sw_3dvis
      VERSION = VD20I
      INSTINDEX = REG_DWORD 0x0000003b
      NAME = Sw 3D Visualization
10:18:03.02 retrieving HKLM\SOFTWARE\Siemens\syngo\COMPONENTS\sw_ve
HKEY_LOCAL_MACHINE\SOFTWARE\Siemens\syngo\COMPONENTS\sw_ver
      VERSION = VD20I
      INSTINDEX = REG_DWORD 0x00000001
      NAME = Sw Version
10:18:03.03 deleting folder /C:\Temp\BaRe3D
10:18:03.10 creating folder C:\Temp\BaRe3D
10:18:03.10 creating folder C:\Temp\BaRe3D\bitmaps
10:18:03.10 contents of folder folder C:\syngo\config\3D\bitmaps:
      volume in drive C is MED_SYSTEM
      volume serial Number is 00B5-63C5

      Directory of C:\syngo\config\3D\bitmaps

04/23/2003  07:15 PM  <DIR>          .
04/23/2003  07:15 PM  <DIR>          ..
06/26/2002  09:37 AM             30,054  ax_gold.bmp
06/26/2002  09:37 AM             30,054  ax_white.bmp
06/26/2002  09:37 AM             30,054  ct_colon.bmp
06/26/2002  09:37 AM             30,054  ct_trachea.bmp
06/26/2002  09:37 AM             30,054  ct_vessel.bmp
01/10/2002  02:03 AM             30,054  Default1.bmp
02/13/2002  08:20 AM             30,054  DefaultFly.bmp
06/26/2002  09:37 AM             30,054  mr_inner_ear.bmp
06/26/2002  09:37 AM             30,054  mr_vessel.bmp
               9 File(s)          270,486 bytes

```

Fig. 111: 3dBackup

This window shows the results of one process.

4.2.5 Remote Diagnostics

HP-Z0000YPR-111005-02-976

This menu entry provides information about the customer log file.

Customer Log

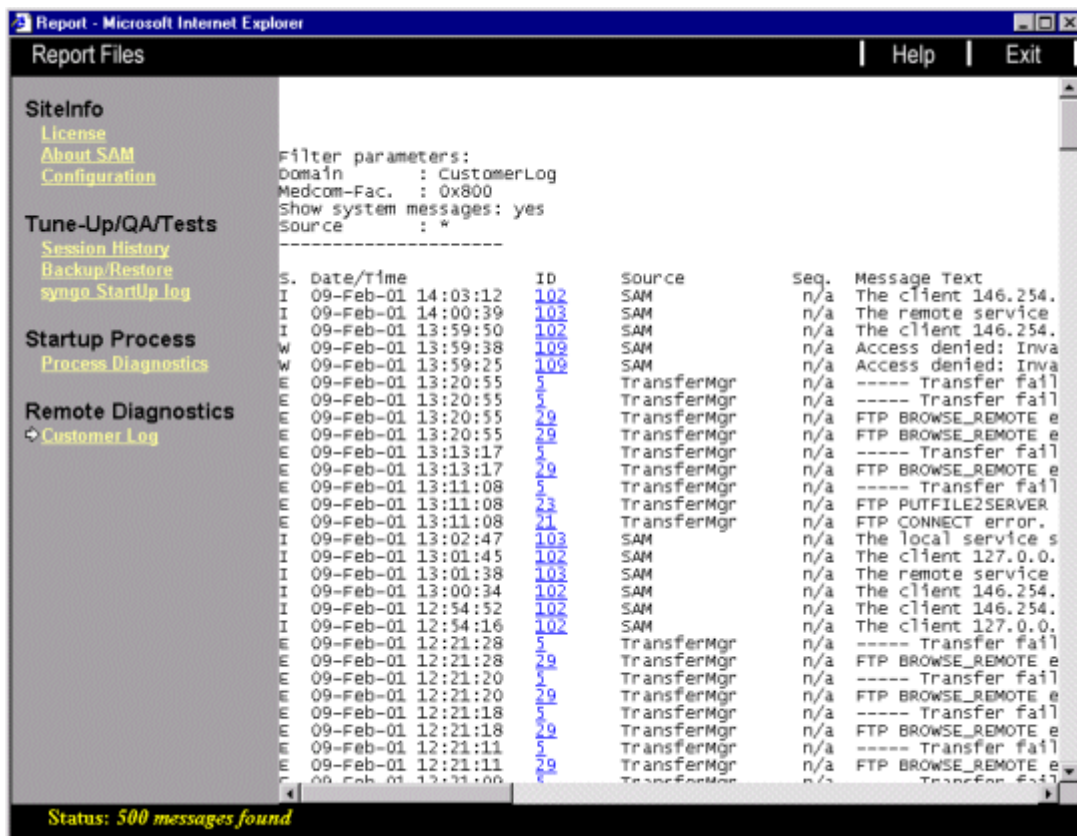


Fig. 112: Customer Log

Use this link to view the Remote Diagnostics customer log, which contains the following:

- Login trails
- Date and time of login
- Date and time of logout
- Who terminated the service session (local or remote service user)
- Service activities as defined within each service program

Click the ID number for more information about the specific message.

Message Description			Documentation	Exit
Message:	E	CAP_AN_30	CAP_AN_IMA_MEDIA_ABSENT	
Text:	Ima Media absent - %1 (Load/FileName/Destination - %2)			
Explanation:	No media is inserted in the drive.			
Action:	Insert media in drive.			
Information:				
Last Update:	Monday, March 21, 2005			

Fig. 113: Message Description

5 Auto Report

HP-Z0000YPR-111005-02-975

n.a.

5.1 General

HP-Z0000YPR-111005-02-974

n.a.

5.1.1 Introduction

HP-Z0000YPR-111005-02-973

Auto Report is used during various work functions:

- Once configured and enabled, Auto Report is automatically activated once the various filter criteria are met. This supports you, the service engineer, during troubleshooting. In particular, sporadic problems can be located and reported to the Service Center.
- In addition, Auto Report can trigger a preferred file transfer to collect certain system reports for further evaluation at the Service Center.
- The Auto Report tool can send a system report to any mail account at Service Headquarters.
- FTP (File Transfer Protocol) is used for file transfers.
- SMTP (Simple Mail Transfer Protocol) is used for mail communication.

5.1.2 Features

HP-Z0000YPR-111005-02-972

Auto Report tool features:

- Sending various notifications, from mail to mail messages including command outputs or file attachments to a particular user at the Service Center.
- Definition of simultaneous, multiple reports.
- Monitoring selectable files, **Time Triggered** and **Event Triggered**.
 - **Time Triggered** means that the Auto Report application searches for a word or a combination of words in a defined ASCII file or in the event log at an adjustable time.
 - **Event Triggered** means that Auto Report becomes active when a defined event occurs, e.g. the DISK_FULL event triggers Auto Report to send a mail to Headquarters, as is the case when a file has changed.
- Collecting system data at an adjustable time and sending it to Headquarters at flexible intervals.

5.2 Functions

HP-Z0000YPR-111005-02-971

n.a.

5.2.1 Enter Auto Report

HP-Z0000YPR-111005-02-970

Auto Report can be started from the Home menu by selecting the Auto Report button. The **REPORT** selection box containing the site-specific reports and the **Enabling Status** is displayed.

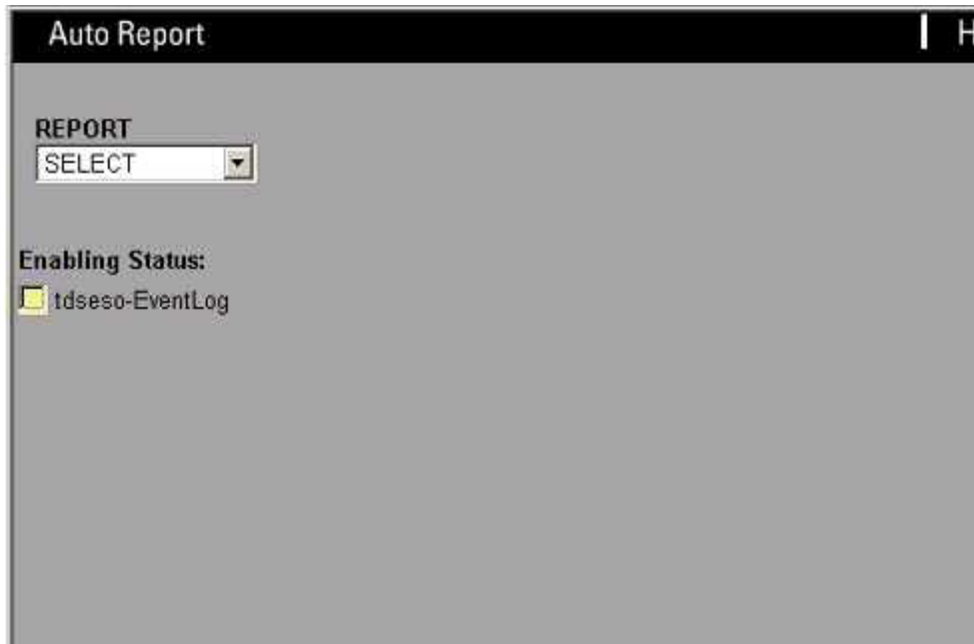


Fig. 114: Start the Auto Report Tool

- **REPORT**

The **Report** selection box contains a list of system-specific reports.

- **Enabling Status**

Enabling Status indicates whether the report is enabled or disabled. To change the status, you must open the report via the **REPORT** selection box and modify the **Active** checkbox. If this box is checked, the report is enabled. By clicking **Save** in the Action bar, the status under **Enabling Status** will be updated.

If a report or a template form has been selected from the **REPORT** selection box, the full status of the report is shown in the function frame.

Fig. 115: Example of the full Auto Report Tool

- **Active**

This checkbox is used to enable or disable a report. The action takes effect when the change is saved via the **Save** button in the Action bar. The status of this checkbox is also displayed in the left-hand frame under **Enabling Status**.

- **Name**

When the **TEMPLATE** has been selected, you must enter the name of the report to be created. If you have selected an existing report and then changed the name, a new report will be created when the **Save** button is clicked.

NOTE

When renaming a report, the “old” report that has been used as a template does not get removed or modified in any way.

- **Target**

Displays the remote computer to which the mail is to be sent. This has to be configured in the Service **Configuration** function. Mail can be sent to a modality-specific account on the Remote Server (e.g. rdiag_ct) or to a specific user (e.g. Dieter.Mustermann@med.siemens.de) at the Service Center.

- **Save**

By clicking **Save** in the Action bar, the changes made to the currently selected report are stored.

- **Delete**

The currently displayed report will be removed by clicking **Delete**.

- **Begin**

The **Begin** selection box allows you to define the start date of the reporting period. When using the **TEMPLATE** report, the default start date is the creation date.

- **End**

The End selection box allows you to define the end date of the reporting period. When using the **TEMPLATE** report, the default end date is one year after the **Begin** date.

5.2.2 Sample source

HP-Z0000YPR-111005-02-969

Sample source? defines the sample source and its sample mechanism. Auto Report enables pattern matching on an ASCII file or an event trigger.



Fig. 116: Sample Source?

- **Source**

You may choose between **Event Log** as the default or any **File** to be monitored. When **Event Log** is selected, all events that are written in the event log / customer log will be sampled and evaluated. When **File** is selected, pattern matching of the file contents will be performed. The file name has to be entered in the **Sample what?** frame. Pattern matching in the file option only works with ASCII files.

5.2.3 Sample what

HP-Z0000YPR-111005-02-968

1) Sample source? is set to Event Log

If the **Event Log** is selected in the **Sample source?** frame, all events that are recorded in the event log files will be sampled and evaluated. Various filter criteria (severity, message ID, source, domain) that are also used in the **Event Log** reader application can be selected. For detailed information on these filters, please refer to the Online Help for the **Event Log** reader.

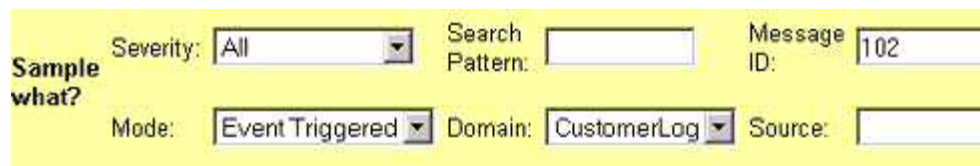
The image shows a screenshot of a web-based form titled "Sample what?". The form is divided into two rows of controls. The first row contains a "Severity:" dropdown menu set to "All", a "Search Pattern:" text input field, and a "Message ID:" text input field containing the value "102". The second row contains a "Mode:" dropdown menu set to "Event Triggered", a "Domain:" dropdown menu set to "CustomerLog", and a "Source:" text input field. The form has a yellow background and a thin border.

Fig. 117: Sample what? (Sample Source: Event Log)

- **Severity**

This is the classification of the error message. Possible selections are Error, Warning, Information, or All.

- **Search Pattern**

When a text string is entered, pattern matching is performed with the message text in the event log file. Wild cards are supported.

- **Message ID**

Filters for specific **Message IDs** in the event log file. In the above example, **Message ID** 102 was entered. It sends a message to Headquarters when the user successfully logs into the Service Software.

- **Mode**

Toggles between Time Triggered and Event Triggered. If Event Triggered is selected, the Auto Report will be activated as soon as the defined event occurs; the **Sample when?** frame displays only **Continuously**. If Time Triggered is selected, the system will perform pattern matching at the time that has been set in the parameters in the **Sample when?** frame.

- **Domain**

Allows you to select events from a certain domain (**Application**, **Customer Log**, **Security**, or **System**).

- **Source**

Allows you to specify different system software components, e.g. CSA_SAM (SAM application).

2) Sample source? is set to File

If File is selected in the Sample source? frame, the Sample what? frame displays fields that are necessary for sampling a defined file with a particular search pattern.

Fig. 118: Sample what? (Source: File)

- **File Name**

Specify the name of the file which is to be sampled. Wild cards are supported.

NOTE

Directory names must be separated with a '\'.

- **Search Pattern**

If no **Search Pattern** is entered, Auto Report will send a mail when the file has been created or changed; otherwise, a mail will be sent when the **Search Pattern** has been found by a sample.

- **Mode**

Toggles between Time Triggered and Event Triggered. If Event Triggered is selected, the Auto Report will become active as soon as the defined event occurs; the **Sample when?** frame displays only **Continuously**. If Time Triggered is selected, the system will perform pattern matching at the time that has been set in the parameters in the **Sample when?** frame.

- **Lines**

When a search pattern is entered and the **Mail: with partial attachment** is selected in the **Send what?** frame, this value defines the number of **Lines** to be added to the mail. The occurrence of the **Search Pattern** is in the center of the added lines.

5.2.4 Sample when

HP-Z0000YPR-111005-02-967

The sample rate (time and day) can be adjusted in the **Sample when?** frame. Parameters are dependent on the mode set in the **Sample what?** frame.

1) Mode: Event Triggered

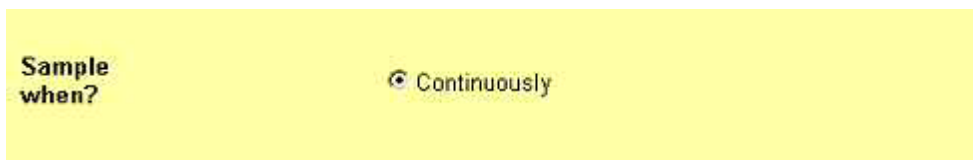


Fig. 119: Sample when? Event Triggered

- **Continuously**

Sampling is performed continuously; Auto Report starts sending the mail when the event defined by the filters in the **Sample what?** frame meets the conditions.

2) Mode: Time Triggered



Fig. 120: Sample when? Time Triggered

- **Continuously**

Specifies a time for continuous sampling and can be adjusted via the **Sample times** parameters.

- **Sample times** (only with Continuously)

The time when the Auto Report starts sampling, in **Hours** (0 - 23) and **Minutes** (1 - 59). If you want the Auto Report to start sampling at 3pm, you must enter 15:00.

NOTE

It is also possible to enter several sample times.

Use a semicolon (;) to separate the values entered, e.g. 14:45;15:00;15:15.



Fig. 121: Sample when? Daily, weekly, and monthly

- **Daily**

Select hours and minutes from the corresponding menus.

- **Weekly**

Select the day of the week and the hours and minutes from the corresponding menus.

- **Monthly**

Select the day of the month and the hours and minutes from the corresponding menus.

5.2.5 Send what

HP-Z0000YPR-111005-02-966

The frame **Send what?** specifies the type of mails, with or without attachments, etc. to be sent by the Auto Report tool.

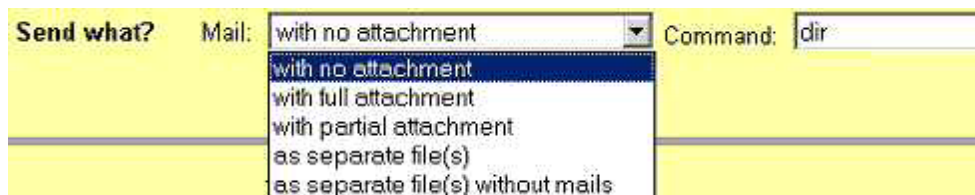


Fig. 122: Send what?

- **Mail**

with no attachment sends only a notification with the name of the file that was entered in the Sample what? frame.

with full attachment sends the whole file attached to the mail.

with partial attachment sends only those lines of the sampled file which meet the filter criteria in the Sample what? frame.

as separate file(s) forwards the selected report file(s) to a remote computer using FTP file transfer, and sends a mail to the remote target. In this case, the FTP target address can be specified in a selection box. If you cannot find the correct FTP target, you must add the FTP target in the Service Configuration function.

as separate file(s) without mails forwards the selected report file(s) to a remote computer using FTP file transfer, and **does not** send a mail to the remote target. In this case, the FTP target address can be specified in a selection box. If you cannot find the correct FTP target, you must add the FTP target in the Service Configuration function.

NOTE

The various Mail options only work with the Sample source? mode File.

- **Command**

The output from the executed application or DOS command entered will be added to the mail message.

5.2.6 Send when

HP-Z0000YPR-111005-02-965

In the **Send when?** frame, you can set the time at which the gathered information should be sent as a mail message or separate file. The settings are analogous to the settings in the **Sample when?** frame:



Fig. 123: Sample when?

- **At once**

The Auto Report is generated and sent immediately to the selected target.

- **Daily**

If **Daily** is selected, the selection boxes **Hour** and **Minute** will appear. Select hours and minutes from the corresponding menus.

- **Weekly**

If **Weekly** is selected, the selection boxes **Day**, **Hour**, and **Minute** will appear. Select the day of the week, hours and minutes from the corresponding menus.

- **Monthly**

If **Monthly** is selected, the selection boxes **Day**, **Hour**, and **Minute** will appear. Select the day of the month, hours and minutes from the corresponding menus.

5.2.7 Create new Auto Report

HP-Z0000YPR-111005-02-964

Fig. 124: Create new Auto Report

- Activate the **Auto Report** function in the Service Home menu.
- In the **REPORT** field, select the **TEMPLATE** entry.
- Create a representative name (e.g. display_crash) for the new job and enter it in the **Name** field.
- You must now specify the various parameters of the Auto Report function for the new job.

NOTE

The forms are fully interactive and the parameters are interdependent.

- Check and, if necessary, adjust the information entered into the **Sample source?**, **Sample what?**, **Sample when?**, and **Send when?** fields according to your requirements.
- Verify the **Enabling Status**, **Begin**, and **End** date of the new job and the notification target.
- Finally, save the modified job by activating the **Save** button. A message box appears to confirm that the data was saved successfully, and the new Auto Report name immediately appears under **Enabling Status** and is marked “Active” by default.
- Close the **Auto Report** by clicking the **Home** button in the Navigation bar.

EDB

EDB provides reports regarding events that are stored in the system event log.

The following settings are required for **EDB** reporting:

1. Target: Select **SRS_mail**.
2. Name: e.g. **edb**.
3. End time e.g. 2016
4. Sample source? Source: Select **File**
5. File name: (Enter the path and file name)
C:\MedCom\service\html\seso\sitedb\export*.edb ↵ Sample what? Mode: Select **Time Triggered**
6. Sample when? Click the **Daily** radio button. **Choose a time when the system is usually in operation!**
7. Send what? Mail: Select **as separate file(s) without mails**, Send what? FTP target: **SRS_ftp**
8. Send when? Click the **At once** radio button

QDB (Quality Assurance Database)

The QA data is transferred to the server at regular intervals for statistic evaluation and trend analysis. The data is gathered in the MR quality database.

Helium level and magnet shield temperature data is transferred to the server at regular intervals. This function provides a brief overview of the current helium levels and magnet shield temperatures for the reporting sites.

Under the following URL in the Siemens Intranet you are able to see data sent by your system: (<https://mr.med.siemens.de/tools/mrquality/mr/format/>)

The following settings are required for **QDB** reporting:

1. Target: Select **SRS_mail**.
2. Name: e.g. **qdb**.
3. End time e.g. 2016
4. Sample source? Source: Select **File**
5. File name: (Enter the path and file name)
C:\MedCom\service\html\seso\sitedb\export\export.xml ↵ Sample what? Mode: Select **Time Triggered**
6. Sample when? Click the **Daily** radio button. **Choose a time when the system is usually in operation!**
7. Send what? Mail: Select **as separate file(s) without mails**, Send what? FTP target: **SRS_ftp**
8. Send when? Click the **At once** radio button

6 File & Image Tools

HP-Z0000YPR-111005-02-963

n.a.

6.1 General

6.1.1 Introduction

HP-Z0000YPR-111005-02-962

The service application **File & Image Tools** provides a functionality for transferring file(s) and medical image(s) from a syngo-based system to a remote computer, e.g. a Remote Server or a service laptop, and vice versa. Raw data can be handled in the same way as images if stored as a DICOM object in the database.

6.1.2 Features

HP-Z0000YPR-111005-02-961

The **File & Image Tools** functionality can be divided into the following main features:

- Transfer file(s) and image(s) from the system to a remote computer and vice versa
- Extract file(s) or image(s) to be transferred
- Display the image header
- Display/modify the transfer list of the extracted file(s) and image(s)

The following transfer modes are available:

- **Transfer to system**, by means of a transfer from a remote computer (e.g. Remote Server) to the system.
- **Transfer from system**, by means of a transfer from the system to the remote computer.
- The **Transfer Manager** mechanism supports two transfer start times: **immediately** (transfer begins when you select a file and then activate the transfer with the 'Go' button) and **preferred** (transfer is scheduled for a later date).

6.2 Functions

HP-Z0000YPR-111005-02-960

n.a.

6.2.1 Start File & Image Tool

HP-Z0000YPR-111005-02-959

The **File & Image Tool** can be started from the Home menu by clicking the **File & Image Tool** button. Once you have started this function, you will see the **File & Image Tool** frame as shown in the following figure. Select an item to be performed from the **Source** selection box.

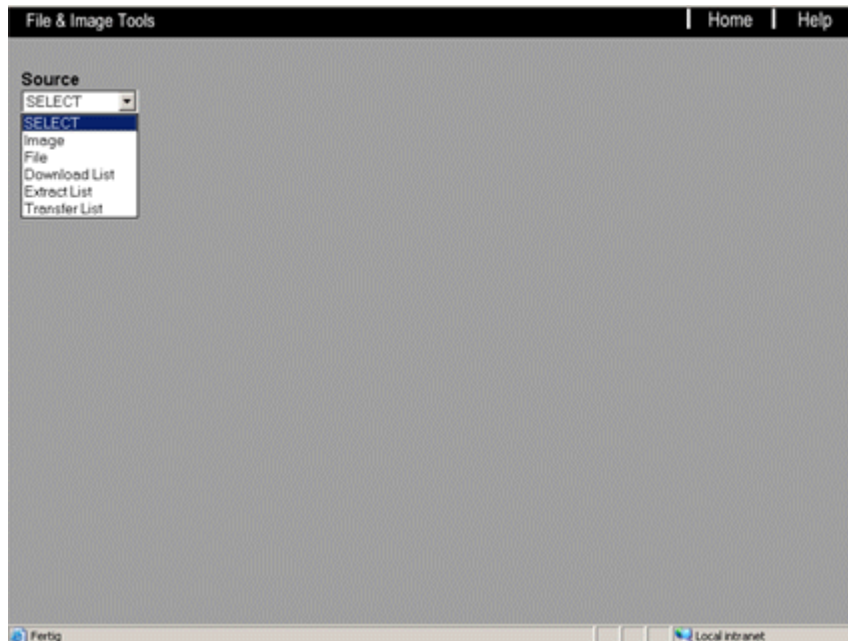


Fig. 125: Start the File & Image Tool

- **Source**

The **Source** selection box contains a list of applications that can be executed within the **File & Image Tools** service function. When you select an item from the **Source** list, the **Action** button is added to the navigation frame.

- **Navigation bar**

- **Home**
Brings you back to the home page.
- **Help**
Starts this Online Help window.

6.2.2 Modes

HP-Z0000YPR-111005-02-958

The **File & Image Tools** function allows a bi-directional transfer of any file from the system to a remote location, such as a Remote Server or PC, or vice versa. A modem and router connection is supported. User interaction takes place via the **Action** item and supports two transfer modes (**Transfer from system / Transfer to system**).

- **Configuration**

Please note that the File & Image Tool requires the **FTP Target** information from the remote computer. When a transfer is initiated, the remote computer data is verified in the target list. This has to be configured in the **Configuration** menu. Please refer to the Online Help of the **Configuration/Service/FTP** service function for more information on adding or modifying FTP targets. Also, refer to the Remote Server documentation (not part of Service Software Online Help) for detailed information regarding the prerequisites for adding a system to the Remote Server.

- **FTP Protocol**

All transfers are based on the **FTP** protocol. Once a transfer is started, the system initiates an FTP session to the remote computer and logs into the **rdiagftp** account. If authentication is successful, the transfer becomes active.

NOTE

When the error message **A connection with the server could not be established** appears in a pop-up window, verify that the **FTP server (receiver)** on the remote computer has been started and configured properly. Also ensure that the **FTP account password** has been configured correctly.

- **File Name**

The transferred file name on the remote computer has the following format:

<modality><serial number>_<original filename>

Example: mr275_AcrobatInstall.log

<material-number>_<serial number>_<original filename>

Example: 01234567_275_AcrobatInstall.log

The transferred DICOM object name on the remote computer has the following format:

<modality><serial number>_<DICOM object Loyd-number>.img

Example: mr275_2.0.21456.img

<material-number>_<serial number>_<DICOM object Loyd-number>.img

Example: 01234567_275_2.0.21456.img

NOTE

Transferred data is not encoded or compressed by the **File & Image Tool** transfer mechanisms.

6.2.3 Start Time

HP-Z0000YPR-111005-02-957

Two transfer start times are supported by the **File & Image Tools**. The local or remote user can submit a transfer start time by selecting the item **When: immediately** or **preferred** in the various transfer menus.

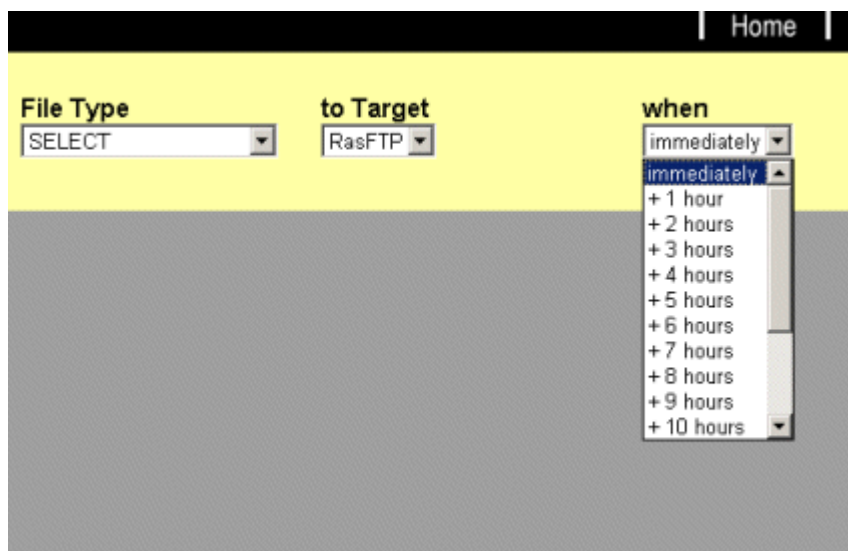


Fig. 126: Start File & Image Tools

- **immediately**

Starts the transfer once you select a file/image and click the **Go** button.

If the transfer is interrupted, e.g. the communication line is hung up, the Transfer Manager will respond by re-attempting to transfer the file/image to the remote computer. In this case, the transfer behaves like a preferred transfer.

When you start the Service Software as a local user, the system first establishes a connection to the remote computer. Afterward, you can select the file to be transferred.

If you are a remote user, the existing connection is used for file exchange.

- **preferred**

Initiates an unattended batch transfer.

Once the user has initiated a preferred transfer time, a batch transfer queue is added to the Transfer Manager, and the system will try to establish a connection to the remote computer at the pre-defined time (local time + selected hour(s)). A transfer will be restarted automatically if it does not complete successfully (e.g. the line was disconnected, or remote access was set to "No Access").

At first, an immediate retry is attempted 5 times. If this is not successful, the retry is queued again after 1, 6, 12, and 24 hours or when the system is booted. The 24 hour retry is performed for a maximum of 7 days.

The **Transfer to system** function is only available in connection with a **Remote Server**. Refer to the Remote Server manual for information on how to add a system on the Remote Server.

6.2.4 Patient Data Access

HP-Z0000YPR-111005-02-956

If you start the **File & Image Tools/Image** remotely and the **Patient Data Access** is disabled, the following window is displayed.

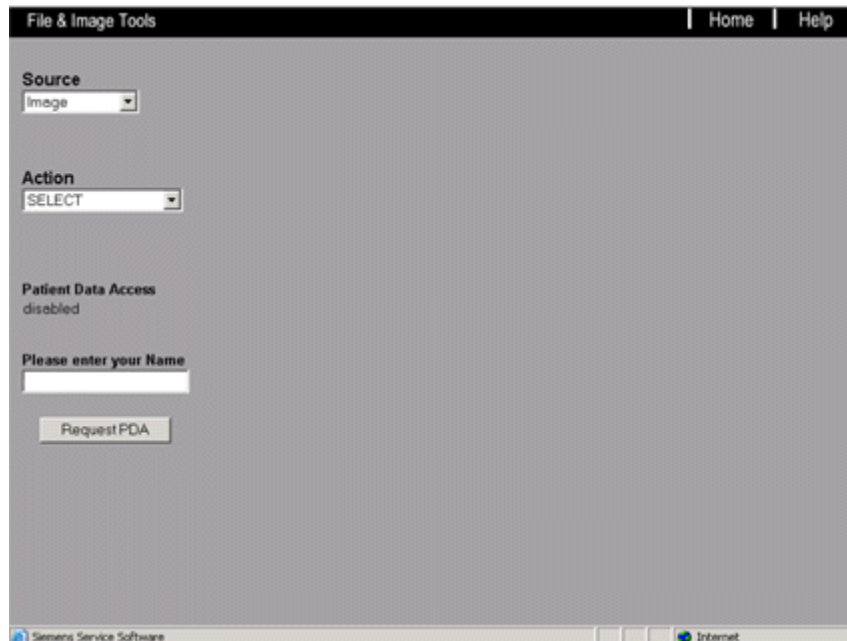


Fig. 127: Patient Data Access Request (PDA)

- **Source**

The **Source** selection box contains a list of applications that can be executed within the **File & Image Tools** service function. When you select an item from the **Source** list, click the **Action** button to add the **Patient Data Access** section to the navigation frame.

- **Action**

The **Action** selection box is context-sensitive and contains all **Action** items available for the selected **Source**. When you select an item from the **Action** list, the contents of the function frame (right-hand screen) will change accordingly.

- **Patient Data Access**

Patient Data Access controls whether the patient name or the Loyd number is displayed for selecting the patient images.

- **Enabled**

The patient name is displayed.

- **Disabled**

The Loyd number is displayed.

- **Please enter your name**

Enter your name.

- **Request PDA**

When you click the **Request PDA** (Patient Data Access) button, the request for accessing and displaying the patient name in the function frame will be sent to the local operator.

- **Navigation bar**

- **Home**

Brings you back to the home page.

- **Help**

Launches this Online Help file.

The following step-by-step instructions show the procedure for enabling **Patient Data Access**.

Request from the remote user

The service person is logged in remotely and needs to transfer images with **Patient Data Access** disabled. If he continues without requesting **Patient Data Access**, the patient images can be selected by the Loyd number. If the correct Loyd number is not available, he has to submit a request for **Patient Data Access** to the local operator by entering his name in the **Please enter your name** field and clicking the **Request PDA** button.

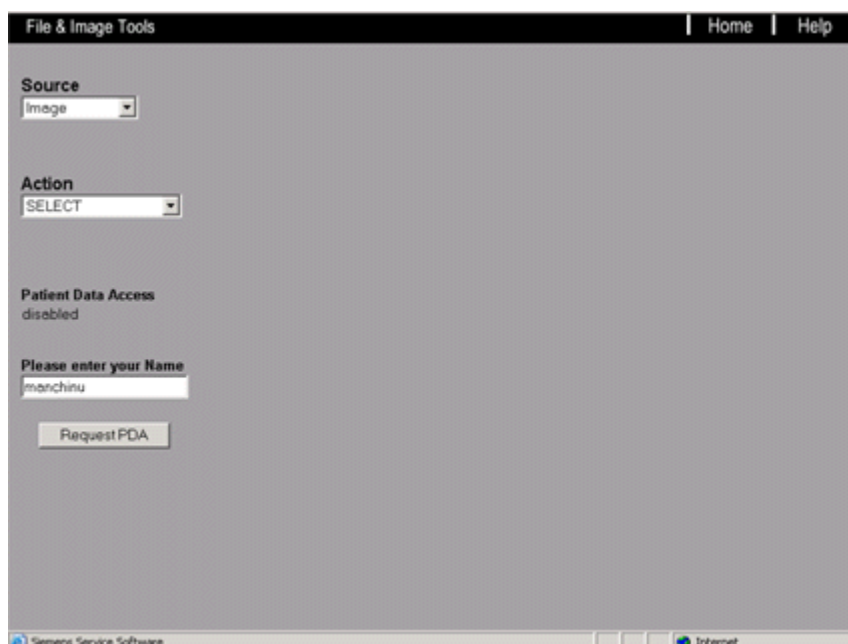


Fig. 128: Patient Data Access Request (PDA)

The following pop-up window is displayed.

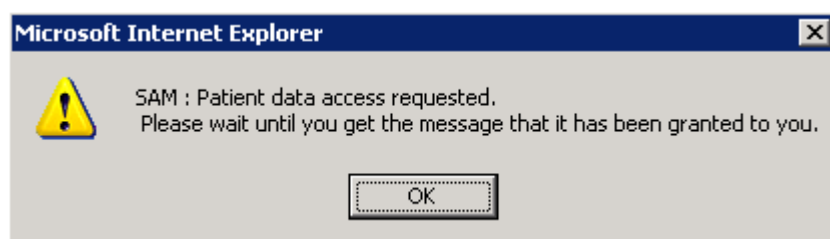


Fig. 129: Patient Data Access Request (PDA)

Acknowledgement from the local operator

The local operator can see the **Patient Data Access** request from the remote user in the **Remote Service Access Control** platform. When you select the **Allow patient data access** box and click the **Apply** button, the status information **Patient data access:** switches from **Disabled** to **Enabled**.

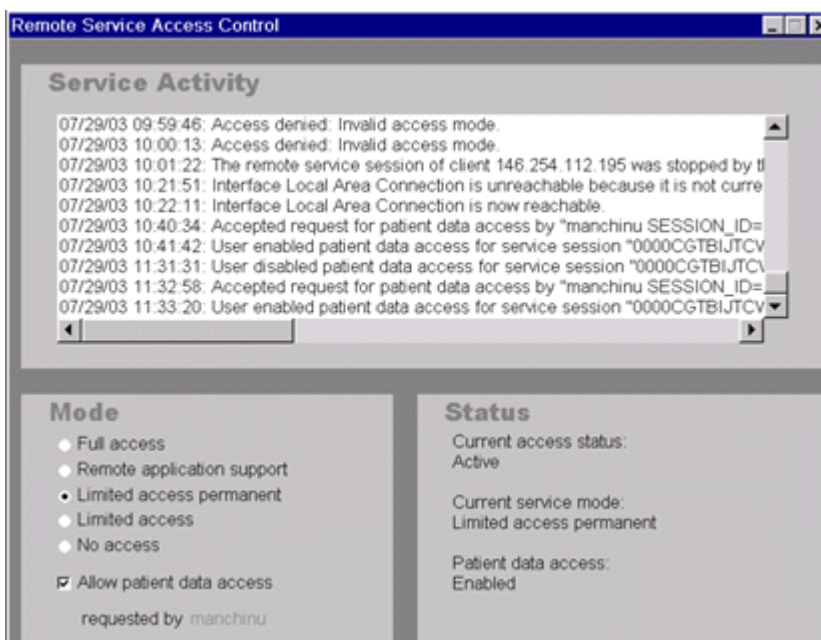


Fig. 130: Patient Data Access acknowledgement

The remote user will see the following pop-up window.

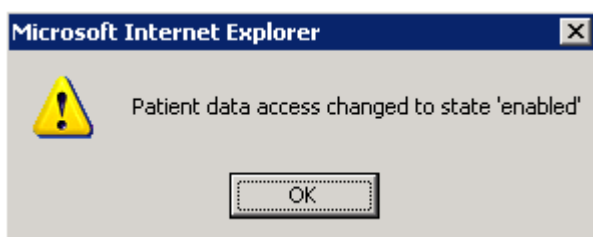


Fig. 131: Patient Data Access acknowledgement

When you confirm the popup, the following **File & Image Tools/Image** window will be automatically displayed.

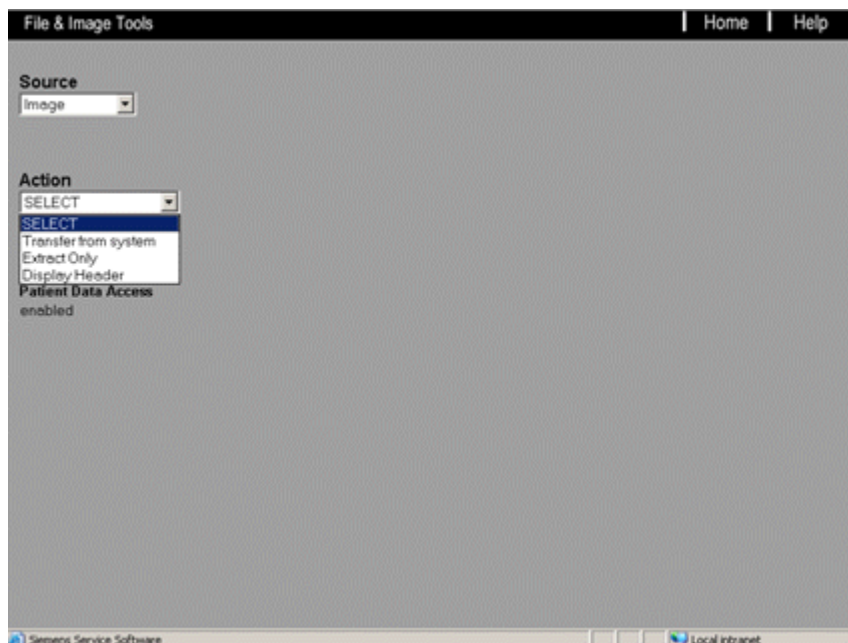


Fig. 132: Action items for enabled

Patient Data Access is now enabled, and patient information will be displayed. The **Patient Data Access** is active for the duration of the remote service session. For example, if you exit the File & Image Tools to perform another service function and then return to the File & Images Tools, the Patient Data Access remains active until the remote user closes the remote service session.

6.3 Source

HP-Z0000YPR-111005-02-955

n.a.

6.3.1 Image

HP-Z0000YPR-111005-02-954

This chapter contains all **Action** items with regard to the transfer of **DICOM Objects** (e.g. images and raw data).

There is **no difference** between extracting and transferring a medical DICOM **image** or a **raw data** object since both DICOM objects are stored in the syngo database and are treated in the same way.

NOTE

Patient information can only be accessed remotely if the customer has enabled **Patient Data Access**. Therefore, the graphical user interface looks different depending on whether the **Patient Data Access** is enabled or disabled (see the chapter entitled **Patient Data Access** for more details).

NOTE

All confidential patient information is removed from the DICOM object header prior to an image/raw data transfer or when you use the **Extract only image** function.

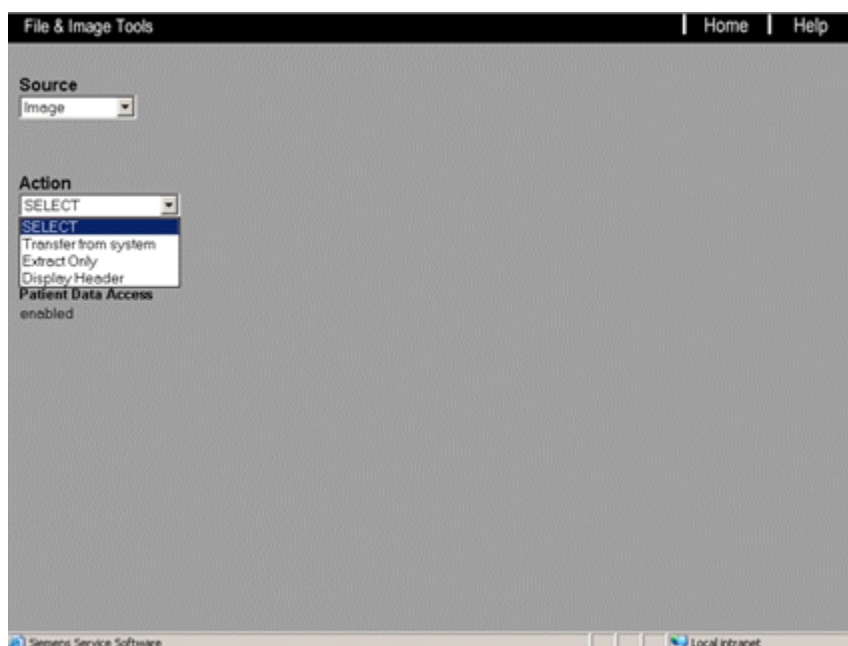


Fig. 133: Action items for enabled

- **Source**

The **Source** selection box contains a list of applications that can be executed within the **File & Image Tools** service function. When you select an item from the **Source** list, click the **Action** button to add the **Patient Data Access** section to the navigation frame.

- **Action**

The **Action** selection box is context-sensitive and contains all **Action** items available for the selected **Source**. When you select an item from the **Action** list, the contents of the function frame (right-hand screen) will change accordingly.

- **Patient Data Access**

When the Files & Image Tools are started locally, Patient Data Access is always **enabled** and cannot be disabled.

- **Navigation bar**

- **Home**
Brings you back to the home page.
- **Help**
Launches this Online Help file.

Transfer from system

The **Transfer from system** mode transfers DICOM objects (images or raw data) from the system to a remote computer, such as a PC or a Remote Server. Please refer to the Start Time chapter for more details on the transfer start time (immediately or preferred).

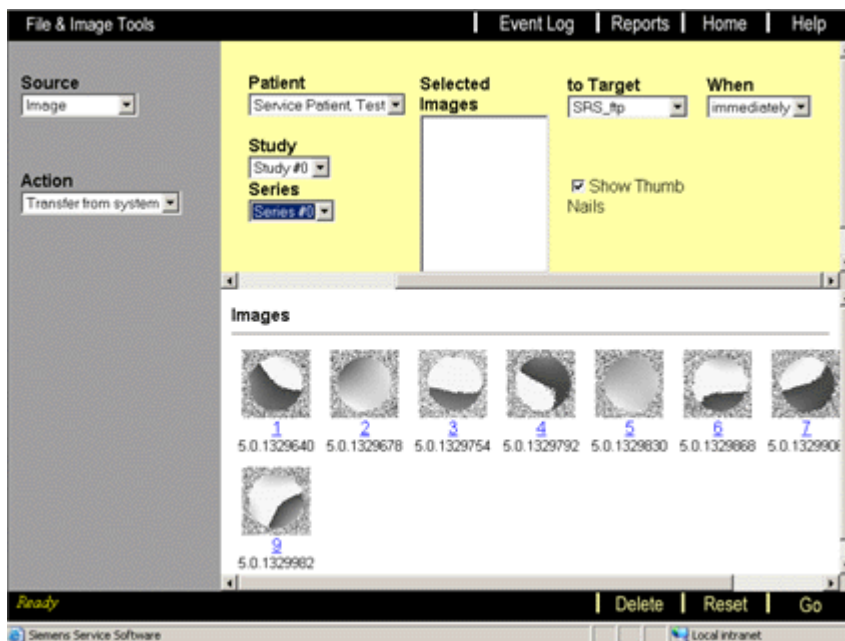


Fig. 134: Image Transfer from system

- **Patient**

The **Patient** selection box contains a list of all patients available on the system. When you select an item from the patient list, the **Study** list is automatically updated.

- **Study**

The **Study** selection box contains a list of all studies available for the selected patient. When you select an item from the Study list, the **Series** list is automatically updated.

- **Series**

The **Series** selection box contains a list of all series available for the selected study. When you select an item from the Series list, the image thumbnails will be displayed automatically if the **Show Thumb Nails** box has been checked.

- **Selected Images**

Here, you will find the images already selected for transfer to the remote computer. You can select an image by clicking on the image icon or the corresponding image number.

- **Images**

The image icons will be displayed here. Below the image, you will find the image number and the Loyd number. The Loyd number identifies the image in the syngo data-base. The **Images** menu also supports multiple selection. To make a multiple selection, press and hold the **CTRL** key and click the images you want to select.

- **to Target**

Identifies the remote computer. This is either the Remote Server identification (if you are logged in through the Remote Server) or the remote PC identification. Targets can be added or modified in the configuration using **Service/FTP**.

- **Show Thumb Nails**

Images are displayed as image thumbnails by default. If you expect too many images and want to save transfer time, you can deselect this option to display the image numbers instead of the images.

Images

Image 13001	Image 14001	Image 15001	Image 16001	Image 17001	Image 18001
3.0.64541	3.0.64572	3.0.64605	3.0.64636	3.0.64667	3.0.64698
Image 21001	Image 22001				
3.0.64791	3.0.64822				

Fig. 135: Image numbers only

- **When**

Select the appropriate transfer start time: Either **immediately** (start now) or **preferred** (start later at 22:00). For more details on the transfer start time, see the Start Time chapter.

- **Action bar**

- **Delete**

The highlighted image in the **Selected Images** menu will be removed from the transfer list once the **Delete** is clicked.

- **Reset**

Removes all entries from the transfer list in the **Selected Images** menu.

- **Go**

When a large file is transferred, a new window is opened that shows the file transfer progress. Each file selection will open another window in order to transfer files in parallel.

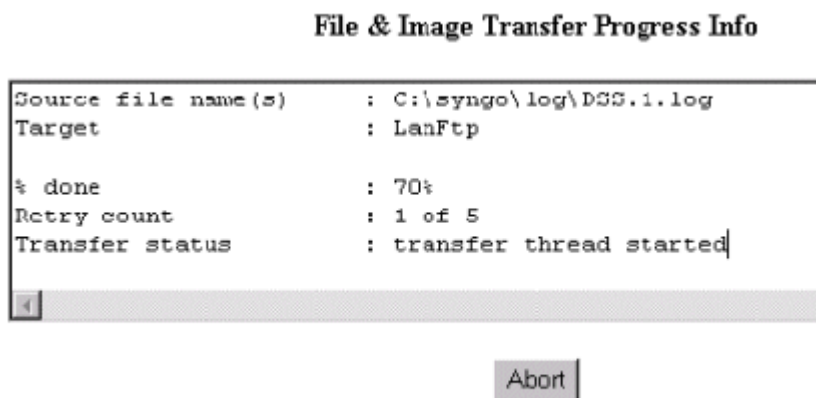


Fig. 136: Transfer progress

- **Source file name(s)**
The path and file name(s) of the file to be transferred to the remote computer.
- **Target**
Identifies the remote computer. This is the Remote Server identification (if you are logged in through the Remote Server) or the remote PC identification. Targets can be added or modified in the configuration using **Service/FTP**.
- **% done**
Indicates the progress of the file transfer in percent.
- **Retry count**
Shows the number of immediate transfer retries. Immediate transfer will be stopped after 5 attempts, and the job will be queued for a preferred transfer. Please refer to the Start Time chapter for more details on the immediately/preferred transfer start time.
- **Transfer status**
Displays the current status of the transfer. This can be an error message or information regarding the transfer itself.
- **Abort**
Aborts the file transfer function and removes it from the Transfer Manager queue.

Extract Only

The **Extract Only** function prepares images for transmission, so that they can be picked up later upon request. For this purpose, the selected images are copied to the directory c:\MedCom\service\extract. Use the **Extract List / Transfer from system** functions to transfer the extracted images (see chapter entitled Extract List), or use the **Transfer list / Show/Delete** function (see the Transfer List chapter) to show or delete the extracted DICOM objects.



Fig. 137: Image Extract only

- **Patient**

The **Patient** selection box contains a list of all patients available on the system. When you select an item from the patient list, the **Study** list is automatically updated.

- **Study**

The **Study** selection box contains a list of all studies available for the selected patient. When you select an item from the Study list, the **Series** list is automatically updated.

- **Series**

The **Series** selection box contains a list of all series available for the selected study. When you select an item from the Series list, the image thumbnails will be displayed automatically if the **Show Thumb Nails** box has been checked.

- **Selected Images**

Here, you will find the images already selected for transfer to the remote computer. You can select an image by clicking the image icon or the corresponding image number.

- **Images**

The image icons will be displayed here. The image number and the Loyd number are displayed below the image. The Loyd number identifies the image in the syngo database. The **Images** menu also supports multiple selection. To make a multiple selection, press and hold the **CTRL key** and click the images you want to select.

- **to Target**

Identifies the remote computer. This is either the Remote Server identification (if you are logged in through the Remote Server) or the remote PC identification. Targets can be added or modified in the configuration using **Service/FTP**.

- **Show Thumb Nails**

Images are displayed as image thumbnails by default. If you expect too many images and want to save transfer time, you can deselect this option to display the image numbers instead of the images.

Images					
Image 13001	Image 14001	Image 15001	Image 16001	Image 17001	Image 18001
3.0.64541	3.0.64572	3.0.64605	3.0.64636	3.0.64667	3.0.64698
Image 21001	Image 22001				
3.0.64791	3.0.64822				

Fig. 138: Image numbers only

- **Action bar**

- **Delete**

The highlighted image in the **Selected Images** menu will be removed from the transfer list once the **Delete** is clicked.

- **Reset**

Removes all entries from the transfer list in the **Selected Images** menu.

- **Go**

When a large file is transferred, a new window is opened that shows the file transfer progress. Each file selection will open another window in order to transfer files in parallel.

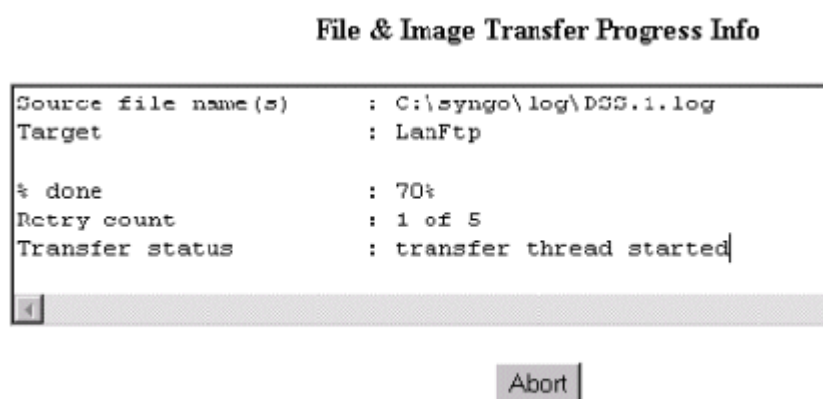


Fig. 139: Transfer progress

- **Source file name(s)**

The path and file name(s) of the file to be transferred to the remote computer.

- **Target**

Identifies the remote computer. This is the Remote Server identification (if you are logged in through the Remote Server) or the remote PC identification. Targets can be added or modified in the configuration using **Service/FTP**.

- **% done**

Indicates the progress of the file transfer in percent.

- **Retry count**

Shows the number of immediate transfer retries. Immediate transfer will be stopped after 5 attempts, and the job will be queued for a preferred transfer. Please refer to the Start Time chapter for more details on the immediately/preferred transfer start time.

- **Transfer status**

Displays the current status of the transfer. This can be an error message or information regarding the transfer itself.

- **Abort**

Aborts the file transfer function and removes it from the Transfer Manager queue.

Display Header

To find system errors without viewing the whole image, only the anonymous header information of an extracted DICOM object (image or raw data) will be displayed. The extracted header information will be displayed in a separate browser window.



Fig. 140: Image Display Header

- **Patient**

The **Patient** selection box contains a list of all patients available on the system. When you select an item from the patient list, the **Study** list is automatically updated.

- **Study**

The **Study** selection box contains a list of all studies available for the selected patient. When you select an item from the Study list, the **Series** list is automatically updated.

- **Series**

The **Series** selection box contains a list of all series available for the selected study. When you select an item from the Series list, the image thumbnails will be displayed automatically if the **Show Thumb Nails** box has been checked.

- **Images**

The image icons will be displayed here. The image number and the Loyd number are displayed below the image. The Loyd number identifies the image in the syngo data-base.

The Images menu also supports multiple selection. To make a multiple selection, press and hold the **CTRL** key and click the DICOM object or the DICOM object number you want to select.

Once you click the image thumbnail or the image number, a separate browser window will be opened with the detailed DICOM header information.

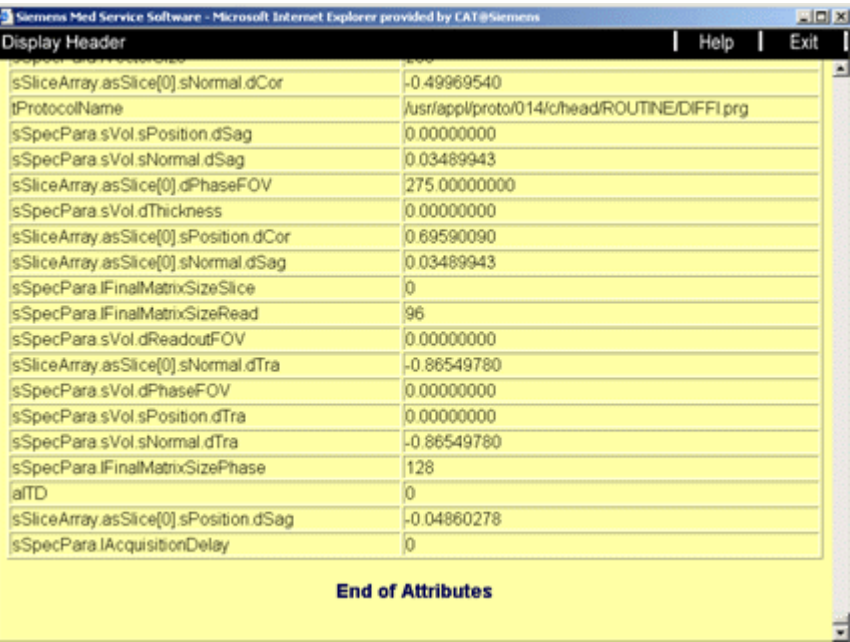


Fig. 141: Detailed DICOM Header information

Use the Exit button in the Navigation bar to close this window.

- **Show Thumb Nail**

Images will be displayed as image thumbnails by default. If you expect too many images and want to save transfer time, you can deselect this option to display the image numbers instead of the images.

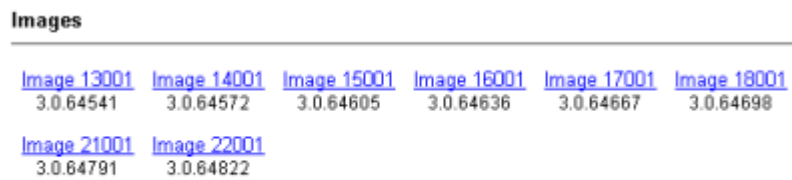


Fig. 142: Image numbers only

Evaluate Image

To evaluate an image, you must select the patient or ID. Proceed as follows: Click the Patient selection list. A list of all patients available in the store is displayed. Select Service Patient or a patient name or ID from the list. Note that you can select only one patient at a time. Scroll the menu up and down to browse through the entire list. To evaluate one image, click the blue link below the image icon

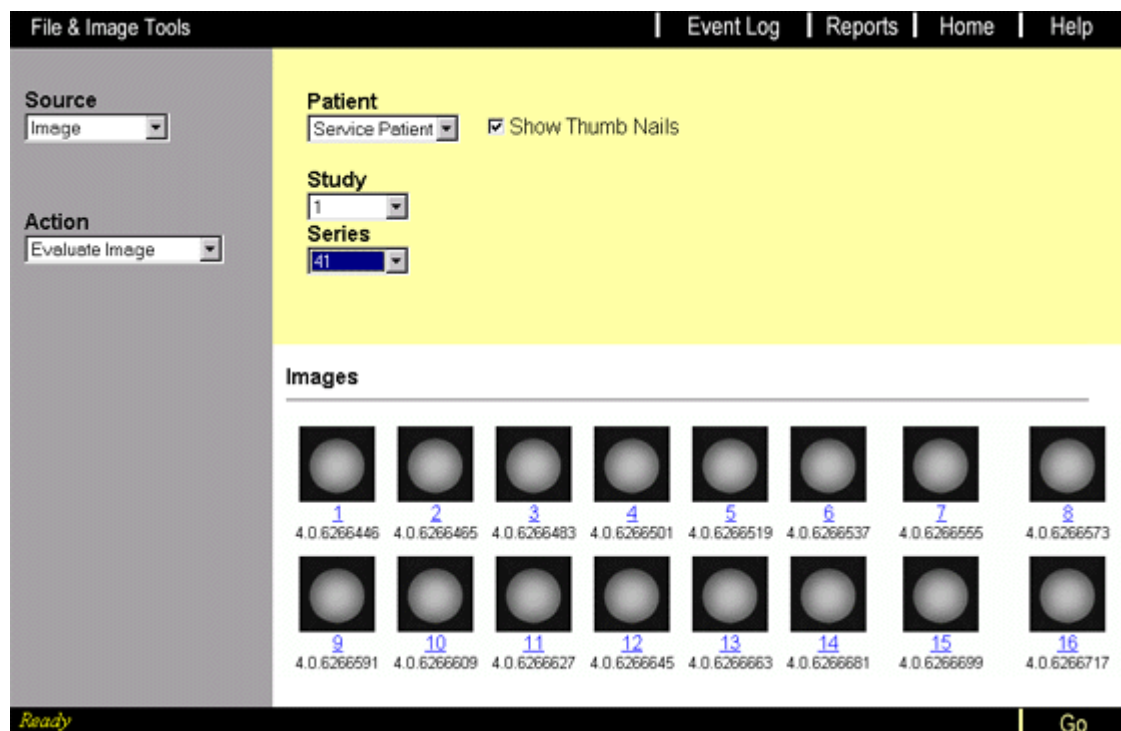


Fig. 143: Example for Series Evaluation, select Patient, Study, Series

To evaluate the series, click **Evaluation Series**

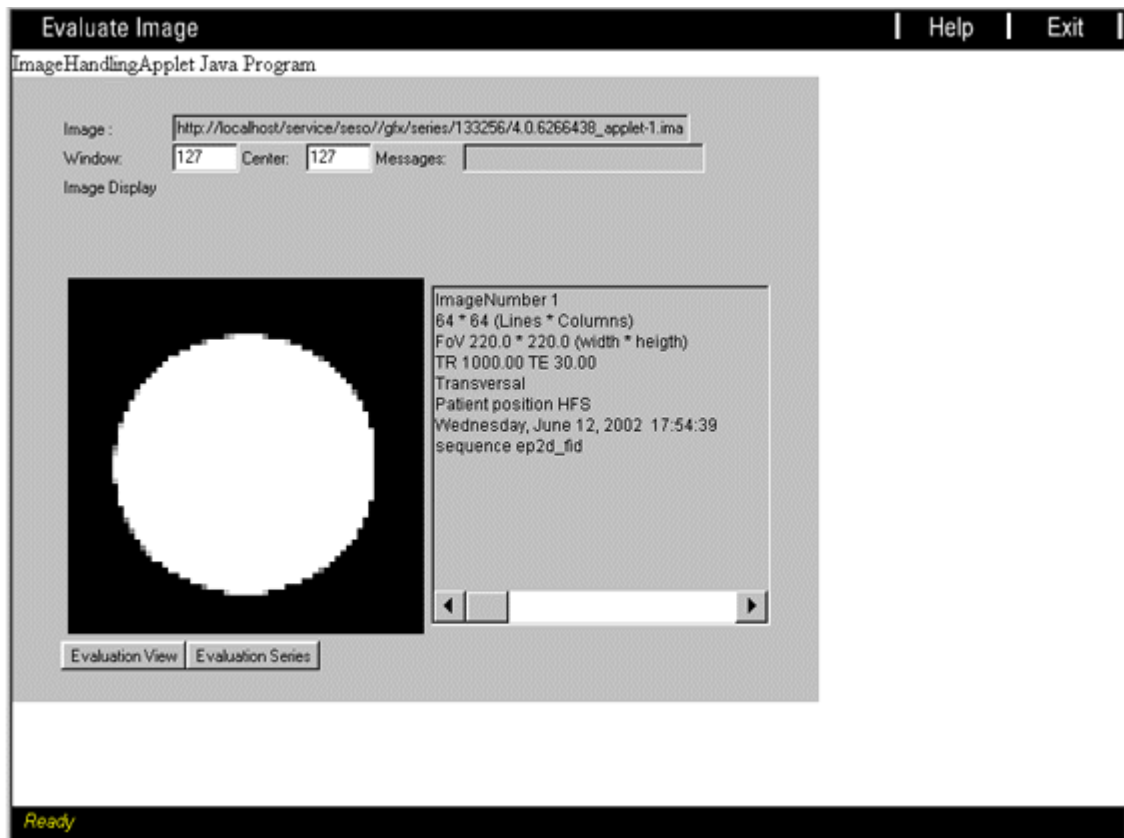


Fig. 144: Evaluation Series

To calculate, click **ROI Series Calculation**.

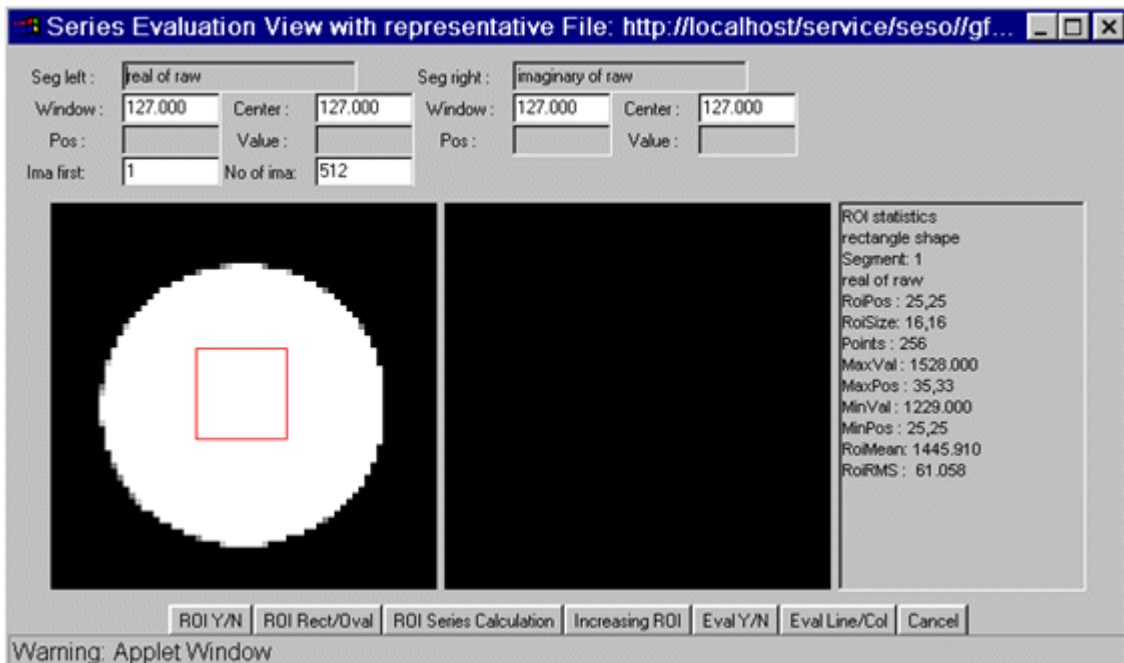


Fig. 145: ROI

⇒ Result of ROI (mean calculation), absolute part of the images.

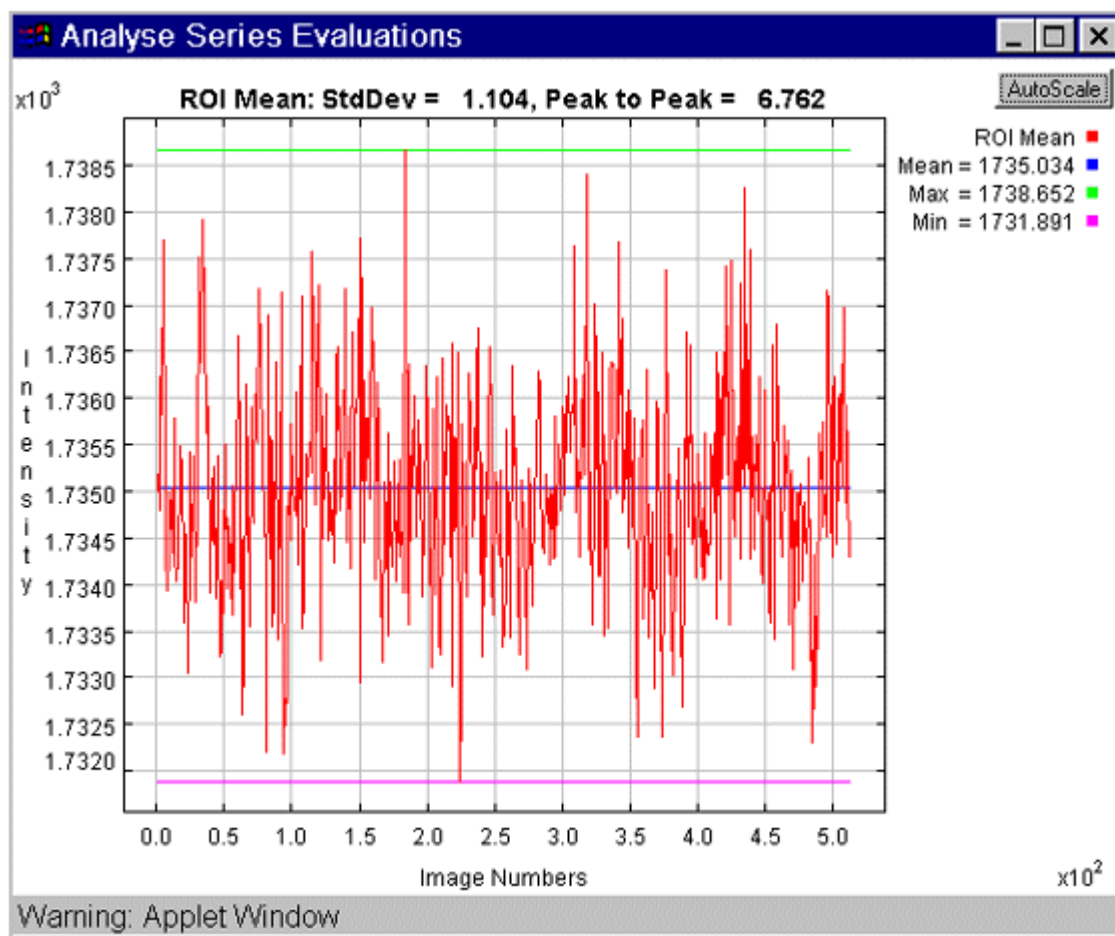


Fig. 146: ROI Mean (%) StdDev Peak to Peak Series Calculation

⇒ Result of ROI (mean calculation), real part of the images.

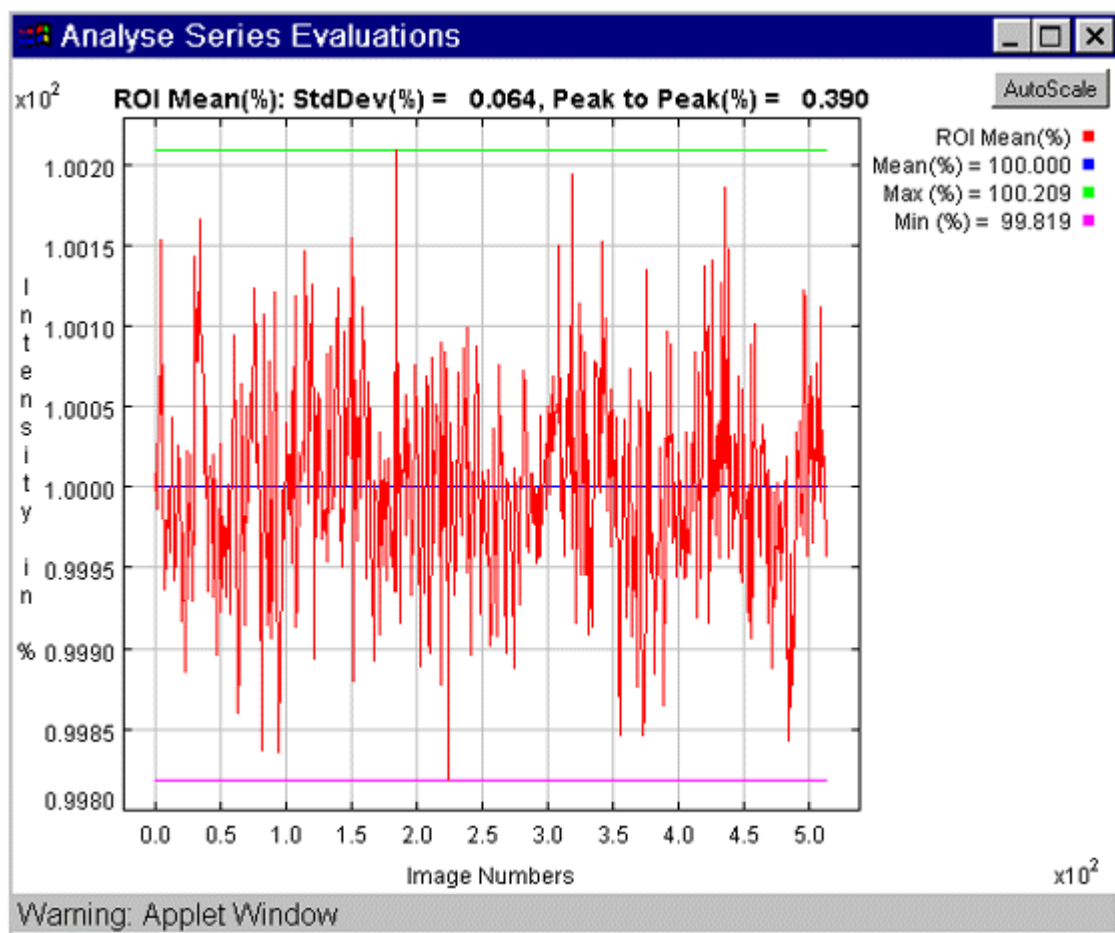


Fig. 147: ROI Mean (%) StdDev (%) Series Calculation

To calculate, click **Increasing ROI**

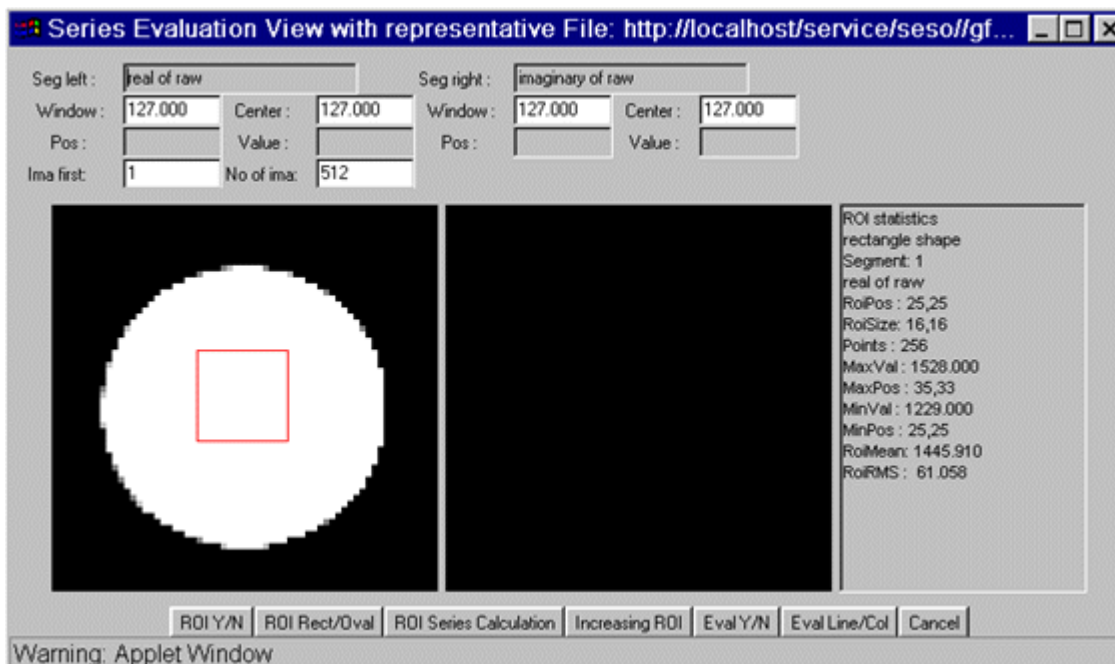


Fig. 148: ROI

⇒ Result of ROI mean calculation (increasing ROI StdDev)

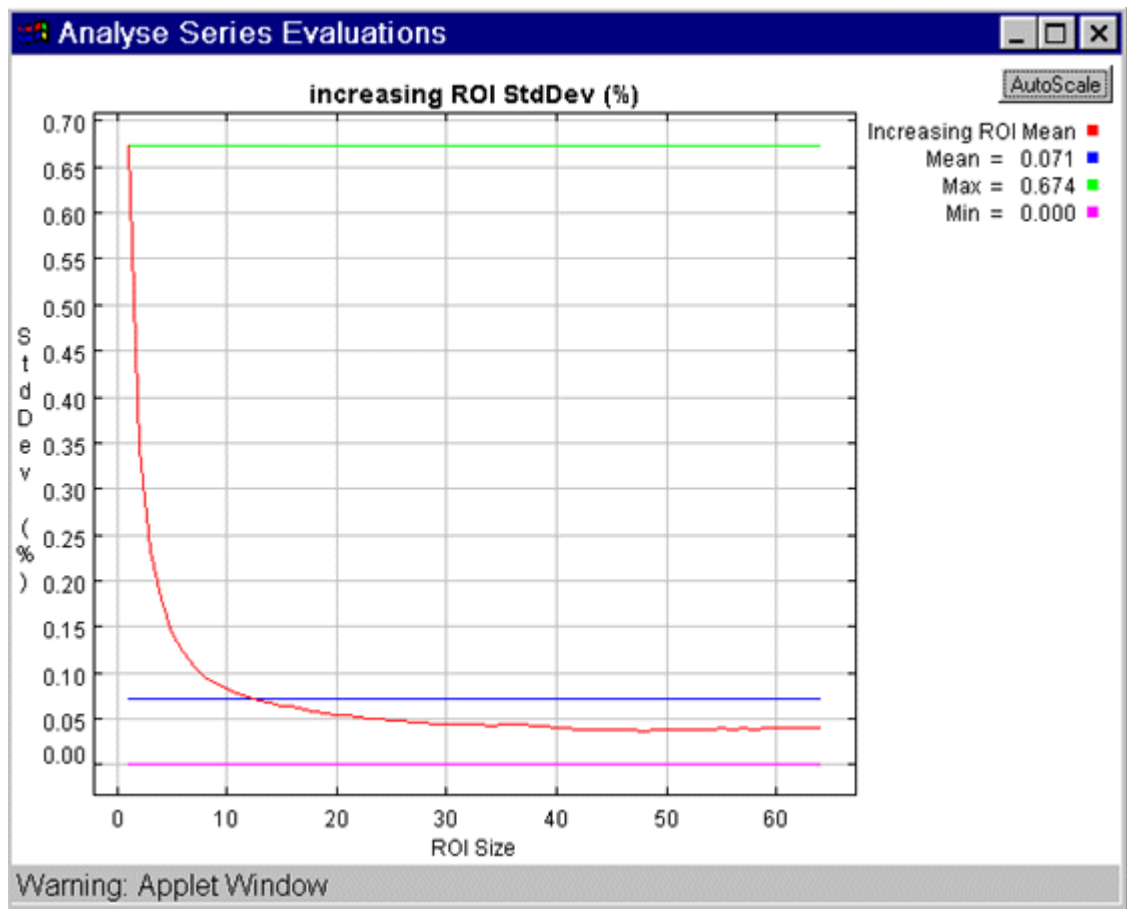


Fig. 149: Increasing ROI StdDev (%) Series Calculation

6.3.2 File

HP-Z0000YPR-111005-02-953

This chapter contains all **Action** items regarding the transfer of a **File**.

Only files that are not allocated by the system can be transferred. If you need to transfer an allocated file, find out which application is allocating the file. Stop that application with the appropriate service tool.

NOTE

Images and raw data cannot be transferred with the File function. Please refer to the Image chapter for more details on the image transfer functions.

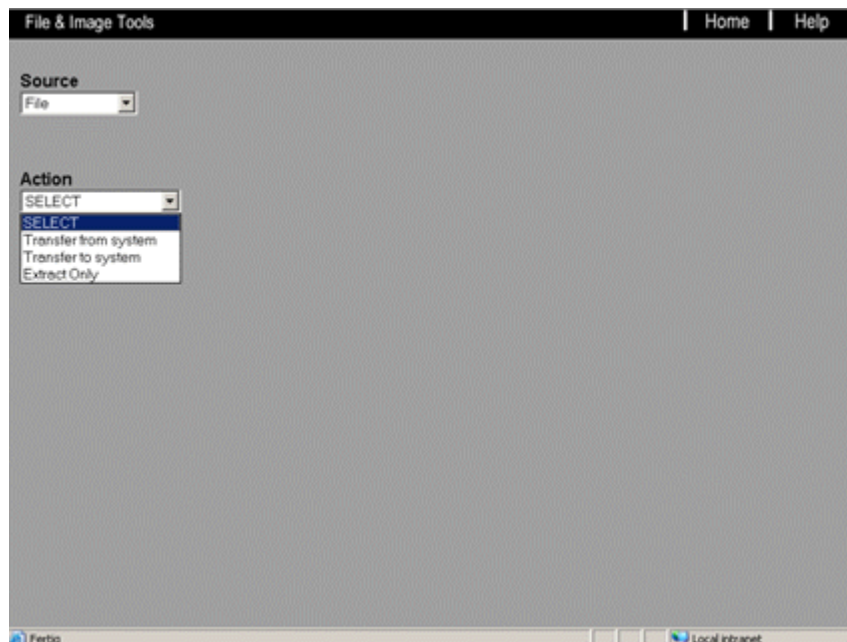


Fig. 150: Action items for File

- **Source**

The **Source** selection box contains a list of applications that can be executed within the **File & Image Tools** service function. When you select an item from the **Source** list, the **Action** button is added to the navigation frame.

- **Action**

The **Action** selection box is context-sensitive; it contains all **Action** items that are available for the selected **Source**. When you select an item from the **Action** list, the contents of the function frame (right-hand screen) will change accordingly.

- **Navigation bar**

- **Home**
Brings you back to the home page.
- **Help**
Starts this Online Help file.

Transfer from system

The **Transfer from system** function transfers files from the system to a remote computer, such as a PC or Remote Server. Please refer to the Modes chapter for more details on the image transfer functions.

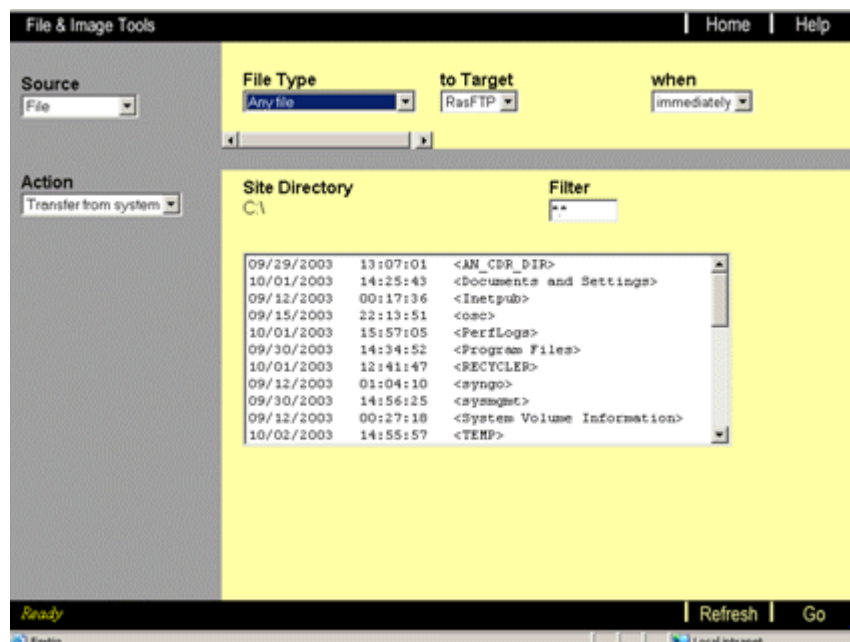


Fig. 151: File transfer from the system

- **File Type (Any File)**

This selection box contains a list of the predefined options. When you select an option, the function frame is adapted automatically.

- **to Target**

Identifies the remote computer. This is either the Remote Server identification (if you are logged in through the Remote Server) or the remote PC identification. Targets can be added or modified in the configuration using **Service/FTP**.

- **When**

Select the appropriate transfer start time: either immediately (start now) or preferred (start later at 22:00). For more details on the transfer start time, refer to the Start Time chapter.

- **Site Directory**

Here you can browse through the directory structure of the system, similar to Microsoft Windows Explorer.

Directories are surrounded by "<>". Click a directory name just **once** to open its contents.

At least one file has to be selected before the **Go** button in the **Navigation bar** becomes functional. To select a file for file transfer, click the file so it is highlighted. **Multiple** file selection is also supported. To select **multiple** files, press and hold the **CTRL** key and click the next file to be selected. To select a **range** of files, select the first file, press and hold the **SHIFT** key, and click the last file to be selected.

- **Filter**

With this dialog box you can filter the files to be displayed; e.g. *.txt: All files with the extension '.txt' are displayed.

- **Refresh**

Refreshes the contents of the **Site Directory**.

Transfer to system

The **Transfer to system** function transfers files from a remote computer to the system. All files will be transferred to the `c:\MedCom\service\incoming` directory.

Make sure that the file(s) you want to transfer is/are located on the remote computer. The file(s) must be located in the `<anonymous FTP>/outgoing/<modality>` (xx is the syngo modality; it is to be replaced with the modality, for example, MR or CT) directory on the remote computer.

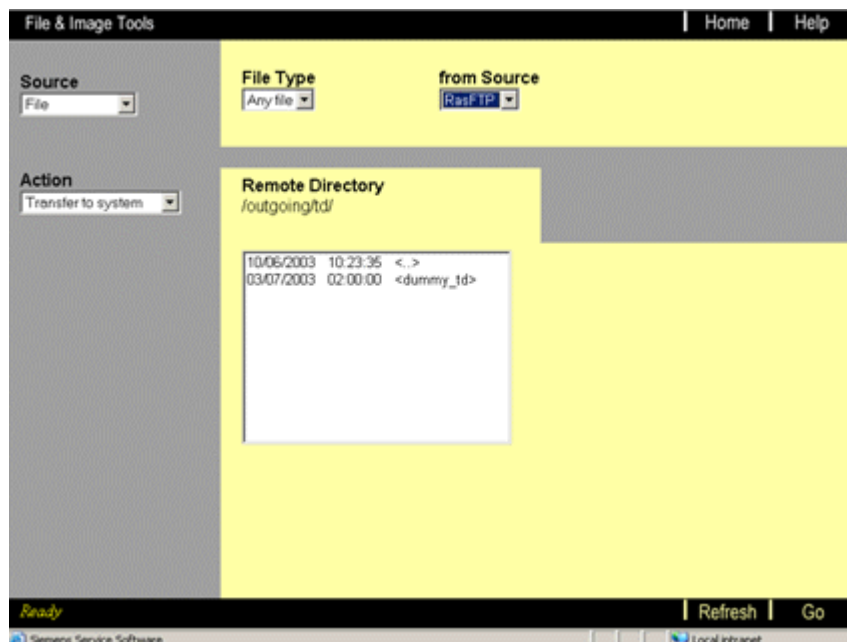


Fig. 152: File transfer to the system

- **File Type**

The selection box is fixed and cannot be changed.

- **from Source**

Identification of the remote computer. This has to be the remote computer from which you want to transfer the files.

- **Remote Directory**

Here you can browse through the directory structure of the remote computer, similar to Microsoft Windows Explorer. The system points to the <anonymous FTP>/outgoing/<modality> (xx is the syngo modality; it is to be replaced with the modality, for example, MR or CT) directory on the remote computer by default.

Directories are surrounded by "<>". Click a directory name just **once** to open its contents.

At least one file has to be selected before the **Go** button in the **Navigation bar** becomes functional. To select a file for file transfer, click the file so it is highlighted. **Multiple** file selection is also supported. To select **multiple** files, press and hold the **CTRL** key and click the next file to be selected. To select a **range** of files, select the first file, press and hold the **SHIFT** key, and click the last file to be selected.

- **Filter**

With the dialog box, you can filter the files to be displayed; e.g. *.txt: All files with the extension '.txt' are displayed.

- **Refresh**

Refreshes the contents of the Remote Directory.

- **Go**

When a large file is transferred, a new window is opened that shows the file transfer progress. Each file selection will open another window in order to transfer files in parallel.

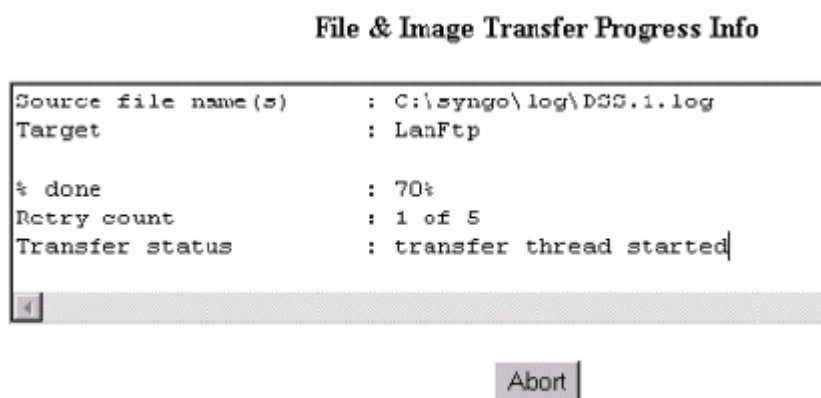


Fig. 153: Transfer progress

- **Source file name(s)**

The path and file name(s) of the file to be transferred to the remote computer.

- **Target**

Identifies the remote computer. This is either the Remote Server identification (if you are logged in through the Remote Server) or the remote PC identification. Targets can be added or modified in the configuration using Service/FTP.

- **% done**

Indicates the progress of the file transfer in percent.

- **Retry count**

Shows the number of immediate transfer retries. Immediate transfer will be stopped after 5 attempts, and the job will be queued for a preferred transfer. Please refer to the Start Time chapter for more details on the immediately/preferred transfer start time.

- **Transfer status**

Displays the current status of the transfer. This can be an error message or information regarding the transfer itself.

- **Abort**

Aborts the file transfer function and removes it from the Transfer Manager queue.

Extract Only

The **Extract Only** function prepares files for transmission, so that they can be picked up later upon request. For this purpose, the selected files are copied to the directory `c:\Med-Com\service\extract`. Use the **Extract List / Transfer from system** function to transfer the extracted images (see chapter entitled Extract List), or use the **Transfer list / Show/Delete** function (see the Transfer List chapter) to show or delete the extracted files.

NOTE

Images and raw data cannot be extracted using the File / Extract Only function. Please use Image / Extract Only (see the Image chapter for more detailed information).

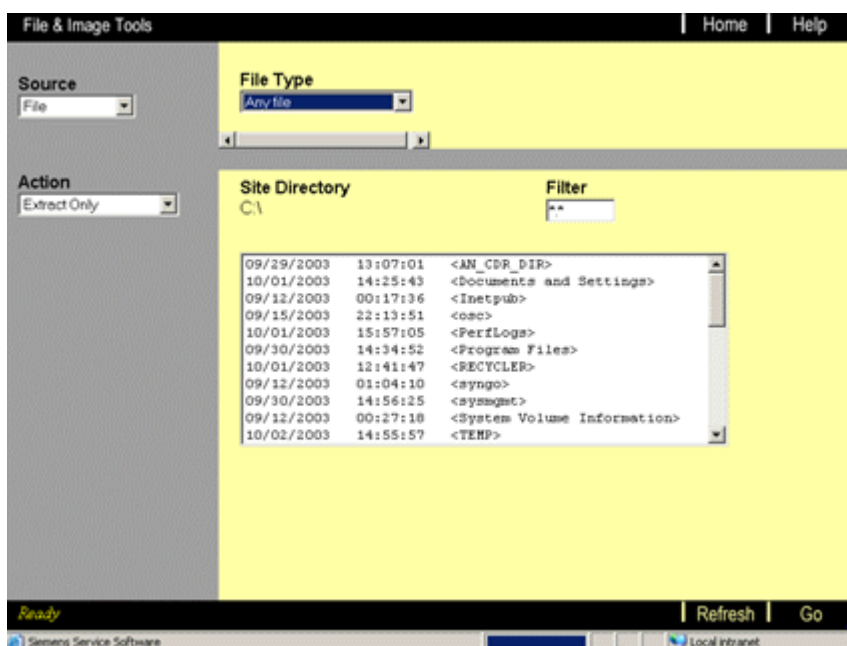


Fig. 154: File Extract Only

- **File Type**

The selection box is fixed and cannot be changed.

- **Site Directory**

Here you can browse through the directory structure of the system, similar to Microsoft Windows Explorer.

Directories are surrounded by "<>". Click a directory name just **once** to open its contents.

At least one file has to be selected before the **Go** button in the **Navigation bar** becomes functional. To select a file for file transfer, click the file so it is highlighted. **Multiple** file selection is also supported. To select **multiple** files, press and hold the **CTRL** key and click the next file to be selected. To select a **range** of files, select the first file, press and hold the **SHIFT** key, and click the last file to be selected.

- **Filter**

With this dialog box, you can filter the files to be displayed; e.g. *.txt: All files with the extension '.txt' are displayed.

- **Action bar**

- **Refresh**

Refreshes the contents of the **Site Directory**.

6.3.3 Download List

HP-Z0000YPR-111005-02-952

This chapter contains all **Action** items to display the files that have been transferred to the system.

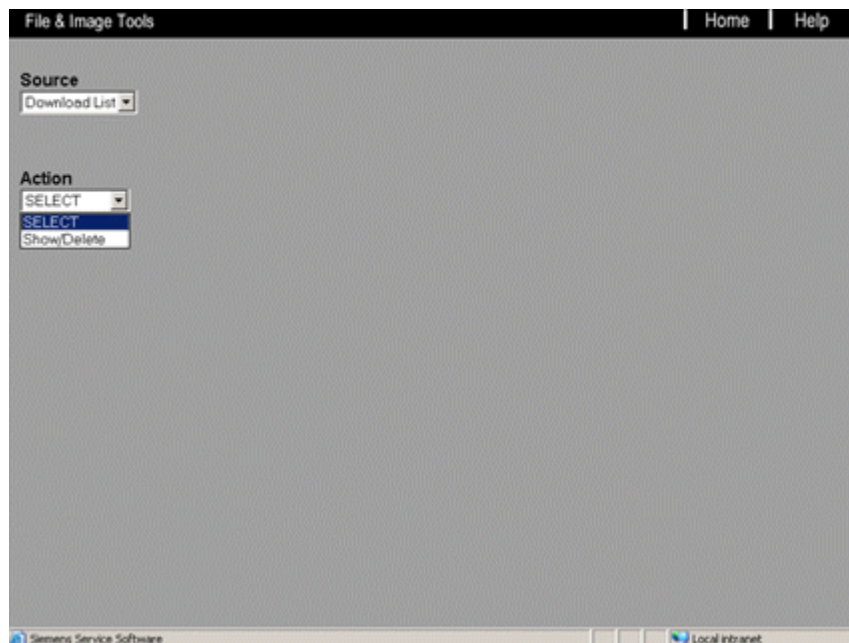


Fig. 155: Action items for Download List

- **Source**
The **Source** selection box contains a list of applications that can be executed within the **File & Image Tools** service function. When you select an item from the **Source** list, the **Action** button is added to the navigation frame.
- **Action**
The **Action** selection box is context-sensitive; it contains all **Action** items that are available for the selected **Source**. When you select an item from the **Action** list, the contents of the function frame (right-hand screen) will change accordingly.
- **Navigation bar**
 - **Home**
Brings you back to the home page. .
 - **Help**
Starts this Online Help file.

Show/Delete

The **Download List / Show/Delete** function displays the files that have been transferred to the system. Please refer to the Transfer to system chapter for more information on how to transfer files to the system.



Fig. 156: Download List Show/Delete

- **Site Directory**

Here you can browse through the directory structure of the system, similar to Microsoft Windows Explorer.

Directories are surrounded by "<>". Click a directory name just **once** to open its contents.

At least one file has to be selected before the **Go** button in the **Navigation bar** becomes functional. To select a file for file transfer, click the file so it is highlighted. **Multiple** file selection is also supported. To select **multiple** files, press and hold the **CTRL** key and click the next file to be selected. To select a **range** of files, select the first file, press and hold the **SHIFT** key, and click the last file to be selected.

- **Filter**

With this dialog box, you can filter the files to be displayed; e.g. *.txt: All files with the extension '*.txt' are displayed.

- **Action bar**

- **Refresh**

Refreshes the contents of the **Site Directory**.

- **Delete**

The highlighted image in the **Selected Images** menu will be removed from the transfer list once Delete is clicked.

6.3.4 Extract List

HP-Z0000YPR-111005-02-951

This chapter contains all **Action** items with regard to the **Source** selection **Extract List**.

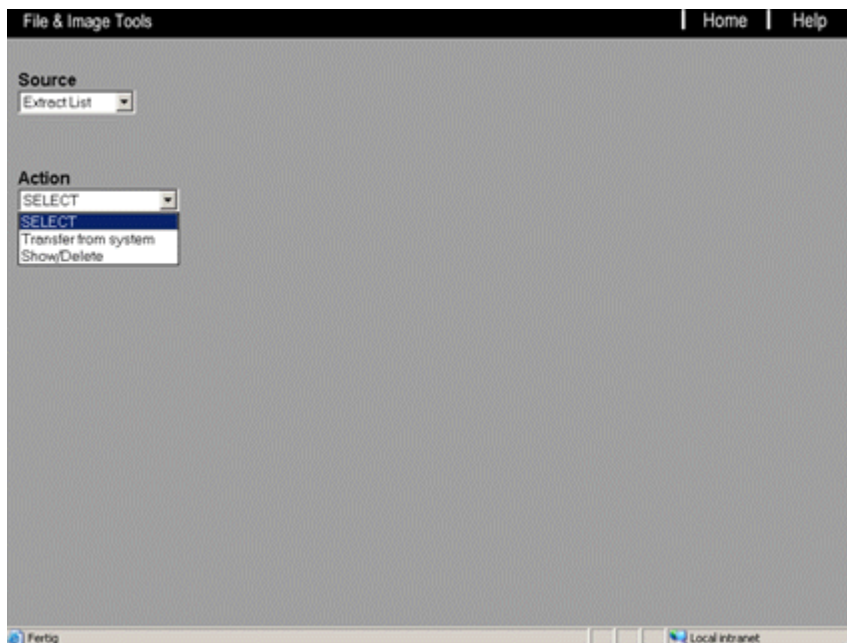


Fig. 157: Action item for Extract List

- **Source**

The **Source** selection box contains a list of applications that can be executed within the **File & Image Tools** service function. When you select an item from the **Source** list, the **Action** button is added to the navigation frame.

- **Action**

The **Action** selection box is context-sensitive; it contains all **Action** items that are available for the selected **Source**. When you select an item from the **Action** list, the contents of the function frame (right-hand screen) will change accordingly.

- **Navigation bar**

- **Home**
Brings you back to the home page.
- **Help**
Starts this Online Help file.

Transfer from system

The **Transfer from system** function allows you to transfer previously extracted images, raw data, and files from the system to a remote computer, such as a PC or Remote Server.

The information previously provided for **Image / Transfer from/to system** and **File / Transfer from/to system** is also applicable here.

All files in the directory `c:\MedCom\service\extract` will be available for selection. Scroll up and down the screen to browse through the entire list.

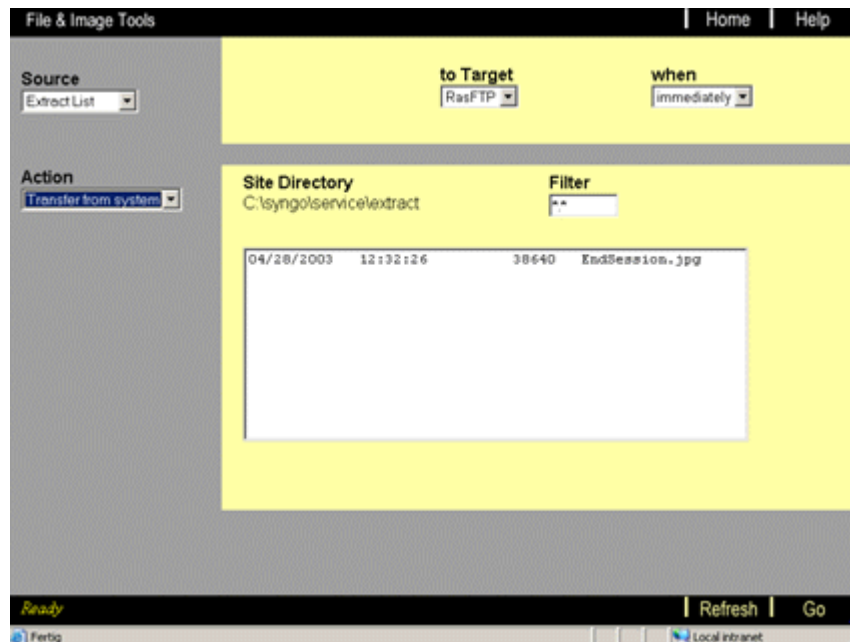


Fig. 158: Extract List Transfer from system

- **to Target**

Identifies the remote computer. This is either the Remote Server identification (if you are logged in through the Remote Server) or the remote PC identification. Targets can be added or modified in the configuration using **Service/FTP**.

- **When**

Select the appropriate transfer start time: Either **immediately** (start now) or **preferred** (start later at 22:00). For more details on the transfer start time, refer to the Start Time chapter

- **Site Directory**

Here you can browse through the directory structure of the system, similar to Microsoft Windows Explorer.

Directories are surrounded by "<>". Click a directory name just **once** to open its contents.

At least one file has to be selected before the **Go** button in the **Navigation bar** becomes functional. To select a file for file transfer, click the file so it is highlighted. **Multiple** file selection is also supported. To select **multiple** files, press and hold the **CTRL** key and click the next file to be selected. To select a **range** of files, select the first file, press and hold the **SHIFT** key, and click the last file to be selected.

- **Filter**

With this dialog box you can filter the files to be displayed; e.g. *.txt: All files with the extension '*.txt' are displayed.

- **Refresh**

Refreshes the contents of the **Site Directory**.

- **Go**

When a large file is transferred, a new window is opened that shows the file transfer progress. Each file selection will open another window in order to transfer files in parallel.

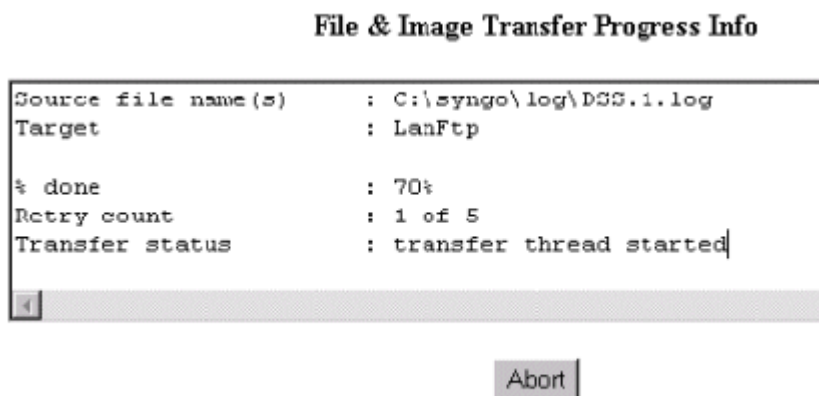


Fig. 159: Transfer progress

- **Source file name(s)**

The path and file name(s) of the file to be transferred to the remote computer.

- **Target**

Identifies the remote computer. This is the Remote Server identification (if you are logged in through the Remote Server) or the remote PC identification. Targets can be added or modified in the configuration using **Service/FTP**.

- **% done**

Indicates the progress of the file transfer in percent.

- **Retry count**

Shows the number of immediate transfer retries. Immediate transfer will be stopped after 5 attempts, and the job will be queued for a preferred transfer. Please refer to the Start Time chapter for more details on the immediately/preferred transfer start time.

- **Transfer status**

Displays the current status of the transfer. This can be an error message or information regarding the transfer itself.

- **Abort**

Aborts the file transfer function and removes it from the Transfer Manager queue.

Show/Delete

With **Show/Delete**, you can display and/or delete previously extracted images, raw data, and other files from the extract list. Refer to **Image / Extract Only** and **File / Extract Only** for more information on extracting DICOM objects and files.

All files in the directory `c:\MedCom\service\extract` will be available for selection. Scroll up and down the screen to browse through the entire list.

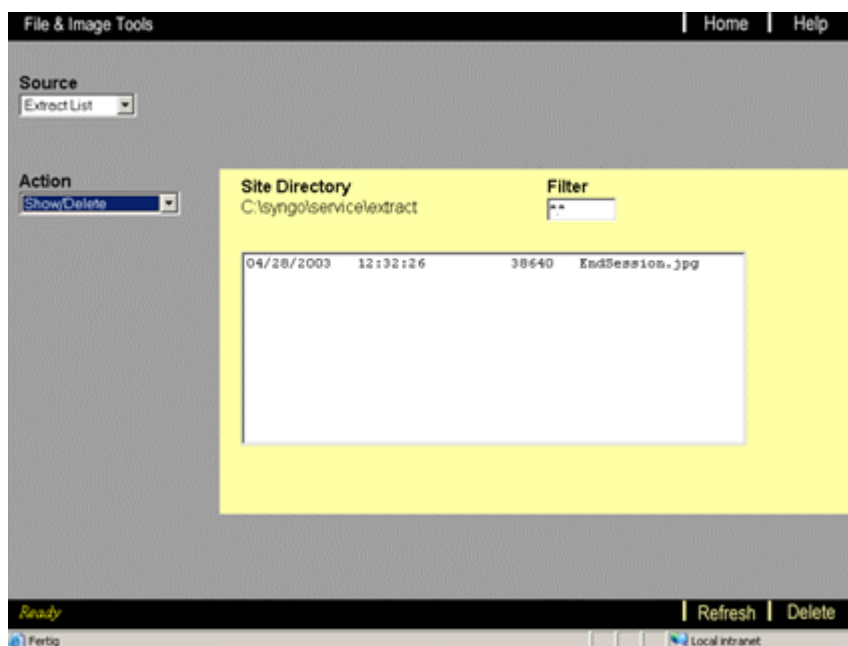


Fig. 160: Extract List Show/Delete

- **Site Directory**

Here you can browse through the directory structure of the system, similar to Microsoft Windows Explorer, beginning at the c:\MedCom\service\incoming directory.

Directories are surrounded by "<>". Click a directory name just **once** to open its contents.

At least one file has to be selected before the **Go** button in the **Navigation bar** becomes functional. To select a file for file transfer, click the file so it is highlighted. **Multiple** file selection is also supported. To select **multiple** files, press and hold the **CTRL** key and click the next file to be selected. To select a **range** of files, select the first file, press and hold the **SHIFT** key, and click the last file to be selected.

- **Filter**

With this dialog box, you can filter the files to be displayed; e.g. *.txt: All files with the extension '.txt' are displayed.

- **Action bar**

- **Refresh**

Refreshes the contents of the Site Directory.

- **Delete**

The highlighted image in the **Site Directory** menu will be removed from the transfer list once Delete is clicked.

6.3.5 Transfer list

HP-Z0000YPR-111005-02-950

This chapter contains all **Action** items with regard to the **Source** Transfer List.

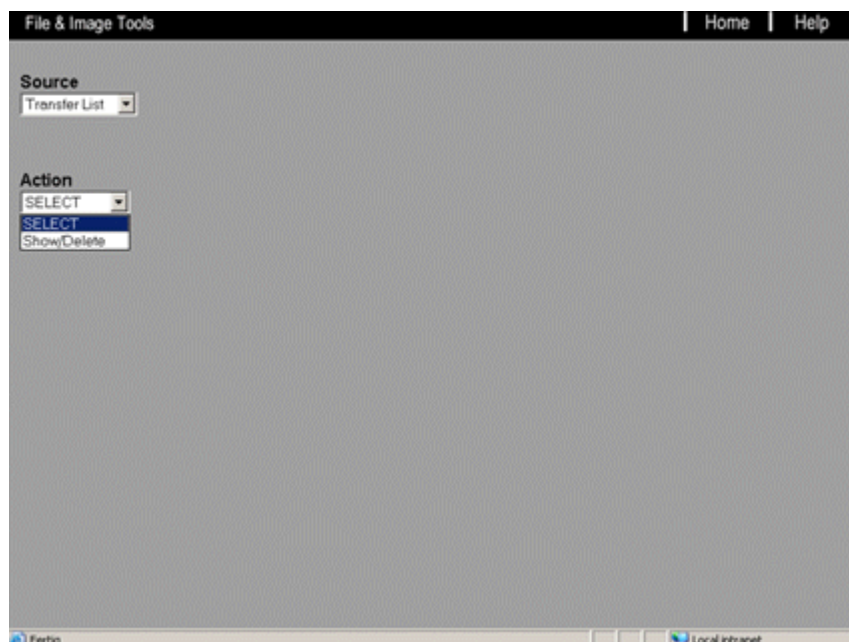


Fig. 161: Action items for Transfer List

- **Source**

The **Source** selection box contains a list of applications that can be executed within the **File & Image Tools** service function. When you select an item from the **Source** list, the **Action** button is added to the navigation frame.

- **Action**

The **Action** selection box is context-sensitive; it contains all **Action** items that are available for the selected **Source**. When you select an item from the **Action** list, the contents of the function frame (right-hand screen) will change accordingly.

- **Navigation bar**

- **Home**
Brings you back to the home page.
- **Help**
Starts this Online Help file.

Show/Delete

Show/Delete displays the Transfer Manager jobs pending for the specified target and may remove these jobs from the transfer list.

To display or remove pending batch jobs for a specific target, select the appropriate pending batch job. Only one pending batch job can be selected at a time.

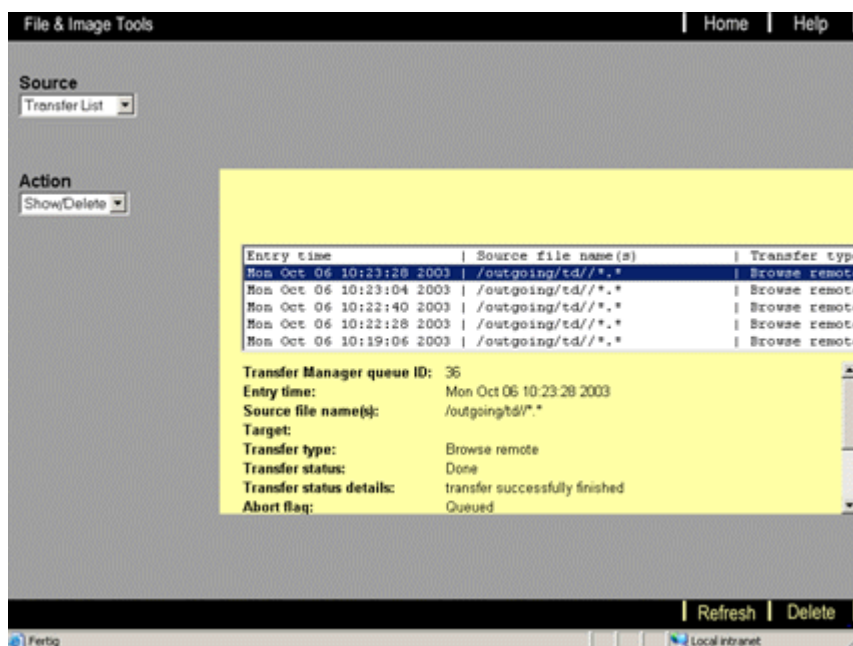


Fig. 162: Transfer List Show/Delete

- **Transfer Manager queue ID**

The unique location of the selected job in the transfer list.

- **Entry time**

Shows the time when the queue was added to the transfer list.

- **Source file name(s)**

Displays the path and the name of the file to be transferred to the remote computer.

- **Target**

Identifies the remote computer. This is either the Remote Server identification (if you are logged in through the Remote Server) or the remote PC identification. Targets can be added or modified in the configuration using **Service/FTP**.

- **Transfer type**

Shows the type of transfer that has to be used to transfer the file.

- **FTP upload**

Transfers the file from the system to the remote computer using the FTP protocol.

- **FTP download**

Transfers the file from the remote computer to the system using the FTP protocol.

- **Browser remote**

This mode indicates that the **Transfer to system** function is “browsing” or retrieving the directory contents of the Remote Server directory.

- **Mail**

Mail is being sent to the remote computer using the SMTP protocol.

- **Transfer status**

Indicates the progress of the transfer.

- **Queued**
Has been added to the Transfer Manager queue.
- **Done**
Has been executed successfully and will be automatically removed from the queue after approximately 5 minutes.
- **Failed**
If the transfer failed, the system will retry automatically. Please refer to the Start Time chapter for more details on the image transfer functions.
- **Cancelled**
If a transfer is aborted, the job will be relegated to cancelled status.
- **Initial**
The Transfer Manager initializes the queue job. When the queue job is successfully initialized, the job will be designated the queued or active status.
- **Active**
The queued job is executing.
- **Dialing**
The system establishes a modem connection to the remote computer.
- **Connecting**
The TCP/IP connection to the remote computer is being established.
- **Transferring**
The file is being transferred (copied) to/from the remote computer.
- **Disconnecting**
The TCP/IP connection to the remote computer is being disconnected.
- **Hanging up**
The modem connection is being disconnected.
- **Transfer status details**
Displays the current status of the transfer. This can be an error message or information regarding the transfer itself.
- **Abort flag**
 - **Queued**
This job is entered into the Transfer Manager queue.
 - **Aborted**
The queued job will be removed from the Transfer Manager queue.
- **Total bytes to transfer**
Amount of bytes that have to be transferred to the remote computer.
- **Actually transferred bytes**
Total amount of bytes transferred to the remote computer.
- **Retry limit**
Shows the number of retries allowed; the number has been fixed to 5 retries.
- **Retry count**
Shows the amount of transfer retries. The transfer will be stopped after 5 attempts.

- **Transfer time**

Shows the time at which the file was transferred.

- **Action bar**

- **Refresh**

Refreshes the contents of the Transfer Manager queue.

- **Delete**

The highlighted selection in the Transfer Manager queue will be removed from the transfer list once Delete is clicked.

7 DICOM Tests

7.1 General

HP-Z0000YPR-111005-02-949

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7.1.1 Remote access

HP-Z0000YPR-111005-02-948

For most functions, DICOM Test Tools require **Full Access Mode** to perform remote service. "Verify Network" and the "Analysis" window can be started in the **Limited Access Mode**.

Full access is necessary because some functions stop or start application processes. However, full access disables user interaction at the customer user interface. Since certain tests require user interaction, the DICOM Test Tools make it possible to **release the customer user interface**. For this purpose, use the buttons **Release host** and **Back to full access**. An appropriate flashing message (displayed in the customer UI) indicates that someone is logged on in "Full Access Mode".

7.1.2 Layout

HP-Z0000YPR-111005-02-947

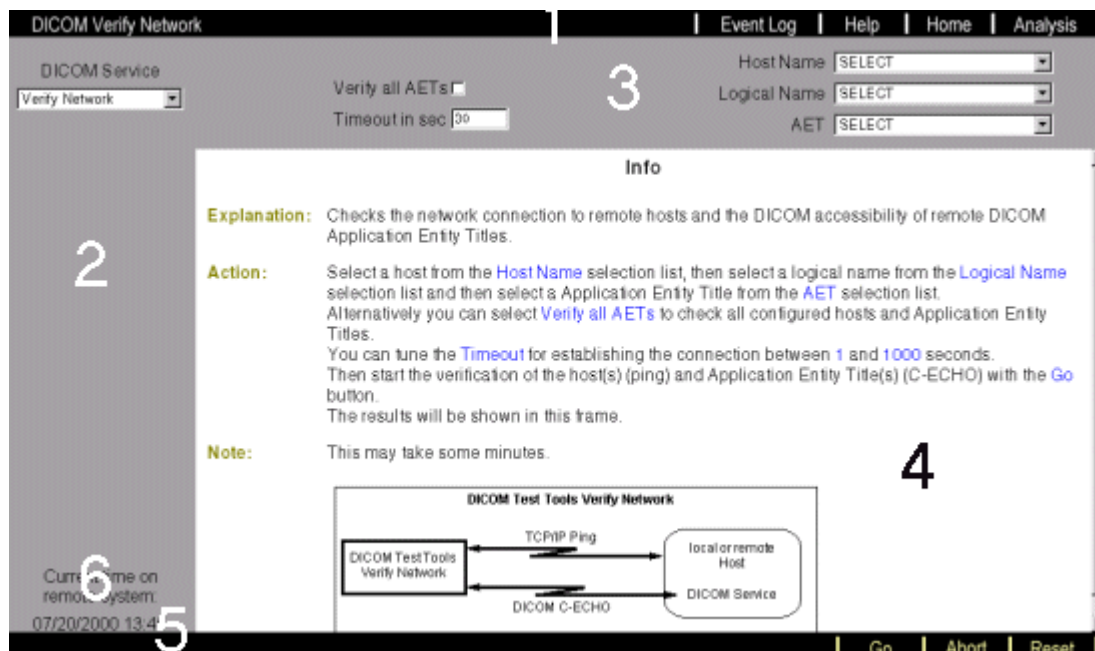


Fig. 163: Layout of the DICOM service menu

1. Navigation bar
2. DICOM Service and Mode selection frame (navigation frame)
3. Specific selection frame (navigation frame)
4. Function frame
5. Action Bar
6. Current date and time on remote system

7.1.3 Features

HP-Z0000YPR-111005-02-946

This chapter provides an overview of the tools supported by DICOM Services.

DICOM Services

- **Verify Network:**

Network connection on TCP/IP level (ping) and on DICOM level (DICOM Verification (C-ECHO)) can be tested either for a certain destination or for all configured DICOM nodes.

- **Storage:**

The trace level for the DICOM log file (merge_stu.log) can be extended to display e.g. the association negotiation (T3), the command set of a DICOM message (T7), and the data set of a DICOM message (T2).

In addition, a Test Receiver may be started to capture original DICOM images on the hard disk. A Test Sender can be used to check the receiving functions of the local or remote DICOM application.

- **Basic Print:**

A test basic print provider may be started and images may be sent to this process. The message transfer and the film may be displayed.

The trace level for the DICOM log file (merge_hc.log) may be extended to display e.g. the association negotiation (T3), the command set of a DICOM message (T7), and the data set of a DICOM message (T2).

- **Query/Retrieve:**

The trace level for the DICOM log file (merge_stu.log) may be extended to display e.g. the association negotiation (T3), the command set of a DICOM message (T7), and the data set of a DICOM message (T2).

- **Storage Commitment:**

The trace level for the DICOM log file (merge_stu.log) may be extended to display e.g. the association negotiation (T3), the command set of a DICOM message (T7), and the data set of a DICOM message (T2).

- **Modality Worklist:**

The trace level for the DICOM log file (merge_HRI.log) may be extended to display e.g. the association negotiation (T3), the command set of a DICOM message (T7), and the data set of a DICOM message (T2).

- **Modality Performed Procedure Steps:**

The trace level for the DICOM log file (merge_HRI.log) may be extended to display e.g. the association negotiation (T3), the command set of a DICOM message (T7), and the data set of a DICOM message (T2).

How to Use the DICOM Service Tools

Choose the required function from the DICOM Service and Mode selection frame (2), and select additional information (if required) from the specific selection frame (3).

Activate the configured process by clicking **Go** on the Action bar (5).

The results are displayed in the function frame (4).

The **Analysis platform** may be activated from the navigation bar (1). The analysis frame layout is identical to the DICOM service platform.

Exit the DICOM Service Tools

To exit the DICOM Service Tools, stop the last test function by clicking the **Abort** button in the action bar. Then select the **Home** button in the navigation bar. An information window appears.

NOTE

If you want to save the captured messages (including images) and log files, you need to click the **Cancel** button.

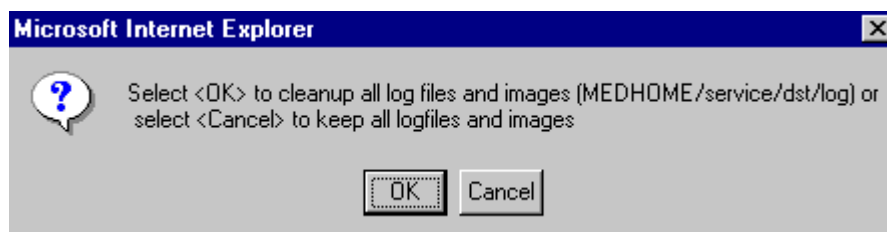


Fig. 164: Exiting the DICOM Test Tools

7.2 Functions

HP-Z0000YPR-111005-02-945

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7.2.1 Dicom Services

HP-Z0000YPR-111005-02-944

Tools are provided for the DICOM Services **Verification, Storage, Storage Commitment, Query/Retrieve, Basic Print, Modality Worklist and Modality Performed Procedure Steps**. These tools may be selected under the menu item **DICOM Service**.

After you have selected a certain tool, you need to **Abort** it before invoking another tool or exiting the DICOM Test Tools.

Verify Network

DICOM Verify Network | Event Log | Help | Home | Analysis

DICOM Service: **Verify Network**

Verify all AETs: ☐ Timeout in sec:

Host Name: Logical Name: AET:

Info

Explanation: Checks the network connection to remote hosts and the DICOM accessibility of remote DICOM Application Entity Titles.

Action: Select a host from the **Host Name** selection list, then select a logical name from the **Logical Name** selection list and then select a Application Entity Title from the **AET** selection list. Alternatively you can select **Verify all AETs** to check all configured hosts and Application Entity Titles. You can tune the **Timeout** for establishing the connection between **1** and **1000** seconds. Then start the verification of the host(s) (ping) and Application Entity Title(s) (C-ECHO) with the **Go** button. The results will be shown in this frame.

Note: This may take some minutes.

Current time on remote system: 12/16/2000 09:47

DICOM Test Tools Verify Network

DICOM TestTools Verify Network ↔ TCP/IP Ping ↔ local or remote Host

DICOM TestTools Verify Network ↔ DICOM C-ECHO ↔ DICOM Service

Go | Abort | Reset

Fig. 165: DICOM Verification of a single destination

Verify Network performs a **TCP/IP ping**. If this ping is successful, the DICOM Service **DICOM Verification** sends a C-ECHO request to a DICOM destination process (AET) on the local or remote system.

DICOM Verification may be performed for a specific DICOM node or simultaneously for all configured DICOM nodes (Verify all AETs).

The test consists of 2 steps: The first step is the TCP/IP ping, and the second step is the C-ECHO command on the DICOM level.

You can stop the test by clicking the **Abort** button in the action bar.

Host	IP	Ping Status	AET	CEcho Status
146.254.120.74		OK	CSYNGO1	C-ECHO successful
cmv10001	146.254.120.75	OK	049SA1DCDADRSP	C-ECHO successful
cmv10001	146.254.120.75	OK	049SA1DCDADQRY	C-ECHO successful
cmv10001	146.254.120.75	OK	049SA1DCDADNES	C-ECHO successful
cvzwlz1	146.254.120.82	OK	CVZWMZ1	C-ECHO successful
cvrt1	146.254.120.76	Timeout	CVRT1	not performed
cease1	146.254.120.83	OK	CEASE1	Unable to open association: Association negotiation failed
cmsxs1	146.254.120.86	OK	049SA1OT07DRSP	Unable to open association: Failed to connect to remote host
cmsxs1	146.254.120.86	OK	049SA1OT07DQRY	Unable to open association: Failed to connect to remote host
cmsxs1	146.254.120.86	OK	049SA1OT07DNES	Unable to open association: Failed to connect to remote host

Fig. 166: DICOM Verification of all AETs

- **Verify all AETs:**

A TCP/IP ping command and a DICOM C_ECHO command are performed automatically for all configured DICOM nodes. If a node is not running or not accessible, you have to wait for 30 seconds (default time).

- **Timeout in sec**

The default timeout for C_ECHO of 30 seconds may be changed here.

- **Host Name**

This field is only displayed if **Verify all AETs** is not selected. Select the host name of the destination device. The device itself (own host name) may be selected as well. After the logical name or AET (next two buttons) is selected, the appropriate host name will be displayed automatically.

- **Logical Name**

This field is only displayed if **Verify all AETs** is not selected. Select the logical name of the destination device. If the AET (next button) has been previously selected, the appropriate logical name will be displayed automatically.

- **AET**

Select the Application Entity Title of the destination device.

Troubleshooting remarks

- A TCP/IP ping may fail for the following reasons:
 - The remote host is turned off or not in the same network, no router is configured, or the routing demon is not started.
 - The TCP/IP address is configured incorrectly.
 - A networking device (router, switch, bridge, etc.) denies access to the other network.

- A C-ECHO will fail for the following reasons:
 - AET or port number is not configured correctly.
 - The DICOM process on the remote host is not running.
 - Certain products check the IP address or AET of the sending system. If the AET or IP address of the local system is not entered correctly, DICOM verification will fail.
 - The remote host does not support DICOM verification as a Service Class Provider (see DICOM Conformance Statement).

Storage

The storage service class determines how DICOM images and other objects (waveforms, LUTs, overlays) are sent (SCU) or received (SCP). The DICOM Standard defines several transfer syntaxes, each of which is a set of encoding rules for communication (e.g. byte ordering, compression).

Extend Trace Level

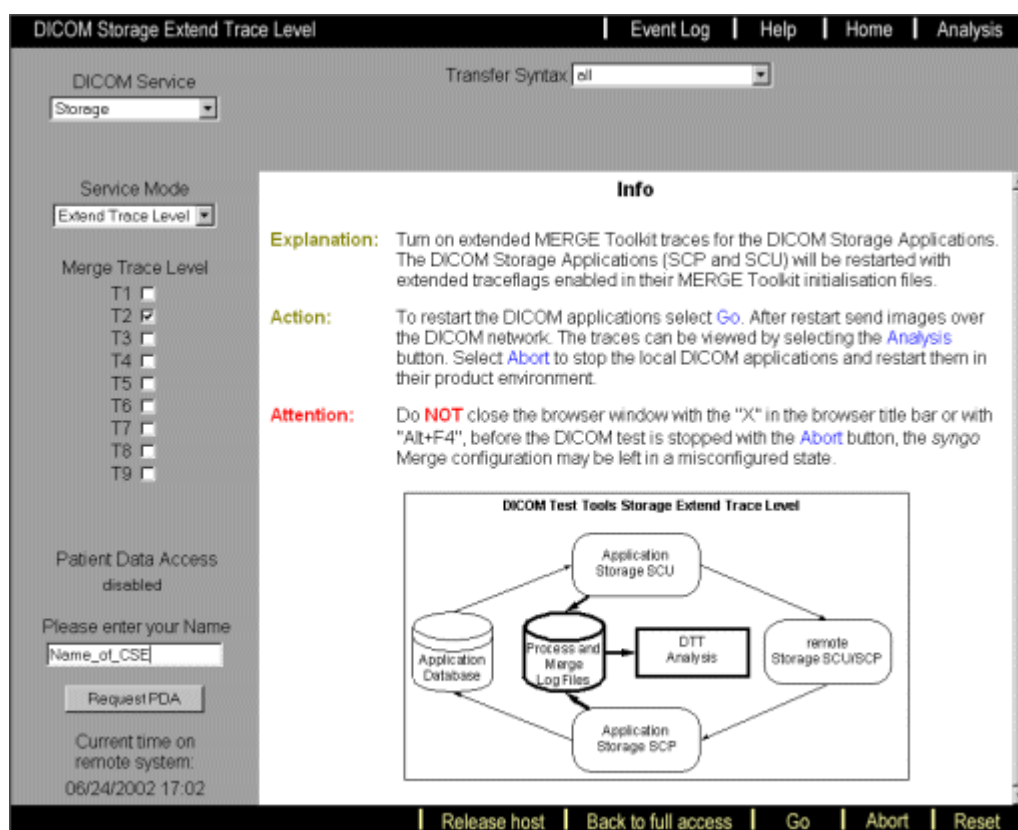


Fig. 167: Storage - Extended trace menu

This DICOM implementation is based on a toolkit from Merge Technologies. This tool kit includes a mergecom.log file designed as a ring buffer. Certain trace levels may be enabled to receive more detailed information regarding the latest DICOM communications. The respective mergecom.log file can be displayed under Analysis at "Examine Logs - merge_stu.log". Basic knowledge of DICOM is necessary for interpreting the log files.

Storage, Storage Commitment and Query/Retrieve use the same mergecom.log file, as they belong to the same syngo service.

C:\%MEDHOME%\log\merge_stu.log Storage, Storage Commitment, and Query/Retrieve

The storage test tools may only be started remotely in “Full Access Mode”. In this case, no user interaction is possible. It is therefore not possible to send / print images or to perform a query request. Since these user interactions are required in the mergecom.log files to receive the appropriate messages, DICOM Test Tools provide the option of enabling user interaction even in full access mode.

- **Release host**

Select this item in the Action bar to work with the user interface (identical to limited access). It is even possible to iconize the “Remote Access Control Window”.

- **Back to full access**

After the images have been transferred or a Query/Retrieve has been performed, you have to disable interactions at the user interface to perform further tasks in the DICOM Test Tools.

- **Transfer syntax**

This drop-down list lets you choose transfer syntaxes used for the association negotiation after a restart of the DICOM processes. It provides additional troubleshooting possibilities, since you can limit the settings for the next associations. Before selecting other DICOM services or functions, or before finishing the DICOM Test Tools, “abort” the Extend Trace Level function. The original Transfer Syntaxes will be set again.

- **Merge trace level**

The following trace levels may be enabled to provide more detailed information in the log file merge_stu.log. This log file may be displayed under “Analysis - Examine Logs”. **Also refer to the example of an association with trace level T2, T3, T7 and T9 in the appendix.**

- **T1** This trace level does not affect the mergecom.log file.

- **T2 data set**

This trace level shows the attributes of the transferred message, e.g. the image header attributes or the attributes within a DICOM message.

NOTE

Trace level T2 also shows patient data in the log file. For this reason, “Patient Data Access” is required for remote access.

- **T3 association negotiation**

Every DICOM communication starts with the association negotiation. The initiator (SCU) sends a list of all supported image objects and transfer syntaxes to the remote system (SCP). The AETs of both systems on the T3 level are also displayed.

The SCP responds with the status (accept or reject) of the image objects and selects one of the proposed transfer syntaxes.

- **T4 rejection of incoming messages**

When incoming associations are automatically rejected (IP address not known), log messages will be displayed at this trace level.

- **T5 regular and extended regulations**

Logs messages relating to regular and extended regulations.

- **T6 mergecom configuration files**

At this trace level the related mergecom.ini, mergecom.pro, mergecom.app, and mergecom.srv files will be written in the mergecom.log file. This may be helpful when analyzing the log file, although it requires a large amount of space in the log file.

- **T7 command set**

Following the association negotiation, a command set (attributes with group 0) will be transferred. Before sending an image, for example, the command set will tell the remote system the type of image object (SOP Class UID such as CT, RF, etc.) to be transferred, including the UID of the image.

The response command set after each image will show the status of the transfer, where status = 0 means a successful transfer.

- **T8 streaming messages**

Log messages relating to the streaming in and out of messages and file objects are displayed on the T8 level.

- **T9 PDU level**

The entire DICOM association is displayed on the PDU level (Protocol Data Unit). Each PDU defined with a maximum size during the association negotiation will be displayed in one line. The association negotiation will be shown in greater detail.

T9 is the only way to display the association release request and response at the end of an association.

Patient Data Access

In case of remote access to the DICOM Service Tools, viewing patient-relevant data (name, ID, etc.) is not permitted. However, if you enable trace level T2, patient data will be displayed in the log file. For this purpose, the customer has to provide additional access rights, allowing the service engineer to view patient-relevant data. To obtain these rights, the service engineer has to enter his name and send a "Patient Data Access" request to the customer's user interface.

- **Please enter your name**

Enter the name of the local service engineer logged in remotely who requires Patient Data Access rights.

- **Request PDA**

When you select this button, an additional selection item (Patient Data Access) is activated in the customer user interface under "Option - Remote Access". Additionally, the name of the service engineer will be displayed. The customer has to select this menu item for trace level T2 to be activated in the log file.

Send Images

This menu item can be used to send DICOM images (stored on the hard disk via the Test Receiver or, e.g. FTP) to a user-specified DICOM receiver. By sending these images to the local DICOM receiver, you are able to import images into the database and display them in the user interface.

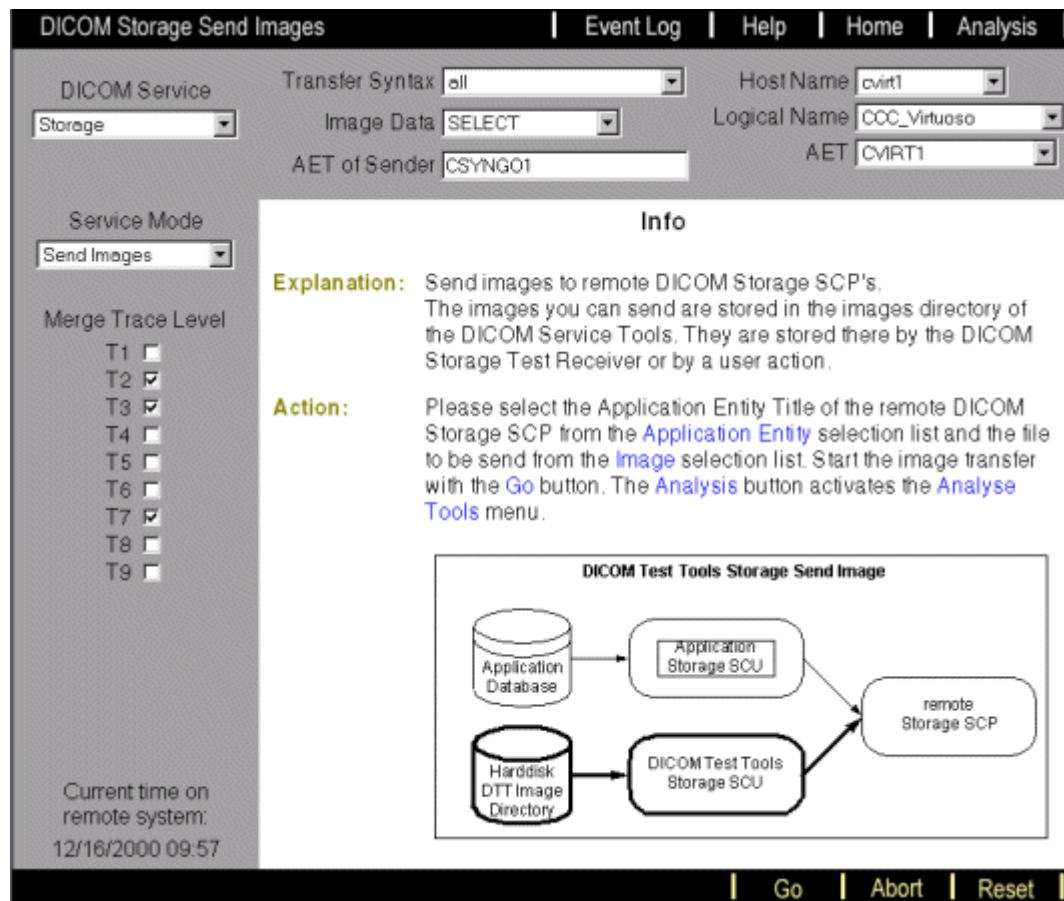


Fig. 168: Storage - Send Images menu

- **Transfer syntax**

This drop-down list lets you choose transfer syntaxes that will be enabled after the DICOM processes are restarted. It provides additional troubleshooting possibilities, as you can limit the settings for the next associations.

Before selecting other DICOM Services or functions, or before finishing the DICOM Test Tools, “abort” the “Send Images” function. The original Transfer Syntaxes will be set again.

- **Image Data**

A list of DICOM images located in the directory %MEDHOME%\service\dst\img is provided. The Test Receivers (see Simulate/Add local/remote Receiver) will copy all received images into this directory. Images copied manually (ftp) into this directory will also be displayed. Certain default DICOM images will be delivered with the DICOM Test Tools for test purposes.

- **AET of Sender**

The sending AET is, by default, the AET of the local storage SCP. Usually it is not necessary to change this value.

- **Host Name**

Host name of the destination system. The system itself (host name) may be selected as well. When selecting the logical name or AET (next two buttons), the appropriate host name will be displayed automatically.

- **Logical Name**

Select the logical name of the destination device. When selecting the AET (next button) first, the appropriate logical name will be displayed automatically.

- **AET**

Select the Storage AET of the destination device. Please note: The Send function will fail if you e.g. select the Query/Retrieve AET of the remote system instead of the Storage AET.

- **Merge trace level**

The following trace levels may be enabled to provide more detailed information in the log file merge_stu.log. This log file may be displayed under “Analysis - Examine Logs“. **Also refer to the example of an association with trace level T2, T3, T7 and T9 in the appendix.**

- **T1**

This trace level does not affect the mergecom.log file.

- **T2 data set**

This trace level shows the attributes of the transferred message, e.g. the image header attributes or the attributes within a DICOM message.

NOTE

Trace level T2 also shows patient data in the log file. For this reason, “Patient Data Access“ is required for remote access.

- **T3 association negotiation**

Every DICOM communication starts with the association negotiation. The initiator (SCU) sends a list of all supported image objects and transfer syntaxes to the remote system (SCP). The AETs of both systems are also displayed at the T3 level.

The SCP responds with the status (accept or reject) of the image objects and selects one of the proposed transfer syntaxes.

- **T4 rejection of incoming messages**

If incoming associations are automatically rejected (IP address not known), log messages will be displayed on this trace level.

- **T5 regular and extended regulations**

Logs messages relating to regular and extended regulations.

- **T6 mergecom configuration files**

At this trace level the related mergecom.ini, mergecom.pro, mergecom.app, and mergecom.srv files will be written in the mergecom.log file. This may be helpful when analyzing the log file, however, please note that analysis requires a large amount of space in the log file.

- **T7 command set**

Following the association negotiation, a command set (attributes with group 0) will be transferred. Before sending an image, for example, the command set will tell the remote system the type of image object (SOP Class UID such as CT, RF, etc.) to be transferred, including the UID of the image.

The response command set after each image will show the status of the transfer, where status = 0 means a successful transfer.

- **T8 streaming messages**

Log messages relating to the streaming in and out of messages and file objects are displayed at the T8 level.

- **T9 PDU level**

The entire DICOM association is displayed on the PDU level (Protocol Data Unit). Each PDU defined with a maximum size during the association negotiation will be displayed in one line.

The association negotiation will be shown in greater detail.

T9 is the only way to display the association release request and response at the end of an association.

Patient Data Access

In case of remote access to the DICOM Service Tools, viewing patient-relevant data (name, ID, etc.) is not permitted. However, if you enable trace level T2, patient data will be displayed in the log file. For this purpose, the customer has to provide additional access rights, allowing the service engineer to view patient-relevant data. To obtain these rights, the service engineer has to enter his name and send a "Patient Data Access" request to the customer's user interface.

- **Please enter your name**

Enter the name of the local service engineer logged in remotely who requires Patient Data Access rights.

- **Request PDA**

Selecting this button initiates the activation of an additional selection item (Patient Data Access) in the customer user interface under "Option - Remote Access". Additionally, the name of the service engineer will be displayed. The customer has to select this menu item for trace level T2 to be activated in the log file.

Before selecting other DICOM Services or functions, or before finishing the DICOM Test Tools, "abort" the "Send Images" function.

Simulate Remote Receiver

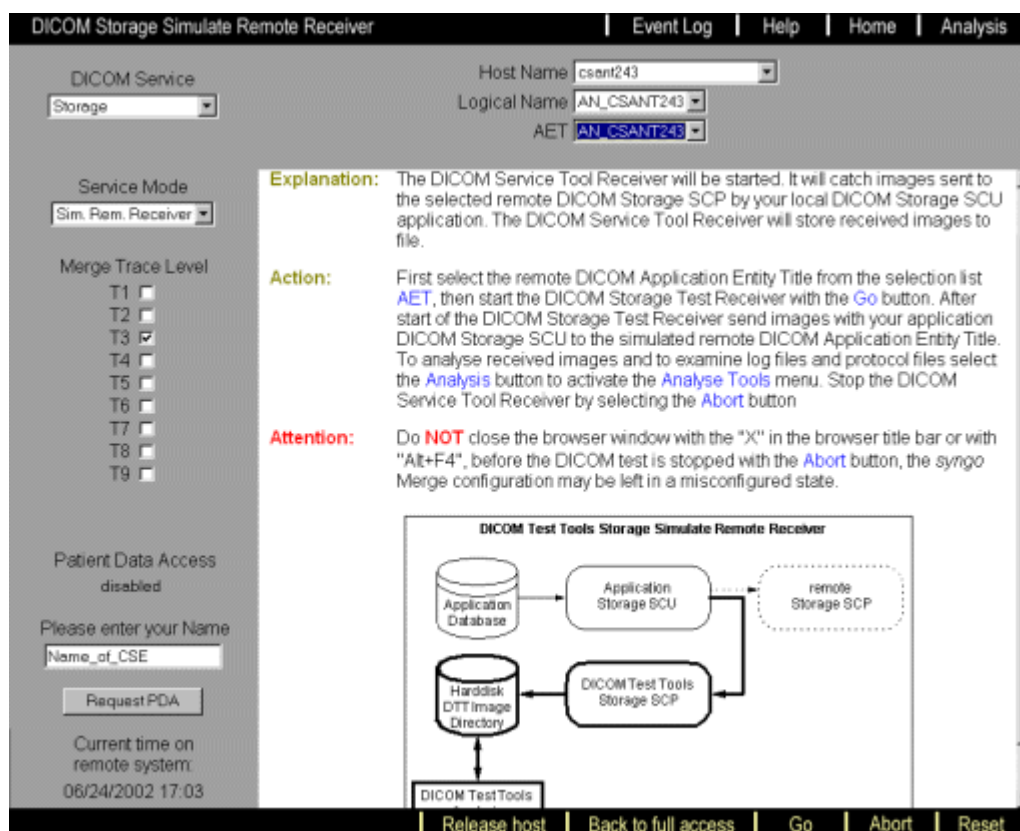


Fig. 169: Storage - Simulate Remote Receiver menu

When this menu item is selected, the Test Receiver provided by the DICOM Test Tools is started to simulate a remote DICOM receiver. You will receive a list of all DICOM AETs configured on your system. Select the AET you would like to have replaced with the Test Receiver. All images sent by the local sender to this destination will be received by the Test Receiver and stored on the hard disk in DICOM format.

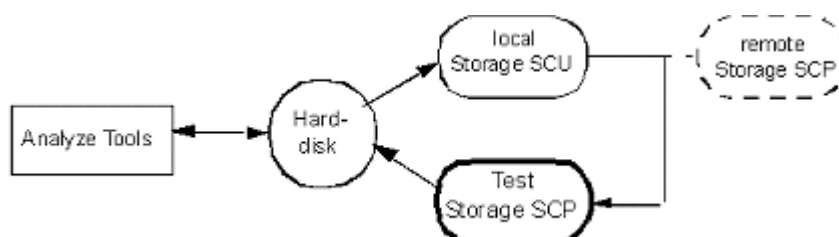


Fig. 170: Storage - Simulate Remote Receiver diagram

Go to Analysis to have the received images evaluated. An image header dump may be performed, or the image can be displayed.

The received images and the log files may be removed when exiting the DICOM Service Tools.

The storage test tools may only be started remotely in "Full Access Mode". In this case, no user interaction is possible, and, therefore, it is also not possible to send / print images or to perform a query request. Since these user interactions are necessary to receive the appropriate messages in the mergecom.log files, the DICOM Test Tools provide the option of enabling user interaction even in full access mode.

- **Release host**

Select this item in the Action bar to work with the user interface (identical to limited access). It is even possible to iconize the “Remote Access Control Window”.

- **Back to full access**

After the images have been transferred or a Query/Retrieve has been performed, this function disables interaction at the user interface to perform further tasks in the DICOM Test Tools.

- **Host Name**

Host name of the destination system. The system itself (host name) may be selected as well. When selecting the logical name or AET (next two buttons), the appropriate host name will be displayed automatically.

- **Logical Name**

Select the logical name of the destination device. When selecting the AET (next button) first, the appropriate logical name will be displayed automatically.

- **AET**

Select the Storage AET of the destination device. Please note: The Send function will fail if you e.g. select the Query/Retrieve AET of the remote system instead of the Storage AET.

- **Merge trace level**

The following trace levels may be enabled to provide more detailed information in the log file merge_stu.log. This log file may be displayed under “Analysis - Examine Logs”.

Also refer to the example of an association with trace level T2, T3, T7 and T9 in the appendix.

- **T1**

This trace level does not affect the mergecom.log file.

- **T2 data set**

This trace level shows the attributes of the transferred message, e.g. the image header attributes or the attributes within a DICOM message.

NOTE

Trace level T2 also shows patient data in the log file. For this reason, “Patient Data Access” is required for remote access.

- **T3 association negotiation**

Every DICOM communication starts with the association negotiation. The initiator (SCU) sends a list of all supported image objects and transfer syntaxes to the remote system (SCP). The AETs of both systems are also displayed at the T3 level.

The SCP responds with the status (accept or reject) of the image objects and selects one of the proposed transfer syntaxes.

- **T4 rejection of incoming messages**

If incoming associations are automatically rejected (IP address not known), log messages will be displayed on this trace level.

- **T5 regular and extended regulations**

Log messages relating to regular and extended regulations.

- **T6 mergecom configuration files**

At this trace level the related mergecom.ini, mergecom.pro, mergecom.app, and mergecom.srv files will be written in the mergecom.log file. This may be helpful when analyzing the log file, however, please note that analysis requires a large amount of space in the log file.

- **T7 command set**

Following the association negotiation, a command set (attributes with group 0) will be transferred. Before sending an image, for example, the command set will tell the remote system the type of image object (SOP Class UID such as CT, RF, etc.) to be transferred, including the UID of the image. The response command set after each image will show the status of the transfer, where status = 0 means a successful transfer.

- **T8 streaming messages**

Log messages relating to the streaming in and out of messages and file objects are displayed at the T8 level.

- **T9 PDU level**

The entire DICOM association is displayed on the PDU level (Protocol Data Unit). Each PDU defined with a maximum size during the association negotiation will be displayed in one line.

The association negotiation will be shown in greater detail.

T9 is the only way to display the association release request and response at the end of an association.

Patient Data Access

In case of remote access to the DICOM Service Tools, viewing patient-relevant data (name, ID, etc.) is not permitted. However, if you enable trace level T2, patient data will be displayed in the log file. For this purpose, the customer has to provide additional access rights, allowing the service engineer to view patient-relevant data. To obtain these rights, the service engineer has to enter his name and send a "Patient Data Access" request to the customer's user interface.

The objects (images) captured with the Test Receiver will normally be anonymized. If patient-relevant data is required for troubleshooting (not anonymized), it is necessary to obtain the "Patient Data Access" rights from the customer's user interface.

- **Please enter your name**

Enter the name of the local service engineer logged in remotely who requires Patient Data Access rights.

- **Request PDA**

When you select this button, an additional selection item (Patient Data Access) will be activated in the customer user interface under "Option - Remote Access". Additionally, the name of the service engineer will be displayed. The customer has to select this menu item for trace level T2 to be activated in the log file.

Before selecting other DICOM services or functions, or before finishing the DICOM Test Tools, "abort" the "Simulate Remote Receiver" function.

Simulate Local Receiver

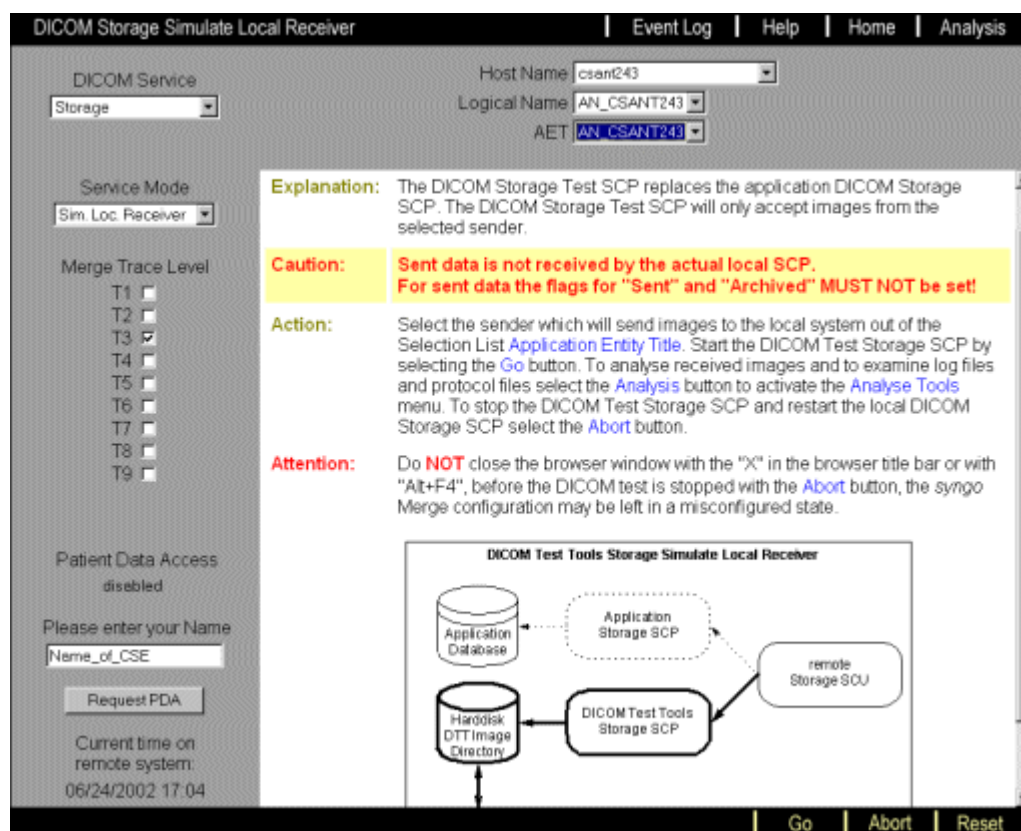


Fig. 171: Storage - Simulate Local Receiver menu

When this menu item is selected, a Test Receiver provided by the DICOM Test Tools is started using the same AET and port number as the application DICOM receiver. The original application receiver will be stopped. All images sent by remote senders will be stored in DICOM format on the hard disk.

NOTE

None of the images received by the Test Receiver will be stored in the database. To avoid losing images, select the AET of the sending system. All other senders will be rejected.

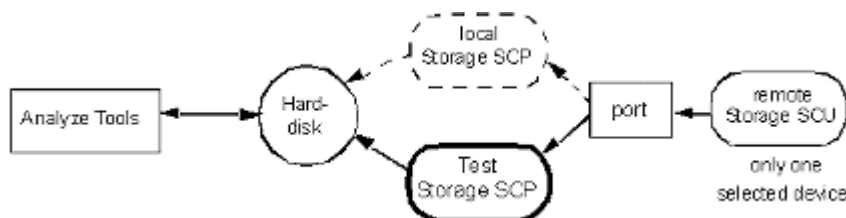


Fig. 172: Storage - Simulate Local Receiver diagram

The received images may be analyzed with the tools provided by "DICOM Analysis".

The received images and the log files may be removed when exiting from the DICOM Service Tools.

- **Host Name**

Host name of the sending system. For data security, only images from this AET will be received by the Test Receiver. All other systems will be rejected. The system itself (host name) may be selected as well. When selecting the logical name or AET (next two buttons), the appropriate host name will be displayed automatically.

- **Logical Name**

Select the logical name of the sending system. When selecting the AET (next button) first, the appropriate logical name will be displayed automatically.

- **AET**

Select the storage AET of the sending system. Please note: The Send function will fail if you e.g. select the Query/Retrieve AET of the remote system instead of the Storage AET.

- **Merge trace level**

The following trace levels may be enabled to provide more detailed information in the log file merge_stu.log. This log file may be displayed under “Analysis - Examine Logs”.

Also refer to the example of an association with trace level T2, T3, T7 and T9 in the appendix.

- **T1**

This trace level does not affect the mergecom.log file.

- **T2 data set**

This trace level shows the attributes of the transferred message, e.g. the image header attributes or the attributes within a DICOM message.

NOTE

Trace level T2 also shows patient data in the log file. For this reason, “Patient Data Access” is required for remote access.

- **T3 association negotiation**

Every DICOM communication starts with the association negotiation. The initiator (SCU) sends a list of all supported image objects and transfer syntaxes to the remote system (SCP). The AETs of both systems are also displayed at the T3 level.

The SCP responds with the status (accept or reject) of the image objects and selects one of the proposed transfer syntaxes.

- **T4 rejection of incoming messages**

If incoming associations are automatically rejected (IP address not known), log messages will be displayed on this trace level.

- **T5 regular and extended regulations**

Log messages relating to regular and extended regulations.

- **T6 mergecom configuration files**

At this trace level the related mergecom.ini, mergecom.pro, mergecom.app, and mergecom.srv files will be written in the mergecom.log file. This may be helpful when analyzing the log file, however, please note that analysis requires a large amount of space in the log file.

- **T7 command set**

Following the association negotiation, a command set (attributes with group 0) will be transferred. Before sending an image, for example, the command set will tell the remote system the type of image object (SOP Class UID such as CT, RF, etc.) to be transferred, including the UID of the image. The response command set after each image will show the status of the transfer, where status = 0 means a successful transfer.

- **T8 streaming messages**

Log messages relating to the streaming in and out of messages and file objects are displayed at the T8 level.

- **T9 PDU level**

The entire DICOM association is displayed on the PDU level (Protocol Data Unit). Each PDU defined with a maximum size during the association negotiation will be displayed in one line.

The association negotiation will be shown in greater detail.

T9 is the only way to display the association release request and response at the end of an association.

Patient Data Access

In case of remote access to the DICOM Service Tools, viewing patient-relevant data (name, ID, etc.) is not permitted. However, if you enable trace level T2, patient data will be displayed in the log file. For this purpose, the customer has to provide additional access rights, allowing the service engineer to view patient-relevant data. To obtain these rights, the service engineer has to enter his name and send a "Patient Data Access" request to the customer's user interface.

The objects (images) captured with the Test Receiver will normally be anonymized. If patient-relevant data is required for troubleshooting (not anonymized), it is necessary to obtain the "Patient Data Access" rights from the customer's user interface.

- **Please enter your name**

Enter the name of the local service engineer logged in remotely who requires Patient Data Access rights.

- **Request PDA**

When you select this button, an additional selection item (Patient Data Access) is activated in the customer user interface under "Option - Remote Access". Additionally, the name of the service engineer will be displayed. The customer has to select this menu item for trace level T2 to be activated in the log file.

Before selecting other DICOM services or functions, or before finishing the DICOM Test Tools, "abort" the "Simulate Remote Receiver" function.

Additional Local Receiver

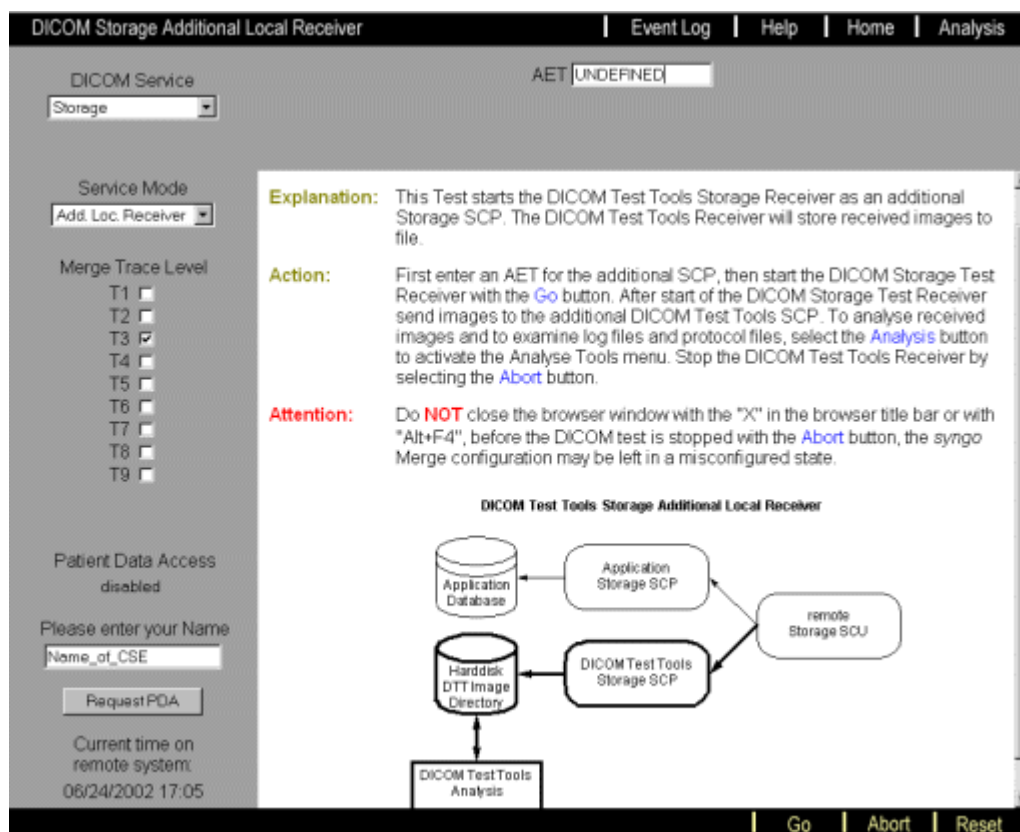


Fig. 173: Storage - Additional Local Receiver menu

This option allows you to select an additional DICOM receiver that is used for test purposes only. The AET of this Test Receiver has to be entered during configuration. The Test Receiver is listening under port **110**. The Test Receiver may be used to receive images from the local application as well as any remote system (as long as this AET is configured there).

You can only send from a certain AET, which needs to be selected. All other remote nodes can still send to the original DICOM receiver.

NOTE

None of the images received by the Test Receiver will be stored in the database. To avoid losing images, select the AET of the sending system. All other senders will still send to the original local DICOM receiver.

To send to the additional Test Receiver, configure this Test Storage SCP (port: 110, AET will be defined in the test tools) at the remote system (SCU).

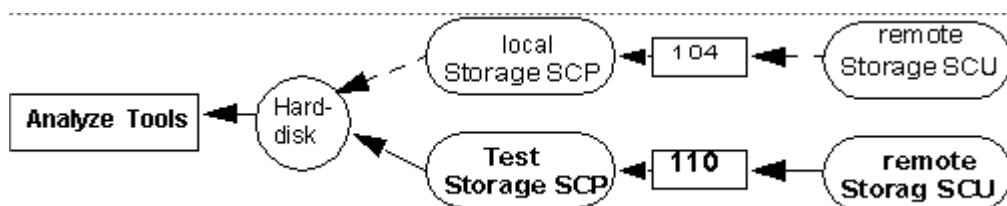


Fig. 174: Storage - Add Local Receiver diagram

The received images may be analyzed with the tools provided by "DICOM Analysis".

The images received and the log files may be removed when exiting the DICOM Service Tools.

- **AET**

You need to define an AET for the additional Test Receiver. At the remote system, you need to configure this AET with port number 110.

- **Merge trace level**

The following trace levels may be enabled to provide more detailed information in the log file merge_stu.log. This log file may be displayed under “Analysis - Examine Logs”. **Also refer to the example of an association with trace level T2, T3, T7 and T9 in the appendix.**

- **T1**

This trace level does not affect the mergecom.log file.

- **T2 data set**

This trace level shows the attributes of the transferred message, e.g. the image header attributes or the attributes within a DICOM message.

NOTE

Trace level T2 also shows patient data in the log file. For this reason, “Patient Data Access” is required for remote access.

- **T3 association negotiation**

Every DICOM communication starts with the association negotiation. The initiator (SCU) sends a list of all supported image objects and transfer syntaxes to the remote system SCP. The AETs of both systems are also displayed at the T3 level.

The SCP responds with the status (accept or reject) of the image objects and selects one of the proposed transfer syntaxes.

- **T4 rejection of incoming messages**

If incoming associations are automatically rejected (IP address not known), log messages will be displayed on this trace level.

- **T5 regular and extended regulations**

Log messages relating to regular and extended regulations.

- **T6 mergecom configuration files**

At this trace level the related mergecom.ini, mergecom.pro, mergecom.app, and mergecom.srv files will be written in the mergecom.log file. This may be helpful when analyzing the log file, however, please note that analysis requires a large amount of space in the log file.

- **T7 command set**

Following the association negotiation, a command set (attributes with group 0) will be transferred. Before sending, for example, an image, the command set will tell the remote system the type of image object (SOP Class UID such as CT, RF, etc.) to be transferred, including the UID of the image. The response command set after each image will show the status of the transfer, where status = 0 means a successful transfer.

- **T8 streaming messages**

Log messages relating to the streaming in and out of messages and file objects are displayed at the T8 level.

- **T9 PDU level**

The entire DICOM association is displayed on the PDU level (Protocol Data Unit). Each PDU defined with a maximum size during the association negotiation will be displayed in one line.

The association negotiation will be shown in greater detail.

T9 is the only way to display the association release request and response at the end of an association.

Patient Data Access

In case of remote access to the DICOM Service Tools, viewing patient-relevant data (name, ID, etc.) is not permitted. However, if you enable trace level T2, patient data will be displayed in the log file. For this purpose, the customer has to provide additional access rights, allowing the service engineer to view patient-relevant data. To obtain these rights, the service engineer has to enter his name and send a "Patient Data Access" request to the customer's user interface.

The objects (images) captured with the Test Receiver will normally be anonymized. If patient-relevant data is required for troubleshooting (not anonymized), it is necessary to obtain the "Patient Data Access" rights from the customer's user interface.

- **Please enter your name**

Enter the name of the local service engineer logged in remotely who requires Patient Data Access rights.

- **Request PDA**

When you select this button, an additional selection item (Patient Data Access) will be activated in the customer user interface under "Option - Remote Access". Additionally, the name of the service engineer will be displayed. The customer has to select this menu item for trace level T2 to be activated in the log file.

Before selecting other DICOM services or functions, or before finishing the DICOM Test Tools, it is necessary to "abort" the "Simulate Remote Receiver" function.

Basic Print

This menu provides tools for testing the Basic Print function of the local DICOM Basic Print process. You may start a Test Basic Print provider which replaces the remote Basic Print provider (usually a laser camera). After printing a film from the user interface to the Test Basic Print process, you can analyze the log files and display the film using the tools provided under Analysis.

Extend Trace Level

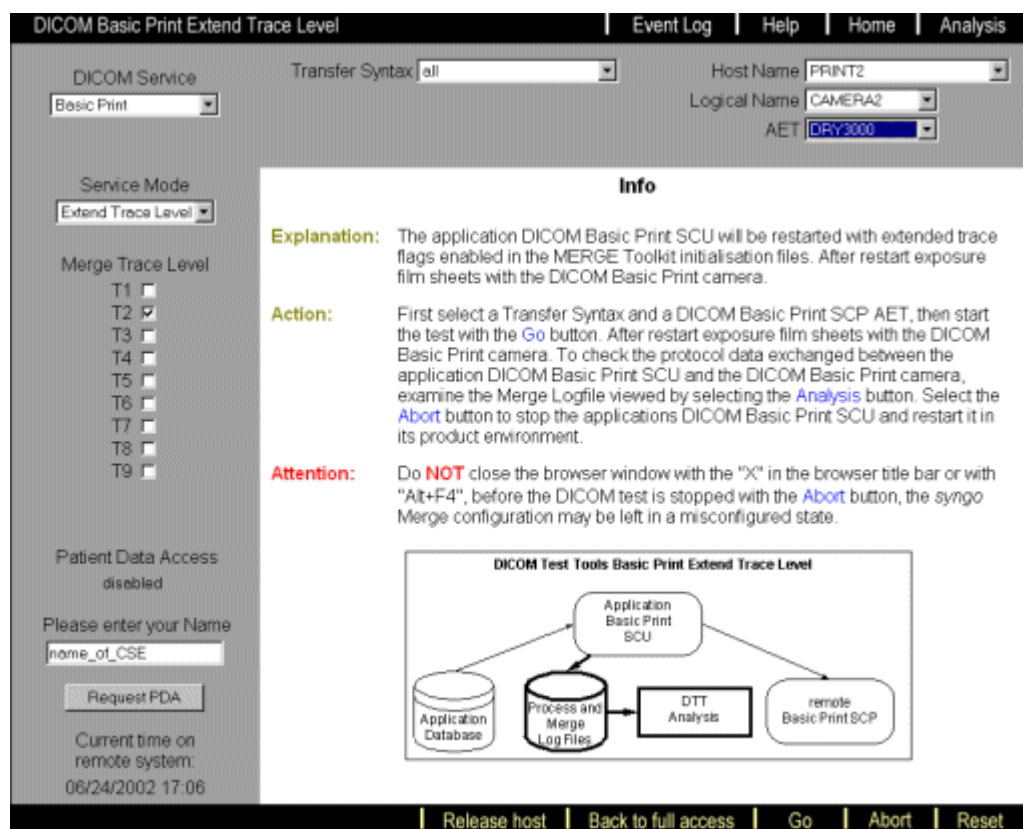


Fig. 175: Basic Print - Extended trace menu

This DICOM implementation is based on a toolkit from Merge Technologies. This tool kit includes a mergecom.log file designed as a ring buffer. Certain trace levels may be enabled to receive more detailed information regarding the latest DICOM communications. The respective mergecom.log file can be displayed under Analysis at "Examine Logs - merge_hc.log". Basic knowledge of DICOM is necessary for interpreting the log files.

Location and name of the mergecom.log file for Basic Print:

C:\%MEDHOME%\log\merge_hc.log Basic Print (hardcopy)

The Print Test Tools may only be started remotely in "Full Access Mode". In this case, no user interaction is possible, and, therefore, it is also not possible to send / print images or to perform a query request. Since these user interactions are necessary to receive the appropriate messages in the mergecom.log files, the DICOM Test Tools provide the option of enabling user interaction even in full access mode.

- **Release host**

Select this item in the Action bar to work with the user interface (identical to limited access). The "Remote Access Control Window" can be iconized.

- **Back to full access**

After the images have been sent to the camera, disable interaction at the user interface to perform further tasks in the DICOM Test Tools.

- **Transfer syntax**

This drop-down list allows you to choose transfer syntaxes that will be used for the association negotiation after a restart of the DICOM processes. It provides additional troubleshooting possibilities, as you can limit the settings for the next associations. Before selecting other DICOM services or functions, or before finishing the DICOM Test Tools, "abort" the Extend Trace Level function. The original Transfer Syntaxes will be set again.

- **Host Name**

Host name of the camera. The system itself (host name) may be selected as well. When selecting the logical name or AET (next two buttons), the appropriate host name will be displayed automatically.

- **Logical Name**

Select the logical name of the camera. When selecting the AET (next button) first, the appropriate logical name will be displayed automatically.

- **AET**

Select the AET of the camera.

- **Merge trace level**

The following trace levels may be enabled to provide more detailed information in the log file merge_stu.log. This log file may be displayed under "Analysis - Examine Logs".

Also refer to the example of an association with trace level T2, T3, T7 and T9 in the appendix.

- **T1**

This trace level does not affect the mergecom.log file.

- **T2 data set**

This trace level shows the attributes of the transferred message, e.g. the image header attributes or the attributes within a DICOM message.

NOTE

Trace level T2 also shows patient data in the log file. For this reason, "Patient Data Access" is required for remote access.

- **T3 association negotiation**

Every DICOM communication starts with the association negotiation. The initiator (SCU) sends a list of all supported image objects and transfer syntaxes to the remote system (SCP). The AETs of both systems are also displayed at the T3 level.

The SCP responds with the status (accept or reject) of the image objects and selects one of the proposed transfer syntaxes.

- **T4 rejection of incoming messages**

If incoming associations are automatically rejected (IP address not known); log messages will be displayed on this trace level.

- **T5 regular and extended regulations**

Log messages relating to regular and extended regulations.

- **T6 mergecom configuration files**

At this trace level the related mergecom.ini, mergecom.pro, mergecom.app, and mergecom.srv files will be written in the mergecom.log file. This may be helpful when analyzing the log file, however, note that analysis requires a large amount of space in the log file.

- **T7 command set**

Following the association negotiation, a command set (attributes with group 0) will be transferred. Before sending an image, for example, the command set will tell the remote system the type of image object (SOP Class UID such as CT, RF, etc.) to be transferred, including the UID of the image. The response command set after each image will show the status of the transfer, where status = 0 means a successful transfer.

- **T8 streaming messages**

Log messages relating to the streaming in and out of messages and file objects are displayed at the T8 level.

- **T9 PDU level**

The entire DICOM association is displayed on the PDU level (Protocol Data Unit). Each PDU defined with a maximum size during the association negotiation will be displayed in one line.

The association negotiation will be shown in greater detail.

T9 is the only way to display the association release request and response at the end of an association.

Patient Data Access

In case of remote access to the DICOM Service Tools, viewing patient-relevant data (name, ID, etc.) is not permitted. However, if you enable trace level T2, patient data will be displayed in the log file. For this purpose, the customer has to provide additional access rights, allowing the service engineer to view patient-relevant data. To obtain these rights, the service engineer has to enter his name and send a "Patient Data Access" request to the customer's user interface.

- **Please enter your name**

Enter the name of the local service engineer logged in remotely who requires Patient Data Access rights.

- **Request PDA**

When you select this button, an additional selection item (Patient Data Access) will be activated in the customer user interface under "Option - Remote Access". Additionally, the name of the service engineer will be displayed. The customer has to select this menu item for trace level T2 to be activated in the log file.

Simulate Camera

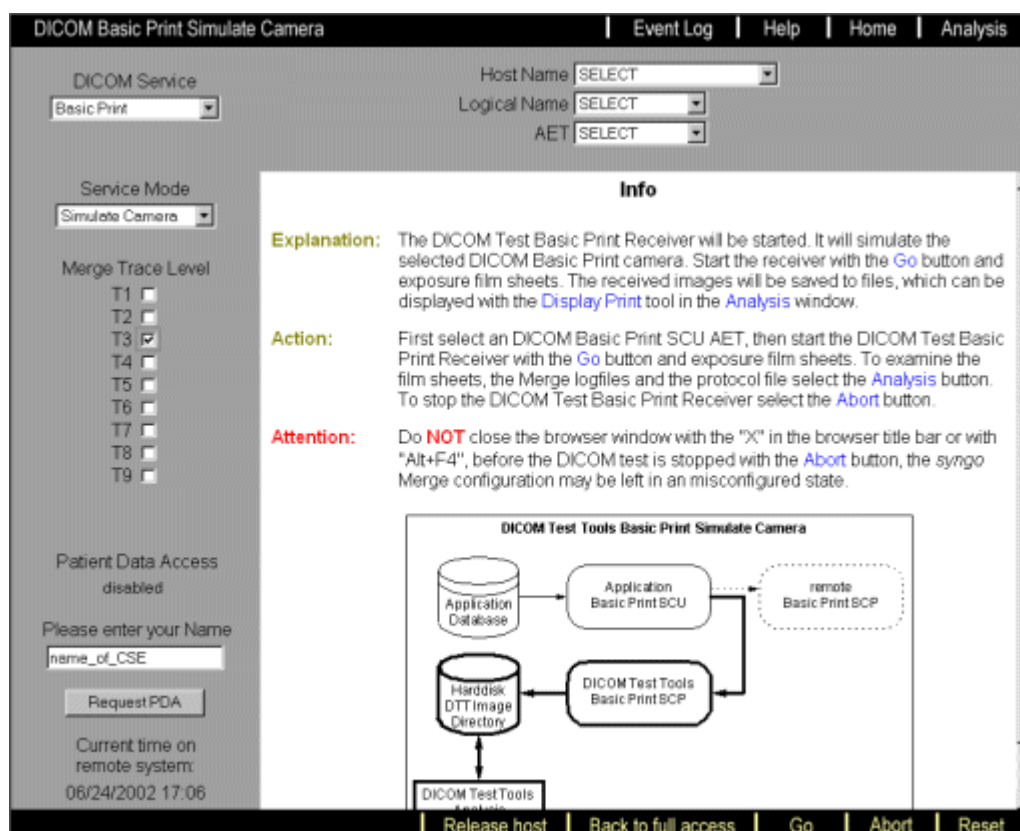


Fig. 176: Basic Print - Simulate Camera menu

When this menu item is selected, the Test Basic Print SCP provided by the DICOM Service Tools is started in place of the camera (remote Basic Print SCP). You will receive a list of all configured Basic Print AETs. Select the appropriate laser camera. All films sent from the user interface to the corresponding laser camera will then be received by the Test Basic Print SCP.

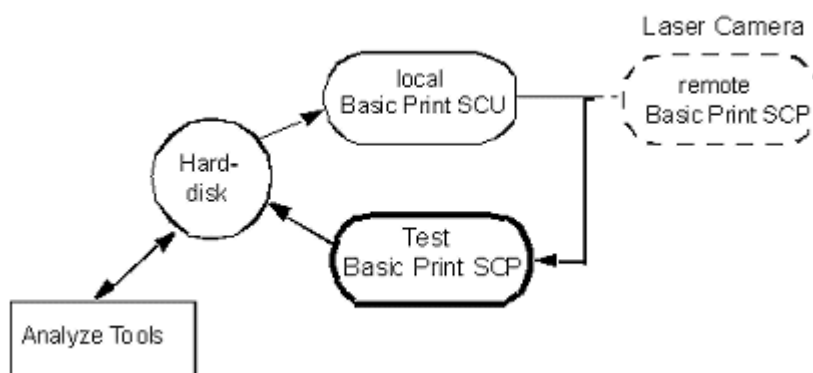


Fig. 177: Basic Print - Simulate Camera diagram

The Test Basic Print SCP is not identical to the laser camera. As a result, its behavior may be different from the remote Basic Print SCP.

The Print Test Tools may only be started remotely in “Full Access Mode”. In this case, no user interaction is possible, and, therefore, it is also not possible to send / print images or to perform a query request. Since these user interactions are necessary to receive the appropriate messages in the mergecom.log files, the DICOM Test Tools provide the option of enabling user interaction even in full access mode.

- **Release host**

Select this item in the Action bar to work with the user interface (identical to limited access). It is even possible to iconize the “Remote Access Control Window”.

- **Back to full access**

After the images have been sent to the camera, disable interaction at the user interface to perform further tasks in the DICOM Test Tools.

- **Host Name**

Host name of the camera. The system itself (host name) may be selected as well. When selecting the logical name or AET (next two buttons), the appropriate host name will be displayed automatically.

- **Logical Name**

Select the logical name of the camera. When selecting the AET (next button) first, the appropriate logical name will be displayed automatically.

- **AET**

Select the AET of the camera.

- **Merge trace level**

The following trace levels may be enabled to provide more detailed information in the log file merge_stu.log. This log file may be displayed under “Analysis - Examine Logs”.

Also refer to the example of an association with trace level T2, T3, T7 and T9 in the appendix.

- **T1**

This trace level does not affect the mergecom.log file.

- **T2 data set**

This trace level shows the attributes of the transferred message, e.g. the image header attributes or the attributes within a DICOM message.

NOTE

Trace level T2 also shows patient data in the log file. For this reason, “Patient Data Access” is required for remote access.

- **T3 association negotiation**

Every DICOM communication starts with the association negotiation. The initiator (SCU) sends a list of all supported image objects and transfer syntaxes to the remote system (SCP). The AETs of both systems are also displayed at the T3 level.

The SCP responds with the status (accept or reject) of the image objects and selects one of the proposed transfer syntaxes.

- **T4 rejection of incoming messages**

If incoming associations are automatically rejected (IP address not known), log messages will be displayed on this trace level.

- **T5 regular and extended regulations**

Log messages relating to regular and extended regulations.

- **T6 mergecom configuration files**

At this trace level the related mergecom.ini, mergecom.pro, mergecom.app, and mergecom.srv files will be written in the mergecom.log file. This may be helpful when analyzing the log file, however, note that analysis requires a large amount of space in the log file.

- **T7 command set**

Following the association negotiation, a command set (attributes with group 0) will be transferred. Before sending an image, for example, the command set will tell the remote system the type of image object (SOP Class UID such as CT, RF, etc.) to be transferred, including the UID of the image. The response command set after each image will show the status of the transfer, where status = 0 means a successful transfer.

- **T8 streaming messages**

Log messages relating to the streaming in and out of messages and file objects are displayed at the T8 level.

- **T9 PDU level**

The entire DICOM association is displayed on the PDU level (Protocol Data Unit). Each PDU defined with a maximum size during the association negotiation will be displayed in one line.

The association negotiation will be shown in greater detail.

T9 is the only way to display the association release request and response at the end of an association.

Patient Data Access

In case of remote access to the DICOM Service Tools, viewing patient-relevant data (name, ID, etc.) is not permitted. However, if you enable trace level T2, patient data will be displayed in the log file. For this purpose, the customer has to provide additional access rights, allowing the service engineer to view patient-relevant data. To obtain these rights, the service engineer has to enter his name and send a "Patient Data Access" request to the customer's user interface.

- **Please enter your name**

Enter the name of the local service engineer logged in remotely who requires Patient Data Access rights.

- **Request PDA**

When you select this button, an additional selection item (Patient Data Access) will be activated in the customer user interface under "Option - Remote Access." Additionally, the name of the service engineer will be displayed. The customer has to select this menu item for trace level T2 to be activated in the log file.

Storage commitment

Storage Commitment sends a list of image UIDs to the configured destination and waits for the response of the remote provider.

For this service, it is only possible to extend the trace level of the appropriate mergecom.log file. Storage Commitment uses the same DICOM process as Storage and therefore also the same log file, which is stu_merge.log. In this response, trace level T2 shows whether or not the images have been successfully stored at the destination.

Extend Trace Level

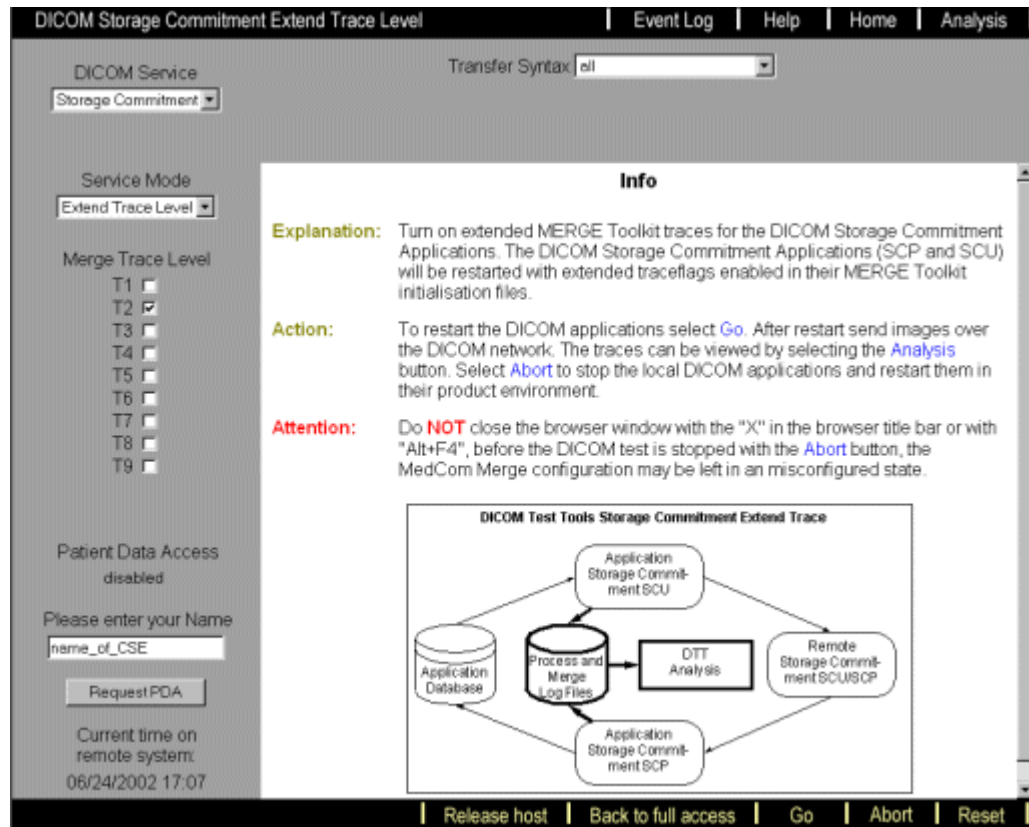


Fig. 178: Storage Commitment - Extend trace menu

This DICOM implementation is based on a toolkit from Merge Technologies. This tool kit includes a mergecom.log file designed as a ring buffer. Certain trace levels may be enabled to obtain more detailed information regarding the latest DICOM communications. The respective mergecom.log file can be displayed under Analysis at "Examine Logs - merge_stu.log". Basic knowledge of DICOM is necessary for interpreting the log files.

Storage, Storage Commitment and Query/Retrieve use the same mergecom.log file, as they belong to the same syngo service.

C:\%MEDHOME%\log\merge_stu. logs Storage, Storage Commitment and Query/Retrieve

The DICOM Test Tools may only be started remotely in "Full Access Mode". In this case, no user interaction is possible, and, therefore, it is also not possible to send / print images or to perform a query request. Since these user interactions are necessary to receive the appropriate messages in the mergecom.log files, the DICOM Test Tools provide the option of enabling user interaction even in full access mode.

- **Release host**

Select this item in the Action bar to work with the user interface (identical to limited access). It is even possible to iconize the "Remote Access Control Window".

- **Back to full access**

After the images have been transferred or a Query/Retrieve has been performed, this function disables interaction at the user interface to perform further tasks in the DICOM Test Tools.

- **Transfer syntax**

This drop-down list allows you to choose transfer syntaxes that will be used for the association negotiation after a restart of the DICOM processes. It provides additional troubleshooting possibilities, as you can limit the settings for the next associations.

Before selecting other DICOM services or functions, or before finishing the DICOM Test Tools, "abort" the Extend Trace Level function. The original Transfer Syntaxes will be set again.

- **Merge trace level**

The following trace levels may be enabled to provide more detailed information in the log file merge_stu.log. This log file may be displayed under "Analysis - Examine Logs".

Also refer to the example of an association with trace level T2, T3, T7 and T9 in the appendix.

- **T1**

This trace level does not affect the mergecom.log file.

- **T2 data set**

This trace level shows the attributes of the transferred message, e.g. the image header attributes or the attributes within a DICOM message.

NOTE

Trace level T2 also shows patient data in the log file. For this reason, "Patient Data Access" is required for remote access.

- **T3 association negotiation**

Every DICOM communication starts with the association negotiation. The initiator (SCU) sends a list of all supported image objects and transfer syntaxes to the remote system (SCP). The AETs of both systems are also displayed at the T3 level.

The SCP responds with the status (accept or reject) of the image objects and selects one of the proposed transfer syntaxes.

- **T4 rejection of incoming messages**

If incoming associations are automatically rejected (IP address not known), log messages will be displayed on this trace level.

- **T5 regular and extended regulations**

Log messages relating to regular and extended regulations.

- **T6 mergecom configuration files**

At this trace level the related mergecom.ini, mergecom.pro, mergecom.app, and mergecom.srv files will be written in the mergecom.log file. This may be helpful when analyzing the log file, however, note that analysis requires a large amount of space in the log file.

- **T7 command set**

Following the association negotiation, a command set (attributes with group 0) will be transferred. Before sending an image, for example, the command set will tell the remote system the type of image object (SOP Class UID such as CT, RF, etc.) to be

transferred, including the UID of the image. The response command set after each image will show the status of the transfer, where status = 0 means a successful transfer.

- **T8 streaming messages**

Log messages relating to the streaming in and out of messages and file objects are displayed at the T8 level.

- **T9 PDU level**

The entire DICOM association is displayed on the PDU level (Protocol Data Unit). Each PDU defined with a maximum size during the association negotiation will be displayed in one line.

The association negotiation will be shown in greater detail.

T9 is the only way to display the association release request and response at the end of an association.

Patient Data Access

In case of remote access to the DICOM Service Tools, viewing patient-relevant data (name, ID, etc.) is not permitted. However, if you enable trace level T2, patient data will be displayed in the log file. For this purpose, the customer has to provide additional access rights, allowing the service engineer to view patient-relevant data. To obtain these rights, the service engineer has to enter his name and send a "Patient Data Access" request to the customer's user interface.

- **Please enter your name**

Enter the name of the local service engineer logged in remotely who requires Patient Data Access rights.

- **Request PDA**

When you select this button, an additional selection item (Patient Data Access) will be activated in the customer user interface under "Option - Remote Access". Additionally, the name of the service engineer will be displayed. The customer has to select this menu item for trace level T2 to be activated in the log file.

Query/Retrieve

For this service, it is only possible to extend the trace level of the appropriate mergecom.log file. Query/Retrieve uses the same DICOM process as Storage and therefore also the same log file, which is stu_merge.log. The Query request and response message attributes may be displayed on trace level T2.

Extend Trace Level

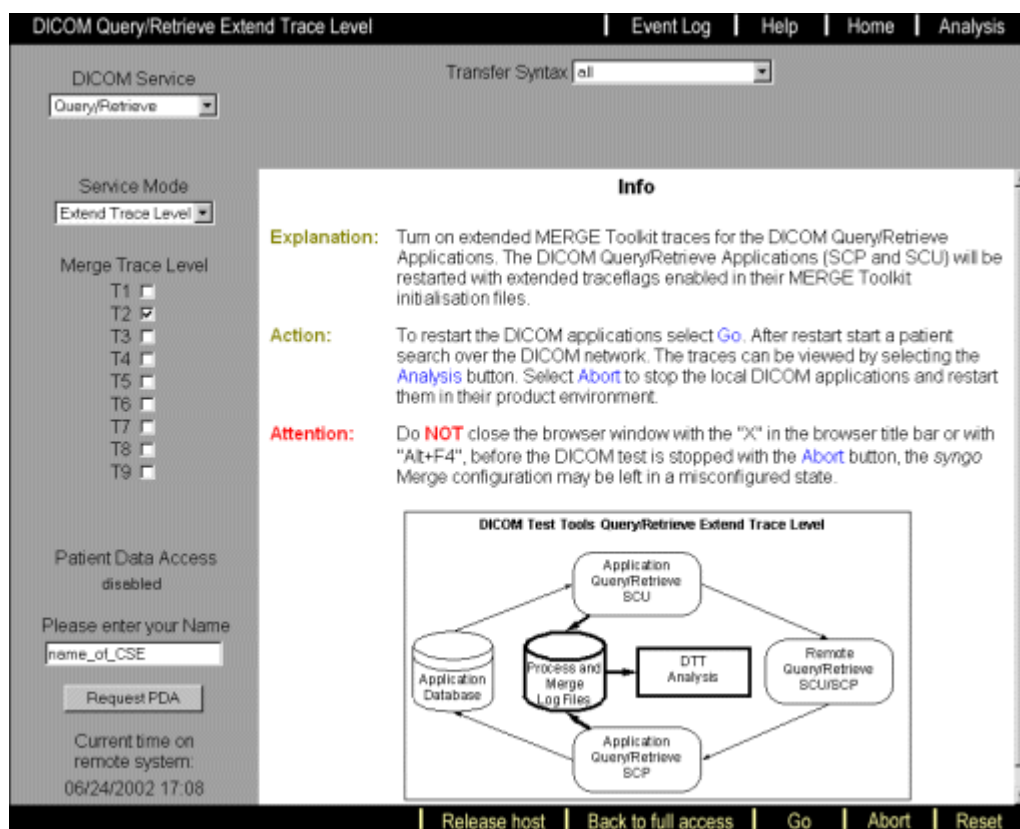


Fig. 179: Query/Retrieve - Extend trace menu

This DICOM implementation is based on a toolkit from Merge Technologies. This tool kit includes a mergecom.log file designed as a ring buffer. Certain trace levels may be enabled to obtain more detailed information regarding the latest DICOM communications. The respective mergecom.log file can be displayed under Analysis at "Examine Logs - merge_stu.log". Basic knowledge of DICOM is necessary for interpreting the log files.

Storage, Storage Commitment and Query/Retrieve use the same mergecom.log file, as they belong to the same syngo service.

C:\%MEDHOME%\log\merge_stu. logs Storage, Storage Commitment and Query/Retrieve

The DICOM Test Tools may only be started remotely in "Full Access Mode". In this case, no user interaction is possible, and, therefore, it is also not possible to send / print images or to perform a query request. Since these user interactions are necessary to receive the appropriate messages in the mergecom.log files, the DICOM Test Tools provide the option of enabling user interaction even in full access mode.

- **Release host**

Select this item in the Action bar to work with the user interface (identical to limited access). The "Remote Access Control Window" can be iconized.

- **Back to full access**

After the images have been transferred or a Query/Retrieve has been performed, this function disables interaction at the user interface to perform further tasks in the DICOM Test Tools.

- **Transfer syntax**

This drop-down list allows you to choose transfer syntaxes that will be used for the association negotiation after a restart of the DICOM processes. It provides additional troubleshooting possibilities, as you can limit the settings for the next associations.

Before selecting other DICOM services or functions, or before finishing the DICOM Test Tools, “abort” the Extend Trace Level function. The original Transfer Syntaxes will be set again.

- **Merge trace level**

The following trace levels may be enabled to provide more detailed information in the log file merge_stu.log. This log file may be displayed under “Analysis - Examine Logs”. **Also refer to the example of an association with trace level T2, T3, T7 and T9 in the appendix.**

- **T1**

This trace level does not affect the mergecom.log file.

- **T2 data set**

This trace level shows the attributes of the transferred message, e.g. the image header attributes or the attributes within a DICOM message.

NOTE

Trace level T2 also shows patient data in the log file. For this reason, “Patient Data Access” is required for remote access.

- **T3 association negotiation**

Every DICOM communication starts with the association negotiation. The initiator (SCU) sends a list of all supported image objects and transfer syntaxes to the remote system (SCP). The AETs of both systems are also displayed at the T3 level.

The SCP responds with the status (accept or reject) of the image objects and selects one of the proposed transfer syntaxes.

- **T4 rejection of incoming messages**

If incoming associations are automatically rejected (IP address not known), log messages will be displayed on this trace level.

- **T5 regular and extended regulations**

Log messages relating to regular and extended regulations.

- **T6 mergecom configuration files**

At this trace level the related mergecom.ini, mergecom.pro, mergecom.app, and mergecom.srv files will be written in the mergecom.log file. This may be helpful when analyzing the log file, however, note that analysis requires a large amount of space in the log file.

- **T7 command set**

Following the association negotiation, a command set (attributes with group 0) will be transferred. Before sending an image, for example, the command set will tell the remote system the type of image object (SOP Class UID such as CT, RF, etc.) to be transferred, including the UID of the image. The response command set after each image will show the status of the transfer, where status = 0 means a successful transfer.

- **T8 streaming messages**

Log messages relating to the streaming in and out of messages and file objects are displayed at the T8 level.

- **T9 PDU level**

The entire DICOM association is displayed on the PDU level (Protocol Data Unit). Each PDU defined with a maximum size during the association negotiation will be displayed in one line.

The association negotiation will be shown in greater detail.

T9 is the only way to display the association release request and response at the end of an association.

Patient Data Access

In case of remote access to the DICOM Service Tools, viewing patient-relevant data (name, ID, etc.) is not permitted. However, if you enable trace level T2, patient data will be displayed in the log file. For this purpose, the customer has to provide additional access rights, allowing the service engineer to view patient-relevant data. To obtain these rights, the service engineer has to enter his name and send a "Patient Data Access" request to the customer's user interface.

- **Please enter your name**

Enter the name of the local service engineer logged in remotely who requires Patient Data Access rights.

- **Request PDA**

When you select this button, an additional selection item (Patient Data Access) will be activated in the customer user interface under "Option - Remote Access". Additionally, the name of the service engineer will be displayed. The customer has to select this menu item for trace level T2 to be activated in the log file.

Basic Worklist

For this service, it is only possible to extend the trace level of the appropriate mergecom.log file. The Worklist request and response message attributes may be displayed at trace level T2.

Extend Trace Level

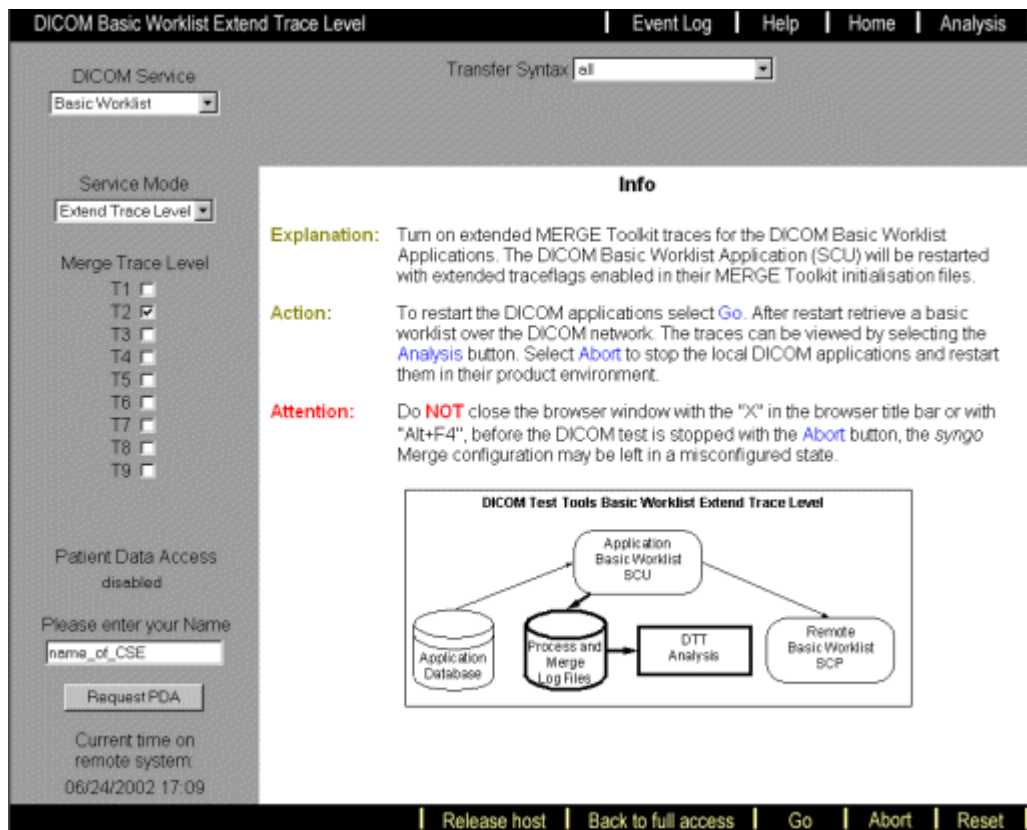


Fig. 180: Basic Worklist - Extend trace menu

This DICOM implementation is based on a toolkit from Merge Technologies. This tool kit includes a mergecom.log file designed as a ring buffer. Certain trace levels may be enabled to obtain more detailed information regarding the latest DICOM communications. The respective mergecom.log file can be displayed under Analysis at “Examine Logs - merge_wl.log”. Basic knowledge of DICOM is necessary for interpreting the log files.

The DICOM Test Tools may only be started remotely in “Full Access Mode”. In this case, no user interaction is possible. It is therefore not possible to perform a Worklist request from the user interface. Since user interaction is necessary to receive the appropriate messages in the merge_HRI.log files, the DICOM Test Tools provide the option of enabling user interaction even in full access mode.

- **Release host**

Select this item in the Action bar to work with the user interface (identical to limited access). It is even possible to iconize the “Remote Access Control Window”.

- **Back to full access**

After the Worklist request has been performed, disable interaction at the user interface to perform further tasks in the DICOM Test Tools.

- **Transfer syntax**

This drop-down list allows you to choose transfer syntaxes that will be used for the association negotiation after a restart of the DICOM processes. It provides additional troubleshooting possibilities, as you can limit the settings for the next associations.

Before selecting other DICOM services or functions, or before finishing the DICOM Test Tools, “abort” the Extend Trace Level function. The original Transfer Syntaxes will be set again.

- **Merge trace level**

The following trace levels may be enabled to provide more detailed information in the log file merge_stu.log. This log file may be displayed under “Analysis - Examine Logs”. **Also refer to the example of an association with trace level T2, T3, T7 and T9 in the appendix.**

- **T1**

This trace level does not affect the mergecom.log file.

- **T2 data set**

This trace level shows the attributes of the transferred message, e.g. the image header attributes or the attributes within a DICOM message.

NOTE

Trace level T2 also shows patient data in the log file. For this reason, “Patient Data Access” is required for remote access.

- **T3 association negotiation**

Every DICOM communication starts with the association negotiation. The initiator (SCU) sends a list of all supported image objects and transfer syntaxes to the remote system (SCP). The AETs of both systems are also displayed at the T3 level.

The SCP responds with the status (accept or reject) of the image objects and selects one of the proposed transfer syntaxes.

- **T4 rejection of incoming messages**

If incoming associations are automatically rejected (IP address not known), log messages will be displayed on this trace level.

- **T5 regular and extended regulations**

Log messages relating to regular and extended regulations.

- **T6 mergecom configuration files**

At this trace level the related mergecom.ini, mergecom.pro, mergecom.app, and mergecom.srv files will be written in the mergecom.log file. This may be helpful when analyzing the log file, however, note that analysis requires a large amount of space in the log file.

- **T7 command set**

Following the association negotiation, a command set (attributes with group 0) will be transferred. Before sending an image, for example, the command set will tell the remote system the type of image object (SOP Class UID such as CT, RF, etc.) to be transferred, including the UID of the image. The response command set after each image will show the status of the transfer, where status = 0 means a successful transfer.

- **T8 streaming messages**

Log messages relating to the streaming in and out of messages and file objects are displayed at the T8 level.

- **T9 PDU level**

The entire DICOM association is displayed on the PDU level (Protocol Data Unit). Each PDU defined with a maximum size during the association negotiation will be displayed in one line.

The association negotiation will be shown in greater detail.

T9 is the only way to display the association release request and response at the end of an association.

Patient Data Access

If the DICOM Service Tools are accessed remotely, viewing patient-relevant data (name, ID, etc.) is not permitted. However, if you enable trace level T2, patient data will be displayed in the log file. For this purpose, the customer has to provide additional access rights, allowing the service engineer to view patient-relevant data. To obtain these rights, the service engineer has to enter his name and send a "Patient Data Access" request to the customer's user interface.

- **Please enter your name**

Enter the name of the local service engineer logged in remotely who requires Patient Data Access rights.

- **Request PDA**

When you select this button, an additional selection item (Patient Data Access) will be activated in the customer user interface under "Option - Remote Access". Additionally, the name of the service engineer will be displayed. The customer has to select this menu item for trace level T2 to be activated in the log file.

Modality Performed Procedure Step

For this service, it is only possible to extend the trace level of the appropriate mergecom.log file. The Worklist request and response message attributes may be displayed at trace level T2.

Extend Trace Level

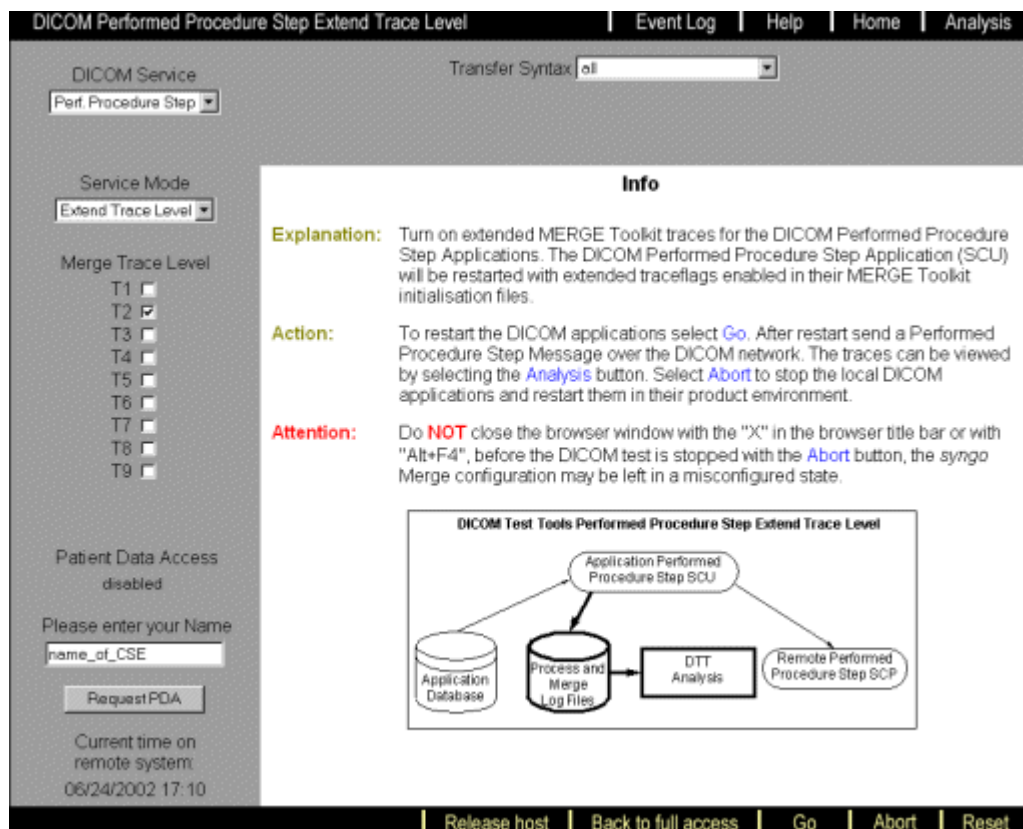


Fig. 181: Modality Performed Procedure Step - Extend trace menu

This DICOM implementation is based on a toolkit from Merge Technologies. This tool kit includes a mergecom.log file designed as a ring buffer. Certain trace levels may be enabled to obtain more detailed information regarding the latest DICOM communications. The respective mergecom.log file can be displayed under Analysis at “Examine Logs - merge_HRI.log”. Basic knowledge of DICOM is necessary for interpreting the log files.

The DICOM Test Tools may only be started remotely in “Full Access Mode”. In this case, no user interaction is possible. It is therefore not possible to perform a Worklist request from the user interface. Since user interaction is necessary to receive the appropriate messages in the merge_HRI.log files, the DICOM Test Tools provide the option of enabling user interaction even in full access mode.

- **Release host**

Select this item in the Action bar to work with the user interface (identical to limited access). It is even possible to iconize the “Remote Access Control Window”.

- **Back to full access**

After the Worklist request has been performed, disable interaction at the user interface to perform further tasks in the DICOM Test Tools.

- **Transfer syntax**

This drop-down list allows you to choose transfer syntaxes that will be used for the association negotiation after a restart of the DICOM processes. It provides additional troubleshooting possibilities, as you can limit the settings for the next associations.

Before selecting other DICOM services or functions, or before finishing the DICOM Test Tools, “abort” the Extend Trace Level function. The original Transfer Syntaxes will be set again.

- **Merge trace level**

The following trace levels may be enabled to provide more detailed information in the log file merge_stu.log. This log file may be displayed under “Analysis - Examine Logs”. **Also refer to the example of an association with trace level T2, T3, T7 and T9 in the appendix.**

- **T1**

This trace level does not affect the mergecom.log file.

- **T2 data set**

This trace level shows the attributes of the transferred message, e.g. the image header attributes or the attributes within a DICOM message.

NOTE

Trace level T2 also shows patient data in the log file. For this reason, “Patient Data Access” is required for remote access.

- **T3 association negotiation**

Every DICOM communication starts with the association negotiation. The initiator (SCU) sends a list of all supported image objects and transfer syntaxes to the remote system (SCP). The AETs of both systems are also displayed at the T3 level.

The SCP responds with the status (accept or reject) of the image objects and selects one of the proposed transfer syntaxes.

- **T4 rejection of incoming messages**

If incoming associations are automatically rejected (IP address not known), log messages will be displayed on this trace level.

- **T5 regular and extended regulations**

Log messages relating to regular and extended regulations.

- **T6 mergecom configuration files**

At this trace level the related mergecom.ini, mergecom.pro, mergecom.app, and mergecom.srv files will be written in the mergecom.log file. This may be helpful when analyzing the log file, however, note that analysis requires a large amount of space in the log file.

- **T7 command set**

Following the association negotiation, a command set (attributes with group 0) will be transferred. Before sending an image, for example, the command set will tell the remote system the type of image object (SOP Class UID such as CT, RF, etc.) to be transferred, including the UID of the image. The response command set after each image will show the status of the transfer, where status = 0 means successful transfer.

- **T8 streaming messages**

Log messages relating to the streaming in and out of messages and file objects are displayed at the T8 level.

- **T9 PDU level**

The entire DICOM association is displayed on the PDU level (Protocol Data Unit). Each PDU defined with a maximum size during the association negotiation will be displayed in one line.

The association negotiation will be shown in greater detail.

T9 is the only way to display the association release request and response at the end of an association.

Patient Data Access

In case of remote access to the DICOM Service Tools, viewing patient-relevant data (name, ID, etc.) is not permitted. However, if you enable trace level T2, patient data will be displayed in the log file. For this purpose, the customer has to provide additional access rights, allowing the service engineer to view patient-relevant data. To obtain these rights, the service engineer has to enter his name and send a "Patient Data Access" request to the customer's user interface.

- **Please enter your name**

Enter the name of the local service engineer logged in remotely who requires Patient Data Access rights.

- **Request PDA**

When you select this button, an additional selection item (Patient Data Access) will be activated in the customer user interface under "Option - Remote Access". Additionally, the name of the service engineer will be displayed. The customer has to select this menu item for trace level T2 to be activated in the log file.

7.2.2 Dicom Analysis

HP-Z0000YPR-111005-02-943

The analysis tools provide the option of displaying and evaluating the messages and images captured by the Test Receiver (Storage, Basic Print) as well as displaying the appropriate mergecom.log files.

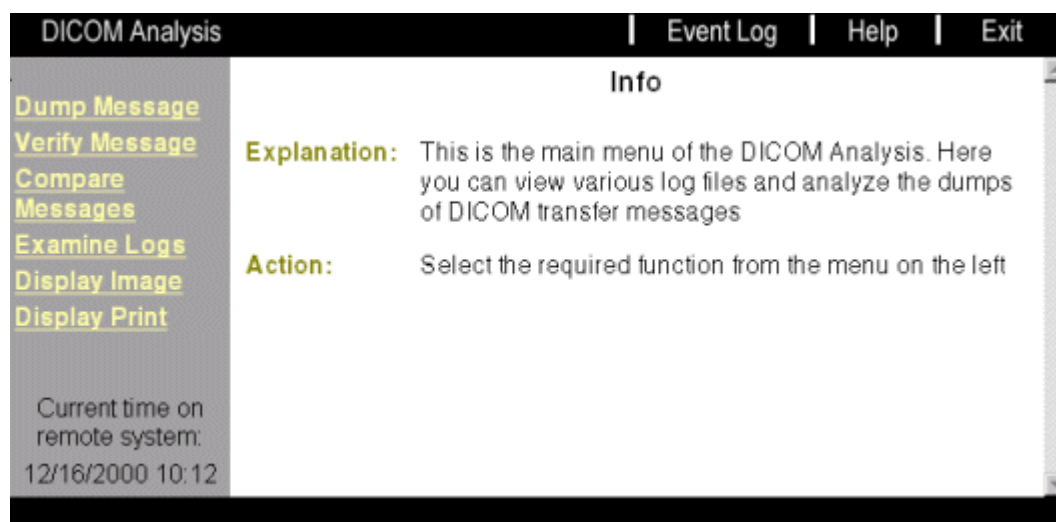


Fig. 182: Dicom Analysis menu

Dump Message

The image header information of images captured by the Test Receiver (simulate local/remote receiver), sent via "Export to Offline" from the customer user interface or copied manually into the directory %MEDHOME%\service\dst\img will be displayed by different header dump tools. The image has to be selected from a given selection list. Please note that all saved files may be removed when exiting the DICOM Test Tools (selectable).



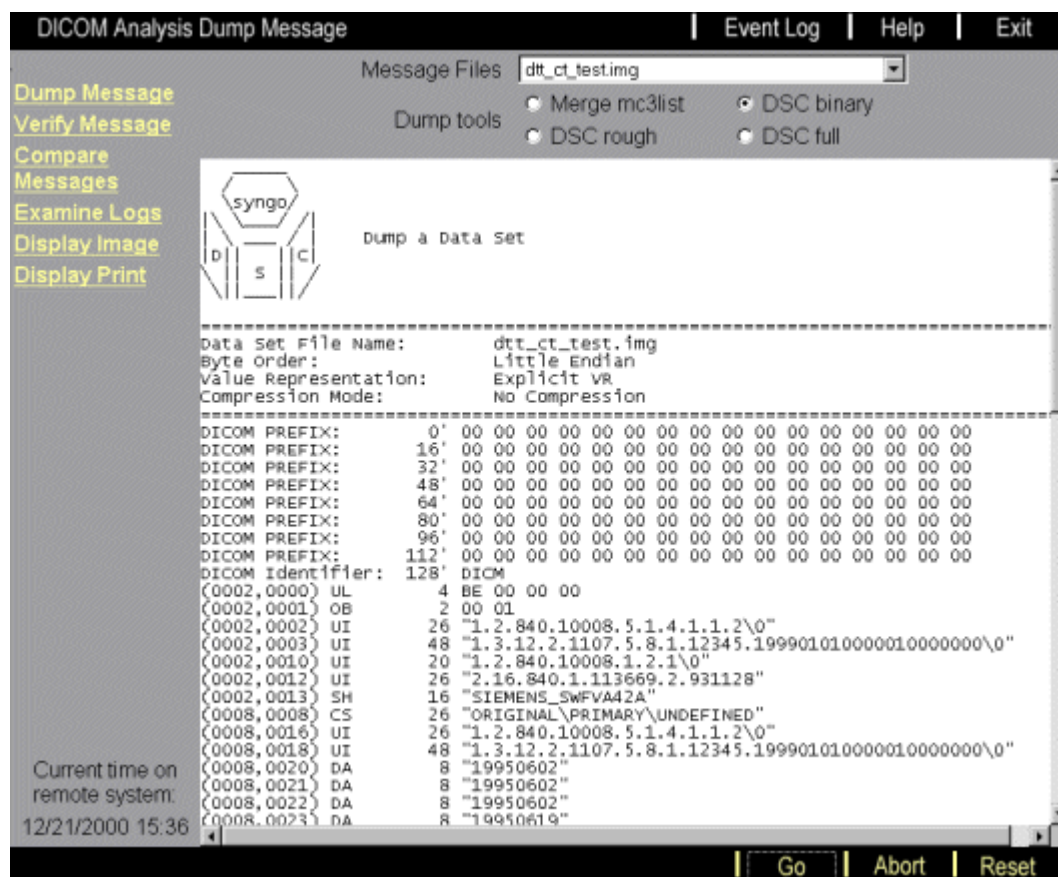


Fig. 184: Dicom Dump Message binary

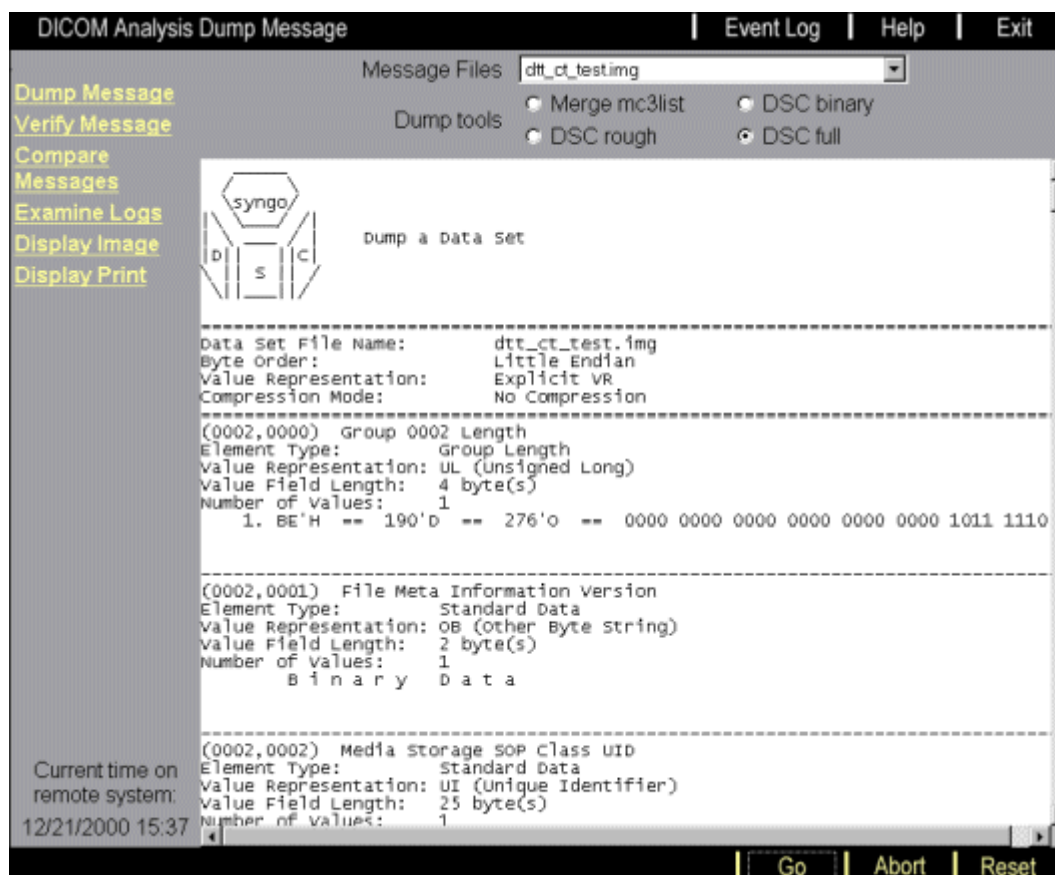


Fig. 185: Dicom Dump Message full

Verify Message

The image header information of images captured by the Test Receiver (simulate local/remote receiver) and sent via “Export to Offline” from the customer user interface or copied manually to the %MEDHOME%\service\dst\img directory will be evaluated for DICOM conformity. Please note that not all image header attributes marked “Error” are in violation of the DICOM Standard. For example, missing or incorrect attributes will be displayed even if they are optional attributes. The tool provides a means of identifying the nature of such “Error” messages. Each indicated header attribute should also be verified manually using the DICOM Standard.



Fig. 186: Dicom Verify Message menu

Compare Message

Two images captured by the Test Receiver (simulate local/remote receiver) and sent via “Export to Offline” from the customer user interface or copied manually to the %MED-HOME%\service\dst\img directory will be compared. The message has to be selected by the user from a given selection list.

This example shows a possible test scenario:

- Send an image to a remote DICOM node.
- Simulate the remote node and resend the same image. This image will be stored in DICOM format on the hard disk.
- Simulate the local receiver and send the image from the remote system to the local system. This image will also be stored in DICOM format on the hard disk.
- Compare the sent image with the received image.

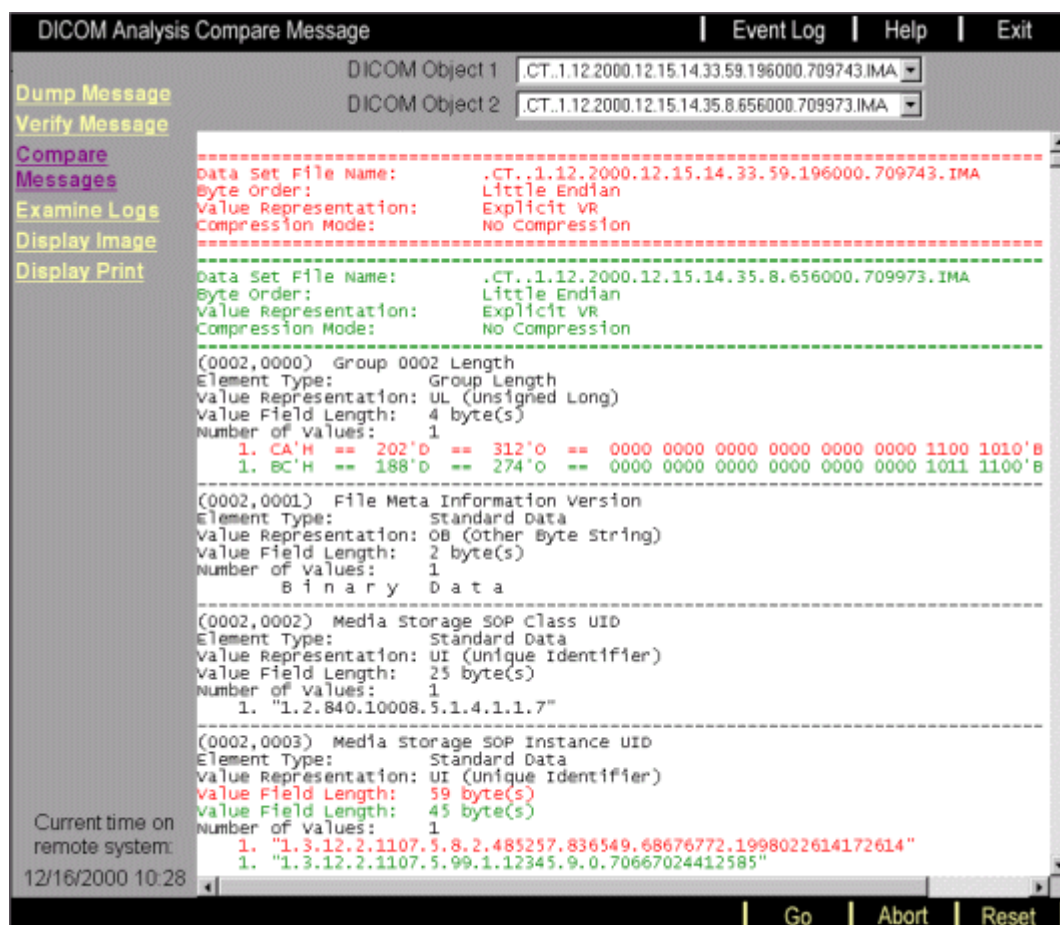


Fig. 187: Dicom Compare Message menu

Examine Logs

This menu item displays the log directory including the mergecom.log files of the appropriate DICOM Services. Choose the process for which you want to see the log file:

Merge log files

Application Storage SCU/SCP

Test Tools Storage SCU

Test Tools Storage SCP

Application Storage Commitment SCU

Application Query/Retrieve SCU

Application Basic Print SCU

Test Tools Basic Print SCP

Application Worklist SCU

Application Modality Performed Procedure

Step SCU

merge_stu.log

merge_dst_sscu.log (process log:
dst_sscu_out.log)

merge_dst_sscp.log (process log:
dst_sscp_out.log)

merge_stu.log

merge_stu.log

merge_hc.log

merge_dst_pscp.log (process log:
dst_pscp_out.log)

merge_HRI.log

merge_HRI.log

The log file is displayed when you click the appropriate file name.



Fig. 188: Dicom Examine Logs menu

Display Image

DICOM images captured by the Test Receiver (simulate local/remote receiver) and sent via "Export to Offline" from the customer user interface or copied manually to the %MED-HOME%\service\dst\img directory will be displayed. The image has to be selected by the user from a given selection list.

NOTE

The viewer is a Java applet that does not support all types of DICOM images. If the images cannot be displayed properly, you need to transfer them to your PC and display them with other viewer applications such as the MV300 Viewer.

The following images cannot be displayed:

- Monochrome 16 bit images with bits of 8, 10, 15, 16 stored
- Palette Color images
- Multiframe images - only the frame in the middle of the sequence will be displayed
- images that do not have the attributes of group 2 (part 10 header for offline images)

The following images are displayed:

- Monochrome 8 bit images and 16 bit images with bits of 12 stored
- RGB images
- all types of overlay data
- JPEG compressed images

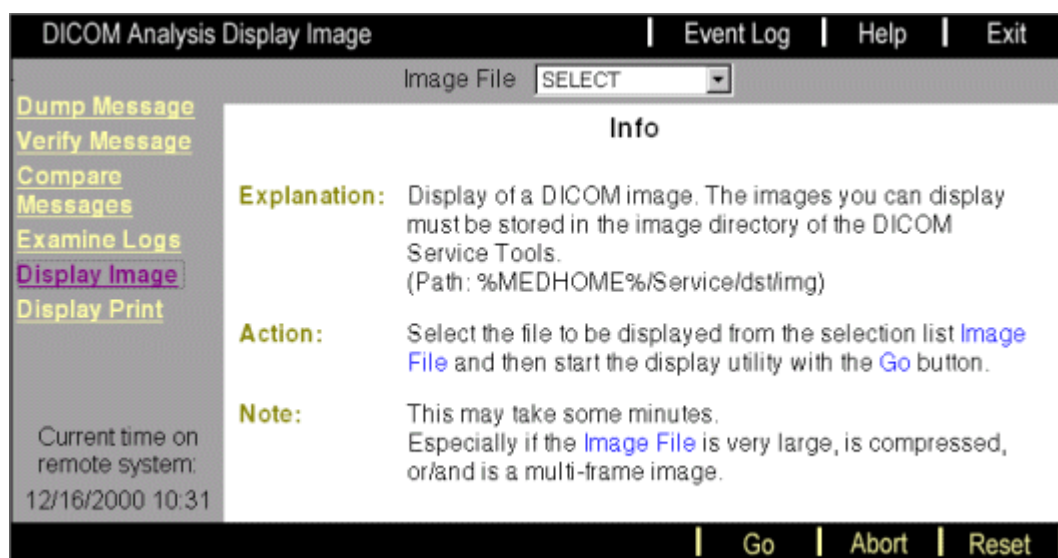


Fig. 189: Dicom Display Image menu

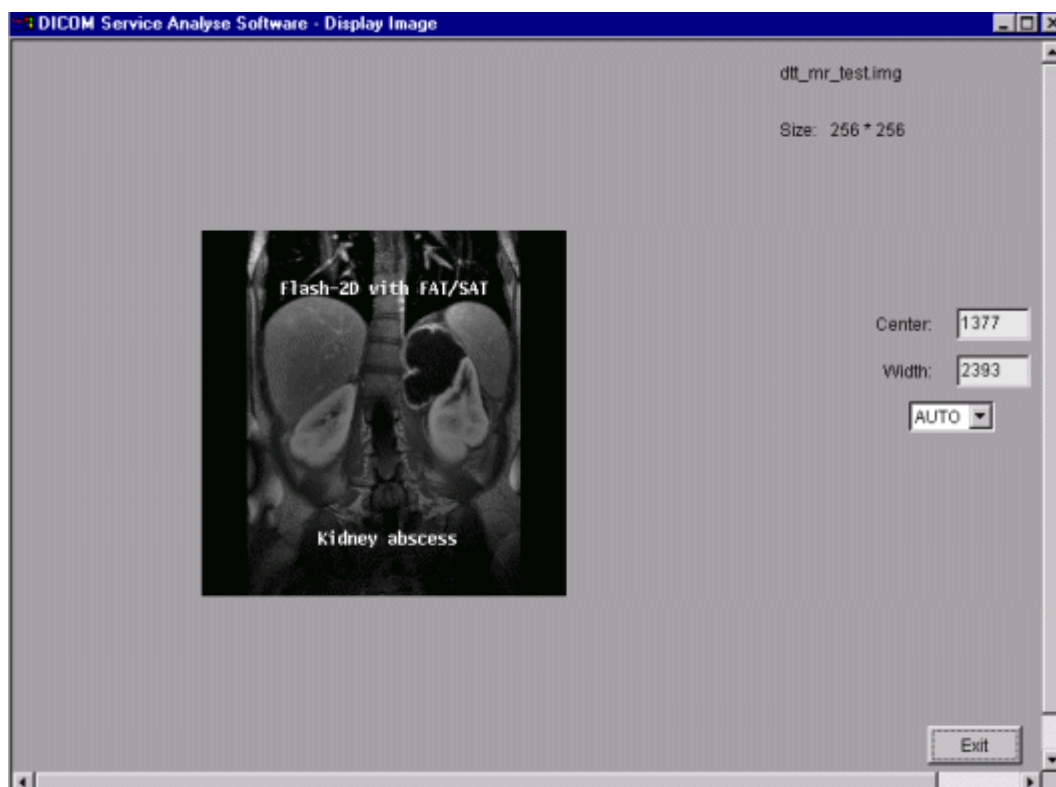


Fig. 190: The Dicom Display Image viewer

Display Print

With this option, the bitmaps generated by the camera simulator will be displayed on screen. Please note that the bitmap is exactly like the scanning array of the laser camera, i.e. the images are mirrored since the image is printed on the reverse side of the film. The pictures are in BMP (Windows bitmap) format. They are also located in the directory %MED-COM%\service\dst\img, and may be copied to a floppy disk for further evaluation.

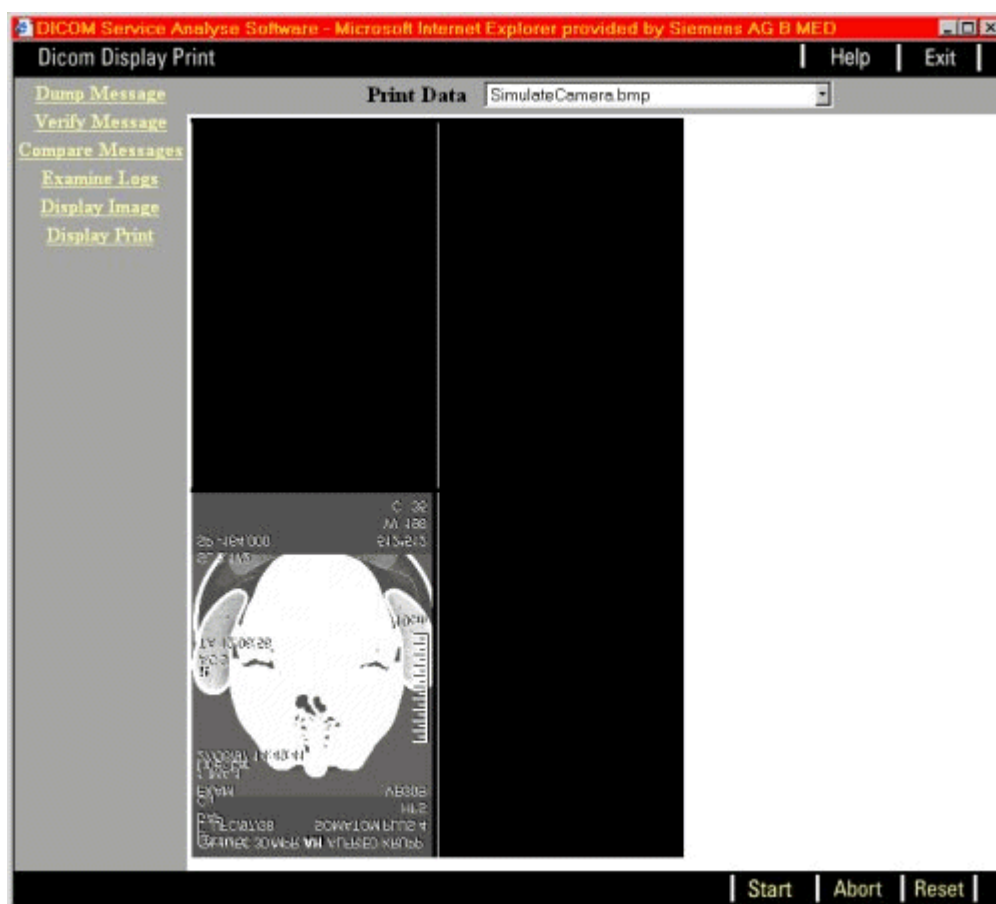


Fig. 191: Dicom Display Print menu

7.2.3 merge log file

 HP-Z0000YPR-111005-02-941

This example shows the merge_stu.log with trace level T2, T3, T7, and T9.

1. Each DICOM communication starts with the association negotiation, which may be displayed on trace level T3. The Service Class User (SCU) sends an association request message to the provider and presents all supported objects with a list of proposed transfer syntaxes (in this case only one). Each object will be identified by a Presentation Context ID. Additionally, the Application Entity Titles (AETs) and the maximum Protocol Data Unit size (PDU) will be transferred.

(312) 07-21 14:58:22.84 MC3 T3: **REQUEST Association Parameters:**

(312) 07-21 14:58:22.84 MC3 T3: | Calling_presentation_address: **cmv10001**
(146.254.120.75)

(312) 07-21 14:58:22.85 MC3 T3: | Called_presentation_address:

(312) 07-21 14:58:22.85 MC3 T3: | **Called_application_title: CSYNGO1**

(312) 07-21 14:58:22.85 MC3 T3: | **Calling_application_title: 049SA1DCDADRSP**

(312) 07-21 14:58:22.85 MC3 T3: | Application_context_name:
1.2.840.10008.3.1.1.1

(312) 07-21 14:58:22.85 MC3 T3: | **PDU_maximum_length: 28672**

(312) 07-21 14:58:22.85 MC3 T3: | Implem_class_uid: 1.3.12.2.1107.5.8

(312) 07-21 14:58:22.85 MC3 T3: | Version_name: SNKIT_1.10

(312) 07-21 14:58:22.85 MC3 T3: | Max_operations_invoked: Not Sent

(312) 07-21 14:58:22.85 MC3 T3: | Max_operations_performed: Not Sent

(312) 07-21 14:58:22.85 MC3 T3: | **PRESENTATION CONTEXTS:**

(312) 07-21 14:58:22.86 MC3 T3: | **(1)Pres_context_id: 1**

(312) 07-21 14:58:22.86 MC3 T3: | **(1)Abstract_syntax:**

1.2.840.10008.5.1.4.1.1.4

(STANDARD_MR)

(312) 07-21 14:58:22.86 MC3 T3: | **(1) Proposed transfer syntaxes:**

(312) 07-21 14:58:22.86 MC3 T3: | **1.2.840.10008.1.2 (Implicit
Little Endian)**

(312) 07-21 14:58:22.86 MC3 T3: | **(1)SCU Role: Notsent(requestorsupportsSCU)**

(312) 07-21 14:58:22.86 MC3 T3: | **(1)SCP Role: Notsent(requestorsupportsSCP)**

(312) 07-21 14:58:22.86 MC3 T3: | **(1)No extended application info**

(312) 07-21 14:58:22.86 MC3 T3: | **(2)Pres_context_id: 3**

(312) 07-21 14:58:22.86 MC3 T3: | **(2)Abstract_syntax:**

1.2.840.10008.5.1.4.1.1.1

(STANDARD_CR)

(312) 07-21 14:58:22.87 MC3 T3: | **(2)Proposed transfer syntaxes:**

(312) 07-21 14:58:22.87 MC3 T3: | **1.2.840.10008.1.2 (Implicit Little Endian)**

(312) 07-21 14:58:22.87 MC3 T3: | (2) SCU Role: Not sent (requestor supports SCU)

(312) 07-21 14:58:22.87 MC3 T3: | (2) SCP Role: Not sent (requestor supports SCP)

(312) 07-21 14:58:22.87 MC3 T3: | (2) No extended application info

(312) 07-21 14:58:22.87 MC3 T3: | **(3) Pres_context_id: 5**

.....

.....

.....

(312) 07-21 14:58:23.01 MC3 T3: | **(21) Pres_context_id: 41**

(312) 07-21 14:58:23.02 MC3 T3: | (21) Abstract_syntax:

1.2.840.10008.5.1.4.1.1.1.2.1

(STANDARD_MG_PROCESS)

(312) 07-21 14:58:23.02 MC3 T3: | (21) Proposed transfer syntaxes:

(312) 07-21 14:58:23.02 MC3 T3: | **1.2.840.10008.1.2 (Implicit Little Endian)**

(312) 07-21 14:58:23.02 MC3 T3: | (21) SCU Role: Not sent (requestor supports SCU)

(312) 07-21 14:58:23.02 MC3 T3: | (21) SCP Role: Not sent (requestor supports SCP)

(312) 07-21 14:58:23.02 MC3 T3: | (21) No extended application info

2. The response of the Service Class Provider shows the objects are supported (accepted) and the proposed transfer syntaxes are used.

(312) 07-21 14:58:23.02 MC3 T3: **RESPONSE Association Parameters:**

(312) 07-21 14:58:23.02 MC3 T3: | Calling_presentation_address: cmv10001
(146.254.120.75)

(312) 07-21 14:58:23.02 MC3 T3: | Called_presentation_address:

(312) 07-21 14:58:23.02 MC3 T3: | **Called_application_title: CSYNGO1**

(312) 07-21 14:58:23.02 MC3 T3: | **Calling_application_title: 049SA1DCDADRSP**

(312) 07-21 14:58:23.03 MC3 T3: | Application_context_name:

1.2.840.10008.3.1.1.1

(312) 07-21 14:58:23.03 MC3 T3: | **PDU_maximum_length: 28672**

(312) 07-21 14:58:23.03 MC3 T3: | Implem_class_uid: 1.3.12.2.1107.5.9.990801

(312) 07-21 14:58:23.03 MC3 T3: | Version_name: SIEMENS_SWFVA47A

(312) 07-21 14:58:23.03 MC3 T3: | Max_operations_invoked: Not Sent

(312) 07-21 14:58:23.03 MC3 T3: | Max_operations_performed: Not Sent

(312) 07-21 14:58:23.03 MC3 T3: | **PRESENTATION CONTEXTS:**

(312) 07-21 14:58:23.03 MC3 T3: | **(1) Pres_context_id: 1**

(312) 07-21 14:58:23.03 MC3 T3: | (1) Abstract_syntax:

1.2.840.10008.5.1.4.1.1.4 **(STANDARD_MR)**

(312) 07-21 14:58:23.03 MC3 T3: | **(1) Result_transfer_syntax:**

1.2.840.10008.1.2 (Implicit**Little Endian)**

(312) 07-21 14:58:23.04 MC3 T3: | (1) SCU Role: Not sent (remote accepts requestor SCU role)

(312) 07-21 14:58:23.04 MC3 T3: | (1) SCP Role: Not sent (remote accepts requestor SCP role)

(312) 07-21 14:58:23.04 MC3 T3: | (1) No extended application info

(312) 07-21 14:58:23.04 MC3 T3: | **(1) Negotiation Result: Accepted**

(312) 07-21 14:58:23.04 MC3 T3: | **(2) Pres_context_id: 3**

(312) 07-21 14:58:23.04 MC3 T3: | (2) Abstract_syntax:

1.2.840.10008.5.1.4.1.1.1

(STANDARD_CR)

(312) 07-21 14:58:23.04 MC3 T3: | (2) Result_transfer_syntax:

1.2.840.10008.1.2 (Implicit**Little Endian)**

(312) 07-21 14:58:23.04 MC3 T3: | (2) SCU Role: Not sent (remote accepts requestor SCU role)

(312) 07-21 14:58:23.05 MC3 T3: | (2) SCP Role: Not sent (remote accepts requestor SCP role)

(312) 07-21 14:58:23.05 MC3 T3: | (2) No extended application info

(312) 07-21 14:58:23.05 MC3 T3: | **(2) Negotiation Result: Accepted**

(312) 07-21 14:58:23.05 MC3 T3: | **(3) Pres_context_id: 5**

.....

.....

.....

(312) 07-21 14:58:23.20 MC3 T3: | **(21) Pres_context_id: 41**

(312) 07-21 14:58:23.20 MC3 T3: | (21) Abstract_syntax:

1.2.840.10008.5.1.4.1.1.1.2.1

(STANDARD_MG_PROCESS)

(312) 07-21 14:58:23.20 MC3 T3: | **(21) Result_transfer_syntax: (Implicit Big Endian)**

(312) 07-21 14:58:23.20 MC3 T3: | (21) SCU Role: Not sent (remote accepts requestor SCU role)

(312) 07-21 14:58:23.20 MC3 T3: | (21) SCP Role: Not sent (remote accepts requestor SCP role)

(312) 07-21 14:58:23.20 MC3 T3: | (21) No extended application info

(312) 07-21 14:58:23.20 MC3 T3: | **(21) Negotiation Result: REJECTED: Abstract syntax**

3. After the association negotiation, the “Command Set” of the first object (e.g. image) will be sent and may be displayed on trace level T7. All attributes within the Command Set belong to group 0000 and indicate which type of object will be sent (context ID and Affected SOP Class UID) and the unique object identification (Affected SOP Instance UID)

(409) 07-21 14:58:23.34 MC3 T7: **Message received on context 5**

(409) 07-21 14:58:23.34 MC3 T7: Group 0x0000 Elements:

(409) 07-21 14:58:23.34 MC3 T7: (0000,0000) Group 0000 Length
 (409) 07-21 14:58:23.34 MC3 T7: (1): 126
 (409) 07-21 14:58:23.34 MC3 T7: **(0000,0002) Affected SOP Class UID**
 (409) 07-21 14:58:23.34 MC3 T7: **(1): 1.2.840.10008.5.1.4.1.1.2**
 (409) 07-21 14:58:23.34 MC3 T7: (0000,0100) Command Field
 (409) 07-21 14:58:23.34 MC3 T7: (1): 1
 (409) 07-21 14:58:23.35 MC3 T7: (0000,0110) Message ID
 (409) 07-21 14:58:23.35 MC3 T7: (1): 0
 (409) 07-21 14:58:23.35 MC3 T7: (0000,0700) Priority
 (409) 07-21 14:58:23.35 MC3 T7: (1): 0
 (409) 07-21 14:58:23.35 MC3 T7: (0000,0800) Data Set Type
 (409) 07-21 14:58:23.35 MC3 T7: (1): 0
(409) 07-21 14:58:23.35 MC3 T7: (0000,1000) Affected SOP Instance UID
 (409) 07-21 14:58:23.35 MC3 T7: (1):

1.3.12.2.1107.5.1.3.24013.4.0.2508718952428

4. On trace level T2, the “Command Set” with the additional information from the message ID is displayed again. This ID shows the transfer syntax used, where

100 is equal to Implicit Value Representation, Little Endian

101 is equal to Explicit Value Representation, Little Endian

102 is equal to Explicit Value Representation, Big Endian

104 is equal to RLE

105 is equal to JPEG Baseline (lossy)

106 is equal to JPEG Extended_2_4 (lossy)

122 is equal to JPEG_Lossless_Hier_14

After the Command Set, the “Data Set” will be shown including all other attributes of the sent object. In the case of image transfer, these attributes are identical to the image header information. In the case of a Query or Worklist request, you can see the attributes sent within these requests.

The value length will be shown within () and the value itself within | |.

Command Set:

(409) 07-21 14:58:26.37 MC3 T2: **Message (STANDARD_CT,C_STORE_RQ) received on context 5**

(409) **Message ID: 100**

(409) **Service Supported: STANDARD_CT (0032)**

(409) **Command Supported: C_STORE_RQ**

(409) **Message Attributes:**

(409) 0000,0000 Group 0000 Length VR: UL VM: 1

(409) (00000004) |126|

(409) **0000,0002 Affected SOP Class UID** VR: UI VM: 1

(409) (0000025) |1.2.840.10008.5.1.4.1.1.2|
(409) 0000,0100 Command Field VR: US VM: 1
(409) (0000002) |1|
(409) 0000,0110 Message ID VR: US VM: 1
(409) (0000002) |0|
(409) 0000,0700 Priority VR: US VM: 1
(409) (0000002) |0|
(409) 0000,0800 Data Set Type VR: US VM: 1
(409) (0000002) |0|
(409) **0000,1000 Affected SOP Instance UID** VR: UI VM: 1
(409) (0000043) |1.3.12.2.1107.5.1.3.24013.4.0.2508718952428|

Data Set:

(409) **0008,0008 Image Type** VR: CS VM: 1-n
(409) (0000035) |ORIGINAL|
(409) |PRIMARY|
(409) |LOCALIZER|
(409) |CT_SOM5 TOPI|
(409) **0008,0016 SOP Class UID** VR: UI VM: 1
(409) (0000025) |1.2.840.10008.5.1.4.1.1.2|
(409) **0008,0018 SOP Instance UID** VR: UI VM: 1
(409) (0000043) |1.3.12.2.1107.5.1.3.24013.4.0.2508718952428|
(409) 0008,0020 Study Date VR: DA VM: 1
(409) (0000008) |20000711|
(409) 0008,0021 Series Date VR: DA VM: 1
(409) (0000008) |20000711|
(409) 0008,0022 Acquisition Date VR: DA VM: 1
(409) (0000008) |20000711|
(409) 0008,0023 Image Date VR: DA VM: 1
(409) (0000008) |20000711|
(409) 0008,0030 Study Time VR: TM VM: 1
(409) (0000013) |143100.140000|
(409) 0008,0031 Series Time VR: TM VM: 1
(409) (0000013) |143145.375000|
(409) 0008,0032 Acquisition Time VR: TMVM: 1
(409) (0000013) |143203.890001|
(409) 0008,0033 Image Time VR: TM VM: 1
(409) (0000013) |143203.890001|
(409) **0008,0050 Accession Number** VR: SH VM: 1

(409) (0000010) IFONo200012 I
 (409) **0008,0060 Modality** VR: CS VM: 1
 (409) (0000002) ICTI
 (409) 0008,0070 Manufacturer VR: LO VM: 1
 (409) (0000007) ISIEMENSI
 (409) 0008,0080 Institution Name VR: LO VM: 1
 (409) (0000011) ISiemens CTE I
 (409) 0008,0081 Institution Address VR: ST VM: 1
 (409) (0000032) IStreet\x0A/27E5E2\x0ADistrict\x0ACountryI
 (409) 0008,0090 Referring Physician's Name VR: PN VM: 1
 (409) (0000061)
 IRefFamNam^RefGiveNam^RefMidNam^RefTitle^RefSuf^RefI
 (409) IPre^RefDegrI
 (409) 0008,1010 Station Name VR: SH VM: 1
 (409) (0000007) IF24013CI
 (409) **0008,1030 Study Description** VR: LO VM: 1
 (409) (0000020) IAbdomen^AbdThinSlicel
 (409) 0008,103E Series Description VR: LO VM: 1
 (409) (0000008) ITopogramI
 (409) 0008,1040 Institutional Department Name VR: LO VM: 1
 (409) (0000000) <null>
 (409) 0008,1050 Performing Physician's Name VR: PN VM: 1-n
 (409) (0000047) ISchedPerfPhNam1^SchedPerfPhNam2^SchedPerfPhNam3I
 (409) 0008,1060 Name of Physician(s) Reading Study VR: PN VM: 1-n
 (409) (0000000) <null>
 (409) 0008,1070 Operators' Name VR: PN VM: 1-n
 (409) (0000012) IDr. GrueningI
 (409) 0008,1080 Admitting Diagnosis Description VR: LO VM: 1-n
 (409) (0000015) ILWS Fehlerbild1I
 (409) **0008,1090 Manufacturer's Model Name** VR: LO VM: 1
 (409) (0000011) IVolume ZoomI
 (409) 0008,2111 Derivation Description VR: ST VM: 1
 (409) (0000000) <null>
 (409) **0010,0010 Patient's Name** VR: PN VM: 1
 (409) (0000040) ICCTJul2000^VZVA20A^SpineSSD^Dr.^CCC_(\xFF\xFF)I
 (409) **0010,0020 Patient ID** VR: LO VM: 1
 (409) (0000009) IPatID_VZ3I
 (409) 0010,0030 Patient's Birth Date VR: DA VM: 1


```

(409)      (0000008) |19550824|
(409) 0010,0032 Patient's Birth Time          VR: TM VM: 1
(409) (0000000) <null>
(409) 0010,0040 Patient's Sex                VR: CS VM: 1
(409) (0000001) |F|
(409) 0010,1000 Other Patient IDs            VR: LO VM: 1-n
(409) (0000000) <null>
(409) 0010,1001 Other Patient Names          VR: PN VM: 1-n
(409) (0000000) <null>
.....
.....
(409) 0018,0010 Contrast/Bolus Agent          VR: LO VM: 1
(409) (0000000) <null>
(409) 0018,0015 Body Part Examined          VR: CS VM: 1
(409)      (0000007) |ABDOMEN|
.....
(409) 0018,1000 Device Serial Number        VR: LO VM: 1
(409) (0000000) <null>
(409) 0018,1020 Software Version(s)        VR: LO VM: 1-n
(409)      (0000007) |VA20A.1|
.....
(409) 0018,5100 Patient Position           VR: CS VM: 1
(409) (0000003) |HFS|
(409) 0020,000D Study Instance UID          VR: UI VM: 1
(409)      (0000043) |1.3.12.2.1107.5.1.3.24013.4.0.2883410072770|
(409) 0020,000E Series Instance UID        VR: UI VM: 1
(409)      (0000043) |1.3.12.2.1107.5.1.3.24013.4.0.2882929363340|
(409) 0020,0010 Study ID                   VR: SH VM: 1
(409)      (0000001) |I|
(409) 0020,0011 Series Number              VR: IS VM: 1
(409)      (0000001) |I|
(409) 0020,0012 Acquisition Number           VR: IS VM: 1
(409)      (0000001) |I|
(409) 0020,0013 Image Number              VR: IS VM: 1
(409)      (0000001) |I|
(409) 0020,0032 Image Position (Patient)     VR: DS VM: 3
(409)      (0000048) |      0|
(409)      |      -398|

```

(409)		-1	
(409)	0020,0037 Image Orientation (Patient)		VR: DS VM: 6
(409)	(0000096)	0	
(409)		1	
(409)		0	
(409)		0	
(409)		0	
(409)		-1	
(409)	0020,0052 Frame of Reference UID		VR: UI VM: 1
(409)	(0000043) 1.3.12.2.1107.5.1.3.24013.4.0.2507320363132		
(409)	0020,0060 Laterality		VR: CS VM: 1
(409)	(0000000) <null>		
(409)	0020,1002 Images in Acquisition		VR: IS VM: 1
(409)	(0000001) 0		
(409)	0020,1040 Position Reference Indicator		VR: LO VM: 1
(409)	(0000000) <null>		
(409)	0020,1041 Slice Location		VR: DS VM: 1
(409)	(0000016)	-1	
(409)	0020,4000 Image Comments		VR: LT VM: 1
(409)	(0000000) <null>		
(409)	0028,0002 Samples per Pixel		VR: US VM: 1
(409)	(0000002) 1		
(409)	0028,0004 Photometric Interpretation		VR: CS VM: 1
(409)	(0000011) MONOCHROME2		
(409)	0028,0010 Rows		VR: US VM: 1
(409)	(0000002) 512		
(409)	0028,0011 Columns		VR: US VM: 1
(409)	(0000002) 512		
(409)	0028,0030 Pixel Spacing		VR: DS VM: 2
(409)	(0000032)	1	
(409)		1	
(409)	0028,0100 Bits Allocated		VR: US VM: 1
(409)	(0000002) 16		
(409)	0028,0101 Bits Stored		VR: US VM: 1
(409)	(0000002) 12		
(409)	0028,0102 High Bit		VR: US VM: 1
(409)	(0000002) 11		
(409)	0028,0103 Pixel Representation		VR: US VM: 1

```

(409)      (0000002) |0|
.....
(409) 0028,1050 Window Center          VR: DS VM: 1-n
(409)      (0000016) |      50|
(409) 0028,1051 Window Width          VR: DS VM: 1-n
(409)      (0000016) |      350|
(409) 0028,1052 Rescale Intercept      VR: DS VM: 1
(409)      (0000016) |      -1024|
(409) 0028,1053 Rescale Slope          VR: DS VM: 1
(409)      (0000016) |      1.0002441|
.....
(409) 7FE0,0010 Pixel Data            VR: OW VM: 1
(409) <Callback routine>
(409) ===== End of Message ID: 100 =====

```

5. After each object (image), a response will be sent from the provider, in which you can see the status of the transfer (0 = successful). A table of the status codes is available in the Intranet at <http://dicom.med.siemens.de> at DICOM - Glossary&Tables.

This “Command Set” message may be displayed on T7.

(409) 07-21 14:58:29.85 MC3 T7: Message (STANDARD_CT,C_STORE_RSP) sent on context 5

```

(409) 07-21 14:58:29.85 MC3 T7: Group 0x0000 Elements:
(409) 07-21 14:58:29.86 MC3 T7: (0000,0000) Group 0000 Length
(409) 07-21 14:58:29.86 MC3 T7: (1): 126
(409) 07-21 14:58:29.86 MC3 T7: (0000,0002) Affected SOP Class UID
(409) 07-21 14:58:29.86 MC3 T7: (1): 1.2.840.10008.5.1.4.1.1.2
(409) 07-21 14:58:29.86 MC3 T7: (0000,0100) Command Field
(409) 07-21 14:58:29.86 MC3 T7: (1): 32769
(409) 07-21 14:58:29.86 MC3 T7: (0000,0120) Message ID Being Responded To
(409) 07-21 14:58:29.87 MC3 T7: (1): 0
(409) 07-21 14:58:29.87 MC3 T7: (0000,0800) Data Set Type
(409) 07-21 14:58:29.87 MC3 T7: (1): 257
(409) 07-21 14:58:29.87 MC3 T7: (0000,0900) Status
(409) 07-21 14:58:29.87 MC3 T7: (1): 0
(409) 07-21 14:58:29.87 MC3 T7: (0000,1000) Affected SOP Instance UID
(409) 07-21 14:58:29.87 MC3 T7: (1):
1.3.12.2.1107.5.1.3.24013.4.0.2508718952428

```

6. When you turn on trace level T9, you will also see the whole association on the Protocol Data Unit level (PDU). This will add a lot more information into the log file which is helpful in some special cases. For example, when the transfer is interrupted you can see how many packets have been transferred successfully.

The Association Release request and response can also only be seen on trace level T9.

(409) 07-21 14:58:29.96 MC3 T9: **PDU Type: 05 (Assoc_Rel_RQ) Received**

(409) 07-21 14:58:29.96 MC3 T9: Reserved Byte: 00

(409) 07-21 14:58:29.96 MC3 T9: PDU length: 0000 0004 (4 bytes)

(409) 07-21 14:58:29.96 MC3 T9: Reserved Bytes (7-10): 0000 0000

(409) 07-21 14:58:29.96 MC3 T9: **PDU Type: 06 (Assoc_Rel_RP) Sent**

(409) 07-21 14:58:29.96 MC3 T9: Reserved Byte: 00

(409) 07-21 14:58:29.96 MC3 T9: PDU length: 0000 0004 (4 bytes)

8 Event log

HP-Z0000YPR-111005-02-942

n.a.

8.1 General

HP-Z0000YPR-111005-02-939

n.a.

8.1.1 Introduction

HP-Z0000YPR-111005-02-940

The Event Log Viewer helps you to isolate hardware and software problems. It is a tool for displaying events from the “Windows event log”. Various filter criteria can be used to obtain a filtered output from the eventlog.

The **Event Log Viewer** mechanism starts automatically when the system is started and can be used at any time in parallel to any service function.

8.1.2 Features

HP-Z0000YPR-111005-02-938

The **Event Log Viewer** can be started from the Service Home menu by clicking the Event Log button in the navigation bar. It is started in a separate window so that it is possible to evaluate the system Event Log in parallel to other service functions. The Event Viewer is divided into two main tab cards, the **Expert Mode** and the **Pattern Mode**. The following main features are provided by the Event Viewer for error evaluation:

- Multiple selection of “Domain” types (Application, Customer Log, Security, and System), which refers to the related Windows event logs (exception Customer Log).
- Multiple selection of “Severity” types (Error, Warning, Information and Success).
- Selection of one of the following “Facility” levels (Service, Developer, Customer, and User-defined).
- Display of extended message information, which is the content of the message binary data used by some system components.
- Search within a certain time period.
- Search for a text pattern within the message text.
- Search for a Source or combinations of it (process name).
- Search for a message ID or combinations of ID's.
- Use of pre-defined filter templates, and creation of individual filter templates.
- Use of pre-defined search patterns to search for message combinations.
- Printing of search results and/or extracting them to a file for delivery via SRS.

8.2 Functions

HP-Z0000YPR-111005-02-937

n.a.

8.2.1 Enter Event Log Viewer

HP-Z0000YPR-111005-02-936

The **Event Log Viewer** can be started from the Main Menu in the service UI by clicking the Event Log button in the navigation bar. The Event Log Viewer is started in a separate window so that it is possible to work in parallel to other service functions.

8.2.2 Domain

HP-Z0000YPR-111005-02-935

Events are collected in groups, also known as "Domains". For a better performance of the Event Log, each Domain has its own log file. Multiple Domains can be selected.

The following Domains are selectable:

- **Application (default)**

Messages from all running applications (e.g.: syngo, database applications, modality-specific applications) are recorded in this log file.

- **CustomerLog**

The customer log shows the service activities to the customer in the Remote Service “**Access Control Platform**” (e.g. if a local or remote service session is active, or a service application has been activated).

- **Security**

Security events (e.g. attempts to log on to the system) are recorded in the security log file.

- **System**

The System log records events of the Windows XP system components. For example, if a driver failed to launch during start-up.

8.2.3 Severity

HP-Z0000YPR-111005-02-934

This is a classification of the errors or messages supplied by the regular Windows XP software. Multiple “Severities” can be selected and the different severities are displayed in different colors.

- **Error (red)**

A function could not be completed because of a problem within the system. The problem cannot be dealt with internally.

- **Warning (blue)**

An event that is not necessarily significant, but may indicate a possible future problem. For example, when the disk space is low, a Warning event will be logged.

- **Information (black)**

An event that describes the successful operation of an application, driver or service. For example, when a network driver loads successfully, an Information event will be logged.

- **Success (green)**

An audited security access attempt is successful. For example, a user's successful attempt to log on to the system will be logged as a Success Audit event.

8.2.4 Facility

HP-Z0000YPR-111005-02-933

The Facility is a more detailed, syngo-based classification of the errors or messages.

- **Developer**

All messages classified as Developer, Service and Customer will be displayed.

- **Service (default)**

All messages classified as Service and Customer will be displayed. Developer messages are not shown.

- **Customer**

Only messages classified as Customer are displayed. Developer and Service messages are not shown.

8.2.5 Extended

HP-Z0000YPR-111005-02-932

With the **Extended** button, an additional line below the Message Text is shown with additional information belonging to the event. Some modalities or system components use the message binary data (Windows event property) to log additional information which is then called up with the **Extended** button. It is hidden again with the **Standard** button.

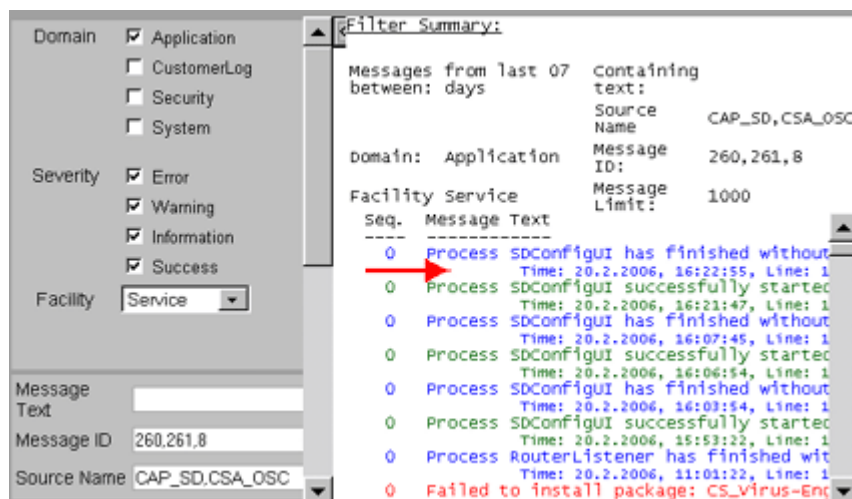


Fig. 192: Extended View

8.2.6 NT Messages

HP-Z0000YPR-111005-02-931

Not available with this version.

8.2.7 Search Pattern

HP-Z0000YPR-111005-02-930

The **Pattern Mode** is entered by clicking the Pattern Mode tab card. The Pattern Mode provides pre-defined search patterns which can be triggered by selecting (checkbox) the pattern name of interest and by executing the query with the GO button. The Pattern tab card is available only if certain search patterns are provided by the system.

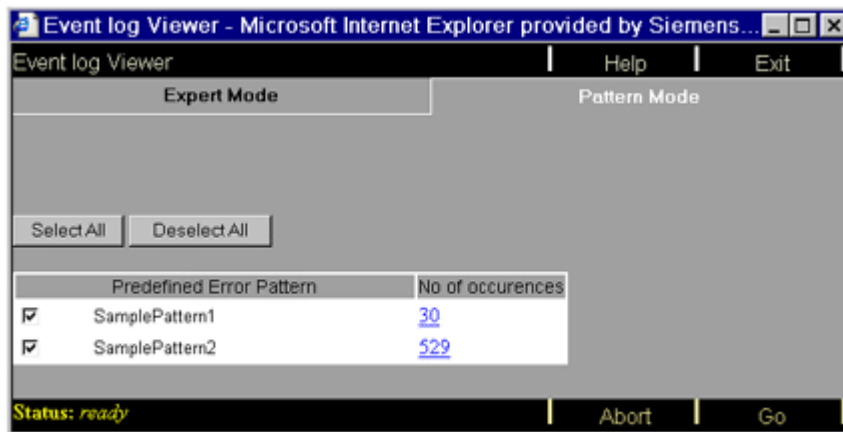


Fig. 193: Tab Card - Pattern Mode

A search pattern is a combination of more than one event, whereby OR and AND relations as well as their sequence of occurrence are supported. Any filter settings made in Expert Mode are not applied for a pattern search. The number of found relations is displayed in the table column "No of occurrences." The shown number is underlined. Clicking it opens a separate window showing the detailed query results with the particular messages found. In this window, the **Extended** information lines can also be displayed. The **Print** and **Extract** functions are also available.

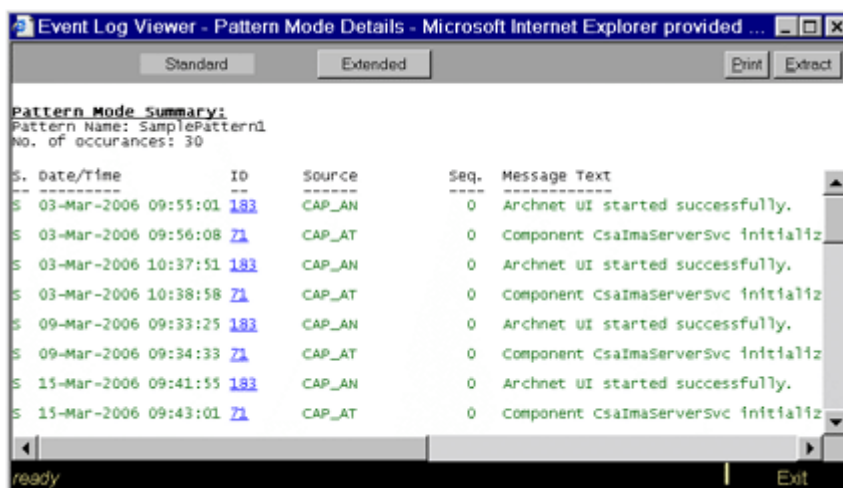


Fig. 194: Pattern Search Result

8.2.8 Source name

HP-Z0000YPR-111005-02-929

The **Source Name** has the format <GG>_<Component> to identify the module that created the event. The syngo filming component, for example, would show up as CAP_FL.

A wild card search is possible; e.g., CAP* will display all messages where the source name begins with CAP.

NOTE

The search string is case sensitive!

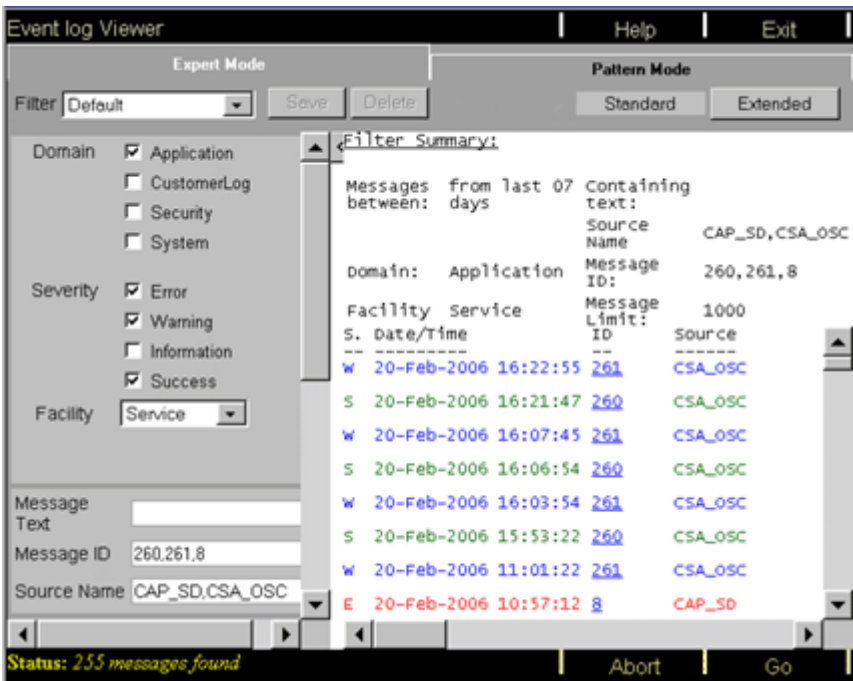


Fig. 195: Event Log Viewer - part 1

8.2.9 Message ID

HP-Z0000YPR-111005-02-928

Use this filter to search for a specific **Message ID** number. Enter the **Message ID** number to look for. This will show you how often a particular **Message ID** occurred in the current event log. Combinations of ID numbers are separated by commas (e.g. 102,23,4)

NOTE

You may find the same Message ID number in different Sources. Enter the filter Source first (e.g. *FL), and then define the Message ID you are looking for.

8.2.10 Search order

HP-Z0000YPR-111005-02-927

Depending on the selection, the most recent or the oldest of all filtered messages is displayed at the top of the message list

8.2.11 Message limit

HP-Z0000YPR-111005-02-926

Defines the maximum number of message lines which will be displayed in the output window.

NOTE

Please note that this value limits the output of the message lines. If you don't find the message that you are looking for, increase the entered value or remove the entry so that all found messages are displayed.

8.2.12 Extract to File

HP-Z0000YPR-111005-02-925

The **Extract** function is used to save the query results in a file on the system, in case the headquarters support center is requesting such an event log extract for error evaluation

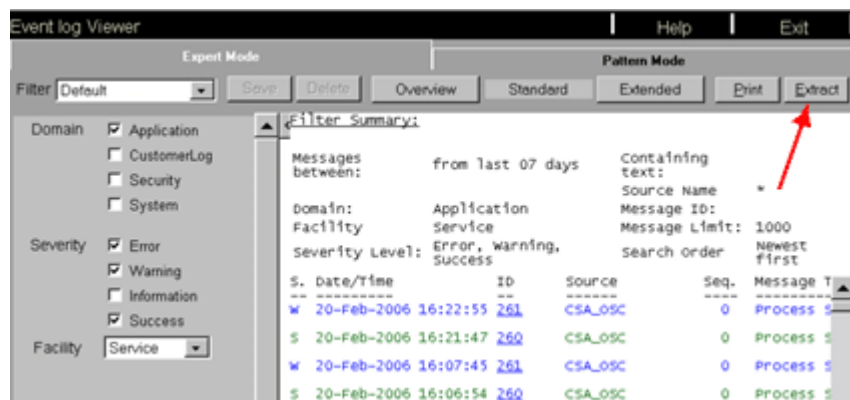


Fig. 196: Extract Button

With a mouse click on the Extract button, an additional window is shown to enter a name for the extract-file, to select in which format the extract will be created (ASCII or XML) and if the content will be compressed to reduce the file size

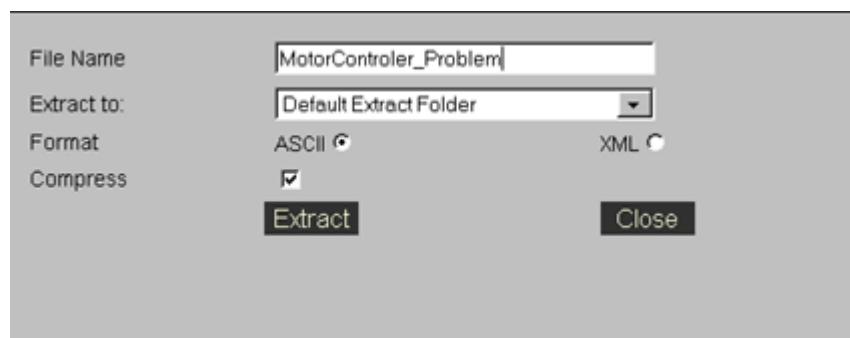


Fig. 197: Extract File Name

The "Default Extract Folder" is the directory `c:\MedCom\Service\Extract`, where it will be stored if no other location is chosen from the select box.

8.2.13 Time Range

HP-Z0000YPR-111005-02-924

This filter limits the output to the number of messages valid for the desired date and time or a relative time range.

- **Absolute Time Range** The Absolute Time Range is a filter showing only events that have occurred within a specific time period by entering the From Date/Time and the To Date/Time.
- **Relative Time Range** This is the default selection if the Event Log Viewer is started with a value of 7 days. It is possible to select between days or hours combined with a free number.

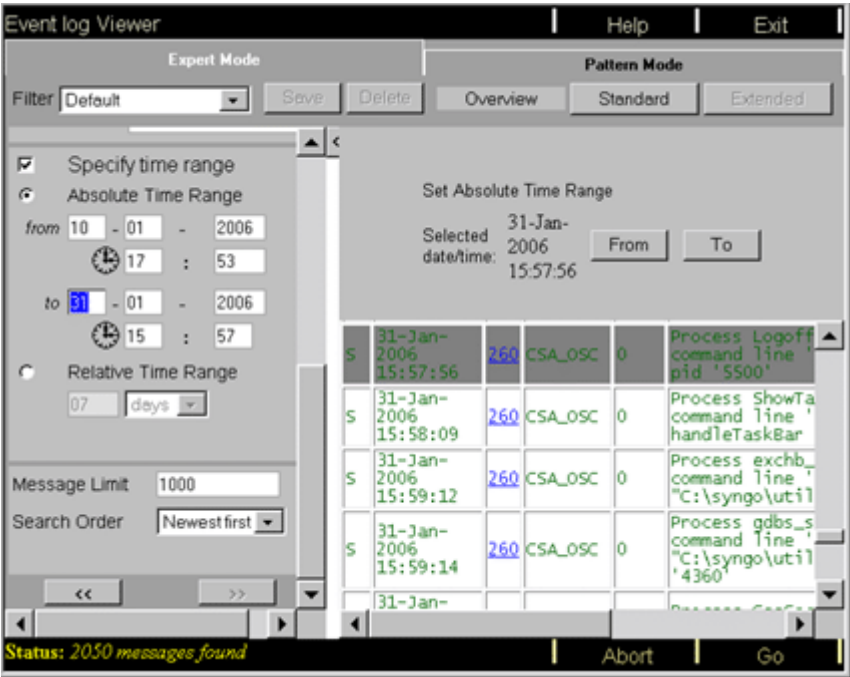


Fig. 198: Feature Overview

8.2.14 Detailed Message Text

HP-Z0000YPR-111005-02-923

Each Message ID in the Event Log output window has a hypertext link (blue Message ID number) to a Detailed Message Text explanation. Here you will find more detailed information on the recommended action and an explanation for rectifying the problem.

To access the Detailed Message Text information, just click on the Message ID and a separate window will open.

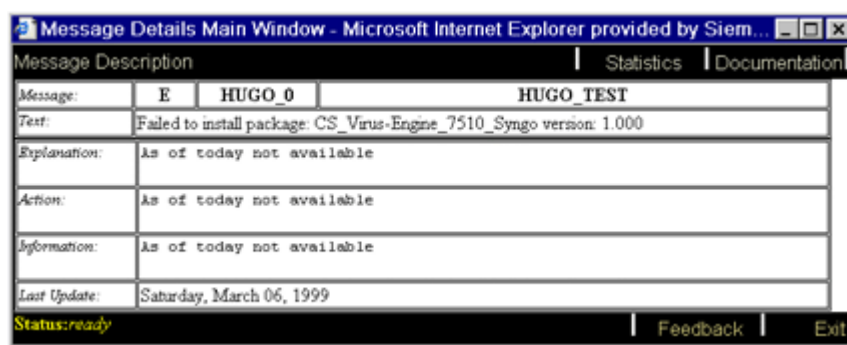


Fig. 199: Detailed Message Window

9 Help

HP-Z0000YPR-111005-02-922

n.a.

9.1 General

HP-Z0000YPR-111005-02-921

n.a.

9.1.1 Introduction

HP-Z0000YPR-111005-02-920

This section describes the Help function of the syngo service software package.
It provides brief information on the Help function.

9.1.2 Features

HP-Z0000YPR-111005-02-919

The **Help** functionality is context-sensitive. This means that when you access the Help function, you will receive information on the function you are currently using.

For example, you are currently using the File & Image Tools service function and you would like to obtain information about the functions supplied in this menu. When you click the **Help** button, a separate window with the appropriate content (in this case, File & Image Tools Help) is displayed.

9.2 Functions

HP-Z0000YPR-111005-02-918

n.a.

9.2.1 Online Help

HP-Z0000YPR-111005-02-917

Online Help can be accessed from every service software window.

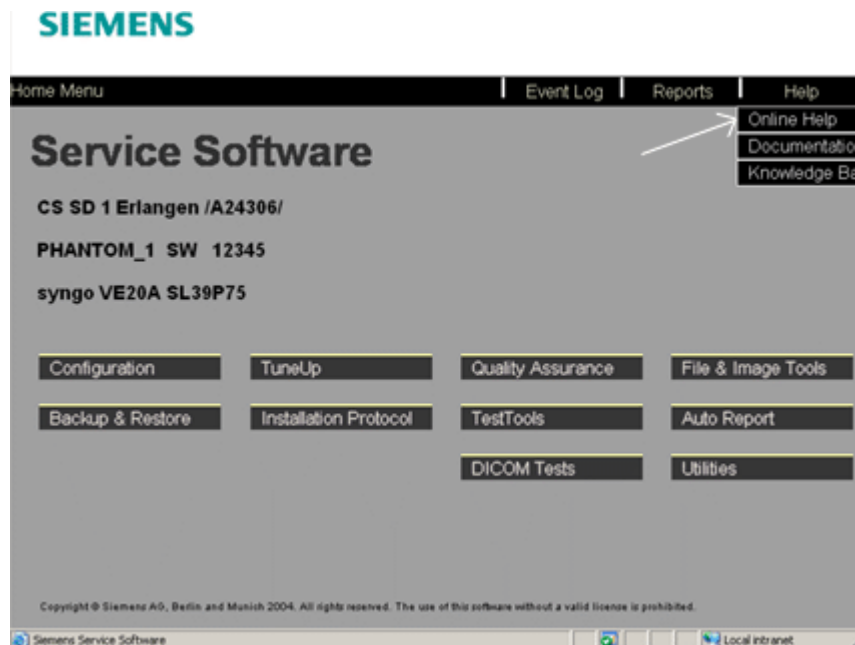


Fig. 200:

The Online Help overview shows all available help entries arranged according to the main functions of the service software. Its structure is similar to that of the service software and is navigated from the left-hand **navigation frame**.



Fig. 201:

9.2.2 Documentation

HP-Z0000YPR-111005-02-916

CB DOC:

A pop-up window appears when you click the **Documentation** button in the navigation bar of the **Help window** or within the "Home" menu under Help. A default URL address for the Online Documentation CD-ROM is displayed. Click the **OK** button to open the home page of the Computer-Based Documentation (CB-Doc) in a separate window.

NOTE

If you are logged in remotely, the default will automatically point to any local drive.

In either case, whether you are logged in remotely or locally at the customer site, you can always change the default address to access any other Web server on the network. To do so, change the URL address from `file:///<CD drive>/default.htm` to `http://<web_address>`.

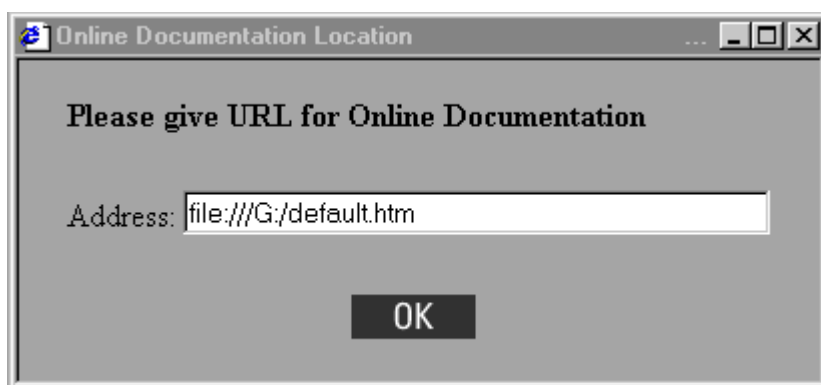


Fig. 202: URL for Online Documentation

9.2.3 Knowledge Base

HP-Z0000YPR-111005-02-915

This button allows access to the Knowledge Base of Medical Solutions (Customer Service) from the system itself, using a successfully established remote connection to the Siemens Remote Server (SRS). It opens an additional browser window with a Knowledge Base login screen. Authentication by the service engineer with his own account is still required.

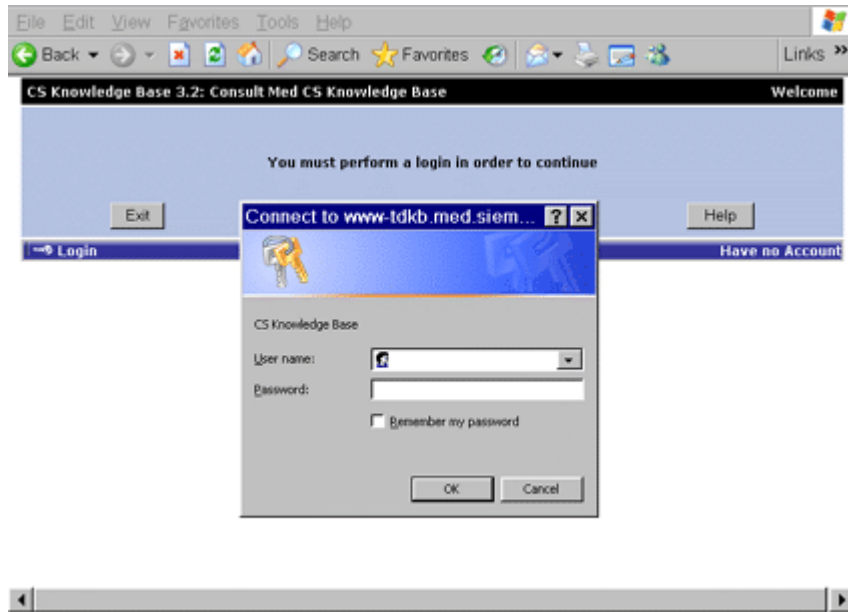


Fig. 203: Knowledge Base Access Window

10 Backup & Restore

HP-Z0000YPR-121005-02-999

n.a.

10.1 General

HP-Z0000YPR-121005-02-998

n.a.

10.1.1 Introduction

HP-Z0000YPR-121005-02-997

This section describes the **Backup & Restore** function.

Backup & Restore provides a basic backup and restore functionality for predefined dynamic data in your system.

The following “Backup & Restore” packages are available:

- **ClonePackage** (for transferring global syngo configuration data to other systems)
- **CustomerProtocols** (Customer protocols for the MR system)
- **Diagnostic** (MRsavelog files)
- **ExamDB** (ExamDB settings for the database)
- **ExternConvert**
- **Master-Backup** (All system-specific backup packages in one)
- **Numaris4** (syngo MR-specific data e.g. “**TuneUp**“, “**QA**“, “**customer protocols**“, and “**Install & Maintenance**“ data)
- **Security-Settings** (Security settings (data that includes system-specific security settings.))
- **SW-OptionIDs**
- **SW-Settings02** (data that includes system-specific data and software and user/customer settings.)

10.1.2 Features

HP-Z0000YPR-121005-02-996

The Backup & Restore function provides the following features:

- Backup of predefined dynamic data to the hard disk (and to the network) or MOD, including configurable pre- and post-backup operations.
- Restoration of predefined dynamic data from the hard disk (also from the network) or MOD, including configurable pre- and post-restore operations.

NOTE

The predefined list of dynamic data and pre-/post-operations for the backup/restore procedure can be changed within the configuration function.

10.2 Functions

HP-Z0000YPR-121005-02-995

n.a.

10.2.1 Handling

HP-Z0000YPR-121005-02-994

As Backup & Restore is started, the left frame contains a selection menu. Make the appropriate command selection and start the corresponding procedure by clicking the **Go** button.

NOTE

All screenshots are only included as examples for offline reading. The screenshots are from a syngo-only console.

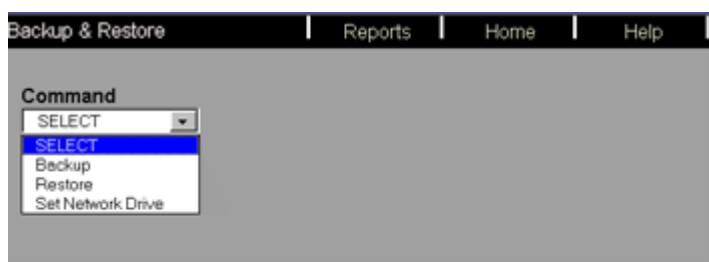


Fig. 204: Select Backup or Restore

Backup

For **syngo MR** systems it is recommended that you save all backup packages at once with the **Master-Backup** package (e.g. during preventive maintenance).

NOTE

Ignore the following error messages during the **Master-Backup**:
CorRea.bru error (only at the MRSC)
ITO errors (If **CA Unicenter** has not been installed)

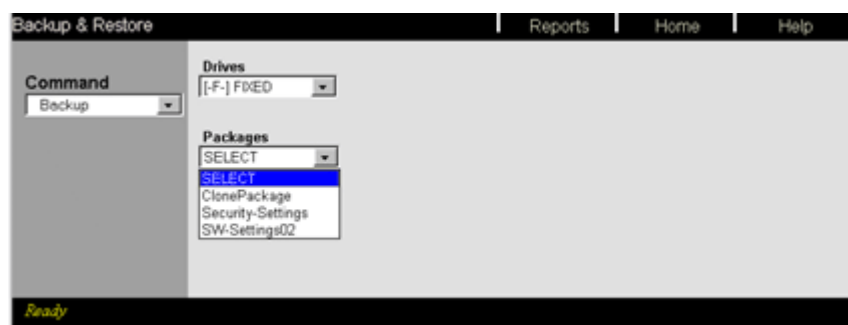


Fig. 205: Backup - e.g. Security Settings

- Firstly enter an appropriate destination **drive** letter for the backup, e.g. drive "F: " to back up the predefined dynamic data to drive "F:".

- Then select the **package** you want to save. You can select SW-Settings 02 ,Security-Settings or the ClonePackage
 - SW-Settings02 is a complete backup of the entire system
 - Security settings are an excerpt containing security-related settings, if enabled under configuration. They can be used for multiple exports to other systems.
 - The ClonePackage consists of configuration items that are not system-specific settings (such as the IP address), and can be used to export general settings to other systems. The DICOM node configurations are changed so that the node from the location in which the **ClonePackage** was created automatically becomes a receiver node in the system to which the settings are copied.
- Then select the **Go** button to start the backup procedure.

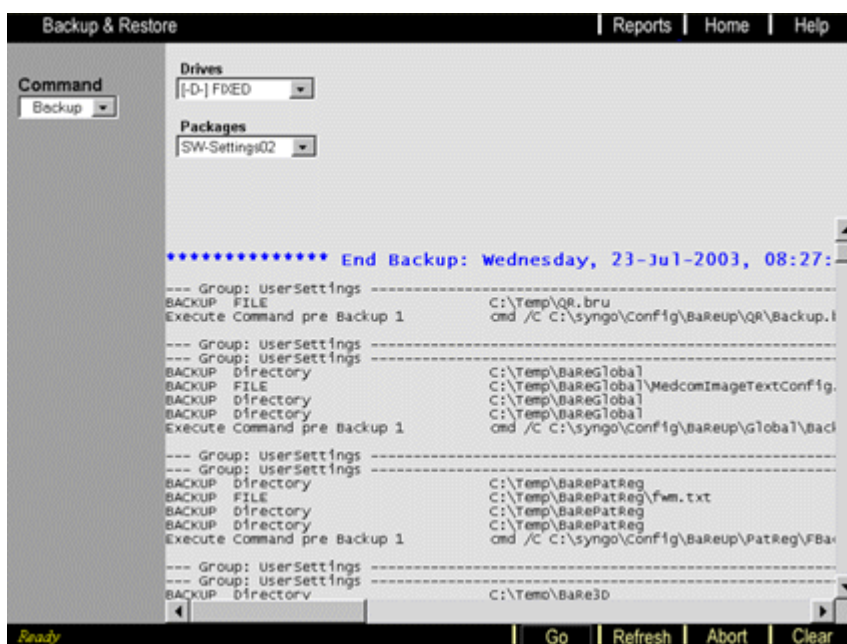


Fig. 206: Backup - SW-Settings02

You will see the status of the backup SW-Settings02 operation in the lower part of the output screen.

Restore

For **syngo MR** systems it is recommended to restore **all** "Groups" from the **Master-Backup** e.g. after re-installing software.

- **Drives**

First, choose an appropriate source **Drive** letter, e.g. D: for restoring previously stored data from drive D.

- **Archives**

Then, choose the **Archive** you want to restore. All available backup packages are visibly stored in the main root of the selected drive.

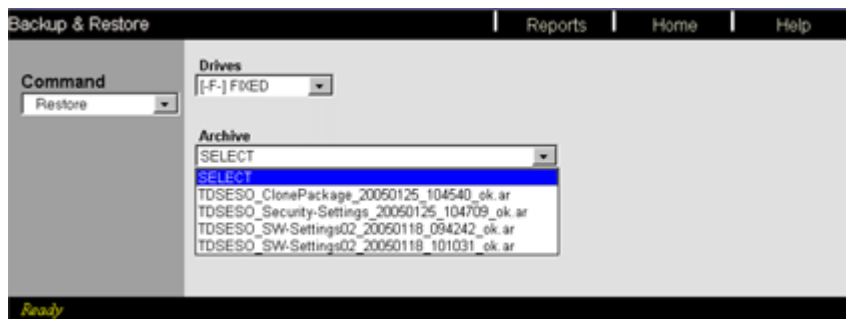


Fig. 207: Restore - Select an Archive

- **Groups**

When you select the Security settings, you will see Group = all (all security setting groups are mentioned).

When you select the SW settings02 or the ClonePackage, you will see several Group entries which you can choose to restore (multiple selections are possible with **Ctrl** or **Shift**).

- **GO**

Select the **Go** button to start the restore procedure. The status of the ongoing restore operation is shown in the lower part of the output screen.

NOTE

Error messages are represented in red characters.

Groups

Security-Settings containing the Security Configuration Data for:

- Users,
- Groups and
- Roles,
- Functional Privileges and
- Data Security Settings

They do not contain the selected settings of the security properties page, nor do they contain the DICOM users (AET users). The DICOM users are restored with the restore operation of the DICOM nodes.

Groups of SW-Settings02

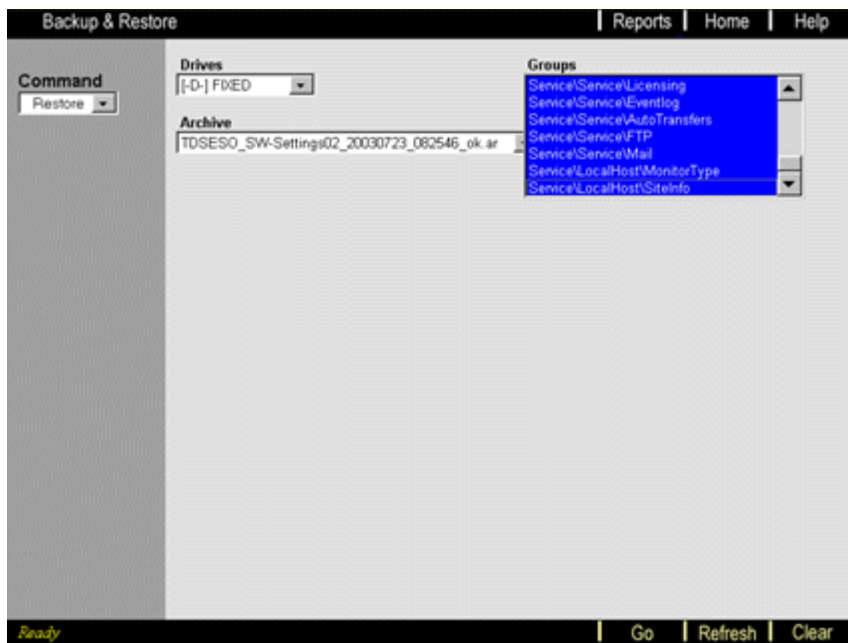


Fig. 208: Restore - Groups of "SW-Settings02"

The following groups are available for "SW-Settings02":

User Settings

Service\Service\SaveSecurityConfig

Service\Localhost\Users

Service\TCPIP\LAN Settings

Service\TCPIP\RAS Settings

Service\Database\SatelliteConsole

Service\Database\DBOffset

Service\Application\WorklistResults

Service\Application\MPPS

Service\Application\CorRea

Service\Application\Viewer

Service\Application\PatientRegistration

Service\EPRI\Server

Service\DICOM\ImportExportDirectories

Service\DICOM\HISRISNodes

Service\DICOM\LUTFiles

Service\DICOM\PrintDevices

Service\DICOM\NetworkNodes

Service\DICOM\Offline Devices

Service\DICOM\General

Service\ExternalDevices\PaperPrinter

Service\Service\SecConfig

Service\Service\ITO
Service\Service\SNMP
Service\Service\Licensing
Service\Service\Eventlog
Service\Service\FTP
Service\Service\Mail
Service\LocalHost\MonitorType
Service\LocalHost\SiteInfo

ClonePackage

Each group represents a set of configuration data for the entire system (see below).

Subsets of those configuration items/groups that are not system-specific settings (such as the IP address) can be used to export global settings to other systems (within a network).

NOTE

The DICOM node configurations are changed so that the node from the location in which the ClonePackage was created automatically becomes a receiver node in the system to which the settings are copied.

Set Network Drive

It is possible to store the backup data to a shared device in the network.

The share has to be defined and saved. Subsequently, the share will be selectable under backup drives.

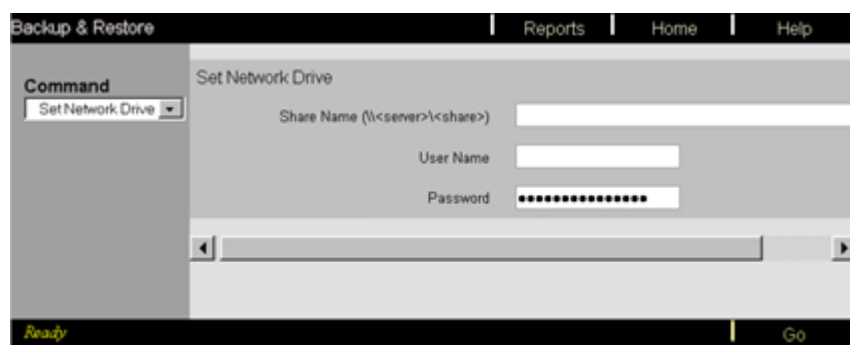


Fig. 209: Set Network Drive

11 Event Log Customer

HP-Z0000YPR-121005-02-993

n.a.

11.1 General

HP-Z0000YPR-121005-02-992

n.a.

11.1.1 Introduction

HP-Z0000YPR-121005-02-991

The **Event Log Viewer** helps you solve hardware and software-related problems. It is a tool used for monitoring events on the system. The events are stored in the Event Log. Various filter criteria can be used to obtain a filtered output from the Event Log.

The **Event Log Viewer** mechanism is launched automatically when the system starts up and can be used at any time in parallel to any service function.

11.1.2 Feature

HP-Z0000YPR-121005-02-990

The Customer Event Log Viewer supports the following filter criteria:

- Selection of one of four domains (Application, Customer Log, Security, and System).
- Selection of one or all three severity levels (Information, Warning, Error).
- Selection of one of two facility levels (Service, Customer).
- Display of NT messages.
- Date and time of evaluation.
- Display most recent or oldest message line firstly.

11.2 Functions

HP-Z0000YPR-121005-02-989

n.a.

11.2.1 Event Log Viewer

HP-Z0000YPR-121005-02-988

There are various service-related options in the **Options menu**. Firstly click **Options** to access the sub-menu, followed by **Event Log**.

NOTE

All options can only be started from the local console.

This chapter will explain the service-related items:

- **Event Log**

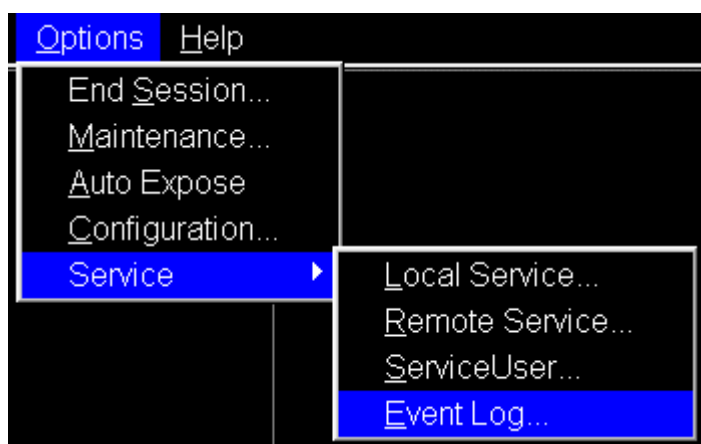


Fig. 210: Menu Options: Service - Event Log

In most cases, an icon will appear on the bottom right of the screen as soon as an application (e.g. Event Log, Remote Service, etc.) is launched. Each application can be minimized so that it appears as an icon by clicking “_” on the title bar. Clicking an icon reopens the associated application.

A new window will appear.

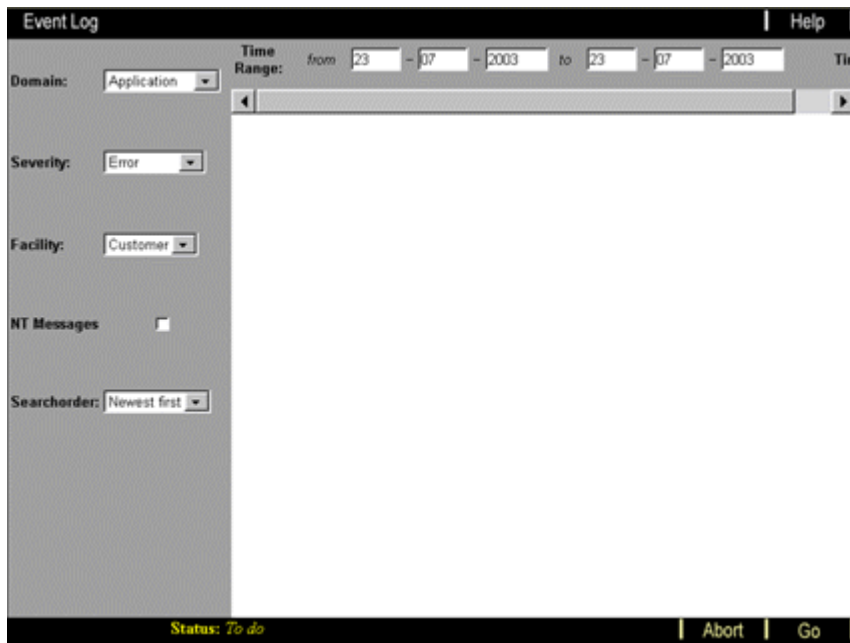


Fig. 211: Event Log application for the customer

- **Go**

The current date is entered as default. Click the **Go** button to display all messages from the current day. The number of messages displayed will be limited to **100 lines** and the most recent entry will be displayed first.

NOTE

The Event Log Viewer window is not updated automatically. To view the most recent messages, click the Go button again.

- **Abort**

Clicking the **Abort** button while messages are being loaded will stop this operation.

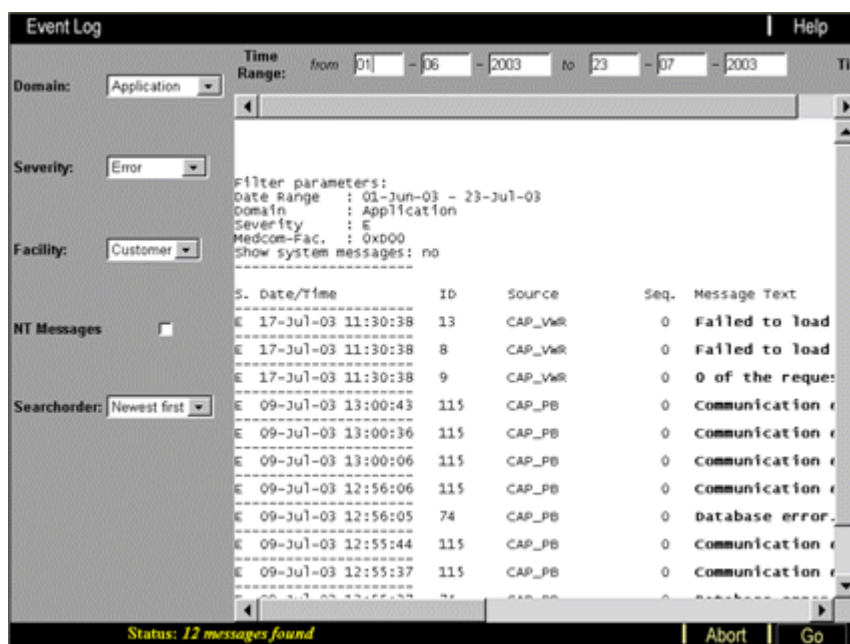


Fig. 212: Event Log results for the customer

Filter parameters are displayed firstly, followed by filtered messages. The items in the table title have the following meaning:

- **S**
Severity (E = Error, W = Warning, I = Information, S = Success)
- **Date/Time**
Date and time when the message occurred and was logged.
- **ID**
Message ID number
- **Source**
Software application which logged the message.
- **seq.**
Sequence number of an event belonging to an event array (from 0 to 31). These events occur in a sequence starting with the root cause (0) and followed by possible events (1 to 31) reported by other modules and caused by the event identified as 0.
- **Message Text**
Message text sent from the application that caused the problem.

11.2.2 Domain

HP-Z0000YPR-121005-02-987

The Event Log Viewer is used to view and manage events. Events are collected in groups. These groups are also known as domains. To optimize event log performance, each domain has its own log file.



Fig. 213: Domain filter

The following domains can be selected:

- **Application (default)**

Messages from all running applications (e.g.: syngo applications, database applications, or modality-specific applications) are recorded in this log file.

- **CustomerLog**

The customer log shows the service activities (e.g. if a local or remote service session is active, or a service application has been activated) to the customer in the **Remote Service Access Control Platform**.

- **Security**

Security events (e.g. attempts to log on to the system) are recorded in the security log file. The security event trigger is usually deactivated. It can be activated via the Windows NT **User Manager** application. Log in as administrator and start the **User Manager** application <Administrative Tools> <User Manager>. Then select the <Policies> <Audit...> option to enable/disable security events.

- **System**

The system log records events associated with Windows NT system components (e.g. a driver failing to launch during start-up).

11.2.3 Severity

HP-Z0000YPR-121005-02-986

This classifies the errors/messages supplied by the standard Windows NT software.

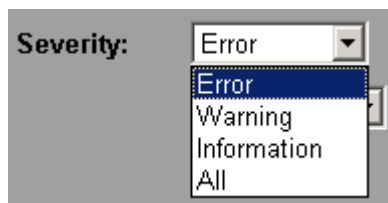


Fig. 214: Severity filter

- **Error (default)**
A function could not be completed successfully due to a problem within the system. The problem cannot be dealt with internally.
- **Warning**
A message to alert the user to a possible error condition. It will be handled internally by the system.
- **Information**
Any additional information which could be useful for troubleshooting the problem and messages confirming that a function has been completed successfully (e.g. status message concerning disk usage).
- **All**
All severities will be selected.

11.2.4 Facility

HP-Z0000YPR-121005-02-985

The facility is an event (message) classification. The default Customer Event Log Viewer setup provides the **Service** and **Customer** facilities, which limit the displayed events according to their classification. Customer events are shown in bold.

NOTE

When neither the System, Security, nor the Customer Log domain is selected, the facility is set to none.



Fig. 215: Facility filter for the customer

- **Service**

All messages classified as Service and Customer are displayed. Developer messages are not shown.

- **Customer (default)**

Only messages classified as Customer are displayed. Developer and Service messages are not displayed.

11.2.5 NT Messages

HP-Z0000YPR-121005-02-984

If this option is selected, messages from the operating system (Win NT) are displayed. These include driver initialization failed, process not started, etc.

NOTE

Dependent upon the domain selected. The NT messages filter is automatically activated for System, Security, and Customer Log.

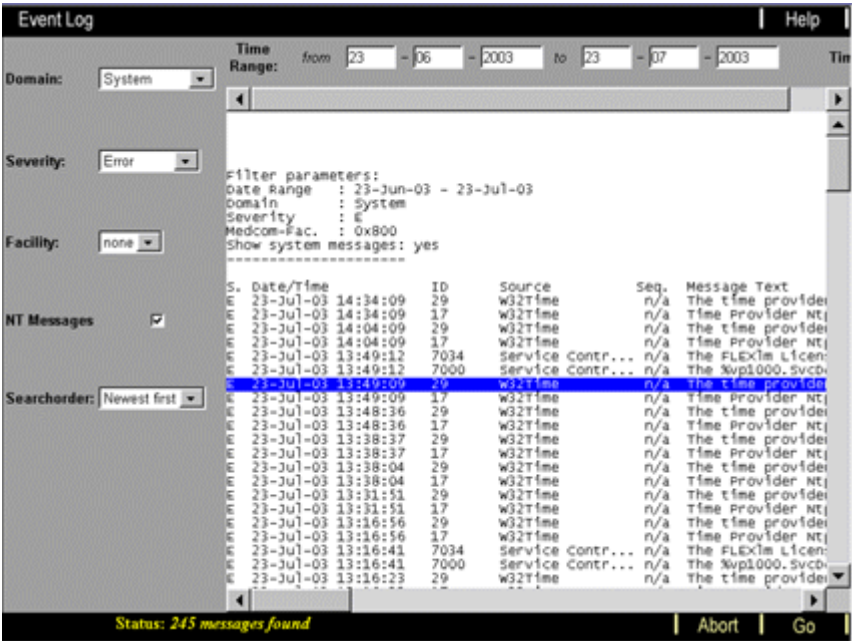


Fig. 216: Event Log query results with NT Message filter for the customer

NOTE

Sequence numbers are not displayed in operating system messages (NT messages). N/A is displayed instead.

- **Message Text**
The message text sent from the application that caused the problem.

11.2.6 Search order

HP-Z0000YPR-121005-02-983

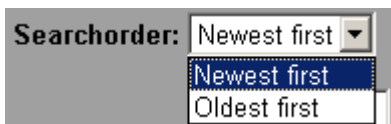


Fig. 217: Searchorder filter

Default: Newest first

- **Newest first**

The most recent of all the filtered messages is displayed at the top.

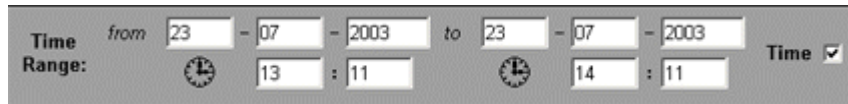
- **Oldest first**

The oldest of all the filtered messages is displayed at the top.

11.2.7 Time Range

HP-Z0000YPR-121005-02-982

This filter restricts the display to the number of messages that are valid for the selected date and time range.



Time Range: from 23 - 07 - 2003 to 23 - 07 - 2003 Time ☒
13 : 11 14 : 11

Fig. 218: Time Range filter

Default: Current day

- **From**
Enter the start date and time of the range.
- **To**
Enter the end date and time for the range.
- **Time (Default: Previous hour)**
To enter a time range, select the **Time** checkbox. The time fields are displayed.

12 Local Service

HP-Z0000YPR-121005-02-981

n.a.

12.1 General

HP-Z0000YPR-121005-02-980

n.a.

12.1.1 Introduction

HP-Z0000YPR-121005-02-979

There are various service-related options in the **Options menu**. These service-related options can be started at the local console only.

12.1.2 Feature

HP-Z0000YPR-121005-02-978

This chapter explains the following service-related items:

- Local Service
- Service User

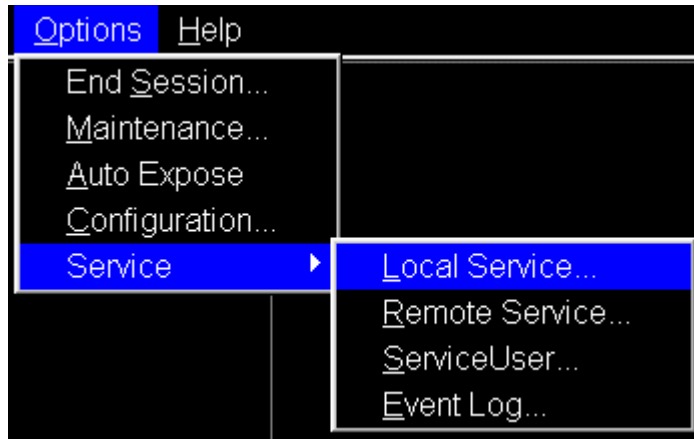


Fig. 219: Menu Options: Service - Local Service

As soon as an application is launched, an icon appears at the bottom right of the screen. Each application can be minimized so that it appears as an icon by clicking “_” on the title bar. Clicking an icon reopens the associated application.

12.2 Functions

HP-Z0000YPR-121005-02-977

n.a.

12.2.1 Local Service

HP-Z0000YPR-121005-02-976

To start the service software UI from the console, select **Local Service** from the **Options** menu in the **System** menu:

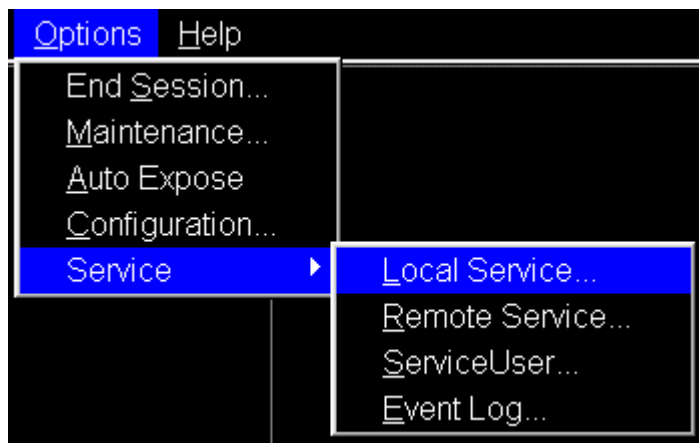


Fig. 220: Menu Options: Service - Local Service

The Internet browser starts up, showing the **Authentication menu** for the service software UI. Next, type in the **variable password**. You can enter a new password (you will need to delete the old password first) or just the last six characters of the existing password (as shown in the example below). The characters will not be visible when typed in. Please refer to **Service Software UI -> Authentication menu** for more details.

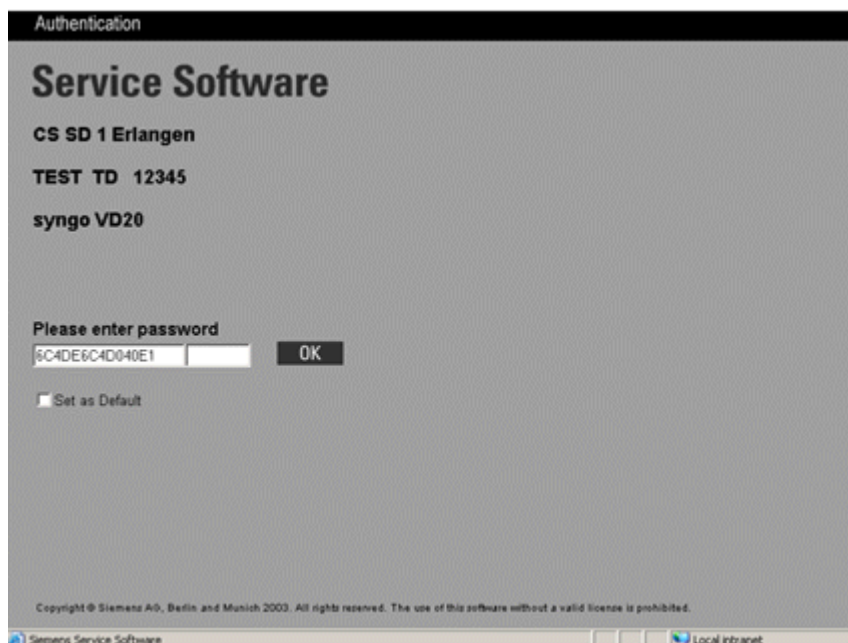


Fig. 221: Authentication Page

Press the **OK** button to submit the password entered.

If the submitted password is not valid, a notification window with a message to this effect will pop up. Click OK to confirm, and then re-enter the password.

If the **Set as Default** box is checked and a valid password has been entered, only the first 14 characters will be saved. The next time you log in, these 14 characters will be displayed, and only the last six characters will need to be entered.

The service software's **Home menu** will then appear.

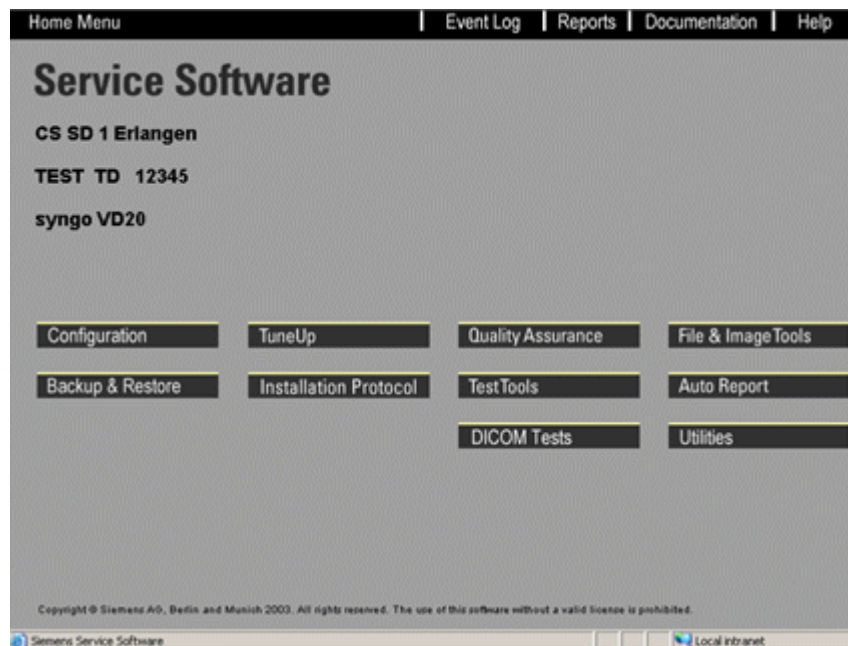


Fig. 222: Home Menu

13 Service User

HP-Z0000YPR-121005-02-975

n.a.

13.1 General

HP-Z0000YPR-121005-02-974

n.a.

13.1.1 Introduction

HP-Z0000YPR-121005-02-973

There are various service-related options in the **Options menu**. These service-related options can be started from the local console only.

13.1.2 Feature

HP-Z0000YPR-121005-02-972

This chapter explains the following service-related items:

- Local service
- Service user

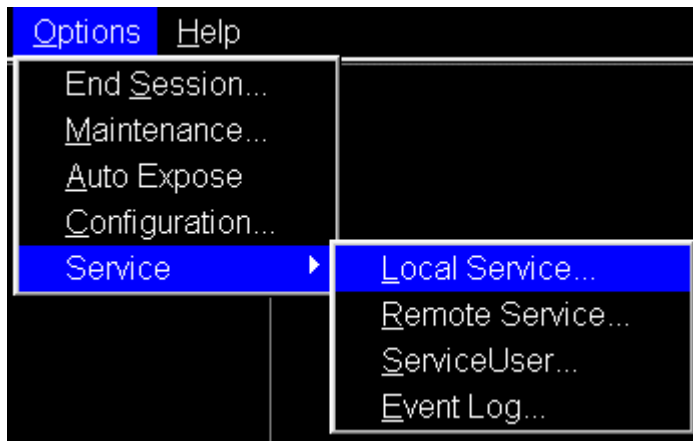


Fig. 223: Menu Options: Service - Local Service

As soon as an application is launched, an icon appears at the bottom right of the screen. You can minimize each application so that it appears as an icon by clicking the “_” on the title bar. Clicking an icon reopens the associated application.

13.2 Functions

HP-Z0000YPR-121005-02-971

n.a.

13.2.1 Service User

HP-Z0000YPR-121005-02-970

If **User Management** is enabled, a new menu entry will appear under **Options/Service** as a new user with the name LocalServiceUser is automatically created without a default password.

Any syngo user (hospital staff) with the rights to generate a service user password is able to generate a temporary password for a service job by opening this menu.

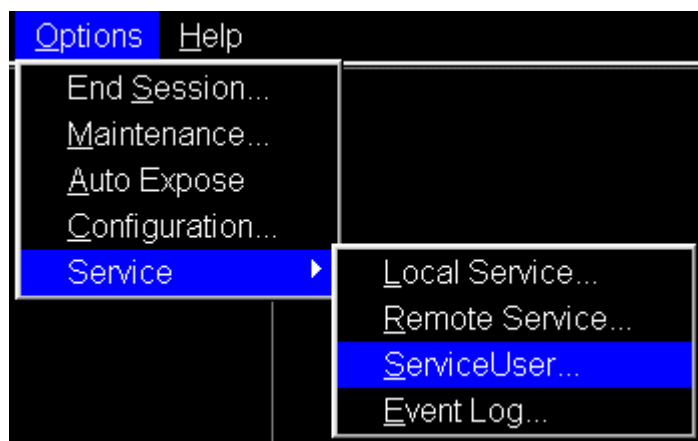


Fig. 224: Options Menu: Service - Service User

In the password generation window, a comment (e.g. "Software Update") has to be entered and the expiration period has to be specified (the default is 2 days).

Click **Apply** to generate the temporary password which will be valid for the period specified.

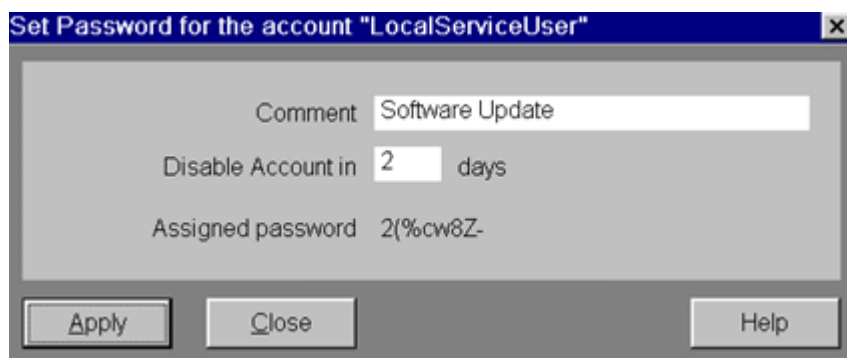


Fig. 225: Service User Grant

NOTE

Please write down the password.

After the current user logs off via **Option -> End Session** with **Log Off<user name>** or **Log In Different User**, the service technician is able to log in with the user name **Local-ServiceUser** and the newly generated password.



Fig. 226: End Session

14 Remote Service

HP-Z0000YPR-121005-02-037

n.a.

14.1 General

HP-Z0000YPR-121005-02-968

n.a.

14.1.1 Introduction

HP-Z0000YPR-121005-02-967

There are various service-related options in the **Options menu**. These service-related options can be started at the local console only.

14.1.2 Feature

HP-Z0000YPR-121005-02-966

This chapter explains the following service-related items:

- Remote service



Fig. 227:

As soon as an application is launched, an icon appears at the bottom right of the screen. Each application can be minimized so that it appears as an icon by clicking “_” on the title bar. Clicking an icon reopens the associated application.

14.2 Functions

HP-Z0000YPR-121005-02-041

System Type:Avanto
Espreo
TrioTim
SymphonyTim
ClinScan

n.a.

14.2.1 Remote Service

HP-Z0000YPR-121005-02-964

To access the system from a remote location, **Remote Access** must be granted via the **Remote Service Access Control** window. Select **Remote Service** from the **Options menu** to open the Remote Service Access control window. The Service Software can be accessed remotely via modem, ISDN router or LAN.

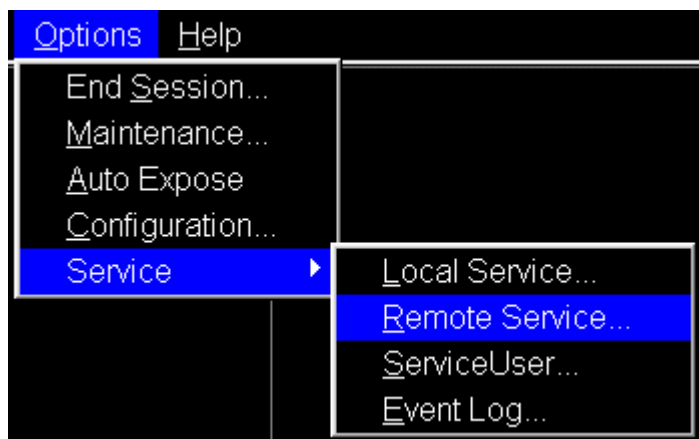


Fig. 228: Menu Options (Service)

When the **Remote Service** option is selected, the **Remote Service Access Control** window will appear.

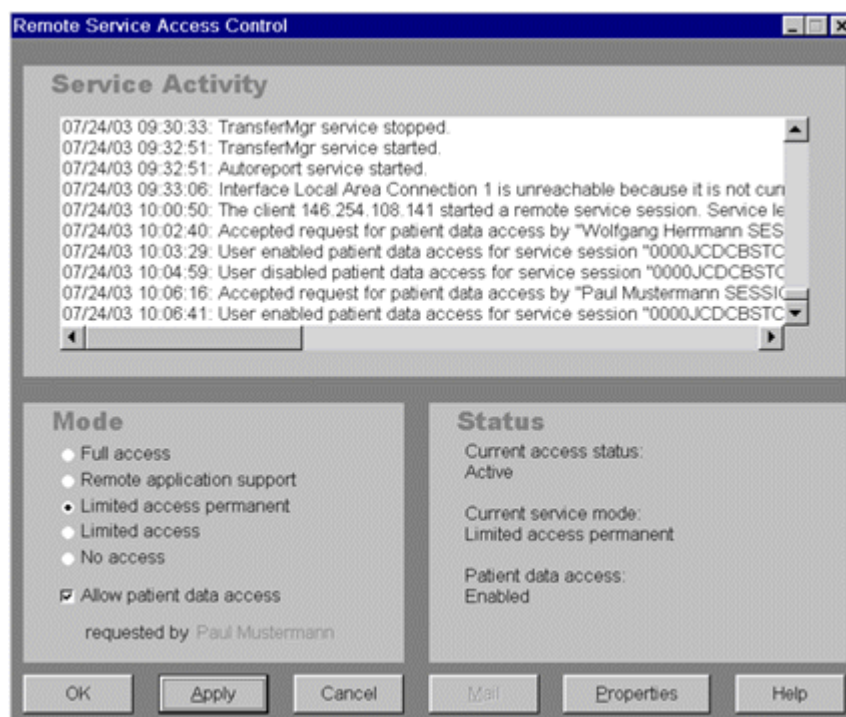


Fig. 229: Remote Service Access Control window

Select the appropriate access mode and click **Apply**. By clicking the **OK** button, the selection made will be applied, and the **Remote Service Access Control** window will be closed.

Modes

Remote access to the system via modem, ISDN router or LAN must first be authorized by selecting the **Remote Service** function from the **Options menu**.

Remote access can be set to:

- **Remote application support** (Not used on syngo MR systems)
- **Full Access**
- **Limited access permanent**
- **No access**
- **Allow patient access**

By default, remote access is set to **No access**.

Remote Application Support

This software version supports the customer option **Remote Assistance located in the Options menu**. If the Remote Assistance menu item is grayed out, the customer license is not available.

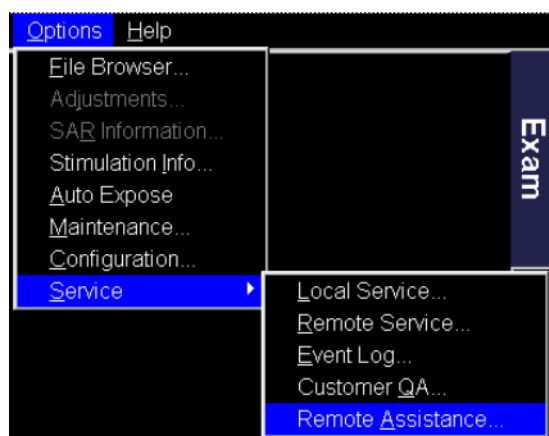


Fig. 230:

NOTE

syngo MR systems supports the customer option **Expert-i** for application support within the customer LAN (Local Area Network).

Full Access

If **Full Access** is granted, all Service Software functions will be available. The system is assigned exclusively to the remote session; i.e. the local user cannot perform any other operations, and the customer user interface is hidden (gray screen).

To enable **Full Access** mode, check the **Full Access** box in the Mode menu.

Once enabled, the modem port will be available for a configurable period of time. If remote login does not occur during this time period, the modem port will be automatically disabled (default approx. 2 hours).

The customer is able to change the access mode from **Full Access** to **Limited Access**, **Limited Access Permanent** or **No Access** (and vice versa) at any time. Changing from **Full Access** to another mode closes all functions within **Full Access** mode.

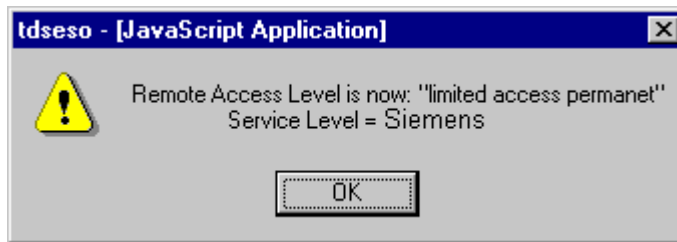


Fig. 231: Selecting Limited Access while RDIAG is active with Full Access

In this case, the remote user will be notified accordingly.

In **Full Access** mode, the Remote Service Access Control window cannot be minimized to an icon or closed with **Exit**.

NOTE

When using DICOM Test Tools, some syngo applications are required and must be executed by the user at the local site. Normally, this is not possible in Full Access mode. However, with DICOM Test Tools temporary access can be given to the user at the local site. This will be indicated to the local site user by red flashing characters in the STATUS field of the Remote Service Access Control window. If the application is minimized to an icon, users will see a flashing Remote Service Access Control icon.

Limited Access (Permanent)

If **Limited Access** or **Limited Access Permanent** mode is granted, only those Service Software functions that do not interfere with the customer's routine work are available. The system will not be locked in any way; i.e. the customer can continue operating the system.

During a remote diagnostics session, an icon informing the customer of the status of remote diagnostics access is displayed in the right-hand corner of the screen.

In **Limited Access** mode, the modem port will be available for a configurable period of time once it has been enabled. If remote login does not occur during this time, the modem port will be disabled automatically.

In **Limited Access Permanent** mode, the modem port is **permanently** available.

When logged onto the system, an icon indicates the status to the customer (see below).

The customer can change the access mode from **Limited Access** or **Limited Access Permanent** to **Full Access** (and vice versa) at any time. The customer has to confirm the end of the current service mode in a pop-up window.

The remote user will be informed of the change of service mode.

With **Full Access**, all Service Software functions are available.

No Access

Remote access is disabled immediately when **No Access** is selected in the **Remote Service Access Control** menu.

When in limited or full mode, remote access remains enabled for an additional **30 minutes** after the remote user has logged off or the connection has been interrupted due to a bad line.

If remote access is disabled while a batch transfer of files is in progress, disabling will be postponed until the transfer has been completed.

Due to a watchdog function provided by the modem, the connection is closed when the modem port has been inactive for 60 minutes. This prevents high telecommunications costs if an idle connection is left running.

Acknowledgement from the local operator

The local operator can see the **Patient Data Access** request from the remote user in the **Remote Service Access Control Platform**. Checking the **Allow Patient Access** box and clicking the **Apply** button switches the **Current Patient Mode** status information from **disabled** to **enabled**.

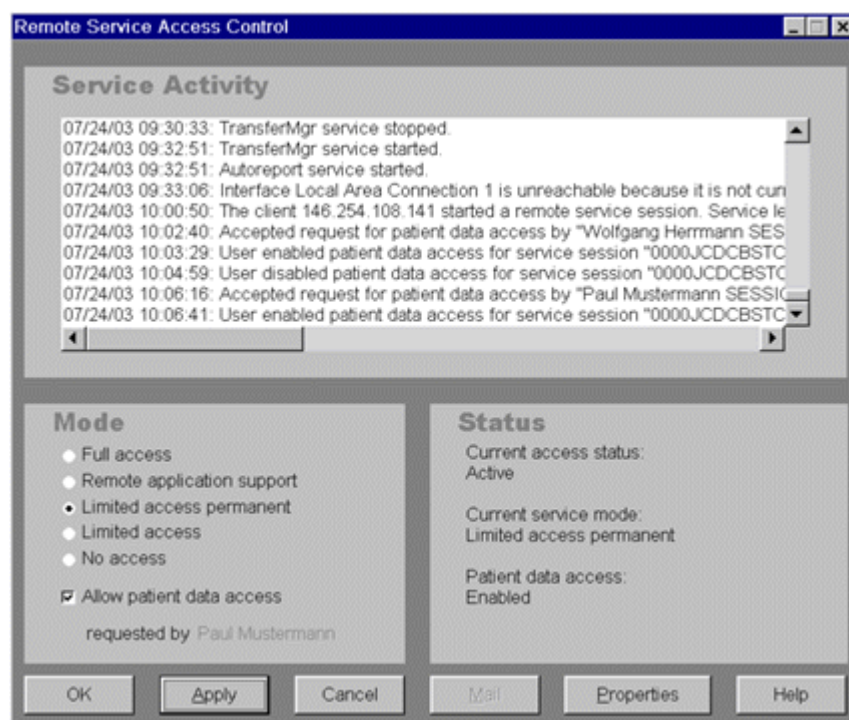


Fig. 232: Patient Data Access acknowledgement

The remote user will see the following pop-up window.

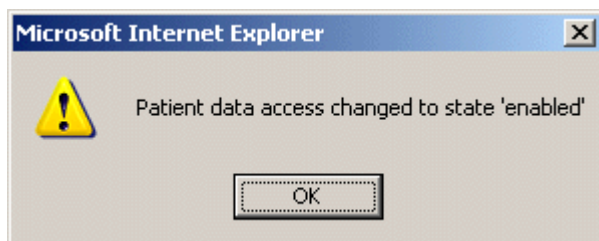


Fig. 233: Patient Data Access acknowledgement

Remote access icons

The first icon shows that remote access is **disabled**.



Fig. 234: Remote Access Icon Disabled

The second icon shows that remote access is **enabled**, although a remote service session has not been started.



Fig. 235: Remote Access Icon Enabled

The third icon shows that remote access is **enabled** and **active**. In other words, the service person is logged in.



Fig. 236: Remote Access Icon Active

Status

The status area shows the current access status and the current service mode.

Service Activity

The upper part of the Remote Service Access Control window shows the **Service Activity**; i.e. a record of all local and remote service activities.

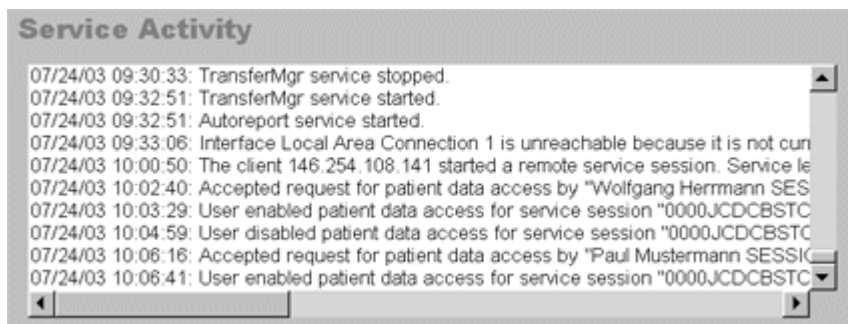


Fig. 237: Monitoring the Service Session

15 Save Log

HP-Z0000YPR-121005-02-043

System Type: Avanto
Espre
TrioTim
SymphonyTim

n.a.

15.1 General

HP-Z0000YPR-121005-02-962

n.a.

15.1.1 Introduction

HP-Z0000YPR-121005-02-961

The **SaveLog** function uses a special key combination to generate a single compressed file (ZIP) for transfer to the Service Center. This file contains a collection of several log and system files for evaluation.

15.1.2 Features

HP-Z0000YPR-121005-02-960

- The **SaveLog** file can be created on the local unit itself or via remote access by the service center.
- Further specific files, required or requested for evaluation, may be manually added to the compressed **SaveLog** file (ZIP).
- Screen captures of current monitor images are automatically added to the **SafeLog**, if they were generated using the screen capture key combination.

15.2 Functions

HP-Z0000YPR-121005-02-959

n.a.

15.2.1 SaveLog as a local user

HP-Z0000YPR-121005-02-958

Press the key combination **Ctrl+Alt+S** to generate the SaveLog file. A pop-up command window shows that the SaveLog script has started; this will close after approx. 2 to 3 minutes.



Fig. 238: Start generating the SaveLog File

The results are stored as a ZIP file in the directory `//syngo/service/extract` with the file name structure `SaveLog_%version%_%computername%_%serialnumber%_%date%_%time%.zip`

(e.g. `SaveLog_VC20B_TDSESO_12345_20021111_1216p.zip`).

This file can be transferred with **file & image** by selecting **Extract List** and **Transfer from System**.

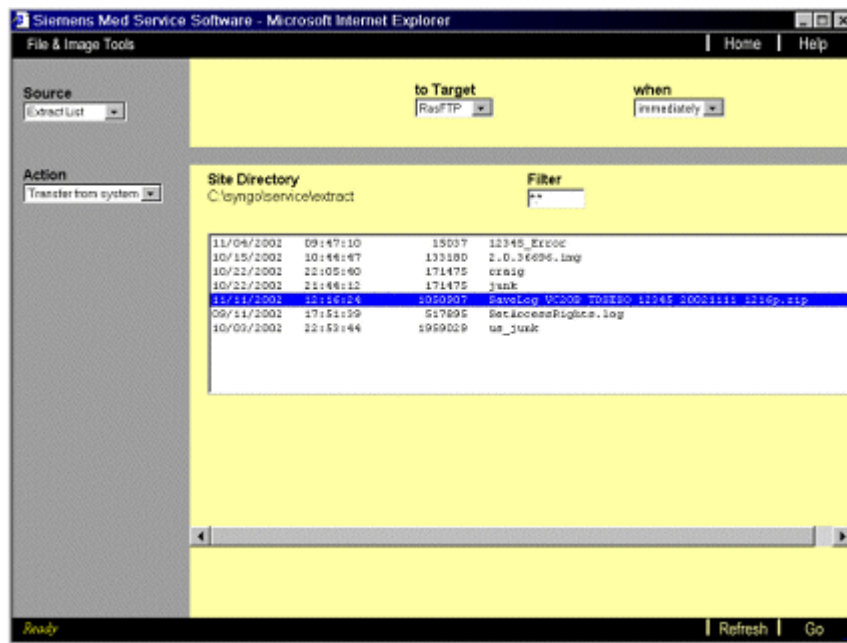


Fig. 239: Transfer with File & Image

Before the transfer takes place, the **target** needs to be selected. The target definition and configuration have to be set under **Configuration, FTP**.

During the **SaveLog** procedure, the stages of the script are recorded in a log file called syngoSaveLog.log. This log file can be viewed with **Reports** and Process Diagnostics.

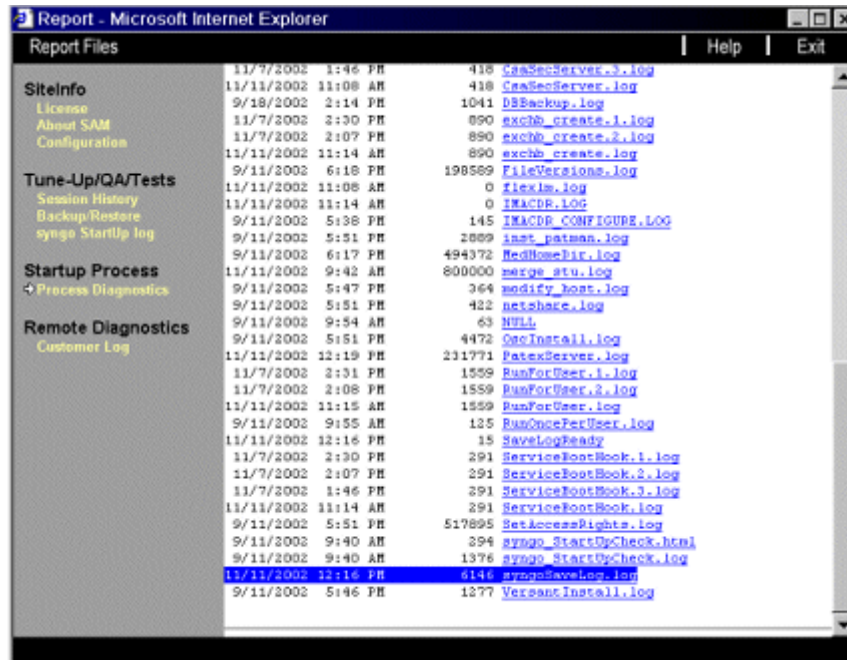
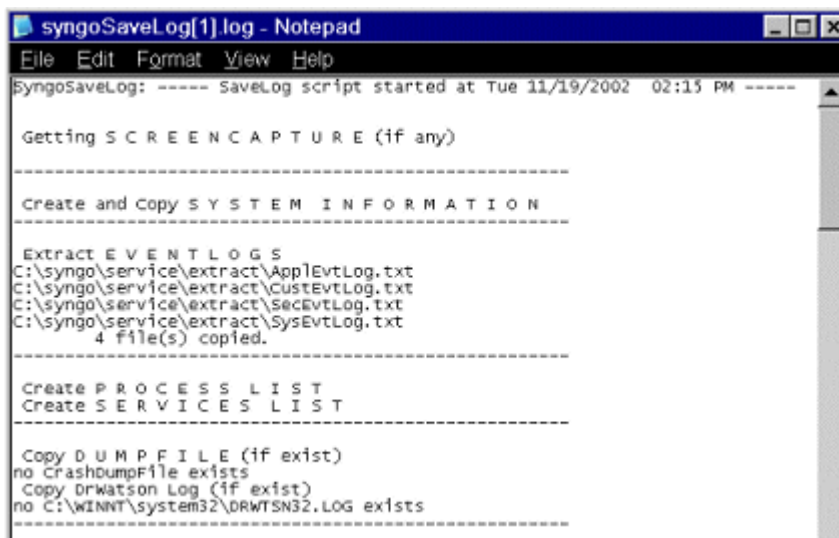


Fig. 240: Report --> Process Diagnostics

When you select the **syngoSaveLog.log**, an additional Notepad window opens to display the content of the syngoSaveLog.log.



```
syngoSaveLog[1].log - Notepad
File Edit Format View Help
SyngoSaveLog: ----- SaveLog script started at Tue 11/19/2002 02:15 PM -----

Getting SCREENCAPTURE (if any)

-----
Create and Copy SYSTEM INFORMATION
-----

Extract EVENT LOGS
C:\syngo\service\extract\ApplEvtLog.txt
C:\syngo\service\extract\CustEvtLog.txt
C:\syngo\service\extract\SecEvtLog.txt
C:\syngo\service\extract\SysEvtLog.txt
4 file(s) copied.

-----
Create PROCESS LIST
Create SERVICES LIST
-----

Copy DUMPFIL (if exist)
no CrashdumpFile exists
Copy DrWatson Log (if exist)
no C:\WINNT\system32\DRWTSN32.LOG exists
-----
```

Fig. 241: Contents of syngoSaveLog.log

15.2.2 SaveLog as a remote user

HP-Z0000YPR-121005-02-957

To create a SaveLog as a remote user, “**Full Access**” mode has to be enabled at the local system via the Application menu bar under **Option --> Service --> Remote Service**.

CAUTION

Please note: The system cannot be used in Full Access mode.

⇒ The monitor screen is blocked.

The remote user launches the Service Software and selects **Utilities** from the Service menu. On the Utility page, (s)he selects “**Escape to OS**” from the selection box, followed by “**Create ServiceSaveLog**” from the command selection box.

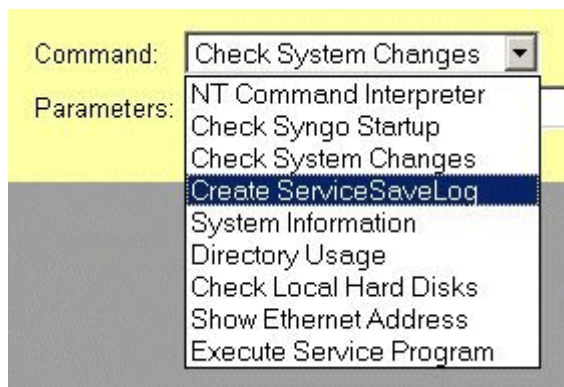


Fig. 242: Escape to OS - Create ServiceSaveLog

Clicking **Go** launches the SaveLog process. Several windows are then displayed for the safety check.



Fig. 243: Patient Data Access is required

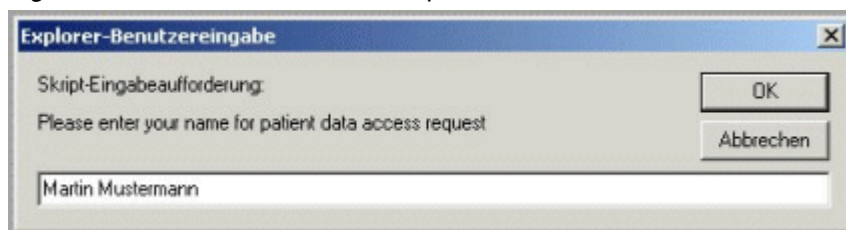


Fig. 244: For Patient Data Access your name is required

The remote user has to enter his or her name in this window.

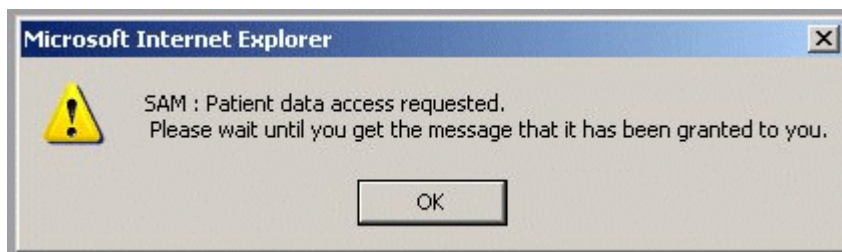


Fig. 245: SAM requests Patient Data Access at the remote system



Fig. 246: Patient Data Access changed to enabled

When the last information window has closed, clicking **Go** again starts the process of generating the SaveLog on the local system. This will take approx. 2 to 3 minutes and ends when the content of **syngoSaveLog.log** is displayed.

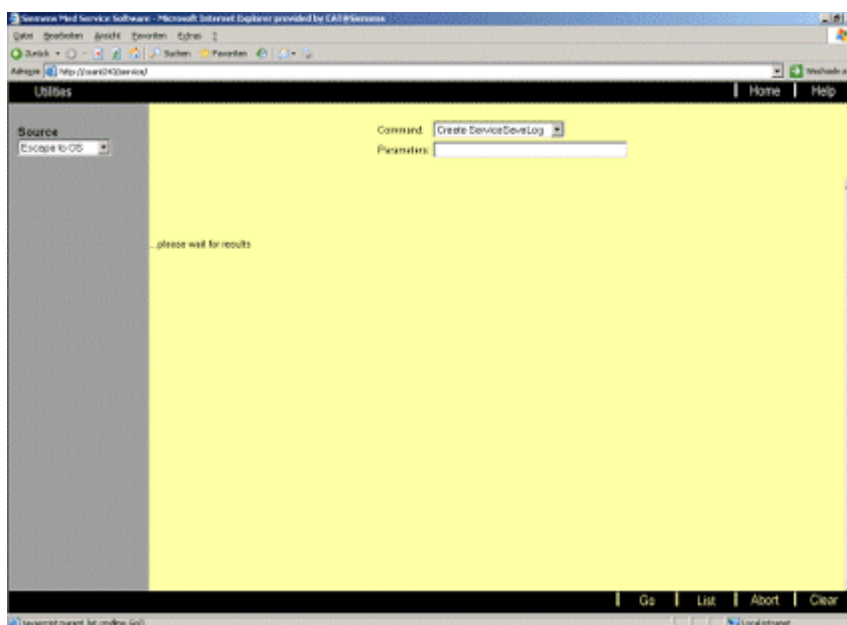


Fig. 247: Remote User Screen

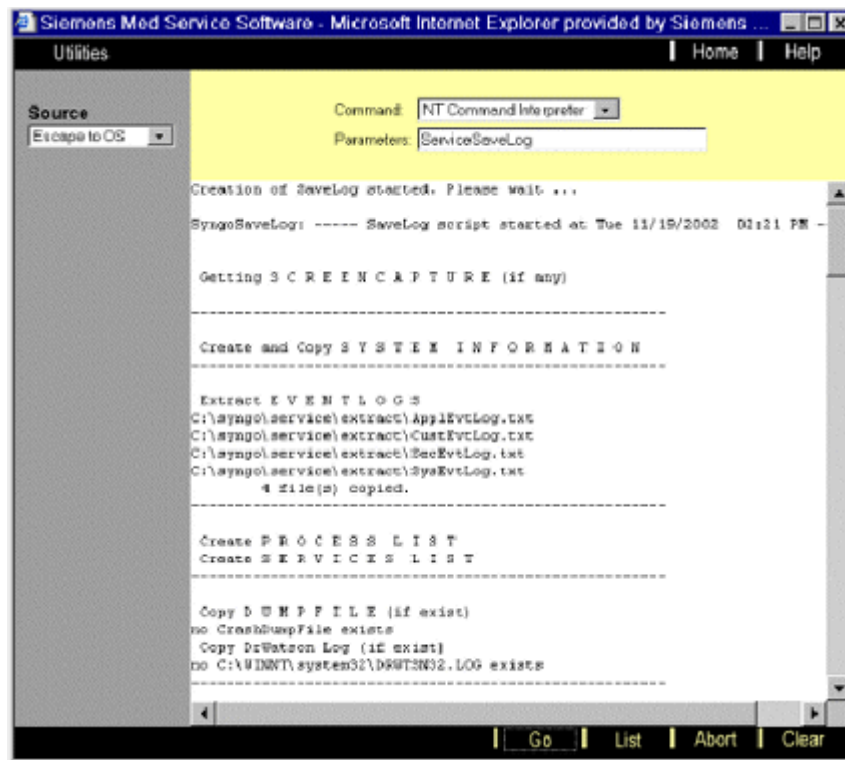


Fig. 248: syngSaveLog.log

Remote "**Full Access**" can be disabled again to restore normal working conditions. The SaveLog can then be transferred with **file & image** by selecting **Extract List** and **Transfer from System**.

16 Status Area

HP-Z0000YPR-121005-02-956

n.a.

16.1 General

HP-Z0000YPR-121005-02-955

n.a.

16.1.1 Introduction

HP-Z0000YPR-121005-02-051

This chapter describes the functionality of the icons at the customer's status area.

16.1.2 Features

HP-Z0000YPR-121005-02-953

The “Resource monitor“

A tool to provide information on the memory / disk capacity.

17 Software Distribution

17.1 General

17.1.1 Introduction

HP-Z0000YPR-300407-02-999

Software Distribution is one of the Siemens **Enhanced Productivity Services** that allows remote system software updates and any other kind of remote software distribution and installation.

Software Distribution is provided by Siemens Remote Service and requires a broadband connection to the SRS, as well as activated **System Management Package (MNP)** on the system.

17.1.2 Features

HP-Z0000YPR-300407-02-998

Software Distribution differs slightly in how the package transfer is initiated. The software package can be automatically **pushed** to each system, triggered by the Software Distribution Server, or it can be **pulled** by the system from the Software Distribution Server, triggered by the **software catalog** within the syngo service software.

The package can not only be automatically transferred to the system, it can also be silently installed on the system without any notification, depending on the modality type. This is allowed only for certain kinds of software packages (e.g. virus pattern updates) and requires a one-time acceptance by the user (UI checkbox), which, of course, can be modified by the user at any time.

The software catalog can be accessed during a remote session. However, “Full Access” is required for downloads and installations.

18 TuneUp

18.1 Procedures

18.1.1 General

HP-Z000100Q-031105-02-032

System Type:Avanto
 Espree
 TrioTim
 SymphonyTim
 ClinScan

TuneUp Function

TuneUp comprises a series of procedures (workflow) for tuning the MR system.

Each individual procedure automatically saves data in the system parameter files, log files and database files.

Normal Mode

All procedures with **To Do** status are selected by default and will be performed. A single procedure can be selected and performed.

Interaction

If user interaction is required (e.g. phantom change), the user will be informed in a message pop-up window.

Coil TuneUp

The TuneUp procedures “**Receive Path Calibration**” replace the former “**Coil TuneUp**”.

Status Field

The **Status field** for each **TuneUp** procedure indicates the current condition of the procedure based on the following criteria:

To Do - indicates that the procedure has not yet been performed or needs to be performed because of a change in the status of another dependent procedure.

Running - indicates that the procedure is presently running.

Not OK - indicates that the procedure ran, but the results are not within specification.

Error - indicates that an error occurred during the last operation. If this procedure is selected and **Go** is performed, the status will immediately change to **To Do**.

Done - indicates that the procedure has been completed successfully.

TuneUp Verification

All **TuneUp** workflow procedures have corresponding verification routines available in the **Quality Assurance** workflow menu.

Dependencies TuneUp/ QA

If a **TuneUp** procedure has changed its status (except for **Expert** status) and is successfully finished, the corresponding procedure(s) in the Quality Assurance workflow menu is/are changed to **To Do**.

If one procedure has a significant impact on another procedure, but only one has been selected, performing the selected procedure will change the status of the other procedure.

TuneUp procedures may affect different QA procedures, e.g. after Phantom Shim, the QA procedures Coil Check, Shim Check, Gradient Sensitivity Check, Fat Saturation have to be performed. If the TuneUp procedure is successfully completed, the corresponding QA procedures will be changed to status **To Do**.

In the same way, TuneUp procedures may depend on prior successfully performed TuneUp procedures. For example, the procedures “Tuning Calibration” and “BC Tuning” are prerequisites for “RF Characteristic”.

Dependencies TuneUp/ TuneUp

In the same way, TuneUp procedures may depend on prior successfully performed TuneUp procedures. For example, the prerequisites for “RF Characteristic” are the procedures “Tuning Calibration” and “BC Tuning.”

Dynamic Data

The TuneUp procedure generates site-specific data. After completing the TuneUp, these data have be saved to a CD. They may be required for later software updates or software reinstallations.

Safety

Observe the safety regulations in the “General Safety Notes - MR-Specific Safety Information” document.

Tuning Calibration

```
----- HP-Z000100Q-041105-02-001 -----
System Type: . . . . .Avanto
              Espree
              SymphonyTim
```

General

Tuning Calibration determines the forward and reflection characteristics of the RF transmission path under specific termination conditions (open, short circuit and 50 W). The calibration measurement includes the following analog/digital parts:

- RCCS
- BCCS
- TALES
- TX module
- RX module (MR SIG1)
- Two terminators (one (1) 50 Ohm and one (1) short circuit)
- RF cable BNC (2m) part no. 87 33 479.

If the specifications of one of these components are not within tolerance, the process will terminate with an error.

The procedure runs in interactive mode. The user has to perform the correct **TALES** set-up according to the on-screen instructions. The TALES is located at the left side of the magnet.

NOTE	Normally, tuning calibration is handled in the factory. If so, this procedure is marked as "Done" in the user interface. Otherwise, you have perform the procedure yourself.
-------------	---

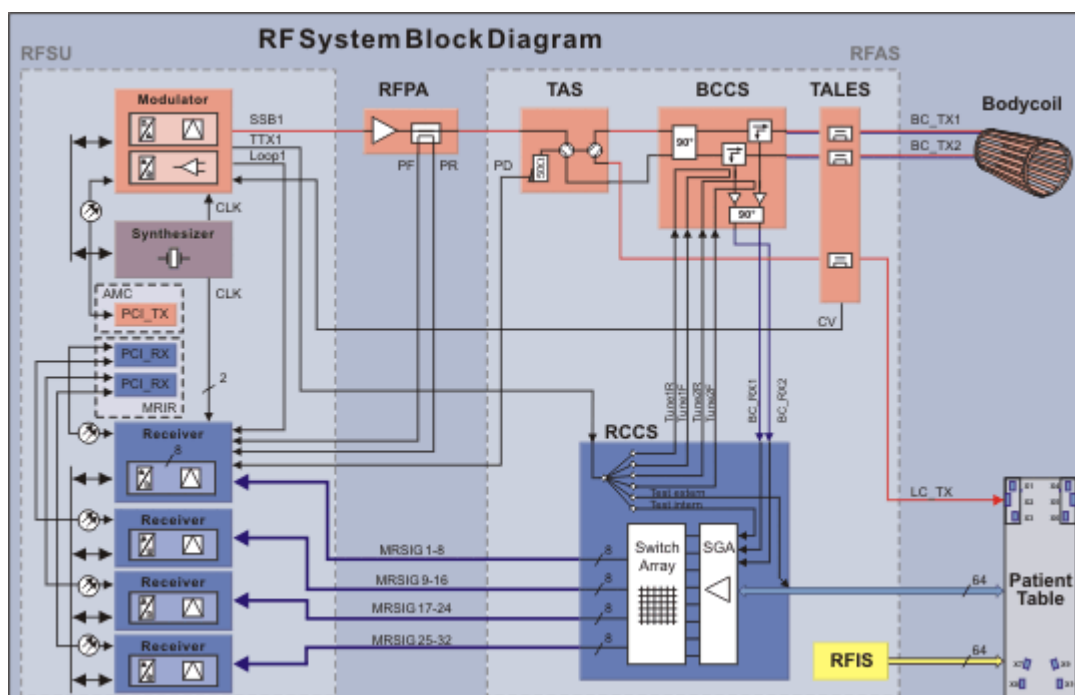


Fig. 249:

Functional Description

The **Tuning Calibration procedure** calibrates the TX path up to the output of the TALE (TALES calibration). The TALE output is terminated using an open, a short circuit and a 50 W terminator. The forward and reflected RF signal is measured under these different conditions by sending a rectangular RF pulse (tuncal sequence).

For each termination condition, the MR frequency is varied through the complete operating range of the particular system. For systems Avanto, Espree and Symphony Tim (63 MHz), the frequency range is 63.45 to 63.75 MHz with a frequency step of 10 kHz or 31 separate measurements.

For systems using a hybrid splitter (CP Body Coil), both the 0° and 90° TALE output is measured. Due to coupling between the two systems, coupling is measured and compensated as well. This is performed by measuring the 90° TX path when sending at 0° and vice versa.

An additional measurement is made by connecting TALE outputs X3 and X4 with cable of known transmission: two raw data sets will be created by sending one time on channel 1 and the second time on channel 2.

Measurement

Location of TALES Connectors

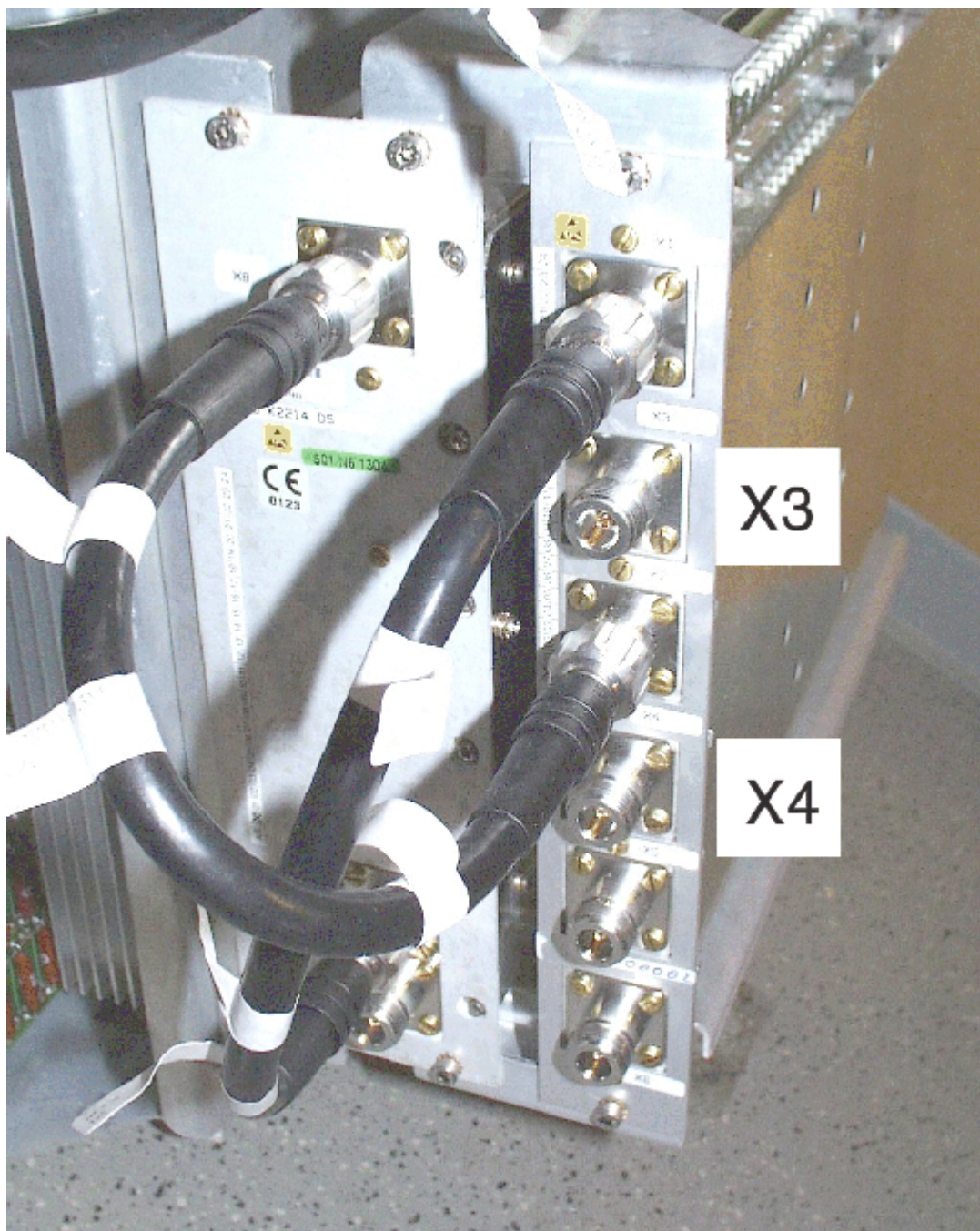


Fig. 250:

Service Tool Kit



Fig. 251:



Fig. 252: BNC cable (8733479) with connected adapters

Troubleshooting

Common Problems

- External cable connections are not OK:
 - Check Transmit path: Modulator - ACC top - Filterplate - RCCS
 - Check Receive path: Receiver/MR Sig 1-32 - ACC top- Filterplate RCCS
 - Check RF cables
 - Check correct cable connection at RCCS:
X61(TU1F), X62(TU1R); X63(TU2F), X64(TU2R)
- Bad 50 Ohm terminators; check with DVM
- Wrong termination during calibration: 50 Ohm - short - open

BC Tuning

HP-Z000100Q-181108-02-007

System Type: SymphonyTim

General

The procedure is used only for systems with a circular polarized body coil.

Physical background

The procedure is used to measure and adjust the decoupling of the circular polarized body coil. The transmission through the body coil or the decoupling of the two body coil systems describes a correlation between the body coil systems. The transmission is measured by sending a signal to the 0° system and measuring the signal in the 90° system. The transmission T is given by:

$$T = 20 \times \log\left(\frac{U_1}{U_2}\right)$$

Fig. 253:

The measurement is performed with fixed Cs/Cp values in a broad frequency range.

The resonance is given by the frequency with the lowest BC reflection. The decoupling is given by the frequency at which the transmission has its maximum value. The maximum transmission should be lower than -16 dB which means that the portion $P = P_0 \cdot (10^{-(16\text{dB}/10\text{dB})})$ or 2.5% of the input power is lost for imaging (P_0 is the input power for the body coil).

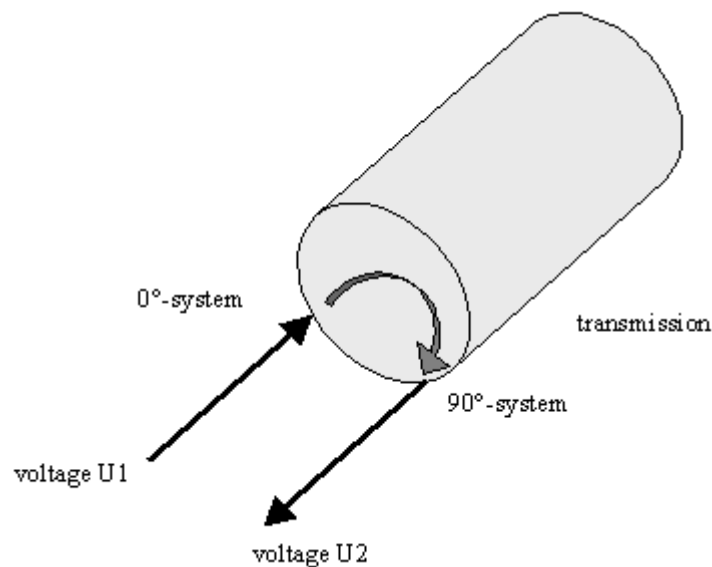


Fig. 254:

Measurement

Prerequisites

Since BC Tuning uses the H and S matrices for calculations, the Tuning Calibration TuneUp procedure has to be completed successfully before starting BC Tuning.

CAUTION
--

If the "BC Tuning" step is outside of specifications, you must adjust the BC manually.

During the manual adjustment of BC Tuning you have to work next to the non-insulated connections of the gradient cables at the gradient coil.

Hazardous voltages may be present at the connections of the gradient cables.

For this reason, you must switch off the gradient amplifier:

- ⇒ Abort BC Tuning.
- ⇒ Start TestTools / Supervision mask and click the radio button: GPA Supervision "OFF"; leave the Supervision mask with "Save". Switch off circuit breaker F6/ GSSU in the ACC.
- ⇒ Wait at least 20 minutes before you start the manual BC Tuning adjustments.
- ⇒ Start the TuneUp step "BC Tuning" separately. When the results are within specifications, switch on the GPA by F6/ GSSU in the ACC. In TestTools/ Supervision mask click the radio button: GPA Supervision "OFF"; leave the Supervision mask with "Save".
- ⇒ Perform "Reboot scanner" and continue with the standard TuneUp workflow.

To gain access to the trimmer capacitors in the back of the body coil, you must remove the back funnel of the magnet cover.

The gradient amplifier must be switched off for at least 20 minutes before performing the BC Tuning procedure, i.e. adjustments to the trimmer capacitors at the body coil. In the meantime you should remove the funnel.

To turn the trimmer capacitors, you need the special anti-magnetic adjustment tool (i.e. long screwdriver), part no. 73 66 771.

Adjustments

The procedure will guide you through iterative fine adjustments. Between the adjustment steps, the resonance and decoupling of both systems (0 and 90 degree) can be varied manually using the three trim capacitors at the coil. The program stops when the resonance frequency and decoupling are within specifications.

NOTE

If the decoupling (transmission) specifications cannot be achieved, it is necessary to mechanically readjust the radial and axial position of the BC.

Reflection (Minimum) Step 1

	Frequency				Factor			
	Value	Low Spec	High Spec	Unit	Value	Low Spec	High Spec	Unit
0°	63.69	63.45	63.75	MHz	0.1	0.0	0.3	
90°	63.88	63.45	63.75	MHz	0.1	0.0	0.3	
Delta	185.0		100.0	kHz				

Transmission (Maximum) Step 1

	Value	Low Spec	High Spec	Unit
Frequency	63.57	63.30	63.90	MHz
Decoupling	-23.99	-80.00	-16.00	dB

Fig. 255:

The trim capacitors T1 and T2 influence the resonance frequencies of system 1 and 2 respectively. The capacitor TD effects the decoupling between the two systems.

You only have to adjust the decoupling if the transmission value is not in spec.

The trim capacitors have 60 turns. Their preset is in the center position. This means you can make 30 turns clockwise or 30 turns counterclockwise.

NOTICE

Be careful when reaching the end position of the trimmers to prevent damage to them.

⇒ Do not use any force.

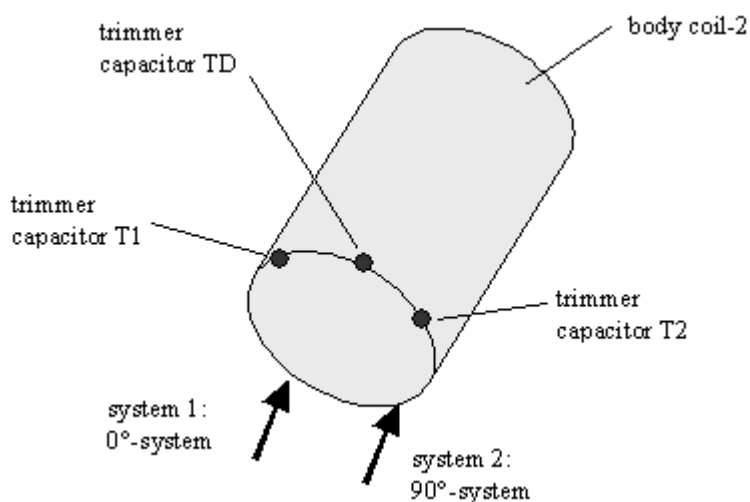


Fig. 256:

Trimmer capacitor	Main influence on	Clockwise (cw)	Counterclockwise (ccw)
T2	Resonance frequency system 1 (0 degree)	Resonance frequency 1, smaller	Resonance frequency 1, higher
T1	Resonance frequency system 2 (90 degrees)	Resonance frequency 2, smaller	Resonance frequency 2, higher
TD	Decoupling of systems 1 and 2	Capacity larger	Capacity lower

If the values are slightly out of spec, there's no need to worry; the system will still work.

Step 1	Reflection adjustment of delta frequency	
	delta < 100 kHz ?	
	Yes	No
	No action necessary	Frequency 0 degree > frequency 90 degrees
		Yes
		No
		Turn T2 2x cw Turn T1 2x ccw
	Go to Step 1	

Step 2	Mechanical adjustment of decoupling (transmission)	
	Transmission 0... -12 dB	
	Yes	No
	Adjust mechanical position of the BC in the gradient coil.	Go to Step 3
	Go to Step 2	

Step 3	Electrical adjustment of decoupling (transmission)		
	Transmission T = -12 dB .. -16 dB		
	Yes		No
Step 3.1	Turn TD 3x cw, press start button on panel		No action necessary
	T lower		
	Yes	No	
Step 3.1.1	Turn TD 3x cw, press start button on panel	Turn TD 3x ccw, press start button on panel	
	T lower -> go to 3.1.1	T lower -> go to 3.1.1	
	No -> undo last step	No -> undo last step	
Step 3.2	Turn T1, T2 2x cw, press start button on panel		
	T lower		
	Yes	No	
Step 3.2.1	Turn T1, T2 2x cw, press start button on panel	Turn T1, T2 2x ccw, press start button on panel	
	T lower -> go to 3.2.1	T lower -> go to 3.2.1	
	No -> undo last step	No -> undo last step	
If necessary repeat Step 3.1			

RF Characteristic

HP-Z000100Q-041105-02-009

System Type:Avanto
SymphonyTim

General

RF Characteristic is a function for measuring the amplitude and phase characteristic of the RF amplifier so that the distortion of the RF envelopes caused by the non-linearity of the amplifier can be corrected.

Physical Background

The exact calibration of the RF transmitting system is important for obtaining optimum image quality. The RF output not only defines the amount of excitation (i.e. flip angle), it is also important for defining the RF pulse profile and as a result the slice profile in the slice-selection procedure.

Distortions of the RF envelopes caused by the non-linearity of the amplifier can produce distortions of the slice profile and slice thickness. The corresponding inverse function (of the amplitude and phase characteristic of the RF amplifier) can be used for predestination in order to achieve a linear output characteristic and to avoid distortions of the slice profile.

The **RF Characteristic** function allows you to calibrate the RF output. This is performed by measuring the output of the RF transmission in terms of the amplitude and phase based on a specific input to the RF amplifier. The measurement includes the entire operating range of the RF amplifier, defining the input/output characteristic of the amplifier. Based on this characteristic, the RF input (DAC signal) pulse can be corrected so that the RF output at the transmitting coil has the desired RF amplitude and profile.

For Avanto and Symphony, A Tim systems the possibly very high power (reflected power from the no-tune body coil), which is reflected to the TAS 50 Ohm terminator, has to be measured. This is necessary to verify the correct function of the TALES and DICO.

Measurement

The RF characteristic is measured as follows:

1. **Calibration Voltage Search:** Starting with 100 Volts output (this is just a nominal value). The TALES readings are used to increase the output until a 566 Volts output is measured.
2. **RF Calibration:** The DICO has to be calibrated since it is used for the final measurement of the characteristic. The DICO value is measured based on the resulting input voltage from step 1. The DICO is calibrated using the known relationship between TALES and DICO. $\text{DICO factor} = \text{TALES voltage} / \text{PF ADC signal (DICO)}$
3. **RF Max Power Search:** By carefully increasing the output voltage, the max. power of 12.2 kwh is searched.
4. **RF Characteristic measurement:** For the following measurement, the RF is sent to the 50 Ohm dummy in the filter plate. 256 pulses with different amplitudes are applied. The first pulse starts with the maximum voltage, decreasing by 0.25 dB for each subsequent pulse. This ensures that the complete characteristic is measured.
5. **The RF Characteristic** is saved in the rfchara.dat file.
6. The next step is the **verify measurement**. This includes the calculation of amplitude and phase deviation and checking if the results are within specifications. The results and graphics can be viewed in the service software under Tune-Up, Report, Tune-Up Results.
7. Additionally for Avanto systems the **TAS dummy conversion factor** which characterizes the attenuation of the path DICO-TALES-TAS, is determined and saved.

TAS dummy conversion factor = Voltage Dummy / Amplitude Dummy

with:

Voltage Dummy = $10^{((\text{TAS-TALES_TAS-Damping}/20\text{dB}) \times \text{Amplitude DICO} \times \text{DICO factor})}$

Amplitude DICO = Voltage, obtained from DICO forward during the TAS dummy calibration measurement

Amplitude Dummy = ADC value, obtained from PD signal during the TAS dummy calibration measurement.

(For additional information, go to the Functional Description/Tune-up).

NOTE

To verify this factor after replacing components in the RF-path, such as BCCS_63, it is necessary to run the QA expert mode function "RF Path Attenuation Check."

TAS-Dummy Calibration result			
	Value	Low Spec	High Spec
TAS-Dummy conversion factor	0.0953	0.0600	0.1200
TAS-Dummy conversion factor error	0.1		
Saving conversion factor for TAS-Dummy			

Fig. 257:

RF Characteristic LC

 HP-Z000100Q-071105-02-019

System Type:Avanto
SymphonyTim

General

RF Characteristic is a function for measuring the amplitude and phase characteristics of the DORAK2204 15kW and CORA 2kW BB-RF amplifier so that the distortion of the RF envelopes caused by the non-linearity of the amplifier can be corrected.

Physical Background

The exact calibration of the RF transmitting system is important for obtaining optimum image quality. The RF output not only defines the amount of excitation (i.e. flip angle), it is also important for defining the RF pulse profile and therefore the slice profile in the slice-selection procedure.

Distortions of the RF envelopes caused by the non-linearity of the amplifier can produce distortions in the slice profile and slice thickness. The corresponding inverse function (of the amplitude and phase characteristics of the RF amplifier) can be used for predistortion in order to achieve a linear output characteristic and to avoid distortions of the slice profile.

The **RF Characteristic** function allows to calibrate the RF output. This is handled by measuring the output of the RF transmission in terms of the amplitude and phase based on a specific input to the RF amplifier. The measurement includes the entire operating range of the RF amplifier, thereby defining the input/output characteristic of the amplifier. Based on this characteristic, the RF input (DAC signal) pulse can be corrected so that the RF output at the transmitting coil has the desired RF amplitude and profile.

System-dependent constellation

The RF-characteristic is measured using the DORA 15kW and CORA BB RF-amplifiers and all available nuclei (H and all X-nuclei) in the local coil (LC) path. The service technician must interact as follows:

The dummy load in the TAS has to be connected to the LC-output of the TALES:

- Connect the service plug (mode switched to TX) to the coil connector #1 at the patient table.
- Disconnect the LC-output cable at TALES X6.
- Disconnect the cable (W1230) at BCCS X7 (connection to TAS). Leave X7 open.
- Connect this cable (W1230) to TALES X6.

Measurement

The RF characteristic is measured as follows:

1. **Calibration Voltage Search:** Starting with 100 Volts output (this is just a nominal value). The TALES readings are used to increase the output until 250 Volts output are measured.
2. **RF Calibration:** The DICO has to be calibrated since it is used for the final measurement of the characteristic. The DICO value is measured based on the resulting input voltage from step 1. The DICO is calibrated using the known relationship between TALES and DICO.
3. **RF Max Power Search:** By carefully increasing the output voltage, the max. power (depending on the nucleus) is being looked for.
4. **RF Characteristic measurement:** For the following measurement, the RF is sent to the 50 Ohm dummy in the filter plate. 256 pulses with different amplitudes are applied. The first pulse starts with the maximum voltage, decreasing it by 0.25 dB for each subsequent pulse. This ensures that the complete characteristic is measured.
5. **The RF Characteristic** is saved in the rfchara.dat file.
6. As a final step, **verify measurement**. This includes the calculation of amplitude and phase deviation. Also check to ensure that the results are within specification. The results and graphics can be viewed in the service software under Tune-Up, Report, Tune-Up Results.

Regulator Adjust

HP-Z000100Q-071105-02-001

System Type:Avanto
 Espre
 TrioTim
 SymphonyTim

General

Regulator Adjust is a procedure to automatically adjust the setting of the gradient PID regulators located in the gradient cabinet.

Physical Background

The Regulator Adjust routine verifies the correct setting of the gradient regulator by looking at several aspects of the actual current output and comparing them to the expected ideal pulse output. The sequence uses a gradient table to generate the gradient pulses so that the selected gradient (depending on the selected orientation) is stepped from the maximum negative to the maximum positive gradient amplitude pulse in linear steps. Each gradient in the table is switched on using a ramp up and ramp down time that conforms to the minimum defined ramp up time (to maximum amplitude) for the system type.

Based on the maximum gradient amplitude pulse, the program calculates the characteristic overshoot that is detected just after the ramp up time. The amplitude of this overshoot is determined as a percentage relative to the nominal gradient amplitude. In addition, the time it takes for this exponential-like overshoot to return to a specified value is calculated. Also, a smoothness calculation is performed to determine the degree of smoothness of the top portion of the maximum amplitude pulse.

Finally, each line in the gradient table is integrated and the resulting mean value for each line is fitted to a linear function. The difference between the data and the linear fit and the maximum deviation are returned to calculate the new regulator settings.

Table Adjustment

```
----- HP-Z000100Q-071105-02-003 -----
System Type: . . . . . Avanto
                  Espree
                  TrioTim
                  SymphonyTim
```

General

The preparation measurement of ECC is used to determine the magnet center.

Measurement

The goal of the **Table Adjustment** is to check for correct positioning of the phantom in the magnet center and for sufficient field homogeneity. This is performed with a double echo sequence. The first echo has no readout gradient. The length of the MR signal is a rough check of the homogeneity. The second echo uses a readout gradient and calculates the phantom position in the readout direction.

The table position will be corrected automatically. The new light marker distance is saved.

Phantom Shim

HP-Z000100Q-071105-02-005

System Type:Avanto
Espre
TrioTim
SymphonyTim

General

This procedure performs a 3D measurement of the phantom and calculates the frequency differences for neighboring volume elements. Based on this information, the spherical shim terms are calculated. The linear terms are compensated using the gradient offsets. If the active shim is installed, the shim channels are used for compensating the second order terms.

In the **Normal** Mode, several iterations are performed for the phantom shim.

Before the 3D measurement is performed, two measurements will check the polarity of all shim channels, if available.

Measurement

The shim check consists of two measurements with a delta applied to the shim currents. The shim currents are calculated based on the measured field changes and compared with the nominal ones. Wrong shim polarity or switched shim channels will produce an error.

The shim optimization is started using the current temporary settings. After the measurement, optimized shim currents are calculated and stored in temporary and permanent parameters. If the changes are outside specifications, another two iterations are added.

Cross Term Compensation

HP-Z000100Q-071105-02-007

System Type: Avanto
Espreo
TrioTim
SymphonyTim

General

The **Cross-Term Compensation** procedure measures eddy currents generated by so-called cross terms. The first order term will be compensated by the predistortion of the gradient waveform.

The Cross-Term Compensation measurement requires the normal body phantom.

Physical Background

The operation of a single gradient, e.g. Z gradient, produces not only eddy currents coupled to the Z gradient, but also small amounts of eddy currents which can be decomposed into X and Y gradient contributions (cross terms), up to 0.2% of the main Z gradient amplitude. These eddy currents decay exponentially in time, in general with only one time constant. The procedure measures the decay of eddy currents caused by the cross terms. Subsequently, it calculates the amplitude and time constant of the eddy current contributions and the parameters of the inverse filter function which compensate the effect. The result is a much cleaner main gradient with very small residual cross terms. This helps improve the final image quality.

Eddy Current Compensation

 HP-Z000100Q-071105-02-009

System Type:Avanto
 Espre
 TrioTim
 SymphonyTim

General

Eddy Current Compensation (ECC) is a procedure that measures the eddy currents of all three gradients. The zero order term will be compensated by changing the synthesizer frequency, the first order term will be compensated by predistortion of the gradient waveforms.

The **ECC** measurement requires the smaller spherical head phantom rather than the normal body phantom. The user is reminded to use the small spherical phantom in a confirmation message.

Physical Background

When a gradient is switched off, the gradient field should immediately drop to zero. However, this is not the case due to the eddy currents. By measuring the time dependency of the gradient after the gradient pulse is switched off, the amplitudes and time constants of the different eddy current contributions can be calculated. Afterward, the parameters of the inverse filter function which compensates this effect can be determined. When the eddy current effect is fully compensated, the gradient fields should be zero after the gradient is switched off.

Measurement

The objective of the **Preparation Measurement** is to check for correct positioning of the phantom in the magnet center and for sufficient field homogeneity. This is performed via a double echo sequence.

The first echo has no readout gradient. The length of the MR signal is a rough check of the homogeneity.

The second echo uses a readout gradient and calculates the phantom position in the readout direction.

Coil Power Losses

HP-Z000100Q-071105-02-011

System Type: Avanto
Espreo
TrioTim
SymphonyTim

General

During **Coil Power Losses** measurements, the power dissipated in the connection lines and in the body coil during a reference pulse is determined. The coil power loss is evaluated by the RFSWD software.

The measurement requires the small (170 mm diameter) spherical phantom rather than the normal body phantom. The user is reminded to use the small spherical phantom in a confirmation message.

Measurement

At first a complete inline adjustment with a non-volume selective frequency adjustment is performed to make sure that a transmitter and DICO adjustment is executed. During the transmitter adjustment the coil power losses are measured. The following values will be logged in the coil file. (Refer to the logfile in Reports/TuneUp results):

- P_{ref} TALES from Adjust Transmitter
- Pickup probe 0 value (if probe available)
- Pickup probe 1 value (if probe available)
- Evaluated Coil Power Losses $P_{CPL} = P_{ref} \text{ TALES} \times \text{Sigma Phantom}$

Sigma Phantom, (refer to the log-file in Reports/TuneUp results) is a term for correction of phantom influences on the measurement.

(The pick-up probes in the Body Coil are connected via attenuators to receiver 2, PF/PR-inputs, if installed).

The subsequent measurement verifies the position and size of the phantom to ensure that the small (170 mm diameter) spherical phantom is used.

The small spherical phantom is necessary because of the reduced losses and dielectric resonances in the phantom.

Gradient Delay

 HP-Z000100Q-071105-02-012

System Type:Avanto
 Espreo
 TrioTim
 SymphonyTim

General

The **Gradient Delay** procedure measures the effective gradient delay in each gradient so that the timing of the gradients can be determined accurately. A save is performed automatically after each measurement.

Physical Background

For all imaging methods, the exact timing of the gradients relative to the RF timing in the imaging sequence is important for optimum image quality. However, a small time delay (about $30\mu\text{s}$) exists between the time the gradient is switched on digitally by the sequencer and the time the actual gradient field is present. Part of this delay is the same for all systems and is due to timing delays in the GCU and the gradient amplifier. However, another part of this delay is specific to the magnet and RF body coil configuration. It therefore depends on the system and the gradient axis. For this reason, it is important to measure the delays for all three gradient axes empirically.

Measurement

Each gradient is measured at three different gradient strengths (e.g. 1.5, 4 and 8 mT/m) giving three gradient delays for each gradient. For an ideal gradient system, all delays should be the same.

The **Max Diff Deviation** is the maximum difference of the delay times with different gradient strengths, i.e. high gradient pulses may be different from low amplitude pulses. If the tolerance is exceeded, this indicates some kind of nonlinearity in the gradient system.

Gradient Sensitivity

```
HP-Z000100Q-071105-02-014
System Type: . . . . . Avanto
                  Espree
                  TrioTim
                  SymphonyTim
```

General

The **Gradient Sensitivity** procedure adjusts the gradient sensitivities so that the system will use the correct gradient strenghts and measure the correct image size in all orientations.

In addition, the Grad Sens program is used to determine the distance from the light marker to the magnet center.

Functional Description

A spherical phantom with a specific diameter is placed into the magnet center using the light marker.

A standard imaging sequence is repeated three times, one measurement for each orientation. The three images are then analyzed to determine the exact location of the center of the phantom and the phantom diameters. By swapping the standard readout and phase encoding directions in one orientation, the center and diameter of the phantom for each gradient direction can be determined, once with the gradient as a readout and again with the gradient as a phase encoder. The average of these two results determines the final value that is used for calibrating the gradient sensitivities.

Measurement

The measured images of the different orientations can be loaded and displayed. Image windowing is performed by clicking and dragging the mouse or by directly entering the image window and center values.

The gradient sensitivity is defined as the gradient strength per ADC unit (e.g. 0.000166 mT/m per 1 ADC unit).

NOTE

The patient table position is corrected automatically. In case of problems (e.g. if the table moves in the wrong direction), you should verify the correct Z gradient polarity.

Adjust Second Lightmarker

HP-Z000100Q-071105-02-016

System Type:Avanto
Espre
SymphonyTim

General

A second lightmarker in back of the magnet is available as an option. The distance between this lightmarker and the magnet center is determined in the tune-up step **Adjust Second Lightmarker**.

The measurement uses the gradsens sequence with a sagittal slice orientation. The position of the phantom center in the image is determined using the gradsens evaluation algorithm. The phantom is centered by moving the patient table. The corrected lightmarker distance (back) is stored in the measperm section.

Rel. Receive Path Calib.

HP-Z000100Q-031105-02-035

System Type: Avanto
Espreo
TrioTim
SymphonyTim

General

For every receive path of the RCCS, a correction factor is determined in the Tune-Up. These factors are used to adjust the amplifications of each input/output channel combination of the RCCS. The image brightness of a coil is then independent from the receive path used.

The **Relative Receive Path Calibration** contains all loop measurements which can be performed without MR measurements.

The scale factors of receiver channels not yet calibrated are adjusted by evaluating raw data amplitudes of the sequence rx_cal (no MR-signal detection). The TX-signal is switched via the tuning MUX in the RCCS one after the other to every input SGA channel and output channel of the RCCS.

In addition to the brightness correction, another factor is calculated for each RCCS input channel by measuring the relation of the raw data signals for high and low receiver gain. This factor is called **Receiver gain calibration factor** and is needed to measure sequences that switch the gain between the measurement of single phase steps. If the amplitudes of these steps are not corrected exactly, image artifacts are produced.

Functional Description

For further information, refer to the functional description/Tune-up/QA.

Testresults/ Evaluation

The following output data will be shown under Reports/Tune-Up Results:

Calibration of RCCS input-matrix:

- Raw data from the reference measurement, low/high gain noise, low/high gain amplitude (graphic representation)
- Raw data from input channel #1, low/high gain noise, low/high gain amplitude (graphic representation)
- Measurement data from input channels (system dependent)

Calibration of RCCS output-matrix:

- Raw data from the reference measurement, low/high gain noise, low/high gain amplitude (graphic representation)
- Raw data from output channel #1, low/high gain noise, low/high gain amplitude (graphic representation)
- Measurement data from output channels (system dependent)

Results:

- Combined RX Channel Correction (Old/New) against specifications
- MUX Channel Correction (Old/New) against specifications
- Receiver Gain Calibration Factor

Troubleshooting

In case of problems (status **Not ok**) check the following items:

- RX cables and connections X01A-X08A (magnet room)
- RX cables and connections W0181-W184 (equipment room)
- Receiver boards (TestTools/MOD/REC)
- RCCS (TestTools/RCCS)

Abs. Receive Path Calib.

HP-Z000100Q-031105-02-036

System Type: Avanto
Espre
TrioTim
SymphonyTim

General

For every receive path of the RCCS, a correction factor is determined in the Tune-Up. These factors are used to adjust the amplifications of each input/output channel combination of the RCCS. The image brightness of a coil is then independent of the receive path used.

The **Absolute Receive Path Calibration** requires an MR measurement.

The absolute brightness calibration of the receiver matrix is achieved by evaluating the brightness of a test image measurement with the Body Coil (Avanto, Espre, Symphony TIM) or Head Matrix Coil (Trio Tim).

Functional Description

For further information, refer to the functional description/Tune-up/QA.

Testresults/ Evaluation

The following output data will be shown under Reports/Tune-Up Results:

- MR-Image with protocol Body_Tra or Headmatrix_Tra_sn
- Image brightness results against specifications
- Combined RX Channel Correction (Old/New) against specifications; the number of RX channels (8, 18 or 32) depends on the individual system.

Troubleshooting

In case of problems (status **Not ok**) check the following items:

- RX cables and connections X01A-X08A (magnet room)
- RX cables and connections W0181-W184 (equipment room)
- Receiver boards (TestTools/MOD/REC)
- RCCS (TestTools RF/RCCS)

18.1.2 Expert

HP-Z000100Q-031105-02-033

System Type:Avanto
Espreo
TrioTim
SymphonyTim

Expert Mode Platforms

Some procedures have an Expert platform. These platforms should be used by qualified persons only, because they may provide additional settings and configurable parameters which require a more detailed knowledge of the underlying physics. In the Expert Mode, an arbitrary coil can be selected for most of the available procedures.

The Expert Mode should be used for troubleshooting only.

In the **Expert Mode**, **TuneUp** is no longer automated. Only one selected procedure is performed at a time.

The **RF Characteristic** procedure in the **Expert Mode** has an additional **Save** button saves the RF characteristic after the measurement.

NOTE

Any adjustments will be used in standard imaging.

The **status** of a procedure performed in general TuneUp, e.g. “Error” or “Not OK” will not be changed by performing the same procedure in expert mode, whereas the result parameters can be changed.

To reach the “Done” status of any tune function, the procedure has to be successfully run in general TuneUp.

Extended Frequency Adjustment

```
----- HP-Z000100Q-031105-02-034 -----
System Type: . . . . . Avanto
                  Espree
                  TrioTim
                  SymphonyTim
```

General

Extended Frequency Adjustment is only available in Expert Mode.

The procedure performs several non-volume selective frequency adjustments with a moving frequency interval. As a result, the entire specified frequency range is covered to find the resonance peak even it is far away from the current frequency.

MR System	Lower frequency (Hz)	Upper frequency (Hz)
1.5 Tesla	63450000	63750000
3 Tesla	123100000	123265000

The function finds the resonance frequency after a ramp down/up and can be used with the array shim device as well.

Tuning Calibration

HP-Z000100Q-041105-02-002

System Type:Avanto
Espre
TrioTim
SymphonyTim

Expert Mode

Step-by-step tuning calibration is possible in the **ExpertMode**, i.e. you can individually select the termination condition of TALES.

Tales		X4		
		open	50 Ohm	short
X3	open		☐	
	50 Ohm	☐	☒	☐
	short		☐	
☐ Short circuit between X3 and X4				

Fig. 258:

RF Characteristic

HP-Z000100Q-041105-02-010

System Type: Avanto

Espre

TrioTim

SymphonyTim

Expert Mode

In the **Expert Mode**, the function operates similar to standard mode except for the following options:

- 1. Option to save the RF Characteristic after measurement.
- 2. Option to modify Maximum of allowed RF power.
- 3. Selection of RF-amplifier, coil path and nucleus, if available.
- 4. Option to switch off the frequency adjustment.
- 5. Option to disable the switching of TAS to dummy load.

Expert Mode does not include RF verification at the end of the measurement.

After selecting the RF Characteristic in the Expert Mode, the Expert Mode platform (shown below as an example) appears. Start the measurement by clicking **“Go”**.

RFPA		RFPAK2204		RFPAK2137	
Path		BC	LC	BC	LC
Nucleus	1H	☒	☐	-	-
	3He	☐	☐	-	-
	19F	☐	☐	-	-
	7Li	-	-	☐	☐
	13C	-	-	☐	☐
	23Na	-	-	☐	☐
	31P	-	-	☐	☐
	129Xe	-	-	☐	☐

RF Maximum Voltage Volt

Settings

☐ No adjustments

☐ No switch to dummy

Fig. 259:

Regulator Adjust

HP-Z000100Q-071105-02-002

System Type:Avanto
 Espre
 TrioTim
 SymphonyTim

Expert Mode

In the **Expert Mode**, the user has greater control over the adjustment as well as the setting of the regulator values. For this reason, only a single gradient operation is possible. By default, the x gradient is selected, but the user can exclusively set one of the three gradients from the **Expert Mode** platform.

Three numerical values (P=proportional portion, I=integral portion, D=differential portion; range 1 to 128) display the regulator settings for the selected gradient. When selecting the gradient for the first time, the existing regulator settings are displayed for this gradient. The values can be modified by the user. When selecting **Go**, these values are loaded into the gradient system and the iterative measurement and adjustment are performed starting with these values. At the end, the new, calculated regulator settings are displayed. These settings are loaded when selecting **Go** the next time.

Table Adjustment

```
----- HP-Z000100Q-071105-02-004 -----
System Type: . . . . . Avanto
                  Espree
                  TrioTim
                  SymphonyTim
```

Expert Mode

In the **Expert Mode**, the procedure **Table Adjustment** is identical to the normal mode.

Phantom Shim

```
----- HP-Z000100Q-071105-02-006 -----
System Type: . . . . .Avanto
              Espree
              TrioTim
              SymphonyTim
```

Expert Mode

In the **ExpertMode**, the **Phantom Shim** procedure operates similarly except that only one iteration is performed. No extra functionality is provided.

Cross Term Compensation

HP-Z000100Q-071105-02-008

System Type: Avanto
Espreo
TrioTim
SymphonyTim

Expert Mode

The **ExpertMode** performs the CTC measurement in individual steps rather than automatically:

- 1. Enter the main axis to be measured (X, Y or Z).
- 2. After each measurement step, the **Cross-Term Compensation** parameters are saved automatically.

Eddy Current Compensation

 HP-Z000100Q-071105-02-010

System Type:Avanto
 Espre
 TrioTim
 SymphonyTim

Expert Mode

The **ExpertMode** performs the ECC measurement in individual steps rather than automatically. The following options are available:

1. Enter the step number. In standard operation, the procedure will start with the 1st step (no compensation). If a number other than 1 is entered, the procedure will use the current **ECC** parameters.
2. **Preparation.** You select either the **Preparation** measurement or one gradient at a time. When you open the Expert platform, it defaults to **Preparation**.
3. **File name display.** This displays the prefix for the data files. The default for the prefix is "ecc".
4. The **ECC** parameters are saved automatically after each measurement step.
5. The parameters of the most successful ECC compensation (normally the last iteration) will be used in standard imaging.

Gradient Delay

HP-Z000100Q-071105-02-013

System Type: Avanto
Espreo
TrioTim
SymphonyTim

Expert Mode

The **Expert Mode, Gradient Delay** operates similar to the normal mode. No extra functionality is provided.

Gradient Sensitivity

HP-Z000100Q-071105-02-015

System Type:Avanto
Espre
TrioTim
SymphonyTim

Expert Mode

The **Expert Mode**, **Gradient Delay** operates similar to the normal mode. No extra functionality is provided.

19 Quality Assurance

19.1 Procedures

19.1.1 General

HP-Z000100Q-031105-02-031

System Type: Avanto
Espreo
TrioTim
SymphonyTim

Quality Assurance Functions

GeneralQuality Assurance is an automated procedure for checking overall system quality.

Unlike with **TuneUp**, system parameters are not changed.

TuneUp Verification

All **TuneUp** workflow procedures have corresponding verification routines available in the **Quality Assurance** workflow menu.

Dependencies TuneUp/ QA

If a **TuneUp** procedure has changed, its status (except for **Expert** status) and is successfully completed, the corresponding procedure(s) in the Quality Assurance workflow menu is/are changed to **To Do**.

If one procedure has a significant impact on another procedure, although only one has been selected, the selected procedure will change the status of the other procedure.

TuneUp procedures may affect different QA procedures, e.g. after Phantom Shim, the QA procedures Coil Check, Shim Check, Gradient Sensitivity Check, Fat Saturation have to be performed. If the TuneUp procedure is successfully finished, the corresponding QA procedures will be changed to status **To Do**.

Normal Mode

All Procedures with the **To Do** status are selected by default and performed. A single procedure can be selected and performed.

Standby

In the standard **QA mode** (non-expert mode), a **Standby** option is available. If the user selects **Standby**, the system will be switched in the standby mode after completion of the standard QA.

In the **Standby** mode all units, except for the host, are switched off. The units **cannot be switched on** again **using remote access**.

Interaction

If user interaction is required (e.g. phantom change), the user will be informed in a message pop-up window.

Status Field

The **Status field** for each **TuneUp** procedure indicates the current condition of the procedure based on the following criteria:

To Do - indicates that the procedure has not yet been performed or needs to be performed because of a change in status of another dependent procedure.

Running - indicates that the procedure is presently running.

Not OK - indicates that the procedure ran, but the results are not within specifications.

Error - indicates that an error occurred during the last operation. If this procedure is selected and **Go** is performed, the status will change immediately to **To Do**.

Done - indicates that the procedure has been completed successfully.

Coil QA

The QA procedure “**Coil Check**” replaces the former “**Coil QA**” for all coils except the one which is used for the tune-up procedures “**Receive Path Calibration**”.

Rel. Receive Path Calib. Check

HP-Z000100Q-031105-02-041

System Type: Avanto
Espreo
TrioTim
SymphonyTim

General

The **Relative Receive Path Calibration Check** verifies the receive path correction matrix and the low/high gain factor of the RCCS.

The procedure contains all loop measurements which can be performed without MR measurements.

Raw data will be measured (service sequence rx_cal) in a loop (RF test signal loop) over all RCCS input channels, switching between high and low gain. The raw data will be corrected by the gain factor and the receive path correction matrix.

The Check evaluates the phase and amplitude of the high and low gain measurements and compares the result against specifications.

Testresults/ Evaluation

The following output data will be shown under Reports/QA Results:

RCCS matrix QA measurement:

- Raw data from reference measurement, low/high gain noise, low/high gain amplitude (graphic representation)
- Raw data from input channel #1, low/high gain noise, low/high gain amplitude (graphic representation)
- Measurement data from input channels (system dependent)

Evaluation:

- Noise Evaluation (Low Gain Noise, Mean, Max/ HighGain Noise, Mean, Max) against specifications
- Difference of high/low gain signals (magnitude, phase difference) against specifications
- Ratio of reference signal/testsignal, with receiver matrix correction against specifications

Troubleshooting

In case of problems (result “**not ok**”) check the following items:

- Repeat the Rel. Receive Path Calib. Check; if not ok, repeat TuneUp/ Rel. Receive Path Calib.
- RX cables and connections X01A-X08A (magnet room)
- Receiver boards (TestTools/ MOD/REC)
- RCCS (TestTools/ RCCS)

Abs. Receive Path Calib. Check

HP-Z000100Q-031105-02-042

System Type:Avanto
Espreo
SymphonyTim

General

The Absolute Path Calibration Check performs MR image measurements with the Body coil in the transverse/coronal/sagittal direction.

The image brightness, SN and inhomogeneity are compared against specifications.

Testresults/ Evaluation

The following output data will be shown under Reports/ TuneUp Results:

- MR image with protocol Body_Tra
- Image brightness results against specifications
- Inhomogeneity results against specifications
- MR image with protocols Body_Cor and Body_Sag
- Brightness/SN measurement and inhomogeneity results against specifications

Troubleshooting

In case of problems (status **not ok**) check the following items:

- RX cables and connections X01A-X08A (magnet room)
- RX cables and connections W0181-W184 (equipment room)
- Receiver boards (TestTools/ MOD/REC)
- RCCS (TestTools/ RCCS)

Image Orientation

HP-Z000100Q-071105-02-020

System Type: Avanto
Espreo
TrioTim
SymphonyTim

General

The **Image Orientation** procedure measures the position of the loaded phantom and the spherical phantom to determine the correct gradient orientation and polarity.

The two images measured are automatically evaluated following the measurement.

With the first image the correct connections of the gradient cables (x/y/z axis) are tested; with the second image the correct polarity of the gradient currents is checked.

Because the image of the loader is not symmetrical with respect to interchanging gradient axes, an exchange of gradient axes (i.e. wrong cable connections) can be detected.

Incorrect gradient polarity can be detected by identifying the asymmetrical cap of the spherical phantom.

In case of incorrect gradient axis orientation, polarity or positioning of the phantom, the procedure is finished with the **Error** status.

The images and error messages can be displayed by clicking **Reports** and selecting the corresponding procedure in the **Session History Log** or by selecting the corresponding log file under **QA Results**.

Test Results

The two images measured can be loaded and displayed in the HTML/XML output page. Image windowing is performed by clicking and dragging the mouse or by directly entering the image window and center values.

Example:

Image 1: Check for swapped gradient axes

Performing interchange check measurement.

Starting 1. of 1 measurements.

Image: Loader and phantom

Image : http://185.20.9.93/service/seso/sitedb/log/quality/general/image_orientation/c
 Window: 256 Center: 128 Messages: Load and Display finished! □□
 Image Display

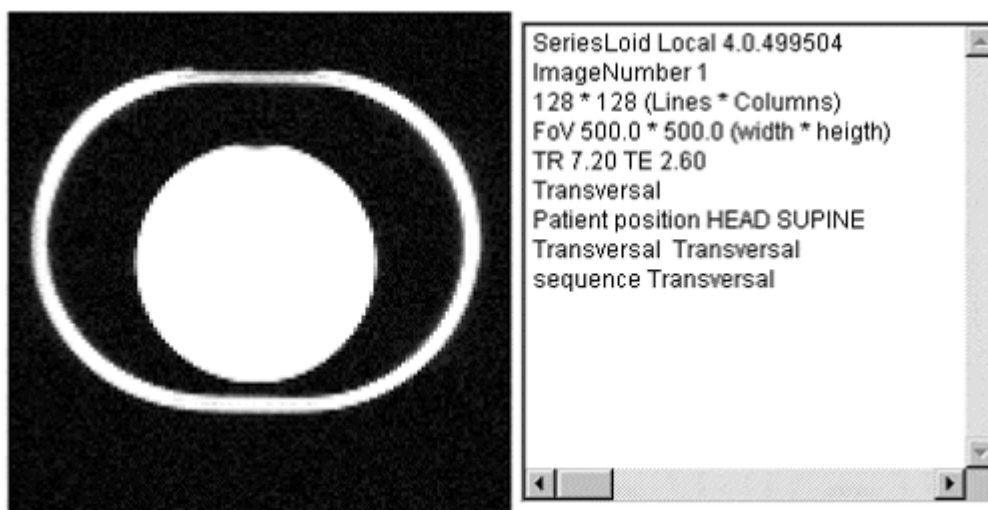


Fig. 260:

Image 2: Check for swapped polarities

Image: Orientation T > S -45°

Image : http://185.20.9.93/service/seso/sitedb/log/quality/general/image_orientation/c
 Window: 348 Center: 242 Messages: Load and Display finished! □□
 Image Display

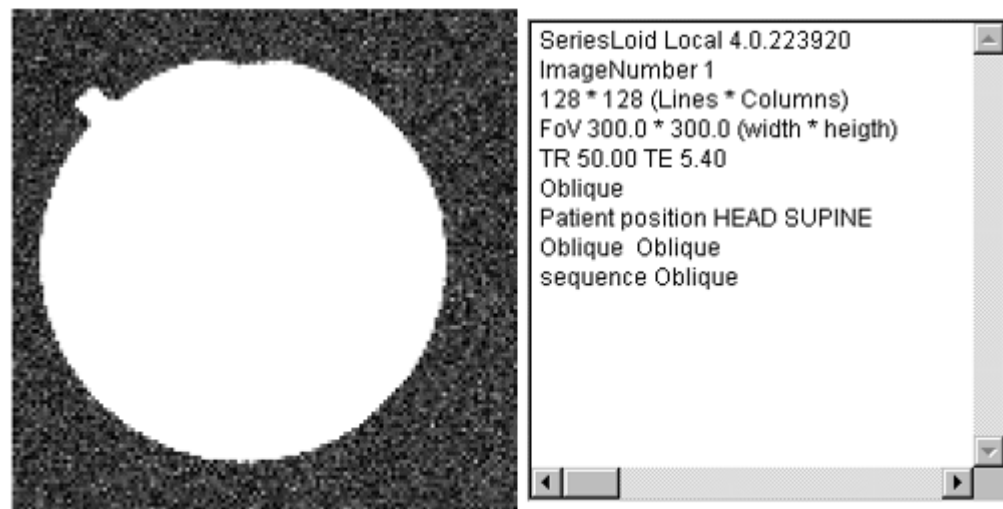


Fig. 261:

Image Orientation check failed!

The location of the phantom filling opening was identified in the wrong direction.

The y Axis might be ok.

The x and z Axis may have wrong polarity.

Results are NOT in Specification!

Program finished successfully!

Fig. 262:

Incorrect gradient polarity can be detected by identifying the asymmetrical cap of the spherical phantom. The example in image 2 shows an incorrect polarity of the X and Z axis. For correct polarity, the cap of the phantom should be on the right side.

BC Tuning Check

 HP-Z000100Q-181108-02-011

System Type:Avanto
 Espre
 TrioTim
 SymphonyTim

General

The procedure is used only for systems with a circular polarized body coil.

Physical background

The procedure is used to measure the decoupling of the circular polarized body coil. The transmission through the body coil or the decoupling of the two body coil systems describes a correlation between the body coil systems. The transmission is measured by sending a signal into the 0 system and measuring the signal in the 90 system. Transmission T is given by:

The measurement is performed with fixed Cs/Cp values in a broad frequency range.

The resonance is given by the frequency with the lowest BC reflection. The decoupling is given by the frequency at which the transmission has its maximum value. The maximum transmission should be lower than -16 dB which means that the portion $P=P_0 \cdot (10^{-(16\text{dB}/10\text{dB})})$ or 2.5% of the input power is lost for imaging (P_0 is the input power for the body coil).

Measurement**Prerequisites**

Since BC Tuning uses the H and S matrices to perform calculations, the tuning calibration and BC Tuning TuneUp procedure have to be completed successfully before starting the BC Tuning check.

Measurement

Only one measurement step will be performed.

Output of the procedure:

- Graphics:
 - Reflection amplitude, reflection phase over frequency
 - Transmission amplitude, transmission phase over frequency

- Table values, reflection:
 - Frequency at minimum reflection (against specs)
 - Reflection factor at minimum (against specs)
 - Delta frequency of 0 degree system and 90 degree system at minimum (against spec)
 - Reflection phase at system frequency
- Table values transmission:
 - Frequency at maximum (against spec)
 - Decoupling at maximum (against spec)

NOTE

If errors occur, you must repeat the TuneUp procedure “BC Tuning”.

RF Verify

 HP-Z000100Q-081105-02-003

System Type:Avanto
 Espreo
 TrioTim
 SymphonyTim

General

RF Verify will perform a verification measurement of the RF Characteristic across the complete voltage range to the maximum output voltage.

The measurement part of the **RF Verify** routine is almost identical to the RF Characteristic measurement routine except for one important difference: The verify routine does not use the linear characteristic but the measured RF Characteristic. Therefore, the **RF Verify** measurement should ideally result in a linear response. Also, only one measurement is performed using the maximum voltage for the system (or the user modified maximum voltage). The actual DICO conversion factor from the RF Characteristic last saved is used for determining the maximum voltage.

Test Results

The value of the **DICO PF ADC Conversion** is calculated as follows:

DICO Conversion = RF voltage output/RF ADC value (e.g.: 651.9/13346 = 0.0488).

The **Amplitude Deviation** is the difference (in %) between the output voltage of the RFPA measured in the QA procedure "RF Verify" and the ideal output voltage measured during the tune-up procedure "RF Characteristic" for every input voltage value.

The **Phase Deviation** is simply the phase difference from mid phase (p) since the phase data has been normalized to p. The maximum phase deviation of the measured points is compared to the allowed specification. It corresponds to a slight frequency drift.

These deviations are caused by drifts of the RFPA components.

If there is little or no compensation for amplitude nonlinearity and phase rotation. The slice profile will be distorted with the following results:

- The signal to noise drops (more crosstalk in multislice imaging)
- The contrast drops (more partial volume effect).

Regulator Check

HP-Z000100Q-081105-02-005

System Type: Avanto
Esprea
TrioTim
SymphonyTim

General

The **Regulator Check** routine verifies the correct setting of the gradient regulator by looking at several aspects of the actual current output and by comparing it to the expected ideal pulse output. The sequence uses a gradient table to generate the gradient pulses so that the selected gradient (depending on the selected orientation) is stepped from the maximum negative to the maximum positive gradient amplitude pulse in linear steps. Each gradient in the table is switched on using a ramp up and ramp down time that conforms to the minimum defined ramp up time (to maximum amplitude) for the system type.

Based on the maximum gradient amplitude pulse, the program calculates the characteristic overshoot that is detected just after the ramp up time. The amplitude of this overshoot is determined as a percentage relative to the nominal gradient amplitude. Also, the time it takes for this exponential-like overshoot to return to a specified value is calculated. In addition, a smoothness calculation is performed to determine the degree of smoothness of the top portion of the maximum amplitude pulse.

Finally, each line in the gradient table is integrated and the resulting mean value for each line is fitted to a linear function. The difference between the data and the linear fit and the maximum deviation are returned as quality parameters.

Test Results

All values measured are listed in a table in the HTML output page together with some graphics.

The current gradient regulator parameters (P = proportional portion; I = integral portion; D = differential, ranging from 1 to 128) are shown in the header row of the table.

The graphic diagrams show:

- The complete gradient current pulse (Pulse Form)
- The enlarged overshoot portion and current ripple (Top Portion)
- The deviation from the linearity of the integrals of the pulse amplitude from the lowest to the highest pulse amplitude (Linear Deviation)

The **Smoothness** is the maximum **positive** deviation of the gradient pulse top portion. It is a measure for the current ripple.

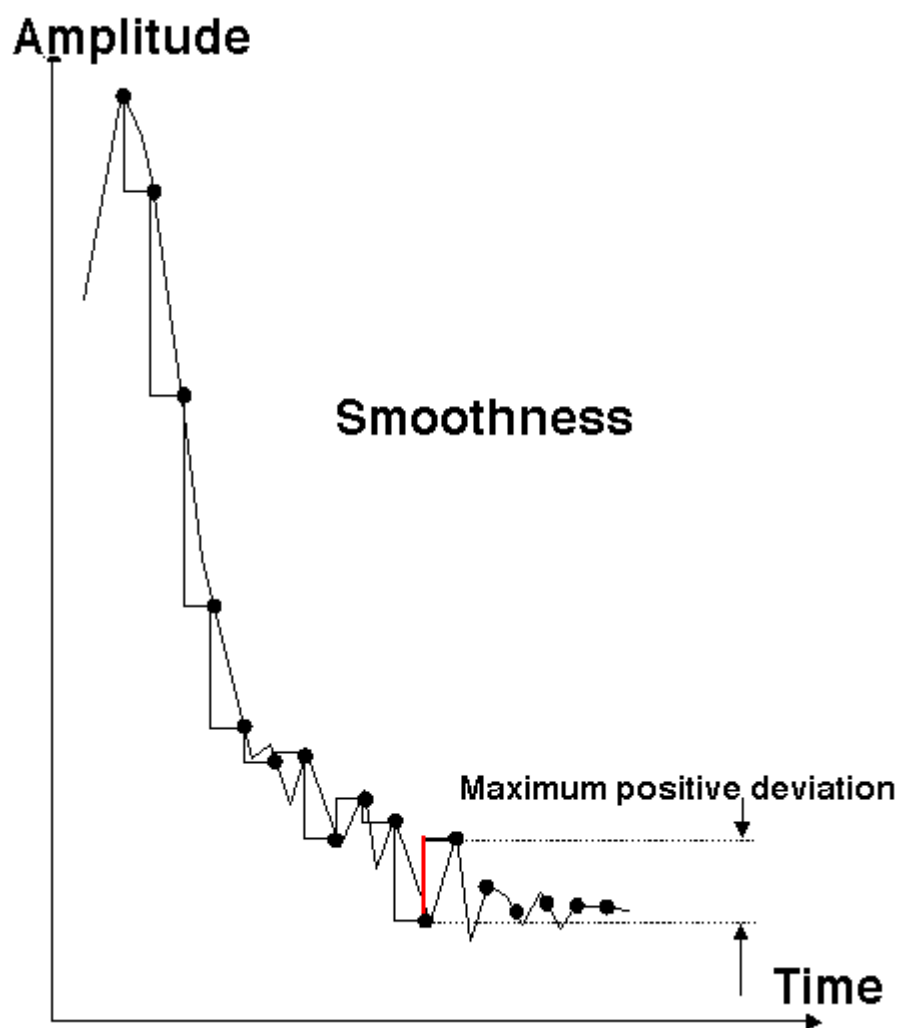


Fig. 263:

Grad. Rise Time Check

HP-Z000100Q-221105-02-001

System Type: Avanto
Espreo
TrioTim
SymphonyTim

General

The **Gradient Rise Time Check** measures the minimal gradient rise time or the so-called “Slew Rate”.

The measurement uses a spin-echo EPI sequence; the tested gradient channel has phase-encoding as well as readout functionality. In the readout interval, develop five echoes because of the bipolar triangle gradient pulses. These echoes should appear according to the pre-dephasing exactly at the pulse spikes. During the measurement, the pulse amplitudes run the range from -Gmax to +Gmax.

Simultaneously with the pulse amplitudes, the ascent of the pulse ramp (with constant pulse ramp time) changes and determines the limits of the gradient system.

If there is a deviation of the echo position from the nominal position, the required slew rate will not be reached.

The limits can be determined and calculated from the raw data (representation of magnitude with corresponding windowing).

Measurement

Sequence of procedures:

- Checking the phantom used
- Data measurement for three gradient axes (without additional load gradient pulses)
- Evaluation of the raw data image and calculation of gradient rise time
- Numerical and graphical output

Testresults/ Evaluation

The following output data will be shown under Reports/ QA Results:

- Graphics of echo maximum versus gradient slew rate, echo 1-5
Statistics of measured values against specifications
- Y/Z/X- Gradient Rise Time Results
- Y/Z/X Gradient Rise Time Multiple Echo Output (raw data image)

Troubleshooting

In case of problems (result “**Not ok**”) check the following items:

- Stability/ value of mains voltage

- Line resistance
- GPA line transformer
- GPA rectifier board
- GPA power stages
- Results of Regulator Check

Phantom Shim Check

```
----- HP-Z000100Q-081105-02-007 -----
System Type: . . . . . Avanto
                  Espree
                  TrioTim
                  SymphonyTim
```

General

This procedure measures the quality of the magnetic field homogeneity within the phantom volume. The field terms can be determined exactly based on the frequency distribution in the volume elements making up the phantom image. The Quality Assurance version of the program differs from the tune-up version in that the pre-existing standard shim terms are used for the single measurement in order to determine the remaining field terms and the overall quality of the homogeneity within the phantom volume. The shim terms are not adjusted.

Before the 3D measurement is performed, two measurements check for correct cabling of the shim coils, for defective shim units (if available in the system) and for correct gradient offsets as follows:

First, the actual shim status of the system is measured.

During the second measurement, defined shim offset currents or reversed gradient offset currents will be applied. The effects on the magnetic field can now be seen and compared to the predicted field changes.

This detects wrong cabling, wrong polarity, open or short coils or defective shim amplifiers. The shim terms are not adjusted.

Test Results

All field terms measured are listed in a table in the HTML output page. Measured values exceeding specifications are displayed with a red background.
Shim terms exceeding specifications are marked with an asterisk in the Log-File.

Troubleshooting

For further information, refer to TSG/TU/QA.

Cross Term Compensation Check

HP-Z000100Q-081105-02-010

System Type:Avanto
Espreo
TrioTim
SymphonyTim

General

The **Cross-Term Compensation Check (CTC)** checks the quality of cross-term compensation. The CTC requires the normal body phantom.

Physical Background

The operation of a single gradient, e.g. Z gradient, produces not only eddy currents coupled to the Z gradient, but also small amounts of eddy currents coupled to the X and Y gradients, resulting in a cross-term of up to 0.2% of the main Z gradient amplitude. These eddy currents decay exponentially over time, in general with only one time constant. The CTC measures the decay of eddy currents caused by the cross-terms. The parameters of the inverse filter function which compensates the effect are determined by the CTC procedure during tune-up.

Eddy Current Compensation Check

HP-Z000100Q-081105-02-012

System Type: Avanto
Espreo
TrioTim
SymphonyTim

General

The **Eddy Current Compensation Check (ECCC)** checks the quality of the eddy current compensation.

Physical Background

Gradient pulses are distorted by eddy currents induced in metallic parts of the magnet. To obtain high image quality, the eddy currents must be compensated for due to predistortion of the gradient currents so that the resulting gradient pulses get the required shape. This eddy current compensation is performed by a signal processor which applies a digital filter to the gradient pulses. The filter parameters are determined by the ECCC procedure during tune-up.

Test Results

The objective of the **Preparation Measurement** is to check for correct positioning of the phantom in the magnet center and for sufficient field homogeneity. This is performed via a double echo sequence. The first echo has no readout gradient. The length of the MR signal is a rough check of the homogeneity. The second echo uses an readout gradient and calculates the phantom position in the readout direction.

Measurements have shown that for slice shifts that are not too large, it is sufficient to divide the eddy field into a 0 order term and a 1st order term.

The **zero order term (B_0)** results from an asymmetry of the gradient coil with respect to the cryo shield and other conductive materials. The field is space independent and therefore present in the complete imaging volume. It adds up to the normal B_0 field. The unit used for the amplitude (x) is mTm / mT; i.e. a gradient ramped up to 1mT/m will produce a 0 order term with an amplitude of x mT. The time constant of the most important part is about 500ms, but there are additional shorter time constant parts as well.

The **first order term (x,y,z gradient)** is the most important term because it has the same symmetry as the gradient field. The amplitude is provided in % of the gradient.

Higher order terms are usually small and negligible - as long as the slice shifts and off-center zooms are not too large. These terms are not compensated.

Troubleshooting

For further information, refer to TSG/TU/QA.

Coil Power Losses Check

HP-Z000100Q-081105-02-014

System Type:Avanto
Espre
TrioTim
SymphonyTim

General

No measurement is performed while checking for Coil Power Losses. It only verifies that the tune-up of Coil Power Loss has been completed successfully. For this purpose, the coil power loss value in the coil file is checked. The default of this value is very low. For this reason, it would be out of specification if no tune-up were performed.

Gradient Delay Check

HP-Z000100Q-081105-02-015

System Type: Avanto
Espreo
TrioTim
SymphonyTim

General

The **Grad Delay** procedure measures the effective gradient delay in each gradient so that the timing of the gradients can be determined accurately.

Test Results

A difference between old and new delays indicates a drift of components involved in the generation of the gradient pulses. As a result, the S/N value of the MR system may be degraded.

Each gradient is measured at three different gradient strengths (e.g. 1.5, 4 and 8 mT/m) giving three gradient delays for each gradient. For an ideal gradient system, all delays should be the same.

The **Max Diff Deviation** is the maximum difference of the delay times with different gradient strengths, i.e. high gradient pulses may differ from low amplitude pulses. If the tolerance is exceeded, some kind of nonlinearity in the gradient system is indicated.

Gradient Sensitivity Check

HP-Z000100Q-081105-02-017

System Type:Avanto
Espreo
TrioTim
SymphonyTim

General

The **Grad Sens Check** evaluates the phantom diameter and checks it against specifications. The phantom diameter is evaluated for each gradient axis.

Calculation Artefacts

HP-Z000100Q-081105-02-019

System Type: Avanto
Espreo
TrioTim
SymphonyTim

General

Calculation Artifacts analyzes the quantity of ghosts/artifacts outside the phantom.

A double-echo sequence with preceding in-line adjustments followed by image reconstruction for the three slice orientations with slice shift 0 is used. One measurement is performed for each selected orientation.

The transverse orientation is performed first, followed by the sagittal and then the coronal measurement.

Gradient orientations:

	TRA	SAG	COR
X	Frequency	Slice	Phase
Y	Phase	Frequency	Slice
Z	Slice	Phase	frequency

Each gradient is used as a phase-encoding gradient. For this purpose, the normal readout and phase-encoding gradients are swapped for the sagittal orientation so that the Z-gradient is used once as a phase-encoding gradient.

Physical Background

Artifacts in MR images are produced by a variety of mechanisms. For example, gradient deficiencies (instability, non-linearity, 50/60Hz ripple, or RF instabilities) produce periodic blurring or ghosting in the direction of the phase encoding gradient.

Test Results

The measured images will be loaded and displayed. Image windowing is performed by clicking and dragging the mouse or by directly entering image window and center values.

Troubleshooting

For further information, refer to TSG/TU/QA.

Spike Check

HP-Z000100Q-081105-02-021

System Type:Avanto
 Espreo
 TrioTim
 SymphonyTim

General

The **Spike Check** is a quick test to check the MR imaging system for "spikes" and RF interference. The raw data are evaluated by a program that detects and identifies spikes. The image data are evaluated for RF interference.

Physical Background

Electrostatic sparks during the measurement may generate some "spikes" within the raw data. "Spike" means individual raw data points or small groups of raw data points with amplitudes above the typical noise level of the MR imaging system. **Spikes** can degrade the MR imaging quality. In this case, the signal-to-noise ratio can be reduced or the image can show a periodic intensity modulation or structure.

One typical source of spikes is the mechanical and electrostatic stress of the MR system caused by the operation of the gradient system.

A simple way to check the MR system for spikes is to run a noise measurement. This is a measurement that will not generate an MR signal, but rather pure statistical noise.

The "spike" sequence generates mechanical and electrostatic stress by switching the gradient system rapidly within the raw data acquisition period with maximum gradient strength. It also runs through tables covering a wide range of different gradient amplitudes. This sequence does not apply RF pulses. As a result, it does not generate an MR signal.

Test Results

The raw data measured and the RF interference images will be loaded and displayed. Image windowing is performed by clicking and dragging the mouse or by directly entering the image window and center values.

The same algorithm used for spike searching is applied to both image data and raw data. In the case of image data (RF-interference in the image), the spikes are RF-interference patterns that have been picked up during the noise measurement. Subsequently they are Fourier-transformed resulting in a spike in the image data, i.e., the RF-interferences will result in pixels of increased intensity in the image data.

Click Log-File in the HTML output page. Additional windows are opened displaying the spike check results.

- Total number of spike positions
- Number of spikes found in the real and imaginary part
- Number of spikes correlated in the real and imaginary part
- Position (column, row) of spikes

- Amplitudes of spikes in the real and imaginary part
- Magnitude of the highest spike
- Threshold for spike discrimination

Troubleshooting

For further information, refer to TSG/TU/QA.

Stability Check

 HP-Z000100Q-081105-02-023

System Type:Avanto
 Esprea
 TrioTim
 SymphonyTim

General

The **Stability Check** evaluates the stability of the MR signal under various sequence conditions. A spin-echo and a gradient echo are measured without phase encoding using fixed sequence parameters. As a result, all raw data lines should look alike. The stability of the echo signal across the lines measured is evaluated. In the standard mode, all three orientations are measured so that each gradient is performed as a readout gradient. In addition, both 0 and +50 mm slice shift positions are measured.

The evaluation consists of determining the variation of the amplitudes, the phases and the positions of the echoes.

Test Results

The **spin echo sequence** is sensitive to instabilities in the range of a few Hz up to 100 Hz which may be caused by a 50/60 Hz line hum.

The **gradient echo/ flash sequence** is sensitive from DC up to a few Hz which may be caused by slow external field interferences, field instabilities, or mechanical vibrations.

The measurement result of the center slice is sensitive to instabilities in the Bo Field. The 50 mm offset slice is more sensitive to gradient instabilities.

Amplitude stability (Spin-echo):

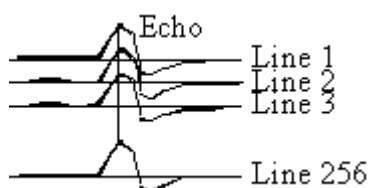


Fig. 264:

The amplitude value in the column of maximum signal is displayed in the graphics output.

Phase stability (Spin-echo):

The **phase shift** of a **spin echo** is determined by the integral across the magnetic field from a 90° to a 180° pulse minus the integral from a 180° pulse to the echo:

$$\varphi = \int_0^{TE/2} \gamma B(t) dt - \int_{TE/2}^{TE} \gamma B(t) dt$$

Fig. 265:

For a constant field or a slowly varying field, both integrals cancel each other. However, for an oscillating field with a period of TE (20 ms = 1/50 Hz), this expression becomes the maximum.

For the **gradient echo** sequence, the phase stability is evaluated by fitting a straight line to the phase data. When you subtract the value of the linear fit from the phase data, the phase stability changes. The slope of the straight line indicates the linear phase drift.

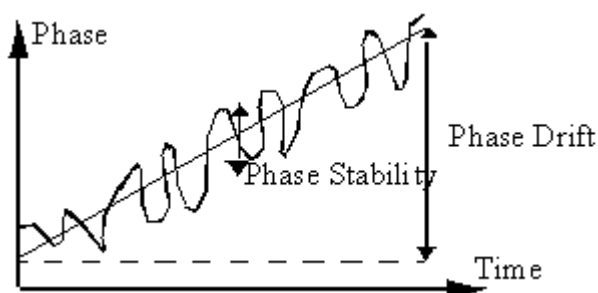


Fig. 266:

Stability of echo position:

The position of the echo peak column is determined for every measurement line and checked against specifications. The cause of an instable echo position may be gradient-like interferences from the environment or from the gradient itself. The sensitivity for the direction of the interference is determined by the orientation of the readout gradient.

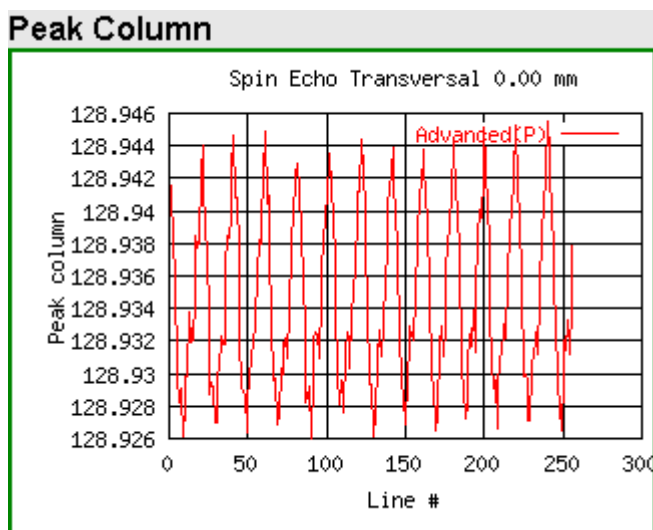


Fig. 267:

Troubleshooting

Remarks on how to use the Stability Check:

Symptom/ Deviation	Possible cause/ Service action
Amplitude	Echo maximum shifting in readout direction/
	RF problem/ Run TestTools/ RF
	Slice profile problem/ RF pulse stability/ Run TestTools/ RF/ MOD/REC/ RFPA
Phase	Magnetic field fluctuation
Depending on orientation	Gradient problem/ check relevant gradient
Independent of orientation	RF system problem/ Run TestTools/ RF
	Magnetic field or external interference
Only slice shift dependent	Gradient problem/ check slice shift gradient
	Spatial instability
Only in gradient echo	DC or slowly varying magnetic field problem
	Compare amplitude with spin-echo, static homogeneity problem

- Switch off components one by one to isolate the cause of instability:
 - Cold head
 - Patient blower
 - Lighting in magnet room
 - Shim PSU
 - Slice selection gradient
 - Readout gradient
 - Complete gradient amplifier
- For periodic instability vary TR and TE to confirm or rule out frequency of interference
 - Vary TR to avoid being in synchronization with the interference
 - Vary TE to become more sensitive to particular frequencies

For further information, refer to the TSG/ TU/QA.

Fat Saturation

HP-Z000100Q-081105-02-026

System Type: Avanto
Espreo
TrioTim
SymphonyTim

General

Fat saturation is used during MR imaging to suppress the fat signal while leaving the water signal unchanged. Fat has an MR frequency 3.5ppm lower than water. This is obtained by applying a low bandwidth RF pulse to the fat frequency followed by spoiling gradients. This requires good homogeneity in the imaging volume in the order of 1-2ppm peak-peak to prevent water saturation as well. In addition, the RF flip angle of the Fat Sat pulse should be about 90 degrees to get optimal results.

Measurement

One requirement is to run the test with the standard water-filled body phantom. Since no fat signal is present, no real fat saturation can be tested. However, physically it makes no difference to saturate water instead of fat. Only the frequency of the RF pulse must be that of water.

First, a reference image is measured without any saturation. This determines the 100% signal level. This is followed by a second measurement with water saturation. The quality of fat saturation is now determined. The signal of the saturation measurement is divided by the signal of the normal measurement, which should be as low as possible.

Test Results /Evaluation

First, the normal image which shows the contours of the phantom is analyzed. Position and diameters are calculated. Then, a region of interest which, for example, is 80% of the circle (defined by status file variable) of the phantom diameter is taken.

This region is now scanned into the normal image and into the water saturated image. The ratio of both signals is calculated point by point. To reduce noise at every point, a 3x3 average over neighboring points is taken. Point by point saturation ratios are calculated, resulting in a mean and a maximum value. These two values are compared to specifications.

The images measured are loaded and displayed. Image windowing is performed by clicking and dragging the mouse or by directly entering the image window and center values.

The degree of Fat Sat depends largely on the shim condition. As a rule, for a mean saturation ratio of 15%, the peak-peak homogeneity of the field has to be better than 4ppm.

Troubleshooting

For further information, refer to TSG/TU/QA.

Synthesizer Check

 HP-Z000100Q-081105-02-027

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System Type: . . . . .Avanto
              Espree
              TrioTim
              SymphonyTim
  
```

General

The **SynthesizerCheck** procedure is an important test to verify proper operation of the synthesizer. It is used to validate the frequency and phase quality of the synthesizer with respect to the accuracy of the slice position.

Measurement

The MR system contains a maximum of 32 receiver channels. 4 receiver channels are connected to 1 double NCO (Numeric Controlled Oscillator), so that we have a total of 16 RX NCO's. Double NCO means that we have two groups of NCO's, so that it is possible to switch the group of 4 receiver channels either to NCO-group 1 or NCO-group 2.

For transmission, we have one double NCO. The transmit path can be switched to NCO-group 1 or NCO-group 2.

During simultaneous sending and receiving, the phase and frequency of the TX-NCO and RX-NCO within one group is the same. In the following QA-procedure asynchronous behaviour of the TX and RX-NCO is needed, this means that one NCO is the stepping phase and frequency while the other one is fixed. For that reason we always are using different groups of NCO's when sending and receiving simultaneously, i.e. TX-NCO is fixed and RX-NCO is stepping.

In a **first step**, the verification of the frequency and phase operation of the RF reference signal is performed by using the **MR signal** itself (nco-sequence). After demodulation of the received MR signal, the reference signal is compared to the MR signal in terms of frequency and phase. In one test, the reference signal is varied in frequency over the whole range of the MR system. A second test modifies the phase in fixed intervals. In this way the entire operating range of the oscillator can be tested. The test is performed for one RX-NCO from group 1 and one RX-NCO from group 2. The two RX-NCO's are validated and can be used for testing the other NCO's.

In a **second** step we perform loop-measurements (nco_loop-sequence) between the tested RX-NCO's and the TX-NCO's. In this way, we approve the correct function of the TX-NCO's.

In additional steps, all other RX-NCO's are tested against the approved TX-NCO's by testing group 1 RX-NCO's and group 2 RX-NCO's separately.

Troubleshooting

For further information, refer to TSG/TU/QA.

SAR Monitor Test

HP-Z000100Q-170107-02-001

System Type: Espree
Avanto
TrioTim
SymphonyTim

General

The purpose of this test is to verify that the SAR (Safe Absorption Ratio) check is activated. It is also tested if the pickup-coils gain a signal during the measurement. Exception: if there is only one receiver cassette present which prevents the measurement of the pickup-coils. (Accordingly, a message will be shown).

Measurement

A location series is taken with the gradsens sequence. This is to ensure that the phantom (with loader) has been inserted and centered. This is required to obtain proper SAR values. The SAR monitor test is not meant to obtain a quantitative calibration of the SAR. It only tests to ensure that the SAR is switched on. However, the values retrieved will be displayed.

Procedure:

- Prior to performing measurements with the gradsens-sequence; the values for the pick-up coils and the SAR value have to be reset.
- Evaluating the series to check for the proper position of the inserted phantom.
- Reading the values for the SAR and two pickup-coil values.
- Displaying the measured values and the ratio of the two pickup coil values.

Temperature Sensor Check

HP-Z000100Q-081105-02-029

System Type:Avanto
Espreo
TrioTim
SymphonyTim

General

The temperature of a patient must not exceed acceptable limits during MR imaging. For this reason, the maximum energy absorbed by the patient during the MR measurement is limited. These limits are provided in guidelines which describe the specific power absorption of the patient. Since these limits are only valid within particular environmental temperature ranges, it is necessary to detect deviations from this range. The Patient Environment Temperature sensors (PETS) are used for this purpose.

Measurement

The procedure checks that the (PETS) are working correctly by measuring the temperature recorded as the sensor is heated and by checking that a 5 degree centigrade rise is obtained within 30 seconds. The test is performed automatically. There is no Expert mode.

The check requires no input from the user. Before repeating the test, you have to wait approximately 15 minutes.

Procedure:

- Measurement of initial temperature and check if it is within reasonable limits
- Start heating and continue until either a 5 degree temperature rise is measured or 30 seconds have passed.
- Output a pass/fail message

RF Verify LC

HP-Z000100Q-081105-02-030

System Type: Avanto
TrioTim
SymphonyTim

General

RF Verify LC will perform a verification of the RF characteristic tune-up in a loop across all RF-amplifiers and nuclei on the local coil RF-path.

The measurement part of the **RF Verify LC** routine is almost identical to the RF Characteristic measurement routine except for one important difference: The verify routine does not use the linear characteristic but rather the measured RF Characteristic. The **RF Verify** measurement should therefore ideally result in a linear response. Also, only one measurement is performed using the maximum voltage for the LC RX-path. The actual DICO conversion factor from the RF Characteristic LC last saved is used for determining the maximum voltage.

NOTE

The dummy load in the TAS_3TM must be connected to the LC-output of the TALES:

- Connect the service plug (mode switched to TX) to the coil connector #1 at the patient table.
- Disconnect LC-output cable at TALES X6.
- Disconnect cable (W1230) at BCCS X7 (connection to TAS_3TM). Leave X7 open.
- Connect this cable (W1230) to TALES X6.

Stability_LongTerm Check

HP-Z000100Q-081105-02-031

System Type:Avanto
 Espree
 TrioTim
 SymphonyTim

General

A new application of MR imaging visualizes activities in the brain. For this purpose the patient will be exposed to a periodical optical, acoustical or other form of stimulation. During the exposition, a series of MR images will be acquired. Subsequently, the images will be evaluated for changes in signal intensity. The regions of the brain activated by the stimulation can be localized by very small variations in signal intensity. These variations can easily be hidden by various instabilities of the system and/or the environment. For this reason, MR system instability and interferences from the environment must be below a specified level.

The **Stability_LongTerm Check** determines the sum of all instabilities while performing a phantom measurement. Typical sources for instabilities are:

- Local B0 field variation caused by temperature variation in the shim unit generates frequency instabilities. The result is a virtual movement of the measured object during acquisitions.
- B1 field variations due to instable RF produce image intensity variations.
- Gradient fields generate frequency and phase variations, also leading to virtual movement of the object.
- Mechanical vibrations, e.g. by the cold head, produce signal intensity variation caused by the movement of the measured object relative to the magnet isocenter.
- Movement of phantom fluid cause phase errors.

Measurement**NOTE**

Wait at least 15 minutes between positioning the phantom and starting the measurement. Small movements in phantom liquid will also affect measurement results.

A series of 512 measurements (EPI_FID sequence) with 11 slices will be performed and repeated once again. The first measurements is performed with a flip angle of 30 degrees (evaluation of stability of B1 field, i.e. instabilities in the TX- chain), the second with a flip angle of 90 degrees on 1 and 1.5T systems and a 70 degree flip angle on 3T systems (minimization of unstable RF, detection of B0 and gradient-like interferences).

Test Results/ Evaluation

The numerical results and graphics can be viewed in the service software under Quality Assurance/Reports/QA results/General/SfpStability_LongTerm Check.

The following graphical and numerical evaluations will be performed:

-
- Mean values of a 15x15 ROI related to the mean value of all measurements as a function of the image number
- Standard deviation of mean values of a 15x15 ROI (linear and logarithmic plot)
- Center position in the X and Y direction of the phantom, dependent on the image number (Expert mode only)
- Measurement protocol values
- Statistics of mean values (for ROI size 15x15 and 25x25):
 - Mean Image Intensity
 - Variation peak-peak
 - Relative variation peak-peak
 - Standard Deviation
 - Total relative linear drift
 - Standard Deviation from linear fit
- Image quality parameters such as SNR and ghosting
- Statistics of Phantom Movement (Expert mode only)

Troubleshooting

For further information, refer to TSG/TU/QA.

19.1.2 Coil

HP-Z000100Q-031105-02-001

System Type:Avanto
Espreo
TrioTim
SymphonyTim

Coil Quality Assurance

The QA procedure “**Coil Check**” replaces the former “**Coil QA**” for all coils except the one which is used for the “**Receive Path Calibration**” tune-up procedures.

Coil Check

HP-Z000100Q-031105-02-002

System Type: Avanto
Espreo
TrioTim
SymphonyTim

General

This **Coil Check** procedure is used to determine the signal to noise (S/N) ratio and to verify image intensity for each coil. For individual coils one or more protocols have been defined. After each protocol is measured, the routine will analyze the resulting image for S/N, image intensity and in some cases for the homogeneity of the intensity profile.

For each configured coil an entry with an easily distinguished (speaking) name followed by “Coil Check” is listed. Coils that are plugged in are indicated accordingly.

The coil/phantom positioning will be displayed in a pop-up window after starting the procedure with the **Go** button.

In the standard mode and for coils where only one phantom position is necessary, the Coil Check performs all required protocols without user intervention. If more than one protocol is required, the procedure measures and evaluates each image from the protocol before continuing with the next protocol.

For coils requiring more than one phantom positioning, the user will be guided through various positioning by displaying an appropriate picture of the coil/phantom setup.

Combined images simultaneously acquired with several coil elements can be evaluated. It is also possible to provide phantom coordinates for the evaluation. This allows the user to evaluate images with more than one phantom, which can be used to decrease the measurement time.

19.1.3 Expert

HP-Z000100Q-031105-02-030

System Type:Avanto
Espreo
TrioTim
SymphonyTim

Expert Mode Platforms

Some QA procedures have an Expert platform. These platforms should be used by qualified persons only. They may provide additional settings and configurable parameters which require a more detailed knowledge of the underlying physics. In the Expert Mode, an arbitrary coil can be selected for most of the procedures available.

Expert Mode should only be used for troubleshooting.

In the **Expert Mode**, **Quality Assurance** is no longer automated. Only one selected procedure is performed at a time. Additional procedures, e.g. **Field Stability**, are available in the Expert Mode.

To reach the status **Done** of one of the tune functions, the procedure has to be successfully run under the general Quality Assurance procedure.

Coil Check

HP-Z000100Q-081105-02-001

System Type: Avanto
Espreo
TrioTim
SymphonyTim

Expert Mode

The **Expert Mode** will provide the option to individually select the available protocols for each coil.

In the Expert Mode, the user can perform the standard Coil Check evaluation with image that are stored in the database. No measurement is performed in this case, only the appropriate Series Loid has to be supplied.

In the Expert Mode it is possible to select the protocols to be performed. A list of the available protocols is displayed. As default, all protocols are selected. Any combination of protocols can be selected. When selecting Go in the QA platform, these protocols will be performed.

Image Orientation

HP-Z000100Q-071105-02-021

System Type:Avanto
Espre
TrioTim
SymphonyTim

Expert Mode

No extra functionality is provided by the Expert Mode.

RF Verify

HP-Z000100Q-081105-02-004

System Type: Avanto
Espreo
TrioTim
SymphonyTim

Expert Mode

In **Expert Mode**, the function operates similar to standard mode except that the following options are available:

1. Option to modify the maximum of allowed RF power.
2. Selection of RFamplifier, coil path and nucleus, if available.
3. Option to switch off the frequency adjustment.
4. Option to disable switching of TAS to the dummy load.

NOTE

If selecting the local coil path (LC), the dummy load in the TAS_3TM must be connected to the LC output of the TALES:

Disconnect the output cable from TALES X6, disconnect the input cable at BCCS X7 and connect it with TALES output X6.

Additionally, the service plug must be connected to coil connector 1 at the patient table.

After selecting RF Verify in the Expert Mode, the Expert Mode platform (shown below as an example) appears. Start the measurement by clicking “**Go**”.

RFPA		RFPAK2204		RFPAK2137	
Path		BC	LC	BC	LC
Nucleus	1H	☒	☐	-	-
	3He	☐	☐	-	-
	19F	☐	☐	-	-
	7Li	-	-	☐	☐
	13C	-	-	☐	☐
	23Na	-	-	☐	☐
	31P	-	-	☐	☐
	129Xe	-	-	☐	☐
RF Maximum Voltage		781.024968		Volt	
Settings		☐ No adjustments ☐ No switch to dummy			

Fig. 268:

Grad. Rise Time Check

HP-Z000100Q-031105-02-038

System Type: Avanto
Espreo
TrioTim
SymphonyTim

General

Gradient Rise Time Check measures the minimal gradient rise time or the so-called “Slew Rate”.

The measurement uses a spin-echo EPI sequence; the tested gradient channel has phase-encoding as well as readout functionality. In the readout interval, develop five echoes because of the bipolar triangle gradient pulses. According to the pre-dephasing, these echoes should appear exactly at the pulse pikes. During the measurement, the pulse amplitudes run the range from -Gmax to +Gmax.

Simultaneously with the pulse amplitudes, the ascent of the pulse ramp (with constant pulse ramp time) changes and determines the limits of the gradient system.

If there is a deviation of the echo position from the nominal position, the required slew rate will not be reached.

The limits can be determined and calculated from the raw data (representation of magnitude with corresponding windowing).

Expert Mode

Additional manual selections are:

- Manual selection of gradient axes
- Manual selection of additional gradient load pulses (load test)

Measurement

Sequence of procedures:

- Checking the phantom used
- Data measurement for three gradient axes (without additional load gradient pulses)
- Evaluation of the raw data image and calculation of gradient rise time
- Numerical and graphical output

Testresults/ Evaluation

The following output data will be shown under Reports/ QA Results:

- Graphics of echo maximum versus gradient slew rate, echo 1-5
Statistics of measured values against specifications
- Y/Z/X- Gradient Rise Time Results

- Y/Z/X Gradient Rise Time Multiple Echo Output (raw data image)

Troubleshooting

In case of problems (result “**not ok**”) check the following items:

- Stability/ value of mains voltage
- Line resistance
- GPA line transformer
- GPA rectifier board
- GPA power stages
- Results of Regulator Check

Regulator Check

HP-Z000100Q-081105-02-006

System Type: Avanto
Espreo
TrioTim
SymphonyTim

Expert Mode

This mode lets the user select the three orientations (TRA, SAG and COR).

Linearity Check

HP-Z000100Q-081105-02-009

System Type:Avanto
 Espreo
 TrioTim
 SymphonyTim

Expert Mode

The gradient amplitude linearity check performs MR signal measurements to determine the linearity of the gradient system. All three gradients are used to produce the spin-echo signals which are evaluated across the complete gradient range. The procedure is only available in the **Expert Mode**. The used coil is shown in the user interface. The gradient axis can be selected via radio buttons.

Physical Background

The Linearity Check is based on the MR method to determine the linearity of the gradient system. All three gradients are used to produce the spin-echo signals which are evaluated across the entire gradient range. An additional evaluation is used to determine whether the gradient linearity or the RF signal is out of specifications.

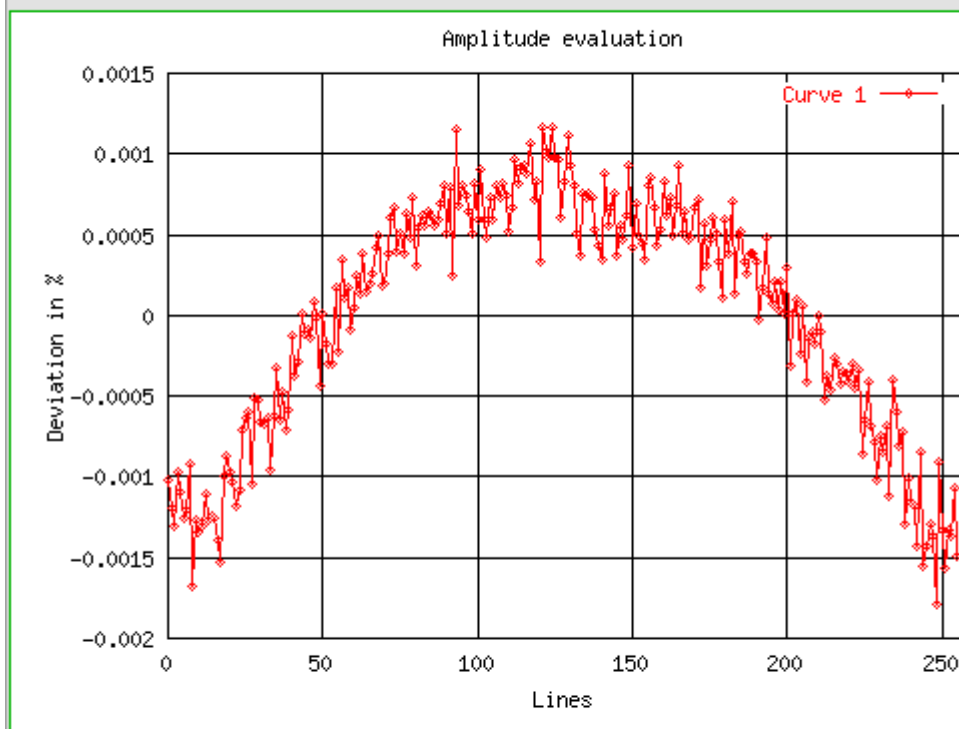
Test Results

The numerical results of the gradient linearity tests and a corresponding graph is displayed in an HTML window (under Report: sfpLinearityCheckExpert). A label identifying the test including slice orientation and gradient axis is printed out. If the result is out of specifications, the respective field in the table appears in color.

Each point of the graph represents one amplitude value of the gradient axis tested (line 0-255). The deviation value means the deviation of the integral value of the gradient pulse from the nominal value, relating to the integral value of the maximum gradient pulse.

If a non-constant pattern is seen, especially at the beginning or end of the plots (near line 0 or 255), this may indicate a problem with the linearity of the gradient output at high amplitudes.

Slice orientation: Sagittal. Performing Y Gradient(readout) Linearity Measurement



Result:

Name	Value	Low_Spec	High_Spec	Unit
Deviation	0.001	0.000	0.010	µm

Fig. 269:

Troubleshooting

In case of errors, check the input voltage of the power stages and the stability of the mains voltage.

- Run the General QA/ Gradient Rise Time Check.
- Swap power stages. Replace defective power stages.
- Check/ replace defective GSSU boards.

Phantom Shim Check

HP-Z000100Q-081105-02-008

System Type:Avanto
Espreo
TrioTim
SymphonyTim

Expert Mode

The **Phantom Shim Check** procedure in **Expert Mode** is similar except that the results are not written to the database.

Cross Term Compensation Check

HP-Z000100Q-081105-02-011

System Type: Avanto
 Espreo
 TrioTim
 SymphonyTim

Expert Mode

The **ExpertMode** performs the CTC measurement in individual steps rather than automatically:

1. Enter the main axis to be measured (X, Y or Z).
2. The results are not written to the database.

Eddy Current Compensation Check

HP-Z000100Q-081105-02-013

System Type:Avanto
Espreo
TrioTim
SymphonyTim

Expert Mode

The **Expert** mode performs the ECC measurement in individual steps rather than automatically. The following options are available:

1. **Preparation.** You can select either **Preparation** or one gradient at a time. When you open the Expert platform, it defaults to **Preparation**.
2. **File name display.** This displays the prefix for the data files. The default for the prefix is "ecc".
3. The results are not written to the database.

Gradient Delay Check

HP-Z000100Q-081105-02-016

System Type: Avanto
Espreo
TrioTim
SymphonyTim

Expert Mode

The only difference in the **Expert Mode** is that data are not written to the database.

Gradients Sensitivity Check

HP-Z000100Q-081105-02-018

System Type:Avanto
Espre
TrioTim
SymphonyTim

Expert Mode

The only difference in the **Expert Mode** is that data are not written to the database.

Calculation Artefacts

HP-Z000100Q-081105-02-020

System Type: Avanto

Espre

TrioTim

SymphonyTim

Expert Mode

The **Expert Mode** lets the user select the slice shift 0mm, 50 mm, and the three slice orientations separately. The **Evaluate** option evaluates any image in the actual segment for artifacts.

Spike Check

HP-Z000100Q-081105-02-022

System Type:Avanto

Espre

TrioTim

SymphonyTim

Expert Mode

The **Spike Check** in the **Expert Mode** is similar, except that results are not written to the database. An additional sequence **Advanced Spike** is offered:

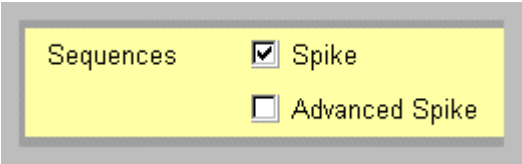


Fig. 270:

In some cases, spike artifacts may be visible with customer protocols, but not with the regular spike check. If so, it is advisable to reproduce spikes and troubleshoot with the same setup (coils, protocols, position of patient table, patient monitoring systems, etc.) which is used by the customer.

If the artifacts cannot be reproduced at all and also all typical spike sources (see Siemens Knowledge Base) are ruled out, contact RSC/ HSC.

NOTICE

The “Advanced Spike” procedure contains a powerful gradient pulse scheme which induces high mechanical stress over a wide frequency range. It shall be therefore used very careful and only when all other means are exhausted.

Risk of hardware damage!

⇒ **DO NOT USE** the “Advanced Spike” procedure without guidance from RSC/ HSC!

RFNoise Check

HP-Z000100Q-031105-02-039

System Type: Avanto
Espreo
TrioTim
SymphonyTim

General

The **RF Noise Check** is only available in the **Expert Mode**.

The procedure is used to measure RF interferences.

The noise measurement is performed with the service sequence rf_noise; the whole receive bandwidth is covered by variations of the center frequency from slice to slice, i.e. 50 slices of 10 kHz cover a -250 to +240 kHz bandwidth.

The evaluation part of the procedure is comparable to the **Spike Check**.

Testresults/ Evaluation

The following output data will be shown under Reports/Tune-Up Results:

- RF Noisy Summary, spike amplitude related to varying center frequencies against specifications

Troubleshooting

In case of problems (status **not ok**) check the following items:

- Refer to TSG/ Tune-up/ QA/ QA Procedures/ Coil Check/ Service Sequence RF_noise.

Stability Check

HP-Z000100Q-081105-02-024

System Type:Avanto
Espreo
TrioTim
SymphonyTim

Expert Mode

The **Expert Mode** allows the user to modify the default parameters. The following selections are possible:

- Spin-echo or gradient echo (Flash) sequence measurement. The spin-echo sequence is selected by default. This setting is exclusive, i.e. for each **Go** performed in the Expert mode, only one sequence can be selected.
- Slice shift of 0 and +50 mm. Center (no shift) is selected as the default. If no selection is made, **Go** is not possible.
- Sagittal, coronal and transverse slice selection. Sagittal slice orientation is selected as the default, but any one, two or all three can be selected. If none is selected, **Go** is not possible.

In addition, the user can modify the TR and TE values from their default settings. If the user selects a TR or TE value that is not valid for the selected sequence, an error message is generated. When selecting Go, the valid input ranges will be displayed.

The **Evaluate** option allows the user to evaluate the stability of the last measurement.

The **Evaluate File** option allows the user to evaluate the stability of an existing raw data file.

Fat Saturation

HP-Z000100Q-081105-02-025

System Type: Avanto
Espreo
TrioTim
SymphonyTim

Expert Mode

Using this mode, the user can select the three orientations (TRA, SAG and COR).

Synthesizer Check

HP-Z000100Q-081105-02-028

System Type:Avanto
Espre
TrioTim
SymphonyTim

Expert Mode

The only difference in the **Expert Mode** is that data are not written to the database.

Field Stability

HP-Z000100Q-081105-02-033

System Type: Avanto

Espree

TrioTim

SymphonyTim

Expert Mode

Principle and Evaluation of MR Measurements

The phase of an MR FID (Free Induction Decay) is determined by the following formula:

$$\varphi(t) = \gamma \int_0^t B(t) dt \approx \gamma B t$$

Fig. 271:

This assumes that the field $B(t)$ does not change much during sampling of the FID. Time 0 is determined by the RF pulse. As a result, the phase at a certain time t directly indicates the field. However, because of hardware imperfections, it is better to take the slope of the phase over time which also determines the field.

To measure the field stability over a certain time, a series of FID measurements is performed. From every FID, the phase over time is calculated and a linear fit is performed with the slope indicating the frequency. The time range of the fit should be long to achieve high accuracy, but it is usually limited by the length of the FID. A good compromise is 20ms, averaging a 50Hz line hum which is of no interest in this measurement.

The total measurement time is determined by the repetition time and number of FIDs.

Evaluation of Field Stability

The result of the field measurement is the field (= frequency) over the measurement time. As a measure of quality, the linear drift over 10min (= typical measurement time of a long imaging sequence) and the maximum deviation from this linear drift are taken.

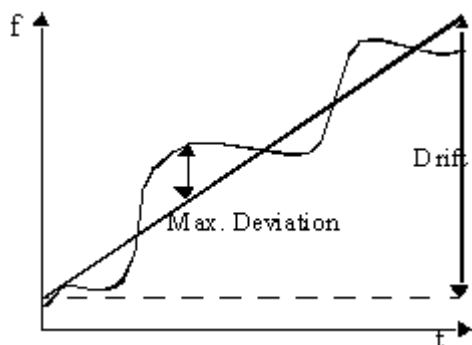


Fig. 272:

When the measurement time is different from 10min, the drift is corrected for 10min.

A drift over 10min should not exceed the pixel bandwidth of a low bandwidth sequence (e.g. 20Hz).

The maximum deviation is important for gradient echo sequences to avoid smearing. To allow long echo times, it should be typically less than 2Hz.

In the user interface you can select repetition time, the no. of acquisition as well as the evaluation time.

☒ **Measurement** Coil Body

Repetition Time (ms) 301

No. Acqu. 1

Eval. Time (ms) 20

☐ **Evaluate File** ☒ Last measurement

☐ Series Loid loidfile

Fig. 273:

The measurement time with default parameters is $512 \times 301\text{s} = 154\text{s}$.

The **Evaluate** option can be used to evaluate the raw data of the last measurement for field instabilities.

The **Evaluate** file option evaluates an existing raw data file for field instabilities.

Test Results

The output in an HTML window (available under Reports/ QAResults/Expert/SfpFieldStabilityExpert) shows the numerical value of the frequency drift (Hz/10min) and the maximum frequency deviation during the measurement time. The graphics show the frequency variation and the drift-corrected frequency variation across the measurement time.

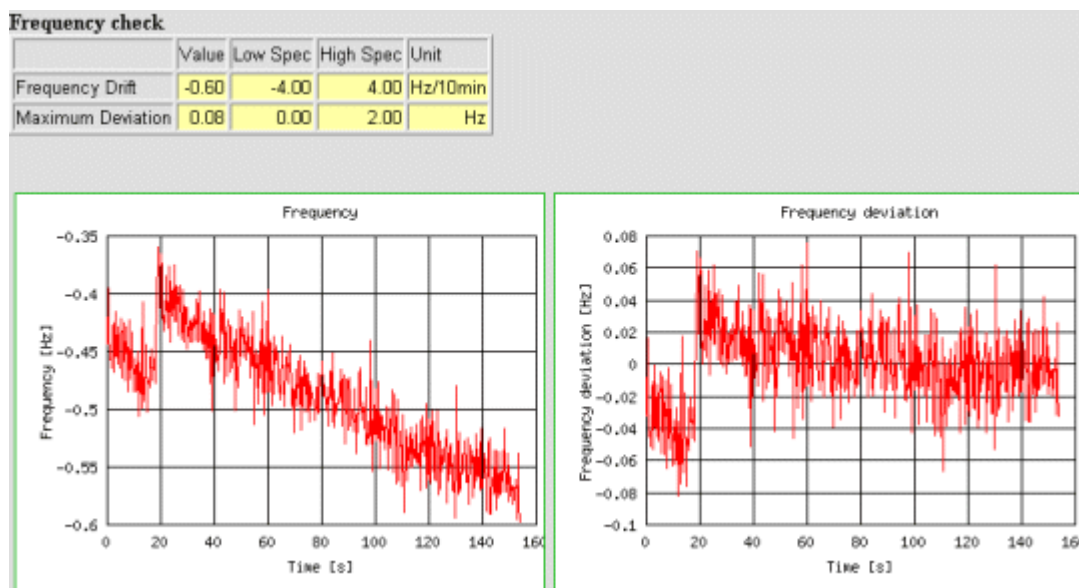


Fig. 274:

The specifications of the magnetic field drift of 0.1ppm/hour (according to 6.4Hz/hour for 1.5T magnets) could be exceeded after ramping the magnet or activating the screen reset. In this case, the frequency drift may be increased up to 200Hz/hour.

Troubleshooting

For further information, refer to TSG/TU/QA.

Stability_LongTerm Check

HP-Z000100Q-081105-02-032

System Type:Avanto
 Espree
 TrioTim
 SymphonyTim

Expert Mode

The function Stability_LongTerm Check/ Expert Mode is available on the system independent of a special license.

The user interface of this function allows you to choose several parameters for the measurement.

Fig. 275:

Additionally, the last measurement or a special Loid File can be evaluated. The parameter **Slice No.** will be used for the evaluation.

Additional numerical output is available in the Expert Mode:

- Statistics of phantom movement:
 - Phantom movement peak-peak in the "Phase" and "Read" direction (Pixel)
 - Standard deviation of phantom movement (pixel)
 - Linear drift of phantom movement (pixel)
 - Standard deviation of phantom movement in relation to the linear drift

Additional graphical output in the Expert Mode:

- Relative peak-peak deviation of the ROI mean values (linear plot)

- Relative peak-peak deviation of the ROI mean values (linear plot)
- SNR in proportion to the mean SNR (ROI size 15x15)
- Ghost intensity in proportion to the mean ghost intensity
- Center position in the readout direction
- Center position in the phase direction

DynFieldMap Check

 HP-Z000100Q-221105-02-002

System Type:Avanto
 Espreo
 TrioTim
 SymphonyTim

General

The purpose of the **dynamic field mapping** function is to measure and analyze the residual field perturbations with sufficient spatial resolution to identify in particular the higher (>2nd) order terms. It supplements the existing tools such that all spatial order terms are detected and checked against specifications. Due to missing compensation facilities for 2nd and higher order terms, the dynamic field mapping function is restricted to the QA procedure.

Expert Mode

Extended capabilities are available within the Expert Mode:

- Each gradient axis can be selected separately and analyzed on its own.
- The measurement can be modified regarding the weighting of the time constants. Besides the standard setting used in normal mode, the weighting may emphasize only the short as well as only the long time constants.

Selection is via the sequence parameters: The duration and timing of the preparing gradient pulses (**Single Pulse** or **Bipolar String**) determine the amount of transient fields and the weighting of the contributing time constants.

Note that the specification values are valid only for the standard setting and are not applicable to any deviant selection.

Physical Background

Gradient pulses are always accompanied by unwanted time-dynamic field perturbations caused by e.g. eddy current or hysteresis effects. These field perturbations comprise multiple terms with different spatial dependencies. The 0th and 1st order terms are analyzed and compensated by ECC and CTC tune-up procedures and checked against specifications by the respective QA procedures.

Measurement

The procedure measures the perturbing transient field distribution using the spherical body phantom.

Test results/ Evaluation

The following output data will be shown under Reports/ QA Results:

- Peak-peak values on ppm
- Spherical harmonics

Troubleshooting

A possible reason for measured values out of tolerance is a defective gradient coil.

Please contact HSC for further actions.

RF Path Attenuation Check

HP-Z000100Q-081105-02-034

System Type:Avanto

SymphonyTim

Expert Mode

General

The reflection for the Body Coil is very high, because of the missing Body Coil tuning. This means that the reflected RF-power is transmitted to the 50 Ohm dummy termination in the TAS. Once there, the voltage is measured by the receiver and evaluated by the RFSWD software.

voltage, the dummy load has to be calibrated during tune-up.

The damping factor of the TAS-TALES-TAS line has to be within a specified range. Therefore, if a component in this line, e.g. the BCCS is replaced in the course of service, the RF Path Attenuation Check has to be performed afterward.

Measurement

The procedure is divided in two steps. Step 1 is a loop via TAS to the dummy load where the RF-amplitude is measured. The path for step 1 is switched automatically. Step two is a loop via TALES to the dummy load where the RF amplitude is measured once more. The ratio between the first and second amplitude provides the damping factor of the TAS-TALES-TAS line.

- ☒ **Step 1** Measurement of RF-signal using loop via TAS-Dummy.
- ☐ **Step 2** Measurement of RF-signal using loop via TALES.
Please terminate TALES output using short circuit termination.

Fig. 276:

Measurement set-up for Step 2:

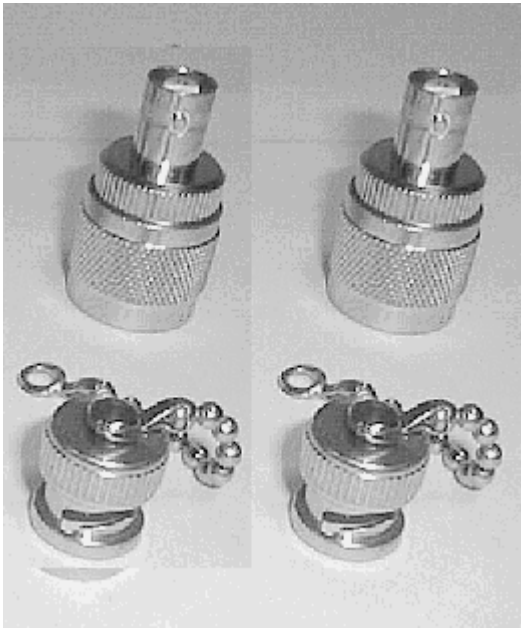


Fig. 277:

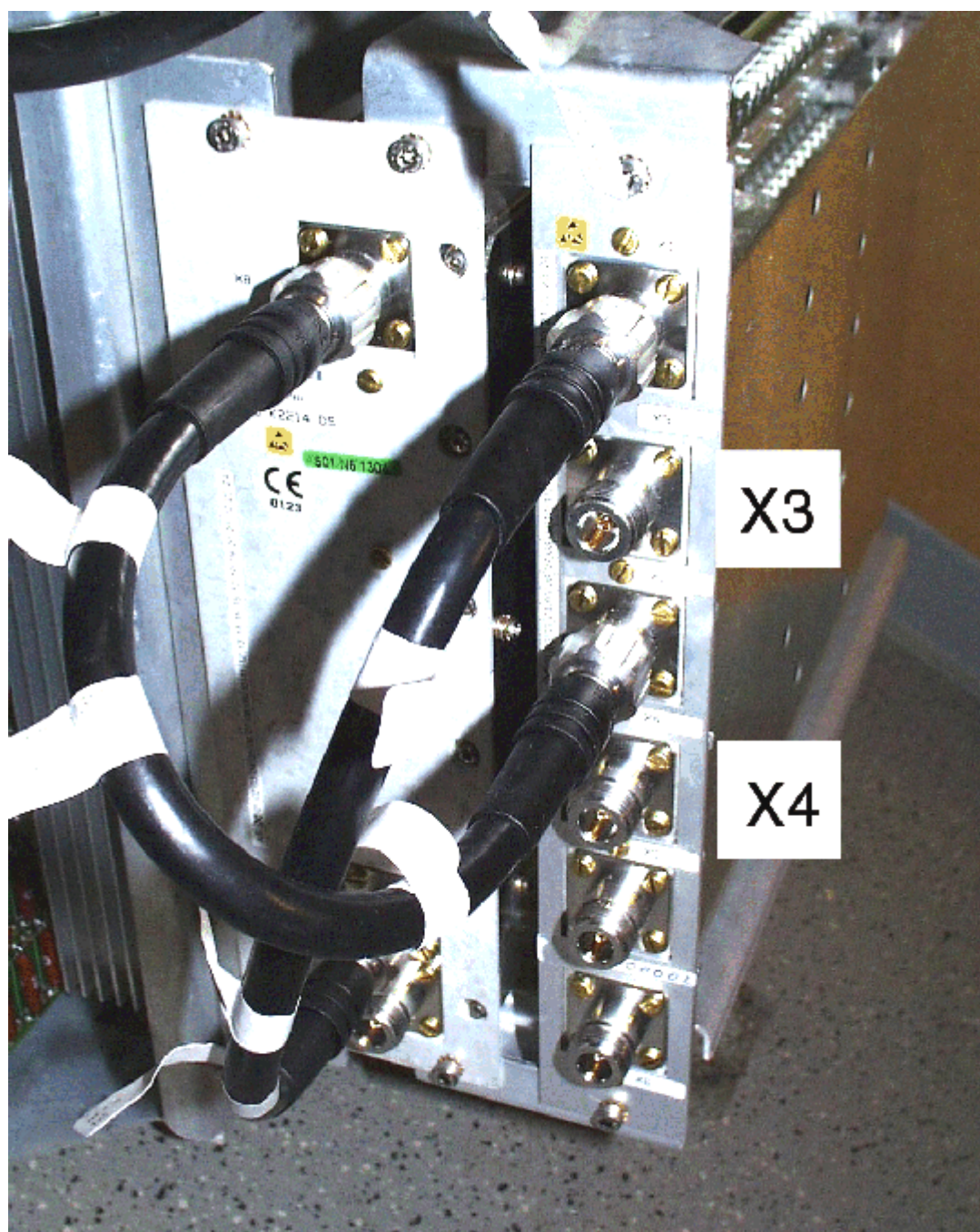


Fig. 278:

Disconnect the RF cables to the Body Coil from TALEs X3/X4 and terminate TALEs X3/X4 using two N to BNC adapters (part no. 3164514) and two BNC short-circuit terminations (part no. 3110392).

20 Shim

20.1 Procedures

20.1.1 General

HP-EH110289-140206-02-001

System Type: Avanto
Espreo
TrioTim
SymphonyTim

Shimming

Superconductive magnets for MR imaging systems must have a homogeneous magnetic field at the required strength. To this end, a shimming technique is used to reduce field inhomogeneity to a minimum.

Physics Background

The primary objective of MR systems is to provide diagnostic images for medical purposes. Images of the body can be derived from the information obtained by exciting protons in a magnetic field using a radio frequency signal. As the protons relax, they release their own radio signal that is detectable and provides information about the chemical contents of the body. The image quality depends on the homogeneity of the magnetic field: the better the homogeneity, the better the image quality. Shimming is an integral part of the magnet installation. It compensates for unit fabrication spread and environmental effects on the system homogeneity.

Shim

HP-EH110289-140206-02-002

System Type:SymphonyTim

Passive Shim Flowchart

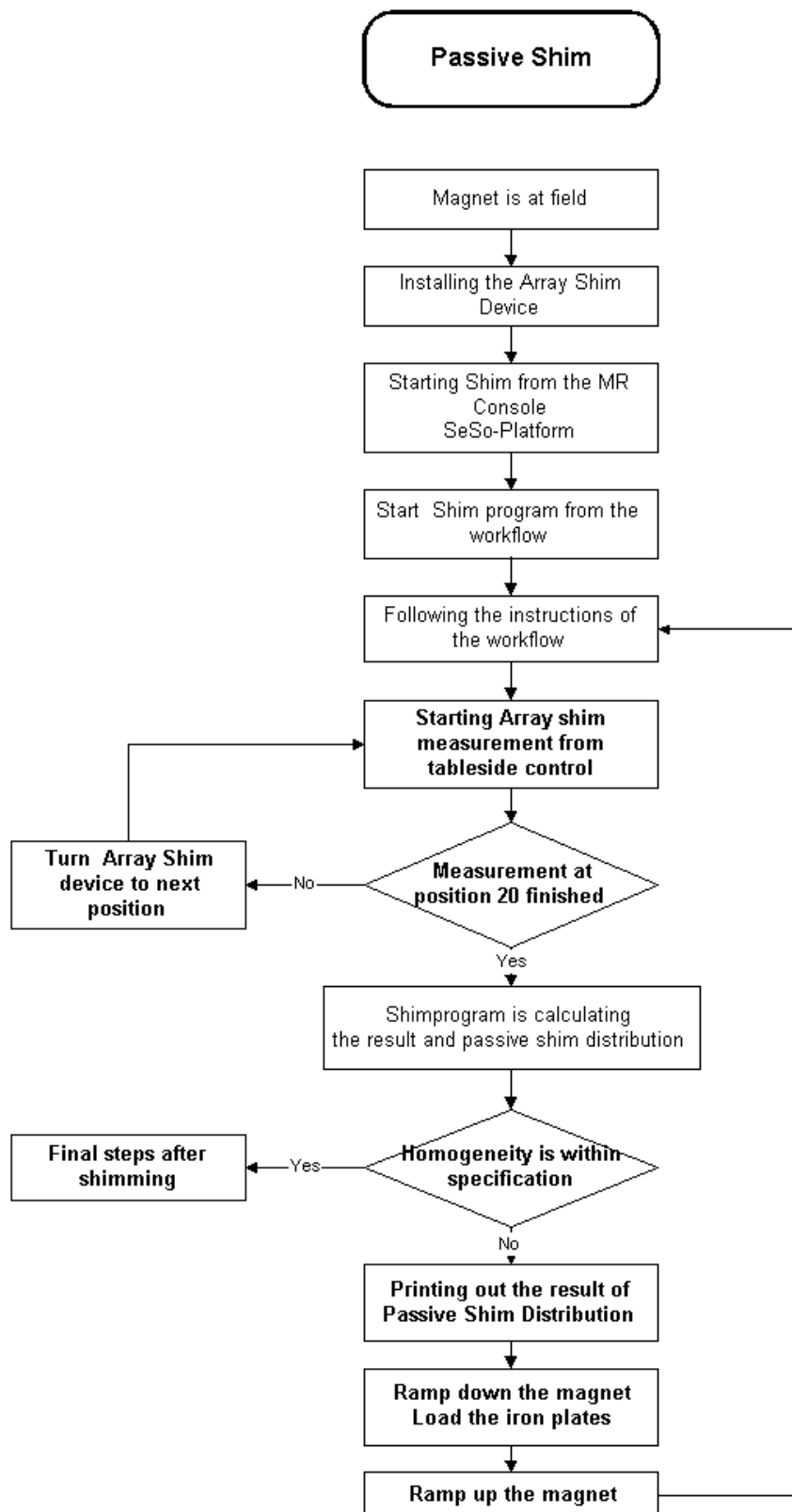


Fig. 279:

Prerequisites for Shimming

NOTE

The iron distribution has to be removed after an upgrade to Symphony, A Tim System. Start shimming from bare.

- System is switched on and magnet is at field.
- Frequency of the system is within following range:
Maximum frequency: 63.700 000 MHz; +0 Hz
Minimum frequency: 63.500 000 MHz

To prevent Software Problems

Reboot the system with GPA and the shim amplifier is on

- Move the table into the top position. It is slightly magnetic and may affect the magnetic field homogeneity .

The shim position for the table has to be the same as the position used later on for patient examinations.

- Switch off the gradient cabinet:

- K2217:**
1. Switch off circuit breaker **F110** at ACC
 2. Switch off circuit breaker **F6** in the ACC (GPA control)

- Switch off the shim amplifier in the ACC (if it exists) **Shim** circuit breaker **F21**

NOTE

After the magnet is ramped up, wait for the time specified below before taking the measurement: At least 30 minutes for the bare and first iteration At least 1 hour for subsequent iterations

Array Shim Device

The array shim device for Harmony and Symphony is not part of the auxiliary tools shipped with the system. Instead it has to be ordered separately from the business unit.

The array shim device consists of 4 parts.

- The mechanical configuration
- The array disk: Consists of 25 probes arranged on the disk, having a radius of 25 cm. The array disk should not be opened. If it is, the coils will be detuned and the shim device is **no longer operational**.
- The multiplexer: The multiplexer contains a transmit/receive switch to enable the Array Shim Device for transmit and receive. During the measurement, the multiplexer switches to the next higher probe after each RF pulse. At the end of the measurement, the multiplexer is reset to the first probe.

- Cable to connect patient table and array shim.

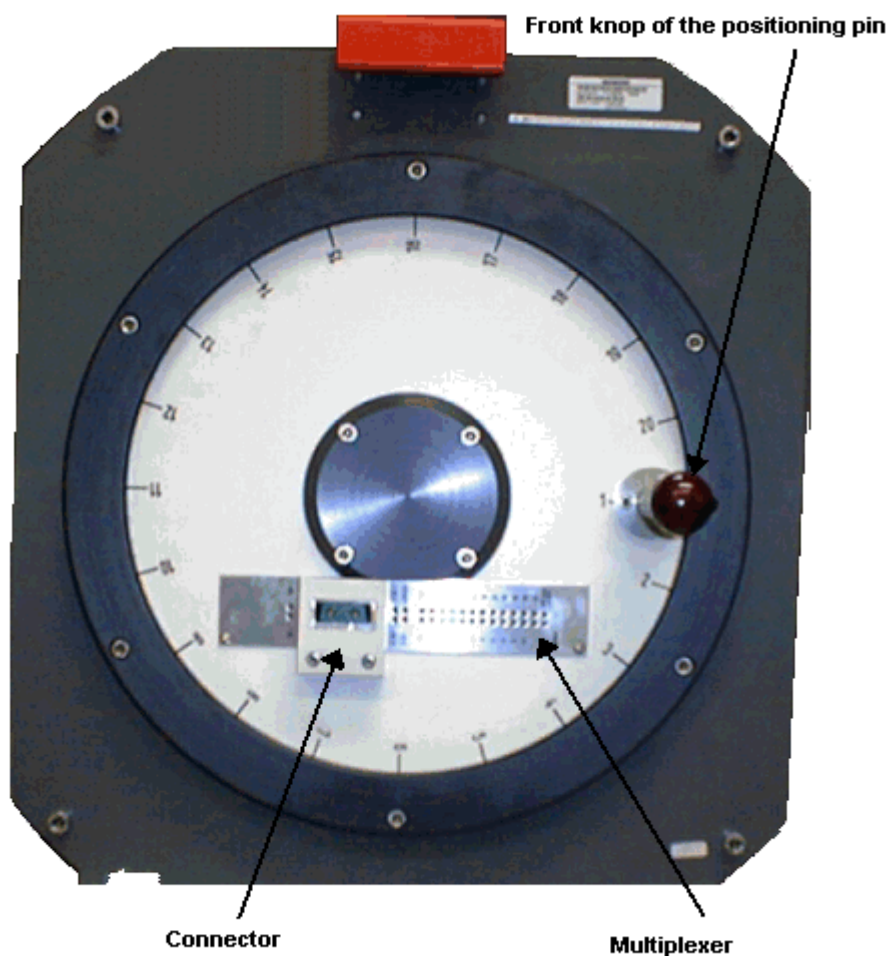


Fig. 280:

The array shim device is shipped in its own crate together with the **additional passive shim kit**. The passive shim kit is packed in a separate small crate and located underneath the array shim device. Use this passive shim kit only when it is not possible to shim the magnet with the Balance kit (shipped with the magnet) and additional shim plates are required due to external influences, i.e. iron shielding or iron girders in the vicinity of the magnet.

The crate is padlocked for additional safety. The combination for the lock is **378**.

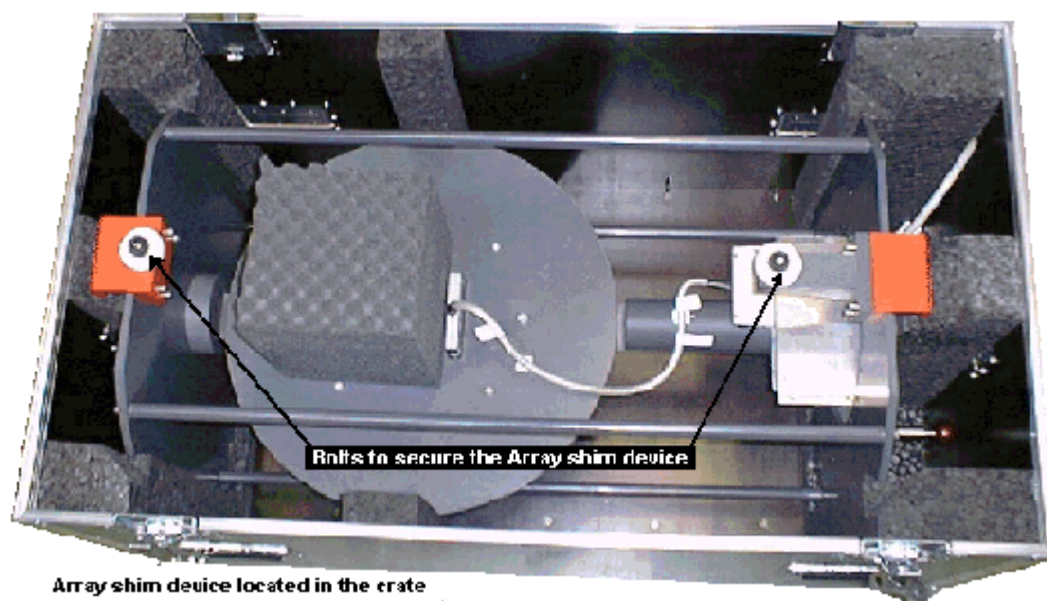
Installation of the Array Shim Device

- Pick up the array shim device by the two red handles, lift it up carefully and remove it from the crate.

NOTE

To prevent damage to the array shim device, use only the two red handles for transport.

Two persons are necessary for lifting the array shim device.



Array shim device located in the crate

Fig. 281:

- Move the patient table into the lowest position.
- Slide the array shim device into the magnet.
Slide it up to the arrow, the label is in front to the left.



Fig. 282:

The array shim device is now positioned in the magnet.

- Secure the array shim device by tightening the upper two bolts in the resonator.

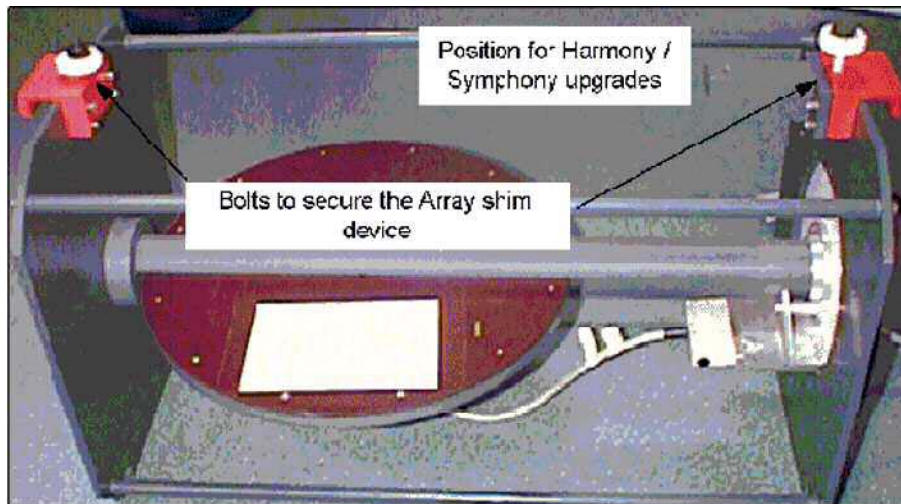


Fig. 283:

- **Move the patient table into the top position.** The table is slightly magnetic and has to be shimmed as well.
- Connect one end of the array shim cable to the multiplexer and the other end to **connector 1** of the patient table.



Fig. 284:

- Set the array shim device to **position 1**, pull out the red front knob of the positioning pin, turn the shim device into position 1.

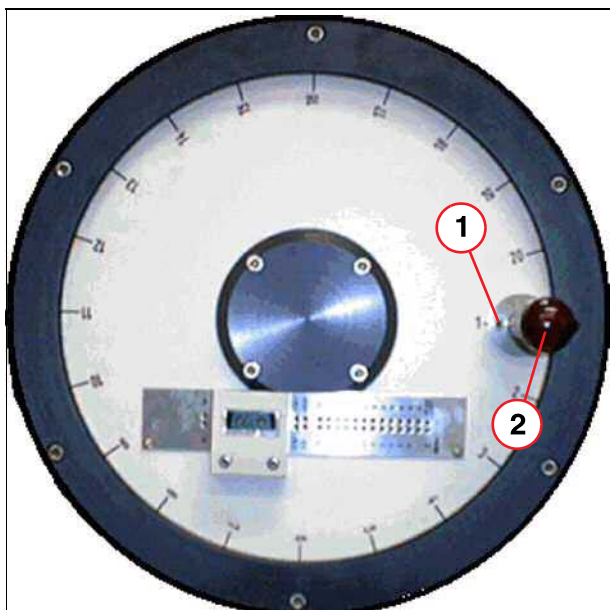


Fig. 285:

Pos. 1 Position 1
Pos. 2 Front knob

Starting the Array Shim Measurement

The array shim measurement consists of two parts:

- The frequency search using the center probe of the array shim device started at the MR console (Frequency range = Center frequency + / - 100kHz)
- The measurement of the 50 cm sphere in 20 steps started at the control panel.

Each measurement consists of 24 measurement points on a 50 cm circle.

The probe signals measured are converted into frequency data and stored into a field data record at the end of the measurement.

This field data record is used for calculating the shim data.

Prerequisites

- The system is switched on and the magnet is at field
- After the magnet is ramped up, wait for the time specified below before taking the measurement:
 - At least 30 minutes for the bare and first iteration
 - At least 1 hour for subsequent iterations
- Move the table into the top position. It is slightly magnetic and may affect the magnetic field homogeneity .

The shim position for the table has to be the same as the position used later on for patient examinations.

- During the array shim measurement, the front funnel can be reinstalled to improve access to the control panel. Refer to "Installing the cover"
- Switch off the gradient cabinet.

- K2217:**
1. Switch off circuit breaker **F110** at ACC
 2. Switch off circuit breaker **F6** in the ACC (GPA-Control)

- Switch off the shim amplifier in the ACC (if it exists)) **Shim** circuit breaker **F21**.

Starting the Service Software

- Start the Service Software UI at the MR console.
- Select **Shim**
- Select the function **Shim** from the workflow
- Start the shim program with **Go**

- The workflow starts with the array shim device positioning.
 - ⇒ The system applies a +1mT/m gradient and measures the frequency at plane 12 and 13
 - ⇒ The system applies a -1mT/m gradient and measures the frequency at plane 12 and 13 again.

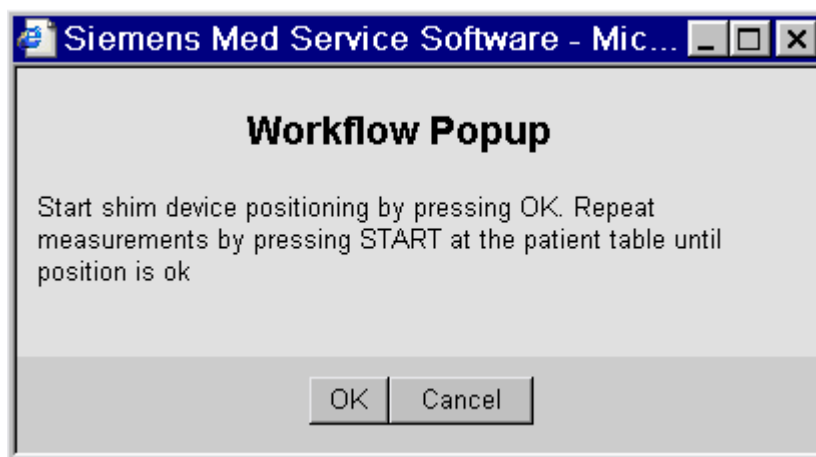


Fig. 286: Workflow

- ⇒ Calculation of the offset from the gradient center
- ⇒ Display of the shift on the display



Fig. 287: Array shim device centering

- ⇒ Example: If the shift is +2.2 mm move the array shim device 2.2mm to the service end.
 - ⇒ Example: If the shift is -2.2mm move the array shim device 2.2mm to the patient end.
 - ⇒ Repeat until the deviation is within +/-1mm
- Switch off the GPA and the Shim Power Supply

- After the frequency adjustment is finished, press OK and start the measurement at the patient table

The following appears at the patient table display:

MEASURE STEP 1

Set the array shim device to position 1 as follows: Pull out the red front knob of the positioning pin up to the end stop with one hand. Use the other hand to turn the array shim device with the positioning pin. Subsequently, release the knob and turn until the positioning pin locks into position 1 as shown in the image below.

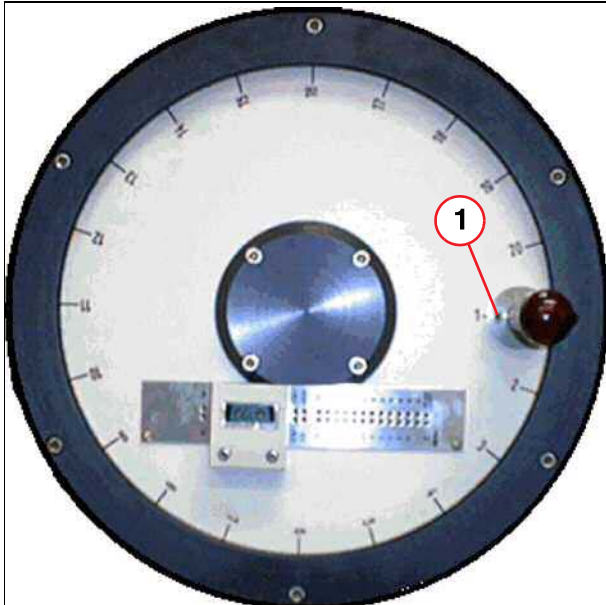


Fig. 288:

Pos. 1 Position 1

- Press the Start/Stop button on the control panel.



Fig. 289:

Pos. 1 Start/Stop

STEP 1 DATA OK

After this, the following message is shown on the patient table display:

MEASURE STEP 2

- Turn the array shim device to position 2

- Start the next measurement by pressing the start/stop button on the control panel.
- Repeat this procedure until the following message appears with position 20:

MEASUREMENT FINISHED

The measurement has been completed. You can now return to the MR operating console. The shim program is checking the specification values and is calculating the passive shim distribution.

After completion the result is saved.

The result can be found in the "Session History" or "Shim Results" folder.

After the calculation is finished, an info box appears

- Select **"Shim Results Harmonics"**
- Check the pk/pk, vrms and second order term specifications.
- Check term a(11.0) for correct positioning of the plotting rig.

NOTE

Values out of specifications are highlighted

Report Files
Purge Files

SiteInfo

- Imprint
- Installation Protocol
- Configuration
- MR Configuration
- License
- About SAM

Tune-Up/QA/Tests

- Session History
- TuneUp Results
- QA Results
- Shim Results
- Test Results
- Export Data
- QA Coil List
- Backup/Restore
- syngo StartUp log

Startup Process

- Process Diagnostics

Remote Diagnostics

- Customer Log

• **Shim Results Harmonics:**

◊ **Specification Check:**

Specification Check

Magnetic Field: 0.948977751183664 T

Sphere Diameter: 0.5 m

	type	current	min	max	unit
specs					
1	pk_pk	245.39		40	ppm
2	v111s	30.148		4	ppm
3	a(2,0)	-4.00	-1	1	ppm
4	a(2,1)	43.06	-1	1	ppm
5	a(2,2)	-23.11	-1	1	ppm
6	b(2,1)	9.62	-1	1	ppm
7	b(2,2)	-63.04	-1	1	ppm
8	a(11,7)	1.27	-4	4	ppm

Fig. 290:

- Check the "Shim Results Iron" folder for "error"

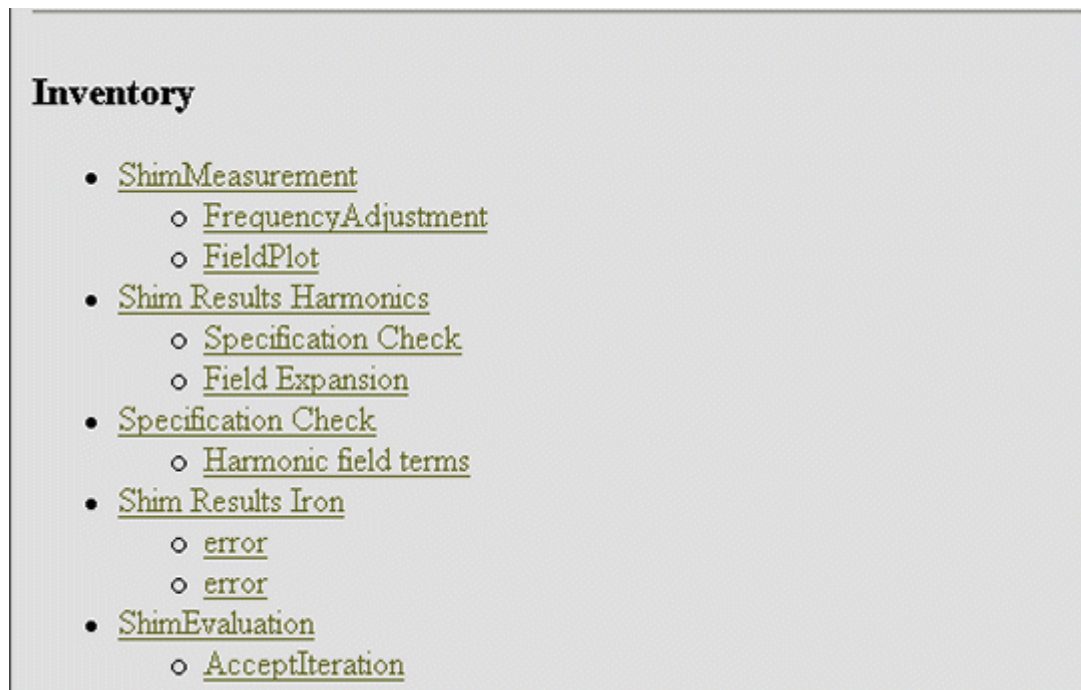


Fig. 291:

NOTE

Details for errors are listed in the chapter "Error handling".

- Select "**Passive Shim Distribution**" if no error appears

Passive Shim Distribution

Iron_to_load ; Shimset: SIEOR24/70-16T15P#1

	Total Mass	Mass Change	Mass Units	Maximum Thickness	Change in Maximum Thickness	Thickness Units
Total	1.369	1.369	kg	3.737	3.737	mm

Total Plates

	Plate 1	Plate 2	Plate 3	Plate 4
Tray	load (total)	load (total)	load (total)	load (total)
All	95 (55)	85 (85)		

Plates in Tray 1

	Thickness	Plate 1	Plate 2	Plate 3	Plate 4
Position	change (total)	load (total)	load (total)	load (total)	load (total)
A	0.000 (0.000)				
B	0.000 (0.000)				
C	0.753 (0.753)	2 (2)	2 (2)		
D	0.000 (0.000)				

OK Cancel

Fig. 292:

- Check the "Passive Shim Distribution" and press "**OK**".

NOTE

OK: The calculated iron will be stored in the database, if the OK button is pressed

Cancel: The calculated iron will not be stored in the database , if the Cancel button is pressed

- Print out the distribution list.
- Count and sort the plates.
- Load the plates.
- Repeat an array shim measurement.

Homogeneity is within Specifications

- If the magnet is shimmed within specifications, the shim is completed.

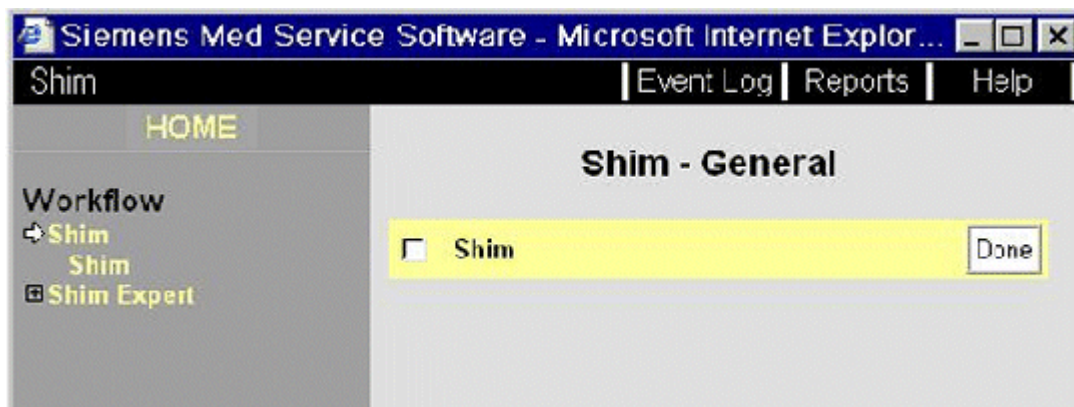


Fig. 293:

Remove the Cover

Removing the Funnel

To maintain the correct position of the array shim device, do not remove it from the magnet during shimming.

Removing the Front Cover Rings

Remove the display to take off the front cover rings.

- Remove the display by loosening the upper two screws and sliding the display toward the front.
- Keep the cable hanging from the display. Push the left and right ring upward until you can remove them.
- After you removed the upper rings, reinsert the display.
- Remove the two mounting screws of the lower ring. Slide the ring upward to the right at approximately 30°.

- Turn the ring toward the front as shown in the image below so that it drops out between cover and funnel.



Fig. 294:

- Prior to removing the funnel: Remove the three bottom nuts at the funnel. Move the table into the lowest position so that you can access the remaining mounting nuts.
- If the system includes an exchangeable patient table, remove it to facilitate taking off the funnel.



Fig. 295:

- Pull out the display again so that you can remove the top mounting nut of the funnel.

- Pull the funnel upward and remove the microphone cable from the microphone pc board located on top in the center of the funnel.
- Remove the funnel carefully from the magnet by slowly pulling it upward. You may have to pull out the cable to the control panels and the display.
- Place the funnel on the floor. Ensure easy access to the shim trays in the gradient coil.

Removing the Shim Trays

⚠ WARNING

The force created by the fringe field of the magnet will pull out the shim plates and move them to a different location in the shim tray.

Adhere to the warning. Otherwise death or serious injury can occur.

⇒ Ramp down the magnet before removing and installing the shim trays.

The shim trays are accessible from the patient end.



Fig. 296:

- Remove the mounting screws of the shim trays.
- Carefully pull the shim trays from the gradient coil.

Each shim tray is identified by a number in front, ranging from **01** to **16**.

When you install the tray, ensure that the number matches the number embossed in the gradient coil. Tighten the trays with the screws.

Installing the Shim iron

Shim kit

The figure shows the shim tray with the position of the trays.

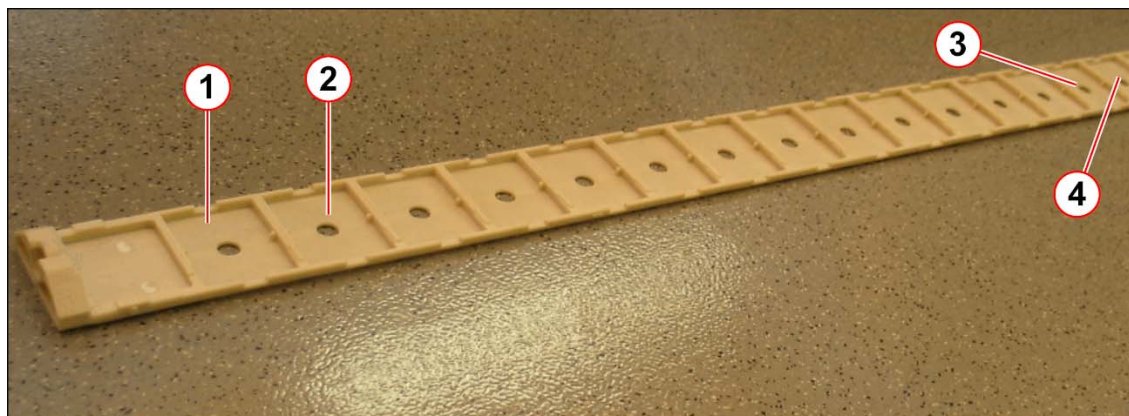


Fig. 297: Shim tray

Pos. 1	Position A
Pos. 2	Position B
Pos. 3	Position P
Pos. 4	Position R

The figure shows the shim plates and the packing shims.

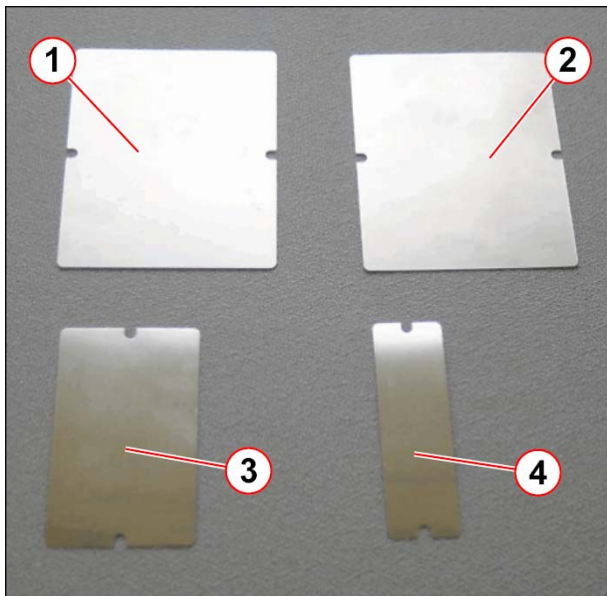


Fig. 298: Shim plates

Pos. 1	Plate 1 (64.8 x 80 x 0.3 mm)
Pos. 2	Plate 2 (64.8 x 80 x 0.1 mm)
Pos. 3	Plate 3 (64.8 x 40 x 0.05 mm)
Pos. 4	Plate 4 (64.8 x 20 x 0.05 mm)

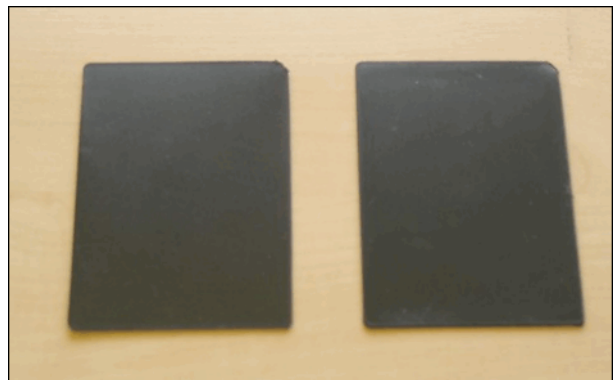


Fig. 299: Packing shims (75 x 55 x 1 mm, polypropylene molding)



Fig. 300: Insulation shim (80 x 64.8 x 0.19 mm, polyester packer)

Shim Plate Distribution

Passive Shim Distribution

iron_to_load ; Shimset: SIEOR24/70-16T15P#1

	Total Mass	Mass Change	Mass Units	Maximum Thickness	Change in Maximum Thickness	Thickness Units
Total	6.057	0.397	kg	3.553	0.000	mm

Total Plates

	Plate 1	Plate 2	Plate 3	Plate 4
Tray	load (total)	load (total)	load (total)	load (total)
All	5 (442)	42 (195)	168 (446)	60 (85)

Plates in Tray 1

	Thickness	Plate 1	Plate 2	Plate 3	Plate 4
Position	change (total)	load (total)	load (total)	load (total)	load (total)
A	0.000 (0.000)				
B	0.000 (0.000)				
C	0.178 (0.178)		1 (1)	3 (3)	1 (1)
D	0.000 (2.263)	0 (8)		0 (1)	
E	0.000 (0.000)				
F	0.120 (1.360)	0 (4)	1 (2)	1 (2)	
G	0.000 (0.642)	0 (2)		0 (3)	0 (1)
H	0.070 (1.004)	0 (3)		3 (7)	
J	0.012 (0.715)	0 (2)	0 (1)	0 (2)	1 (1)
K	0.035 (1.103)	0 (3)	0 (2)	1 (2)	1 (2)
L	0.012 (0.508)	0 (1)	0 (2)	0 (1)	1 (1)
M	0.000 (1.240)	0 (4)	0 (1)	0 (1)	
N	0.000 (1.617)	0 (5)	0 (2)	0 (1)	
P	0.000 (0.000)				
R	0.000 (0.000)				
Total Plates		0 (32)	2 (11)	8 (23)	4 (6)

Fig. 301: Passive Shim Distribution

The table shows the number of plates:

The number without brackets is the number of plates that have to be loaded for the actual iteration

The number in brackets shows the total number of plates.

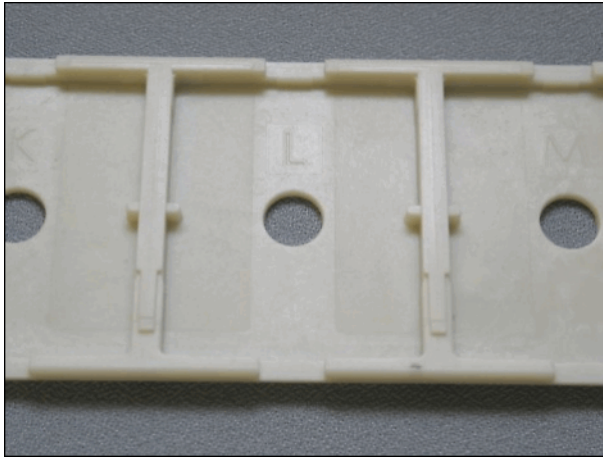
The first position of the shim trays (mounting location at the shim tray) is identified with **A** followed by **B, C, D, E, F, G, H, J, K, L, M, N, P, R**.

NOTE

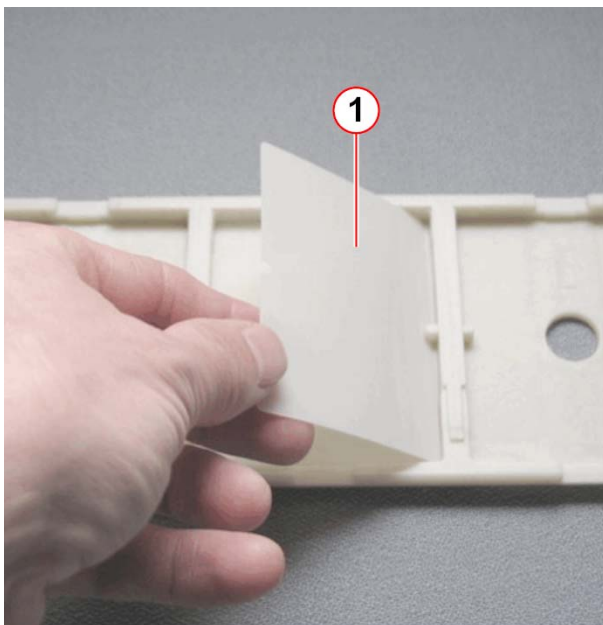
Empty pockets should not be covered. The covers of empty pockets may make noise.

Store remaining covers on-site for re-shimming.

- Install the shim plates as shown in the following figures.



- Select the correct shim tray number and the correct pocket.



- Install an insulation shim first.

Fig. 302: Insulation shim

Pos. 1 Insulation shim

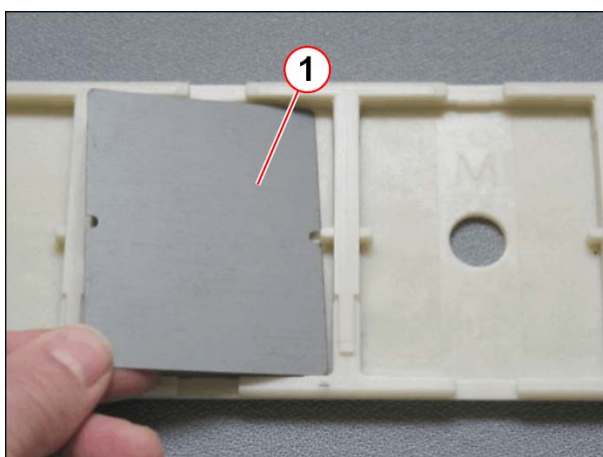


Fig. 303: Shim tray

Pos. 1 Shim plate

- Install the required plate size
 - ⇒ Use one layer of insulation shim between each shim plate.

NOTE

Use one insulation shim between each shim plate.

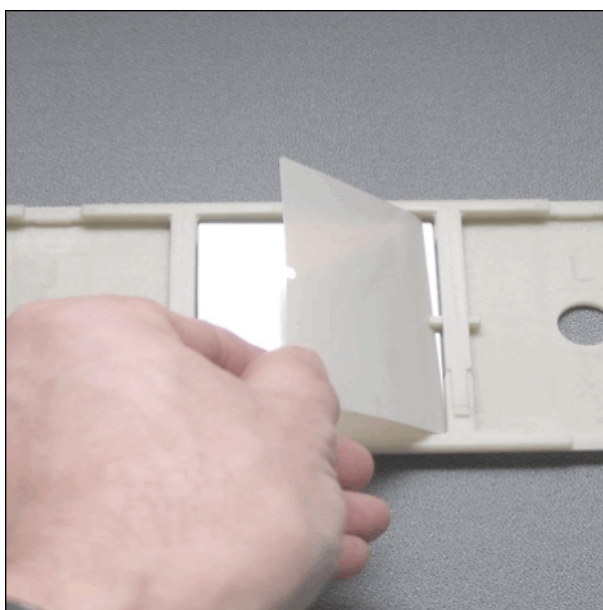
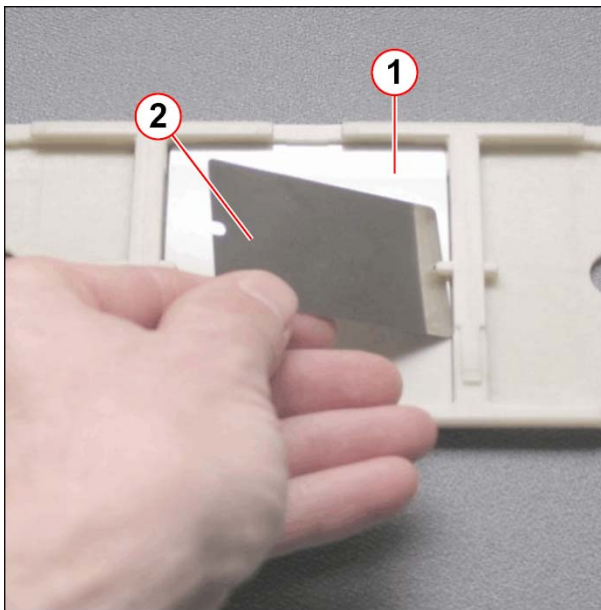


Fig. 304:

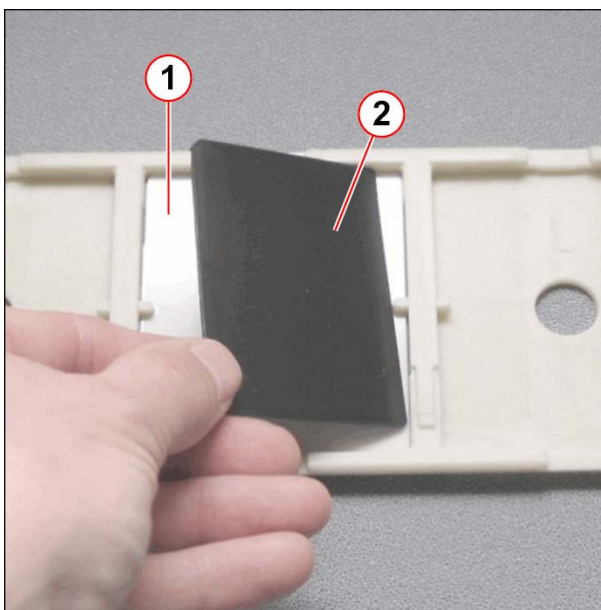
- Install one insulation shim between each shim plate.



- Install the required shim plate.

Fig. 305: Shim tray

Pos. 1 Insulation shim
Pos. 2 Shim plate



- Cover the last shim plate with an insulation shim.
- Fill the space between the shim plates and the cover with packing shims.

Fig. 306: Shim tray

Pos. 1 Insulation shim
Pos. 2 Packing shim

NOTE

Depending on the space between the shim plates and the cover, use 1 mm packing shims to keep the shim plates from moving in the shim trays.

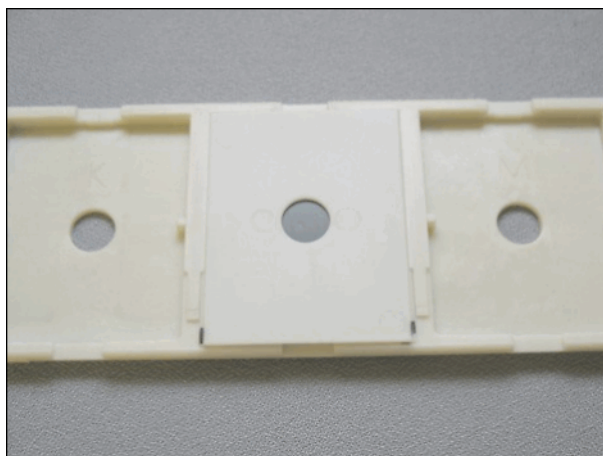
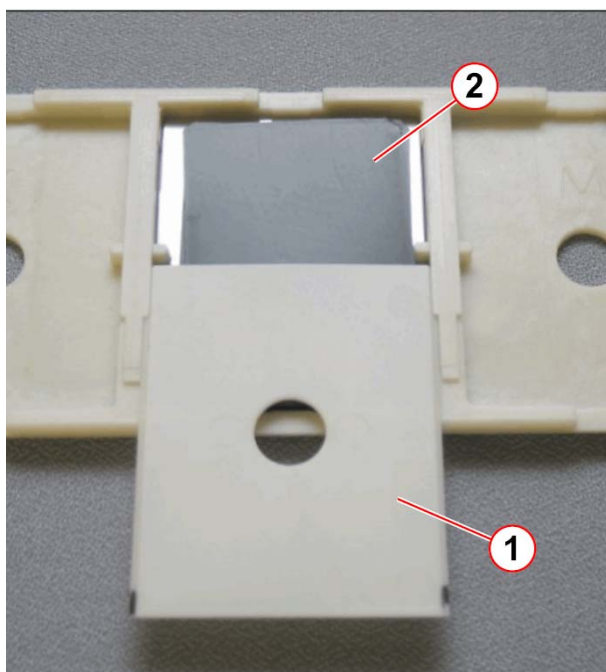


Fig. 307: Shim tray

Pos. 1 Cover

Pos. 2 Packing plate

- Close the pocket with the cover.

NOTE

Ensure that the shim plates are securely seated in the shim trays. Loose shim plates can cause spikes.

- Install the shim trays in the gradient coil.
- Tighten the shim tray with the screw.

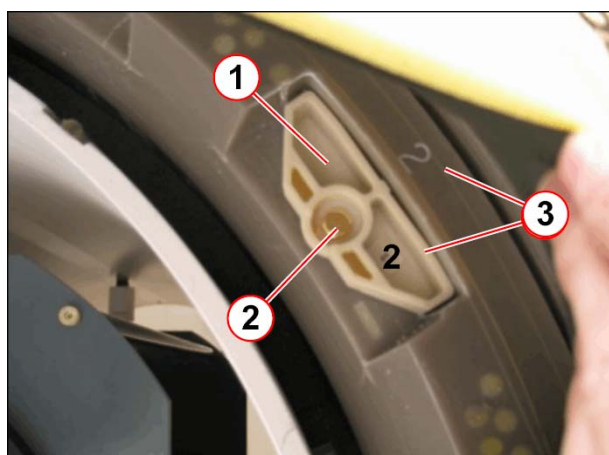


Fig. 308: Shim Tray

Pos. 1 Shim tray
Pos. 2 Screw
Pos. 3 Number



NOTE

When you install the shim trays, ensure that the tray number matches the number embossed in the gradient coil.

Installing the Shim iron (old type)

Old type of Shim Tray

The first position of the shim trays (mounting location at the shim tray) is identified with A followed by **B, C, D, E, F, G, H, J, K, L, M, N, P, R**

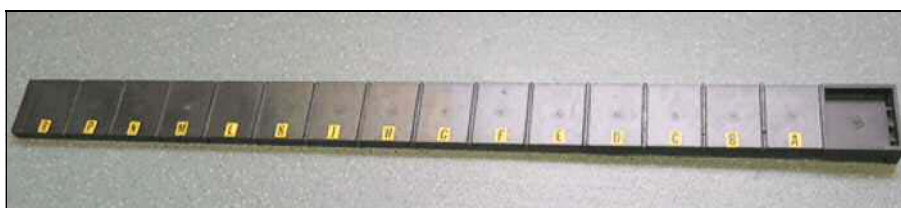


Fig. 309:

- Install the shim plates as shown in the following figures.

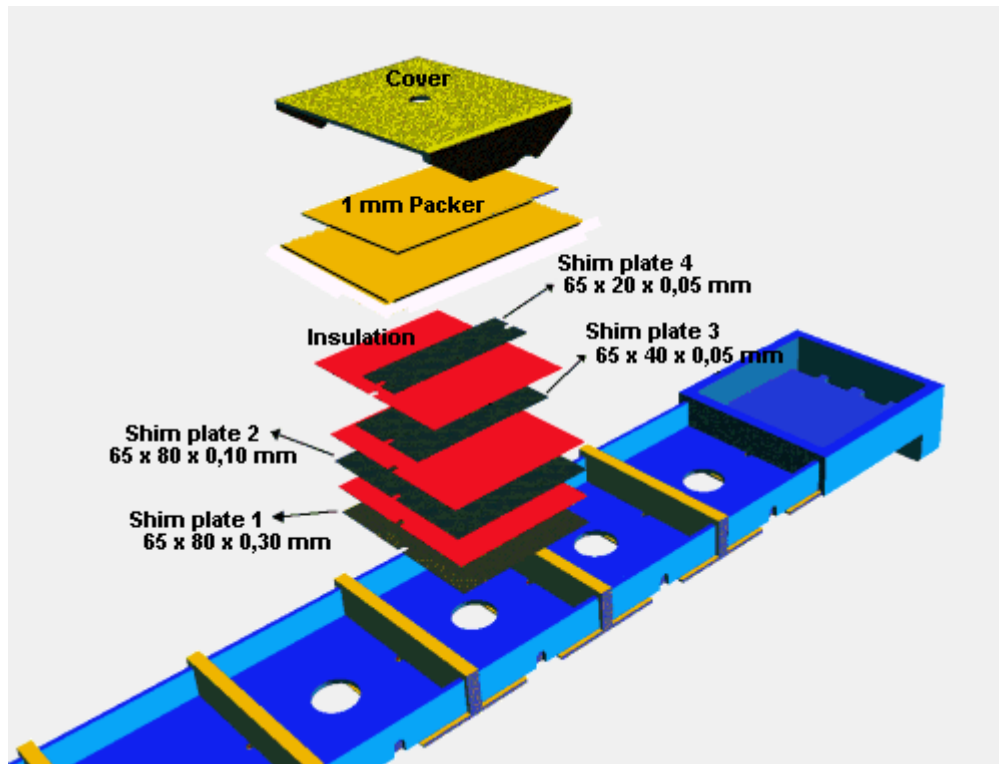


Fig. 310:

NOTE

Ensure that there is a layer of insulation between each shim plate

Depending on the space between the shim plates and the cover, use 1 mm packing plates to keep the shim plates from moving in the shim trays. Rubber pads are no longer used.

NOTE

Ensure that the shim plates are securely seated in the shim trays. Loose shim plates cause spikes.

Installing the Cover

Installing the Funnel

During the array shim measurement, the front funnel can be reinstalled to improve access to the control panel.

- Move the funnel carefully to the magnet.
- Connect the cable to the microphone pc board located on top in the center of the funnel.
- Insert the pins of the front funnel into the holes of the Body Coil
- Pull the cable from the control panels to the patient table control unit.
- Install the display.
- Secure the front funnel only with a few nuts, if a next shim iteration is possible.

Setting the Magnet Current

To compensate for the additional iron and to ensure that the predefined frequency range of (+/- 100 kHz) is maintained for the array shim, a slightly higher magnet current is set when the magnet is ramped for the first time.

This value is the **Operating current/Measured value** listed under **Operating Parameters** in the **Magnet Acceptance Protocol** by **SMT**.

Magnet Current Setting

1. Iteration

Use the value listed as Operating current/Measured value under Operating Parameters in the Magnet Acceptance Protocol by SMT.

2. Iteration and following Iterations

To ensure that the magnet has to be ramped at 4 year intervals only, the magnetic field should lie 100 kHz above the center frequency of the system, given an average frequency drift of approximately 50 kHz a year and a system RF bandwidth of 200 kHz.

NOTE

Maximum frequency for "Tuning Calibration"

Symphony: 63.700000 MHz

- Use the following mathematical formula to compute the current for maximum field strength. The formula takes the tolerances of the different magnet power supplies into consideration

Magnetom Symphony: $I_{\text{final}} = (63.68 \text{ MHz} \times I_{\text{actual}}) / f_{\text{re actual}}$

NOTE

Lower limit of the frequency is 63.50 MHz

Formula to convert the Frequency into Tesla

$$2 * P * f = g * B_0$$

$P = 3.14159$

$f = \text{Frequency in Hz}$

$g = \text{Gyromagnetic coefficient of proton, } 2.67522128 * 10^8 \text{ s}^{-1} \text{ T}^{-1}$

$B_0 = \text{Magnetic field in Tesla}$

Work after shimming

Remaining Shim Plates

The packaged shim plates that are not required should be stored as backup plates in the array shim device kit. These extra shim plates can be used if the delivered shim plates do not suffice.

Only store shim plates in their original packaging in the array shim device kit.

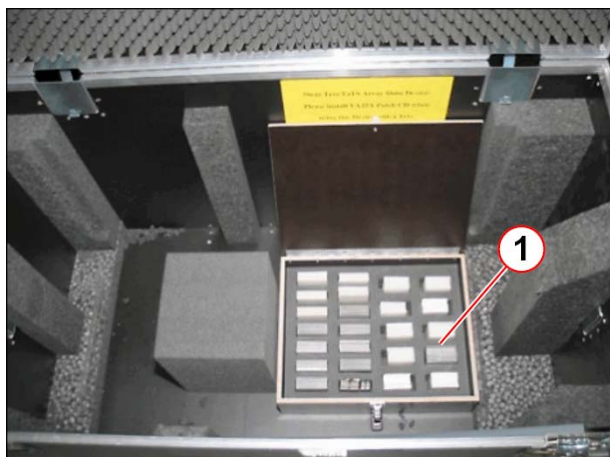


Fig. 311: Array shim device kit
Pos. 1 Shim plates



Fig. 312: Packaged shim plates

Error Handling

What to do if the Frequency Adjustment failed

The following test should be performed if the frequency adjustment failed.

1. Check the shim output for error messages.
 - ➡ “Session History / Magnet Shim”
2. Perform a frequency adjustment using the TX/RX Head Coil or the Extremity Coil
3. Check the RF transmit path

Required tool: Service plug 7715720

Open TestTools and select the following tests:

 - a) RCCS plug1
 - b) RFIS
 - c) RFAS
 - d) RCCS

If the test fails, refer to the RF Troubleshooting Guide.

If the test is OK:

- ➡ Check the magnet current.

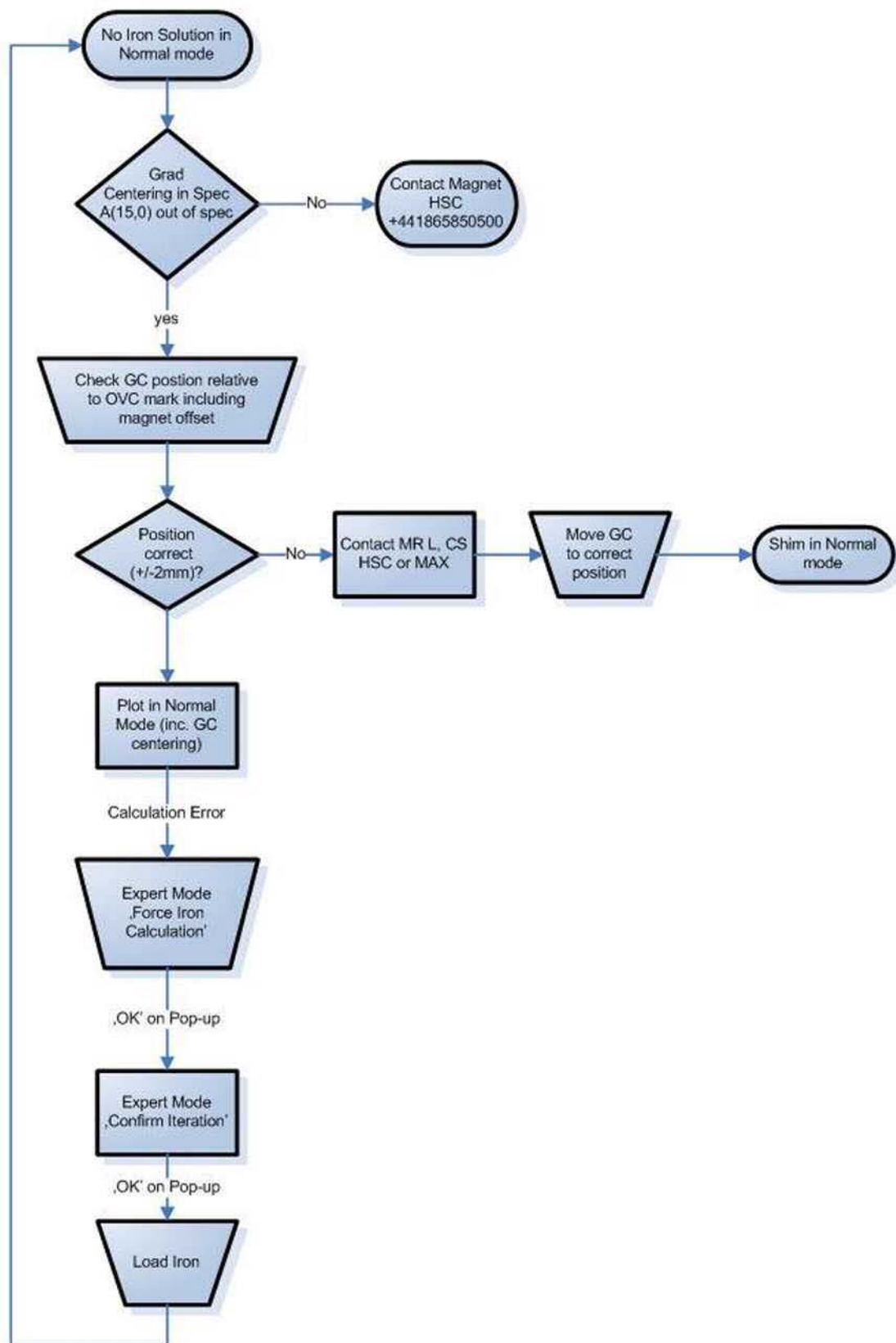
⇒ Check the array shim device.

No Iron Solution in Normal Mode

Check the “StatusLog.txt” file.

⇒ Log file folder: C:\MedCom\service\html\seso\sitedb\log\shim

Use the following workflow if an iron solution was not calculated in the normal mode after the shim device positioning.



Error Messages

The following error messages are provided by the shim program.

The error messages are displayed in the "Shim Results Iron" folder.



Fig. 313:

Error message	Action:
A(11.0) outside +/-8 ppm - Plotting rig is off center. Over + 8 move 2mm to patient end, under -8 move 2mm to service end.	Check the position of the array shim device. Over +8 move device 2 mm to patient end. Under -8 move device 2 mm to service end.
A(21.0); A(22.0); A(23.0) term is too high: Check the MUX cable connection and rig centering.	Check cabling between multiplexer and array disk.
Design Harmonic Check A(12.0) failed - check rig centering"	Design harmonic A(12.0) max -10 ppm Design harmonic A(12.0) min -14ppm Check rig centering.

Error message	Action:
Optimization failed. A solution within spec's could not be found, despite using the maximum mass of iron.	There seems to be too much iron in the environment (room shielding). Please contact Magnet HSC!
Error processing - shim settings file. Cannot find file shim_settings.xml in current directory	Software error. Please contact Magnet HSC!

Further information about troubleshooting is provided in the Troubleshooting Guide.

Shim Support

Send the shim folder as a ZIP-file to the following address to get shim support:

⇒ <mailto:magnet.wscshsc.healthcare@siemens.com>, Magnet HSC)

Shim file folder:

⇒ C:\MedCom\service\html\seso\sitedb\log\shim

20.1.2 Expert

Shim

HP-EH110289-140206-02-003

System Type: SymphonyTim

Expert mode

The expert shim platform exclusively starts the individual workflow steps.



Fig. 314:

Measurement

- Perform a Shim device positioning
 - Position the Array Shim device
 - Select “Shim Device Positioning” and press GO
 - Perform the following workflow
- Perform a field measurement (run the array shim)
- Perform a specification check (harmonics and DSV values)
- Perform a shim calculation (iron calculation)
- Confirm an iteration

Reset to bare

- Remove all iron from the magnet and set the loaded iron in the database to zero and shim the magnet from bare.

Undo last Iteration

- Remove the last iteration from the magnet and the current iron count in the database.

Import SMT Solution File

- Import an externally produced solution (iqshim_output.xml) file.
- Select “Import OMT Solution File”.



Fig. 315:

- Select the externally produced solution file.

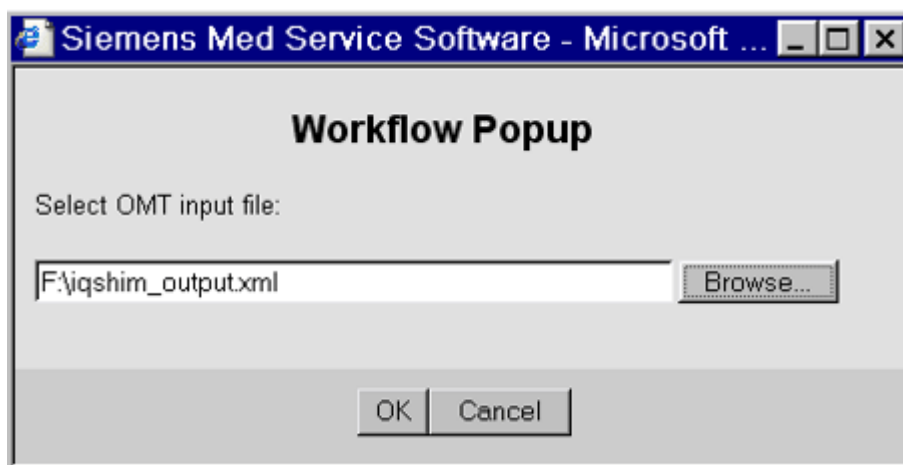


Fig. 316:

"OK" will import the file to the correct directory.

⇒ The solution is displayed. The iteration data are saved to the database.

NOTE

Do not copy the file to the shim directory via Windows Explorer.

Perform an Iron Subtraction Calculation

- Minimize the loaded iron

Force an Iron Calculation

- Ignore any plot checks and attempt to perform an iron calculation

21 Magnet & Cooling

21.1 General

```
----- HP-Z0000XUT-210109-02-001 -----  
System Type: . . . . . TrioTim  
SymphonyTim
```

The function enters parameters for the MSUP, Cooler, and CCS.

21.1.1 Disclaimer

HP-Z0000XUT-210109-02-002

System Type:TrioTim

SymphonyTim

NOTE

The installation and service of equipment described are to be performed by qualified personnel who are employed by Siemens or one of its affiliates or who are otherwise authorized by Siemens or one of its affiliates to provide such services.

Assemblers and other persons who are not employed by or otherwise directly affiliated with or authorized by Siemens or one of its affiliates are directed to contact one of the local offices of Siemens or one of its affiliates before attempting installation or service procedures.

Document revision level

The document corresponds to the version/revision level effective at the time of system delivery. Revisions to hardcopy documentation are not automatically distributed.

Please contact your local Siemens office to order current revision levels.

21.1.2 Introduction

HP-Z0000XUT-210109-02-003

System Type: SymphonyTim
TrioTim

The various parameters for the magnet, the water cooling system and the inside cabinet temperature are configured and displayed in the **Magnet & Cooling** main menu.

In a local service session, the user has access to all available functions.

In a remote service session, the user has access to specific functions only:

- **Limited Access:**

- ⇒ Cooling Status

- **Full Access:**

- ⇒ Cooling Status

- ⇒ Magnet Status

- ⇒ Initialization

- ⇒ Alarm Status

21.2 User interface

21.2.1 Platform Layout

HP-Z0000XUT-210109-02-004

System Type: TrioTim
SymphonyTim

The Magnet & Cooling platform consists of a navigation bar and navigation frame:

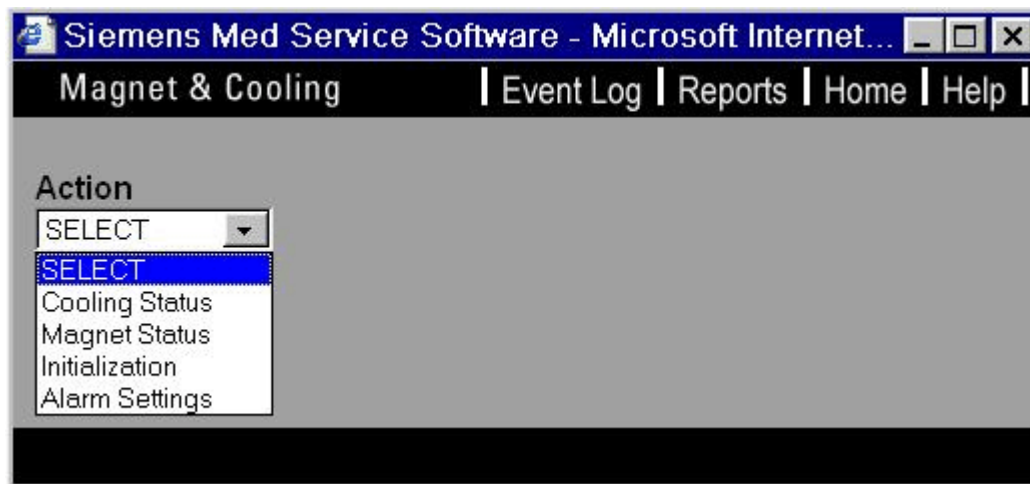


Fig. 317:

21.2.2 User Selection

HP-Z0000XUT-210109-02-005

System Type: TrioTim
SymphonyTim

The Magnet & Cooling main menu provides four actions:

- Magnet Status
- Cooling Status
- Initialization
- Alarm Settings

21.3 Functions

21.3.1 Cooling Status

HP-Z0000XUT-210109-02-012

System Type:SymphonyTim

The actual cooler values are displayed in the Magnet & Cooling UI by selecting “Cooling Status.”

The Cooling Status menu provides the actual values of:

- the ACC cabinet air temperature.
- the GPA cabinet air temperature.
- the water level of the secondary water in the RCA cooler.
- the primary water flow.
- the secondary water flow
 - ⇒ Water flow to the Gradient coil.
 - ⇒ the water flow to the ACC/GPA cabinet.
- the water temperature of the primary water
- the water temperature of the secondary water

The values will be updated by selecting “Update”

21.3.2 Magnet Status

HP-Z0000XUT-210109-02-010

System Type: TrioTim
SymphonyTim

The magnet status menu provides actual values and status information of the magnet, ERDU battery and air temperatures.

The following values of the magnet are displayed:

- Date and time of the measurement.
- Helium fill level of helium probe 1 and helium probe 2.
- Temperature of the 20 Kelvin shield, link and bore.
- Temperature of the 80 Kelvin shield, link and bore.
- Status information.

The measurement is performed once a day. The result are stored in the memory.

The time of measurement is set under “Initialization”.

By selecting **Status** or **Update** in the user interface, a helium measurement is initiated.

The following value of the ERDU is displayed:

- ERDU Battery Voltage.

Display of temperature values:

- Air temperature in the RF room
 - ⇒ The temperature is measured in the air temperature measurement unit (ATMU)
The ATMU is located in the patient air fan.
- Air temperature of the RFIS
 - ⇒ Temperature is measured in the RFIS
The sensor is located on the CAN & Control board D2

21.3.3 Initialization

HP-Z0000XUT-210109-02-013

System Type:SymphonyTim

This menu allows to enter different parameters.

Magnet parameter

Calibration of the Helium Level Meter

- Resistance of helium level probe 1.
 - ⇒ 0% and 100%
- Resistance of helium level probe 2.
 - ⇒ 0% and 100%

Time setting

- Time setting for a magnet supervisory test

Magnet Current

- Display of the Magnet current

Cooler (RCA) setting

The set point of the secondary water temperature can be changed.

Buttons of the Platform

- “Save Settings”
 - ⇒ Saves the changed parameters
- “Default”
 - ⇒ Changes the adjustable parameters back to the default values.
- “Upload”
 - ⇒ Saves the magnet values from the HOST to the magnet supervisory.
Necessary after changing the magnet supervisory.
- “Download”
 - ⇒ Saves the magnet values from the magnet supervisory to the HOST.
Necessary after Start-up.

21.3.4 Alarm Settings

HP-Z0000XUT-210109-02-014

System Type: SymphonyTim

This menu shows the alarm and warning parameters.

The default values are set by the factory. You should only change them in case of problems.

Alarm Setting

Magnet

- Helium level warning.

Cabinet

- Air temperature warning of the GPA and ACC.

Warning levels of the cooler

- Water flow of the primary water circuit.
- Water flow of the Grad. Coil circuit
- Water fow of the ACC/GPA circuit
- Water temperature of the primary circuit, min temp.
- Water temperatureof the primary circuit, max temp.

Buttons on the Platform

- “Save Settings”
 - ⇒ Saves the changed parameters.
- “Default”
 - ⇒ Resets the values to the default values.

22 Customer Quality Assurance

22.1 Procedures

22.1.1 Calculation Artefacts

HP-Z000100Q-140206-02-001

System Type:Avanto
Espreo
TrioTim
SymphonyTim

General

Quality Assurance Function

Customer Quality Assurance is an automated Image Quality procedure for checking image artifacts and the signal-to-noise ratio as well as the signal uniformity of the individual coils

User Interface

Platform Layout

The **Customer Quality Assurance** platform is divided into 4 areas:

- Header Bar for general functions
- Function Frame for input and status fields
- Action Bar for platform-specific functions
- Navigation Frame for coil selection

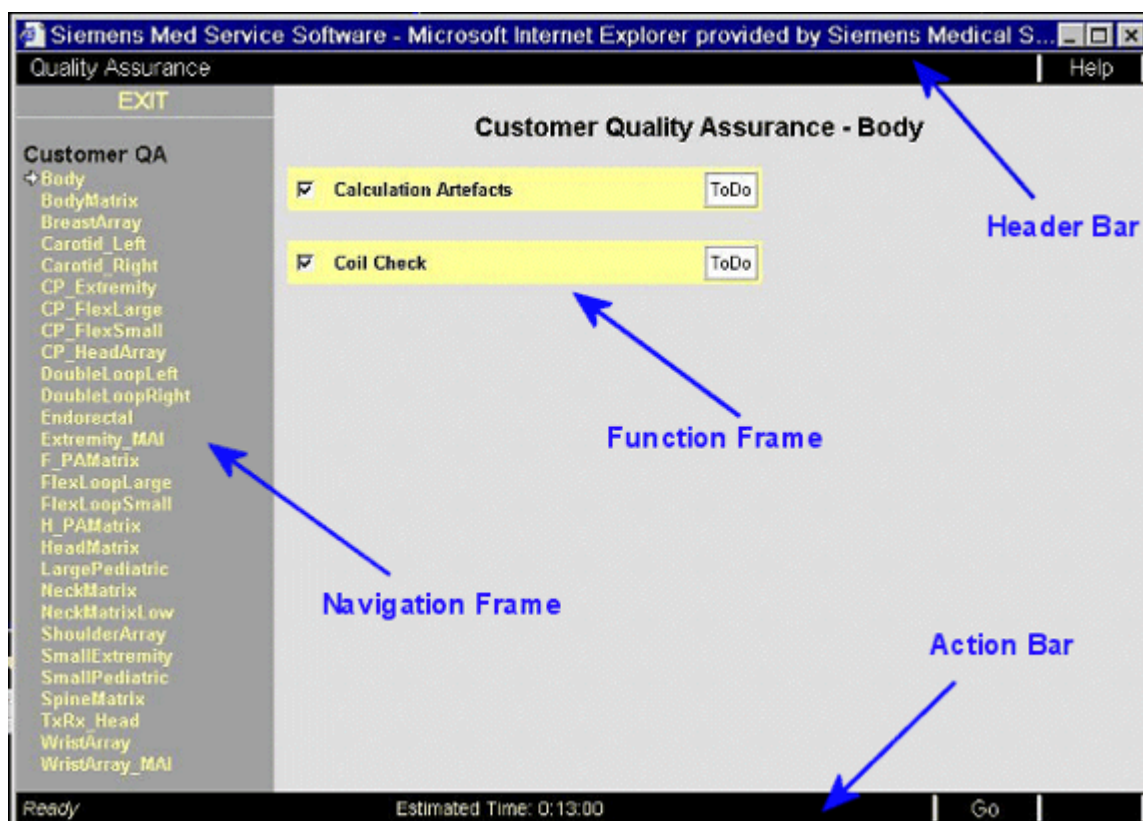


Fig. 318:

NOTE

The number of coils presented in the navigation frame depends on the system used. This means it depends on the system configuration performed during system installation, or the installation of an additional coil.

Status Field

ToDo

Fig. 319:

The **Status field** for each **Quality Assurance** measurement indicates the current condition of the measurement based on the following criteria:

To Do - indicates that the Quality Assurance measurement has not yet been performed or needs to be performed.

Running - indicates that the Quality Assurance measurement is presently running.

Not OK - indicates that the Quality Assurance measurement ran, but the results are not within specifications.

Error - indicates that an error occurred during the last operation. If this Quality Assurance measurement is selected and **Go** is performed, the status will immediately change to **To Do**.

Done - indicates that the Quality Assurance measurement has been completed successfully.

Estimated Time

Estimated Time: 0:58:00

Fig. 320:

The Estimated Time field shows the time the system needs to perform the selected Quality Assurance measurements.

This field is updated after each selection in the Quality Assurance platform and after each Quality Assurance measurement has been completed.

User Selection

Exit

Click **Exit** in the navigation frame to return to the MR main menu.

If a Quality Assurance measurement is running and **Exit** is selected, the measurement continues to run. However, if the Quality Assurance measurement is finished, the measurement is aborted and the user is returned to the MR main menu.

Coil Selection



Fig. 321:

In the navigation frame the user can select the coil. Depending on the coil, the Coil Check and the Calculation Artifacts measurement are displayed in the function frame to start the measurement. All coils that are configured are listed under Customer QA.

All Quality Assurance measurements with To Do status are selected by default and will be performed. Selection box



Fig. 322:

A single measurement can be performed by clicking the selection box.

Dependencies

After a selected measurement has been completed, its status is updated and it is no longer selected.

Go

Initiates the selected measurement(s).

If no measurement is selected when **Go** is activated, an error message indicates that at least one measurement must be selected.

If **Go** is already in progress and is selected again, a message will appear indicating that **Go** is not possible. The same occurs when a standard MR measurement is in progress.

Abort

The **Abort** button is only active when **Go** is in progress.

When you click **Abort**, the current measurement as well as any remaining measurements selected will be aborted.

When a running measurement is aborted, the status will be changed to the previous status.

Help

Help provides general information about the **Customer Quality Assurance** function.

Procedure

Quality Assurance Function

The following procedures provides information regarding image quality assurance measurements.

Selecting the Coil

All coils configured by Service are displayed in the navigation frame (coil selection list).

- Select the coil for quality measurement from the selection list.

Configuring the Quality Measurement

Use the Body Coil to check the system for image artifacts.

- The **Quality Assurance** dialog window is open.
- The coil is selected.
- Select the **Calculation Artifacts** selection box if not already selected by default.

Starting the Quality Measurement

- The **Quality Assurance** dialog window is open.
- The quality measurement is configured.
- Click the **Go** button.

A help window will appear with information (e.g. phantom position) about the quality measurement for the selected coil.

- Follow the instructions in the help window
- Connect the selected coil
 - At this stage of the procedure, the quality measurement can be canceled by clicking the **Cancel** button.
- Click the **OK** button in the help window.
 - ⇒ The help window will close and the measurement will start.

If the quality measurement does not provide perfect results:

- Optimize the position of the phantom and the coil.
- Optimize the phantom position with the help of the laser light localizer.
- Repeat the measurement.
- If the repeated quality measurement does not provide satisfactory results, notify Siemens Service.

Completing the Quality Measurement

- The quality measurement is completed.
- Click the Exit button.
 - ⇒ The Quality Assurance dialog window closes.

22.1.2 Coil Check

HP-Z000100Q-140206-02-002

System Type: Avanto
Espreo
TrioTim
SymphonyTim

General

Quality Assurance Function

Customer Quality Assurance is an automated Image Quality procedure for checking image artifacts and the signal-to-noise ratio as well as the signal uniformity of the individual coils.

User Interface

Platform Layout

The **Customer Quality Assurance** platform is divided into 4 areas:

- Header Bar for general functions
- Function Frame for input and status fields
- Action Bar for platform-specific functions
- Navigation Frame for coil selection

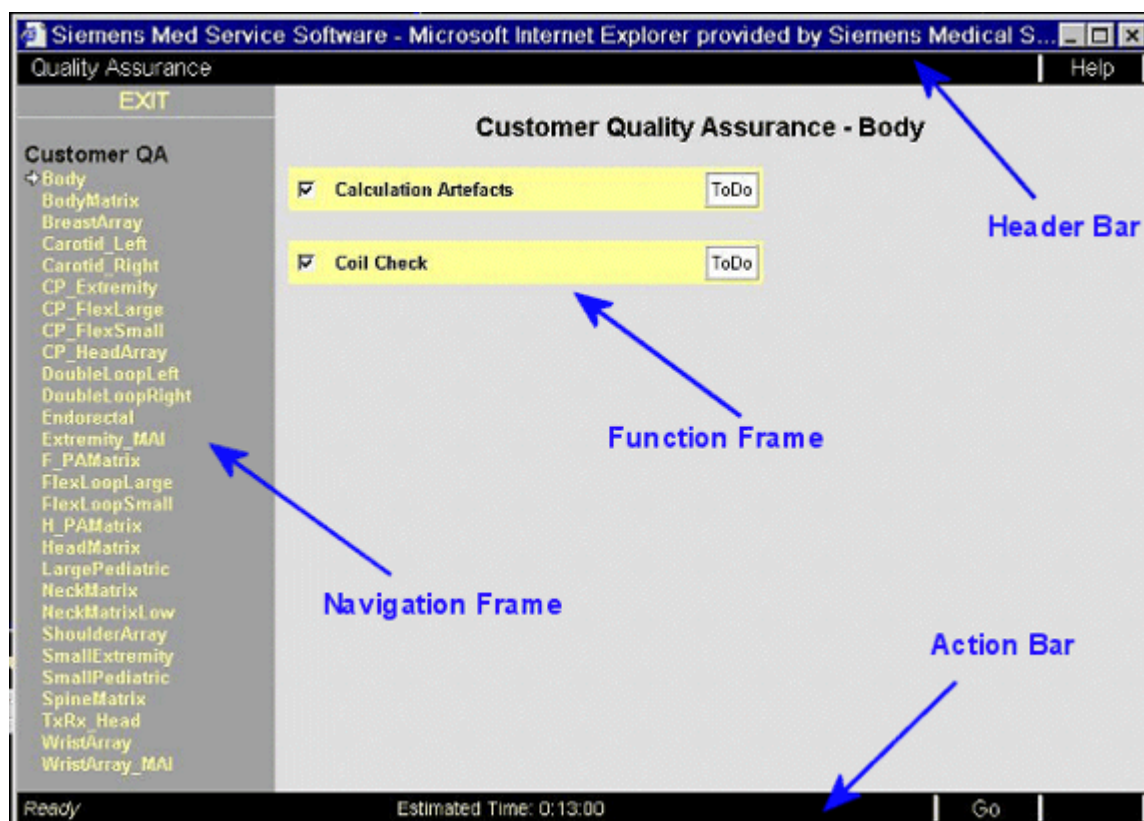


Fig. 323:

NOTE

The number of presented coils in the navigation frame depends on the system used. This means it depends on the system configuration performed during system installation, or the installation of an additional coil.

Status Field

Fig. 324:

The **Status field** for each **Quality Assurance** measurement indicates the current condition of the measurement based on the following criteria:

To Do - indicates that the Quality Assurance measurement has not yet been performed or needs to be performed.

Running - indicates that the Quality Assurance measurement is presently running.

Not OK - indicates that the Quality Assurance measurement ran, but the results are not within specifications.

Error - indicates that an error occurred during the last operation. If this Quality Assurance measurement is selected and **Go** is performed, the status will immediately change to **To Do**.

Done - indicates that the Quality Assurance measurement has been completed successfully.

Estimated Time

Estimated Time: 0:58:00

Fig. 325:

The Estimated Time field shows the time the system needs to perform the selected Quality Assurance measurements.

This field is updated after each selection in the Quality Assurance platform and after each Quality Assurance measurement has been completed.

User Selection

Exit

Click **Exit** in the navigation frame to return to the MR main menu.

If a Quality Assurance measurement is running and **Exit** is selected, the measurement continues to run. However, if the Quality Assurance measurement is finished, the measurement is aborted and the user is returned to the MR main menu.

Coil Selection



Fig. 326:

In the navigation frame the user can select the coil. Depending on the coil, the Coil Check and the Calculation Artifacts measurement are displayed in the function frame to start the measurement. All coils that are configured are listed under Customer QA.

All Quality Assurance measurements with To Do status are selected by default and will be performed. Selection box



Fig. 327:

A single measurement can be performed by clicking the selection box.

Dependencies

After a selected measurement has been completed, its status is updated and it is no longer selected.

Go

Initiates the selected measurement(s).

If no measurement is selected when **Go** is activated, an error message indicates that at least one measurement must be selected.

If **Go** is already in progress and is selected again, a message will appear indicating that **Go** is not possible. The same occurs when a standard MR measurement is in progress.

Abort

The **Abort** button is only active when **Go** is in progress.

When you click **Abort**, the current measurement as well as any remaining measurements selected will be aborted.

When a running measurement is aborted, the status will be changed to the previous status.

Help

Help provides general information about the **Customer Quality Assurance** function.

Procedure

Quality Assurance Function

The following procedures provides information regarding image quality assurance measurements.

Selecting the Coil

All coils configured by Service are displayed in the navigation frame (coil selection list).

- Select the coil for quality measurement from the selection list.

Configuring the Quality Measurement

The Quality Assurance Coil Check measures the signal-to-noise ratio and the signal uniformity of the individual coils.

- The **Quality Assurance** dialog window is open.
- The coil is selected.
- Select the **Coil Check** selection box if not already selected by default.

Starting the Quality Measurement

- The **Quality Assurance** dialog window is open.
- The quality measurement is configured.
- Click the **Go** button.

A help window will appear with information (e.g. phantom position) about the quality measurement for the selected coil.

- Follow the instructions in the help window
- Connect the selected coil
 - At this stage of the procedure, the quality measurement can be canceled by clicking the **Cancel** button.
- Click the **OK** button in the help window.
 - ⇒ The help window will close and the measurement will start.

If the quality measurement does not provide perfect results:

- Optimize the position of the phantom and the coil.
- Optimize the phantom position with the help of the laser light localizer.
- Repeat the measurement.
- If the repeated quality measurement does not provide satisfactory results, notify Siemens Service.

Completing the Quality Measurement

- The quality measurement is completed.
- Click the Exit button.
 - ⇒ The Quality Assurance dialog window closes.

23 TestTools

23.1 General Information

HP-Z0000YPR-220206-02-010

The following detailed help is available:

- **Introduction**

General explanations about TestTools and its user interface.

- **Expert Mode**

Use of the expert mode of TestTools.

- **Tools**

Explanations of the different items available in the Tools platform.

- **Trouble Reset**

Items offered in the Trouble Reset platform. Additionally, a separate help page is available within the service software navigation.

- **Supervision**

Explanations of the different items available in the Supervision platform, and about the AutoSupervision functionality.

- **RHP**

Explanations of the possibilities in the Info platform. Additionally, a separate help page is available within the service software navigation.

- **Info**

Explanations of the possibilities in the Info platform.

- **Test procedures**

Descriptions of the test programs.

- **AutoDiagnoser**

Explanation of the AutoDiagnoser functionality.

Detailed information for the items is available by navigating in the TestTools navigation frame.

Additional information for some items is also available in the service software navigation frame.

23.1.1 Introduction

HP-Z0000YBM-220206-02-001

General

TestTools combines all hardware tests into fully automated procedures. This means that a full system function test can be performed including stability and linearity tests. However, you have the option of selecting individual components for the test run. Testing is FRU/component/spare parts-oriented, i.e. if possible the result of the test is the defective component only.

Test results are permanently saved to a log that can be viewed with the **Report** functions **Session History** and **Test Results**.

The TestTools platform lets you select the following:

- Empty and predefined test selections (navigation “TestTools” and “Problems”, left side)
- Individual components or assemblies (component/FRU section, right side)
- Single tests within these components (navigation “TestTools Expert”, left side)

To save time, individual Test Types can be deselected, provided that the source for error has been narrowed down to a practical level.

After starting the test programs, a browser window with the test results (OK, FAILED, etc.) and detailed outputs including graphics is displayed.

Detailed test information is readily available via hyperlinks.

The window can be closed with the **Exit** button at the top right, or left open for further tests. The contents can be updated with the **Update** button at the bottom right part of the platform.

In case a test fails, a diagnostic table is displayed recommending the next steps.

NOTE

It is not possible to run a test or even a normal sequence when the licenses for the Imager, GPA, and RFPA are invalid or missing.
If it is not possible to start any test, please check for the above.

NOTE

Please ensure that the system is switched on for at least 20 minutes before running any functional or interactive tests to stabilize the temperature in the control cabinet.
Additionally, the doors of the cabinets have to be closed for linearity and stability measurements.

NOTE

If you have doubts as to whether a test is running correctly or a general problem exists with the test, please refer to the Knowledge Database for further information. Usually the entries of interest either start with or contains the phrase "TestTools", so it is possible to search for this word.

NOTE

If the test is aborted with "No results available," it is possible that the source of the problem is not in the component currently tested. Please refer to the event log for more details.

NOTE

After the end of the test run, some tests offer an Expert_TestReport for further information. You will find additional error messages in the report.

When you return defective parts from the measurement control unit, please attach the corresponding Expert_TestReport to the defective component. This log holds information that is of great value to the repair technicians in the factory.

In general all available information (error description, eventlog, etc.) are interesting for the factory and will help improve the quality of the spare parts in stock.

NOTE

While running RF-tests or any test that uses rf_pulse-sequences (e.g. the gradient system pulses with low amplitudes). This is noticeable through the "knocking" of the gradient coil and is completely normal

In case you have problems both with RF and the gradient system, please keep this in mind because the RF tests may not work. The workaround involves changing the supervision for the gradient system in case AutoSupervision is not activated via the Tools platform.

NOTE

Many measurement results are internally stored as float values but displayed as integer. This may cause irritations in tables and graphics.

Structure

Overview

The TestTools platform consists of a Navigation Frame (left), a Function Frame (center), and two Action Bars (top and bottom).

Action Bar - Top

The upper action bar offers the following buttons:

- **Event Log**

The Event Log button opens a separate platform where you can select information from the error logger. Each message ID has a hyperlink to an Extended Error Message explanation providing more information with regard to recommended actions.

- **Report**

The Report button opens a separate platform. You can select the corresponding output log file of the last system tests and step through all data files.

- **Help**

Help provides general information (online help) about the TestTools platform.

Beginning with VB13A, this online help is generated/checked during system boot time and configured so that only the information relevant for this system is displayed (e.g. no help for options that have not been installed). As a result, unnecessary information is no longer displayed.

Additionally, the **Tools**, **Supervision**, **RHP**, and **Info** buttons are located on this bar. They have separate entries in the help-navigation, please refer to these entries.

Action Bar - Bottom

The bottom action bar contains the following fields and buttons:

- **Status** (left)

In this part of the action bar, the status of the current test is displayed. In this way the user always knows the status of the test session and the single tests even if the output window is empty or no new data are displayed.

- **Estimated Time** and **Remaining Time** (center)

The Estimated Time part of the action bar shows the time the system will need to perform the tests for the components and other tests selected.

NOTE

Especially with interactive tests, this time is not very precise. The time the user needs for interactions cannot be calculated or determined precisely. The given time is a minimum time and may be longer when the test is actually performed.

When a measurement is running next to this field, the remaining measurement time is displayed in brackets and acts like a timer. The program begins with a count down when the measurement time exceeds approx. 10 seconds. Otherwise, the timer is not started at all. This application software for the customer proceeds in the same way.

- **Go** (right)

Click the **Go** button to start the selected tests. An output window opens.

- **Abort** (right)

is used to abort the tests or measurements in progress. This button is only visible when a test session is running.

NOTE

It may take a while until the tests are really stopped (approx. 50 seconds, up to two minutes) - please be patient.

Do not abort the MPCU tests, the measurement control may remain in an undefined state as the test loadware is not unloaded.

This may occur with other tests as well depending on the test step being aborted. However, this happens very rarely.

In these cases, a reboot of the Scanner is recommended and necessary.

Also, please avoid aborting imager tests. The imager may stay in a simulation mode and only a power off may help.

- **Default** (right)

Deselects all settings in the platform.

NOTE

Please note that contrary "to previous software versions the available pre-selection in the problems" navigation are separate pages.

Selecting "Default" there will reset the selection to the corresponding preselection. However, it will not deselect all.

By selecting the page in the navigation again, the selection of tests will be reset.

Navigation Frame

The navigation frame provides navigation within the TestTools:

- **Home**

The **Home** button returns you to the Service Home menu. This is the only way to close the TestTools window safely. The "close" icon of the window should not be used. Otherwise, you may not be able to terminate the test programs in progress. The host will have to be rebooted so you can start TestTools again.

- **Navigation**

Allows navigation between the different TestTools modules. Normal and Expert Mode, as well as problem-related preselected test set-ups and the AutoDiagnoser can be started.

Function Frame

The function frame is divided into the following sections:

- **Component/FRU** (top)

This section lets you select the components/assemblies/FRUs to be tested. By selecting "parent" groups, e.g. gradient or RF, all components allocated to this group are selected automatically. Each individual component can still be selected or deselected. If one button is deselected, the parent is deselected as well because the whole group is no longer selected. This works vice versa as well: When you select all FRUs, you will automatically select the parent as well

The different tests are executed by selecting one or several components.

To save time, you can deselect test types as well.

NOTE

You have to select a component first before you deselect one of the test types. This ensures that the component remains selected even if not all tests continue to be selected.

However, if you retroactively change your selection of components, the types of tests selected will be reset and already deselected test types may be selected again.

- **Repeat Count** (center/bottom)

This field is used to set the number of test cycles. Its default value is "1" (one). A new value is taken over. Either click somewhere in the TestTool window (not a button or link of course), or press <return>.

NOTE

The repetition number is valid for one test run only and has to be set again before the next "Go".

Additionally, interactive tests are performed only one time even if the repeat counter is larger than one.

- **Test Types** (bottom)

After selecting one or more components, the tests types available for the selection are automatically activated. If the error source has been narrowed down (e.g. a communication problem) or interactive tests should not be performed (e.g. due to remote diagnostics), you can deselect the corresponding test type(s).

Not all test types are available for every test and at every system. The displayed test types do not really reflect the tests to be started. For further information about which component supports which test type, please refer to the Info-button in the upper action bar.

First you have to select a component before you deselect one of the test types. This ensures that the component remains selected even if not all tests continue to be selected.

After deselecting one or more test types, the selection of these test types is kept even if additional components are selected or other ones are deselected. In this way it does not matter if the test type (de-)selection is performed after all components to be tested

are selected first, or if the selection of components and test types is mixed. There is only one restriction: You have to start with a component. It is not possible to start with test types.

This behavior is only reset after a new TestTools page is selected or when the "Default" button is pressed. Afterwards the old behavior is active again (test types change when components are selected/deselected).

The following test types are available:

- **Functional**

Tests that run fully automatic after invoking the test session. "Fully automatic" includes the necessary positioning of the large spherical phantom and the body loader. Every other interaction is covered in the

- **Comm/Status** (for **Communication/Status**)

checks the communication links and prints out the status of the different components.

- **Interactive**

tests. The necessary instructions are displayed directly before the test starts.

- **Multi Nuclei**

tests are only available when the system is equipped with the Multi Nucleus option (spectroscopy, X- nucleus).

NOTE

These tests are performed when RF tests are selected and the system is equipped with the option. If this is not intended or necessary, please do not forget to deselect the test type "Multi Nuclei".

Every time you perform this type of test, it is necessary to select the nucleus/frequency prior to starting the test. If the desired nucleus is set, it will remain until it is changed again. This should be remembered during troubleshooting.

Nucleus selection

Next to the Multi Nuclei test type, the nucleus selection is displayed if the system is equipped with the multi-nucleus option. Here the nucleus to be tested has to be selected. Of course, this selection has to be performed before the test session starts. It affects only RF tests. Identical to the test type, the selected nucleus is kept after the end of a test session.

Generally the RFPA (RFPA1, RFPA2) used is displayed in the pull-down menu to facilitate selection of the correct component/FRU.

Every nucleus corresponds to a certain frequency which will affect, e.g., the automatic selection of the correct RFPA.

MAGNETOM Avanto and Symphony, A Tim System: The nuclei ^3He , ^{19}F , and of course ^1H are using the DORA amplifier, all other nuclei use the second RFPA CORA (^7Li , ^{13}C , ^{23}Na , ^{31}P , and ^{123}Xe) which was also used in the MAGNETOM Open/Concerto.

MAGNETOM Trio, A Tim System and Verio: Here it is different, because different amplifiers with different frequency ranges are used. Here the Astex amplifier is in charge of ^1H only, whereas the Dressler amplifier supplies all frequencies, including ^1H . Additionally, it is capable of ^{17}O , which is not supported with 1.5 T systems.

MAGNETOM Espree: For this system type the Multi Nucleus option is not available.

- **Long Time**

allows you to enable/disable lengthy tests which are necessary for, e.g., BOLD-imaging. These tests allow you to thoroughly test the linearity and stability.

- **Standby**

The check box "Standby after end of workflow" works in the same way as within Tuneup and QA.

By selecting this box the system is set to standby mode after all selected tests are executed.

NOTE

Be aware when using the Standby functionality that it is not possible anymore to switch on the system remotely.

Test Session

After starting the test session using the **Go** button, the system checks for the necessary system conditions/prerequisites, the so-called pre-conditions.

Error messages may also appear for the following pop-ups and have to be confirmed.

Patient table

- The table should be in the magnet isocenter (position "0000"):
"Position loader with the large spherical phantom in the magnet center (use light marker)."

The corresponding error message is:

"The system cannot detect that the patient table is in the center of the magnet. Please position loader with the large spherical phantom in the magnet center (use light marker)."

- The table should be in the home position (completely out of the magnet):
"Retract patient table out of the magnet into home position by use of the menu or the buttons at the magnet. Note: Do not use the table stop button, otherwise the test suite will abort!"

The corresponding error message is:

"The system cannot detect that the patient table is at the home position. Please retract patient table out of the magnet into home position by use of the menu or the buttons at the magnet. Note: Do not use the table stop button, otherwise the test suite will abort!"

Service Plug

- "Insert Service Plug to coil plug 1. Ensure that the mode selector of the Service Plug is set to TX."

The corresponding error message is (no or incorrectly connected service plug):

"The system cannot detect the Service Plug (in TX mode) at coil plug 1. The following test(s) require the Service Plug. Please insert Service Plug to coil plug 1 and ensure that the Mode selector of the Service Plug is set to TX."

Output Window

After starting one of the test programs, another browser window is opened to display the test results (OK, FAILED, and so on) and the detailed outputs including graphics. Detailed test information is readily available via hyperlinks, including diagnostic tables in case a test fails.

NOTE

The test results can be viewed via the Reports button of the TestTools platform.

In these reports, pdf documents of the displayed information can be generated. In this format, the document facilitate printing and storing them accordingly.

The Output Window is subdivided into two frames (index frame and result frame) and one action bar (bottom):

- **Navigation or Index Frame**

The navigation frame contains an index of all executed tests, including their results (OK, failed, etc.). The links let you jump to the detailed information in the result frame.

At the top of the navigation, the **Home** button will close the window. If needed, the window can be kept open for further tests. It will be updated when the next test session starts.

In case a test fails, a diagnostic table is displayed in the result frame directing you to the possible source of error and recommending next steps.

After the first test results are available, an additional entry becomes visible in the navigation. The “Session Report” basically shows the same information as in previous software versions (including an index and diagnostic texts).

Everything below “Iteration 1” will just show the information selected: component/FRU and its test type will only show the test results of a component/FRU.

The next level will show only the results of the single tests within that component/FRU, including an “Ok” or “failed” at the end of that line.

NOTE

Please be aware that beginning with VB19A the style of the output differs from the style that is used in the result frame.

No border lines of tables are displayed.

All these levels are hyperlinks showing different HTML/XML pages. These pages also can be accessed in sequence by use of the “<” and “>” buttons at the end of the navigation: When you press this button, you scroll to the previous/next page.

When you select **Add Comment** in the navigation frame, you can add a comment to the test session currently running. This is important when you send the test results to a third party (colleague, USC, HSC / Customer Care Centers) or when you have to remember what was done during the tests.

You get a print-out of the comment at the end of the test results.

NOTE

The contents of the comment entry field are erased when the test session is finished. In this case the test results will be displayed automatically, erasing the comment entry.

Please save the comment to be entered as quickly as possible. As long as the test session is running, it is always possible to add an additional comment.

The Update button will not bring up the latest test results, but instead will jump to the top item in the navigation frame. Pressing Update again will display the information.

- **Result Frame**

Here the complete output information of the tests is displayed, including graphics, if available.

At times even more detailed test information is available by offering an additional hyperlink. This information is very helpful, e.g., for the factory.

NOTE

The right mouse button is disabled in normal user mode. It is no longer possible to use the pop-up menu when you want to jump back to the previous location. Instead the <backspace>- or <ALT>-<arrow-left>-buttons have to be used.

You are able to print in the browser. Press <ctrl>-<p>.

Alternatively, you can start the advanced user mode before the start of the service software platform. In this case the pop-up menu is enabled again and can be used.

- **Action Bar - bottom**

During the test run use the **Update** button in the bottom right part of the platform to display the currently available results. An automatic update is performed only at the end of the tests. While the tests are running, you are able to scroll inside the window without any restrictions.

NOTE

The Update button doesn't work at times. If this happens, simply press the button again or close the output window with the "window close" button located to the right on top of the window. The window will reappear as soon as test results are available for display.

Keep in mind that as long as the system is not ready to display the results, it will not help pressing the update button.

Configuration

The features AutoDiagnoser and AutoSupervision made it necessary to add some configuration possibilities to the "**Configuration**" main menu. You can access it via the main service software platform.

Under **Service > TestTools**, the AutoDiagnoser and AutoSupervision can be enabled and disabled.

NOTE

The system will reboot when you switch the AutoDiagnoser from enabled to disabled or vice versa.

The complete TestTools workflow is rebuilt during the reboot.

Additionally, AutoSupervision can be enabled/disabled in the **Tools** platform (please refer to this). No system reboot is necessary in both the configuration and the Tools platform.

Changes

VB13A

Refer to the Online help for further information about:

- Change from HTML to XML, results in different user interface and Tune-Up/QA-like navigation
- Expert mode
- AutoDiagnoser
- AutoSupervision
- Online help is configured to the system, unnecessary information is no longer displayed
- It is possible to add comments to a test session
- Change in Tools platform: enable/disable 10 V-PreAmp error reset, enable/disable AutoSupervision, new terminology
- Generating pdf documents in reports
- Support features of VB13A: new imager tests, MDSD, AMC service boot, Multi Nucleus, long time stability
- Different organization of functional and interactive RCCS tests
- Integrated repeat platform in standard user interface, no longer separate platforms

VB15A

Editorial changes.

VB17A

Refer to the Online help for further information about:

- Editorial changes
- Correction of the PIN Splitter test and new texts for MAGNETOM Verio
- Reworked Diagnostic Messages for all tests
- Shorter test duration for interactive RCCS test. Reduction from > 2 hours to approx. 20 minutes.

VB19A

Refer to the Online help for further information about:

- Editorial changes

- Adaption of **Trouble Reset** platform due to new Magnet Supervisory (MSUP) behaviour
- Adaption of MPCU description as now also the MPCU VI is supported, not only the MPCU-V
- Introduction of a **Standby** button like with the VD software line to set the system into standby after the end of the TestTools workflow.

Service Plug

Quick Reference

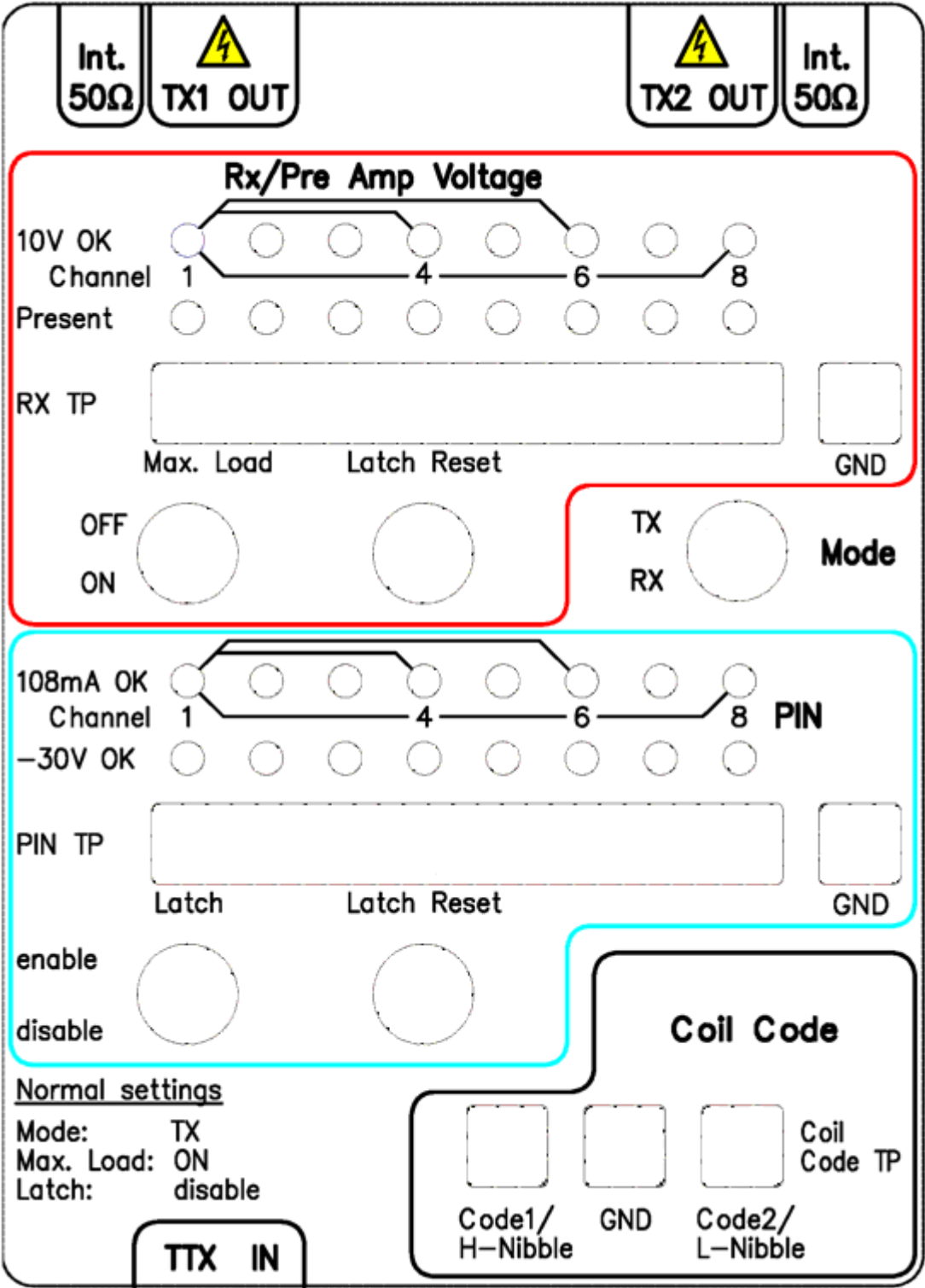


Fig. 328: Front Plate Service Plug

Rx/Pre Amp Voltage

10V OK	Green LEDs to indicate that the PreAmp voltage of the corresponding RX-channel is in specification.
Present	Yellow LEDs to indicate that some PreAmp voltage of the corresponding RX-channel is present. No specification check, lower voltage results in less brightness. LEDs start to glow at approx. 3 V and are bright between 5 and 7 V.
RX TP, GND	Sockets/test points for voltmeter, measured values between 7.9 V and 11.0 V are OK.
Channel 1 .. 8	Label to identify the correct channel.

NOTE

The 10 V of RX-channel 1 supplies the Rx/PreAmp section with power. When the channel has no voltage, the complete Rx/PreAmp section will be "dark".

The 10 V of RX-channel 2 supplies the PIN section with power. When RX channel 2 has no voltage, the complete PIN-section will be "dark" as well.

PIN

108mA OK	Yellow LED's to indicate the bias current of the corresponding PIN-channel is within specifications.
-30V OK	Green LED's to indicate that the reverse bias voltage of the corresponding PIN-channel is in specification.
PIN TP, GND	Sockets/test points for voltmeter. For the bias current values measured between 0.65 V and 0.87 V are OK, for the reverse bias voltage -27.7 V and -31.3 V are OK.
Channel 1 .. 8	Label to identify the correct channel.

Mode

TX	Normal mode of the Service Plug, for both transmit and receive with the Service Plug. Results in coil code "CC".
RX	RX mode of the Service Plug, for transmit with Body Coil and receive with the Service Plug/Pick-Up coil. Results in coil code "CD", which is currently not used (but measurements are possible, e.g. with a pick-up or loop coil).

Coil Code

Code 1 / High Nibble	Sense line for first part of the coil code (high nibble, with Service Plug in TX and RX-mode, this is the left "C").
Code 2 / Low Nibble	Sense line for second part of the coil code (low nibble, with Service Plug in TX-mode this is the right "C", and in RX-mode the "D").
Coil Code TP, GND	Sockets/test points for voltmeter.

Controls

The following controls are available at the Service Plug:

Max. Load	Toggles the load condition of the 10 V-supplies in the RCCS between "normal" (10 - 20 mA) and "maximum" (approx. 70 mA).
Latch Reset (Rx and PIN section)	Resets the error condition latches of the corresponding section (Rx/PreAmp or PIN).
Latch	This switch in the PIN-section allows you to enable or disable the error latch. This is necessary for testing the RFIS via TestTools which would not be possible otherwise.
Mode	Coil code "CC" behaves like the old Service Plug and simulates a TxRx coil (both transmit and receive with the Service Plug), whereas coil code "CD" simulates an Rx-coil only (transmit with Body Coil and receive with the Service Plug).

NOTE

The 10 V of RX-channel 1 supplies the Rx/Pre Amp Voltage section with power, RX-channel 2 supplies the PIN-section. A normal load condition is not possible there. Toggling between "Max. Load" and normal load will not have a considerable influence on channels 1 and 2.

Additionally, the voltage measured via RX-TP will always be quite low.

Connections

The following connections are available at the Service Plug:

TX1 Out, TX2 Out	Possibility to connect an external 50 Ohm power attenuator to the LC-TX-path to increase the maximum allowed power.
------------------	---

Int. 50 Ohm	The internal resistor allows RF up to an average power of approx. 2 W for one sequence. If more power is necessary while troubleshooting, an external 50 Ohm termination has to be used.
TTX IN	For TTX test signal, necessary for TestTools.

Notes on use

NOTICE

Never connect both the 6/8-channel plug and the 4-channel plug at the same time.

⇒ **Irreversible damage to the service plug could result if the plugs are connected for a long time. There is no problem if it is for a few seconds (up to one minute).**

- Never have external cables (e.g. for multimeter, oscilloscope) connected while performing RF-measurements with the Service Plug. The results may be incorrect.
- The normal setting of the control is "TX mode", "Latch disabled" and "Max. Load ON" or "Max. Load OFF" (doesn't matter). The "RX mode" is currently not used.
- The test results with the RCCS-plug tests differ from system to system depending on the length of the RF-cables (different attenuation). In addition, the results from the head and the feet end of the patient table differ by approx. 10 %, again due to a difference in attenuation. The absolute values are not relevant, the change from one channel to the next has to be approx. 50 %.
- In case you measure the PIN-diode voltages and currents using the RFIS-test, you can keep the test running as long as necessary on the same coil connector.

23.1.2 Expert Mode

HP-Z0000YBM-220206-02-002

The expert mode of TestTools enables the service engineer to select/deselect single or multiple tests with a component/FRU. As compared to the export modes of Tune-Up and QA, it can be considered an advanced mode since no additional tests are available.

In the normal mode, all tests of a component/FRU are performed automatically, depending on the selected test types. In the expert mode it is possible to reduce the number of tests performed, decreasing the troubleshooting time after the defective component has been narrowed down.

Tests are only possible within one “parent” (AMC, RF, GPA, etc.). It is not possible to select tests in more than one parent because all parents are separate HTML/XML pages. When you select another page/parent in the navigation, the program will reset the previous page(s).

Press **Go** to invoke only the tests of the currently visible page.

NOTE

Tests are only possible within one “parent” (AMC, RF, GPA, etc.). When you select another page/parent in the navigation, the program will reset the previous page(s).

The combination of tests within a parent can be various and is not limited. Most of the tests of the single components cannot be separated in the AMC section. The various sub-tests of PCI-TX and PCI-MON have to be performed in a specific sequence and in a block.

Non-limitation makes it necessary to use a completely different “organization” for the tests within a section/parent since all possible tests have to be displayed. To this end, a column and row-oriented display has been selected. The columns show the possible test types and the rows show the tests available. A less than optimal side-effect is that one test alone for components/FRUs does not show the difference between the normal mode and the expert mode. Here the expert mode does not provide any advantages (e.g. for AMC, RCCS plug tests, or RFIS).

The display of the columns and rows is dynamically adapted to the width of the window.

The repeat field is located on top of each page. As with the normal mode, it is possible to repeat the tests according to the number entered.

NOTE

In case the system is equipped with the Multi-Nucleus option (spectroscopy), the nucleus drop-down menu for selecting the nucleus to be tested is always displayed underneath the repeat input. This applies to the parents as well, e.g., AMC and GPA. Here the selection is not relevant. It is only used for RF tests.

23.1.3 Tools

HP-Z0000YBM-220206-02-003

The Tools button enables you to access some special settings of the measurement system. Previously, this was only possible via separate passwords and the operating system level:

NOTE

It is possible that the message "Invalid value for Adjustments mode: Othermode" is displayed when entering the platform for the first time during a test session.

This "error message" is correct since other modes exist besides the listed modes for the Service Software . However, these modes cannot be displayed or selected. These other modes are used, e.g., in Tune-Up and QA.

Please set the new mode according to your needs.

The following radio buttons are available:

- The watchdog programs for RF (RFSWD), gradient (GSWD), and the adjustments can be set to
 - Patient mode (called "enabled" in former software versions)
 - Service mode (called "disabled" in former software versions)
 - "Disabled" (only RFSWD, to this extent, not available in former software versions)

NOTICE

When you set the RFSWD to "disabled" , any protection for the components is completely switched off. This is normally not necessary.

Incorrect use of this setting may damage the system or parts of it.

⇒ **"Disabled" should not be used during normal troubleshooting! It should be used only if advised by the MR-Hot-line (HSC MR).**

The setting of the RFSWD (Patient mode/Service mode) has no influence in the Test-Tools measurements as the tests already check the necessary prerequisites and set the mode accordingly.

- RF characteristic can be toggled between the measured characteristic and the linear one. However, this will affect only a few tests and is mainly intended for expert use.
- It is possible to register or unregister the Service Patient so that it is not necessary to run a test or QA to have this patient registered for further troubleshooting or service.
- When a 10 V preamplifier error occurs, the behavior of the system can be changed from "one time reset" to "no reset" for troubleshooting ("Autorecovery RCCS 10V Error"). With the customer, the 10 V error is normally automatically reset one time to avoid system reboots. It may be necessary to disable this functionality for troubleshooting.

NOTE

The setting of this functionality is reset to patient mode with each system reboot.

It may be helpful to change these settings during troubleshooting.

In the bottom action bar, the buttons **Save** (save the settings) and **Default** (reset to normal) are available, in the top it is possible to close the window via **Exit**.

23.1.4 Trouble Reset

HP-Z0000YBM-220206-02-004

The Trouble Reset button opens a separate platform where it is possible to:

- Reboot the host

NOTE

In that case the remote connection is cut!

- Reboot the imager
- Reboot (normal or forced) the scanner, switch it on from standby mode, or bring it to standby mode

NOTE

With the new MSUP K2306 it is not possible to switch on the scanner again.

If the system is in standby mode no choice is offered. Instead only "Not available at the moment" is printed out.
--

This is most interesting for Remote Diagnostics since the System Control is not reachable when sitting at the user console.

For this the **"Remote Service Access Control"** level has to be set to **Full Access**.

To close the window use the Home button in the navigation

With **Go** in the bottom action bar, the requested task is performed. **Update** will update the current status of the task.

Additionally, a **Help** button is available in the upper action bar.

Further help can be found in the main Trouble Reset Help.

23.1.5 Supervision

HP-Z0000YBM-220206-02-005

Platform

A click on the Supervision button within the upper TestTools action bar opens a new browser platform for modifying the supervision status of the CAN-components.

The supervision status should be changed whenever individual components should not be monitored during troubleshooting.

The dimmed entries are switched together with the main entries, e.g. the alarm box together with the Magnet Supervisorion, Cabin-UI (control panels at magnet) and MDSD (Move During Scan Device/Drive) with the patient table.

The following entries are possible:

- **ON**

The component is supervised and the error messages from the peripheral units are monitored (normal setting).

- **OFF**

The error messages from the peripheral units are not monitored, but the data transfer and the acknowledge from the component are still supervised.

Use this mode to suppress incoming error messages from the peripheral units.

- **Real**

The component is installed and supervised (normal setting).

- **Virtual**

The component is treated as existent, but the data transfer to and the acknowledge from the component is only simulated.

This mode should be used for troubleshooting when a test measurement must be started without supervision of a certain component.

Click the **Default** button to display the default configuration. Click **Save** to activate the settings temporarily. The default configuration will be restored after syngo MR is restarted.

To close the platform, choose **Exit** in the upper action bar.

AutoSupervision

Additionally, the control of the AutoSupervision functionality is located in this platform (AutoSupervision for TestTools programs: enable/disable).

AutoSupervision means that TestTools automatically sets the supervision mask according to the test(s) invoked. Components not used or not necessary for a specific test are switched to "Off" at the beginning of the test and reset at the end. As a result multiple defects causes less problems in running the tests (e.g. defective GPA disables any measurements including RF-tests).

The AutoSupervision can also be enabled and disabled under **Configuration > Service > TestTools**, starting at the main service software platform.

23.1.6 RHP

HP-Z0000YBM-220206-02-006

The **RHP** button opens a separate platform for generating and evaluating raw data.

The **Rawdata Handling Package (RHP)** provides facilities to:

- Enable/Disable raw data generation
- Display of raw data
- Apply different mathematical operations (statistics, FFT, etc.) to the raw data set or parts of it for evaluation and troubleshooting
- Remove raw data files from the file system

The tool is intended for expert use only.

Further help can be found in the main RHP Help.

23.1.7 Info

HP-Z0000YBM-220206-02-007

A click on the Info button will bring up a separate window with a list of possible tests including diagnoses for the tests.

The display is organized similar to Windows Explorer. A plus sign (“+”) in front of an entry indicates that there is another level of information. To open this branch click the plus sign.

Additionally, it is possible to toggle between “open all branches” (“**Show**”) and “close all branches” (“**Hide**”) at the top.

NOTE

Contrary to previous software versions, the information about the tests and their diagnostic texts is not narrowed down in case of a test selection within the TestTools user interface. It is always possible to display all available tests.

The window is closed with the **Exit** button in the top action bar.

23.2 Procedures

23.2.1 Problems

Communication

HP-Z000100Q-071205-02-001

Predefined Test

In this predefined test workflow, the communication tests and status request will be performed for all components with a single click.

Component Test	Subcomponent Tests (Comm/Status)
AMC	MPCU, LWL-INT
RF	RFPA/RFPA1 (includes RFPA2), RCCS, RFIS
Gradient	GPA
Imager	Imager
Other	MSUP, SHIM, ACS, PTAB, MDSD, PMU

CAN-Bus Com.

HP-Z000100Q-071205-02-002

Predefined Test

With this predefined test, the AMC/cPCI-CAN (Comm/Status) test can be performed.
The CAN-nodes will be polled in their physical sequence in the bus chain (refer to the description for the cPCI-CAN test).

Image Quality

HP-Z000100Q-071205-02-003

Predefined Test

With this predefined test, all functional and interactive tests of the RF and gradient will be selected with single click of the mouse.

Component Test	Subcomponent Test (Functional/ Interactive)
RF	All available functional and interactive tests
Gradient	All available functional and interactive tests

BC Tuning

HP-Z000100Q-071205-02-004

Predefined Test

With this predefined test , all functional and interactive tests regarding components used for the tuning of the Body Coil can be selected (main Transmit-/Receive-path) with a single click of the mouse.

Component Test	Subcomponent Tests (Functional/ Interactive)
RF	MOD/REC, RFPA/RFPA1, RCCS, RFAS, BCCS/RCCS, BC

RF Chara

HP-Z000100Q-071205-02-005

Predefined Test

With this predefined test, all functional and interactive tests regarding components reliable for RF Chara can be selected (main Transmit-/Receive path) with a single click of the mouse.

Component Test	Subcomponent Tests (Functional/ Interactive)
RF	MOD/REC, RFPA/RFPA1, RFPA2 (if applicable), BCCS/RCCS, RFIS, RFAS, BC

Local TX Coil

HP-Z000100Q-081205-02-001

Predefined Test

With this predefined test, all Communication/Status, functional and interactive tests regarding components responsible for Local TX coils can be selected via a single click of the mouse.

Component Test	Sub-component Tests (Functional/ Interactive)
RF	RFIS, RFAS

Multi Nuclei

HP-Z000100Q-081205-02-002

Predefined Test

With this predefined test, all Multi Nuclei tests regarding components used for Multi Nuclei/spectroscopy measurements can be selected with a single click of the mouse (RF tests that are capable of other frequencies than for ^1H).

Component Test	Subcomponent Tests (Functional/ Interactive)
RF	MOD/REC, RFPA1, RFPA2, RCCS, RCCS Switch, RFAS, Cable LC

The Multi Nuclei selection is only displayed in case the system is equipped with a multi-nucleus option (spectroscopy, X-nucleus).

NOTE

When using the predefined test “Multi Nuclei” both RFPAs are selected automatically.

Please deselect the RFPA that is not used for the frequency in the test.

If this is not done and the RFPA does not support the selected nucleus, a corresponding error message is displayed and the tests will be marked as “failed”.

Coil Plugs

HP-Z000100Q-081205-02-003

Predefined Test

With this predefined test , all interactive tests regarding components used for coil plugs/ local coils can be selected with a single click of the mouse (to troubleshoot problems with the coil sockets or the cabling to the sockets).

Component Test	Subcomponent Tests (Functional/ Interactive)
RF	RCCS plug x, RFIS

System test

HP-Z000100Q-081205-02-004

Predefined Test

With this predefined test a full system test without the interactive, Multi Nuclei, and Long Time tests are selectable; only the Comm/Status and functional test are performed.

This is actually not a "problem". With this button a full system test **without** the interactive tests is easily selectable, only communication/status and functional tests are performed. If the interactive tests are also necessary, please select the test type **Interactive** or **Multi Nucleus** in the Test Types section accordingly.

The main objective is to perform the TestTools via Remote Diagnostics.

23.2.2 TestTools

AMC

HP-Z000100Q-241105-02-004

This section describes the AMC hardware tests. The AMC (General/ Expert) tests allow you to diagnose the FRUs of the **Advanced Measurement Control**:

MPCU, CPCI-CAN Interface, PCI-TX, PCI-MON and LWL-INT. The testing is FRU-based, i.e. the tests results in a list of FRU's with possible defects.

NOTE

When you return AMC FRUs, please attach the corresponding Expert Test Report/ Auto Diagnoser Report to the defective FRU. This log holds information that is of great value to the repair technicians in the factory.

The total test time for the AMC tests is approximately **30 minutes**.

General

To diagnose AMC components, select the **AMC checkbox** or individual checkboxes in **TestTools - General** or select only particular or all checkboxes in **TestTools - Expert - AMC**. Click **Go** to execute all selected AMC tests.

The tests are performed in a logical sequence, allowing for time-efficient testing and correct test results. If a test fails and the tested function is essential for subsequent tests, the test run is **stopped**.

NOTE

Do not abort any AMC-test as this may leave the measurement system in an undefined state (testloadware is not removed). If you have to abort the tests, reboot the MR-scanner afterward.

NOTE

Since the test sequence matters, some test results are only valid when all AMC tests are selected for the test run.

You can select individual FRUs for testing. If you choose to do so, make sure that the pre-requisites are met.

The test results including a list of FRUs with possible defects will be displayed in a separate window. In addition, the AMC test results are permanently saved to a log file that can be viewed using the **Reports** function.

NOTE

In the Event Log you will find additional information in case of errors: Click left on the error identity number in the Event Log list to get more information about the error and the necessary service action.

NOTE

In some very rare cases, a "reboot scanner" or the MPCU-Reload inside the AMC-tests doesn't work. In this case please use the hardware-reset button at the LWL-INT to reboot the AMC.

MPCU

HP-Z0000YBM-111012-02-001

The MPCU is tested using two communication/status tests.

As now both the MPCU-V and the MPCU-VI are used the further descriptions are valid for both components.

MPCU (Comm/Status)

MPCU Communication tests the communication between host and the MPCU. For this purpose, the boot status of the MPCU is requested.

Description:

The following tests will be performed by the BIOS during start up:

- RAM test
- CPU instruction test
- DMA test
- Ethernet test (internal loop back)
- Serial interface test (internal loop back)

LatestMPCU Selftest Status evaluates the boot status of the MPCU, i.e. the MPCU self-test results executed during power on of the MR system or warm restart of the MPCU:

MPCU_STATUS:	Post manager	=	ok
MPCU_STATUS:	Total checksum	=	ok
MPCU_STATUS:	TIME	=	ok
MPCU_STATUS:	Disk access	=	ok
MPCU_STATUS:	RAM	=	ok
MPCU_STATUS:	Bios Configuration	=	ok
MPCU_STATUS:	BIOS EEPROM checksum	=	ok
MPCU_STATUS:	Power	=	ok

Communication from the host to the MPCU is a prerequisite for this test.

NOTE

Beginning with VB19A contrary to previous software versions the result of the selftest is not displayed at the parallel port anymore (X26 at the backplane).

Service Boot

HP-Z000100Q-301105-02-001

With the procedure **Service Boot(Interactive)** the following FRU's can be booted step-by-step:

- MPCU
- cPCI-CAN
- PCI_TX
- PCI-MON

Defects on the FRUs will be detected during booting and error messages will be saved in a report file. Information about the defective FRU can be seen by the service engineer.

NOTE

Only after a successful Service Boot further AMC tests are meaningful.

So perform this test and follow the workflow instructions in case the other tests fail and are not specific enough to decide if or which component is defective.

Service Boot can be started in the Expert UI:

- TestTools Expert / AMC/ Service Boot/ Interactive

cPCI-CAN

HP-Z000100Q-301105-02-002

The cPCI-CAN test consists of one functional test and several CAN-communication tests to the CAN-components of the system.

cPCI-CAN (Functional)

The test checks if at least one component from a distinct bus 0,1, 2, (3 only with Symphony A Tim System) responds. If this is the case, the corresponding channel of LWL-INT is ok. If no component responds, the LWL-INT is probably defective.

NOTE

The normal output/result here is "online".

In case the supervision mask for a component was changed there will be a different output, indicating that the displayed status is not the correct one.

If the RFPA on CAN bus 2 does not respond, also check the RFPA and its FOC connection lines; the RFPA is the only component on CAN bus 2.

CAN bus 0 (open, mute, electrical/optical)	(MPS, ABX,) MSUP, RCCS, RFIS, PTAB
CAN bus 1 (open, active, electrical)	LWL-INT, (SHIM,) ACS, GPA (GPA not with Symphony, A Tim System)
CAN bus 1 (open, active, optical)	MDSD
CAN bus 2 (optical)	RFPA
CAN bus 3 (prop, electrical, optical)	GPA, RCA (both with Symphony, A Tim System only)

Conclusion

If the communication test of a component fails, the power supply of the component, the cable connections or the component/ SLIO may be defective.

cPCI-CAN (Comm/Status)

The communication to the following CAN components is tested and the status will be displayed.

- MSUP
- TALES (Trio, A Tim System only)

- RCCS
- RFIS
- PTAB
- MDSD
- LWL-INT
- SHIM
- ACS
- GPA
- RFPA1/2, RFPA/ Status
- RCA (Symphony, A Tim System only)

LWL-INT

HP-Z000100Q-301105-02-003

LWL-INT (Comm/Status)

LWLINT Communication tests the communication between LWL-INT and the MPCU.

LWLINT Status returns the status of the LWL-INT.

LWLINT_STATUS:	Imager 1 Fan	=ok
LWLINT_STATUS:	Imager 2 Fan	=ok
LWLINT_STATUS:	Imager 3 Fan	=ok
LWLINT_STATUS:	Imager 4 Fan	=ok
LWLINT_STATUS:	SLOW_TRIG_IN	= not active
LWLINT_STATUS:	DOOR	= closed
LWLINT_STATUS:	AMC FAN	=ok
LWLINT_STATUS:	AMC Power Supply (Degrade)	=ok
LWLINT_STATUS:	AMC Power Supply (Power failure)	=ok
LWLINT_STATUS:	-12 V failure ?	=ok
LWLINT_STATUS:	Gradient Filter over temperature ?	=ok

NOTE

Depending on the installed boards one or more entries might not be meaningful anymore.

LWL-INT (Interactive)**LED-Test**

The test outputs a bit pattern to the TX LEDs (U1 to U3 and U14 to U27) in the LWL interface (running LED light).

First you have to remove all FOCs except U6 to U13 (PMU and CAN bus).

The test runs as follows:

1. All LEDs U1 to U3 and U14 to U27 light up.
2. All LEDs U1 to U3 and U14 to U27 will be switched off.
3. Beginning with U1, the LEDs will be switched on and off one after the other.
4. All LEDs light up.
5. All LEDs are switched off.

Diagnosis:

If one of the LEDs does not light up or the LEDs are significantly dimmed, replace the LWL-INT board. Also check the PCI-TX board.

NOTE

The PCI-TX board can be interchanged with the PCI-MON board to localize the error.

PCI-TX

HP-Z000100Q-301105-02-004

PCI-TX (Functional)

- **RFPA_Power_Off**

With this function the RFPA can be switched off.

- **GPA_Power_Off**

With this function the GPA can be switched off.

- **PaliSimMode**

With this function the PALI Sim Mode can be switched on or off. The PALI has to be switched in the Simulation mode for the following tests.

- **GcSeqClk On**

With this function the Sequence Clock can be switched on. This is necessary for special tests on other DSPs. Before starting these tests, the SeqClk has to be activated. After the tests the SeqClk should be switched off. For example the Modulator/NCO Test needs the SeqClk. Test functions which run on the GC-DSP and need the SeqClk are able to switch the SeqClk on and off internally.

- **Tx Ext Mem Test**

This test checks to ensure that the external memory of the DSP is working properly.

- **TxMemTest**

This test checks to ensure that the memory of the DSP is working properly.

- **TxDualPortRam**

This function is a combination of the external memory test via Pci bus (MPCU) and PCI-TX and a bus arbitration test.

- **Tx AddrTest**

This function checks the address decoding.

- **Tx Data Bus**

Test of the data bus with each other's inverse test patterns (0x5a5a5a5a/0xa5a5a5a5).

- **Tx Addr Bus**

Test of the data bus with each other's inverse test patterns (0x5a5a5a5a/0xa5a5a5a5).

- **TxTlk Local Loop**

In a local loop, a sine-shaped series of test patterns is written in the TLK interface and read back.

- **TxTlk Remote Loop**

In a remote loop including the modulator, a sine shaped series of test patterns will be written on the TLK interface and read back.

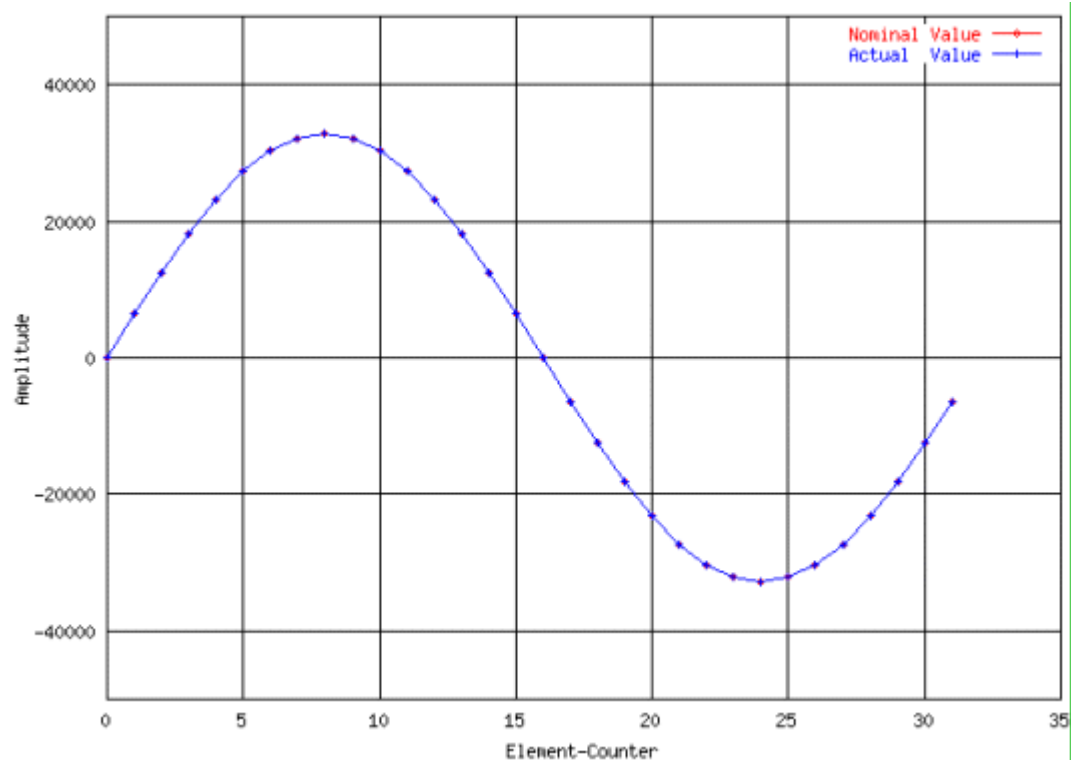


Fig. 329: Sine-shaped test pattern (TxTlkLocalLoop)

- **Tx Mod Nco**

First a complex sine function will be generated; the real part corresponds to the cosine, the imaginary part corresponds to the sine function. Afterward an FFT is performed. The result will be analyzed for number and height of the peaks in relation to the maximum peak. If due to the analyses of the FFT, a deviation from a pure sine function is detected, an error message will be generated.

Before running this test, the sequence clock must be enabled and subsequently disabled again.

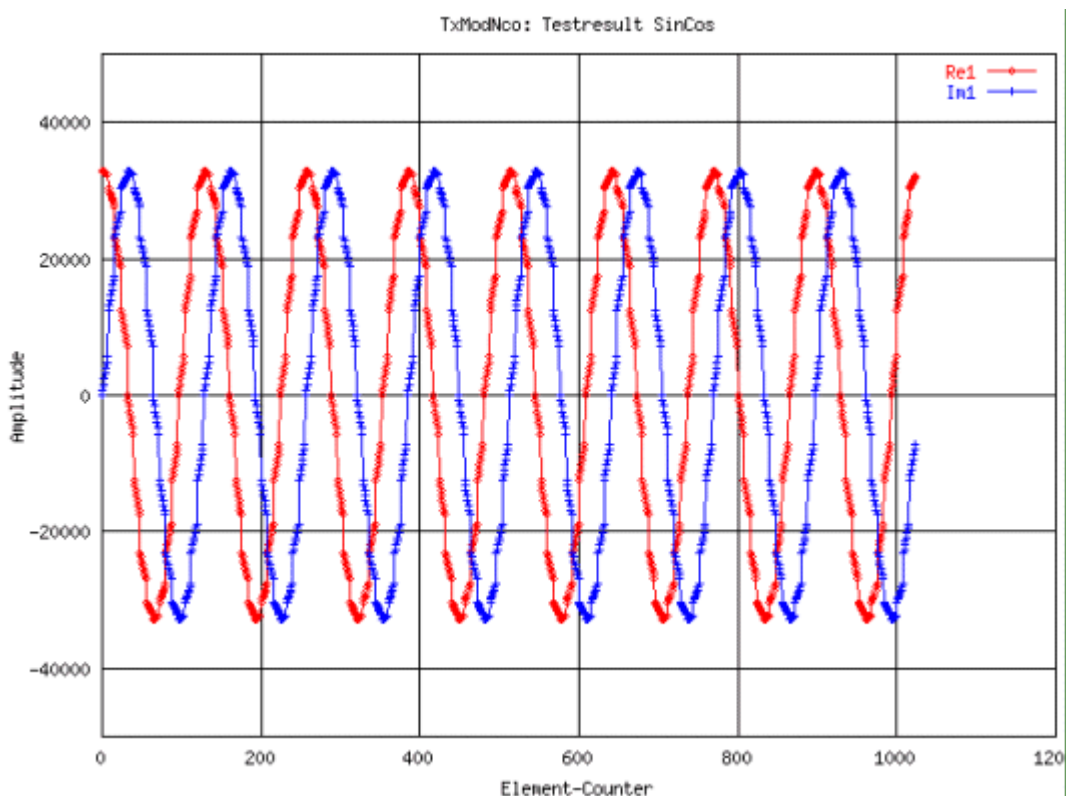


Fig. 330: Complex sine function (TxModNco)

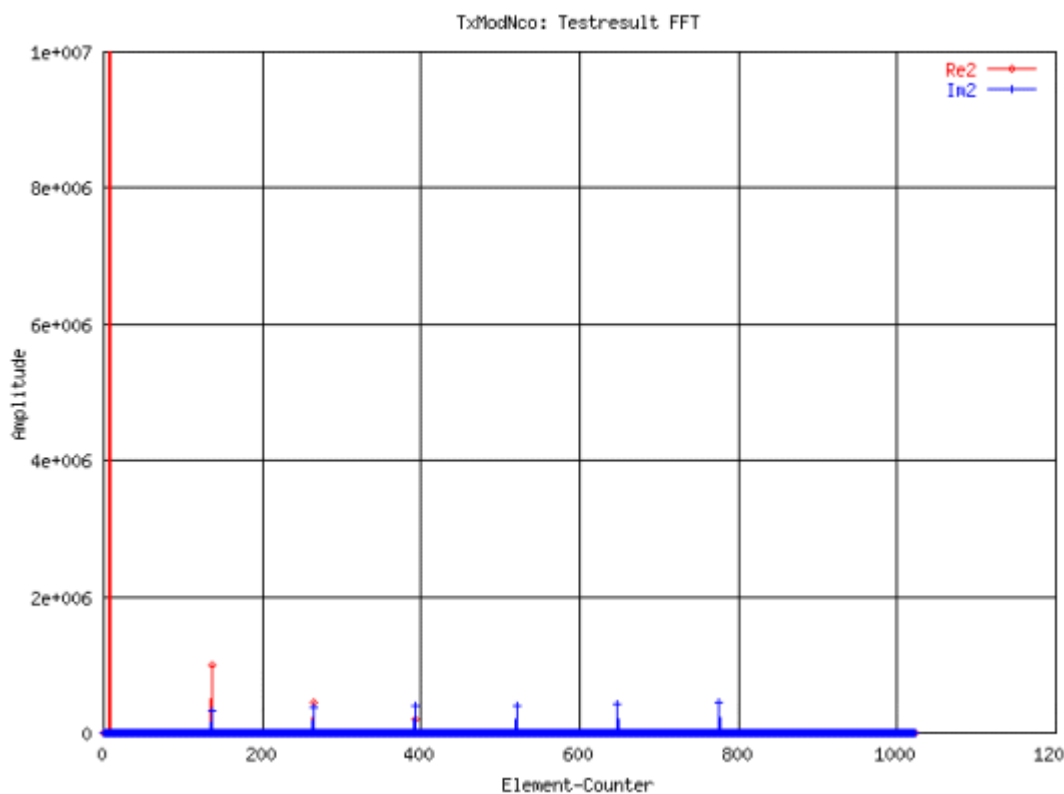


Fig. 331: FFT of the sine function

- **Gc ExtMem Test**

This test checks to ensure that the external memory of the DSP is working properly.

- **GcMem Test**

This test checks to ensure that the memory of the DSP is working properly.

- **GcDualPortRam**

This function is a combination of the external memory test via PCI bus (MPCU) and PciGc and a bus arbitration test.

- **Gc Addr Test**

This function tests the address decoding.

- **Gc Data Bus**

Test of the data bus with each other inverse test patterns (0x5a5a5a5a/0xa5a5a5a5).

- **Gc Addr Bus**

Test of the data bus with each other's inverse test patterns (0x5a5a5a5a/0xa5a5a5a5).

- **Gc Rxif Local Loop**

In a local loop a test pattern will be written in the RX interface and read back.

- **Gc Rxif Remote Loop**

In a remote loop including the RX interface, a test pattern will be written on the first PCI-RX component in the imager (MRIR) and read back.

- **GcCw Local Loop**

In a local loop a test pattern will be written in the CW interface and read back.

- **GcCw RemoteLoop**

In a loop including the LWL-INT, a test pattern will be written in the CW interface and read back.

- **GcGradLocalLoop**

In a local loop, a test pattern will be written in the gradient interface and read back.

- **GcGradRemoteLoop**

In a loop including the gradient small signal module, a test pattern will be written in the gradient interface and read back.

- **GcComRemoteLoop**

In a remote loop including PCI-TX and PCI-MON boards, several test patterns will be written on the Com interface and read back.

- **Gpa Inter Lin (Linearity test) X, Y, Z**

First a positive, trapezoid gradient pulse (100 % amplitude), second a negative trapezoid gradient pulse (100 % amplitude) on a selected axis will be generated. Using the STIMO-interface, the read signals on all three axes will be checked. A signal should be present on the chosen axis only. If no error was detected at this time, the pulse will be stepped from 0 % to 100 % amplitude and will be measured in the flat positive and negative region. The maximum deviation from the expected value below (rMin) and above (sMax) will be calculated. The deviation may not be higher than $\pm 0.05\%$ of the maximum value. The test runs for several seconds.

The read back signals will be represented in the external DSP memory (shared memory). If an unexpected signal has been measured on an axis, the values for the three axes are approximately the range ± 30000 . If the test was successful or if the read back values are out of specification, the values in the shared memory are between ± 1.0 for the axis under test and zero for the axes that were not measured.

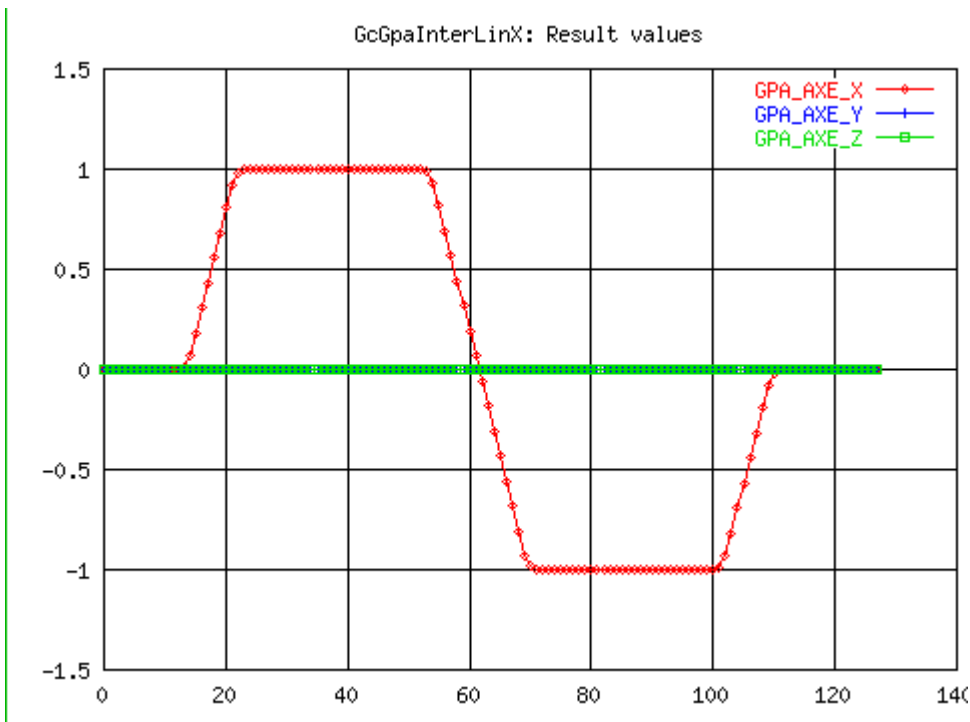


Fig. 332: Results GpaInterLin (example X axis)

- **Gpa Inter Offset (Offset test) X,Y, Z**

The gradient signal will be set to zero on the selected axis. With the STIMO-Interface, the 128 measured values will be read back. The maximum and minimum values will be detected. The absolute values of the maximum and minimum should not be higher than 0.4 % of the maximum value. The absolute maximum will be measured with sign.

Note: This test does not check if a signal is generated on a wrong axis or on more than one axis. Therefore the linearity test should be run first.

- **Gpa Inter Noise (Noise test) X,Y, Z**

First a positive-negative-positive gradient pulse with 100 % amplitude will be generated. The delay will be calculated from the signal read back from the STIMO interface. The delay must be between 1 and 16 pick up samples. The gradient pulses will be overlaid. After correction, the standard deviation will be calculated, which must not be higher than ± 0.1 % of the maximum value.

Note: This test does not check if a signal is generated on a wrong axis or on more than one axis. Therefore the linearity test should be run first.

The read back values will be represented in the shared memory in the range ± 1.0 for the axis under test.

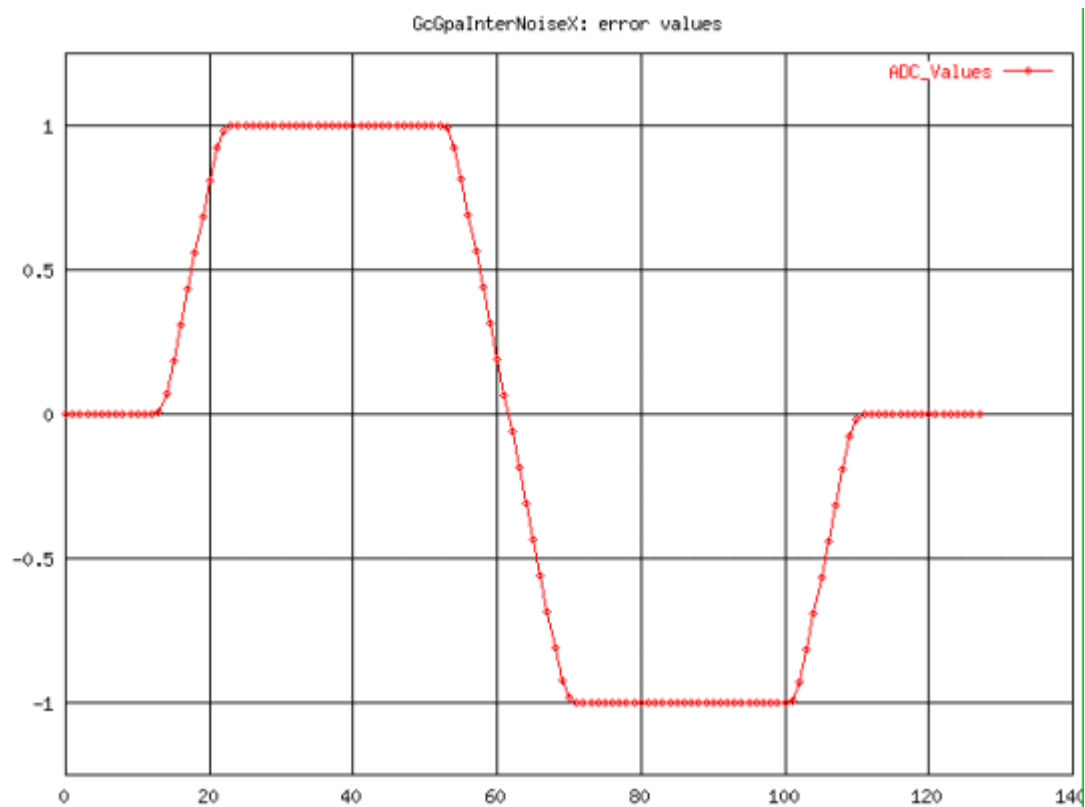


Fig. 333: Results GpaInterNoise (example X axis)

- **MPCU Reload**

With this function the MPCU is reloaded.

NOTE

If especially several remote loops (e.g. GcCw RemoteLoop, etc.) fail and the test TxModNco shows "no result available", check the 10 MHz clock signal (sine shaped) from RFSU/ synthesizer D10 at X7/PCI-TX.

If the clock signal is missing, red LEDs V10 at PCI-TX and/or V5 at PCI-MON are ON.

NOTE

The PCI-TX board can be interchanged with the PCI-MON board to localize the error.

PCI-MON

HP-Z000100Q-301105-02-005

PCI-MON (Functional)

- **RFPA_Power_OFF**
- **GPA_Power_OFF**
- **PaliSimMode**
- **GcSeqClk On**
- **Stimo ExtMem Test**

This test checks to ensure that the external memory of the DSP is working properly.
- **Stimo Mem Test**

This test checks to ensure that the external memory of the DSP is working properly.
- **StimoDualPortRam**

This function is a combination of the external memory test via PCI-bus (MPCU) and PciStimo and a bus arbitration test.
- **Stimo Addr Test**

This function tests the address decoding.
- **Stimo Data Bus**

Test of the data bus with each other's inverse test patterns (0x5a5a5a5a/0xa5a5a5a5).
- **Stimo Addr Bus**

Test of the address bus with each other's inverse test patterns (0x5a5a5a5a/0xa5a5a5a5).
- **StimoAdcRemoteLoop 5A/A5**

In a remote loop a 5 bit test pattern is written via cable W3522 and connector X15 at the AMC backplane to the GSSU/D60 and read back.
- **StimoAdcRemotePattern 1/2**

In a remote loop a test pattern will be written via cable W3522 and connector X15 at the AMC backplane to the GSSU/D60 and read back.
- **Pali ExtMem Test**

This test checks to ensure that the external memory of the DSP is working properly.
- **Pali Mem Test**

This test checks to ensure that the memory of the DSP is working properly.
- **Pali DualPortRam**

This function is a combination of the external memory test PCI bus (MPCU) and PciPali and a bus arbitration test.
- **Pali Addr Test**

This function tests the address decoding.

- **Pali Data Bus**

Test of the data bus with each other's inverse test patterns (0x5a5a5a5a/0xa5a5a5a5).

- **Pali Addr Bus**

Test of the data bus with each other's inverse test patterns (0x5a5a5a5a/0xa5a5a5a5).

- **MPCU Reload**

NOTE

If especially several remote loops (e.g. StimoAdcRemoteLoop 5A/A5, etc.) fail, check the 10 MHz clock signal from the synthesizer at X7/PCI-TX.

If the 10 MHz clock signal from synthesizer is missing, red LEDs V10 at PCI-TX and/or V5 at PCI-MON are ON.

NOTE

The PCI-TX board can be interchanged with the PCI-MON board to localize the error.

Dynamic Test

HP-Z0000YBM-030507-02-001

System Type: Avanto

Espre

SymphonyTim

AMC Dynamic Test (Functional)

This test runs a super-fast sequence that tests the dynamic behavior of the entire AMC, all boards are involved.

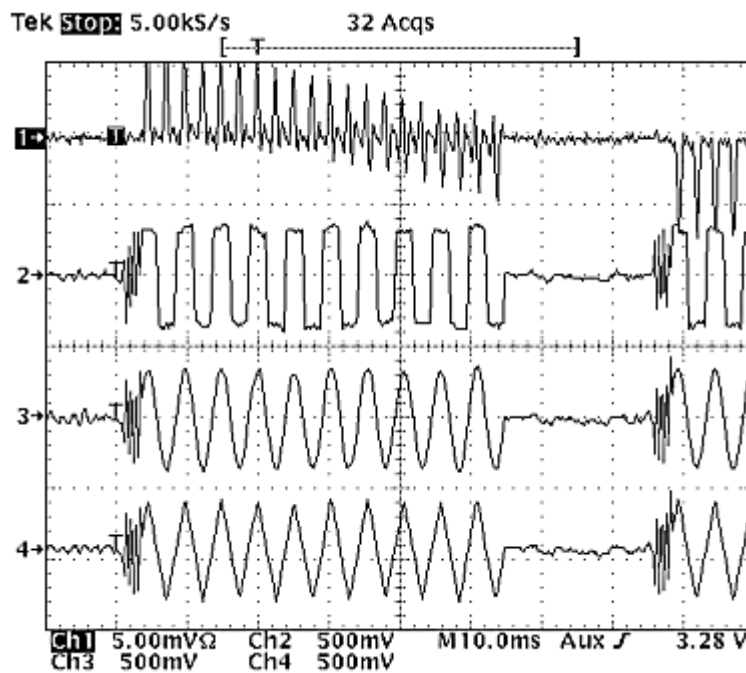


Fig. 334: Overview pulse diagram

Pulse sequence:

- Row 1: RF 20 RF pulses, pulse width 2.5 ms, pulse distance 20 μ s
- Gradient x, y, z per axis 8 triangular pulses with ramp 160 μ s
- Row 2: x-gradient 20 rectangular pulses, pulse width 2.42 ms, ramp 160 μ s
- Row 3: y-gradient 20 sine pulses, pulse width 2.57 ms
- Row 4: z-gradient 20 triangular pulses, pulse width 2.57 ms

Total scan duration 80 ms with TR=30 ms.

FPGA Versions

HP-Z000100Q-301105-02-007

FPGA Versions (Functional)

- PaliSimMode
- GcSeqClk On
- PCI-TX-FPGA Version
- PCI-MON-FPGA Version
- Modulator-FPGA Version
- Imager PCI-RX-FPGA Versions
- MPCU Reload

The function "PaliSimMode" running on PCI-TX is the prerequisite for checking the FPGA versions.

RF

HP-Z000100Q-011205-02-003

System Type: SymphonyTim
Avanto

RF - Avanto/ Symphony, A Tim System

NOTE

This section describes the MAGNETOM Avanto and Symphony, A Tim system.

The RF tests consists of a set of tests designed to verify the performance of the RF system and to assist you in locating defective components.

All displayed graphs can be enlarged by clicking into the image and selecting (dragging) the region of interest. With "autoscale", the original view of the graph is restored.

General

The RF System hardware has been designed with several built-in loops to allow automated testing of RF components. In addition, level measurements are performed via the directional couplers (DICO) in the RFPA, the TALES-C and the TAS-3TM.

The sequence used is common to all RF loop tests except the Body Coil tests. It is called rf_loop and contains six RF pulses of which the first two are relevant for TestTools and the others have zero amplitude. The first pulse is in sinc (function $\sin(x)/x$) form with constant amplitude and is used for level measurements and stability evaluations. The second pulse is also in sinc form and increases from 0 up to the maximum amplitude. It is used in linearity evaluations.

NOTICE

The 50 Ohm termination of the RFPA1 (K2204 DORA) is performed automatically. The dummy load is located in the TAS-3TM at the filter plate and is water-cooled in series with the gradient coil cooling.

When using the RFPA2 (K2137 CORA) or the RFPA1 (K2204 DORA) in LC/ Multi Nucleus mode, the cable to BCCS/X7 must be connected to TALES/X6 while a coil or the service plug must be connected to coil plug 1.

⇒ To prevent damage to the dummy load, do not perform RFPA tests without proper cooling.

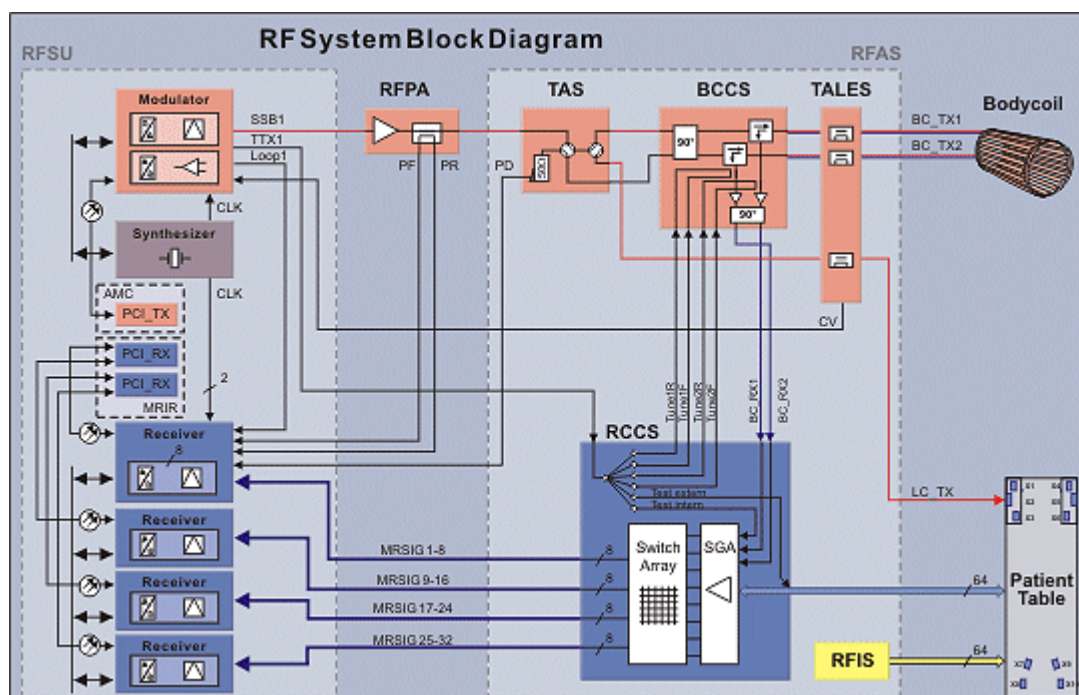


Fig. 335: RF System Block Diagram

Overview RF Tests

Test	Functional	Comm/Status	Interactive	Multi Nucleus	LongTime Stab
MOD/REC	X	-	-	X	X
RFPA1	X	X	-	X	X
RFPA2	-	X		X	-
RCCS	-	X	X		-
RCCS plug i	-	-	X	-	-
RCCS Switch	-	-	X	X	-
BCCS/RCCS	X	-	X	-	X
RFIS	-	X	X	-	-
RFAS	X	X	X	X	-
BC	X	-	X	-	-
Cable BC/LC	-	-	X	-	-

Note:

x= 1 to 10.

Stability

The first pulse of the service sequence rf_loop with a constant RF amplitude is used. The resulting magnitude data is the evaluation of the stability of the center column which is located in the center of the RF pulse. The numerical result is the maximum minus the minimum value divided by the mean value. Instabilities normally cause phase ghosting or poor S/N values. An overlaid graphic represents an average of the samples close to the center column. This allows to differentiate between noise and instabilities.

Linearity

For linearity evaluations, the second RF amplitude is stepped from 0 to a positive pulse amplitude. The center column of each line (maximum of the single pulses) is fitted to a linear function. The numerical result is the maximum deviation from the linear fit divided by the maximum value making up the data set. Non-linearities result in poor slice profiles and reduced S/N.

NOTE

With many tests the first measurements are not used for the final evaluation. So don't be worried if the graphics display and the numerical differ.

Conclusions

With respect to the signal path and data acquisition, the test is identical to the Modulator-Receiver loop. The only difference is that only one raw data line is measured and only the first pulse will be evaluated to check the symmetric signal processing of the receivers.

Reference Voltage (TALES)

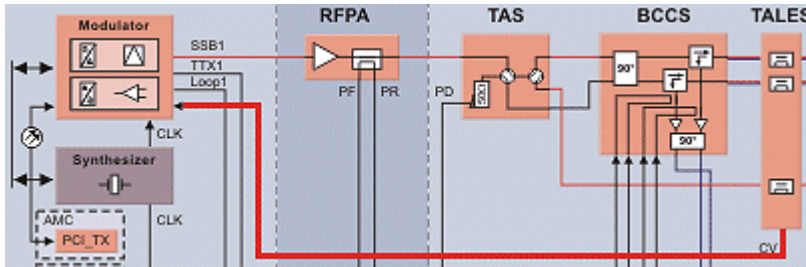


Fig. 338: Reference voltage

Signal Path

The TALES output signals are fed via the modulator (CV-Input) to the AMC/PCI-TX.

Evaluation

Verification that the TALES reference voltage is within 460 to 510 mV.

Conclusions

The TALES continuously supplies all internal directional coupler (DICO) signals triggered by the controller. In addition, it supplies the signal of an integrated reference voltage source. The DICO signals and the reference voltage are multiplexed and fed to the modulator CV input. The reference voltage test provides information regarding correct control of the TALES, intact cabling to the modulator/AMC and signal processing during data acquisition. In case of error, the reference voltage signal can be measured with a digital voltmeter. For further information, refer to the Troubleshooting Guide.

RFPA

HP-Z000100Q-021205-02-006

System Type: Avanto
SymphonyTim

- RFPA Communication/Status
- RFPA Power
- RFPA Power (max)
- RFPA Loop
- RFPA Loop (linear characteristic)
- RFPA-TALES
- RFPA-TALES (linear characteristic)
- RFPA Long Time Stability

NOTE

The functional tests use the BC-transmit path. The 50 Ohm termination of the RFPA (K2204 DORA) is performed automatically. While the RFPA Power, Loop and TALES tests can also be used in the MultiNucleus mode; the LC-transmit path is used.

NOTICE

Using the Multi Nucleus tests, the RFPA (K2204 DORA) transmits to the LC-path (low mode).
⇒ For proper 50 Ohm termination, the cable to BCCS/X7 must be manually connected to TALES/X6 and a coil or the service plug must be connected to coil plug 1.

RFPA Communication/Status

The communication test involves sending a command (status and version query) to an individual component which, if successfully executed, verifies communication with this component.

The status test queries the status of a particular component. It quickly determines whether the component status is OK or shows an error. A faulty status is indicated by three exclamation marks next to the respective status message.

RFPA Power

NOTE

RFPA Power Test: The test requires a valid RF characteristic.

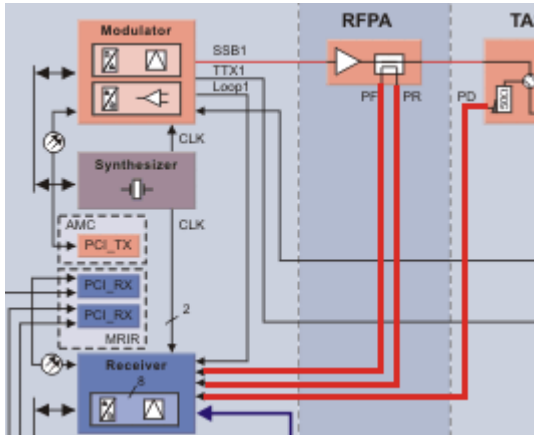


Fig. 339: RFPA Power Test

Signal Path

Functional tests, BC-transmit path:

The output signal SSB1 of the modulator is fed via the RFPA to the dummy load of the TAS-3TM. The Forward_Power Signal is measured via the directional coupler in the RFPA and at the dummy load TAS-3TM, which is displayed on the diagram as "TAS".

Multi Nucleus tests, LC-transmit path:

The output signal SSB1 of the modulator is fed via the RFPA to the TAS-3TM and via SAMI to TALEX X5. From TALEX X6 the signal is routed back to the dummy load TAS-3TM via cable W123 (disconnected from BCCS X7). The Forward_Power Signal is measured via the directional coupler in the RFPA and at the dummy load TAS-3TM, which is displayed on the diagram as "TAS".

The signal amplitude used for these measurements is the maximum amplitude read from the RF (BC/LC) characteristic data.

Evaluation

- The actual forward voltage (measured via directional coupler and TAS-3TM attenuator) is compared to the nominal value read from the RF characteristic measurement.
- The graphical output shows the power forward and power reflected measured by DICO (PF/PR) as well as by TAS-3TM (PD), measured over the total voltage range.
- The ratio of the reflected voltage to the forward voltage (measured via directional coupler) is calculated.

Conclusions

The objective of the test is to ensure that the RFPA operates at peak performance and that the reflected voltage remains below 20% of the forward voltage at a dummy load of 50Ohm.

Peak performance can no longer be achieved when the total gain (signal processing in the RFSU and gain in the RFPA) and, as a result, the characteristic curve has changed due to aging components. If the deviation lies outside the acceptable tolerance, the RF characte-

ristic must be reexamined. If peak performance still cannot be achieved, the output level of the modulator and the directional coupler in the RFPA should be checked to determine the source for error. For further information, refer to the Troubleshooting Guide.

If there is an increase in the reflected voltage, check all cable connections from the RFPA to the Receiver and the transmission cable to the TAS-3TM assembly. You should also check with the LEDs if the TAS-3TM relay is set correctly. Refer to the Function Description. A defective dummy load would cause high voltage reflection as well. In this case the measured voltage by TAS-3TM would be correspondingly low.

RFPA Loop

The RFPA Loop test is performed in two steps. The first measurement is performed with the valid RF characteristic, the second measurement with a predefined linear characteristic, but with different transmitter amplitudes and specifications. If there is no valid characteristic available, i.e. after the software has been installed and before the tune-up has been performed, the linear characteristic is used instead of the valid characteristic for the first measurement as well. In this case, the deviation may be significantly above specifications or the test may even abort.

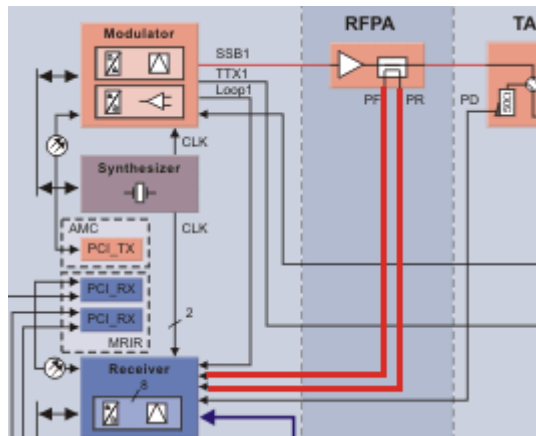


Fig. 340: RFPA Loop

Signal Path

Functional tests, BC-transmit path:

The output signal of the modulator is fed via the RFPA to the dummy load of the TAS-3TM. The forward signal is measured via the directional coupler in the RFPA (PF/PR).

Multi Nucleus tests, LC-transmit path:

The output signal SSB1 of the modulator is fed via the RFPA to the TAS-3TM and via SAMI to TALES X5. From TALES X6 the signal is routed back to the dummy load TAS-3TM via cable W123 (disconnected from BCCS X7). The forward signal is measured via the directional coupler in the RFPA (PF/PR).

Evaluation

The raw data of the first sinc-pulse are evaluated for stability. The raw data of the second sinc-pulse are evaluated for linearity.

The test consists of two measurements. One RF pulse is transmitted for each. In the first measurement, the signals are decoupled by the directional coupler (DICO/PF and PR) in the RFPA as well as by the TAS-3TM (reflected voltage from Body Coil/PD) and converted to voltages using a specific factor. The second measurement acquires the signals from the TALES.

Evaluations

- Forward voltage from the TALES is within tolerance
- Reflected voltage from the TALES is below the limit
- Deviation of the forward voltage from the TALES and the DICO
- Ratio of the reflected and forward voltage from the DICO

Conclusions

This test checks almost the entire transmission component of the system.

The results of the forward voltage at the TALES provide information on whether the actual value acquired by the TALES corresponds to the nominal value determined by the RF characteristic.

The value of the reflected voltage at the TALES is a gauge for the 50 Ohm termination through the Body Coil/ TAS-3TM.

The forward voltage measured via DICO must approximately match that of the TALES. Since the correlation between the DICO value and the TALES voltage only occurs when the characteristic curve is generated (determination of the factor for conversion from receiver ADC values into voltage values), a valid characteristic curve must be present.

The ratio of the reflected and forward voltages measured via DICO in the RFPA is generally used as a gauge for the 50Ohm termination of the RFPA. On the one hand, it is determined by the coil tuning, and on the other hand by the characteristics of the components that follow the RFPA in the transmit path, such as the TX cable, BCCS_123, TALES, etc.

RFPA Long Time Stability

The test is identical to the RFPA Loop test, but runs approximately 10 minutes to test the stability of the RFPA over longer measurement intervals.

RFPA1

HP-Z000100Q-021205-02-007

System Type:Avanto
SymphonyTim

- RFPA1 Communication/Status
- RFPA1 Power
- RFPA1 Power (max)
- RFPA1 Loop
- RFPA1 Loop (linear characteristic)
- RFPA1-TALES
- RFPA1-TALES (linear characteristic)
- RFPA1 Long Time Stability

NOTE

The functional tests use the BC-transmit path. The 50 Ohm termination of the RFPA1 (K2204 DORA) is performed automatically.

While the RFPA1 Power, Loop and TALES tests can also be used in the MultiNucleus mode; the LC-transmit path is used.

NOTICE

Using the Multi Nucleus tests, the RFPA1 (K2204 DORA) transmits to the LC-path (low mode).

⇒ For proper 50 Ohm termination, the cable to BCCS/X7 must be manually connected to TALES/X6 and a coil or the service plug must be connected to coil plug 1.

RFPA1Communication/Status

The communication test involves sending a command (status and version query) to an individual component which, if successfully executed, verifies communication with this component.

The status test queries the status of a particular component. It quickly determines whether the component status is OK or shows an error. A faulty status is indicated by three exclamation marks next to the respective status message.

RFPA1 Power

NOTE

RFPA Power Test: The test requires a valid RF characteristic.

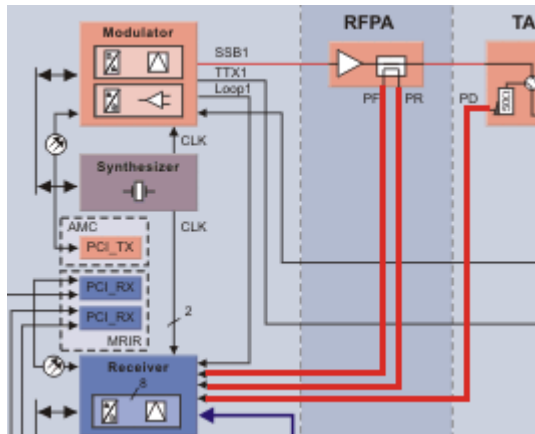


Fig. 342: RFPA Power Test

Signal Path

Functional tests, BC-transmit path:

The output signal SSB1 of the modulator is fed via the RFPA1 to the dummy load of the TAS-3TM. The Forward_Power Signal is measured via the directional coupler in the RFPA1 and at the dummy load TAS-3TM, which is displayed on the diagram as "TAS".

Multi Nucleus tests, LC-transmit path:

The output signal SSB1 of the modulator is fed via the RFPA1 to the TAS-3TM and via SAMI to TALEX X5. From TALEX X6 the signal is routed back to the dummy load TAS-3TM via cable W123 (disconnected from BCCS X7). The Forward_Power Signal is measured via the directional coupler in the RFPA1 and at the dummy load TAS-3TM, which is displayed on the diagram as "TAS".

The signal amplitude used for these measurements is the maximum amplitude read from the RF (BC/LC) characteristic data.

Evaluation

- The actual forward voltage (measured via directional coupler and TAS-3TM attenuator) is compared to the nominal value read from the RF characteristic measurement.
- The graphical output shows the power forward and power reflected measured by DICO (PF/PR) as well as by TAS-3TM (PD), measured over the total voltage range.
- The ratio of the reflected voltage to the forward voltage (measured via directional coupler) is calculated.

Conclusions

The objective of the test is to ensure that the RFPA1 operates at peak performance and that the reflected voltage remains below 20% of the forward voltage at a dummy load of 50Ohm.

Peak performance can no longer be achieved when the total gain (signal processing in the RFSU and gain in the RFPA1) and, as a result, the characteristic curve has changed due to aging components. If the deviation lies outside the acceptable tolerance, the RF characteristic must be reexamined. If peak performance still cannot be achieved, the output level of the modulator and the directional coupler in the RFPA1 should be checked to determine the cause of the error. For further information, refer to the Troubleshooting Guide.

If there is an increase in the reflected voltage, check all cable connections from the RFPA1 to the receiver and the transmission cable to the TAS-3TM assembly. You should also check with the LEDs if the TAS-3TM relay is set correctly. Refer to the Function Description. A defective dummy load would cause high voltage reflection as well. In this case the measured voltage by TAS-3TM would be correspondingly low.

RFPA1 Loop

The RFPA1 Loop test is performed in two steps. The first measurement is performed with the valid RF characteristic, the second measurement with a predefined linear characteristic, but with different transmitter amplitudes and specifications. If there is no valid characteristic available, i.e. after the software has been installed and before the tune-up has been performed, the linear characteristic is used instead of the valid characteristic for the first measurement as well. In this case, the deviation may be significantly above specifications or the test may even abort.

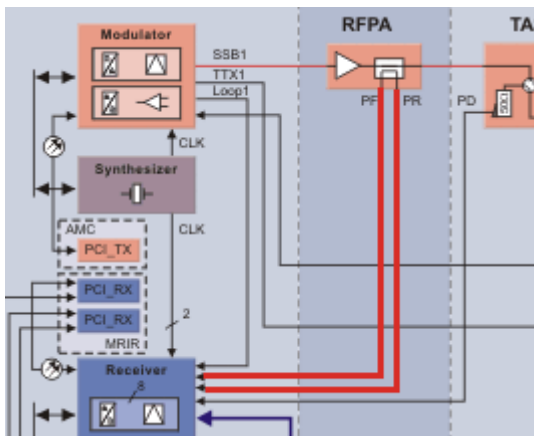


Fig. 343: RFPA Loop

Signal Path

Functional tests, BC-transmit path:

The output signal of the modulator is fed via the RFPA1 to the dummy load of the TAS-3TM. The forward signal is measured via the directional coupler in the RFPA1 (PF/PR).

Multi Nucleus tests, LC-transmit path:

The output signal SSB1 of the modulator is fed via the RFPA1 to the TAS-3TM and via SAM1 to TALES X5. From TALES X6 the signal is routed back to the dummy load TAS-3TM via cable W123 (disconnected from BCCS X7). The forward signal is measured via the directional coupler in the RFPA1 (PF/PR).

Evaluation

The raw data of the first sinc-pulse are evaluated for stability. The raw data of the second sinc-pulse are evaluated for linearity.

Conclusions

This test is used primarily to check the stability and linearity of RFPA1. Modulator control, data acquisition via receiver and RF characteristic acquisition must be intact and correct to evaluate the results.

RFPA1 - TALES

The RFPA1-TALES test is performed in two steps. The first measurement is performed with the valid RF characteristic, the second measurement with a predefined linear characteristic, but with different transmitter amplitudes and specifications. If there is no valid characteristic available, i.e. after the software has been installed and before the tune-up has been performed, the linear characteristic is used instead of the valid characteristic for the first measurement as well. In this case, the deviation may be significantly above specifications or the test may even abort.

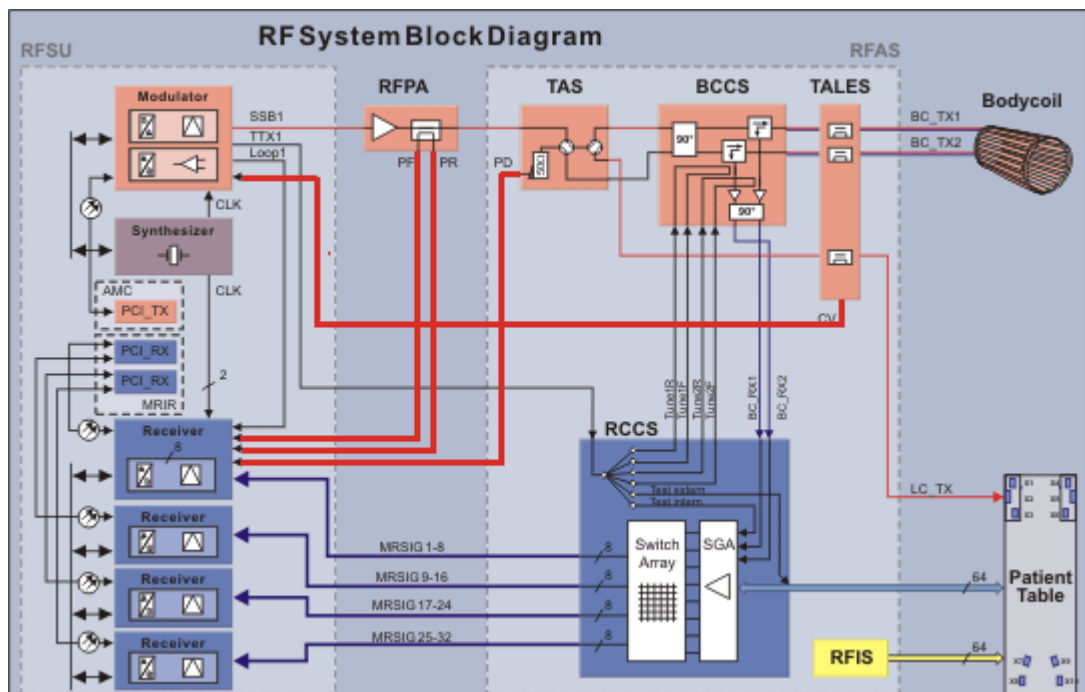


Fig. 344: RFPA TALES

Signal Path

Functional tests, BC-transmit path:

The output signal of the modulator SSB1 is fed via the RFPA1 - TAS-3TM - BCCS_63 - TALES to the Body Coil.

Multi Nucleus tests, LC-transmit path:

The output signal of the modulator SSB1 is fed via the RFPA1 - TAS-3TM - SAMI - TALES to TAS-3TM.

The test consists of two measurements. One RF pulse is transmitted for each. In the first measurement, the signals are decoupled by the directional coupler (DICO/PF and PR) in the RFPA1 as well as by the TAS-3TM (reflected voltage from Body Coil/PD) and converted to voltages using a specific factor. The second measurement acquires the signals from the TALES.

Evaluations

- Forward voltage from the TALES is within tolerance
- Reflected voltage from the TALES is below the limit
- Deviation of the forward voltage from the TALES and the DICO
- Ratio of the reflected and forward voltage from the DICO

Conclusions

This test checks almost the entire transmission component of the system.

The results of the forward voltage at the TALES provide information on whether the actual value acquired by the TALES corresponds to the nominal value determined by the RF characteristic.

The value of the reflected voltage at the TALES is a gauge for the 50 Ohm termination through the Body Coil/ TAS-3TM.

The forward voltage measured via DICO must approximately match that of the TALES. Since the correlation between the DICO value and the TALES voltage only occurs when the characteristic curve is generated (determination of the factor for conversion from receiver ADC values into voltage values), a valid characteristic curve must be present.

The ratio of the reflected and forward voltages measured via DICO in the RFPA1 is generally used as a gauge for the 50Ohm termination of the RFPA1. On the one hand, it is determined by the coil tuning, and on the other hand by the characteristics of the components that follow the RFPA1 in the transmit path, such as the TX cable, BCCS_123, TALES, etc.

RFPA1 Long Time Stability

The test is identical to the RFPA1 Loop test, but runs approximately 10 minutes to test the stability of the RFPA1 over longer measurement intervals.

RFPA2

HP-Z000100Q-021205-02-008

System Type: Avanto

SymphonyTim

- RFPA2 Communication/Status
- RFPA2 Power
- RFPA2 Power (max)
- RFPA2 Loop
- RFPA2 Loop (linear characteristic)
- RFPA2-TALES
- RFPA2-TALES (linear characteristic)

NOTICE

Using the Multi Nucleus tests, the RFPA2 (K2137 CORA) transmits to the LC-path.

⇒ For proper 50 Ohm termination, the cable to BCCS/X7 must be manually connected to TALES/X6 and a coil or the service plug must be connected to coil plug 1.

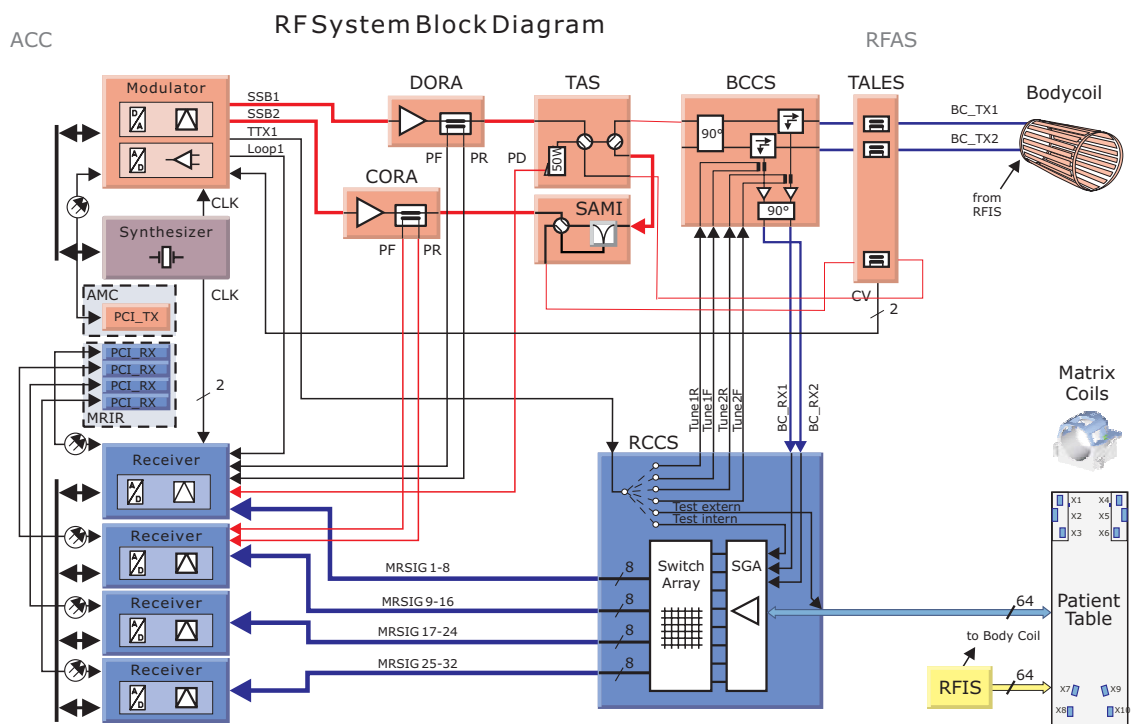


Fig. 345: RFPA2 Multi Nuclei tests

Evaluation

- The actual forward voltage (measured via directional coupler and TAS-3TM attenuator) is compared to the nominal value read from the RF characteristic measurement.
- The graphical output shows the power forward and power reflected measured by DICO (PF/PR) as well as by TAS-3TM (PD), measured over the total voltage range.
- The ratio of the reflected voltage to the forward voltage (measured via directional coupler) is calculated.

Conclusions

The objective of the test is to ensure that the RFPA2 operates at peak performance and that the reflected voltage remains below 20% of the forward voltage at a dummy load of 50Ohm.

Peak performance can no longer be achieved when the total gain (signal processing in the RFSU and gain in the RFPA2) and, as a result, the characteristic curve has changed due to aging components. If the deviation lies outside the acceptable tolerance, the RF characteristic must be reexamined. If peak performance still cannot be achieved, the output level of the modulator and the directional coupler in the RFPA2 should be checked to determine the source for error. For further information, refer to the Troubleshooting Guide.

If there is an increase in the reflected voltage, check all cable connections from the RFPA2 to the Receiver and the transmission cable to the TAS-3TM assembly. You should also check with the LEDs if the TAS-3TM relay is set correctly. Refer to the Function Description. A defective dummy load would cause high voltage reflection as well. In this case the measured voltage by TAS-3TM would be correspondingly low.

RFPA2 Loop

The RFPA2 Loop test is performed in two steps. The first measurement is performed with the valid RF characteristic, the second measurement with a predefined linear characteristic, but with different transmitter amplitudes and specifications. If there is no valid characteristic available, i.e. after the software has been installed and before the tune-up has been performed, the linear characteristic is used instead of the valid characteristic for the first measurement as well. In this case, the deviation may be significantly above specifications or the test may even abort.

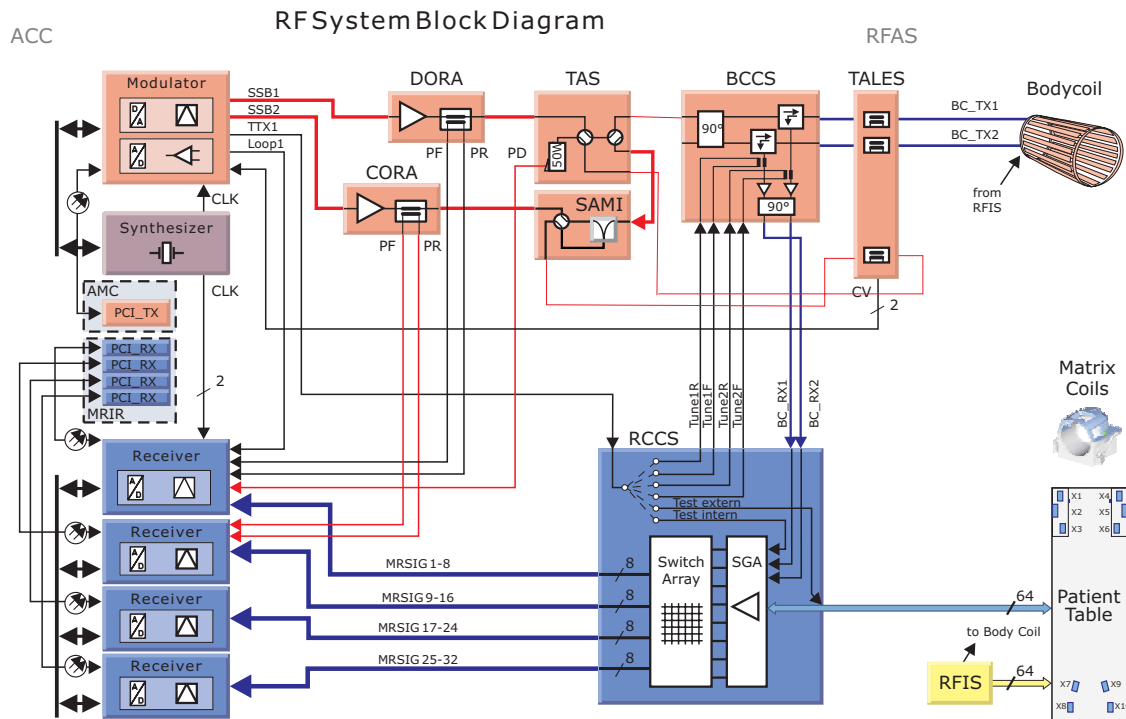


Fig. 347: RFP2 Multi Nuclei tests

Signal Path

Multi Nucleus tests, LC-transmit path:

The output signal SSB2 of the modulator is fed via the RFP2 to the TAS-3TM and via SAMI to TALES X5. From TALES X6 the signal is routed back to the dummy load TAS-3TM via cable W123 (disconnected from BCCS X7). The forward signal is measured via the directional coupler in the RFP2 (PF/PR).

Evaluation

The raw data of the first sinc-pulse are evaluated for stability. The raw data of the second sinc-pulse are evaluated for linearity.

Conclusions

This test is used primarily to check the stability and linearity of the RFP2. Modulator control, data acquisition via receiver and RF characteristic acquisition must be intact and correct to evaluate the results.

RFP2 - TALES

The RFP2 -TALES test is performed in two steps. The first measurement is performed with the valid RF characteristic, the second measurement with a predefined linear characteristic, but with different transmitter amplitudes and specifications. If there is no valid characteristic available, i.e. after the software has been installed and before the tune-up has

been performed, the linear characteristic is used instead of the valid characteristic for the first measurement as well. In this case, the deviation may be significantly above specifications or the test may even abort.

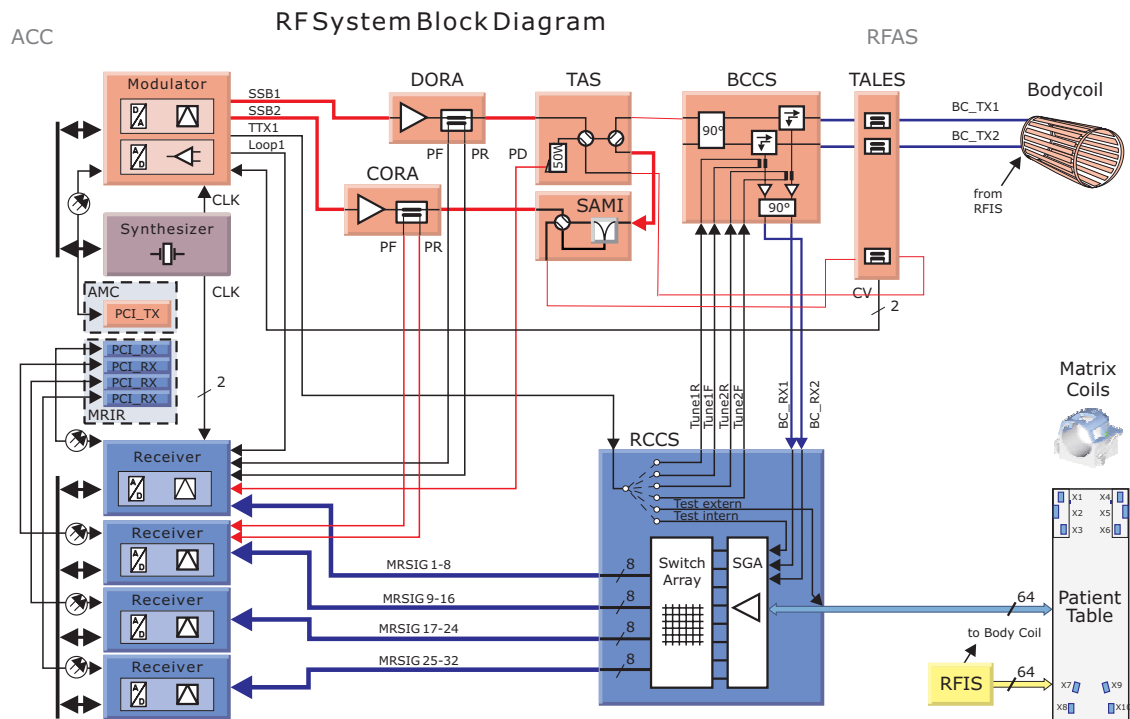


Fig. 348: RFPA2 Multi Nuclei tests

Signal Path

Multi Nucleus tests, LC-transmit path:

The output signal of the modulator SSB2 is fed via the RFPA2 - TAS-3TM - SAMI - TALES to TAS-3TM.

The test consists of two measurements. One RF pulse is transmitted for each. In the first measurement, the signals are decoupled by the directional coupler (DICO/PF and PR) in the RFPA2 as well as by the TAS-3TM (reflected voltage from Body Coil/PD) and converted to voltages using a specific factor. The second measurement acquires the signals from the TALES.

Evaluations

- Forward voltage from the TALES is within tolerance
- Reflected voltage from the TALES is below the limit
- Deviation of the forward voltage from the TALES and the DICO
- Ratio of the reflected and forward voltage from the DICO

Conclusions

This test checks almost the entire transmission component of the system.

The results of the forward voltage at the TALEs provide information on whether the actual value acquired by the TALEs corresponds to the nominal value determined by the RF characteristic.

The value of the reflected voltage at the TALEs is a gauge for the 50 Ohm termination through the Body Coil/ TAS-3TM.

The forward voltage measured via DICO must approximately match that of the TALEs. Since the correlation between the DICO value and the TALEs voltage only occurs when the characteristic curve is generated (determination of the factor for conversion from receiver ADC values into voltage values), a valid characteristic curve must be present.

The ratio of the reflected and forward voltages measured via DICO in the RFPA2 is generally used as a gauge for the 50Ohm termination of the RFPA2. On the one hand, it is determined by the coil tuning, and on the other hand by the characteristics of the components that follow the RFPA2 in the transmit path, such as the TX cable, BCCS_123, TALEs, etc.

RCCS

HP-Z000100Q-021205-02-009

- RCCS (Communication/Status)
- RCCS SGA Loop (High gain) (Functional)
- RCCS Matrix Test ("Interactive")
- RCCS plug 1-10 (Interactive, no "RCCS plug 5" with MAGNETOM Trio Tim and Verio)

RCCS Communication/Status

The communication test involves sending a command (status and version query) to an individual component which, if successfully executed, verifies communication with this component.

The status test queries the status of a particular component. It quickly determines whether the component status is OK or shows an error. A faulty status is indicated by three exclamation marks next to the respective status message.

```
RCCSK2254_STATUS:  FPGA                      = ok
RCCSK2254_STATUS:  CLOCK CONTROL              = active
RCCSK2254_STATUS:  Volt (sum signal)          = ok
RCCSK2254_STATUS:  10 Volt for local coils = on
```

Conclusion

If the communication test of the RCCS fails, the power supply of the RCCS (same as RFIS), the cable connections or the RCCS/SLIO may be defective.

RCCS SGA Loop (High gain) (Functional/ Multi Nuclei)

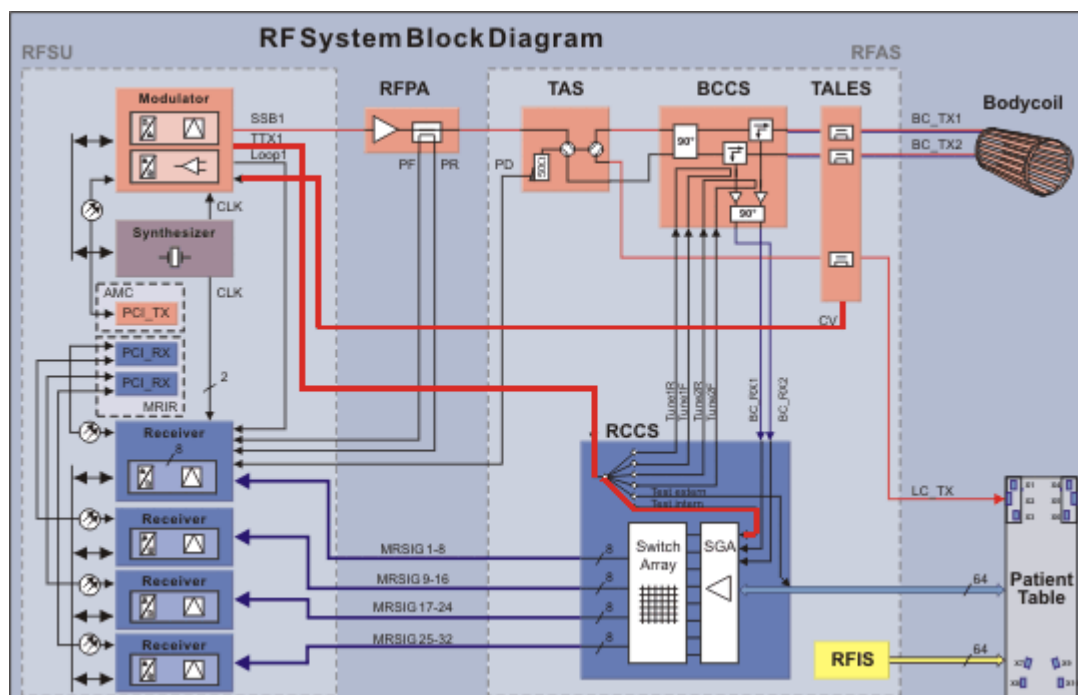


Fig. 349: RCCS SGA Loop (high gain)

Signal Path

The TTX output signal of the modulator is fed to the RCCS. The RCCS input channels (with SGAs on high gain) are switched to the output in the following way:

Input channel 1-8, 9-16, 17-24, 25-32, 33-40, 41-48, 49-56, 57-64 to output channel 1-8 and the Body Coil channels 65-72 to output channel 1-8 once after another.

Evaluations

With every measurement, 64 lines (except the first 7 lines) will be checked for stability and linearity against specifications.

Conclusion

If this test fails, the corresponding SGA's of the RCCS may be defective.

RCCS Matrix Test (Interactive/ Multi Nuclei)

The RCCS Matrix Test checks the 2048 + 8 switch matrix nodal points (64 inputs LC and 8 inputs BC, at maximum 32 outputs) in the RCCS.

The total measurement time for a 32 receive channel system is approximately 20 minutes (selection of complete interactive test), and increases with the selection of the Multi Nuclei tests.

Beside this you can select particular input channels to the RCCS matrix corresponding to the BC-channels or to the nine local coil plugs on the patient table.

Evaluations

The raw data of the first sinc-pulse are evaluated for stability, the raw data of the second sinc-pulse are evaluated for linearity.

Conclusions

The RCCS SGA Loop (High gain) test is used primarily to check the stability and linearity of the preamplifiers in the RCCS. Modulator control and data acquisition via the receiver must be intact to evaluate the results. The SGA amplifiers (high gain und low gain) are located in the RCCS (input channels).

In the RCCS Matrix Test, the measured signal must be within a certain range. If only single measurements fail (with too low signal), the RCCS is probably defective. If a total row of nodal points next to each other fail (e.g. input #1 to #64 to output #5), the corresponding cable connection RCCS output channel (#5 in this case) to the corresponding receiver may be defective.

RCCS plug1

HP-Z000100Q-021205-02-012

System Type:SymphonyTim
Espreo
Avanto

RCCS plug 1-10 - RCCS Service Plug , pos i (interactive)

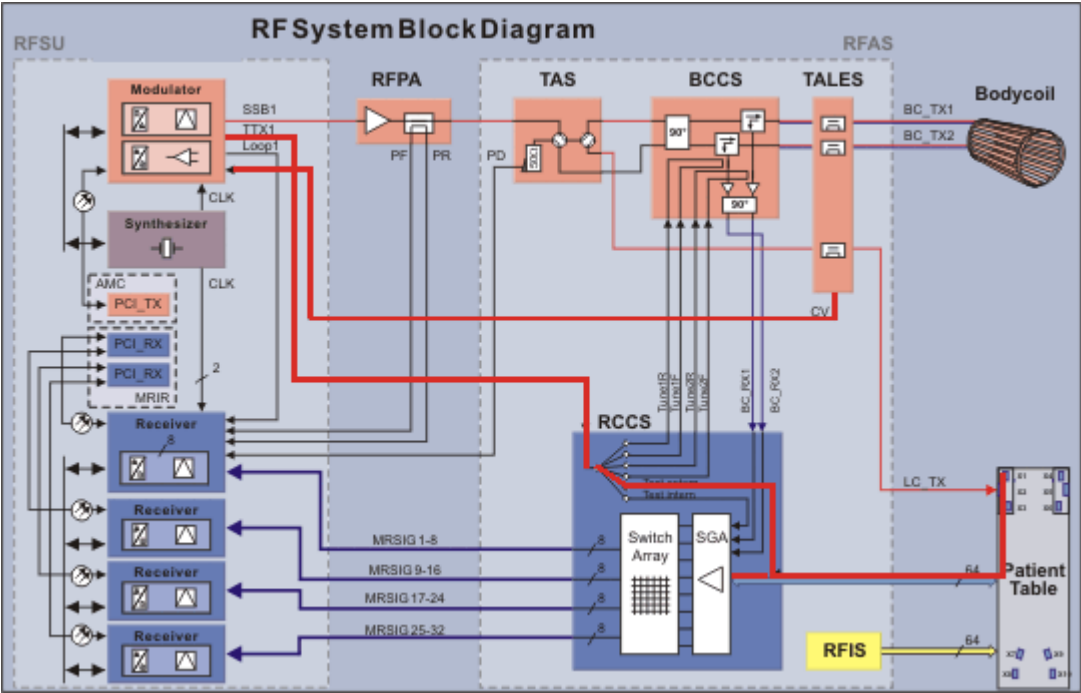


Fig. 350: RCCS interactive test

Preparation

The Service Plug must be connected to a coil connector 1 to 10 at the patient table. Follow the instructions provided during the test run.

Signal Path

The TTX output signal of the modulator is fed via RCCS to coil plug 1 and via auxiliary cable to the Service Plug (attenuator network); the signal goes back via the coil plug under test and cabling to the RCCS and (via switching matrix) to the receiver module.

NOTE

When testing coil connector #1, the auxiliary cable has to be connected to coil connector #4 and the normal cable to connector #1.
When testing all other coil connectors, the auxiliary cable has to be connected to coil connector #1 and the normal cable to the coil connector under test.

In the process, the signal path in the RCCS is selected so that the receiver module 1, channels 1 to 8 according to the number of channels of the plug under test, are used.

Evaluation

The TTX signal is attenuated in the attenuator network of the Service Plug in such a way that the signal level of the first channel is 100%, the other always approximates half of the preceding channel. The signal levels are checked against specifications. The graduated levels allow testing of the cabling from the local coil plug to the RCCS.

Conclusions

In case of an error, the RCCS could be the source. However, it could also result from incorrect or defective cabling from the local coil receptacle to the RCCS (movable cable in the patient table) and the receiver module.

The cable from the coil receptacle to the RCCS and/or the coil receptacle may be defective, if the 10 V preamplifier supply voltage is missing. The corresponding 10 V green and yellow test LED's of the Service Plug are off.

A 10 V error (caused by overload or a short) will be latched in the RCCS and red LED's, corresponding to the input channels, light up. In this case it is necessary to reboot the scanner after eliminating the error.

If the 10 V preamplifier voltage is ok, a preamplifier or switch matrix node of the RCCS may be defective.

On 8 channel local coil receptacles, the value of channel eight may be out of tolerance. Because of the low signal level, this is no proof of an error.

Before replacing the RCCS, you should test the RF receive path first (local coil receptacle - RCCS).

Table of signal lines and connectors:

Coil connector - RX Pin		Connector at RCCS	RCCS # Input Channel or 10V Error LED
LC 1 (8 channel)	A1-A8	X311, 1- 8	1 - 8
LC 2 (4 channel)	1, 13, 18, 10	X311, 9 and X312, 1 - 3	9 - 12
LC 3 (6 channel)	A1-A6	X312, 4 - 9	13 - 18
LC 4 (8 channel)	A1-A8	X321, 1 - 8	19 - 26
LC 5 (4 channel)	1, 13, 18, 10	X321, 9 and X322, 1-3	27 - 30
LC 6 (6 channel)	A1-A6	X322, 4 - 9	31 - 36
LC 7 (6 channel)	A1-A6	X331, 1 - 6	37 - 42
LC 8 (8 channel)	A1-A8	X331, 7 and X332, 1 - 7	43 - 50
LC 9 (6 channel)	A1-A6	X341, 1 - 6	51 - 56
LC 10 (8 channel)	A1-A8	X341, 7 and X342, 1 - 7	57 - 64

Coil connector - RX Pin	Connector at RCCS	RCCS # Input Channel or 10V Error LED
TX-signal - Pin		
TX	Pin A9 on coil connector 1 (from TALES X6)	
TTX	Pin A1 on coil connector 1 (input RCCS X60)	

RCCS connector	Cable	PTAB connectors	Local coil connector
X311, X312	W5112	X102 (feet)/ X105 (head)	1 - 3
X321, X322	W5113	X103 (feet)/ X106 (head)	4, 6
X331, X322	W5110	X101 (feet)	7, 8
X341, X342	W5114	X104 (feet)	9, 10

Test of Preamplifier 10 VDC supply voltage

The voltage supply of the internal RX preamplifier (+10 V; 9.4 to 10.9 VDC) of the local coils can be checked and measured at the Service Plug. The voltage is continuously present independent of a test in progress.

According to the number of channels 4, 6 or 8 green LED's are on => voltages are within tolerance limits, yellow LEDs are on => voltages are present but may be out of tolerance.

If no LED is on, the cable from RCCS to the coil plug may be shorted or the RCCS may be defective. Check the error LED's at the RCCS.

The voltage is influenced by the PIN diode voltage switching. For this reason, abort a RFIS PIN diode test for an accurate measurement of the preamplifier supply voltage with a DVM.

The voltage can be tested with load on (approx. 70 mA) or off (10 - 20 mA). Press the latch reset button to start the test.

If measured with a DVM, a voltage between + 7.9 VDC and + 11 VDC is acceptable. (The voltage measured at the Service Plug sockets is about 0.8 V lower than the voltage at the local coil connector pins because of the voltage drop on the PIN diodes in the Service Plug).

Conclusion

Measurement at the Service Plug	
Green LED OFF 10 VDC (LC pre-amplifier) not OK Measured voltage is lower than 9.4 V:	1. Broken or shorted cable between RCCS and coil connector or short in coil plug. A lighted red LED at RCCS output indicates the corresponding channel. 2. 15 V to 10 V conversion on RCCS defective. Check voltage LED's at RCCS and error messages!

RCCS plug2

HP-Z000100Q-051205-02-001

Please refer to the Online Help for **RCCS plug 1**.

RCCS plug3

HP-Z000100Q-051205-02-002

Please refer to the Online Help for **RCCS plug 1**.

RCCS plug4

HP-Z000100Q-051205-02-003

Please refer to the Online Help for **RCCS plug 1**.

RCCS plug5

HP-Z000100Q-021205-02-010

System Type:Avanto
Espre
SymphonyTim

Please refer to the Online help for **RCCS plug 1**.

RCCS plug6

HP-Z000100Q-051205-02-004

Please refer to the Online Help for **RCCS plug 1**.

RCCS plug7

HP-Z000100Q-051205-02-005

Please refer to the Online Help for **RCCS plug 1**.

RCCS plug8

HP-Z000100Q-051205-02-006

Please refer to the Online Help for **RCCS plug 1**.

RCCS plug9

HP-Z000100Q-051205-02-007

Please refer to the Online Help for **RCCS plug 1**.

RCCS plug10

HP-Z000100Q-051205-02-008

Please refer to the Online Help for **RCCS plug 1**.

RCCS Switch

HP-Z000100Q-051205-02-010

This is an interactive test with a user interface that allows you to select a combination of input channel and output channel of the RCCS. In addition it is possible to select a few fixed measurement times (quick, normal and long time stability).

Signal path

The modulator TTX signal is fed to RCCS: via an internal test path to the selected Input, through the switch matrix to the selected output and then to the receiver .

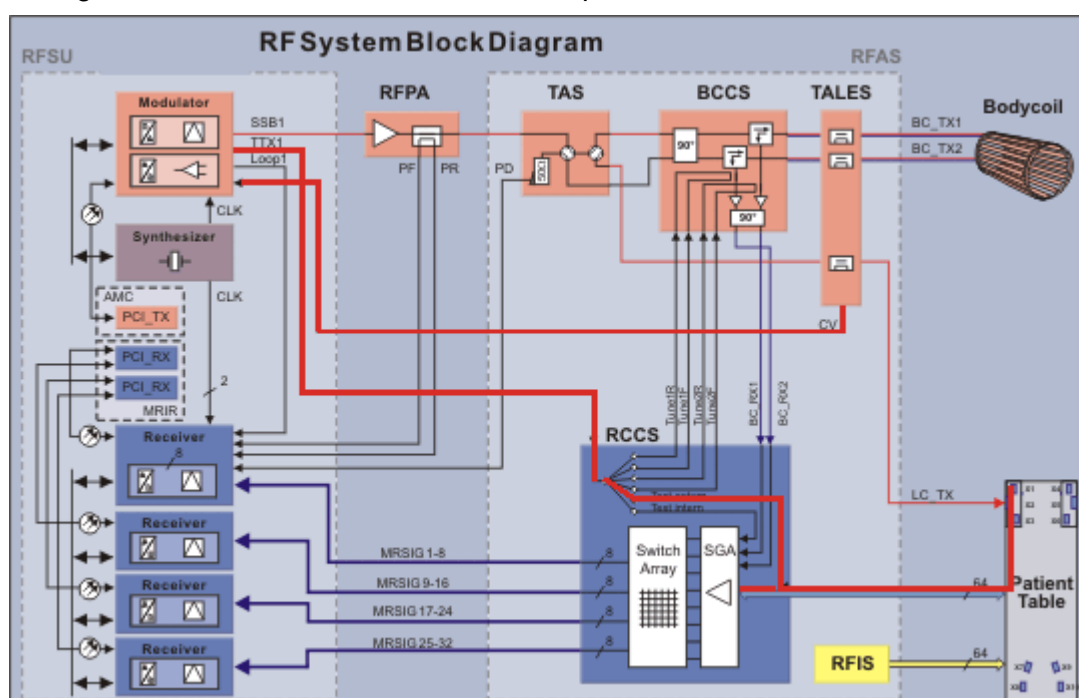


Fig. 351: RCCS interactive test

NOTE

Not all possible combinations make sense, e.g. the BC-channels are “hardwired”.

The test cannot automatically differentiate between systems with 8 or 32 receive channels. Please choose the output channel according to your system otherwise the test will fail.

Evaluation

- Evaluation of the raw data (ADC-value). The maximum magnitude is tested against specifications.
- Linearity check of magnitude data

- Stability check of magnitude data
- Check of phase stability

TALES

HP-Z000100Q-051205-02-011

System Type:Avanto

Esprea

SymphonyTim

n.a.

BCCS/RCCS

HP-Z000100Q-061205-02-004

System Type: Avanto
Espreo
SymphonyTim

- BCCS/RCCS (Functional)
 - Transmitter-Receiver Switch
 - BCCS_63 Hybrid
- BCCS_63/RCCS Loop tests (Interactive/ Long Time Stability)
 - BCCS(0 degree forward) RCCS (high gain)
 - BCCS(0 degree forward) RCCS (low gain)
 - BCCS(0 degree reflected) RCCS (high gain)
 - BCCS(90 degree forward) RCCS (high gain)
 - BCCS(90 degree reflected) RCCS (high gain)

Transmitter-Receiver-Switch (Functional)**Signal Path**

Two measurements with different TX-paths will be performed using the service sequence graddelay with the RF-amplitude set to zero:

- First measurement RFPA to the TAS-3TM/dummy load
- Second measurement RFPA to the Body Coil.

The receive path is always via the TX/RX-switch on BCCS_63 to RCCS/SGA and finally to the receiver.

Evaluation

Two noise measurements will be compared. During the first measurement, the output of the RFPA is disconnected from the TX/RX-switch. During the second measurement, the output of the RFPA is connected to the TX/RX-switch and its noise (even unblank is activated) will be received if the switch does not work properly.

Conclusions

The TX/RX-switch or the switch control (FOC from AMC/LWL-INT/U21 to BCCS_63/U1) is probably defective if the additional noise from the RFPA during the second measurement exceeds specifications.

BCCS_63 Hybrid (Functional)

NOTE

Ensure that the loader with the spherical phantom is positioned in the Body Coil.

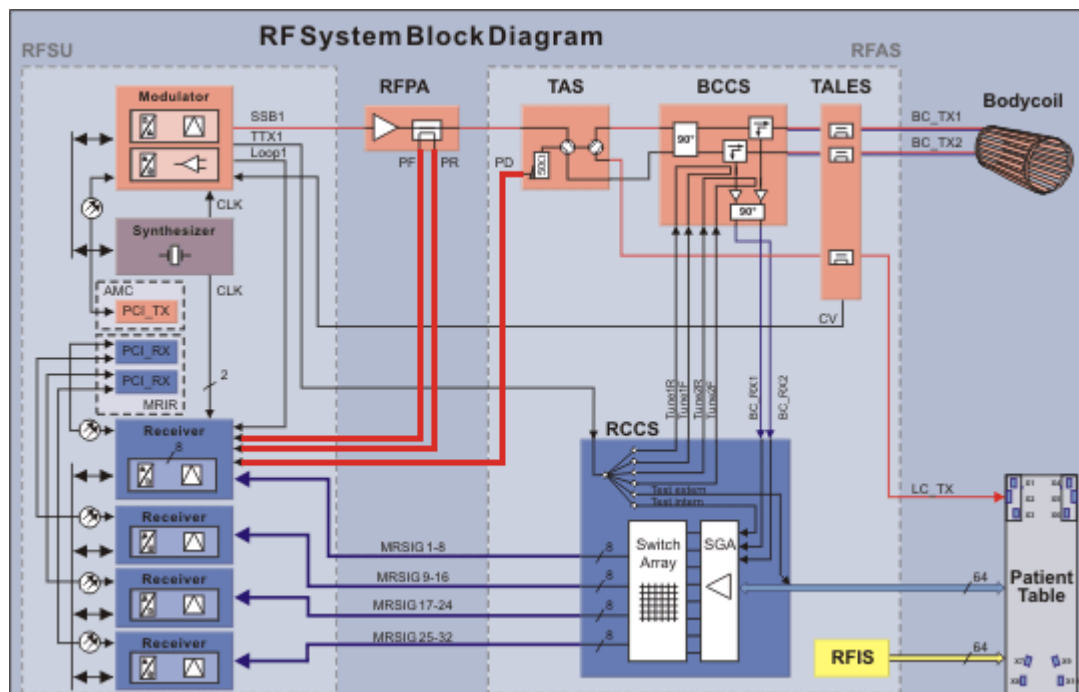


Fig. 352: BCCS_63 Hybrid

Signal Path

The output signal of the modulator is fed via the RFPA - TAS-3TM - BCCS_63 - TALES to the Body Coil.

Evaluation

- The forward voltages of both transmitter channels (90° and 0°) measured at the TALES are compared.
- Ratio of the reflected and forward voltage from the TALES.

Conclusions

On the one hand, the deviation of the forward voltage for 90° and 0° serves as a gauge for symmetrical tuning of the 90° and 0° systems of the Body Coil. On the other hand, the test results should provide information about the BCCS_63 functionality (Hybrid). If the results are not within specifications, first check the tuning of the Body Coil on both systems (tuning ratio of both systems), run Tune-up/BC Tuning.

BCCS_63 - RCCS Loop tests (Interactive/ Long Term Stability)

NOTE

During these tests the RF-cables from the TALEs to the Body coil must be removed at TALEs X3 and X4 in order to get defined conditions for the reflected signals at the open TALEs outputs.

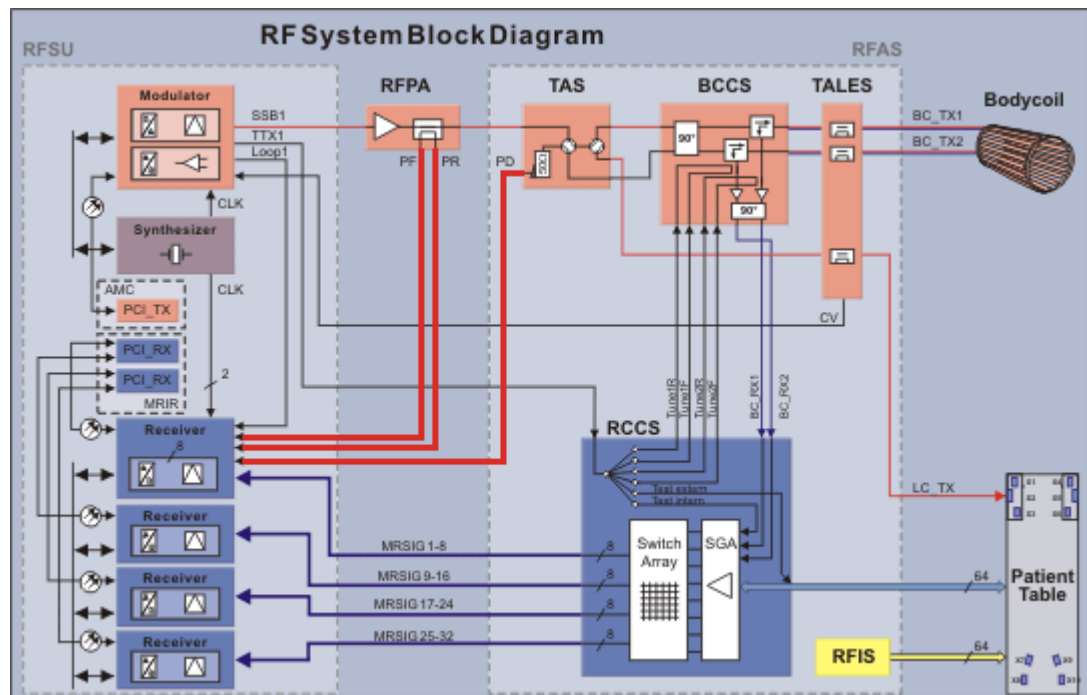


Fig. 353: BCCS_63 - RCCS Loops

Signal Path

- BCCS_63 (0 degree forward) - RCCS Loop (High gain) (MrSig 1/ MrSig 2)

The TTX output signal of the modulator is fed via the RCCS forward to the 0 degree system Body Coil preamplifier input in the BCCS_63. The signal is amplified in the BCCS_63 and fed back via BC_RX1 and in a second measurement via BC_RX2 to RCCS SGA high gain to receiver 1.

(RCCS X61 - <W 5154> - BCCS X4 - BCCS X3 - <W 5153> - RCCS X41 and RCCS X61 - <W 5154> - BCCS X4 - BCCS X9 - <W 5152> - RCCS X42)

- BCCS_63 (0 degree forward) - RCCS Loop (Low gain) (MrSig 1/ MrSig 2)

This set of measurements uses the low gain setting of RCCS SGA $\frac{1}{2}$ s with the same path.

- BCCS_63 (0 degree reflected) - RCCS Loop (High gain) (MrSig 1/ MrSig 2)

In this measurement set the TTX output signal of the modulator is fed via the RCCS **backward** to the **0 degree system** Body Coil preamplifier input in the BCCS_63. The signal is amplified in the BCCS_63 and fed back via BC_RX1 and in a second measurement via BC_RX2 to RCCS SGA high gain to receiver 1. (RCCS X62 - <W 5155> - BCCS X5 - BCCS X3 - <W 5153> - RCCS X41 and RCCS X62 - <W 5155> - BCCS X5 - BCCS X9 - <W 5152> - RCCS X42)

- BCCS_63 (90 degree forward) - RCCS Loop (High gain) (MrSig 1/ MrSig 2)

The TTX output signal of the modulator is fed via the RCCS **forward** to the **90 degree system** Body Coil preamplifier input in the BCCS_63. The signal is amplified in the BCCS_63 and fed back via BC_RX1 and in a second measurement via BC_RX2 to RCCS SGA high gain to receiver 1. (RCCS X63 - <W 5156> - BCCS X10 - BCCS X3 - <W 5153> - RCCS X41 and RCCS X63 - <W 5156> - BCCS X10 - BCCS X9 - <W 5152> - RCCS X42)

- BCCS_63 (90 degree reflected) - RCCS Loop (High gain) (MrSig 1/ MrSig 2)

In this measurement set the TTX output signal of the modulator is fed via the RCCS **backward** to the **90 degree system** Body Coil preamplifier input in the BCCS_63. The signal is amplified in the BCCS_63 and fed back via BC_RX1 and in a second measurement via BC_RX2 to RCCS SGA high gain to receiver 1. (RCCS X64 - <W 5157> - BCCS X11 - BCCS X3 - <W 5153> - RCCS X41 and RCCS X64 - <W 5157> - BCCS X11 - BCCS X9 - <W 5152> - RCCS X42)

Evaluations

The raw data of the first sinc-pulse are evaluated for stability, the raw data of the second sinc-pulse are evaluated for linearity.

Conclusions

This test is used primarily to check the stability and linearity of the preamplifier in the BCCS_63. Modulator control and data acquisition via the receiver must be intact to evaluate the results.

Additionally, the cables and connections between BCCS_63 and RCCS are checked.

BCCS/RCCS Long Time Stability

The test is identical to the BCCS/RCCS Loop test, but runs approximately 10 minutes to test the stability over longer measurement intervals.

RFIS

HP-Z000100Q-051205-02-014

System Type: Avanto
Espreo
SymphonyTim

- RFIS Communication/Status
- RFIS PIN Diodes Service Plug, pos n (interactive)
- RFIS PIN Diodes DVM (interactive)

RFIS Communication/Status

The communication test involves sending a command (status and version query) to an individual component which, if successfully executed, verifies communication with this component.

The status test queries the status of a particular component. It quickly determines whether the component status is OK or shows an error. A faulty status is indicated by three exclamation marks next to the respective status message.

```
RFISK2254_STATUS: Config          = done
RFISK2254_STATUS: Power failure   = not detected
RFISK2254_STATUS: coil change     = not detected
RFISK2254_STATUS: FPGA clock      = enabled
```

Conclusion

If the communication test of the RFIS fails, the power supply of the RFIS, the cable connections or the RFIS/SLIO may be defective.

RFIS K2254 Service_Plug_(pos i) Test (Interactive)

The Service Plug is needed for this test. Refer to the Online Help TestTools/General/Service Plug.

Preparation

Connect the Service Plug to the coil plug under test, the auxiliary plug is not needed. The test can be run just one time or in a loop (break with "Abort").

Test description

The 4, 6 or 8 green and yellow LEDs on the Service Plug (depending on the number of channels of the coil plug) must light up in the following way (latch disabled):

Step 1:

According to the coil plug (number of channels), four, six or eight green LED's ($-31\pm\frac{1}{2}V$ ok, static tuned) light up (idle state).

Step 2:

All green LED's go out, yellow LED's ($108\pm\frac{1}{2}mA$ ok, static detuned) light up.

Step 3:

All yellow LED's go out, green LED's light up.

Step 4:

Once after another yellow LED's light up, green LED's go out.

Step 5:

All yellow LED's go out, green LED's light up (idle state).

Evaluations

During the test, the PIN diode tune voltages and static detune current can be measured with a DVM at the Service Plug.

Green LED means that the tuning voltage (reverse voltage, UR) corresponds to $-29.8 V$ ($-27.7 \dots -31.3 VDC$) is ok.

Yellow LED means detune current (forward current, IF) corresponding to $108 mA$ ($101 \dots 115 mA$) is ok.

At the end of html-window output of the test, the actual temporal sequence of the tested coil connectors will be shown:

Example Output

	Status	Tested PTAB Plugs
GetRFISTestResults	ok	2, 6, 8

Fig. 354: Actual temporal sequence of tested coil connectors (2 - 6 - 8)

Conclusions

This test verifies the function of the RFIS and the cables between RFIS and coil connector under test (PIN diodes bias voltage). If defective PIN diode channels are detected during the test (i.e. corresponding LED's will not light up or switching between UR and IF is not ok), the measurements with the voltmeter according to the following description will provide more detailed information regarding the possible source for error.

U_R green LED	I_F yellow LED	Conclusion
ON	ON	OK

OFF	ON	Error RFIS if measurement is out of tolerance.
ON	OFF	
OFF	OFF	Error RFIS or cabling if voltage is out of tolerance.

If the sequence shown of the tested coil connectors does not match with the real sequence of tests, the following connectors at the RFIS could be interchanged:

RFIS X28 (-> coil connector 1,2,3) interchanged with RFIS X29 (-> coil connector 4,5,6),

RFIS X26 (-> coil connector 7, 8) interchanged with RFIS X27 (-> coil connector 9,10).

RFIS PIN diodes DVM measurements (interactive)

The voltage drop U_R and voltage drop **U_{measured}** according to IF is measured using a DVM connected to sockets on the Service Plug.

The IF corresponding to the measured voltage can be calculated as:

$$I_F = 16.6 + 123.2 \times U_{\text{measured}}$$

- Reverse voltage (U_R) at PIN diode: - 27.7 ... - 31.3 VDC, typically -29.8 VDC
- Voltage drop $U_{\text{measured}}(I_F)$: + 0.65 VDC... + 0.87 VDC, typically 0.74 VDC (corresponding to 108 mA)

To localize the defective DYSCON_16, exchange the boards. If the error moves to another coil plug, the corresponding board is defective.

Measurement at RFIS

To differentiate between RFIS or a cable error, measure the voltages at connectors X26 - X29 at RFIS, as well:

-> Voltage mode: -31 V -> Current mode: typically +14 V (108 mA).

Connectors at the RFIS

There are four modules DYSCON_16 (D1); the contribution of the modules via the four connectors to the coil PIN diode channels is shown in the table below.

Trio RFIS connectors

	Pin X28/D14	Pin X29/D14	Pin X26/D14	Pin X27/D14
Pin_sig/(Pin GND) to LC1	1 - 8 / (9 - 12)			
Pin_sig/(Pin GND) to LC2	21 - 24 / (25, 26)			
Pin_sig/(Pin GND) to LC3	33 - 38 / (39 - 41)			
Pin_sig/(Pin GND) to LC4		1 - 8 / (9 - 12)		
Pin_sig/(Pin GND) to LC5		21 - 24 / (25, 26)		
Pin_sig/(Pin GND) to LC6		33 - 38 / (39 - 41)		
Pin_sig/(Pin GND) to LC7			1 - 6 / (9 - 11)	
Pin_sig/(Pin GND) to LC8			18 - 25 / (26 - 29)	
Pin_sig/(Pin GND) to LC9				1 - 6 / (9 - 11)
Pin_sig/(Pin GND) to LC10				18 - 25 / (26 - 29)

RFIS connectors to PAT Coil connectors

RFIS connector	X28/D14	X29/D14	X26/D14	X27/D14
PAT Coil connector/ Pin_signal/ (Pin_GND)	1 / 1 - 8 / (9 - 12)	4 / 1 - 8 / (9 - 12)	7 / 1 - 6 / (9 - 11)	9 / 1 - 6 / (9 - 11)
	2 / 2, 4, 7, 9 / (15, 16)	5 / 2, 4, 7, 9 / (16, 15)	8 / 1 - 8 / (9 - 12)	10 / 1 - 8 / (9 - 12)
	3 / 1 - 6 / (9 - 11)	6 / 1 - 6 / (9-11)		

Dyscon_16 modules to PAT Coil Connectors

DYSCON_16 # (left->right)	module 1	module 2	module 3	module 4
PAT Coil connector # (channels)	1 (1 - 8)	3 (5, 6)	6 (3 - 6)	8 (7, 8)
	2 (1 - 4)	4 (1 - 8)	7 (1 - 6)	9 (1 - 6)
	3 (1 - 6)	5 (1 - 4)	8 (1 - 6)	10 (1 - 8)
		6 (1, 2)		

LEDs at RFIS

yellow LEDs LC x_PIN y UR_ON	Desired state
ON	U _R (-31 V) on, voltage mode
OFF	I _F (108 mA) on, current mode

RFIS BC PIN Diodes Test

For this test refer to Body Coil (functional) BC Check Tuning.

This test will be performed automatically on every "Reboot Scanner".

Status LEDs at RFIS

LED	Meaning if LED "ON"	Meaning if LED "OFF"
RESET	Reset button pressed. Reset of memory and of both COS-LIO's	
SETTING_SELECT1	LC/BC settings	
SETTING_SELECT2		
SETTING_SELECT3		
SETTING_SELECT4		
DYN_CTRL	Dynamic switching of PIN Diodes of LC and BC	
N_SEQUENCE_ON	Sequence off; all channels in dynamic voltage mode (-31 V)	Sequence on; status of channels determined by DYN_CTRL, setting and configuration of selected setting
PICKUP_SELECT	Pick-up coil selected	Pick-up coil deselected
RESERVE	Not used	
CAN_TX1	CAN_BUS TX1 active (LED flashing)	CAN_BUS TX1 deactivated
CAN_TX2	CAN_BUS TX2 active	CAN_BUS TX2 deactivated
CAN_RX1	CAN_BUS RX1 active	CAN_BUS RX1 deactivated
CAN_RX2	CAN_BUS RX2 active (LED flashing)	CAN_BUS RX2 deactivated
CLOCK_DISABLED	Clock for FPGA deactivated (sleep mode); no communication between COSLIO's and FPGA.	Clock activated; communication between COSLIO's and FPGA possible.

LED	Meaning if LED "ON"	Meaning if LED "OFF"
VOLT_OK	Supply voltages ok	At least one of the supply voltages is missing.
CONF_DONE	Configuration of FPGA ok	RFIS not working
CODE_INTERRUPT	Lights up shortly on coil change	No coil change

RFAS

HP-Z000100Q-061205-02-005

System Type: Avanto
 Espree
 SymphonyTim

- RFAS (Communication/Status)
- RFAS (Functional)
 - Reference Voltage TALES
 - RFPA -TALES
 - BC-TX path (reflection)
 - BC-TX path (voltage)
- RFAS (Interactive/ Multi Nucleus)
 - Local Coil TX Path

RFAS Communication/Status

The RCCS-controls the TAS-3TM component in order to verify communication to the RCCS as well.

The communication test consists of sending a command (status and version query) to an individual component which, if successfully completed, verifies communication with this component.

The status test queries the status of a particular component. It quickly determines whether the component status is OK or shows an error. A faulty status is indicated by three exclamation marks next to the respective status message.

Reference Voltage (TALES)

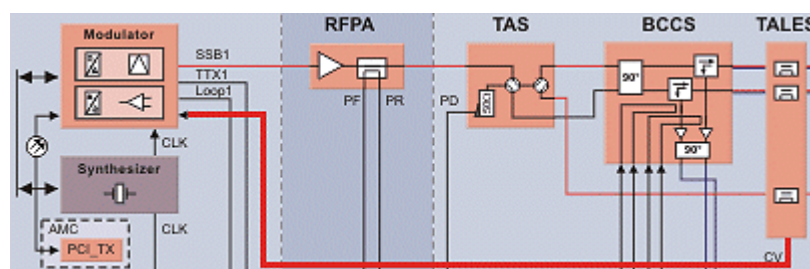


Fig. 355: Reference voltage (TALES)

Signal Path

The TALES output signals are via the modulator (CV-Input) to the AMC/PCI-TX.

Evaluation

Verification that the TALES reference voltage is $500\sqrt{2}\text{mV}$.

Conclusions

The TALES continuously supplies all internal directional coupler signals triggered by the controller via fiber optic cable. In addition, the TALES supplies the signal of an integrated reference voltage source. The directional coupler signals and the reference voltage are multiplexed and fed to the modulator CV input via a coaxial cable. The reference voltage test provides information regarding correct control of the TALES, intact cabling to the modulator/AMC and signal processing during data acquisition. In case of error, the $500\sqrt{2}\text{mV}$ reference voltage signal can be measured with a digital voltmeter. For further information, refer to the Troubleshooting Guide.

RFPA - TALES

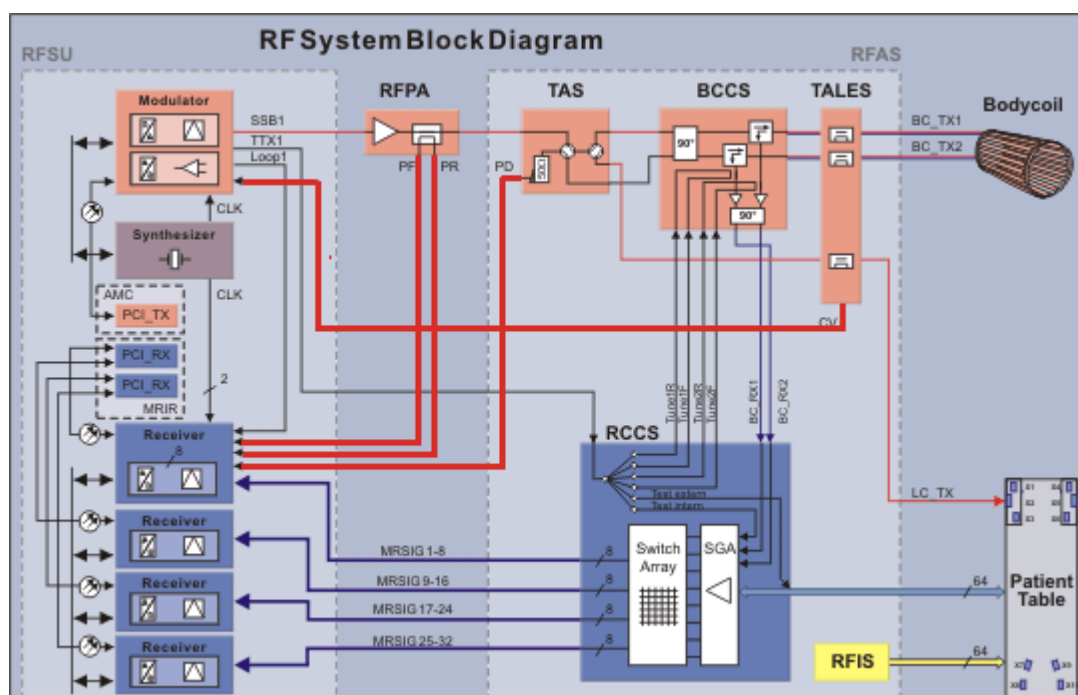


Fig. 356: RFPA TALES

NOTE

Ensure that the loader with spherical phantom is positioned in the Body Coil.

Signal Path

The output signal of the modulator SSB1 is fed via the RFPA - TAS-3TM - BCCS_63 - TALES to the Body Coil.

The test consists of two measurements. One RF pulse is transmitted for each. In the first measurement, the signals are decoupled by the directional coupler (DICO/PF and PR) in the RFPA as well by the TAS-3TM (reflected voltage from Body coil/PD) and converted to voltages using a specific factor. The second measurement acquires the signals from the TALEs.

Evaluations

- Forward voltage from the TALEs is within tolerance
- Reflected voltage from the TALEs is below the limit
- Deviation of the forward voltage from the TALEs and the DICO
- Ratio of the reflected and forward voltage from the DICO

Conclusions

This test practically checks the entire transmit path of the system.

The results of the forward voltage at the TALEs provide information on whether the actual value acquired by the TALEs corresponds to the nominal value determined by the RF characteristic.

The value of the reflected voltage at the TALEs is a gauge for the 50 Ohm adjustment through the Body Coil.

The forward voltage measured via DICO must approximately match that of the TALEs. Since the correlation between the DICO value and the TALEs voltage occurs only when the characteristic curve is generated (determination of the factor for conversion from receiver ADC values into voltage values), a valid characteristic curve has to be present.

The ratio of the reflected and forward voltage measured via DICO in the RFPA is generally used as a gauge for the 50 $\pm$ 1/2 Ohm adjustment of the RFPA. On the one hand, it is determined via coil tuning, and on the other hand via the characteristics of the components that follow the RFPA in the transmit path such as the TX cable, BCCS_63, TALEs, etc.

BC-TX Path (reflection)

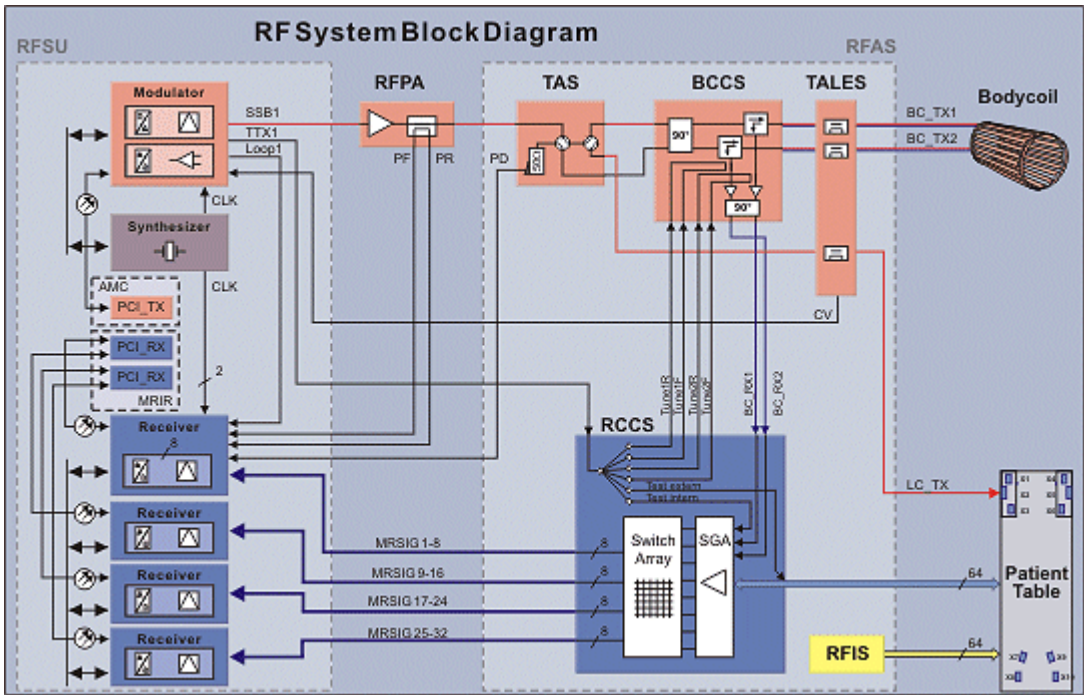


Fig. 357: BC-TX path (reflection)

NOTE

Ensure that the loader with spherical phantom is positioned in the Body Coil.

Signal path

The output signal of the RFPA is fed to the TX coil. The forward and reflected power (PF/PD) is measured via the directional coupler in the RFPA. Additionally, the reflected voltage is measured by the TAS-3TM (PD). The signal amplitude used for this measurement is the maximum amplitude read from the RF-characteristic data, varying from the minimum to the maximum.

Evaluation

- The actual forward and reflected voltage (measured via directional coupler) at maximum reflection and the reflected voltage measured by TAS-3TM is displayed.
- The maximum ratio of the reflected voltage to the forward voltage (measured via directional coupler) is calculated.
- The first diagram shows the measured values (voltage) of power forward and power reflected over the transmitter voltage.
- The second diagram shows the DICO reflection (power reflected/power forward (%)) over the transmitter voltage.

Example Output

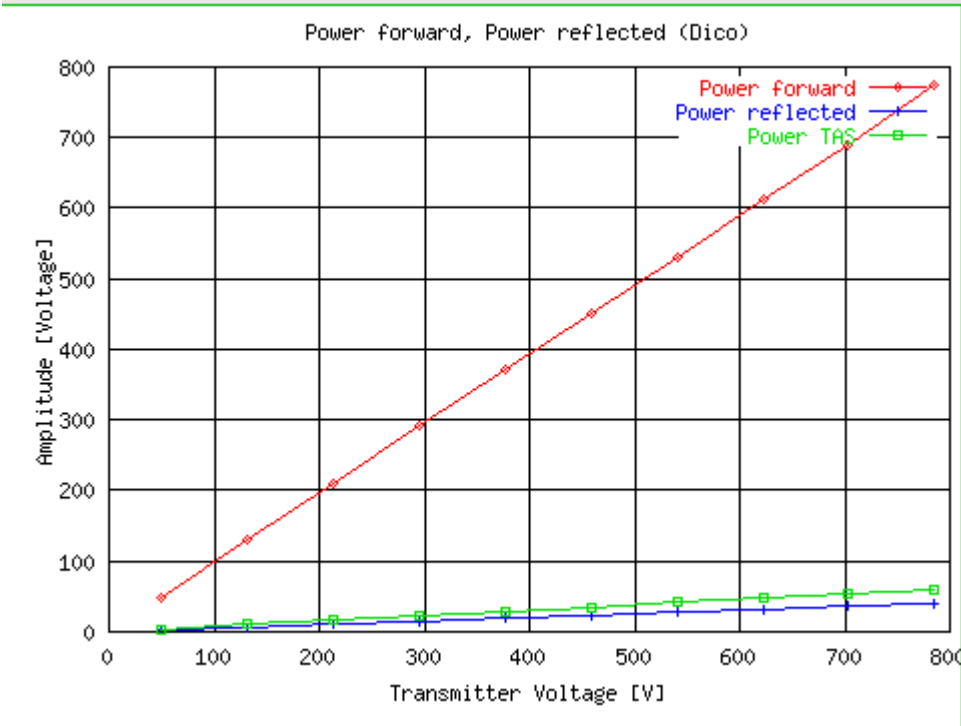


Fig. 358: Power forward, reflected (DICO): Power reflected TAS_3TM
Example Output

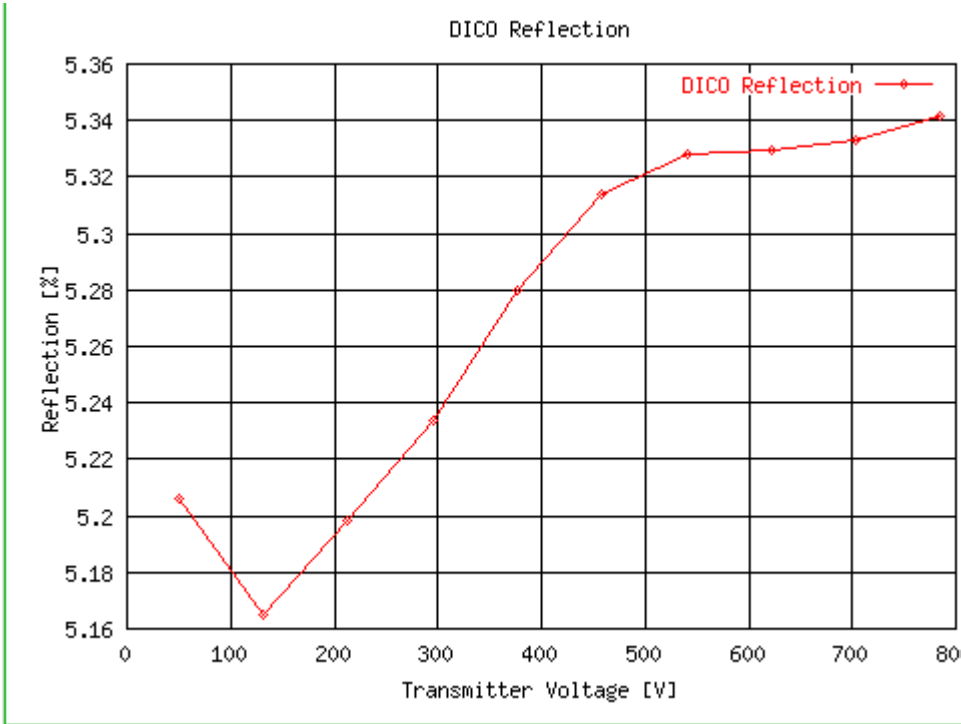


Fig. 359: DICO Reflection

Conclusions

The objective of the test is to ensure that the TX-path works properly without excess reflections caused by defective connectors and cables. If this test fails, perform the interactive cable test BC-TX path (reflection) (for experts only).

BC-TX Path (voltage)

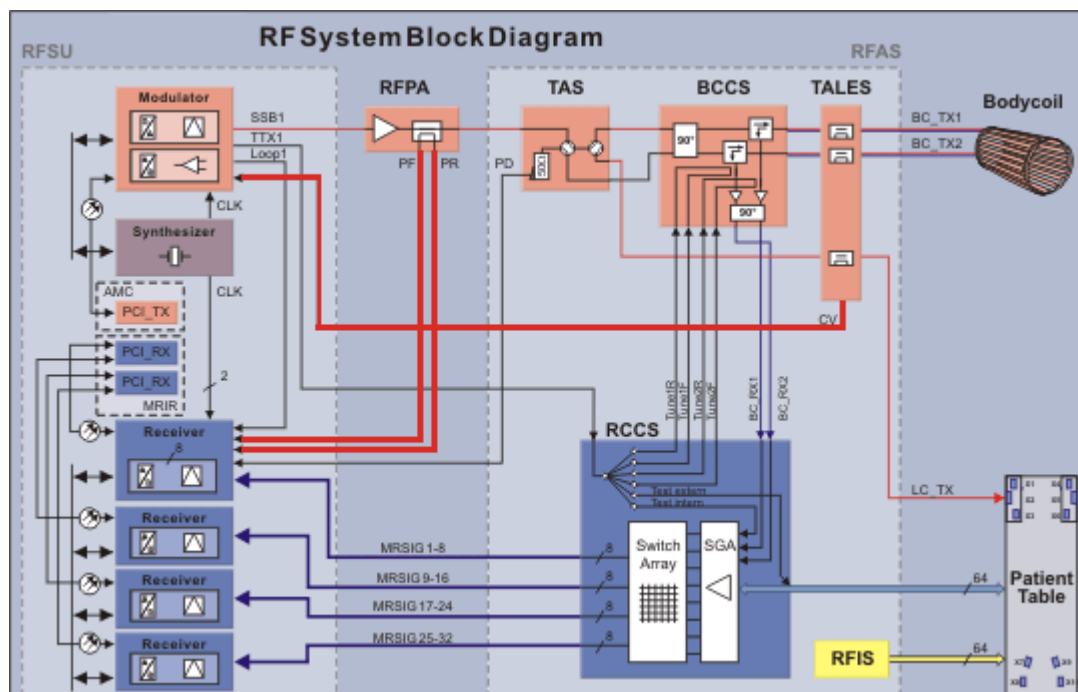


Fig. 360: BC-TX path (voltage)

NOTE

Ensure that the loader with spherical phantom is positioned in the Body Coil.

Signal path

The output signal of the RFPA is fed to the TX-coil. The forward and reflected power is measured via the directional coupler in the RFPA. The signal amplitude used for this measurement is the maximum amplitude read from the RF-characteristic data, varying from minimum to maximum.

Evaluation

- The actual forward and reflected voltage (measured via DICO and TALES) at maximum deviation is displayed.
- The maximum deviation of the forward voltage (measured via DICO and TALES) is calculated.

- The first diagram shows the measured power forward (voltage, measured via DICO and TALES) over the transmitter voltage.
- The second diagram shows the deviation (%) of DICO and TALES power forward over the transmitter voltage.

Example Output

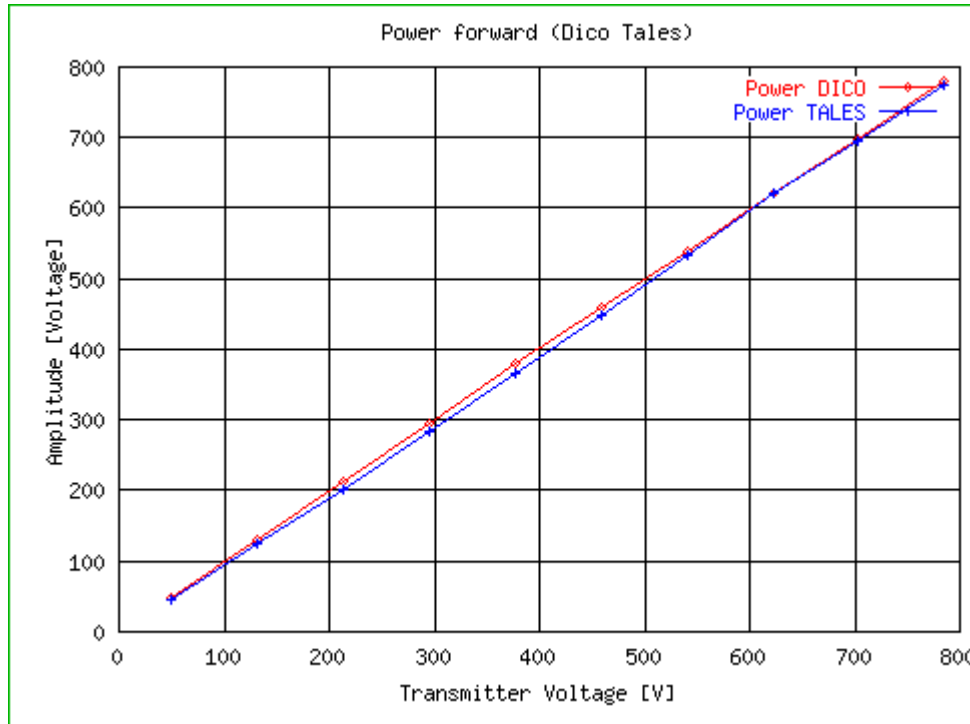


Fig. 361: Power forward measured by DICO and TALES

Example Output

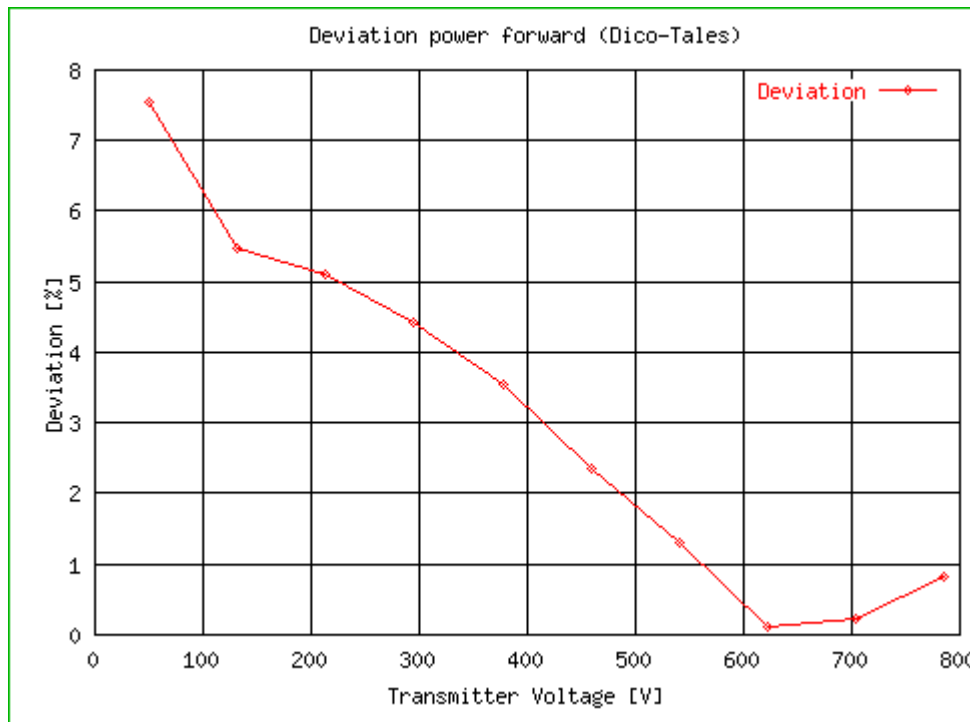


Fig. 362: Deviation between DICO and TALES

Conclusions

If the deviation of DICO and TALES voltages is out of tolerance, first repeat the RF characteristic measurement.

Significantly higher voltage values measured by TALES could be caused by RFPA oscillations on other than the MR-frequency: the DICO measurement is selective at the MR-frequency, but the TALES is sensitive to a wider frequency band.

Perform the interactive **Cable** test BC-TX-path (voltage) (for experts only).

Local Coil TX Path (Interactive/ Multi Nuclei)

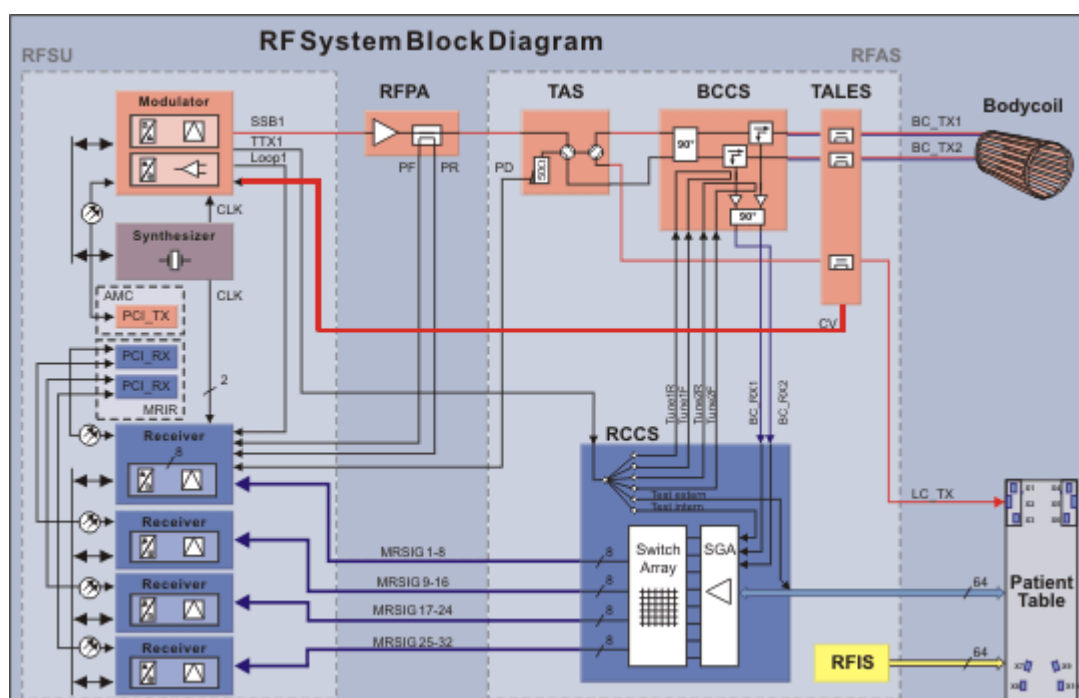


Fig. 363: Local Coil TX Path

Preparation

The Service Plug must be connected to coil connector #1 at the patient table (connector for the Extremity Coil). Follow the instructions provided during the test run.

Signal Path

The output signal of the modulator is fed to the RFPA - TAS-3TM - TALES - Local Coil (local coil connector #1). The TX path of the TAS-3TM is switched to local coil mode.

This test requires that the transmission cable to the extremity coil be terminated at 50Ohm. The connection must be terminated using the 50 Ohm resistor in the Service Plug.

The test consists of two measurements. One RF pulse is transmitted for each measurement. In the first measurement, the signals are decoupled by the directional coupler (DICO) in the RFPA and converted to voltages using a specific factor. The second measurement acquires the signals from the TALES.

Evaluation

- Deviation of the forward voltage from the DICO and the TALEs
- Ratio of the reflected and forward voltage from the TALEs

Conclusions

The forward voltage measured via DICO must approximately match that of the TALEs. Since the correlation between the DICO value and the TALEs voltage occurs only when the characteristic curve is generated (determination of the factor for conversion from receiver ADC values into voltage values), a valid characteristic curve has to be present.

With regard to the linear RF-characteristic, i.e. after the software has been installed and before the tune-up has been performed, the deviation may be significantly above specifications.

The ratio of the reflected voltage and the forward voltage measured via DICO in the RFPA is generally used as a gauge for the 50 Ohm termination of the RFPA. In this test, it is determined by the 50 Ohm resistor in the Service Plug. If the ratio lies outside specifications, the cause is probably the cable connection from the TAS-3TM via Tales to coil connector #1 (movable cable in the table) or in the TAS-3TM itself (for this test, the TAS-3TM is switched to extremity coil path).

If the forward voltage does not exceed a certain level (approx. 100 V), the RF safety watchdog evaluates the measured values within the most sensitive range of the TALEs. In this range, the reflected voltage of the TALEs can no longer be produced and the value 0 is displayed. In this case, the evaluation of return / forward voltage is no longer valid.

Pin Splitter

HP-Z000100Q-181108-02-018

System Type:Avanto
SymphonyTim
Esprea

N.a.: The PIN Splitter is not used on this system type.

BC

HP-Z000100Q-140206-02-003

System Type: Avanto

Espree

SymphonyTim

- Body Coil (Functional)
- Body Coil)Interactive)

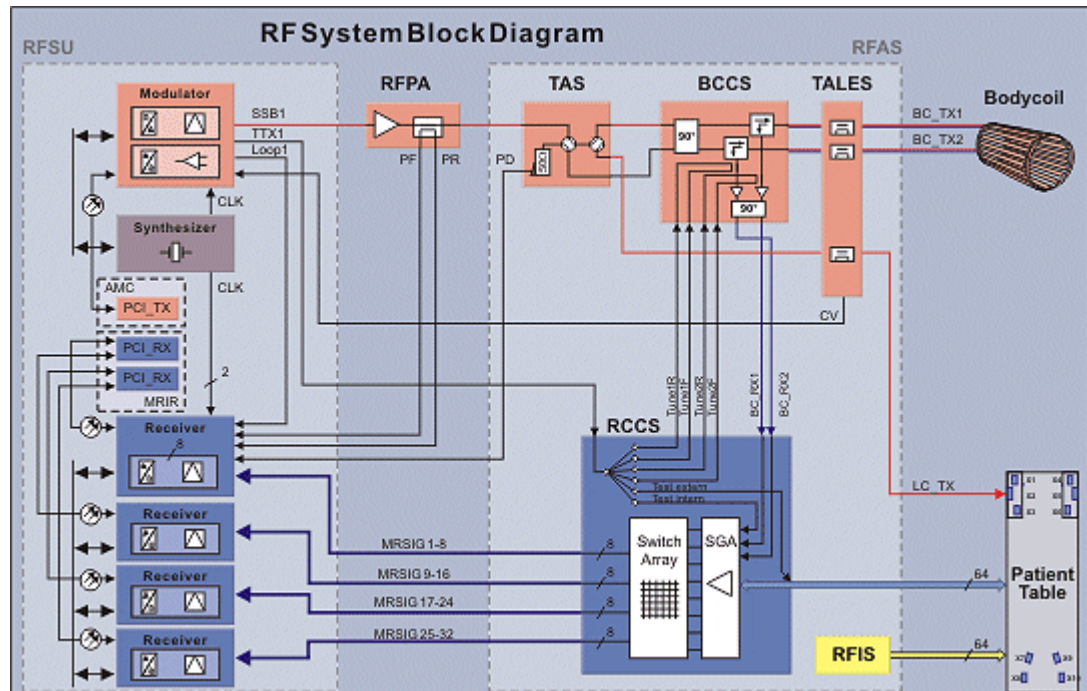


Fig. 364: Body Coil

Body Coil (Functional/ with load)

NOTE

Ensure that the loader with the large spherical phantom is accurately positioned in the Body Coil.

- Reflection Measurement
- Transmission Measurement
- BC Power Test
- BC Check Tuning

Evaluation

The Body Coil test is based on the analysis of reflective voltage factors that depend on the frequency. The test simulates a network analyzer that provides the frequency for the smallest reflection factor and the resonance curve of the Body Coil as results. The available fre-

quency range of approximately 1 MHz is stepped through in 10 kHz steps. The respective reflection factors are produced and entered in a frequency reflection factor diagram. The frequency with the smallest reflection factor is checked against specifications. The Q-factor of the coil will be evaluated and checked against specification.

The decoupling of the Body Coil elements (transmission measurement) is evaluated.

A power test with increasing transmitter amplitude is performed. The TALES reflection values over the transmitter amplitudes are graphically displayed.

Additionally, the **BC Check Tuning** checks the BC Tuning/PIN diodes control (Tune current, detune voltage and static detune voltage) against specifications (IF= 5000 +/-325mA and UR= -360 ..-460V).

If this test fails, check the **Event Log** for **RFIS error messages**. Double-click on the Error ID to get further information about the error and corresponding service actions.

BC tuning test results

	Value
Tune current	ok
Detune voltage	ok
Static 500V	ok

Fig. 365: Example BC: Check Tuning Output

Example Output

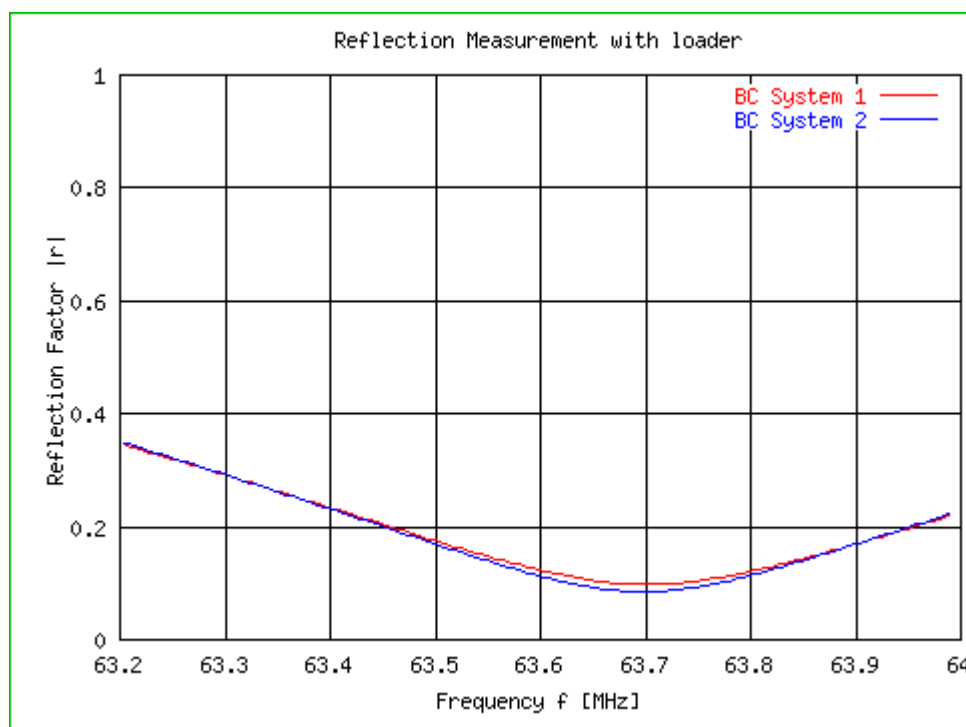


Fig. 366: BC - Reflection Evaluation

Example Output

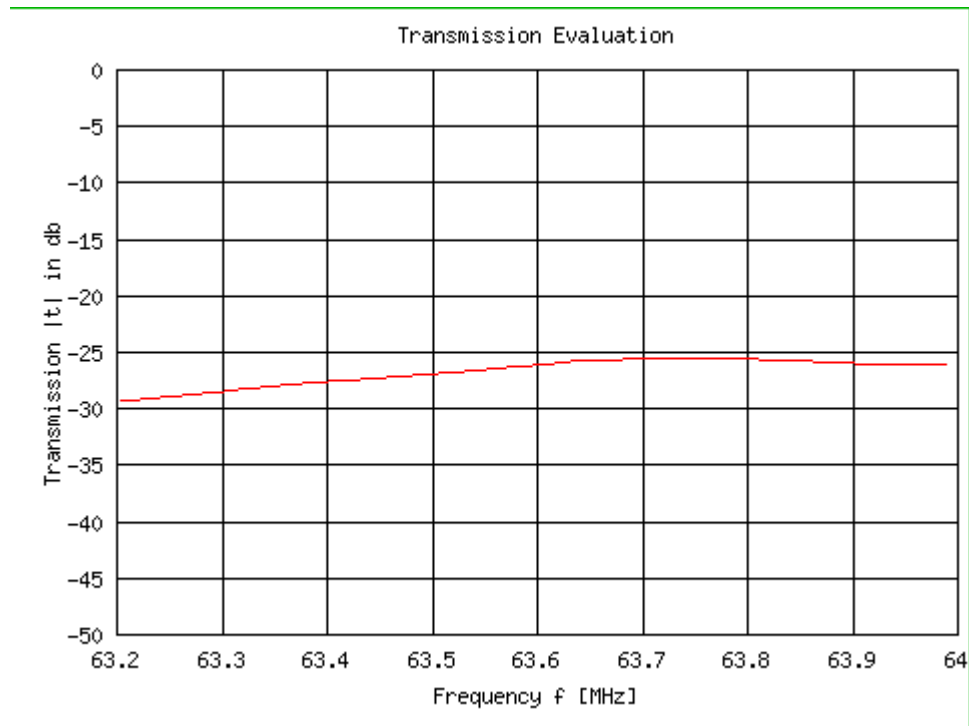


Fig. 367: BC - Transmission Evaluation

Example Output

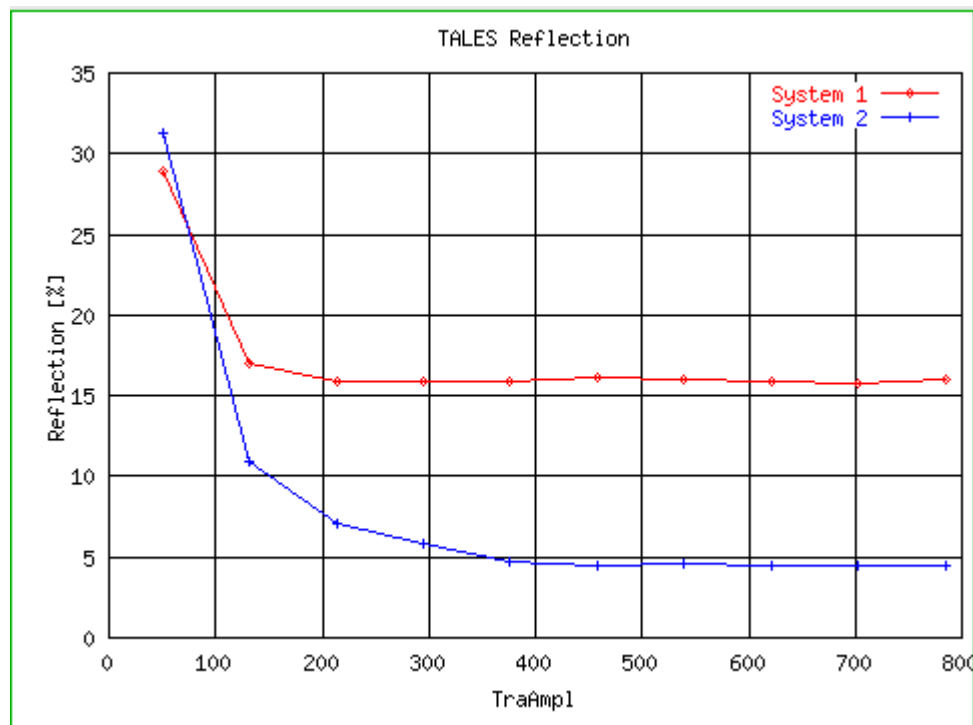


Fig. 368: BC power test - TALES reflection with load

Body Coil (Interactive/ without load)

Load-free, i.e. the table with loader and phantom must be outside the magnet.

- Reflection Measurement

- Transmission Measurement
- BC Power Test

Example Output

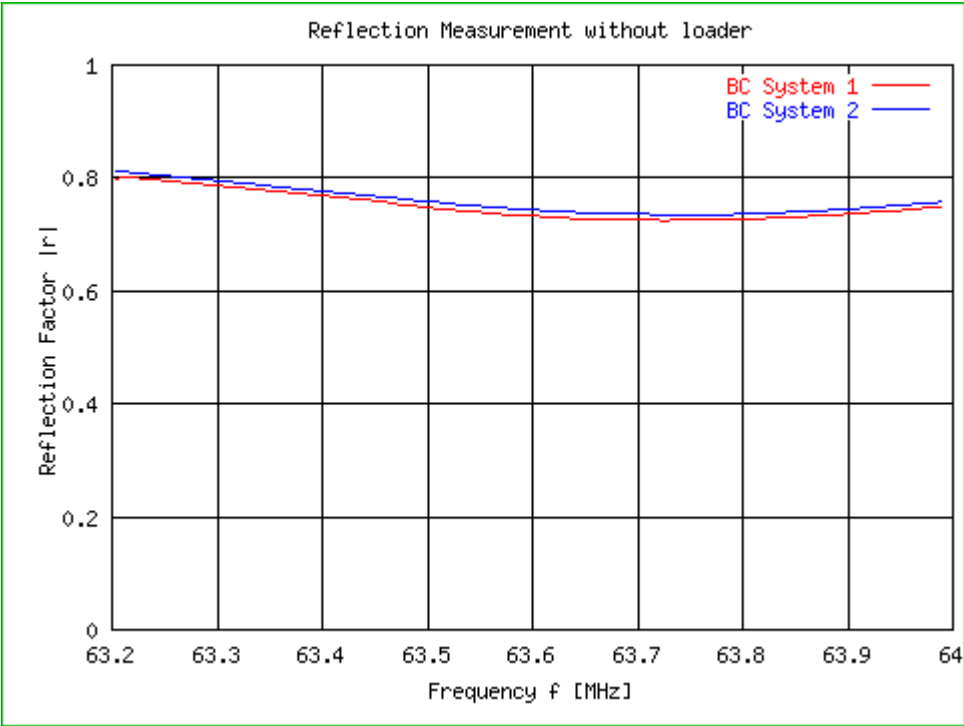


Fig. 369: BC - Reflection Evaluation

Example Output

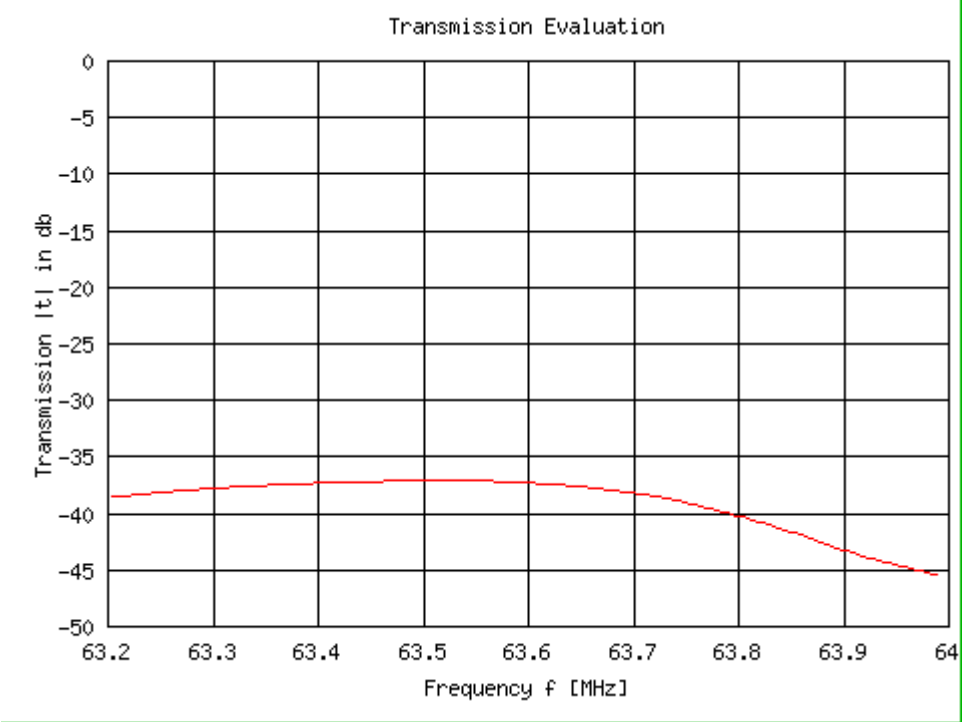


Fig. 370: BC - Transmission Evaluation

Also a power test with increasing transmitter amplitude is performed. The TALES reflection values over the transmitter amplitudes are graphically displayed.

Example Output

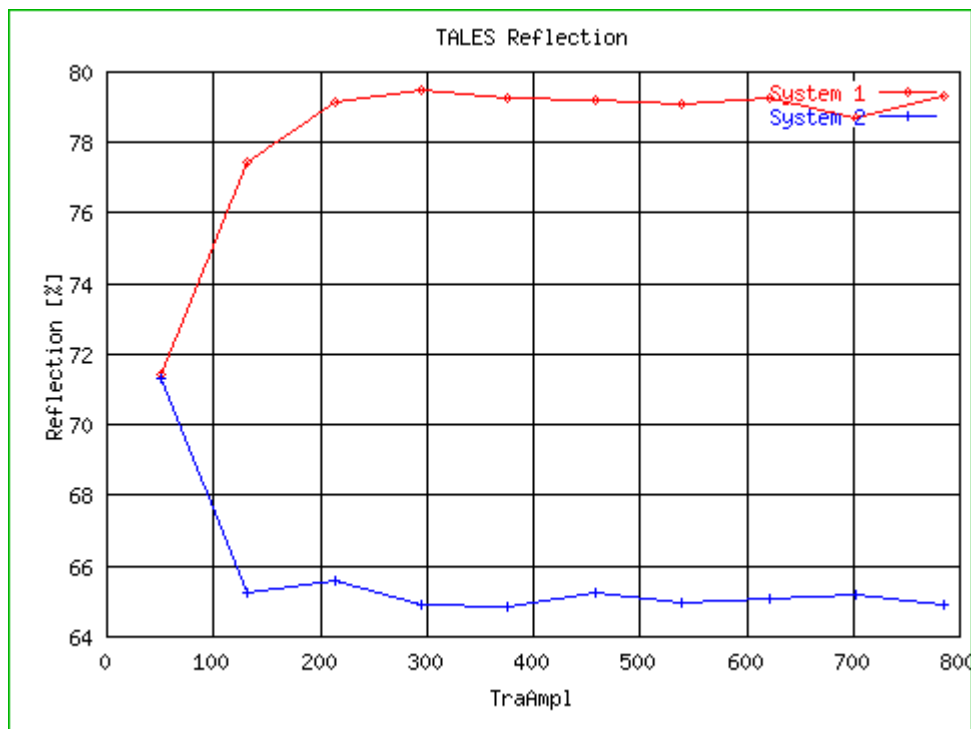


Fig. 371: TALES reflection without load

Conclusion

If the tolerances of the reflection measurement evaluation are exceeded, the problem is probably located in the BC.

The power test checks the disruptive strength of the BC. The test should run without shut-down to the maximum voltage. The reflection of TALES should be approximately constant at voltages higher than 200 V. Refer to the figure below.

The power test without load is the worst case, i.e. the reflected voltages are significantly higher.

NOTE

If the results exceed specifications, please contact the MR-Hotline (HSC) for further instructions.

Cable BC

HP-Z000100Q-061205-02-009

System Type:Avanto
Espre
SymphonyTim

- BC-TX path (refl.)
- BC-TX path (volt.)

NOTE

These tests are for experts only!

Additionally, special equipment is necessary for the tests.

These advanced tests enable you to check the complete TX path to the BC and provide for systematic troubleshooting in the TX path by connecting a power 50 Ohm terminator at various locations in the TX path and/or configuration of particular loops.

The test should be used by experienced service engineers only. Special tools (power dummies, RF-cables, connectors, adapters and attenuator) are needed.

Pop-up windows inform you about the actual measurement configuration.

NOTE

The tests require a valid RF characteristic or the linear RF characteristic.

Ensure that the spherical phantom is positioned in the Body Coil.

NOTE

When using the linear characteristic, it may be possible that the transmitter amplitude used exceeds the allowed range resulting in an aborted measurement. This is normal and not a problem with the system.

Additionally, in these cases only the graphics can be evaluated. The automatic evaluation of the numerical values in the charts cannot be trusted as the specification values used are intended for use with a measured RF characteristic only.

NOTE

For the first measurement of each test, the reflection factors are also evaluated and analyzed. If the reflection exceeds 90%, the test is aborted with the error message

Error: DICO reflection out of spec for first measurement.

In this case check the cabling and the Service Plug/50 Ohm power dummy (if used).

Safety



High RF voltage

Contact with RF conductors during transmitting of the RFPA will result in death or serious physical injury.

- ⇒ Before disconnecting RF transmit cables or connectors in the transmit path, disconnect the RF input cable at X1 and the unblank line U1 from the RFPA.

BC-TX path (refl.) (Interactive)

The objective of the test to check the components of the TX-path to the Body Coil with a maximum transmitter amplitude for disruptive strength step-by-step. Several measurements with an increasing amplitude starting from the minimum (50V) to the maximum voltage (read from the RF characteristic data) are performed.

Signal path

The normal TX path to the Body Coil is used.

The TX path can be opened at several locations and terminated by one or two 50Ω ½Ohm dummies. In the appearing pop-up window you can select the desired TX path.

The selection of the location of the interruption in the TX path has no influence on the measurement but selects the heading of the HTML output table. This is, e.g., helpful when sending the results to the HSC.

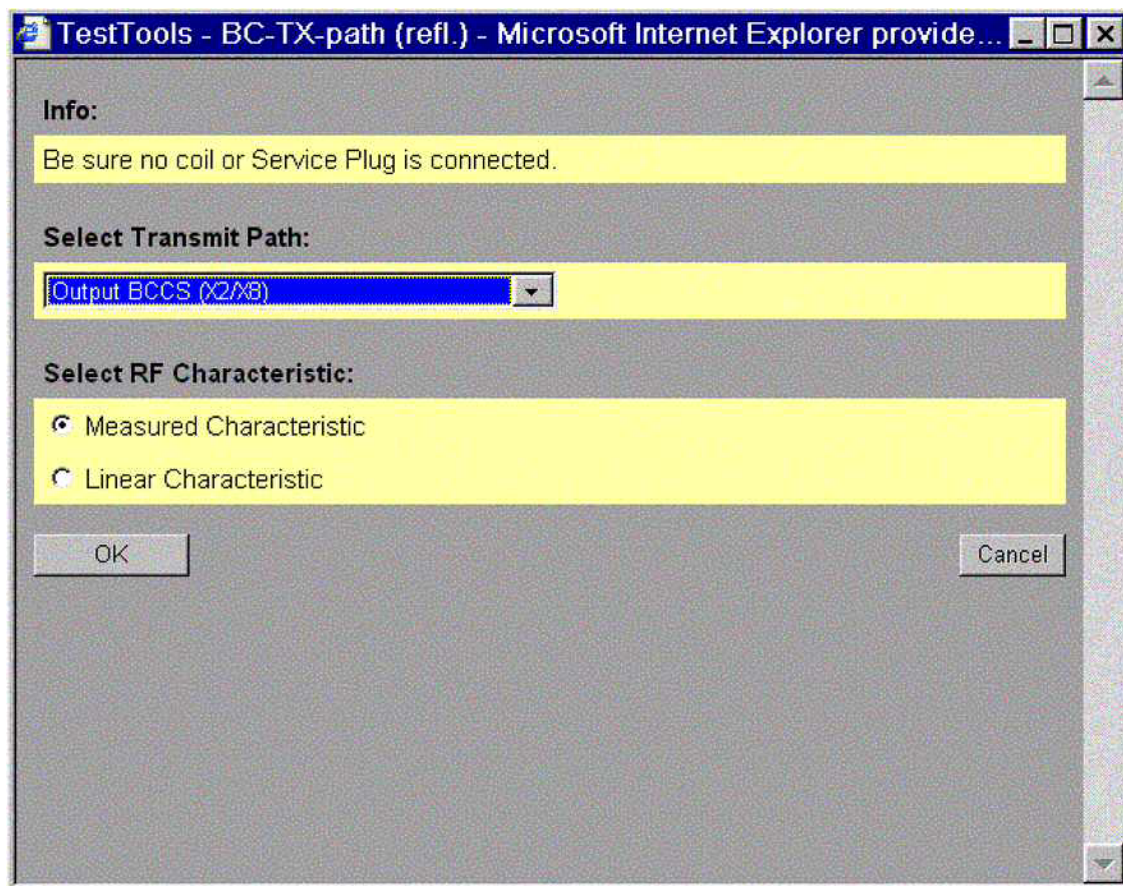


Fig. 372: Pop-up window BC-TX-path (refl.) - Selection of RF Characteristic used

The measurement is started by clicking "OK" or skipped by selecting "Cancel" in the pop-up window.

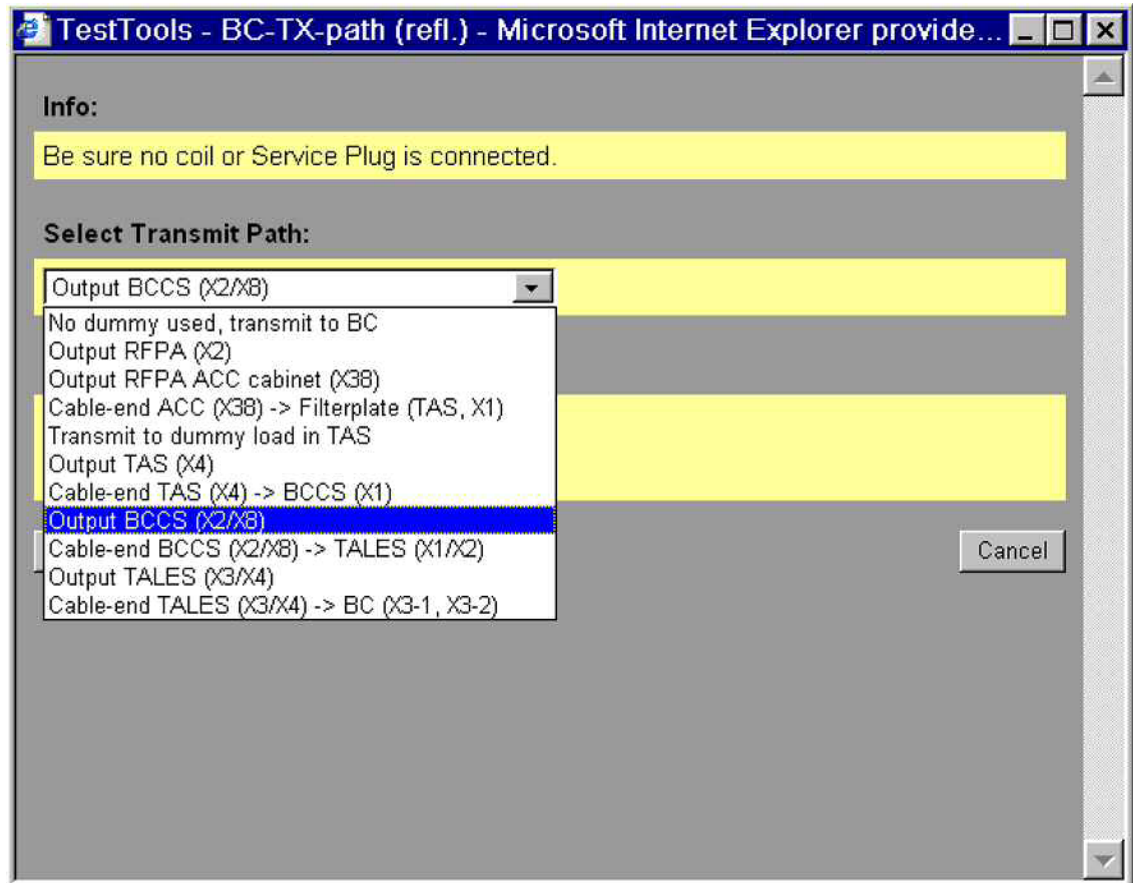


Fig. 373: Pop-up window BC-TX path (refl.) Example: Selection of TX-path termination using 50 Ohm dummies

Evaluation

HTML output table:

- PF (DICO) at maximum reflection (Volt)
- PR (DICO) at maximum reflection (Volt)
- PR/PF (DICO) Reflection at maximum ratio (%)

Graphical output:

- PF, PR (DICO/Volt) over TRA voltage
- PR/PF DICO Reflection (%) over TRA voltage

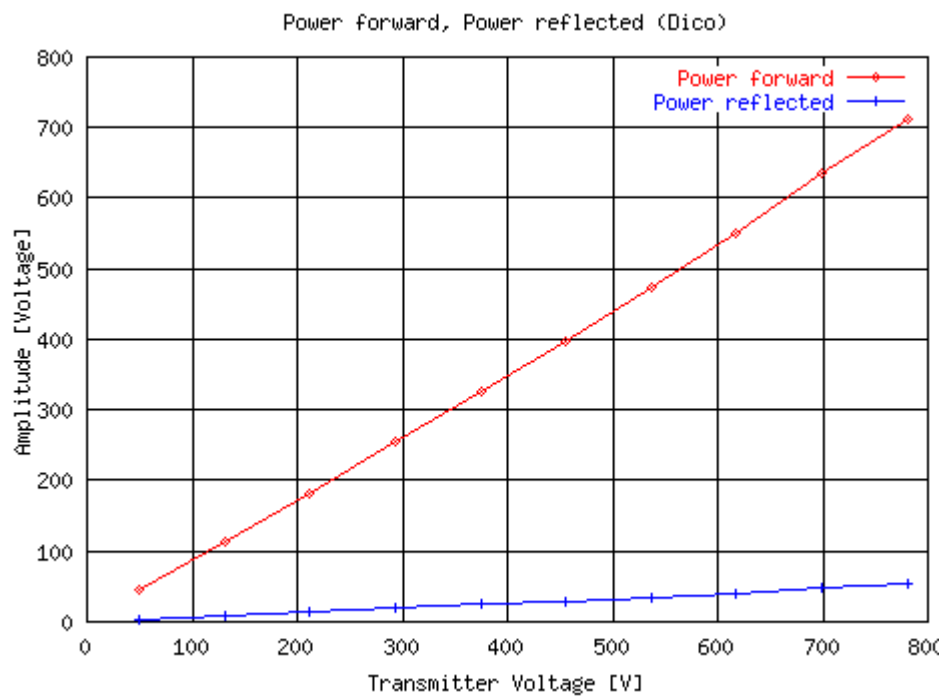


Fig. 374: Example of graphics output

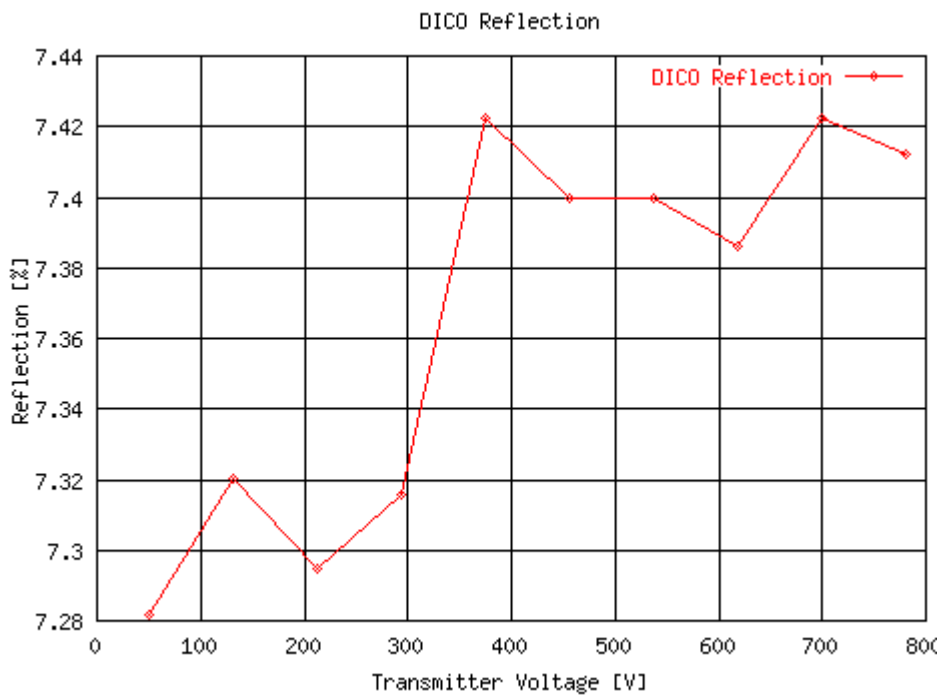


Fig. 375: Example of graphics output

Conclusions

If the reflections in the BC-TX path are too high, you can locate the defective component in the TX path by repeating the test after connecting the dummy load to a component before its current position. Check for loose connectors, in particular.

Examples for errors are low power of RFPA, arcing of capacitors in the BC, defective cables and/or connectors.

The DICO reflection should be below 50%. Reproducible high reflections may be caused by arcing in the Body Coil.

BC-TX path (volt.) (Interactive)

NOTE

When transmitting to the Body Coil, ensure that the loader with spherical phantom is positioned in the Body Coil.

The test requires a valid RF characteristic or the linear RF characteristic.

The objective is to detect oscillations of the RFPA. Several measurements with an increasing amplitude starting from the minimum ($50\text{ i } \frac{1}{2}\text{ V}$) to the maximum voltage (read from the RF characteristic data) are performed.

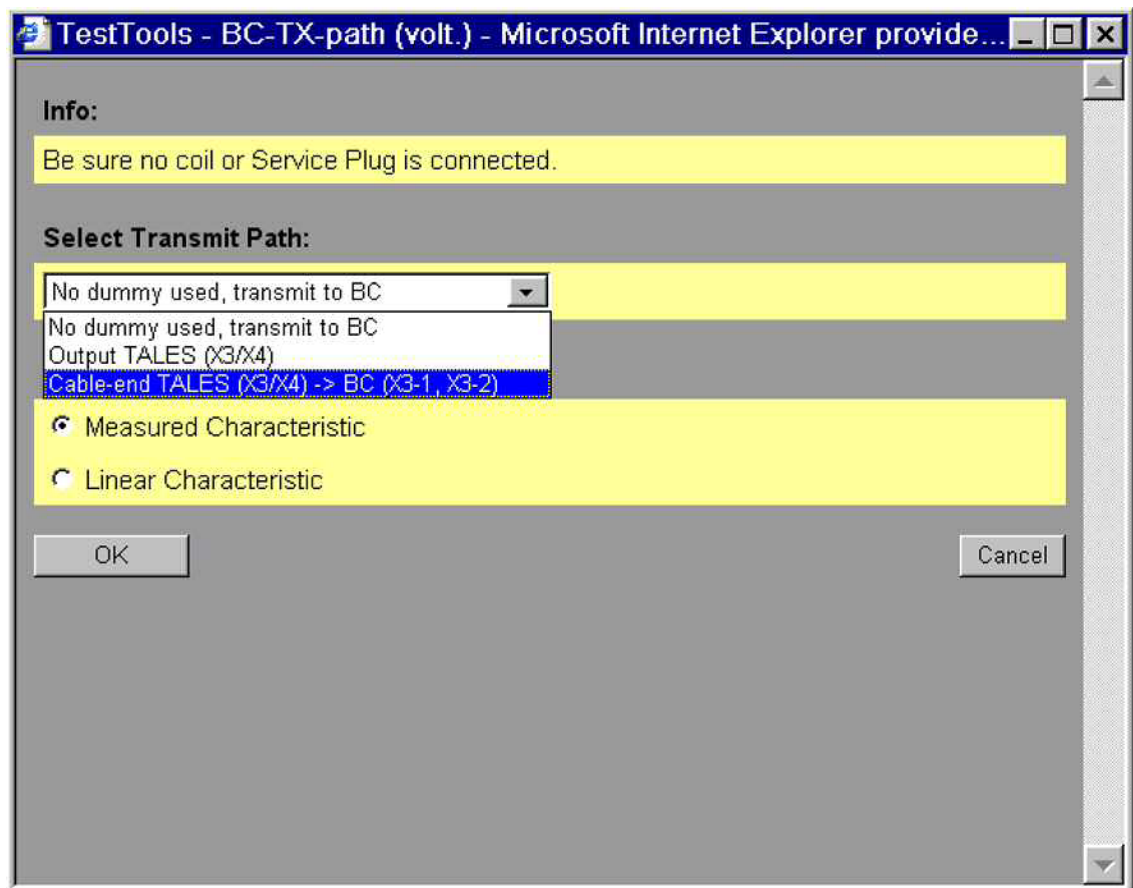


Fig. 376: Pop-up window BC-TX path (refl.) Example: Selection of TX-path termination using 50 Ohm dummies

Signal path

The normal TX path to the Body Coil is used.

The TX path can be opened at locations between TALES and BC and terminated by (one or two) dummy loads. In the appearing pop-up window you can select the desired TX path. The measurement is started by clicking "OK" or skipped by selecting "Cancel" in the pop-up window.

The selection of the location of the interruption in the TX path has no influence on the measurement but selects the heading of the output table. If the interruption is between TALES and RFPA, the TALES values cannot be measured and displayed.

Evaluation

HTML output table:

- PF (DICO) at maximum deviation (Volt)
- PF (TALES) at maximum deviation (Volt)
- $(PF(TALES) - PF(DICO)) / PF(TALES)$ maximum deviation (%)

Graphical output:

- PF(DICO/Volt) and PF(TALES/Volt) over TRA voltage
- $(PF(TALES) - PF(DICO)) / PF(TALES)$ deviation (%) over TRA voltage

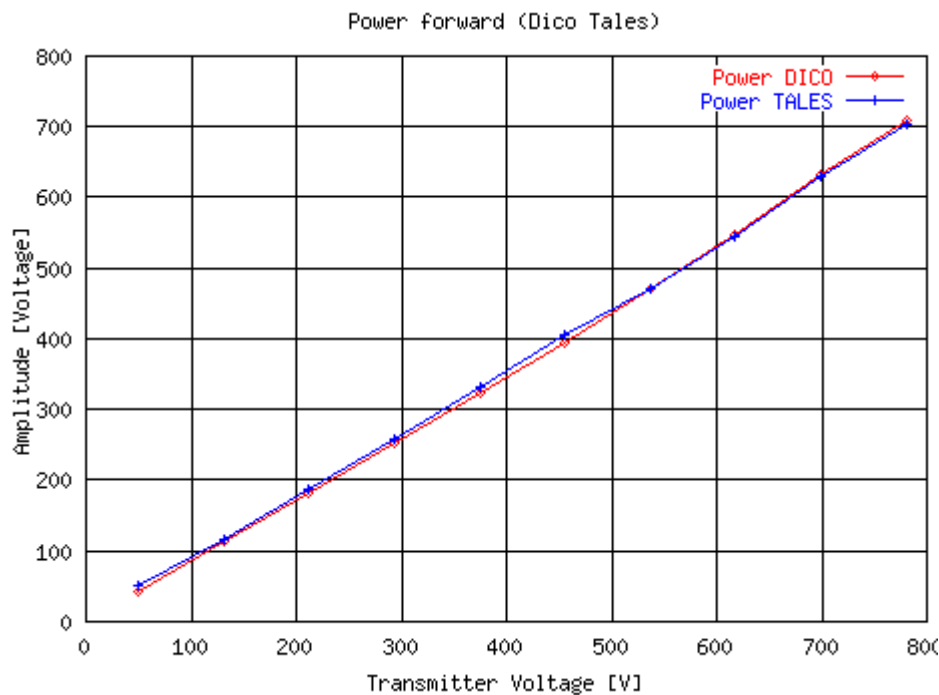


Fig. 377: Example of graphics output

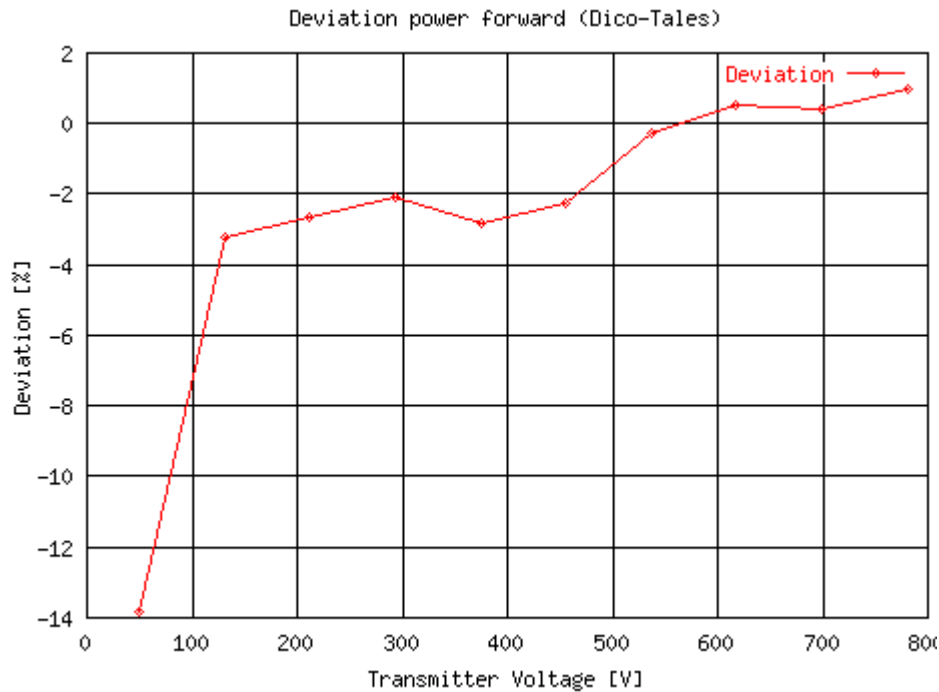


Fig. 378: Example of graphics output

Conclusions

If the deviations between DICO and TALES are out of specification, repeat the RF characteristic measurement.

Locate the defective component in the TX-\ path by repeating the test after connecting the dummy load to a component before its current position.

The reason of significantly higher reflections as measured by TALES in comparison to DICO can be oscillations of the RFPA, transmitted on another frequency besides the MR frequency.

Cable LC

```
HP-Z000100Q-061205-02-010
System Type: . . . . .Avanto
               Espree
               SymphonyTim
```

- LC-TX path (Interactive/ Multi Nuclei)

NOTE

These tests are for experts only!
Additionally, special equipment is necessary for the tests.

These advanced tests enable you to check the complete TX-path to LC and perform systematic troubleshooting in the TX-path by connecting a power 50 Ohm terminator at various locations in the TX-path and/or configuration of particular loops.

The test should be used by experienced service engineers only. Special tools (power dummies, RF-cables, connectors, adapters and attenuator) are needed.

Pop-up windows inform you about the actual measurement configuration.

NOTE

The tests require a valid RF characteristic or the linear RF characteristic. Ensure that a phantom is positioned in the TX-coil.

NOTE

When using the linear characteristic, it may be possible that the transmitter amplitude used exceeds the allowed range resulting in an aborted measurement. This is normal and not a problem with the system.
Additionally, in these cases only the graphics can be evaluated. The automatic evaluation of the numerical values in the charts cannot be trusted as the specification values used are intended for use with a measured RF characteristic only.

NOTE

For the first measurement of each test, the reflection factors are also evaluated and analyzed. If the reflection exceeds 90%, the test is aborted with the error message
Error: DICO reflection out of spec for first measurement.
In this case check the cabeling and the Service Plug/50 Ohm power dummy (if used).

Safety

⚠ WARNING**High RF voltage**

Contact with RF conductors during transmitting of the RFPA will result in death or serious physical injury.

- ⇒ Before disconnecting RF transmit cables or connectors in the transmit path, disconnect the RF input cable at X1 and the unblank line U1 from the RFPA.

LC-TX path (Interactive/ Multi Nuclei)

NOTE

When transmitting to the Local Coil transmit path, ensure that either the Service Plug or a local TxRx-coil is connected to coil plug#1.

The test requires a valid RF characteristic or the linear RF characteristic.

The objective of the test is to check the components of the LC-TX path. Several measurements with an increasing amplitude starting from the minimum (50 V) to the maximum voltage (95 V for Service Plug and maximum local coil voltage read from the coil file) are performed.

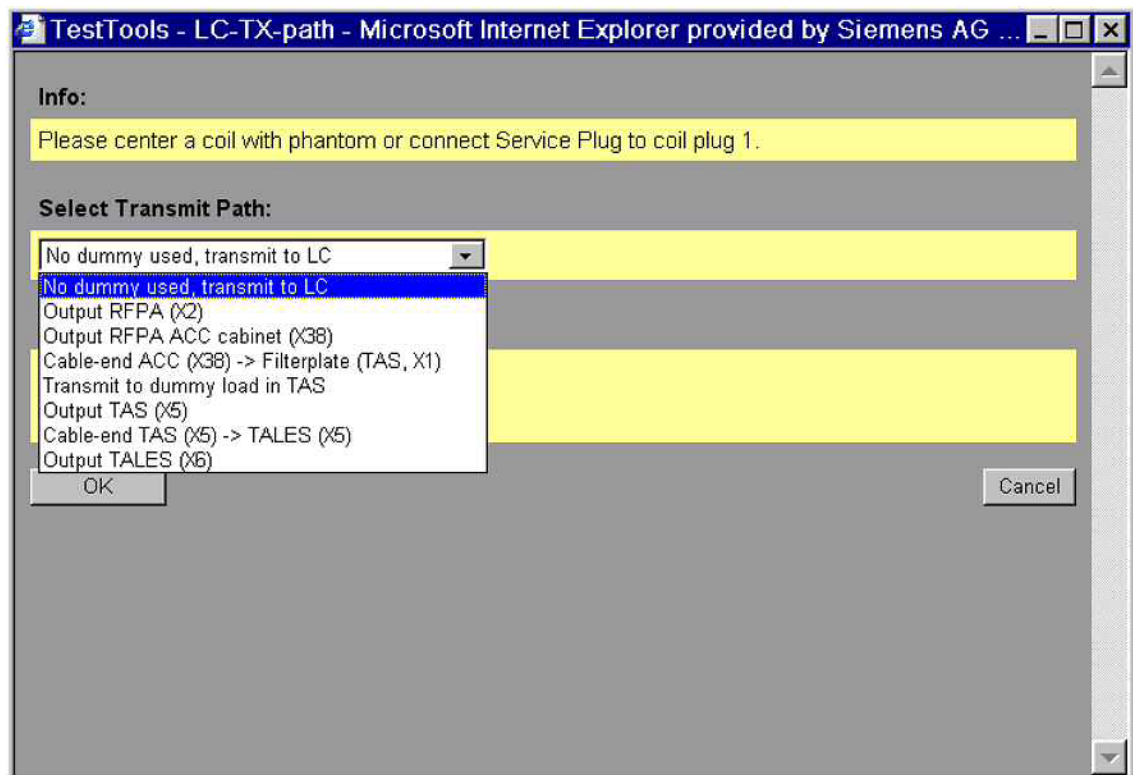


Fig. 379: Pop-up window LC-TXpath - Example: Selection of TX-path termination using an LC

Signal path

The TX path for local coils is used. The RFPA (CORA/DORA) used depends on the selection of the nucleus.

The TX path can be opened at locations and terminated by a 50 Ohm dummy. In the appearing pop-up window you can select the desired TX path (refer to Fig. 74). The measurement is started by clicking "OK" or skipped by selecting "Cancel" in the pop-up window.

The selection of the location of the interruption in the TX path has no influence on the measurement but selects the heading of the output table. If the TX path is terminated by a 50 Ohm dummy between TALES and RFPA, the TALES values cannot be measured and displayed.

NOTE

Whenever a dummy load is used to interrupt the TX-path BEFORE the TALES, the comparison of DICO and TALES-values will not be successful since there are no TALES-values available and, as a result, the test will fail.

This is normal and not a problem with the system.

Evaluation

HTML output table:

- PF (DICO) at maximum deviation (Volt)
- PF (TALES) at maximum deviation (Volt)
- $(PF(TALES) - PF(DICO)) / PF(TALES)$ maximum deviation (%)

Graphical output:

- PF(TALES/Volt) and PF(DICO/Volt) over TRA voltage
- PR/PF DICO reflection (%) and PR/PF TALES reflection (%) over TRA voltage

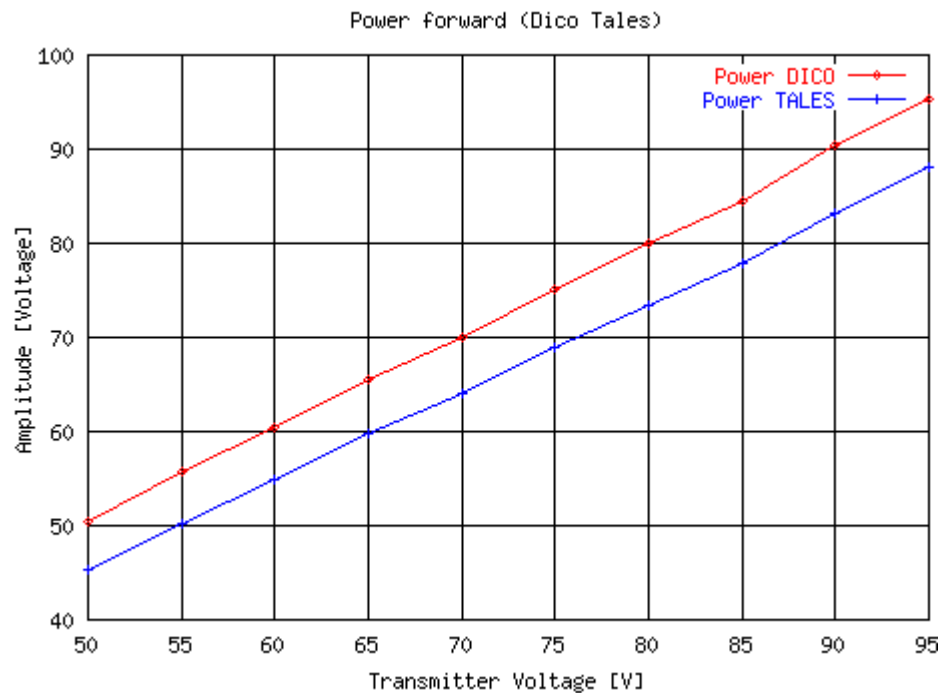


Fig. 380: Example: Transmit to Service Plug

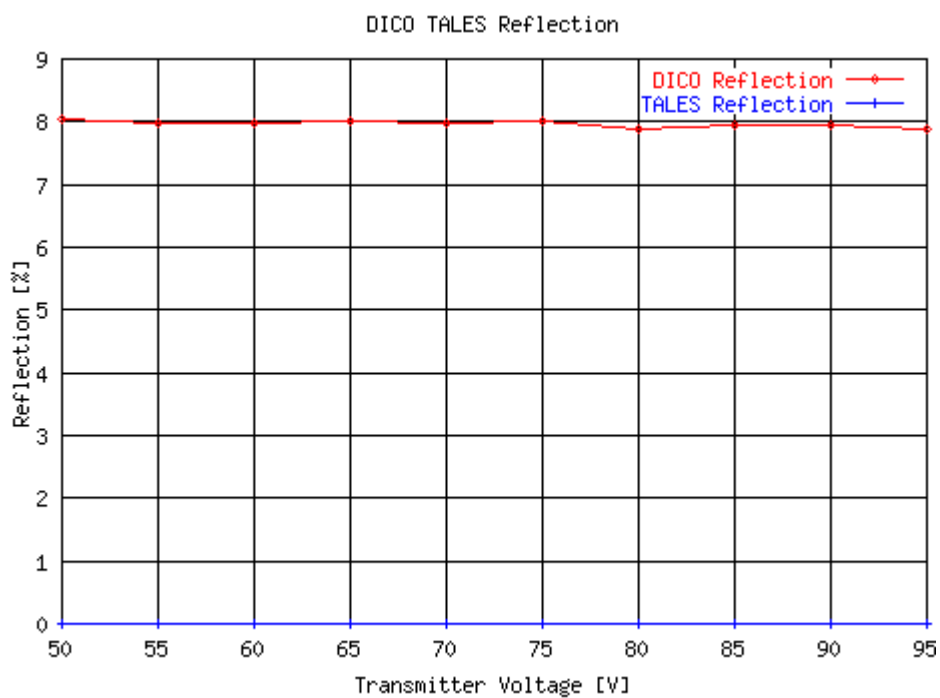


Fig. 381: Example: Transmit to Service Plug

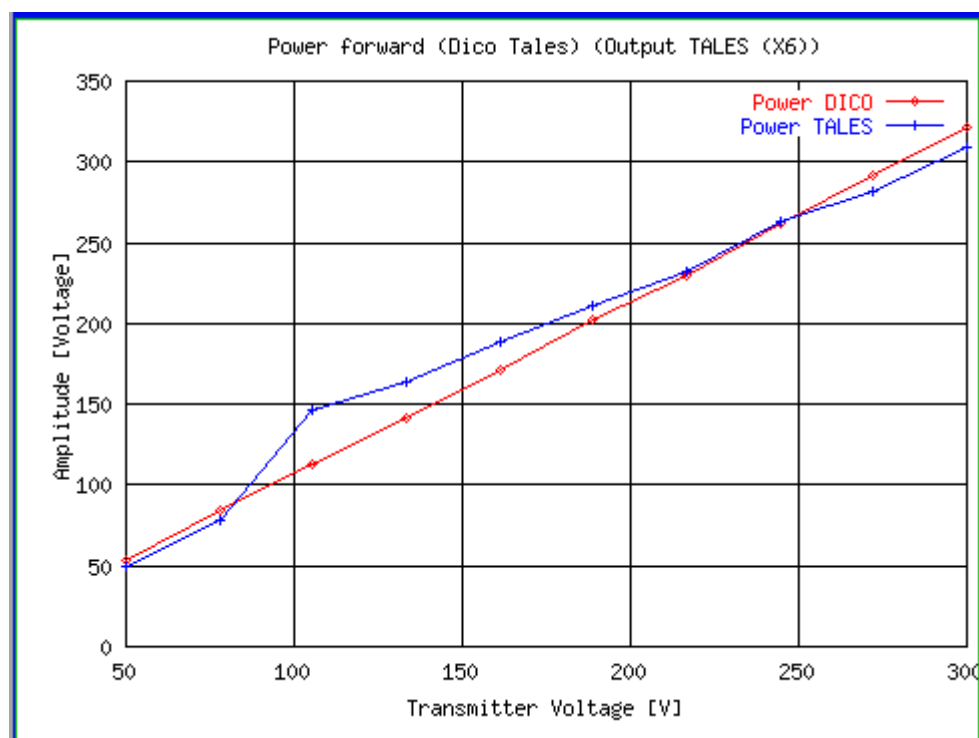


Fig. 382: Example: Transmit to Extremity Coil (Dummy connected to TALES X6)

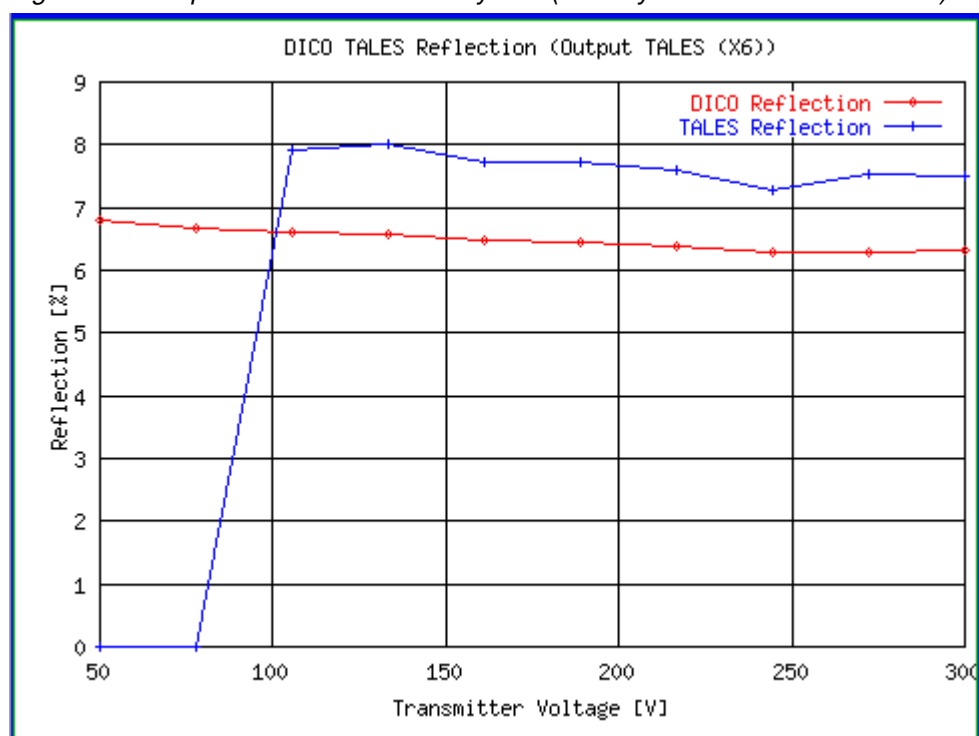


Fig. 383: Example: Transmit to Extremity Coil (Dummy connected to TALES X6)

Conclusions

NOTE

For reflected voltages below 100 V the measured/displayed TALEs value is 0 V.

If the deviations between DICO and TALEs (for voltages higher than 100 V) are out of specification, repeat the RF characteristic measurement for the actual nucleus.

Locate the defective component in the TX path by repeating the test after connecting the 50 Ohm dummy to a component before its current position.

Gradient

HP-Z000100Q-301105-02-008

The stability of the gradient system is essential to the image quality obtained with the MR system. The gradient tests are a tool for verifying the performance of the gradient system as well as assisting you in locating the source for errors.

Before running the gradient tests, perform the AMC test to ensure that the measurement control is functioning as specified.

General

The gradient tests have been developed for locating the source of instability/nonlinearity in the gradient system. For example, when the instability/nonlinearity measured using the Stability Check or CalcAr is a function of the orientation, there is every indication that the instability/nonlinearity is caused by gradient instability/nonlinearity rather than by RF or magnetic field instability/nonlinearity.

In preparation for the tests, the communication of the GPA with the AMC will be tested using the GPA status test.

The results of the tests will be saved automatically in the **Report File**.

Test Evaluation

Both evaluations, Stability and Linearity, analyze two sets of data that are extracted from the raw data. One set of data is the **center column**, which is located in the middle of the gradient pulse plateau. The second set is the **integral of each line of data** of the pulse. The integral is the summation of the data in each line divided by the number of data points.

Stability

The gradient amplitude is constant. The **center column result** is an evaluation of the stability of the center column which is located in the middle of the gradient pulse plateau. The numerical result is the maximum minus the minimum value divided by the mean value. The **integral result** is calculated by using the mean value of each line instead of the center column and using this data to evaluate the stability. Again, the numerical result is the maximum minus the minimum value divided by the mean value. Both center column and integral results are displayed in percent (%).

Linearity

To evaluate the linearity, the gradient amplitude is stepped from a negative to a positive pulse amplitude. The **center column data** and **integral data** of each line are fitted to a linear function. The numerical result is the maximum deviation from the linear fit divided by the maximum value making up the data set. Both center column and integral results are displayed in percent (%)

Test Results/Graphical Output

The numerical results of the Gradient Tests will be displayed in the right frame of the window which appears after completion of the tests. A label identifying the test, the center column result, the specification, the integral result and its specification is printed out. If one or both of these results are out of specifications, the respective field in the table appears in color.

In addition to the numerical results that are displayed for each test, two graphs are plotted. The one graph is a plot using the center column data, the other graph is using the integral data.

With respect to the stability evaluation, the two graphs are simply the plot of the two sets of data (with subtracted mean value), one for the central column data, the other for the integral data. Usually, no pattern is seen in the data, and the data show characteristics typical for noise or ADC sampling errors (changing between a few amplitude values due to the limits of the ADC resolution). However, if a non-constant pattern is seen, this may be indicative of interference at the gradient output.

For the linearity test, the two graphs are the difference between the data and the linear fit that is determined from the data. The first graph uses the central column data, and the other uses the integral data. Ideally, the difference should be zero since the gradient is linearly stepped for each line in the raw data. Usually, no pattern is seen in the graphs, and the data shows characteristics typical of noise or ADC sampling errors (changing between a few amplitude values due to the limits of the ADC resolution). However, if a non-constant pattern is seen, this may be an indication of a problem in the linearity of the gradient output.

GPA status

 HP-Z0000YBM-070812-02-001

System Type:SymphonyTim

GPA (Communication/Status)

- GPA Communication Test
- GPA SetPowerSwitch
- GPA Status

GPA Communication

The communication between AMC/cPCI-CAN and GSSU/D16 K2217 service board via the CAN bus will be tested. In case of error, the system cancels all other tests selected.

If the test fails, run the AMC/cPCI-CAN test and check the cable connection between AMC/cPCI-CAN and GSSU/D16 K2217, followed by the GSSU/D16 service board (with CAN module D1).

GPA Status

The MPCU requests that the GPA provides its present (error) status.

The test provides you with a listing of the present status of the GPA, i.e., all possible error/status conditions, which will be recognized and stored by the local application software of the GPA CAN unit. If an error condition exists, the status of the GPA will be "not ok". Refer to the system error report (service software event log or System/Report) and to the Troubleshooting Guide charts for the respective error conditions. In case of an error, the system cancels all other tests selected.

If the status is "not ok", search for the corresponding error messages. For corrective actions, refer to the system error report details and the Troubleshooting Guide (GPA charts).

The following GPA status messages cause "Status of GPA not ok", i.e., not ready for measurement:

- All messages marked with "!!!" represent errors
- X and/or Y and/or Z Modulator inhibit = yes

If a non-persistent error marked with "!!!" exists in the first run of GPA status, the SetPowerSwitch function resets the GPA to "ok" status.

The test is initiated a second time after the SetPowerSwitch to clear possible previous error messages that are not from the GPA. Check the second status test; if it is okay the first test was most probably affected by previous tests only and the GPA is not defective.

GPA SetPowerSwitch

This function, performed from the MPCU via the CAN bus, initiates a reset of the GPA status "GPAK2217 not ok". All non-persistent error messages will be reset.

The first run of GPA status should be evaluated for the error analysis. Running the AMC tests before the GPA status test will always indicate a stimulation danger error (no real error). This error will be reset by the SetPowerSwitch function.

DAC

HP-Z0000YBM-070812-02-002

System Type:SymphonyTim

DAC (functional)

- DAC (nominal value) test for X, Y, and Z Axis (stability/linearity)

Select all tests using the DAC checkbox or

select the X, Y, and Z Axis checkboxes and high/low amplitude.

Nominal Value X/Y/Z (High/Low Ampl)

The measured signal is the nominal value for the gradient current used to regulate the actual value. The signal is measured at the output of the DACs of D11 K2217, routed to D139 K2217, there selected by the loopmux, digitized, and sent to the PCI-MON. The raw data received (ADC data) will be evaluated.

Selection of a single axis and high/low amplitude (maximum and medium gradient strength) is possible. In case of an error condition, the remaining tests (DAC, X, Y, and Z Axis) can be performed.

Conclusions

In case the signal is too low (stability/center column), check the QLA cable connections between GSSU/D139, X4 and AMC backplane D5, X15 (PCI-MON).

Perform the AMC tests.

Stim.Monitor

HP-Z000100Q-301105-02-011

Stimulation Monitor Test (functional)

TestMode Measurement

With the stimulation monitor switched to test mode, one measurement is performed for each gradient axis. Because of the high gradient amplitudes used, the GPA must be switched off by the stimulation monitor.

Conclusions

If the GPA is not switched off, the PCI-MON or the connection between D60 (GPA) and X15 at the AMC backplane may be defective.

Perform the AMC/PCI-MON test. Check the cabling and connections between D60 (GPA) and X15 at the AMC backplane.

NOTE

Internal errors can cause the GPA to switch off as well, e.g. wrong regulator adjustment (-> resulting error: GPA regulation error). Therefore, check all error messages from the GPA.

ServiceMode Measurement

One measurement is performed in the service mode (GPA switch off by stimulation monitor not possible) with each gradient axis. The maximum values of the stimulations will be recorded and compared to specifications.

Conclusions

If the stimulation values are out of specifications, the stimulation monitor may be defective. Run the AMC/PCI-MON test. If zero stimulation values are measured, the connection to the stimulation monitor may be defective. If the stimulation values are too small, check the jumper setting on D20 (regulator board) and D40 (modulator board). Check for other GPA error messages and refer to the corresponding Troubleshooting Guide charts.

X Axis

HP-Z000100Q-301105-02-019

System Type:SymphonyTim

Actual Value X/Y/Z (High/Low Ampl) (functional)

- Actual value test for X, Y, and Z Axis (stability/linearity)

X, Y, Z Actual Value test

The measured signal is the actual value of the gradient current sent to the gradient coil. The signal is measured by the current sensors (SENS) and routed via D17 K2217 (I/O board) to D12 K2217 (regulator board). After amplification on D12, it is routed to D139 K2217, there selected by the loopmux, digitized, and sent to the PCI-MON via the cable connection. The raw data received (ADC data) will be evaluated.

Selection of a single axis and high/low amplitude is possible. In case of an error condition, all remaining tests (X-, Y-, and Z Axis Functional) can be performed.

Conclusions

In case the signal is too low (stability/center column), check the cable connections between GSSU/D139 K2217 and X15 at the AMC backplane.

If some of the tests fail, check D12 K2217 (regulator board) of the corresponding axis using the "Regulator Check". Check D139 K2217 (service board), the cables to the AMC backplane, and the PCI-MON.

Perform the AMC tests.

Also check the following GPA components:

D14 K2217 (modulator board) and cabling (GPA K2217) to the power stages.

In addition, check for error messages concerning the GSSU power supply and the power stage supply. Check the output voltage of the power stage supply in case linearity tests/high amplitude fail. At a minimum, check the current sensors (SENS) and the power stages. Check the line voltage and the output voltage of rectifier board D32 (GPA K2217).

In case you find error messages from the GPA in the **System/Report** or **GPA_Status list**, refer to the corresponding Troubleshooting Guide GPA charts.

Y Axis

HP-Z000100Q-301105-02-020

System Type: SymphonyTim

Actual Value X/Y/Z (High/Low Ampl) (functional)

- Actual value test for X, Y, and Z Axis (stability/linearity)

X, Y, Z Actual Value test

The measured signal is the actual value of the gradient current sent to the gradient coil. The signal is measured by the current sensors (SENS) and routed via D17 K2217 (I/O board) to D12 K2217 (regulator board). After amplification on D12, it is routed to D139 K2217, there selected by the loopmux, digitized, and sent to the PCI-MON via the cable connection. The raw data received (ADC data) will be evaluated.

Selection of a single axis and high/low amplitude is possible. In case of an error condition, all remaining tests (X-, Y-, and Z Axis Functional) can be performed.

Conclusions

In case the signal is too low (stability/center column), check the cable connections between GSSU/D139 K2217 and X15 at the AMC backplane.

If some of the tests fail, check D12 K2217 (regulator board) of the corresponding axis using the "Regulator Check". Check D139 K2217 (service board), the cables to the AMC backplane, and the PCI-MON.

Perform the AMC tests.

Also check the following GPA components:

D14 K2217 (modulator board) and cabling (GPA K2217) to the power stages.

In addition, check for error messages concerning the GSSU power supply and the power stage supply. Check the output voltage of the power stage supply in case linearity tests/high amplitude fail. At a minimum, check the current sensors (SENS) and the power stages. Check the line voltage and the output voltage of rectifier board D32 (GPA K2217).

In case you find error messages from the GPA in the **System/Report** or **GPA_Status list**, refer to the corresponding Troubleshooting Guide GPA charts.

Z Axis

HP-Z000100Q-301105-02-021

System Type:SymphonyTim

X Axis

Actual Value X/Y/Z (High/Low Ampl) (functional)

- Actual value test for X, Y, and Z Axis (stability/linearity)

X, Y, Z Actual Value test

The measured signal is the actual value of the gradient current sent to the gradient coil. The signal is measured by the current sensors (SENS) and routed via D17 K2217 (I/O board) to D12 K2217 (regulator board). After amplification on D12, it is routed to D139 K2217, there selected by the loopmux, digitized, and sent to the PCI-MON via the cable connection. The raw data received (ADC data) will be evaluated.

Selection of a single axis and high/low amplitude is possible. In case of an error condition, all remaining tests (X-, Y-, and Z Axis Functional) can be performed.

Conclusions

In case the signal is too low (stability/center column), check the cable connections between GSSU/D16 K2217 and X15 at the AMC backplane.

If some of the tests fail, check D12 K2217 (regulator board) of the corresponding axis using the "Regulator Check". Check D139 K2217 (service board), the cables to the AMC backplane, and the PCI-MON.

Perform the AMC tests.

Also check the following GPA components:

D14 K2217 (modulator board) and cabling (GPA K2217) to the power stages.

In addition, check for error messages concerning the GSSU power supply and the power stage supply. Check the output voltage of the power stage supply in case linearity tests/high amplitude fail. At a minimum, check the current sensors (SENS) and the power stages. Check the line voltage and the output voltage of rectifier board D32 (GPA K2217).

In case you find error messages from the GPA in the **System/Report** or **GPA_Status list**, refer to the corresponding Troubleshooting Guide GPA charts.

Grad. linearity

HP-Z000100Q-301105-02-022

System Type: SymphonyTim

Linearity Check (functional)

- Gradient amplitude linearity check using MR signal measurements to determine the linearity of the gradient system. All three gradients are used to produce the spin echo signals which are evaluated over the complete gradient range.
- Position the large spherical phantom with loader in the magnet isocenter. Select the test using the Grad.linearity checkbox.

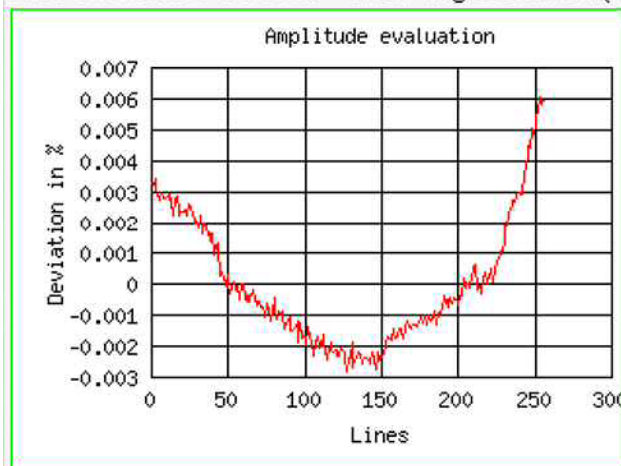
Test results for gradient linearity/ Graphical HTML Output

The numerical results of the gradient linearity tests and a corresponding graph will be displayed in the HTML window after the tests have been completed. A label identifying the test, including slice orientation and gradient axis, is printed out. If the result is out of tolerance, the respective field in the table appears in color.

Each point of the graph represents one amplitude value of the gradient axis tested (line 0-255). The deviation value means the deviation of the integral value of the gradient pulse from the nominal value, relating to the integral value of the maximum gradient pulse.

If a non-constant pattern is seen, especially at the beginning or end of the plots (near line 0 or 255), this may be an indication of a problem with the linearity of the gradient output at high amplitudes.

Slice orientation: Coronal. Performing Z Gradient(readout) Linearity Measurement



Result:

Deviation in [um]		
Value	Low Spec	High Spec
0.002	-	0.010

Fig. 384:

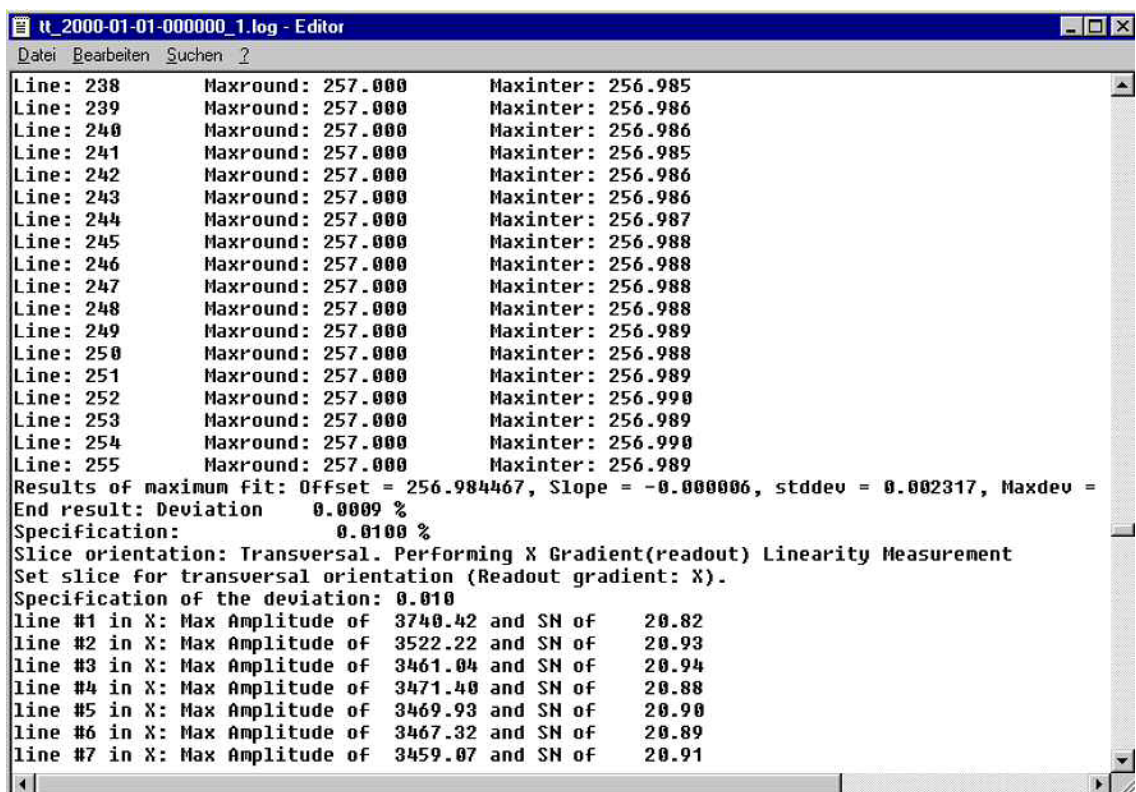
Log File (section of the HTML output page)

Clicking Log-File in the HTML output page lets you open another window displaying all measured maximum gradient amplitudes and some additional information. The example below shows only a section of this window.

NOTE

Beginning with VB11A, the right mouse button is disabled in normal mode. Returning to the previous location can no longer be performed using the pop-up menu. Instead, use the <backspace> or <ALT>-<left arrow> keys.

Alternatively, the advanced use mode can be enabled before the start of the service software platform. In this case, the pop-up menu is enabled again and can be used.



```

tt_2000-01-01-000000_1.log - Editor
Datei Bearbeiten Suchen ?

Line: 238      Maxround: 257.000      Maxinter: 256.985
Line: 239      Maxround: 257.000      Maxinter: 256.986
Line: 240      Maxround: 257.000      Maxinter: 256.986
Line: 241      Maxround: 257.000      Maxinter: 256.985
Line: 242      Maxround: 257.000      Maxinter: 256.986
Line: 243      Maxround: 257.000      Maxinter: 256.986
Line: 244      Maxround: 257.000      Maxinter: 256.987
Line: 245      Maxround: 257.000      Maxinter: 256.988
Line: 246      Maxround: 257.000      Maxinter: 256.988
Line: 247      Maxround: 257.000      Maxinter: 256.988
Line: 248      Maxround: 257.000      Maxinter: 256.988
Line: 249      Maxround: 257.000      Maxinter: 256.989
Line: 250      Maxround: 257.000      Maxinter: 256.988
Line: 251      Maxround: 257.000      Maxinter: 256.989
Line: 252      Maxround: 257.000      Maxinter: 256.990
Line: 253      Maxround: 257.000      Maxinter: 256.989
Line: 254      Maxround: 257.000      Maxinter: 256.990
Line: 255      Maxround: 257.000      Maxinter: 256.989
Results of maximum fit: Offset = 256.984467, Slope = -0.000006, stddev = 0.002317, Maxdev =
End result: Deviation      0.0009 %
Specification:              0.0100 %
Slice orientation: Transversal. Performing X Gradient(readout) Linearity Measurement
Set slice for transversal orientation (Readout gradient: X).
Specification of the deviation: 0.010
line #1 in X: Max Amplitude of 3740.42 and SN of 20.82
line #2 in X: Max Amplitude of 3522.22 and SN of 20.93
line #3 in X: Max Amplitude of 3461.04 and SN of 20.94
line #4 in X: Max Amplitude of 3471.40 and SN of 20.88
line #5 in X: Max Amplitude of 3469.93 and SN of 20.90
line #6 in X: Max Amplitude of 3467.32 and SN of 20.89
line #7 in X: Max Amplitude of 3459.07 and SN of 20.91
  
```

Fig. 385:

Conclusion

In case of an error, check the input voltage of the power stages and the stability of the main voltage. Swap power stages. Replace defective power stages. Check/replace defective GSSUs.

Imager

HP-Z0000YPR-170206-02-001

The imager tests include communication, raw data handling, and image calculation PCI-RX boards and PCI-RX DMA tests.

Imager

HP-Z0000YPR-220206-02-002

Imager Functional

The following functions/components are involved in the test:

- ***Feedback Loop Imager-AMC***
- ***Data Rate Measurement***

The ***Data Rate Measurement*** tests the Imager ***HD-RAID*** for write / read performance.

- ***Data Rate Evaluation***

The tested data rate depends on the configuration of the system:

- 8-channel system: 37,500 MByte/second or more.
- 18-channel system: 84,375 MByte/second or more.
- 32-channel system: 150,000 MByte/second or more.

Imager Comm/Status

The following functions/components are involved in the test:

- Imager PCI-RX Communication
- Imager PCI-RX Status

PCI-RX 1

HP-Z0000YPR-210206-02-005

The following functions/components are involved in the PCI-RX test:

List of **sub tests**

- ***Fifo RAM test***
- ***Echo Ram test***
- ***Mdh Buffer test***
- ***Correction Ram test***
- ***RxOn test***
- ***RxOn Bypass Filter1 test***
- ***RxOn Bypass Filter2 test***
- ***Mofi test***
- ***Nco test***
- ***Static Intern Links test***
- ***Dynamic Intern Links test***
- ***Dynamic Remote Links test***

PCI-RX 2

HP-Z0000YPR-210206-02-007

The following functions/components are involved in the PCI-RX test:

List of ***sub tests***

- ***Fifo RAM test***
- ***Echo Ram test***
- ***Mdh Buffer test***
- ***Correction Ram test***
- ***RxOn test***
- ***RxOn Bypass Filter1 test***
- ***RxOn Bypass Filter2 test***
- ***Mofi test***
- ***Nco test***
- ***Static Intern Links test***
- ***Dynamic Intern Links test***
- ***Dynamic Remote Links test***

PCI-RX 3

HP-Z0000YPR-210206-02-008

The following functions/components are involved in the PCI-RX test:

List of **sub tests**

- ***Fifo RAM test***
- ***Echo Ram test***
- ***Mdh Buffer test***
- ***Correction Ram test***
- ***RxOn test***
- ***RxOn Bypass Filter1 test***
- ***RxOn Bypass Filter2 test***
- ***Mofi test***
- ***Nco test***
- ***Static Intern Links test***
- ***Dynamic Intern Links test***
- ***Dynamic Remote Links test***

NOTE

This component does not exist at systems with 16 channel PCI-RX boards.

PCI-RX 4

HP-Z0000YPR-210206-02-009

The following functions/components are involved in the PCI-RX test:

List of **sub tests**

- ***Fifo RAM test***
- ***Echo Ram test***
- ***Mdh Buffer test***
- ***Correction Ram test***
- ***RxOn test***
- ***RxOn Bypass Filter1 test***
- ***RxOn Bypass Filter2 test***
- ***Mofi test***
- ***Nco test***
- ***Static Intern Links test***
- ***Dynamic Intern Links test***
- ***Dynamic Remote Links test***

NOTE

This component does not exist at systems with 16 channel PCI-RX boards.

PCI-RX DMA

HP-Z0000YPR-220206-02-003

The **PCI-RX DMA (Direct Memory Access)** test transfers data between the Imager and the **PCI-RX** boards via the **PCI bus**.

List of sub-test(s)

- **PCI-RX DMA test**

The test checks the maximum data transfer rate via a special sequence. Normally the sequence will run without any problems. In case the sequence aborts abnormally, the specified data could be achieved.

Host**Network**

HP-Z0000YPR-210206-02-002

The **Network** test displays global TCP/IP and Net BIOS parameters that are read from the NIC's (Network Interface Card).

Network connectivity

Use the following commands to perform the elementary checks of network connectivity (Service level 7 and full access is needed for the following tests).

- Select **Utilities**, **Escape to OS** in the service platform

Command	Note
ARP -A ↵	Displays current ARP entries by querying the current protocol data. If inet_addr is specified, the IP and Physical addresses are displayed. (Only nodes in the local network can be listed)
IPCONFIG/ALL ↵	Display the local network configuration.
PING magic1.hospital.org ↵	To check low-level connectivity. Try using the name first. If this fails, try the IP address. If it does not work with the IP address, suspect the NIC (Network Interface Card) drivers or routing. If ping works, but something high-level does not work, test with NBTSTAT -RN
NETSTAT -rn ↵	To determine whether the local and remote hosts have a default gateway specified.
NSLOOKUP magic1.hosp.org ↵	If DNS is used, find out the corresponding Internet address; reverse name lookup works as well (find the host name for an IP address).
ROUTE PRINT ↵	All locally configured network routes are listed.
TRACERT 155.2.42.117 ↵	Trace routing to a destination host (over a maximum of 30 hops).

Dongle

HP-Z0000YPR-210206-02-003

The **Dongle** test performs multiple reads of the dongle-ID

Enter your dongle-ID, click **OK** (To cancel this test enter 0, click **OK**). Please note that the test itself is not case-sensitive and e.g. **8-5E7124AF6789** is treated the same as **8-5e7124af6789**.

The multiple dongle readout test will be repeated 1000 times. In case an error is detected, the test is aborted immediately because no read errors are allowed.

In case of an error, both the entered dongle-ID and the read ID from the dongle are displayed at the end of the test. In this way mistakes when entering the dongle-ID can be detected much more easily.

Other

MSUP

```
----- HP-EH110289-200206-02-001 -----
System Type: . . . . .Avanto
              Espree
              TrioTim
              SymphonyTim
              Verio
```

General

The Magnet Supervision (MSUP) test consists of a communication / status request only.

MSUP Communication

During the communication test, a command is sent to the MSUP. Successful completion of this test assures satisfactory communication with these components.

The communication test returns the following messages:

- MSUP Communication "OK" or "FAILED"

If the test "failed", check the cable connections and the communication board with the CAN module.

MSUP Status

During the status test, the status of the MSUP is queried. The status test is a fast and reliable way to determine whether the self-test of the respective component failed or not.

This test provides you with a listing of the present status of the MSUP including all possible "ok/not ok" conditions which will be recognized and stored by the local application software of the MSUP CAN-unit.

In case the status is "not ok", perform a "fault" search for the corresponding error message in the system report. For remedy actions, refer to the Troubleshooting Guide.

An "error" status is identified by 3 exclamation marks to the right of the status message. The overall status is therefore "NOT OK" .

NOTE

The MSUP status also contains the status information from the alarm box since this is not an independent CAN-unit. For this reason, it cannot be tested separately.

SHIM

HP-EH110289-200206-02-002

System Type: Avanto
Espreo
TrioTim
SymphonyTim
Verio

SPS

The optional Shim Power Supply (SPS) test consists of a communication / status request only.

SPS Communication

During the communication test, a command is sent to the SPS. Successful completion of this test assures satisfactory communication with these components.

The communication test returns the following messages:

- SHIM Communication "OK" or "FAILED"

If the test "failed", check the cable connections and the communication board with the CAN module.

SPS Status

During the status test, the status of the SHIM power supply is queried. The status test is a fast and reliable way to determine whether the self-test of the respective component failed or not.

This test provides you with a listing of the present status of the SHIM power supply including all possible "ok/not ok" conditions which will be recognized and stored by the local application software of the SHIM CAN-unit.

In case the status is "not ok", perform a "fault" search for the corresponding error message in the system report. For remedy actions, refer to the Troubleshooting Guide.

An "error" status is identified by 3 exclamation marks to the right of the status message. The overall status is therefore "NOT OK" .

ACS

HP-Z0000YBM-070812-02-003

System Type:SymphonyTim

This component is not available for Symphony, A Tim System.

RCA

HP-Z0000YBM-030309-02-001

System Type: SymphonyTim

General

The RCA test consists of a communication/status request only.

Communication

During the communication test, a command is sent to the RCA. Successful completion of this test ensures satisfactory communication with this component.

The communication test returns one of the following messages:

- RCA Communication "OK"
- RCA Communication "FAILED"

If the test "failed", check the CAN module cable connections and communication board.

Status

During the status test, the RCA status is queried. The status test is a fast and reliable way to determine whether the self-test of the respective component failed.

This test provides you with a listing of the present status of the RCA, including all possible "OK/NOT OK" conditions which will be recognized and stored by the local application software of the RCA CAN unit.

If the status is "NOT OK", search for the corresponding error message in the system report. For remedial actions, refer to the Troubleshooting Guide.

An "error" status is identified by 3 exclamation marks to the right of the status message. The overall status is therefore "NOT OK".

NOTE

If no RCA status is reported, check the CAN Bus connection to the component.

PTAB

 HP-EH110289-170206-02-001

System Type:Avanto
 Espreo
 TrioTim
 SymphonyTim
 Verio

General

The patient table test consists of a communication/status request only.

NOTE

The QA contains the check for testing the temperatur sensor for the patient air.

NOTE

The OR-table (special patient table for surgery rooms) is currently not supported by the software.

Any error message from that table is shown on the displays at the magnet.

Communication

The test provides you with a listing of the present status of the selected component, i.e. all possible error/status conditions which will be recognized and stored by the local application software of the component CAN-unit.

An "error" status is identified by 3 exclamation marks to the right of the status message. The overall status of the communication test is therefore "NOT OK".

Status

This test provides you with a listing of the present status of the patient table, i.e. the power supplies, switch status, movement control, display and control panels, including all possible "ok/not ok" conditions which will be recognized and stored by the local application software of the CAN-unit.

In case the status is "not ok", perform a "fault" search for the corresponding error message in the system report. For remedy actions, refer to the Troubleshooting Guide.

An "error" status is identified by 3 exclamation marks to the right of the status message. The overall status is therefore "NOT OK" .

NOTE

If no PTAB status is reported, check the CAN-bus connection to the patient table controller.

NOTE

The QA contains the check for testing the temperatur sensor for the patient air.

MDSD

 HP-EH110289-200206-02-003

System Type:Avanto
 Espreo
 TrioTim
 SymphonyTim
 Verio

General

The MdsTestRoutine test function checks the functionality of the table drive independent of measurements or the table UI.

The test can be used within the framework of the service software

NOTE

This test will automatically operate the patient table. Please make sure that the complete operating range of the patient table is free of obstacles.

For safety reasons, persons may not be present in the magnet room.

Test

During the test sequence, the following steps are performed.

Test sequence

- The table moves into the HOME position.
- The MDSD drive is requested to move
- The actual position is read via the MDSD Proxy.
- In addition, the current value of the MDSD increment decoder is read from the controller (assembly PciGc)
- After reaching its maximum travel range, the table returns to the HOME position.

The values determined, nominal position, real position and AMC increment are evaluated and shown in a table. The maximum deviation is determined by the hardware.

After confirmation via the tester, the table moves into the "HOME position"

PMU

HP-EH110289-170206-02-002

System Type: Avanto
Espreo
TrioTim
SymphonyTim
Verio

General

The optional PMU tests consists of a communication/status request and a transmit (interactive) test.

Communication

The communication test involves sending a command (status and version query) to an individual component which, if successfully executed, verifies communication with this component.

Status

This test provides you with a listing of the present status of the PMU, i.e. the supply voltage, data receiving, if the sensors are connected, including all possible "ok/not ok" conditions which will be recognized and stored by the local application software.

In case the status is "not ok", perform a "fault" search for the corresponding error message in the system report. For remedy actions, refer to the Troubleshooting Guide.

An "error" status is identified by 3 exclamation marks to the right of the status message. The overall status is therefore "NOT OK" .

Transmit (interactive)

A physiological submenu is shown on the PMU display at the magnet front and on the physiological display at the console. After a set-up time, the test signal will show up on the PMU display and on the physiological display. In addition, you will hear a beep from the intercom (if activated with the trigger switch on the left side of the intercom).

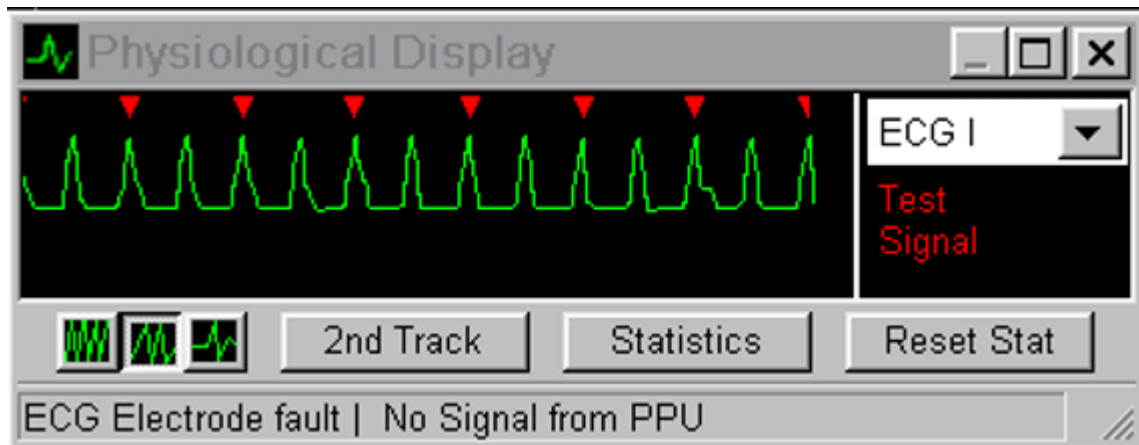


Fig. 386:

Signal Path

At the PERU (pulse and respiratory sensor) a test signal is generated and sent to the AMC via the PDAU (PMU electronics).

Evaluation

After the respective action, two consecutive message windows appear and should be acknowledged by clicking OK.

23.2.3 Auto Diagnoser

HP-Z0000YBM-220206-02-008

System Type: Avanto
Espreo
TrioTim
SymphonyTim
Verio

General

The AutoDiagnoser automatically analyses multiple test results, makes a summary diagnosis of the test results and presents a diagnosis based on these results.

The purpose of TestTools is to help the service technician to locate defective hardware so that it can be replaced. To this end, TestTools provides many individual tests. The problem is that it is usually impossible to identify a defective component based on the results of a single test. As a result, the service engineer has to run several tests and draw conclusions from the test results on the basis of his own experience:

- Our systems are complex
- Multiple components are involved in the simplest tests

The AutoDiagnoser will be helpful in this case. It enables the service engineer to automatically get the data analyzed which he retrieved from the TestTools sessions. This will provide him with additional troubleshooting hints, test recommendations and even direct finger-pointing to a defective component based on the results of one or more than one (multiple) TestTool session(s).

Summary:

- The (new) navigation frame has an AutoDiagnoser item: “Diagnosis” with “AutoDiagnoser”.
- The AutoDiagnoser analyzes multiple test results and automatically provides a summary diagnosis.
- The ability to reliably point out defective boards increases considerably.
- Further tests can be recommended. Their results provide valuable information for a conclusive diagnosis.

Function

For this purpose, the AutoDiagnoser uses two different models. In principle this is the same as the so-called “expert systems” in information technology. Here the software (TestTools, test model) and the hardware (system components, system model) are combined to build such an expert system.

Test Model

In this model, the basic functionality of all tests is described: All components and cables necessary for the test:

- all signal paths used in the test
- all communication paths
- power supply paths

In this way, it describes the hardware dependencies for the test programs.

The purpose of the test model is to describe the hardware on which the results of each test rely. The test model also specifies the conditional probabilities used for the Bayesian network model.

The test model has two main parts. One part describes the hardware dependencies for each test program. The other part describes rules that extend the test model for each program. With the rules, it is possible to describe implicit dependencies. For example, when a test depends on a specific component, it probably also relies on the power cable to the component. If the power cable is used, then the power supply is also used. If the power supply is used, then a specific fuse is used. Using rules to describe such implicit dependencies make the test models for each program much more compact.

System Model

The system model describes the system and its hardware at the board/component (FRU, Field Replaceable Unit) level, including all spare parts and all cables between the components, however, in general not the internal structure of the FRUs. It basically reflects the schematic drawings.

It is basically “only” a machine-readable version of the Top Sheet circuit diagrams.

The model can be configured for multiple systems and different configurations.

Most of the hardware is identical on systems based on the same platform. To simplify the management of multiple systems, and different system configurations, the system model can be specified in variants. The system model is customized for a specific system and configured each time the host starts up (dynamically generated at each host boot).

NOTE

In case the configuration was changed since the last boot, the system model has to be newly generated.

This may increase the reboot time of the system by up to 45 minutes. As an average, 20 minutes after changing the setting, can be expected.

The advantage is that, being model based, there is very little, if any, maintenance required for new program versions:

- If the hardware in a system is changed, only the system model must be changed.
- If there are new or modified test programs, only the test model must be changed.

Principle

Basic Idea

The figure below shows two tests and the dependencies between the test results and the hardware. The first test transmits from the TX cassette to the RX(1) cassette via the Loop1 cable. The dependency diagram for this test says that the test will fail if at least one of the Loop1, TX and RX(1) is defective. The probability that the test will fail can be separately specified for each cable and unit.

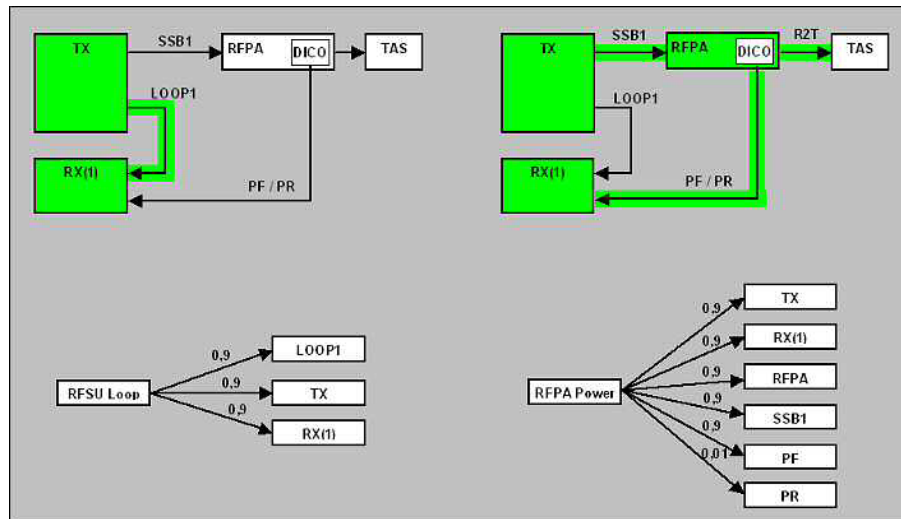


Fig. 387: AutoDiagnoser Basic Idea

=> It is impossible to determine which component is defective just by using a single test result! The AutoDiagnoser combines the test models for all test results into one big dependency model, as shown in the figure below.

On the right side are the test results (error found or not), in the middle are the known dependencies from the individual test models and on the right are the cables and units used. The task for the AutoDiagnoser is to compute the unknown defect probabilities for each of these. A Bayesian network model algorithm is used for this purpose. It can be explained that the algorithm computes the (unknown) defect probabilities that best explain the (known) test results.

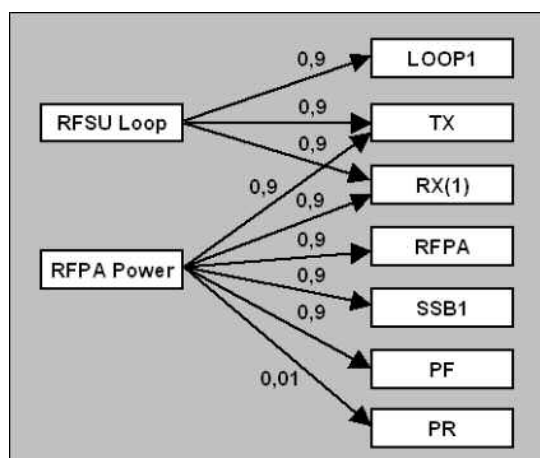


Fig. 388: AutoDiagnoser Network

In the best case, one component has a defect probability close to one while all others have a defect probability close to zero. In this case we can report the component as defective. In most cases it is not possible to isolate a defect so easily. Often several components have a significant defect probability. Then several possible defects must be reported.

Next, the AutoDiagnoser can figure out which tests would return the most information when run. These tests are then recommended. The basic idea is to look for tests that depend only on one or a few, but not all, of the set of possibly defective components. If such a test is negative, the components involved can be eliminated from the set of possibly defect components.

Summary:

- The AutoDiagnoser combines the test models for all test results into one big dependency model.
- The task for the AutoDiagnoser is to compute the unknown defect probabilities for each of these.
- A Bayesian network model algorithm is used for this purpose. It can be explained that the algorithm computes the (unknown) defect probabilities that best explain the (known) test results.
- Another way to describe how the AutoDiagnoser works is that it automatically analyzes the dependencies between executed tests and the system hardware. The two models describe the dependencies and the AutoDiagnoser exploits the knowledge.

Realization

The basic idea is very simple. However, a great deal of software is needed:

- Loading and interpreting the configuration files.

The system model and the test model are optimized to be compact, easy to understand, and easy to maintain. The price that has to be paid is for a complex configuration software.

- Loading and interpreting the test results
- Applying the Bayesian network algorithm
- Interpreting the defect probabilities
- Finding further tests to recommend
- Reporting the results

Configuration**Software Background**

The AutoDiagnoser application relies on certain configuration files. These are interpreted in a configuration step and stored in files. The stored files are newly built each time one of the input configuration files changes. The configuration step may need up to 45 min CPU time to complete. If not performed previously, this configuration step will be performed

when the AutoDiagnoser is invoked at the end of a test-run. To avoid this delay, the configuration step is especially invoked at system start-up. It then runs in background within a low priority process. A synchronization mechanism is used for detecting if the background process is running when the AutoDiagnoser is invoked.

NOTE

The AutoDiagnoser must then wait until the configuration has completed. This means that in case the AutoDiagnoser is in automatic mode, it will not be possible to use TestTools during this time!

Enabling/Disabling the AutoDiagnoser

The base configuration of the AutoDiagnoser is that it can be enabled or disabled. For this navigate to **Configuration > Service > TestTools**, starting from the main service software platform. There AutoSupervision can be enabled and disabled as well.

Summary:

- The automatic start of the AutoDiagnoser can be disabled if the user doesn't want its help.
- Normally it should be set to "enabled".

NOTE

In case the configuration was changed since the last boot, the system model has to be newly generated.

This may increase the reboot time of the system by up to 45 minutes. As an average, 20 minutes after changing the setting can be expected.

As long as the host was not rebooted, the entry will be marked with an asterisk ("*"). This indicates that the displayed setting was not taken over and is not yet active for TestTools.

Specify Start Time

The AutoDiagnoser analyzes test results from previous tests sessions. The starting test session corresponds to the start time of the AutoDiagnoser. Beginning with this start time, all test results are analyzed.

The starting test sessions can be configured via the AutoDiagnoser item in the TestTools navigation.

There the following items are available:

- **History**

Possibility to select a recent TestTools session as the start time.

- **User defined**

Possibility to select any start time, specified by a date and a time.

With **Go** in the bottom action bar, the AutoDiagnoser can be started manually.

The **Default** button in the bottom action bar sets the start time to the current time.

Summary:

- The start time of the test sessions or tests used can be specified.
- Two different possibilities to specify the start time are available.
- Only tests started after the start time will be considered for the diagnosis.

Reports

The AutoDiagnoser analyzes test results from previous tests. The test results are read from the DataXML database. From the results of this diagnosis, a report is produced as it is with any other service function program.

For this basically two different reports will be generated, depending on the mode in which the AutoDiagnoser is used: automatic or manual.

The AutoDiagnoser will come to its end when either a single component could be identified or the program cannot find any additional test. This is usually the case when the diagnosis has the possible defect narrowed down to the combination “transmitter - communication link (cable) - receiver” of a signal. Without a redundant connection, which is normally the case, further testing is usually not possible.

Further tests and diagnoses are of course possible but will not make any sense.

NOTE

In automatic mode, at the end of the test session a summary will be printed out.

"None of the usable test has failed" may be displayed there. This is somewhat misleading. The summary only refers to the tests that can be used for the auto diagnosis. This is not the case with all tests. Some of the tests are not relevant for the AutoDiagnoser.

As a result, the message might be displayed even though some tests failed.

Report with automatic mode

If enabled, the AutoDiagnoser is automatically started at the end of each test session. In this case the report from the AutoDiagnoser is appended at the very end of the normal test report, just before the diagnosis report.

The report in automatic mode has the following features:

- It consists of the number of tests, failed tests, and the test duration.
- It includes hyperlinks to the failed test results, "searching" is no longer necessary.
- The diagnosis is based on the result of all tests of the current test session, as well as results from previous test sessions (see ->Configuration).
- It identifies possibly defect components (including cables) and can also recommend further tests. If further possible tests are detected by the AutoDiagnoser, the TestTools user interface will be already filled with the appropriate checkmarks at the components to be tested. In this case, just a **Go** is necessary to start the tests.
- At the end of this test session, the AutoDiagnoser will start its work again.

Report in manual mode

If no automatic diagnosis is enabled, the AutoDiagnoser can be started manually as well. In this case the resulting diagnosis is displayed in a pop-up window.

The report in manual mode has the following features:

- The diagnosis is based on the result of all tests of the current test session, as well as results from previous test sessions (see ->Configuration).
- It identifies possibly defect components (including cables) and can also recommend further tests. These tests are listed with checkboxes in the result pop-up. There the displayed checkboxes of further possible tests are selected by default,. However, they can also be deselected by the CSE. The selected tests are started by pressing the **OK** or **Go** button.
- At the end of this test session, a manual analysis of the test results by the AutoDiagnoser is possible.

NOTE

If no additional tests are recommended, no further diagnosis is possible. In this case, the window can be closed with the OK and Cancel buttons.

24 RHP

24.1 General

HP-Z000100Q-190907-02-003

Product Perspective

Whenever image defects (e.g. artifacts, distortions, a.s.o.) occur, the raw data of the defective image must be available for troubleshooting.

The **Rawdata Handling Package (RHP)** allows you to:

- Enable/disable raw data generation
- Display raw data
- Apply different mathematical operations (statistics, FFT, etc.) to the raw data set or parts of them for evaluation and troubleshooting
- Remove raw data files from the file system

It is intended for expert use only.

24.1.1 Disclaimer

HP-Z000100Q-170907-02-001

NOTE

The installation and service of equipment described herein is to be performed by qualified personnel who are employed by Siemens or one of its affiliates or who are otherwise authorized by Siemens or one of its affiliates to provide such services. Installation engineers and other persons who are not employed by or otherwise directly affiliated with or authorized by Siemens or one of its affiliates are directed to contact one of the local offices of Siemens or one of its affiliates before attempting installation or service procedures.

Document Revision Level

The document corresponds to the version/revision level effective at the time of system shipment. Revisions to hardcopy documentation are not automatically distributed.

Please contact your local Siemens office to order current revision levels.

24.1.2 Introduction

HP-Z000100Q-160907-02-002

This section describes the **Raw Data Handling Package (RHP)** function.

Refer to the corresponding help items for a description of the user interface and the handling of RHP.

24.1.3 Features

HP-Z000100Q-180907-02-003

RHP provides tools to:

- Enable/disable raw data generation
- Display raw data
- Apply different mathematical operations (statistics, FFT, etc.) to the raw data set or parts of it for evaluation and troubleshooting
- Remove raw data files from the file system

It is intended for expert use only.

24.2 User Interface

HP-Z000100Q-231105-02-013

System Type:Avanto
 Espree
 TrioTim
 SymphonyTim

When you click the **RHP** button in TestTools, the UI displays the **RHP user interface window**



Fig. 389: RHP HTML User Interface

and loads the **RHP Main Menu platform**.

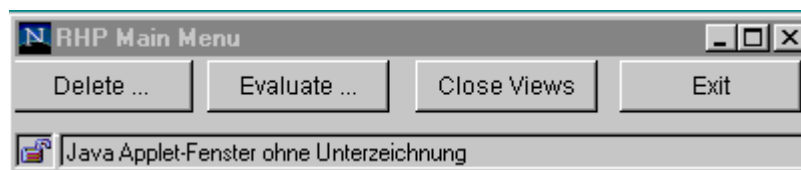


Fig. 390: RHP Main Menu

24.2.1 Structure and Elements

HP-Z000100Q-231105-02-004

System Type: Avanto
Espree
TrioTim
SymphonyTim

RHP platform

The RHP platform consists of a **Navigation Bar**, an **Action Bar** plus a **Function Frame**.

The **Navigation Bar** has two buttons: Help and Exit:

- Click **Exit** to close the menu.
- Click **Help** to display the Help window.

The **Action Bar** provides four buttons:

- Click **Rawdata On** to enable raw data generation.
- Click **Rawdata Off** to disable raw data generation.
- Click **Delete** to delete the selected series of images.
- Click **Go** to execute the command selected in the Trouble Reset application and to display the results in the Function Frame.

The **Function Frame** contains an image data selection menu.



Fig. 391: RHP HTML User Interface

RHP Main Menu Platform

The RHP Main Menu platform lets you access all raw data functions and exit **RHP**.

Click the Evaluate button to display the corresponding window, click the **Exit** button to close RHP. The **Close Views** button closes all open evaluation windows.

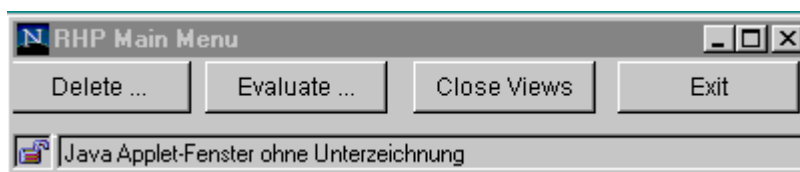


Fig. 392: RHP Main Menu

24.3 Handling

HP-Z000100Q-210907-02-012

RHP Platform



Fig. 393: RHP HTML User Interface

The following can be initiated from this platform:

- Switch raw data generation on/off.
- Copy and delete raw data from the database.

NOTE

By activating the “Raw Data On” button before the measurement, the image data of the measurement will be saved in the last series and the raw data in the second to last series.

To extract the raw data with the “Go” button, you have to manually select the second to the last series of the measurement.

If the “Save Uncombined” feature (saving of individual images of coil elements) was activated in the measurement protocol, the raw data will be saved in the third to last series.

-> Select the series accordingly in the RHP platform.

RHP Main Menu Platform

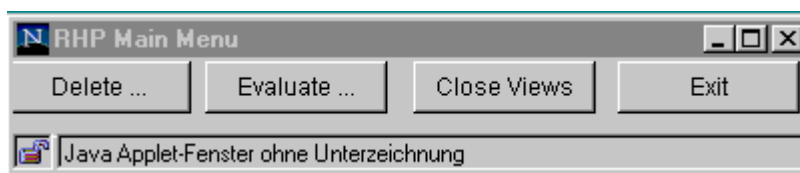


Fig. 394: RHP Main Menu

The following can be initiated from this platform:

- File selection/ deletion -> Button **Delete...** starts the **Delete UI / Data Explorer window**.
- Data evaluation -> Button **Evaluate...** starts the program **ReadRaw** for raw data evaluation.
- “Close Views” closes evaluation views.
- “Exit” terminates RHP.

24.3.1 RHP platform

HP-Z000100Q-200907-02-005

Raw Data On / Raw Data Off

By clicking **Raw Data On**, you enable raw data generation. To disable the function, click **Raw Data Off**. If raw data generation is enabled, it will stay active until it is explicitly deactivated or a new patient is registered.

Copying and Deleting Raw Data from the Database

The **RHP user interface** lets you browse the database and select a patient, study and series. If nothing is selected, the raw data from the last measurement is used. The user can specify a name which is used as a prefix for the copied raw data sets. Clicking **Go** copies all raw data sets of this series from the database to the path specified.

- **Delete**

Deletes the selected series or the series of the last measurement.

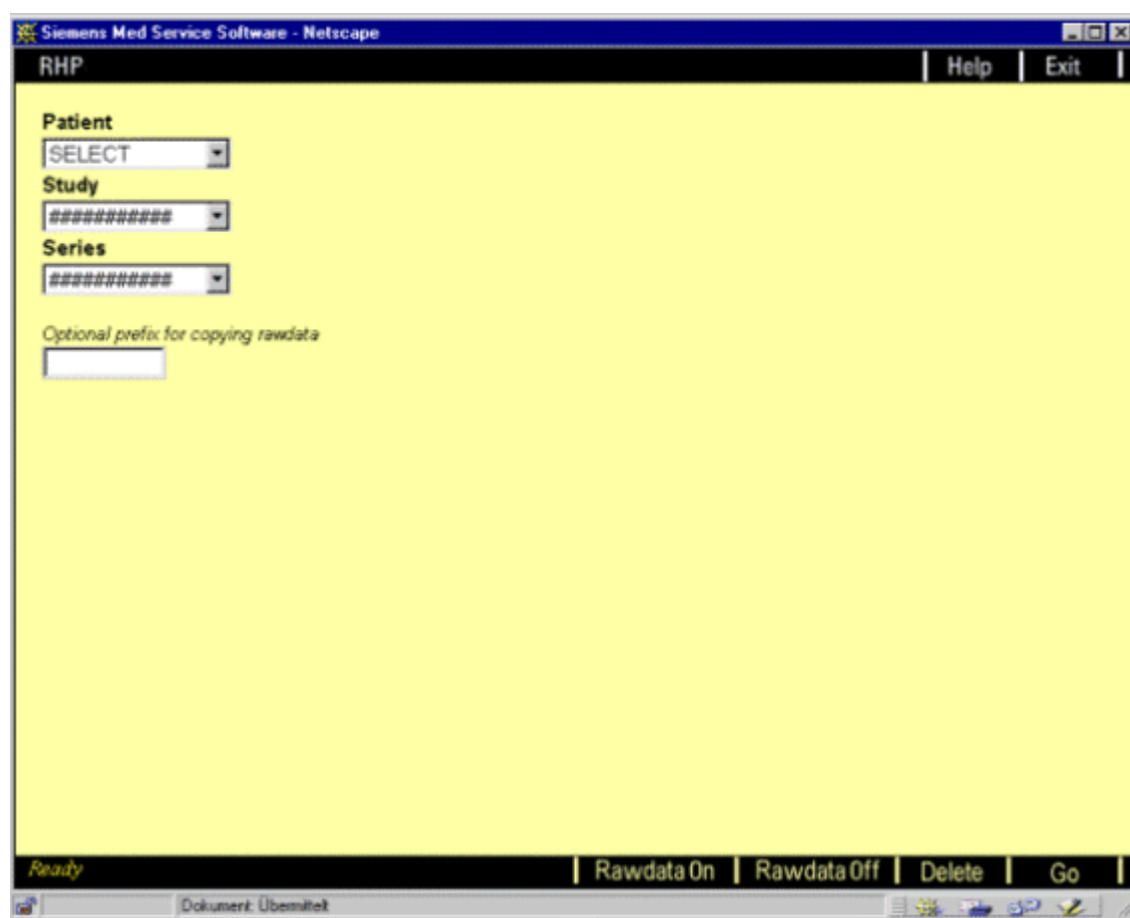


Fig. 395: RHP HTML User Interface

24.3.2 RHP Main Menu platform

HP-Z000100Q-200907-02-006

File Select/Delete Mechanism

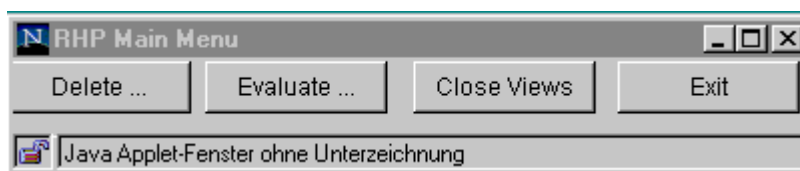


Fig. 396: RHP Main Menu

When you click the **Delete** button in the **RHP Main Menu** the **Delete UI is displayed**. This panel lets you specify the name and path of the file to be deleted. A default file (the last file selected) will be displayed upon start-up. Wildcards (*) can be used to delete more than one file. Click the **Go** button to delete the specified data (you will not be asked for confirmation).

By clicking the **Browse** button in the **Evaluate platform** or in the **Delete UI**, the **Data Explorer** window is opened. The scrollable list contains all subdirectories and raw data files of the current directory in alphabetical order. Subdirectories are marked with a "+".

When you select a file name and click **Select**, the window is closed and the corresponding panel (Evaluate, Delete) is updated with the selected file information.

Clicking **Cancel** discards the selection and closes the window.

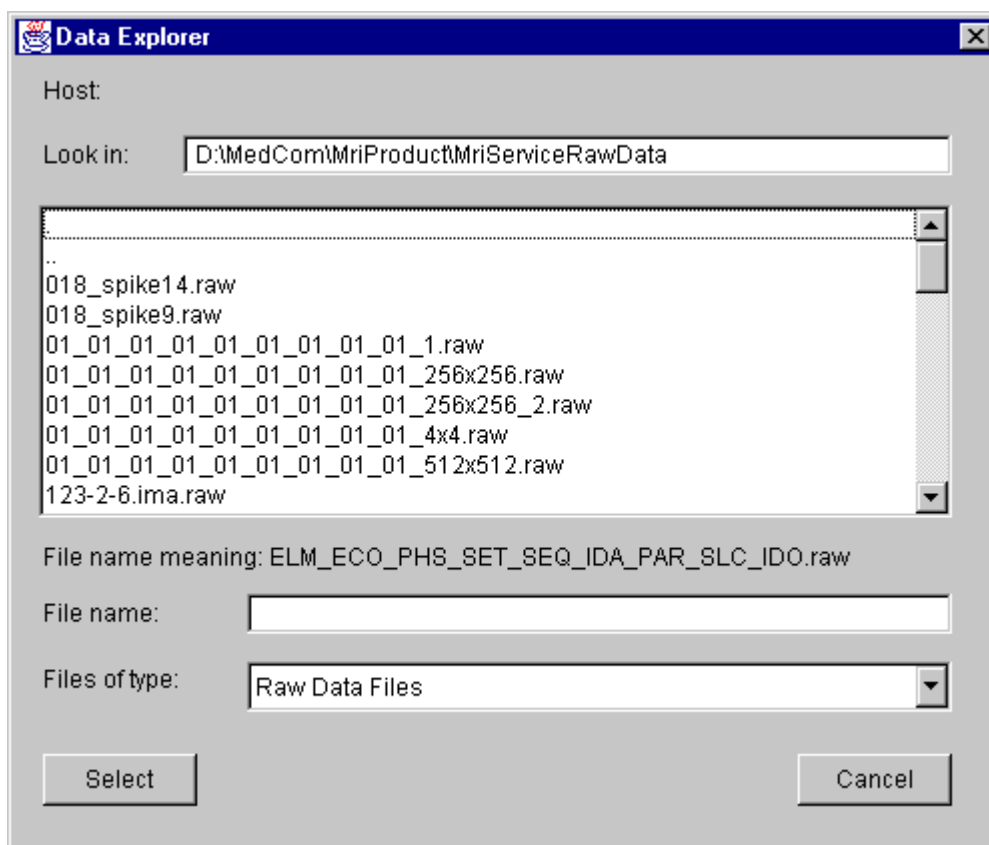


Fig. 397: Data Explorer window

Evaluating Raw Data (ReadRaw)

ReadRaw is a program for evaluating and displaying MR measurement data (raw data). It is only used as a service tool.

ReadRaw is accessed by clicking the **Evaluate** button in the **RHP main menu platform** which opens the **Main Evaluation UI** window. A default file (the last file selected) will be displayed upon start-up. To load a different raw data file, you must specify the path and name in the corresponding text fields or click the **Browse** button to open the **File Select UI**. Using the Evaluation menu of the ReadRaw platform, you can select various sub-platforms for evaluation.

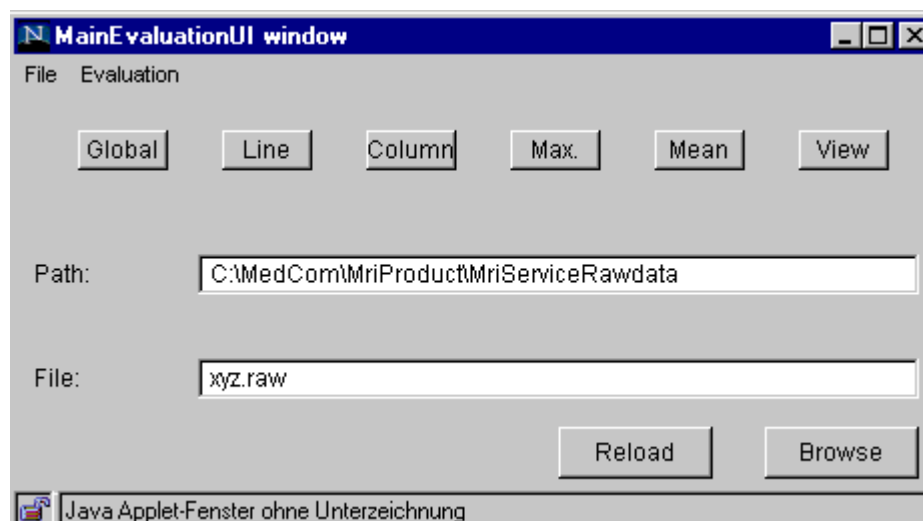


Fig. 398: Main Evaluation window

24.3.3 Global

HP-Z000100Q-200907-02-007

The entire raw data set is included in the **Global Information UI**. It can be used to display the following:

- Raw data header
- Global statistical information
- Global histogram
- Print histogram

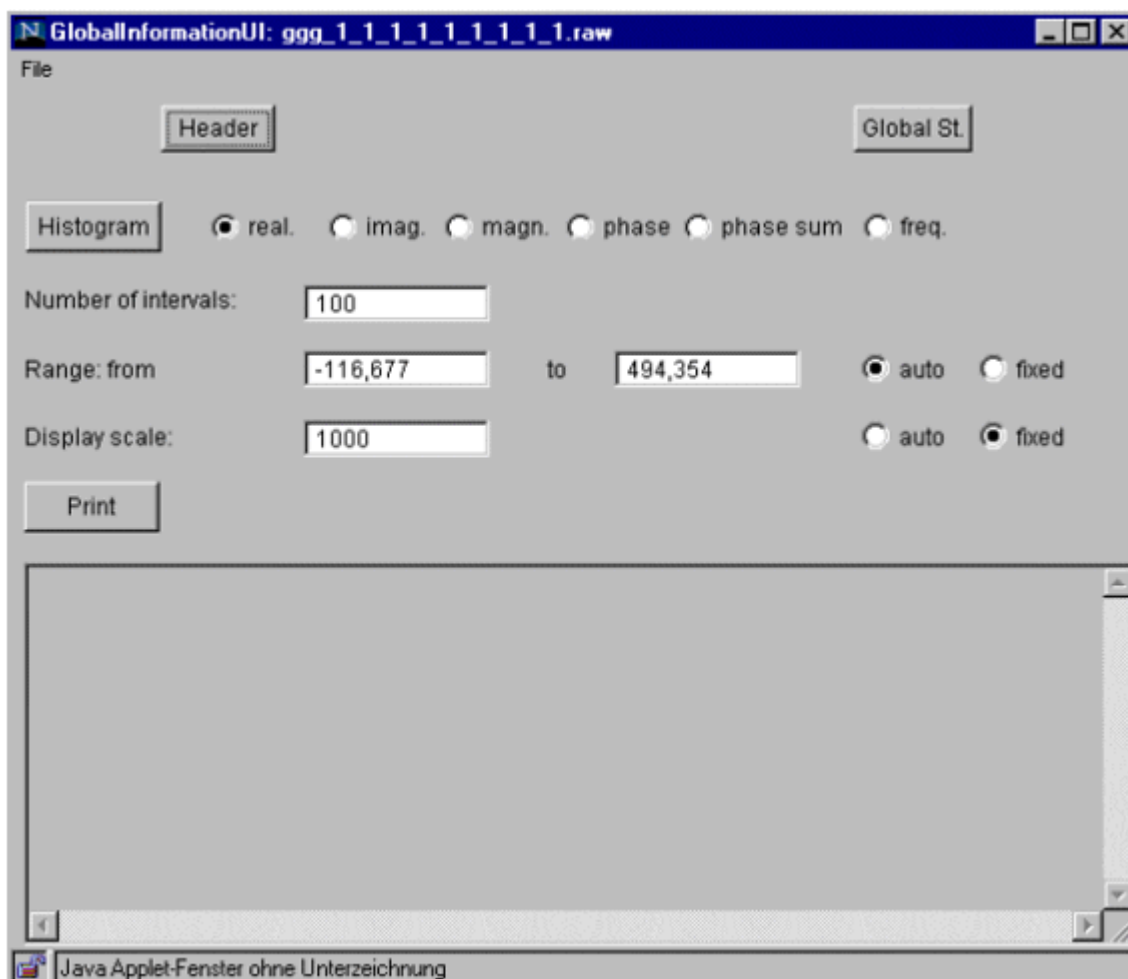


Fig. 399: Global window

Header

Displays parts of the raw data header. This currently includes the header size, data size, number of lines and columns, dwell time and TR time.

Global St.

Statistically evaluates the entire raw data matrix. The magnitude and position of the maximum and minimum raw data values are evaluated and displayed in the message window. The line (nnn) and column numbers (mmm) of the maximum and minimum values are indicated in Lnnn, Cmmm format.

In addition, the difference between maximum and minimum, the mean effective value, the rms value and the standard deviation are calculated.

Since the raw data matrix contains complex data, the statistical evaluation is performed separately for the real part, the imaginary part, the magnitude and the phase.

Histogram

Since the raw data matrix contains complex data, you must select whether the real part, the imaginary part, the magnitude, the phase, the phase sum or the frequency of the complex data are used for histogram evaluation. (The "phase sum" and "frequency" values are calculated in the read or lines direction.) The distribution of the selected part of the data in the raw data matrix is computed. The entire or the selected data value range is divided into a selectable number of intervals and the frequency of the data value occurring per interval is counted. The histogram is displayed within a graphics window.

Print

Displays the histogram output in decimal format.

24.3.4 Line/Column Evaluation

HP-Z000100Q-210907-02-008

This function enables you to select a line or a column of the complex raw data matrix for further evaluation. A line of the raw data matrix corresponds to the readout direction and a column corresponds to the phase-encoding direction during MR data acquisition. An example of the Line Evaluation window is shown below.

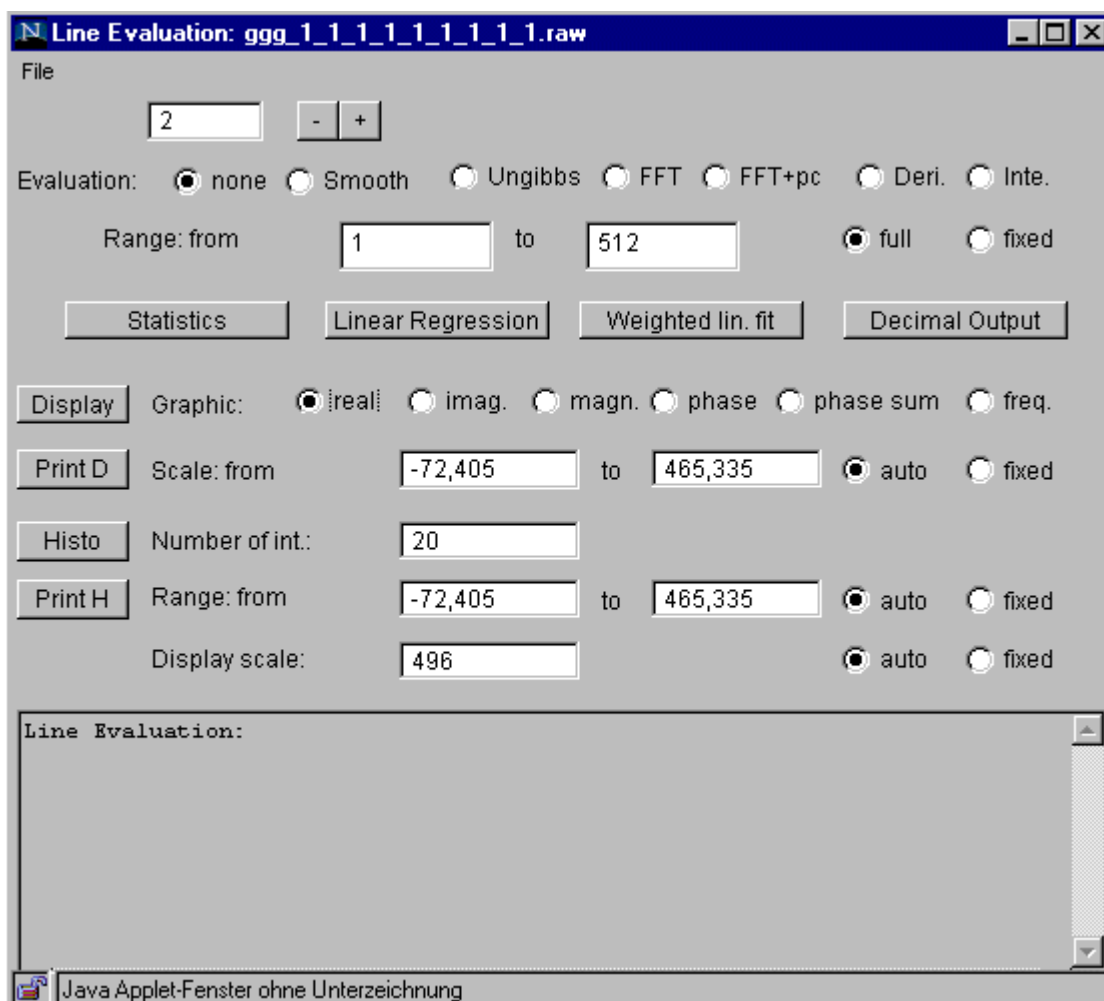


Fig. 400: Line Evaluation window

Smooth

The selected or calculated data vector is numerically smoothed. Smoothing uses a 3 point (Hanning) filter: $[(A[i-1] + 2A[i] + A[i+1])/4]$. The first and last points are handled differently. The results are stored in the actual data vector for further evaluation.

Ungibbs

Numerically removes Gibbs ringing caused by digital (rectangular) filtering of raw data.

NOTE

Ringing occurs if there is a step in the raw data. Due to wrap-around, a step is also the difference between the first and last point in a raw data sample. The latter step is corrected by this function. The Gibbs ringing is calculated for this step and subtracted. Ringing is determined by a convolution of a sinc function with a step function: $a[n] = \sum (1/2 * \text{sinc}(\text{Pi}/2 * i), i=n..\text{inf})$. The results are stored in the actual data vector for further evaluation.

FFT

Calculates the Fourier transformation of the selected or calculated complex data vector. The results are stored in the actual data vector for further evaluation. During transformation, the program checks whether the zero point of the (raw) data is in the center (echo data) or at the beginning (FID) of the data vector.

NOTE

The Fourier transformation can only be calculated for data vectors which are a power of two and longer than 8.

FFT+pc

Calculates the Fourier transformation and zero order (= frequency independent) phase correction of the selected or calculated complex data vector.

Deri

Numerically derives the selected or calculated data vector.

Inte

Numerically integrates the selected or calculated data vector.

Statistics

Statistically evaluates the data vector. The magnitude and position of the maximum and minimum data vector values are evaluated and displayed in the message window. In addition, the difference between maximum and minimum, the mean effective value, the rms value and the standard deviation are calculated.

Since the data vector contains complex data, the statistical evaluation is performed separately for the real part, the imaginary part, the magnitude, the phase and the phase sum.

Linear Regression

A straight line can be fitted to the selected or calculated data using a least-square fit. The two parameters of the straight line equation, the standard deviations and the regression factor **r** are displayed in the message window.

In this case, the standard deviation is the square root of the mean square deviation of the data values from the fitted straight line.

Since the data vector contains complex data, the statistical evaluation is performed separately for the real part, the imaginary part, the magnitude, the phase and the phase sum.

The fitted straight line is plotted in addition to the data (in the graphics window). Also, the calculated deviation between data and numerical fit (linear regression) is plotted in a second graphics window.

Weighted lin. fit

A straight line can be fitted to the selected or calculated data using a weighted (least-square) fit by minimizing the chi-square merit function used here for a self weighted fit to data, assuming the standard deviations of the data points are proportional to their magnitudes. This corresponds to a self weighted fit to data by minimizing the mean square relative deviation. The two parameters of the straight line equation and the chi-square **chi2** are displayed in the message window.

Since the data vector contains complex data, the statistical evaluation is performed separately for the real part, the imaginary part, the magnitude, the phase and the phase sum.

The fitted straight line is plotted in addition to the data (in the graphics window). Also, the calculated deviation between data and numerical fit (linear regression) is plotted in a second graphics window.

Decimal Output

The real, the imaginary, the magnitude, the phase and the phase sum part of the selected or calculated data vector can be displayed numerically in the message window.

Display

Since the data vector contains complex data, you must select whether the real part, the imaginary part, the magnitude, the phase, the phase sum or the frequency of the complex data should be displayed in the graphics window. The "phase sum" is the summed or continued phase value. (Only valid if the oscillation frequency of the MR signal is low against the spectral bandwidth.)

The "frequency" is the numerical derivation of the "phase sum" and shows the momentary frequency. (Only valid in the line or readout direction.) When displaying the phase, the phase angle is given in degrees. The frequency is given in Hz. The data scale factor is usually established automatically so that the value range of the data is within the display range of the graphics. A fixed scale factor can be used (to facilitate comparisons). In this case, you have to determine the value range to be displayed.

Print D

The real, the imaginary, the magnitude, the phase and the phase sum part of the selected or calculated data vector can be displayed numerically in the message window depending on which part is selected in the user interface.

Histo

Since the data vector contains complex data, you must select whether the real part, the imaginary part, the magnitude, the phase, the phase sum or the frequency of the complex data are used for histogram evaluation. The distribution of the selected part of the raw data is computed. The entire or the selected data value range is divided into a selectable number of intervals and the frequency of the data value occurring per interval is counted. The histogram is displayed in a graphics window.

24.3.5 Maximum Evaluation

HP-Z000100Q-210907-02-009

The positions of the maximum magnitude of each line are determined. These values are stored as integer numbers (= column number of the maximum) in the real part of the resulting actual column vector for further evaluation. In addition, the interpolated positions of the maximums are calculated. Again, these values are stored as double numbers in the imaginary part of the resulting actual column vector. Possible evaluations are the same as for line and column.

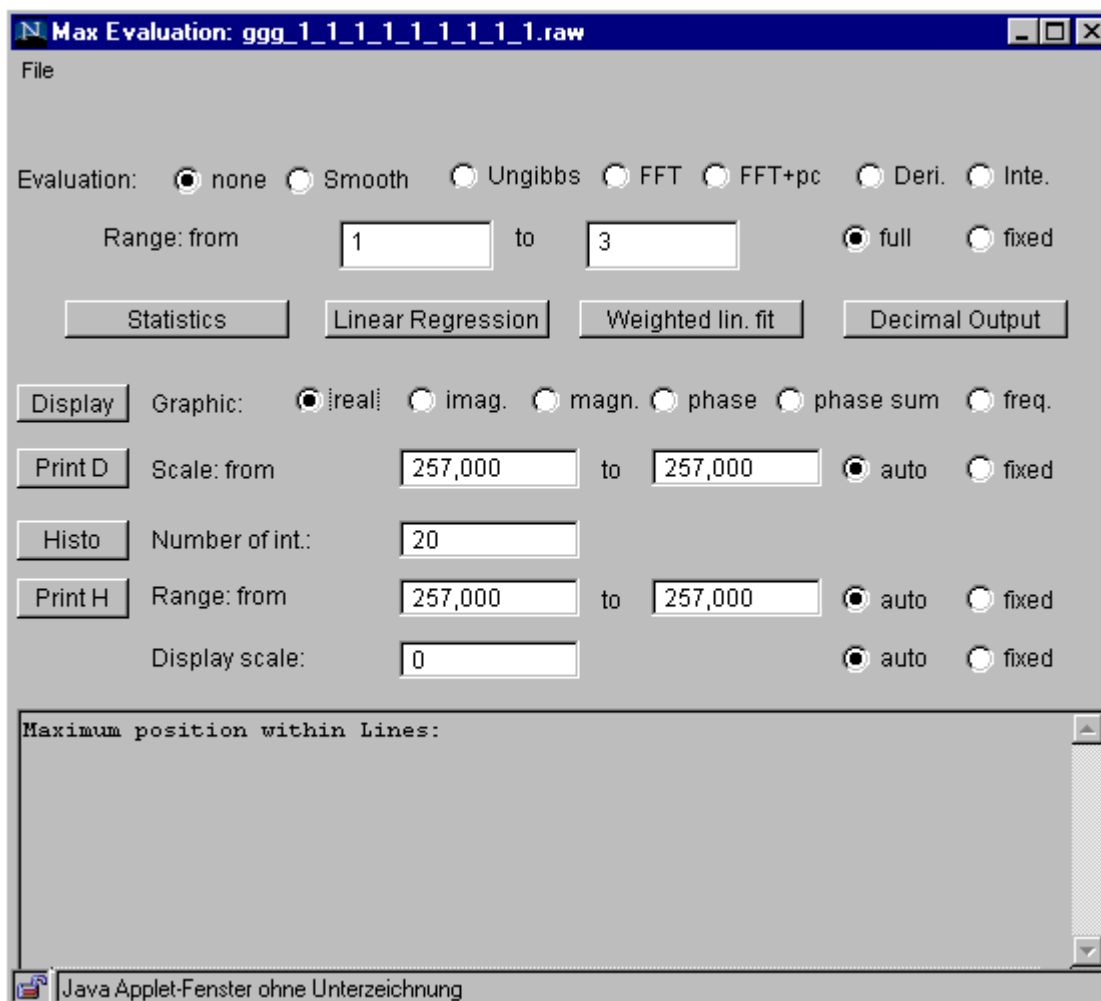


Fig. 401: Maximum Evaluation window

24.3.6 Mean Evaluation

HP-Z000100Q-210907-02-010

The mean values (real and imaginary) of each line of the complex raw data matrix are calculated. They are stored as the actual column data vector for further evaluation. Possible evaluations are the same as for line and column.

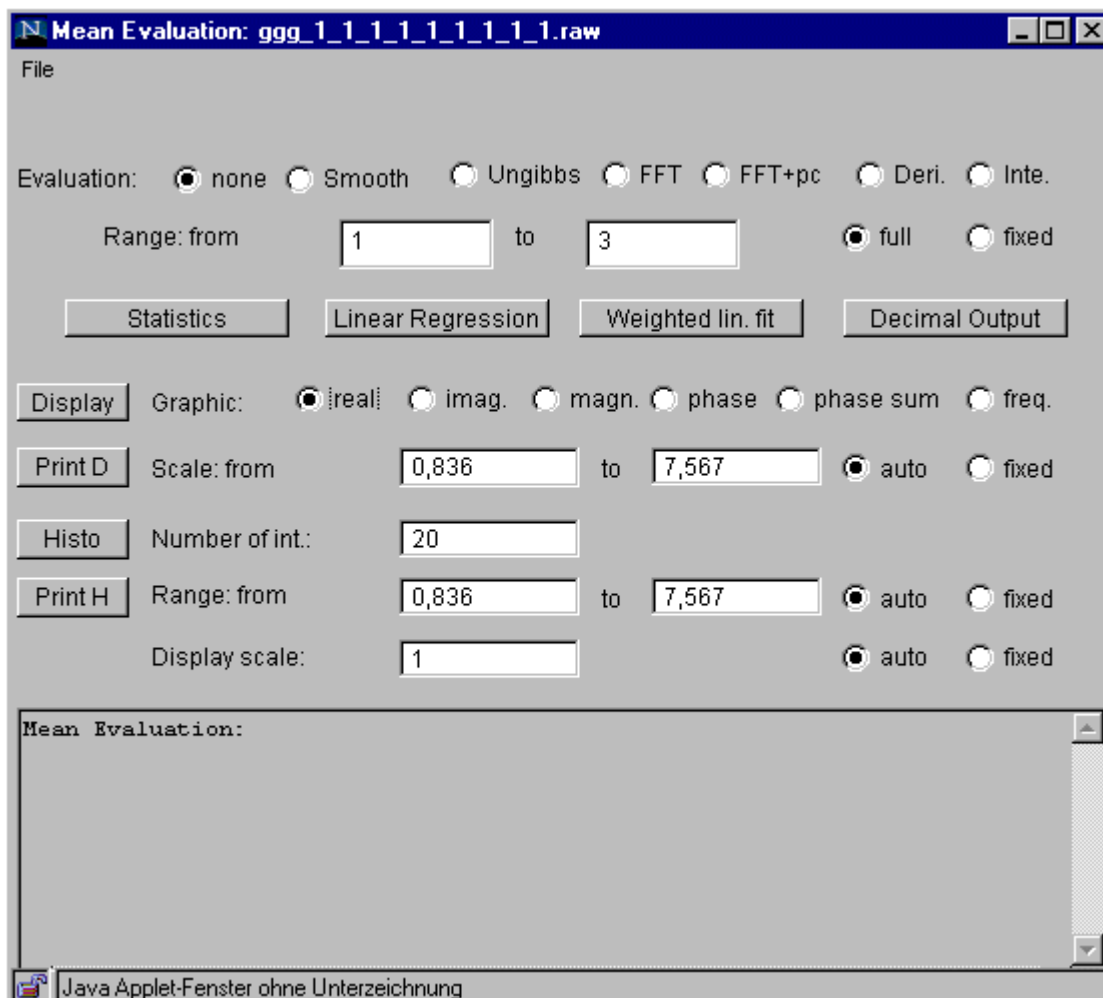


Fig. 402: Mean Evaluation window

24.3.7 View

HP-Z000100Q-210907-02-011

The entire raw data set can be displayed. Since the raw data consists of complex values, you must specify whether you want to view the **real/imaginary part** or the **magnitude/phase**. A toggle button enables you to switch between these displays. The text field on the right side displays header information.

You can zoom in and zoom out using the **Zoom+/-** button. In standard mode, 256×256 pixels are used for the display. If the raw data set is larger than 256 (in at least one direction), the area in the center of the set is used for the display. In zoom mode, 512×512 pixels are used for displaying the raw data. If the raw data set is smaller than the pixel matrix used (in at least one direction), the output is centered in the middle and the unused area is left gray.

Windowing is performed by entering the center and window values or by clicking and dragging the mouse within the graphic output.

By clicking the left mouse button (without dragging) in the graphic output window, the current position (line and column) in the raw data matrix is displayed.

Click the **ROI Y/N** button to enable/disable the display of ROIs. If ROI display is enabled, clicking and dragging the mouse defines the size and position of the ROI. In this mode, the windowing of raw data using the mouse is disabled.

The **ROI Rect/Oval** button toggles between rectangular and oval ROIs. In the ROI mode, the statistical evaluation of the ROI is displayed in the right text field. In addition to size and position, the maximum and minimum values and positions as well as mean and RMS values within the ROIs are displayed.

The two-dimensional FFT is also supported. The **FFT Y/N** toggle button enables you to display the original or the FFT data set (MR image).

Click **Cancel** to exit the platform.

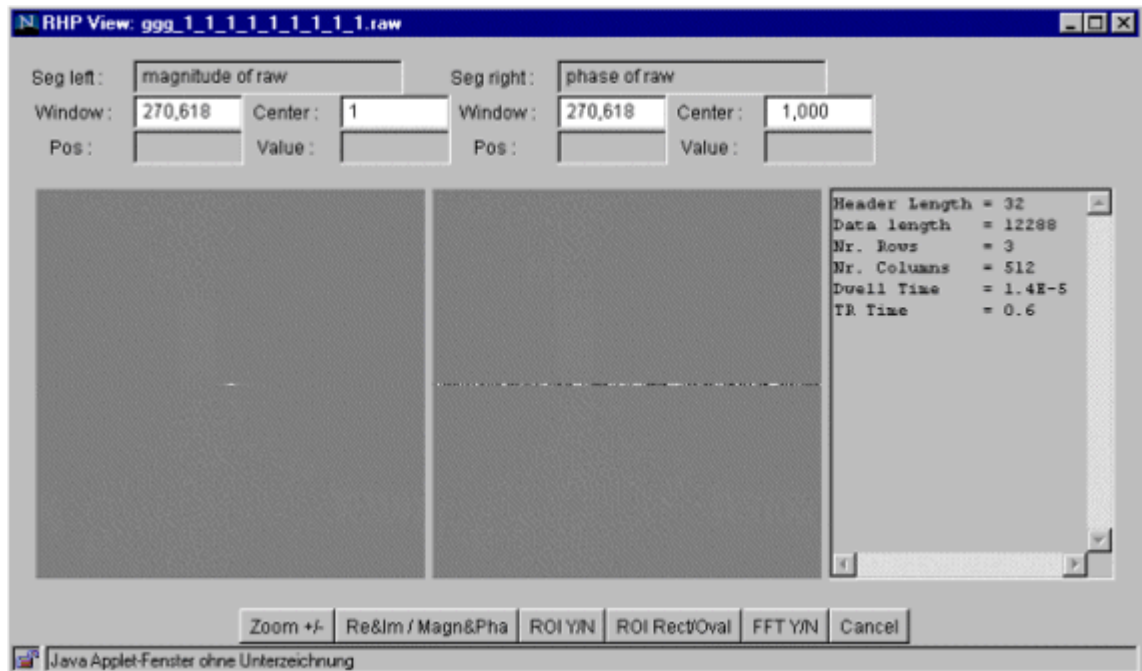


Fig. 403: RHP View window

25 Trouble Reset

25.1 General

HP-Z0000YPR-230206-02-999

System Type: Verio
SymphonyTim
TrioTim
Espreo
Avanto

The Trouble Reset button opens a separate platform where it is possible to:

- Reboot the host

NOTE

In that case the remote connection is cut!

- Reboot the imager
- Reboot (normal or forced) the scanner, switch it on from standby mode, or bring it to standby mode

NOTE

With the new MSUP K2306 it is not possible to switch on the scanner again.
If the system is in standby mode no choice is offered. Instead only “Not available at the moment” is printed out.

This is most applicable to Remote Diagnostics as the System Control can only be used while sitting at the user control.

25.1.1 Disclaimer

HP-Z0000YPR-230206-02-998

System Type: Verio
SymphonyTim
TrioTim
Espree
Avanto

The installation and service of equipment described herein is to be performed by qualified personnel who are employed by Siemens or one of its affiliates or who are otherwise authorized by Siemens or one of its affiliates to provide such services. Installation engineers and other persons who are not employed by or otherwise directly affiliated with or authorized by Siemens or one of its affiliates are directed to contact one of the local offices of Siemens or one of its affiliates before attempting installation or service procedures.

25.1.2 Introduction

```
HP-Z0000YPR-230206-02-003
System Type: . . . . . Verio
              SymphonyTim
              TrioTim
              Espree
              Avanto
```

This section describes the Trouble Reset function.

The Trouble Reset allows you to:

- Reboot the host

NOTE

In that case the remote connection is cut!

- Reboot the imager
- Reboot (normal or forced) the scanner, switch it on from standby mode, or bring it to standby mode

NOTE

With the new MSUP K2306 it is not possible to switch on the scanner again.

If the system is in standby mode no choice is offered. Instead only “Not available at the moment” is printed out.

This is most interesting for Remote Diagnostics since it is not possible to use the System Control when you are not sitting at the user console.

For this the “**Remote Service Access Control**” level has to be set to **Full Access**.

25.1.3 Features

```
HP-Z0000YPR-230206-02-996
System Type: . . . . . Verio
                SymphonyTim
                TrioTim
                Espree
                Avanto
```

The Trouble Reset button opens a separate platform with the following options:

- Reboot the host
 Reboot the host
- Reboot the imager
- Reboot (normal or forced) the scanner, switch it on from standby mode, or bring it to standby mode

NOTE

With the new MSUP K2306 it is not possible to switch on the scanner again.

If the system is in standby mode no choice is offered. Instead only “Not available at the moment” is printed out.

This is most applicable to Remote Diagnostics as the System Control can only be used while sitting at the user control.

To close the window use the Home button in the navigation

Use **Go** in the bottom action bar to perform the requested task, **Update** will update the current status of the task.

25.2 User Interface

25.2.1 Structure and Elements

HP-Z0000YPR-230206-02-995

n.a.

25.2.2 Handling

HP-Z0000YPR-230206-02-994

n.a.

26 Install & Maintenance

26.1 General Information

HP-EH102468-250907-02-088

Please select a sub-item ...

26.1.1 Introduction

HP-EH102468-181108-02-001

System Type: Avanto
Espreo
TrioTim
SymphonyTim
Verio

The Installation & Maintenance Protocol contains the Installation Protocol, Safety-Related Test Protocol (Report), and Maintenance Protocol.
For each protocol, an interactive online checklist is provided with a description for each checkpoint.

Safety note



Please follow the general safety guidelines as well as the MR-specific safety information!
Failure to comply with the safety guidelines may result in death or serious injury.
⇒ **Please read both the safety information in the general safety guidelines and the MR-specific safety information carefully prior to starting work. Compliance with the information provided is mandatory at all times.**

Structure

- The Installation & Maintenance Protocol consists of the following parts:
- "Summary fill in"
 - "Installation"
 - "Safety-Related Tests"
 - "Maintenance"
 - "Preview"
- To obtain detailed information regarding the sub-platforms, select the corresponding 'Help' entry.

Interactive checklist

Each maintenance step (task) features **OK - Not OK** - buttons, as well as a 'Procedure' button and a date specification for the 'Previous Maintenance' and the 'Next Maintenance'

'OK' - 'Not OK' button

Click the **'OK' button** if you have successfully performed the maintenance step. After you click the **'Save'** button, the check is no longer displayed.

If you detect a deficiency which cannot be eliminated, click the **'Not OK' button**. After you click the **'Save'** button, the check is no longer displayed and the 'Not OK' field turns to red.

Enter date

For a new installation:

The current date will be set and saved automatically if you click the **OK- Not OK** button.

For SW updates to VB13 or higher:

Condition: Installation, maintenance, or safety-related tests have been performed during previous service work. At that time, the online Installation & Maintenance Protocol was not part of the Service Software. Check the maintenance and safety-related test 'hardcopy' protocols filled out previously and archived in the 'System Owner Manual'. Copy the dates from the 'hardcopy' protocols into the 'Previous Maintenance' field and click the **'OK'** button.

Regular maintenance:

Click the **'OK - Not OK'** button and the current date will be set automatically.

'Next Maintenance' date

The 'Next Maintenance' date will be automatically calculated based on the maintenance interval.

Procedure button

Use the **'Procedure'** button to obtain a detailed description of the maintenance step. The procedure descriptions are based on the technical 'Safety-Related Test Procedure' and the 'Maintenance Instructions' documentation. The print number of the documents is imprinted on the HTML protocol. You can also use the 'hardcopy' or 'CB-DOC' version to perform maintenance. Check the print number if you prefer the 'hardcopy' or 'CB-DOC' version.

Save button

Save your input with the **'Save'** button before you select another menu. After you click the 'Save' button, the check mark is no longer displayed.

If you switch to another menu without saving your input, it will be lost. A pop-up window prompts you to **'Save'**.

26.1.2 Summary fill in

HP-EH102468-181108-02-002

Enter general information about the system, customer, office, contract, etc. in the 'Summary fill-in' platform.

This information will be **reused** in the protocols for Installation, Safety-Related Tests, and Maintenance.

The entries remain saved for future service sessions. Change the entries during the next session (e.g. measurement devices, district office, ...), if required.

The 'Summary fill-in' platform contains the following parts:

- General
 - Customer site
 - MR system
 - District office
- District office
- Contract
- Measurement devices

-To obtain detailed information regarding the sub-platforms, select the corresponding '**Help**' entry.

26.1.3 Installation

HP-EH102468-181108-02-003

System Type:Avanto
Espreo
TrioTim
SymphonyTim
Verio

All MR components are tested in the factory and then shipped to the site where they are assembled into a complete system for the first time. The adjustments and checks performed during the TuneUp and quality assurance procedures are the **final system test**.

The 'Installation Protocol' is the final test certificate for the system. The completed 'Installation Protocol' is our verification that the system including all coils and options was properly installed and is operating according to manufacturer's specifications.

The protocol must be filled out correctly. Protocols with major mistakes cannot be accepted as proof of quality and must be corrected. The person who completes the protocol is personally responsible for ensuring that it is filled out correctly.

26.1.4 Safety Related Tests

HP-EH102468-181108-02-400

System Type: SymphonyTim

Safety Information



Please follow the general safety guidelines as well as the MR-specific safety information!

Failure to comply with the safety notes may result in death or serious injury.

- ⇒ **Read the safety notes contained in the general safety notes as well as the MR-specific safety information in detail prior to starting work. Adhere to the information provided at all times.**

General remarks

Based on the risk analysis and the clinical evaluation, the manufacturer is obligated to determine the type and frequency of repair activities as well as the inspection measures required for products to ensure continuously safe and proper operation.

The safety-related tests

- are recurring tests. The scope and schedule of these tests are defined by the manufacturer.
- should determine whether the affected product is in good operating condition, ensuring the safety of patients, users, and other individuals.
- do not replace maintenance activities.

Country-specific laws and regulations have to be taken into consideration, e.g. in Germany the “Medizinprodukte-Betreiberverordnung (MPBetreibV)” (Medical Device Operator Regulations), IEC 62353 / DIN EN 62353, etc.

These safety-related tests correspond to the technical safety checks according to “MPBetreibV” (Medical Device Operator Regulations) “Sicherheitstechnische Kontrolle” including IEC 62353 / DIN EN 62353.

The safety-related checks have to be performed **during installation** and at the **intervals** defined in the time schedule.

Time schedule

Activity	Interval (months)	Time required (min)	Comments
General tests	12	165	
Electrical tests			
System	24	30	
MRSC (option)	12	10	
Mechanical tests	24	70	
Function tests	12	210 - 260 ¹	
Annual average including all options is 8.2 hours			

1. The required time depends on the installed options.

Prerequisites

The following service tools and equipment are needed to perform the safety-related tests:

Quantity	Name
1	Loaded body phantom
1	Large spherical phantom with holder
1	Small spherical phantom with holder
1	Service plug if MNO is installed
1	ERDU test load
1	Tamper-proof seal for the magnet stop buttons
1	TALES (every two years)
1	Field service grounding kit (ESD)
1	Set of standard tools
1	Heat gun
1	Flashlight
1	Helium leakage tester

Quantity	Name
1	Smoke wick(p/n 10684293) or test spray p/n 10684091
1	Protective conductor resistance tester e.g. SECUTEST S3 ®, manufacturer Gossen-Metrawatt (recommended)

The following documents are needed to perform the safety-related tests:

Document	Required for
Replacement of Parts > System	Magnet cover dismounting / mounting

Documentation

A protocol of the safety-related tests has to be generated. All fields have to be completed. If a field does not apply to the system or if there is no information to be entered, enter 'n.a.' in the field.

Each test performed successfully has to be confirmed in the test protocol by placing a check mark next to "O.K." Tests that cannot be performed (e.g. options) or which are not performed (e.g. interval is more than one year) should be marked with "n.a." If an error occurs which cannot be eliminated, mark the respective test with "not O.K."

The protocol of the installation "First measurement" serves as a reference document for subsequent repeat measurements. For this reason, it has to be archived by the operator in the System Owner Manual.

The MR quality department needs the protocol for legal reasons. Send a copy of the original to:

Siemens AG
Healthcare IM
MR QP
P.O. BOX 32 60
91050 Erlangen
GERMANY

The protocols of repeat measurements should be filed in the System Owner Manual as well until subsequent safety-related tests have been performed.

We recommend that the country organizations also compile such documentation to ensure optimal operation of products e.g. in the event damages occur.

Abbreviations

Abbrev.	Explanation
SI	Safety Inspection

26.1.5 Maintenance Tests

HP-EH102468-181108-02-200

System Type: SymphonyTim

General

NOTE

These Maintenance Instructions comply with EC guideline CE 93/42/EWG.

WARNING

Please follow the General Safety Notes as well as the MR-Specific Safety Information.

Non-compliance with safety notes may result in death or serious injury.

- ⇒ Carefully read the safety notes in the General Safety Notes as well as the MR-specific Safety Information before starting work. Safety warnings must be observed at all times.

Time frame for the maintenance schedule

Activity	Interval (month)	Time required (min)	Note
Cooling	12	10	
GPA, ACC cabinet	12	10	
Patient fan	12	10	
Gradient filter	12	10	
RCA	12	20	
CCS	12	15	
Magnet system	12	30	
Adsorber	24/36 ¹	70	
TFT Monitor	12	15	
Phantoms	12	15	
RF room door	12	15	
Patient table	12	15	
Option: comfort kit	24	5	
Option: trolley	12	10	

Activity	Interval (month)	Time required (min)	Note
Software	12	30 - 60 ²	
QA	12	70 - 120 ³	Recommended every 3 months
Annual average including all options is 6.1 h			

1. The interval depends on the compressor type (calculation with 24-month interval)
2. Depending on data quantity (calculation with 30 min.)
3. Depending on installed options (calculation with 120 min.)

Activity	Interval (month)	Time required (min)	Note
Displacer replacement	_1	240	If defective, see description in 'Replacement of Parts'
Refilling helium (e.g. 2 x 240 L)	_2	300	Refer to 'Start-up'

1. Interval is variable
2. Interval is variable.

Prerequisites

Service tools or equipment needed for preventive maintenance:

Tab. 5 Preventive maintenance

Quantity	Name	Material number
1	Adsorber - APD HC-10 type (every two years)	7461127
	Adsorber - Leybold Coolpak type (every two years)	4763707
	Adsorber - Sumitomo F70 type (every three years)	7461127
1	Open-end spanner 37 mm (commercially available)	n.a.
1	Set of standard tools	n.a.
1	Bucket 5 to 10 L	n.a.

Quantity	Name	Material number
1	Vacuum pump filter (for optional comfort kit only; every two years)	10496420
2	Batteries (for optional patient trolley only; type Alkaline-Manganese "AA")	n.a.

Service tools or equipment needed for software maintenance:

Tab. 6 Software maintenance

Quantity	Name
1	CD, recordable Note: Use only medical grade CD-Rs, such as Siemens Medical CD-R 80 or TDK Medical CD-R.

Service tools or equipment needed for QA:

Tab. 7 QA

Quantity	Name
1	Phantom setup
1	Service plug if MNO is installed

Documents needed for maintenance:

Tab. 8 Documents

Document	Required for
Replacement of Parts > System	Magnet cover dismounting / mounting

Online Install & Maintenance Protocol

The Install & Maintenance Protocol is integrated into the MR Service Software.

The Install & Maintenance Protocol contains the Installation Protocol, the Safety – related Tests Protocol, and the Maintenance Protocol. For each protocol, an interactive online checklist is provided with a description for each check point. The Install & Maintenance Protocol schedules the next maintenance tasks as well.

You will find the Install & Maintenance Protocol on the Service Software platform under 'Install and Maintenance'. Refer to the Online Help for detailed information regarding the Install & Maintenance Protocol.

PDA maintenance checklist

This information is for countries using the Maintenance Checklist on PDA.

In certain countries, a PDA maintenance checklist is also used. This is not inconsistent. The PDA checklist and "Install&Maintenance" complement each other.

What is to be done on the system?

Complete the safety-related tests and the maintenance record on the system. Save the protocols and generate a backup. The protocols are part of the dynamic data. You also have the option to show, copy or print out the protocol if the customer submits a request to do so.

What is to be done on the PDA?

Choose 'short report' on the PDA. The maintenance task list will not be displayed. Write a note in the PDA checklist, that the test records are successfully completed on the system.

Acronyms:

Abbr.	Explanation
SI	<i>Safety Inspection</i>
SIE	<i>Safety Inspection Electrical Safety</i>
SIM	Safety Inspection Mechanical Safety
PM	Preventive Maintenance
PMP	Preventive Maintenance Preventive Parts Exchange, External Inspection, etc.
PMA	Preventive Maintenance Adjustments
PMF	Preventive Maintenance Function, Operating Value Check
Q	Quality Check
QIQ	Quality Check Image
QSQ	Quality Check System
SW	Software Maintenance

26.1.6 Preview

```
----- HP-EH102468-181108-02-006 -----
System Type: . . . . . Avanto
                  Espree
                  TrioTim
                  SymphonyTim
                  Verio
                  ClinScan
```

Preview provides an overview of the protocols.

- Check to ensure that the protocols are completed in full and are correct.
 - ⇒ If values entered in the protocol are incorrect, they must be corrected at the corresponding 'Install&Maintenance' platform.
- Click the **Save** button to save the protocol. The protocol will be frozen and is then part of the '**Site-specific Data**'.
- Perform '**Backup & Restore**' to save site-specific data.

Use the '**Reports**' platform to print the protocols.

26.1.7 Printing Protocols

HP-EH102468-181108-02-007

Printing protocols

Preconditions: The protocols have to be available, i.e. the **Previews** have been saved.
If there is no printer installed, an external medium (e.g. USB drive) is required.

Printer is installed

Navigate to the 'Install & Maintenance **Reports**' platform.

- Click > **Reports** > SiteInfo > **Install & Maintenance** and select the required protocol.
- Click > **Show as PDF**.
- Click **Print** and the protocol will be printed.
 - ⇒ The HTML protocol will be converted to a PDF protocol.

Printer is not available

With an external media (e.g. USB drive), it is possible to save the protocol on the service laptop or print out the protocol from another computer that has a printer.

NOTE

Please check the external medium first with the virus scanner of the service PC. Be sure that the external medium used is free of viruses.

- Connect a medium to the USB port.
- Navigate to the 'Install & Maintenance **Reports**' platform.
- Click > **Reports** > SiteInfo > **Install & Maintenance** and select the required protocol.
- Click > **Show as PDF**.
 - ⇒ The HTML protocol will be converted to a PDF protocol.
- Click the **Save** icon
 - ⇒ The pop-up window opens ' Save a Copy ...'.
- Navigate to the drive where the medium is connected.
- Save the protocol on the medium.
 - ⇒ Now you can copy the protocol to your service laptop or print out the protocol from another computer that does have a printer.

26.1.8 Archiving Protocols

HP-EH102468-181108-02-008

Archiving protocols

The protocols serve as a quality record of the tests performed.

After completing the test activities, fill out the protocols in full and sign them.

Installation Protocol

The original protocol must be filed in the **System Owner Manual**, which is kept with the system. Also, a copy of the original has to be sent to Erlangen immediately after the installation and TuneUp and after the quality assurance and safety-related tests have been completed.

Send a copy of the original to:

Siemens AG
Healthcare
MR QP
P.O. BOX 32 60
91050 Erlangen
GERMANY

We recommend that the country organizations also compile such documentation to ensure optimal operation of products, on general principles or in case damage occurs.

Safety-Related Test Protocol

If required by country-specific regulations, either the operator or a representative has taken note of the result of the evaluation of the system condition.

The "initial measurement" installation protocol serves as a reference document for subsequent repeat measurements. It must therefore be archived by the operator in the System Owner Manual.

The MR quality department requires the initial protocol for legal reasons. Send a copy of the original to:

Siemens AG
Healthcare
MR QP

P.O. BOX 32 60

91050 Erlangen

GERMANY

The protocols of repeat measurements should also be filed in the System Owner Manual until subsequent safety-related tests have been performed.

We recommend that the country organizations also compile such documentation to ensure optimal operation of products, on general principles or in case damage occurs.

Maintenance Protocol

If required by country-specific regulations, either the operator or a representative has recorded the results for evaluating the system conditions.

The protocol is a document and has to be archived accordingly. After completing the test, it must be filed in the corresponding register in the "System Owner Manual" binder. If required, a copy can be handed to the customer.

We recommend that the country organizations also compile such documentation to ensure optimal operation of products, on general principles or in case damage occurs.

26.2 Procedures

26.2.1 Summary fill in

General

Customer site

HP-EH102468-030206-02-009

Please fill out all input fields.

Tab. 9 Costumer Site Example

Customer/Hospital Name	Filled in automatically
Address	Street ,City
Country	USA
Department	Rad 123
Contact Person	John Q. Public
Phone Number	111 111
Site Modem Number	222 222
Customer Number	333 333

MR System

HP-EH102468-030206-02-010

MR System is the second part of the **General** menu.

System-specific data will be taken over, except for the SAP order number. Please enter the SAP order number.

District Office

HP-EH102468-030206-02-011

Please fill in all input fields.

Tab. 10 District Office - example

SIEMENS Office	Siemens, MR1
Address	Street, Number, City
Region	West 1
Country	Germany
Service Representative	J.Q. Public
Phone Number	222 222

Contract**Contract**

HP-EH102468-030206-02-012

If the customer has a service / maintenance contract, click the **OK** button and enter the contract number and the type of contract in the input fields.

If the customer does not have a service / maintenance contract, click the **Not OK** button.

Measurement Devices

HP-EH102468-050207-02-008

Enter the type and serial numbers of all measurement devices used to install or maintain the system.

Enter the 'Next Calibration Due' date, if requested.

26.2.2 Start-up**Contacts**

HP-EH102468-240907-02-001

Enter the names of the project manager and the sales representative.

Dates

HP-EH102468-210206-02-002

Enter the installation dates.

Document Version

HP-EH102468-210206-02-003

Please enter the complete print number in the corresponding fields. The print number is located at the bottom of the cover page in the folder.

Please follow the syntax of the print number.

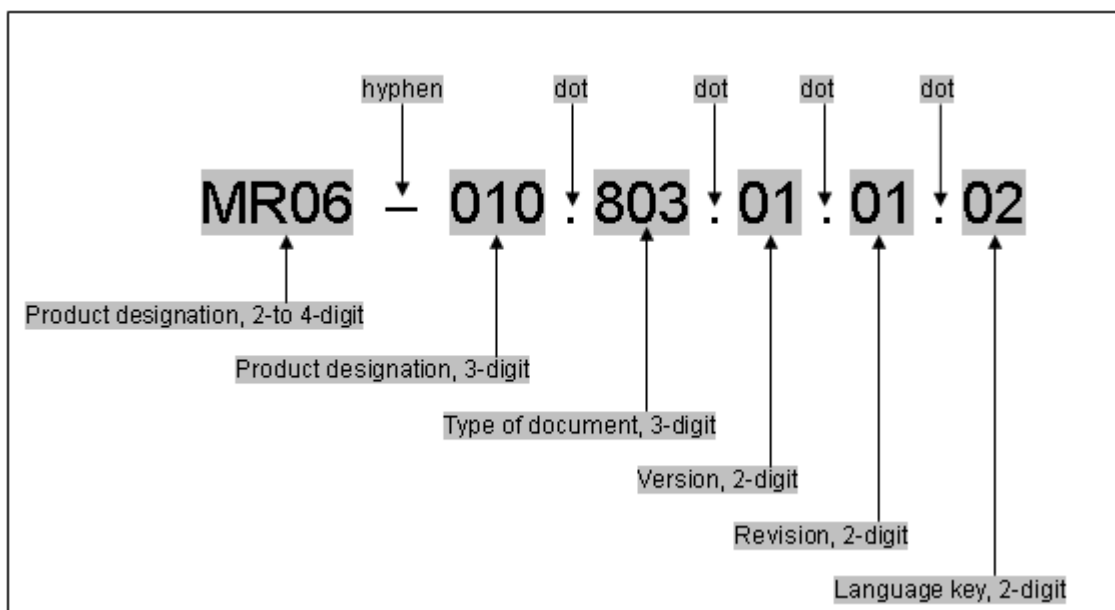


Fig. 404: Syntax print number

Quality Assurance

Magnet Homogeneity

HP-EH102468-210206-02-006

System Type: Avanto
Espre
TrioTim
SymphonyTim
Verio

The value has been taken over from the system.

General QA

HP-EH102468-121012-02-001

System Type:SymphonyTim

Note for using the online 'Install&Maintenance Protocol'

NOTE

The interactive 'Install & Maintenance' protocols take the status from the 'General QA'. If all general QA steps show the 'Done' status, then the 'Install & Maintenance' protocols display the OK status. If one or more steps have the 'ToDo' or 'NotOK' status, then the 'Install & Maintenance' protocols display the 'NotOK' status.

CAUTION: When the Open QA button is pressed, the QA platform opens automatically and all QA steps change to the 'ToDo' status. Please note: the "Open QA" button is only one way to open the QA platform. This notice also applies if the QA platform is opened by different means. The data will be taken over even if you do not use the "Open QA" button.

Quality Assurance (QA)

Required time: 70 - 120 min depending on installed options.

Required tools: QA Phantom Setup, Service Plug (if Multi Nucleus Spectroscopy Option is installed)

The quality measurement of the body coil is a fully automated procedure. The results are entered in the on-site database.

NOTE

Use the Online Help to obtain detailed information regarding, e.g., table control, phantom positioning, or specific items and procedures. Press the 'Help' button in the header bar of the corresponding platform.

Preliminary steps**Starting the service platform**

All actions of the quality assurance procedure are performed via the service platform.

- Start the Service platform > **Options** > **Service** > **Local Service**.

SI General Quality Assurance

Go to the Home menu and press the corresponding button to access the **Quality Assurance** sub platforms.

- Start the measurement by clicking the "Go" button.
 - ⇒ The **General Quality Assurance** check is successfully completed when every individual QA step indicates the status < **done** >.

NOTE

If the results are out of tolerance, perform “Tune-Up” or troubleshooting, if necessary. Repeat “Quality Assurance”.

Coils

HP-EH102468-210206-02-007

System Type:Avanto
Espreo
SymphonyTim
ClinScan

The Coils menu lists all coils that have been connected to system or entered in the configuration menu.

The coil serial number has to be entered in the Serial Number field to add the coil to the installation protocol.

The appropriate quality assurance tests have to be performed before you start printing a new installation protocol.

If you remove a coil from the system or if a coil is no longer available to the system, the serial number must be deleted from the configuration menu so that this particular coil is removed from the installation protocol.

Shoulder Coil

For the shoulder coil, the serial and part numbers of the base carrier have to be recorded. The serial and part numbers of the shoulder cups are not recorded.

Large FoV Adapter

The installation protocol includes a field for each coil used in combination with the Large FoV adapter. These fields should be used to enter the serial and part number of the Large FoV adapter used.

Further Coil Actions

HP-EH102468-080312-02-001

System Type: Espree

Avanto

TrioTim

SymphonyTim

Verio

If a quality assurance test of one or more local coils is out of specification enter in the field the further measures; for instance: head coil out of specification. Defect coil is sent back. Customer / project manager informed and new head coil is ordered; order number / ticked number: 0123456.

Final Checks

The gradient connections at the connection plate on the magnet have been checked successfully

HP-EH102468-010310-02-004

System Type: SymphonyTim
Avanto
Espree
TrioTim
Verio

- Ensure that the gradient connections at the connection plate on the magnet have been checked successfully during start-up, please refer to the 'Start-up' document.
 - ⇒ The gradient connections at the connection plate on the magnet have been checked successfully - click '**OK**'.

The system **cannot be turned over** if "**Not OK**" has been checked.

- If corrective measures were necessary, enter these measures in the comment field (for example: X and Y connection were interchanged, corrected, washer replaced, torque verified, etc.).

The gradient connections at the filter plate (examination room side) have been checked successfully

HP-EH102468-010611-02-001

System Type: SymphonyTim

Avanto

Espreo

TrioTim

Verio

- Ensure that the gradient connections at the filter plate (examination room side) have been checked successfully during start-up, please refer to the 'Start-up' document.
 - ⇒ The gradient connections at the filter plate (examination room side) have been checked successfully- click 'OK'.

The system **cannot be turned over** if "Not OK" has been checked.

- If corrective measures were necessary, enter these measures in the comment field (for example: X and Y connection were interchanged, corrected, washer replaced, torque verified, etc.).

The gradient connections at the filter plate (equipment room side) have been checked successfully

HP-EH102468-010611-02-002

System Type:SymphonyTim

Avanto

Espreo

TrioTim

Verio

- Ensure that the gradient connections at the filter plate (equipment room side) have been checked successfully during start-up, please refer to the 'Start-up' document.
 - ⇒ The gradient connections at the filter plate (equipment room side) have been checked successfully - check off '**OK**'.

The system **cannot be turned over** if "**Not OK**" has been checked.

- If corrective measures were necessary, enter these measures in the comment field (for example: X and Y connection were interchanged, corrected, washer replaced, torque verified, etc.).

HP-EH102468-010611-02-003

Avanto

TrioTim

Espre

Verio

- The system **cannot be turned over** if "Not OK" has been checked.

- If corrective measures were necessary, enter these measures in the comment field (for example: X and Y connection were interchanged, corrected, washer replaced, torque verified, etc.).

Installation, start-up and tune-up of the system is completed

HP-EH102468-210206-02-008

System Type:Avanto
Espreo
TrioTim
SymphonyTim
Verio

Ensure that the installation, start-up and tune-up of the system have been completed.

- Check off 'OK'.

The system **cannot be turned over** if "Not OK" has been checked.

Installation, start-up and tune-up of all options is completed

HP-EH102468-210206-02-009

System Type: Avanto
 Espreo
 TrioTim
 SymphonyTim
 ClinScan
 Verio

Ensure that the installation, start-up and tune-up of all options have been completed.

- Check off 'OK'.

Options **cannot be turned over** if "Not OK" has been checked.

The safety related tests were performed successfully

HP-EH102468-210206-02-011

System Type:Avanto
Espreo
TrioTim
SymphonyTim
ClinScan
Verio

- Ensure that '**Safety-related Tests**' have been completed successfully.
- Check off '**OK**'.

The system **cannot be turned over** if "Not OK" has been checked.

Send back the following signed documents to the MRQ Department

HP-EH102468-180713-02-001

System Type: Avanto
 Espreo
 TrioTim
 SymphonyTim
 Verio

- Start-up Report
- Safety Related Test Report
- Quench Line Acceptance Protocol
- Pictures and sketch of magnet room

Tab. 11 Address MR Quality Department

Siemens AG
 Healthcare
 MR Quality Department
 P.O. BOX 3260
 91050 Erlangen
 Germany

mail to: immrqpininstallations.healthcare@siemens.com

Start-up Report

The 'Start-up Report' is the final test certificate for the system. The completed 'Start-up Report' is our verification that the system including all coils and options was properly installed and is operating according to manufacturer's specifications.

Safety Related Test Report

The protocol of the installation "First measurement" serves as a reference document for subsequent repeat measurements.

The MR quality department needs the protocol for legal reasons.

Quench Line Acceptance Protocol

Siemens as the manufacturer of MRI systems has the obligation to verify that risk mitigation measures - such as the correct installation of the quench line – are implemented and prove the verification by checking (and archiving) written evidence of the installation procedure.

The Quench Line Acceptance Protocol has to be filled out completely and signed by the project manager. Please refer to the Planning Guide. Collect the Quench Line Acceptance Protocol from project manager and send it to the MR Quality Department.

Pictures and sketch of magnet room

A minimum of 4 pictures have to be prepared during the start-up, prior the first ramping. The pictures and the sketch of the magnet room have to be attached to the start-up report. Please refer to the start-up procedure.

Installation and start-up of the gradient supervision is completed. The function tests of the gradient supervision were performed successfully.

HP-EH102468-130711-02-010

System Type:Avanto
Espreo
TrioTim
SymphonyTim
Verio

- Ensure that the installation, start-up and test of the gradient supervision have been completed successfully, please refer to the 'Start-up' document.
 - ⇒ The installation, start-up and test of the gradient supervision have been completed successfully, check off 'OK'.

The system **cannot be turned over** if "Not OK" has been checked.

26.2.3 Safety Related Tests

Measurement Type

HP-EH102468-210206-02-013

System Type: Avanto
Espre
TrioTim
SymphonyTim

The protocol of the “initial measurement” installation serves as a reference document for subsequent repeat measurements. For this reason, it has to be archived by the operator in the System Owner Manual.

First protocol during installation, click > **Initial Measurement**

Repeat protocol during service / maintenance, click > **Repeat Measurement**

General Tests

System

Visual inspection of system

HP-EH102468-060206-02-001

System Type:Avanto
Espreo
TrioTim
SymphonyTim
Verio

System

Required time: 20 min

Required tools: n.a.

SI Visual inspection of system

- Perform the general visual inspection of the system.
 - ⇒ The system may not contain damages that impair safety.
 - ⇒ There must be no undefined parts inside the magnet room. They could be magnetic. Call the user to clarify before any movement of those parts.
- Correct visible defects with respect to safety prior performing further service works.

SI Inspecting visible cables and cable guides

- Check visible cables and cable guides for damage.

Inspecting visible cables and cable guides

HP-EH102468-060206-02-002	
System Type:	Avanto
	Espreo
	TrioTim
	SymphonyTim
	Verio

SI Inspecting visible cables and cable guides

- Check visible cables and cable guides for damage.

Visual inspection of the quench tube

HP-EH102468-110608-02-001

System Type: Espree
 Avanto
 TrioTim
 SymphonyTim
 Verio

Quench tube

Quench tube inspection

Required time: 40 min

Required tools: If missing from the quench tube the following spare parts are available from the SPC and must be fitted:-

- Adhesive tape for Quench tube.
- Quench outlet warning sign.

SI Visual inspection of the quench tube

- Visually inspect the condition of the quench tube.

Internal quench tube inspection

Check the following (visible inspection from the ground only, it is not necessary to remove ceiling tiles etc):

- Check that no unauthorized changes to the design of the quench tube **have been made**, both inside or outside the RF room.
- Check the **quench tube** and **insulation** for **damage**.
- Check that the **connection screws** to the magnet quench valve are all fitted and tight.
- Check the warning **label or tape** is attached to the **tube insulation**.
 - **On long tube sections** the warning labels or tape should be at regular distances of **between 2 - 3 meters**.



Fig. 405: Warning label example- Vent tube



Example of new warning tape supplied with magnets. This will provide a consistent warning message.

Fig. 406: Warning tape- Quench tube

External quench tube inspection

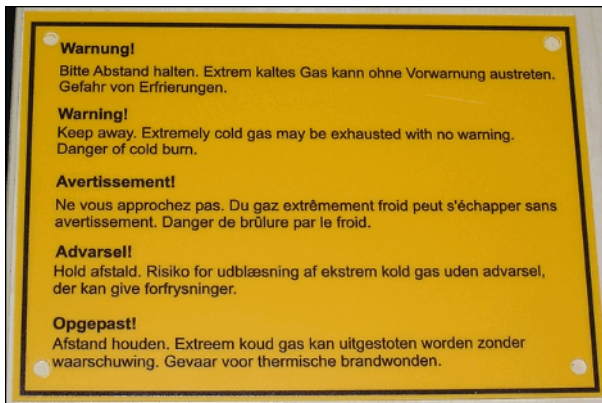
NOTE

Warning Signs

The outlet warning sign must be in English and also the National language.



Fig. 407: Warning sign example - Quench tube outlet



Example of a new warning sign that provides a consistent message in several languages.

Fig. 408: Warning sign example - Quench tube outlet

- Check the **warning sign** is fixed and clearly visible next to the **outlet** of the quench tube.
- Check the **outlet** of the quench tube for **obstructions** (e.g. bird's nest).
- Check for a **protective mesh** on the **quench tube outlet** and its integrity.
- Check a **protective rain cover** is fitted to the **vertical outlets**.
 - The cover must not be **damaged** or restrict the gas flow from the quench tube.

The **outlet** tube cannot be situated where, in case of a quench:-

- Helium gas can be drawn into an **air inlet**.
- Helium gas can enter **open windows**.

To avoid risk of **injury** from cold burns and asphyxiation, access to the quench tube must be restricted by:-

- 3m on each side and below. ([Quench tube - example of venting to the outside](#))
- 6m vertically above the exit.

NOTE

Helium is lighter than air, even when very cold. The cold gas must not be allowed to blow directly onto a window.

- Check that the restricted area has not changed.
- Check that **venting gas** does not represent a hazard to people. For example:-
 - Subsequent building of windows
 - Air conditioning system intake or exhaust pipes
 - Erection of new or temporary buildings
 - Temporarily storing containers.
- If windows are present in the restricted access area, they must be **sealed** and **permanently closed**. Means of opening the windows must be **removed**.
- Inform the operator to initiate corrective action **immediately** in case you find obvious **defects** or problems which could restrict the flow of gas from the quench **tube**.

- Record your findings in the **Protocol**.

NOTE

More information about quench tube specification is available in the Site Planning Guide in the Quench Tube section.

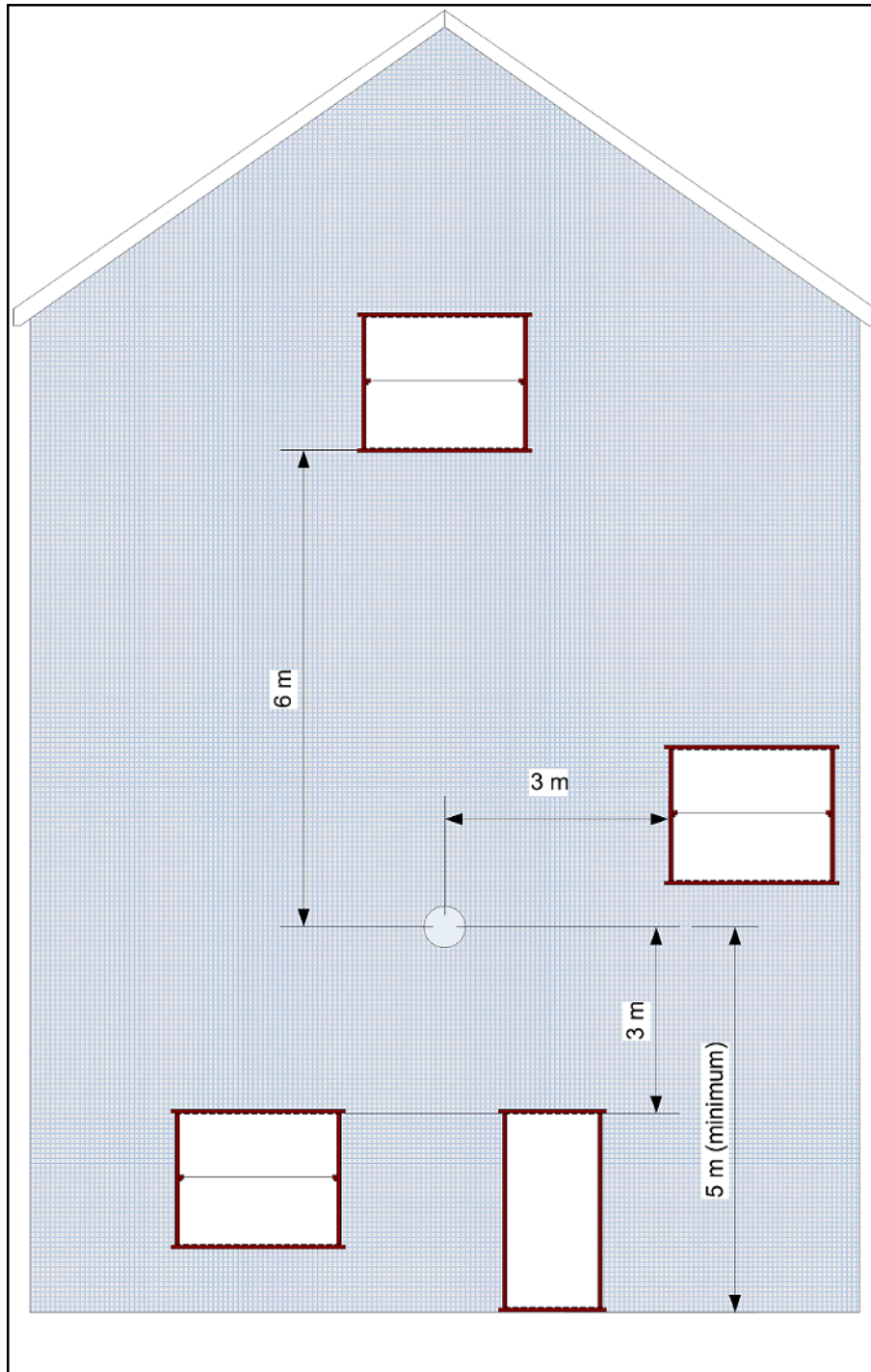


Fig. 409: Quench tube - example of venting to the outside

Check for water in the quench valve

HP-EH102468-060206-02-011

System Type:SymphonyTim

Quench valve

SI Check for water in the quench valve

1. Remove the turret cover.
2. Remove the drip tray screws (2/Quench valve).
3. Remove the drip tray

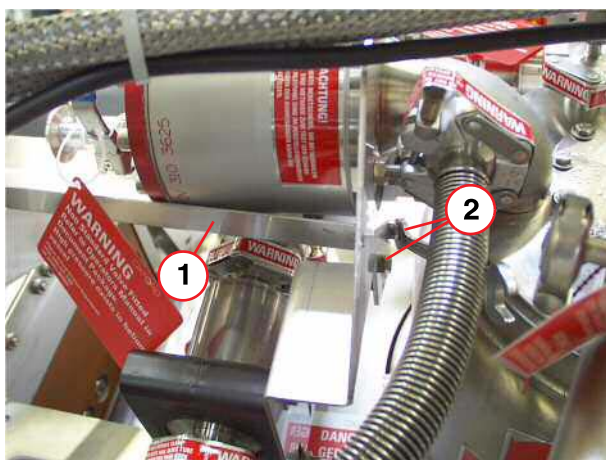


Fig. 410: Quench valve

Pos. 1 Drip tray
Pos. 2 Screws 2 x 10 mm

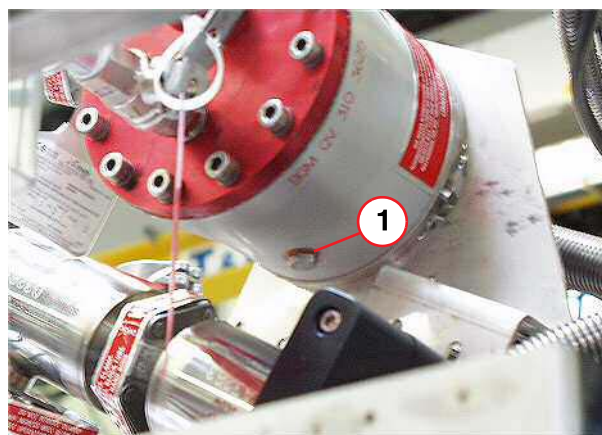


Fig. 411: Quench valve

Pos. 1 Drain screw

- Open the 13 mm drain screw and check for water (1/Quench valve).
 - ⇒ There must not be any water in the quench valve.
 - ⇒ In case there is water, look for the cause (e.g. damaged, missing, or improperly constructed protective rain cover or missing tube insulation). Inform the customer to initiate corrective actions immediately.

Checking the magnet pressure values

HP-EH102468-140711-02-005

System Type: SymphonyTim

Magnet pressures

There are 2 different overpressure valves for the magnet venting system.

Version A: Older systems have 2 1/3-psi overpressure valves in-line.

⇒ Nominal operating pressure 2/3 psi (green field).

Version B: Newer systems with marked revision state 31 are equipped with an 0.5-psi overpressure valve.

⇒ Nominal operating pressure 0.5 psi (green field).

SI **Checking the magnet pressure values**

- Check the magnet pressure at the pressure gauge on the service turret.

 CAUTION

There must always be a positive pressure in the system.

Internal “icing” of the magnet may result, causing the magnet to quench and the helium vessel to rupture.

⇒ **Follow the leakage check procedures during all service operations.**

- Check for leaks if the pressure is low.

If the system pressure remains at zero but there is no apparent leak, check the quench valve integrity. “Refer to the Troubleshooting Guide”

Check for ice formation at the magnet service turret and venting system

HP-EH102468-060206-02-018

System Type:TrioTim
SymphonyTim

Ice formation check

- ⇒ In general, the service turret and venting system must be free of ice formation (except after ramping up or filling).
- Empty out the water from the condensation reservoir located at the electronics side of the magnet.

SI Check for ice formation at the magnet service turret and venting system

NOTE

Note for Trio A Tim System

Always use the magnet service platform when working on the service turret. For correct installation and usage of the service platform, refer to the ‘Magnet Troubleshooting Guide > Service Platform Installation’

⚠ CAUTION

Protective clothing, gloves, and eye protection must be worn during the following procedure.

Personal injury may occur.

⇒ Only trained personnel shall perform the following.

- Depressurize the magnet if necessary. (Do not forget to close the bypass valve after depressurizing).
- Remove the DCL blanking plug (baffle assembly) and replace it with the inspection window.
- Secure the window with the ring nut (hand tighten only)
- Allow the mist to clear and inspect the neck for ice, using a non-magnetic flashlight.
 - ⇒ If the DCL socket is clearly visible, possibly with a light frosting of ice, the system is not considered to be iced (DCL connector visible = OK). =ok



Fig. 412: DCL connector visible = OK

- ⇒ If the DCL socket is not visible, the system is iced ((Too much ice = not OK) and (Too much ice = not OK)) = not ok. The ice will have to be removed to ensure continuous safe operation of the magnet system. **If the system is iced, escalate the problem immediately to your responsible service center (USC, HSC) so that you get the magnet de-iced as soon as possible.**



Fig. 413: Too much ice = not OK



Fig. 414: Too much ice = not OK

CAUTION

If the inspection glass has been in place for a considerable time, some pressure may have built up in the system.

Personal injury may occur.

⇒ Take care when removing the inspection glass, and de-pressurize if necessary.

- De-ice and dry DCL blanking plug (baffle assembly) using a heat gun.
- Remove the inspection glass and then replace the DCL port blanking plug without delay.
 - ⇒ A new copper seal is must be used when replacing the DCL port blanking plug.
- Check for leaks on the port after the system has pressurized.
- If there was ice in the system, check for leaks on the entire turret (Leak Check).
- Ensure all 25-mm clamps at the service turret are tight
- Write down your findings in the SRT Protocol.

Magnet system leak check

HP-EH102468-090206-02-001

System Type: SymphonyTim
TrioTim

Leakage check

SI Magnet system leak check

An electronic helium leak tester should be used to search for leaks around all of the joints of the service turret. Follow the manufacturer’s instructions for correct use of the electronic detector.

The leak rate using an electronic helium leak tester shall not exceed 1 x 10⁻⁴ mbar liter/sec.



WARNING

Observe caution when using an electronic leak detector with the magnet at field.

Accidents to personnel and equipment damage will occur.

⇒ **Check for magnetic parts before use.**

NOTE

Note for Trio A Tim System

Always use the magnet service platform when working on the service turret. For correct installation and usage of the service platform, refer to the ‘Magnet Troubleshooting Guide > Service Platform Installation’

NOTE

The magnet must be run to Zero before checking inside the quench valve.

- The turret, quench valve and venting system must be warmed to room temperature and dried before commencing the leak check. Remove the condensation and water with paper.
- The specific areas for leakage check are:
 - Clamps and seals on vent lines
 - HDCL and syphon seals
 - System pressure gauge
 - Turret top plate
 - Syphon end cap
 - The electrical turret connections
 - Flexible tubing

- Where leaks are detected do the following:
 - Tighten or replace clamps, if damaged.
 - Check that the correct type of seals is fitted.
 - Check seals faces for damage. Replace, if damaged.
 - Tighten the syphon end cap and the central retaining screw for the chain.
 - Tighten syphon/HDCL seals. If leakage persists, replace the copper seals.
 - Check quench valve burst disc and O-ring. Replace, if necessary.
 - Make sure that bolted flanges are currently torqued.
 - Seals covered in vacuum grease must be cleaned and inspected. Refit the seal or O-ring without using vacuum grease.

If problems persist with leaks at the turret or venting, contact your local Uptime Service Center.

NOTE

Do not use any form of vacuum grease on O-ring or seals.

It is recommended that O-ring or seals wiped clean and checked for damage prior to fitting. Vacuum grease dries over time and can cause a seal to leak.

NOTE

Replace used high load seals.

The high load seals (copper (CU) or Heli-flex types) must be replaced with a new one each time the joint is broken.

Metric torque settings

Recommended torque settings for specific bolt sizes must be used. Overtightening of bolts can cause leaks and reduce the life of seals.

Use these torque settings for any bolts on the system turret, particularly when replacing the quench valve burst disk.

Tab. 12 Torque settings

Nominal bolt diameter	4	5	6	8	10	12
Torque setting (NM)	2.1	6.3	7.2	17.6	35	61

User documentation available and legible

HP-EH102468-130206-02-002

System Type: Avanto
Espreo
TrioTim
SymphonyTim
Verio

User documentation

Required time: 5 min
Required tools: n.a.

SI User documentation available and legible

- Ensure that the System Manual is available, legible and valid (e.g. software version).

Checking user icons and button labels

HP-EH102468-130206-02-003

System Type:Avanto
Espreo
TrioTim
SymphonyTim
Verio

User icons, button labels**Required time:** 5 min**Required tools:** n.a.**SI Checking user icons and button labels**

- Inspect the button labeling and user icons for legibility/recognizability (for example “symbol keypad”, “**Magnet Stop button**”, “Patient Table Stop”, etc.).

Identification of the controlled access area (0.5 mT area)

HP-EH102468-130206-02-008

System Type: SymphonyTim

Identification of the controlled access area

Identification of the controlled access area (0.5 mT area)

Required time: 10 min

Required tools: n.a.

Refer to the Spare Parts Catalogue for the order number for the set of warning signs.

NOTE

On newer Systems (since 09 / 2010) a Tesla label (T) with the corresponding field strengths of the system is affixed at the Warning sign "Strong Magnetic Field".

SI Identification of the controlled access area (0.5 mT area)



Fig. 415: Warning sign "Strong Magnetic Field"

NOTE

Customer signs can be used also, but must cover the same safety aspects as the Siemens signs.



Fig. 416: Warning sign "Cardiac Pacemaker"

Prior to initial Startup, the project manager defines the 0.5 mT area.

The warning sign “cardiac pacemaker” is, e.g., for identifying the 0.5mT line. The warning sign “strong magnetic field” is, e.g., for identifying the entrance doors into the examination room or the 0.5mT line.

- Ensure that the warning signs identifying the controlled access area (0.5 mT zone) have been posted at the entrance to the examinationroom as well as all other areas leading to the 0.5 mT zone. The signs have to be posted in a way that it is visible even when the door to the controlled area is opened.

SI SIEMENS warning signs installed

SI Customer signs installed (customer signs cover all safety aspects as listed on SIEMENS signs)

The fringe field extends three-dimensional beyond the magnet and can be reduced through shielding. The field lines represent the ideal flux density distribution in air, which may be distorted by steel reinforcement in buildings.

Fringe field distribution diagram

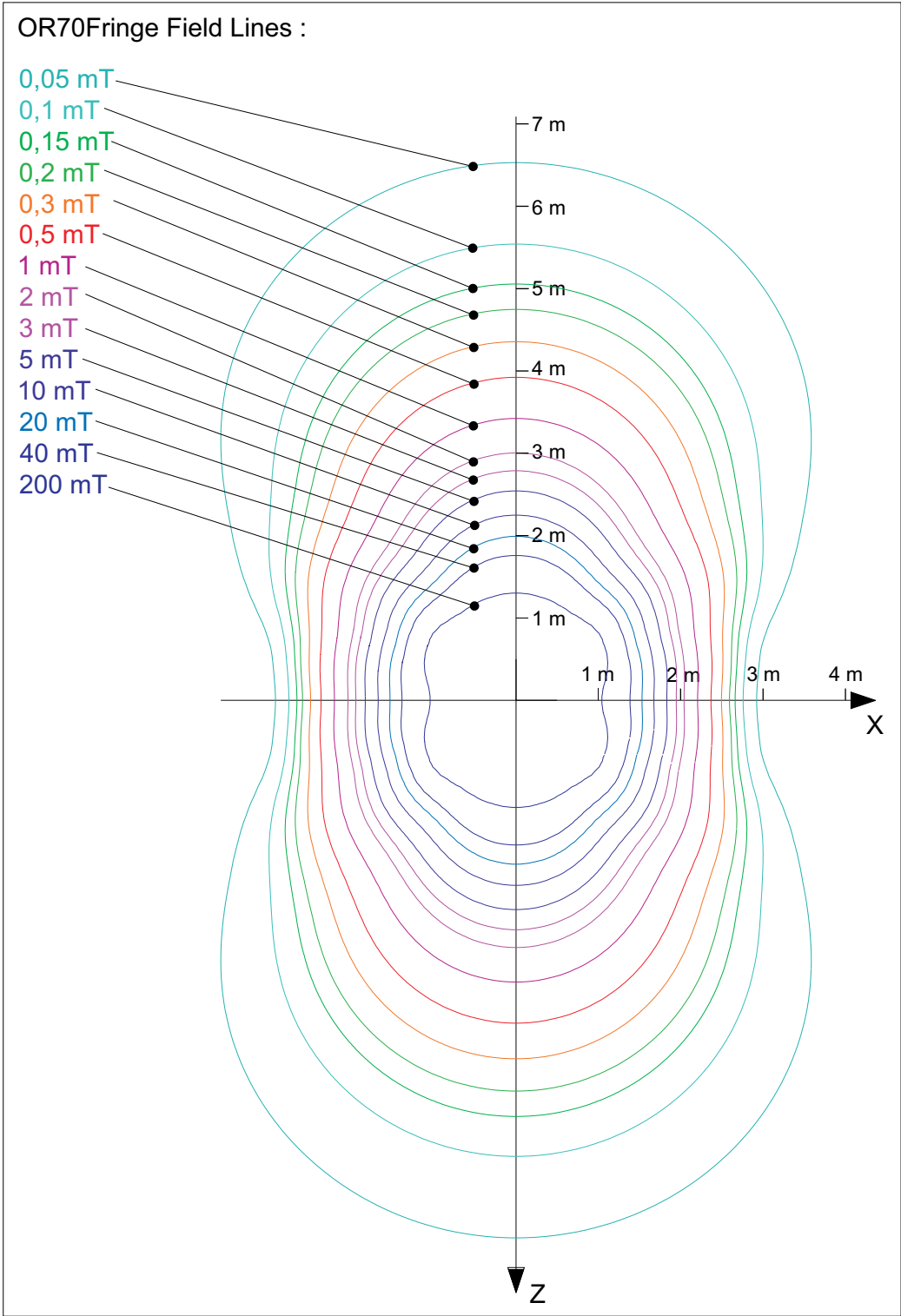


Fig. 417: Magnet fringe field

Checking the laser warning labels

HP-EH102468-140206-02-001

System Type:Avanto
Espreo
TrioTim
SymphonyTim

Laser warning labels

Required time: 5 min

Required tools: n.a.

SI Checking the laser warning labels

- Ensure that the correct warning labels are affixed to the cover.
If the option “control unit magnet rear” is installed, ensure that the correct warning labels are affixed to the cover.
⇒ The labels must be legible.
- Replace missing labels immediately.



Fig. 418: e.g. Avanto magnet cover / laser label

Label configuration for the US Certification label, warning logo type and aperture label



Fig. 419: Laser label configuration USA

Pos. 1 Aperture label, warning logo type

Pos. 2 Certification label

Label configuration for all other countries in local language or alternative language.

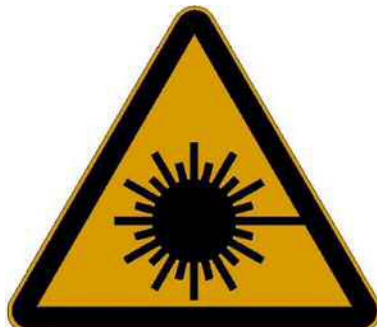


Fig. 420: Laser label



Fig. 421: Laser label 1 (new version)



Fig. 422: Laser label 1 (old version)

Checking the warning signs of the patient table

HP-EH102468-170114-02-002

System Type: SymphonyTim

Warning signs of the Patient Table

Required time: 5 min**Required parts and tools:** Sign (Warning sign 'bruising hazard') if missing. Sign is part of the label set part number 7546430

Fig. 423: Warning sign 'bruising hazard'

SI Checking the warning signs of the patient table

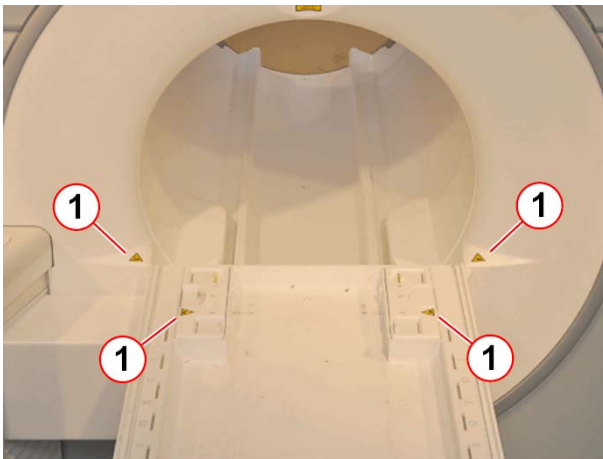


Fig. 424: Patient table

Pos. 1 Warning sign 'bruising hazard'

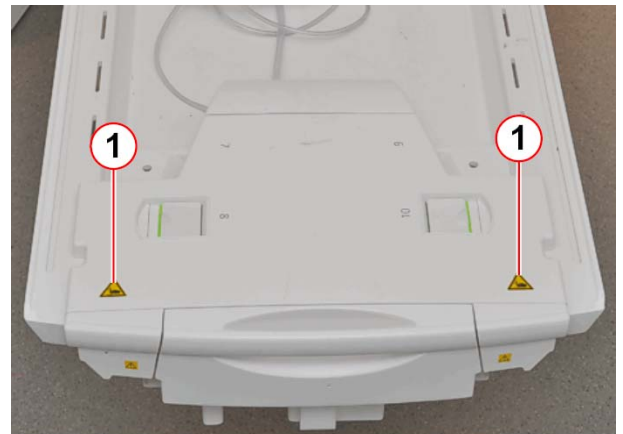


Fig. 425: Patient table

Pos. 1 Warning sign 'bruising hazard'

- Ensure that the warning labels are affixed to the patient table and to the magnet bore .
- Replace missing labels immediately.

Checking the hearing protection sign

HP-EH102468-140206-02-002

System Type:Avanto
Espreo
TrioTim
SymphonyTim
Verio

Hearing protection sign

Required time: 5 min

Required tools or parts: if missing hearing protection sign. The hearing protection sign is part of the warning sign sets, please refer to SPC.

SI Checking the hearing protection sign

- Ensure that the hearing protection sign has been posted at all entrances to the examination room.



Fig. 426: Hearing protection sign

- Install missing signs immediately

Coils

HP-EH102468-060206-02-005

System Type: Avanto
Espreo
TrioTim
SymphonyTim
Verio

Surface coils

SI Visual inspection of the surface coils

- Check housing, cables, and connectors.
- Cables and connectors must not be damaged.
- Remove dust or metal shavings from inside/outside the connectors.

In order to document which coils have been inspected enter the material and the serial number of the inspected options in the Safety-Related Tests Report (option and accessory list).

Options / Accessories

Satellite Console

HP-EH102468-060206-02-003

System Type:Avanto
 Espree
 TrioTim
 SymphonyTim
 Verio

Options and accessories

Required time: 20 min

Required tools: n.a.

SI Visual inspection of options and accessories

- Perform the visual inspection of the options and accessories.
- In conjunction with the operator / user, determine whether additional options / accessories (e.g., a trolley from an old system) are in use.
 - ⇒ Options and accessories may not have damages that impair safety.
- Correct visible defects with respect to safety prior to performing further service works.

In order to document which options have been inspected enter the material and the serial number of the inspected options in the Safety-Related Tests Report (option and accessory list).

Patient trolley (optional)

SI Visual inspection of patient trolley

- Thoroughly inspect the trolley frame, the wheels, interlocking mechanism, the brake system and the hydraulic system for damage, and take corrective action, if necessary.

In order to document which trolley has been inspected enter the material and the serial number of the inspected options in the Safety-Related Tests Report (option and accessory list).

Surface coils

SI Visual inspection of the surface coils

- Check housing, cables, and connectors.
- Cables and connectors must not be damaged.
- Remove dust or metal shavings from inside/outside the connectors.

In order to document which coils have been inspected enter the material and the serial number of the inspected options in the Safety-Related Tests Report (option and accessory list).

Trolley

HP-EH102468-060206-02-004

System Type: Avanto
Espreo
TrioTim
SymphonyTim
Verio

Options and accessories

Required time: 20 min

Required tools: n.a.

SI Visual inspection of options and accessories

- Perform the visual inspection of the options and accessories.
- In conjunction with the operator / user, determine whether additional options / accessories (e.g., a trolley from an old system) are in use.
 - ➡ Options and accessories may not have damages that impair safety.
- Correct visible defects with respect to safety prior to performing further service works.

In order to document which options have been inspected enter the material and the serial number of the inspected options in the Safety-Related Tests Report (option and accessory list).

Patient trolley (optional)

SI Visual inspection of patient trolley

- Thoroughly inspect the trolley frame, the wheels, interlocking mechanism, the brake system and the hydraulic system for damage, and take corrective action, if necessary.

In order to document which trolley has been inspected enter the material and the serial number of the inspected options in the Safety-Related Tests Report (option and accessory list).

Surface coils

SI Visual inspection of the surface coils

- Check housing, cables, and connectors.
- Cables and connectors must not be damaged.
- Remove dust or metal shavings from inside/outside the connectors.

In order to document which coils have been inspected enter the material and the serial number of the inspected options in the Safety-Related Tests Report (option and accessory list).

Electrical Tests

Testing ground wire (system)

HP-EH102468-110608-02-011

System Type:SymphonyTim

Note for 'Install&Maintenance' protocol

NOTE

First enter the measured values in the table.

Then click the OK button.

And then the Save button.

If you switch to another menu without saving your input, it will be lost.

Measurement of the system ground wire (every 2 years)

Required time: 30 min

Required tools: Ground test meter

Before starting the test, switch off the entire system. Ensure that it cannot be switched on inadvertently during the procedure, and check that the ground wires are connected correctly.

Before performing the protective conductor test, read the operator manual for your measurement device. For some measurement devices, the resistance of the measurement cable must be subtracted from the measured results!

WARNING

The ground test meter is ferromagnetic! Do not bring it into the examination room while the magnet is at field!

Failure to comply may cause the ground test meter to become a projectile due to the magnetic force, resulting in serious injury or even death.

⇒ **Perform protective conductor tests using measurement cables of a sufficient length or after the magnet has been ramped down.**

⇒ The protective conductor resistance must not exceed **0.3 ohm**.

- Use the **ground test meter** and measure the **ohmic resistance** between the ground wire bar in the on-site power distribution cabinet and a non-insulated metallic part of the:
- ACC MR electronics cabinet (ground bolt - power distribution box)

SI ACC - line distribution box ground bolt (limit value $\leq 0.3 \Omega$)

Tab. 13 Measured value

Measured value:

- RF-filter plate (ground bolt)

SI RF-filter plate - ground bolt (limit value $\leq 0.3 \Omega$)

Tab. 14 Measured value

Measured value:

- Option RCA (metal frame)

SI RCA - metal frame (limit value $\leq 0.3 \Omega$)

Tab. 15 Measured value

Measured value:

- Magnet cold head compressor (housing)

SI Magnet cold head compressor - housing (limit value $\leq 0.3 \Omega$)

Tab. 16 Measured value

Measured value:

- Magnet (magnet housing)

SI Magnet - magnet housing (limit value $\leq 0.3 \Omega$)

Tab. 17 Measured value

Measured value:

- RF components on the magnet (mounting plate)

SI RF components on the magnet - mounting plate (limit value $\leq 0.3 \Omega$)

Tab. 18 Measured value

Measured value:

- Patient table (screw, table underside)

SI Patient table - screw for table underside (limit value $\leq 0.3 \Omega$)

Tab. 19 Measured value

Measured value:

- MRC host (housing screw)

SI MRC-Host - housing screw (limit value $\leq 0.3 \Omega$)

Tab. 20 Measured value

Measured value:

- MRC TFT monitor (housing screw at the bottom ([TFT monitor](#)))

SI MRC TFT monitor - housing screw at the bottom (limit value $\leq 0.3 \Omega$)

Tab. 21 Measured value

Measured value:

- Intercom operating unit (housing screw)

SI Intercom operating unit -housing screw (limit value $\leq 0.3 \Omega$)

Tab. 22 Measured value

Measured value:

- Intercom (housing screw)

SI Intercom - housing screw (limit value $\leq 0.3 \Omega$)

Tab. 23 Measured value

Measured value:

- Power distribution for the console components (ground screw)

SI Power distribution for the console components - ground screw (limit value $\leq 0.3 \Omega$)

Tab. 24 Measured value

Measured value:

- All other options such as **UPS**, **patient monitor**([TFT monitor](#)), **In-room MRC**, **Digi-cam**, etc.

SI Option 1 (limit value $\leq 0.3 \Omega$)

Tab. 25 Description, material and serial number

Description:	
Material No.:	
Serial No.:	

Tab. 26 Measured value

Measured value:

SI Option 2 (limit value $\leq 0.3 \Omega$)

Tab. 27 Description, material and serial number

Description:	
Material No.:	
Serial No.:	

Tab. 28 Measured value

Measured value:

SI Option 3 (limit value $\leq 0.3 \Omega$)

Tab. 29 Description, material and serial number

Description:	
Material No.:	
Serial No.:	

Tab. 30 Measured value

Measured value:

NOTE

Enter N.A. in the measured value field if the component / option is not present.

- Enter the values measured for each component in the Protocol.

⚠ WARNING

Exceeding threshold values!

In case of electrical fault, exceeding a threshold value may result in death or serious injury.

⇒ Test to determine what exceeds the threshold value. Implement actions immediately to reduce the value prior to continuing work.



Fig. 427: TFT monitor

Pos. 1 Measuring point

NOTICE**Option: In-room MRC**

Do not touch the trackball with the probe of the ground test meter.

The high current used for measuring the resistance may damage the small contact springs of the trackball during the measurement.

⇒ **To perform the measurement, use the uncoated screws at the back of the top cover of the In-Room MRC.**

Tab. 31 Maintenance performed

Maintenance of this section was performed by:

Signature: _____

Date:

Name:

Testing ground wire (MRSC Option)

HP-EH102468-140206-02-011

System Type: Avanto
Espre
TrioTim
SymphonyTim

Measurement of the MRSC Protective Conductor

NOTE

First enter the measured values in the table.
Then click the OK button.
And then the Save button.
If you switch to another menu without saving your inputs, they will be lost.

Measurement of the MRSC ground wire (annually)

Required time: 10 min
Required tools: Ground test meter

Tab. 32 Component is option: the section always has to be concluded with the -Maintenance performed- table

Option present: Yes: No:

Signature: _____

Date: Name:

Use the ground test meter to measure the ohmic resistance between the ground pin at the plug of the power distribution for the MRSC components and a non-insulated metallic part of the:

- MRSC host (housing screw)
- SI MRSC Host - housing screw (limit value ≤ 0.3 Ω)**

Tab. 33 Measured value

- Measured value:
- TFT monitor (housing screw at the bottom)
- SI TFT monitor housing screw at the bottom (limit value ≤ 0.3 Ω)**

Tab. 34 Measured value

- Measured value:
- Option desktop MOD, external

SI Option external desktop MOD (limit value $\leq 0.3 \Omega$)

Tab. 35 Measured value

Measured value:

⇒ The protective conductor resistance must not exceed **0.3 ohm**.

- Enter the values measured for each component in the Safety-Related Tests Protocol.

Tab. 36 Maintenance performed

Maintenance of this section was performed by:

Signature: _____

Date:

Name:

Mechanical Tests

Replacing TALES

HP-EH102468-140206-02-018

System Type: SymphonyTim

TALES

Required time: approx. 70 min

Required tools and parts: Screwdriver, ESD kit, TALES, Service Plug (if multi nucleus spectroscopy option is installed)

Prerequisite: Service access covers are removed, system is switched off

SI Replacing TALES (every 2 years)

Tab. 37 Replacement Date

Startup Date:

Date of Last Replacement: DD MMM YYYY

NOTE

Do not replace the assembly for a new installation
The initial replacement in this case occurs in 2 years.



Fig. 428: TALES

Preliminary steps

1. Switch the system to standby mode by selecting **System > Control** in the main menu.
2. Click the task card MR Scanner.
3. Click **stand-by**.
4. Switch off ACC/F24.
5. Remove the left service access cover from the magnet.

Disassembling / Assembling



1. Disconnect all cables from the TALES.
2. Remove the old TALES and install a new one (use ESD kit for electrostatically sensitive devices).
3. Reconnect all cables. Firmly tighten the N- connectors by hand (1 - 2 Nm).
4. Switch on ACC/F24.
5. Click **System > Control** in the main menu.
6. Click the task card MR Scanner.
7. Click **System ON**.

Adjustments:

1. Start the Service platform.
2. Open the Tune-up platform by pressing the corresponding button.
3. Perform the following Tune-up steps:
 - a) ***Tuning Calibration***
 - b) ***BC Tuning***
 - c) ***RF Characteristic***
 - d) ***Coil Power Losses***
 - e) ***Abs. Rec. Path Calibration***
 - f) ***RF Characteristic LC - if multi-nucleus spectroscopy is installed (optional)***

NOTE

Use the Online Help to obtain detailed information regarding, e.g., table control, phantom positioning, or specific items and procedures. Press the 'Help' button in the header bar of the corresponding platform.

Final checks:

1. Open the Quality Assurance platform by pressing the corresponding button.
2. Perform the following Quality Assurance steps:
 - a) **Coil Check**
 - b) **RF Verify**
 - c) **RFVerify LC - if multi-nucleus spectroscopy is installed (optional)**
 - d) **Coil Power Losses**Check
 - e) If the system has a local TX coil installed, perform QA **Coil Check** for this specific coil.
 - f) **Expert mode > RF Path Attenuation Check**
3. Label the return spare part "Tales": safety maintenance."
4. Return the replaced TALES to Erlangen.
5. Install the left service access cover.

Function Tests

Emergency shutdown buttons

 HP-EH102468-211108-02-006

System Type:SymphonyTim

Emergency buttons

Required time: 15 min

Required tools: n.a.

SI Emergency shutdown buttons

Prerequisites:

The system contactor is switched on.

The system is not booted (system shutdown or press **F2** during boot).

- Press the **emergency shutdown button** (EPO).
- Check that the system contactor in the electronics cabinet is de-energized.
- Check that all system components are voltage-free (e.g. imager, host computer).
- If a UPS (Uninterruptible Power Supply) is installed, it also must be switched off by pressing the EPO.
- Check that the “**Line**” LED at the alarm box is off.
- If there is an audible signal, press the **Alarm Silence** button on the alarm box to stop the signal.
 - ⇒ If the RF-room has an electric or pneumatic door, it has to be possible to open it in the event of a power error.
- Check if the RF-room door can be opened when the system is “powered off”.

Release the emergency shutdown button and reactivate the system by switching on the main power.

- Perform this test on all emergency shutdown buttons in the equipment room, control room, and the examination room.

Checking the safety switches

HP-EH102468-160206-02-002

System Type: Avanto
Espreo
TrioTim
SymphonyTim

Safety switches

Required time: 10 min

Required tools: Wooden block (delivered with the system beginning 2008)

If the wooden block is not on hand, one should be procured locally.

Recommended minimum size for softwoods: 89 mm x 89 mm; length approximately 540 mm.

SI Checking the safety switches

- Move the table all the way up.
- Position the wooden block under the support frame. Pay attention to the hole and to the hand crank on the underside of the support frame. The load on the wooden block and floor is approximately 200 kg. Place a board underneath the wooden block if the flooring is sensitive.

WARNING

Crushing hazard!

There is a potential crushing hazard when you lower the patient table.

⇒ **Pay attention to your fingers when moving the patient table.**



Fig. 429: Example of timber block



Fig. 430: Patient table / switch - S30 test

Pos. 1 Timber block

- Move the patient table carefully down until the support frame is resting on the wooden block and the spindle nut is mechanically disengaged.
 - ⇒ The spindle nut switch S30 opens and the magnet display shows ERROR 63.
 - ⇒ At the MRC, the pop-up window opens < Patient Table Warning: Table Carrier Collision / Lift up table by using.....>, press **OK**



Fig. 431: Magnet display

Pos. 1 ERROR (message) 63

- ⇒ If ERROR 63 does not occur, troubleshooting is necessary.
- Move the table up by pressing the **Up/Inward key** on the control panel.
- Move the table all the way out by pressing the **Down/Outward key** on the control panel.
- Lift the patient table.



Fig. 432: "Lifting up" the patient table plate



Fig. 433: Magnet display

Pos. 1 Error (message) 64

- ⇒ The magnet display shows **ERROR 64** while the tabletop plate is lifted .
- ⇒ At the MRC, the pop-up window opens < Patient Table Warning: ...> , press **O.K.**
- ⇒ If **ERROR 64** does not occur, troubleshooting is necessary.

Checking the patient table stop function

 HP-EH102468-160206-02-003

System Type:Avanto
 Espreo
 TrioTim
 SymphonyTim

STOP function

Required time: 5 min

Required tools:n.a.

SI Checking the patient table stop function

- Activate and then deactivate the light marker, and then press the **center move** button. While the table is moving toward the isocenter, press the **STOP** button at the **left** side of the magnet aperture:

Verify the patient table stop procedure:

- ⇒ The patient table stops; the display indicates the table stop (arrows blinking).
- Make sure that table movement is not possible.
 - ⇒ On the MRC, the pop-up window opens < Patient table notice: STOP button pressed by user.....>, press **OK**
- Deactivate the table **STOP** button by pressing the **IN** and **OUT** buttons in sequence.
- Press the center move button again to continue table movement. Move the table and press the **STOP** button on the **right** side of the magnet:
 - Repeat the steps above to verify the patient table stop procedure.
- Perform the function test for the **third control panel**, if present.
- Press the table **STOP button** on the **intercom at the MRC console**.
 - Perform to verify the patient table stop procedure.

Checking the patient table emergency movement

HP-EH102468-120608-02-011

System Type: Avanto
Espreo
SymphonyTim

Emergency movement

Required time: 5 min

Required tools:n.a.

SI Checking the patient table emergency movement

There are different types of drive mechanisms for the patient tables (exception: Trio A Tim System has only the whole body type).

The whole body drive has an emergency unlocking device. It is used to move the patient table manually.

The partial body patient drive requires no emergency unlocking device. It is possible to move the patient table manually.

Refer to the patient table ([1/Patient table](#)) to differentiate the table type.

Partial body drive

- Move the table into center position by pressing the **center move** button.
- Press the patient table **STOP** button.
- Pull the table out of the magnet.
 - ⇒ It must be possible to pull the table manually out of the magnet. The force required to move the tabletop plate depends on the speed. The slower you move, the less force you need.
- Deactivate the table **STOP** button by pressing the **IN** and **OUT** buttons in sequence.
- Confirm the output on the monitor.

Whole body drive

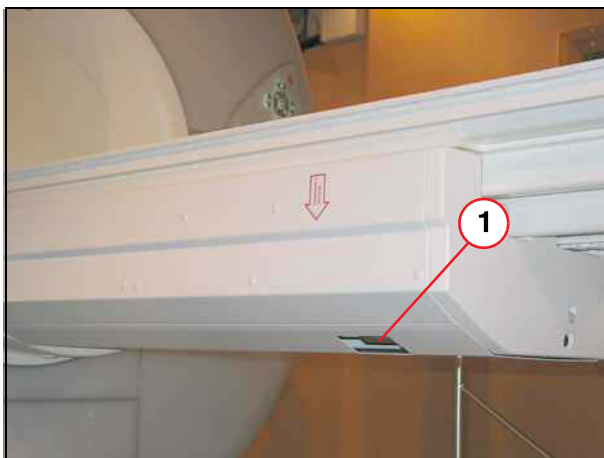


Fig. 434: Patient table
Pos. 1 Emergency release

- Move the table into center position.

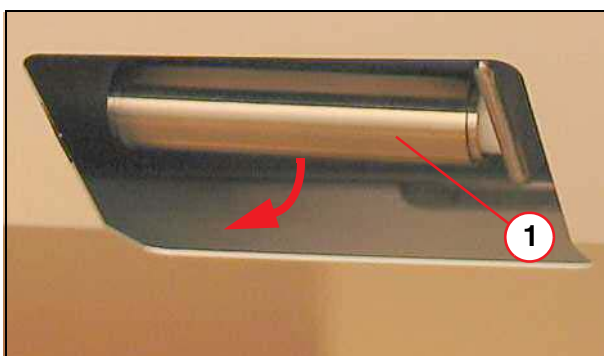


Fig. 435: Emergency release
Pos. 1 Handle

- Pull the emergency release handle on the patient table.
⇒ It must be possible to pull the table manually out of the magnet.

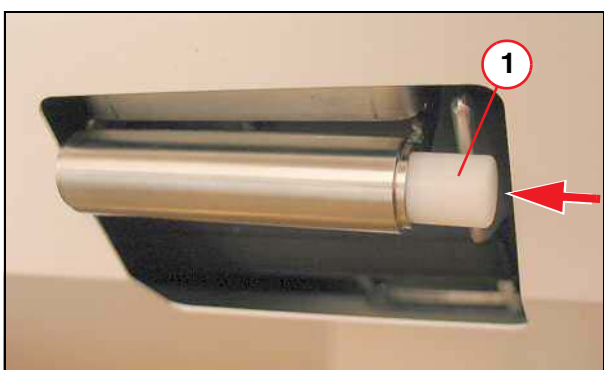


Fig. 436: Emergency release
Pos. 1 Release button

- Press the release button (1) on the emergency release.
- Move the table manually, approximately 10 cm, until the table locks into place. And the handle returns to its original position.
- Confirm the output on the monitor.
- To reset the table, alternately press the IN and OUT buttons on the scanner control panel.
- Press the home button to move the table back into the home position.

Checking the distance between the patient table and the wall in the RF-room

HP-EH102468-160206-02-007

System Type: Avanto
Espreo
TrioTim
SymphonyTim
Verio

Service end

SI Checking the distance between the patient table and the wall in the RF-room

Required time: 5 min**Required tools:** bruising hazard sign p/n 3861718, tape p/n 3474913; if required

- Pull the emergency release handle on the patient table (whole body type) or press the patient table **Stop** button (partial body type).
- Manually move the table all the way back (toward the magnet service end) until the table stops in the horizontal end position.
 - ⇒ If the distance (2/Bruising hazard - controlled access area) between the patient table and the wall (service end) of the RF-room is **greater than** 500 mm, no further actions are necessary. Perform final steps.
 - ⇒ If the distance (2/Bruising hazard - controlled access area) between the patient table and the wall (service end) of the RF-room is **less than** 500 mm, the bruising hazard sign (1/Bruising hazard) has been posted in back of the patient table and the controlled access area has been protected with a yellow-and-black tape (1/Service end area).

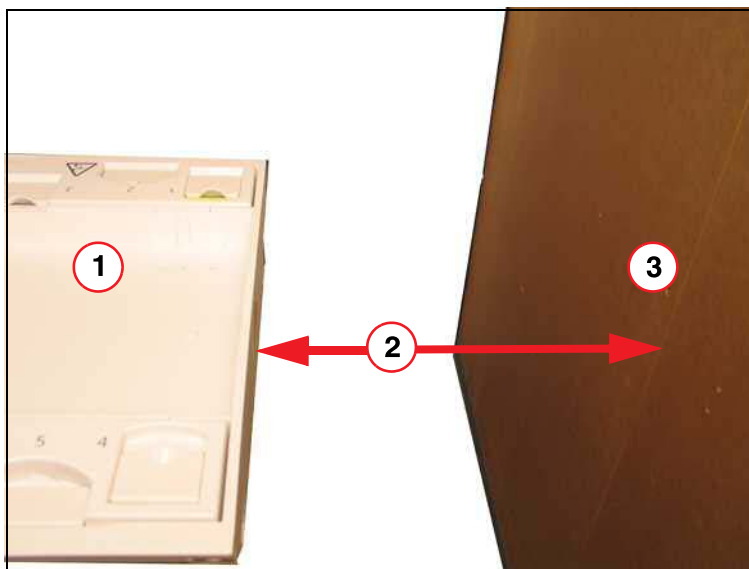


Fig. 437: Bruising hazard - controlled access area

- Pos. 1 Patient table back in horizontal end position
Pos. 2 Distance
Pos. 3 Wall of the RF room

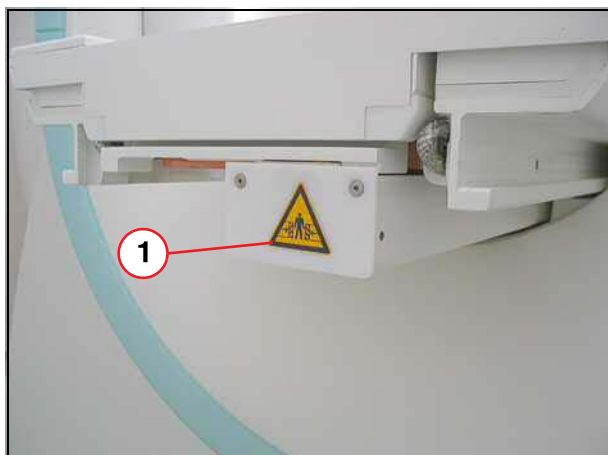


Fig. 438: Bruising hazard

Pos. 1 Label



Fig. 439: Bruising hazard sign

- Ensure that the bruising hazard sign (1/Bruising hazard) has been posted in back of the patient table if the distance between patient table and wall (service end) is less than 500 mm.
- Install missing sign immediately

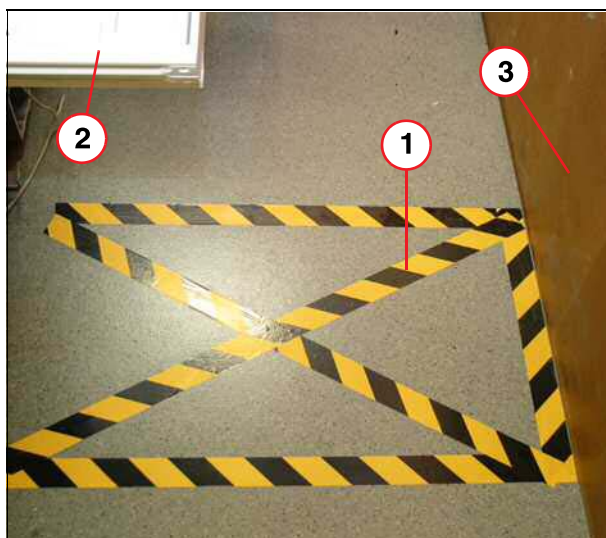


Fig. 440: Service end area

Pos. 1 Marking on the floor

Pos. 2 Patient table

Pos. 3 Wall of the RF room

- Ensure that the controlled access area has been protected with a yellow-and-black tape, if the distance between patient table and wall (service end) is less than 500 mm.

- Install missing tape immediately.

Final steps:

1. Slide the table back into the magnet.
2. Press the release button on the emergency release (whole body type).
3. Move the table manually, approximately 10 cm, until the table locks into place and the handle returns to its original position(whole body type).
4. Confirm the output on monitor.
5. To reset the table, alternately press the IN and OUT buttons on the control panel of the scanner.
6. Press the home button to move the table back into the home position.

Checking the distance between the patient table and the cover

 HP-EH102468-170306-02-001

System Type:Avanto
 Espreo
 TrioTim
 SymphonyTim
 Verio

Cover

SI Checking the distance between the patient table and the cover

Required time: 5 min

Required tools:n.a.

NOTE

The translucent front cover may not be exactly vertical. As a result, the translucent front cover and table may not be in parallel. In this case, do not try to compensate for the angle between the table and the cover using the X-radial adjustment procedure. The procedure also changes the travel path of the tabletop plate inside the magnet bore.

NOTE

The following pictures provide general information on how to perform the distance checks. The cover design differs slightly among Tim systems.

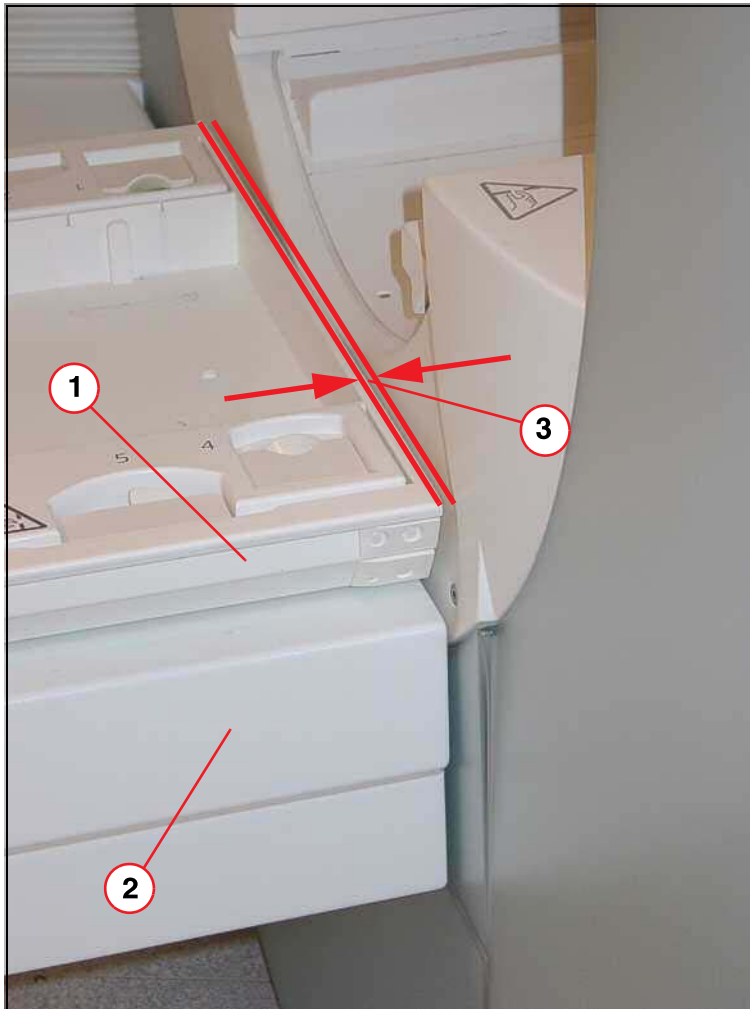


Fig. 441: Clearance, vertical gap

- Pos. 1 Tabletop
Pos. 2 Support frame
Pos. 3 Clearance < 8 mm between table and magnet cover

1. Move the patient table up to the highest position and then lower it about 80 mm.
2. Measure the distance of the vertical clearance (3/Clearance, vertical gap) between the tabletop plate and the magnet cover.
 - ⇒ The distance between the tabletop plate and the magnet cover (3/Clearance, vertical gap) must be within the range of < 8 mm.
 - ⇒ If the distance is not within range, the patient table must be adjusted. Refer to Installation > Start-up > Ramping the magnet > Check the mechanical adjustment of the patient table.

The patient table shall run in parallel to the body coil and covers with a sufficient clearance across the entire tabletop plate to prevent scratching.

NOTE

There is always clearance between the replaceable tabletop plate and the table carrier, and between the table carrier and the slide way. For this reason, the replaceable tabletop plate must be pushed as described.

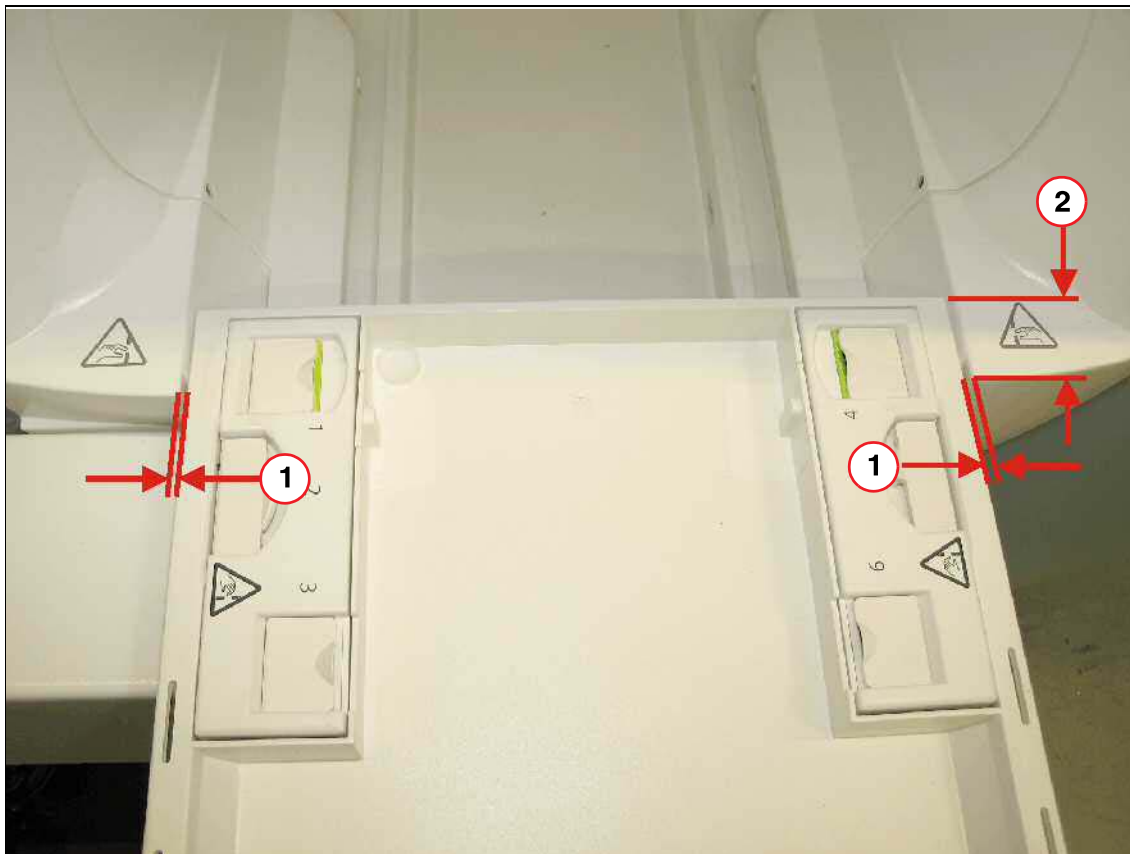


Fig. 442: Case 1: Checking the distance between patient table and cover (patient end)

Pos. 1 Distance to be measured

Pos. 2 Approx. 50 mm

Case 1:

1. Move the patient table approximately 50 mm (for Symphony, A Tim System 400 mm) into the magnet.

2. Push the tabletop plate (replaceable tabletop plate) to the **left** side.

Check the distance of < 8 mm between the patient table and cover on the **right** side, at the front of the magnet.

3. Repeat the procedure by pushing the tabletop plate to the **right** side

Check the distance of < 8 mm on the **left** side, at the front of the magnet.

- ⇒ The distance in both tests has to be within the range of < **8 mm**.
- ⇒ If the distance is not within the range of < **8 mm**, the patient table must be adjusted. Refer to Installation > Start-up > Ramping the magnet > Check the mechanical adjustment of the patient table.

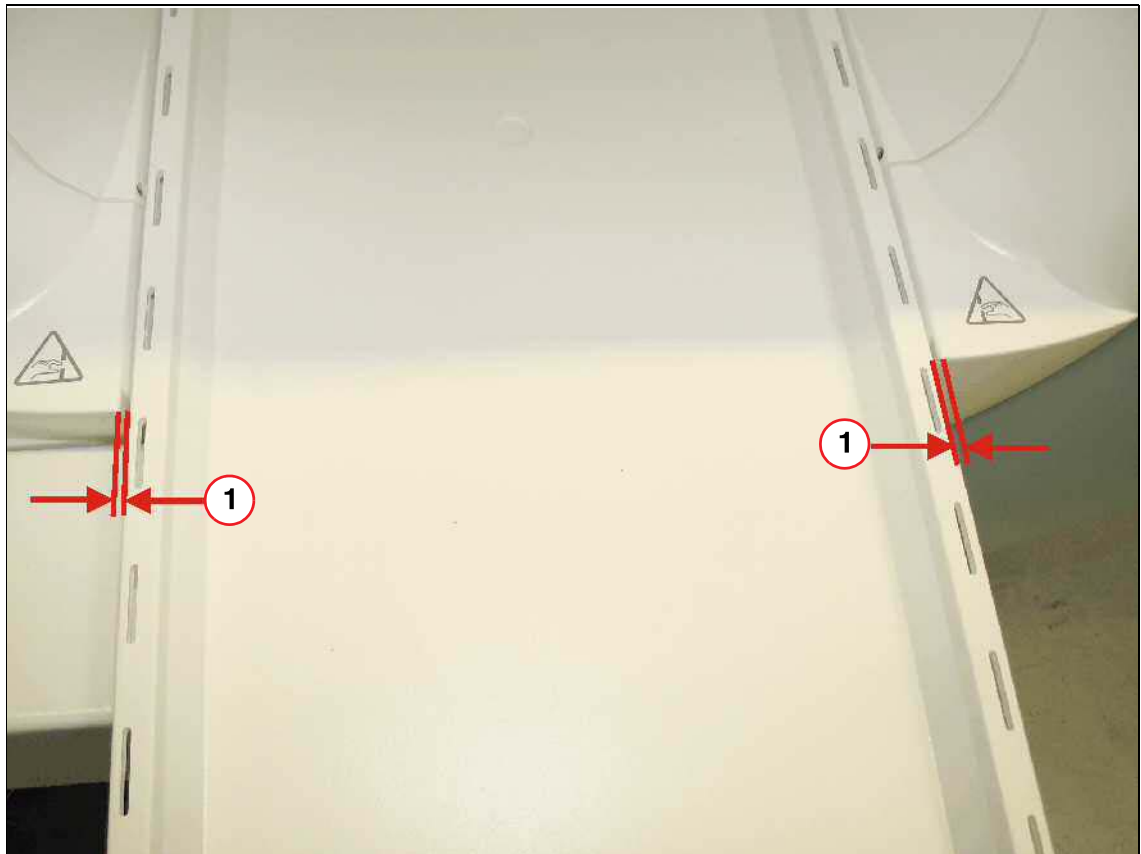


Fig. 443: Case 2: Checking the distance between patient table and cover / body coil (patient end)
Pos. 1 Distance to be measured

Case 2:

1. Move the patient table approximately 1200 mm into the magnet.
2. Push the tabletop plate (replaceable tabletop plate) to the **left** side.
Check the distance of < 8 mm between the patient table and cover on the **right** side, at the front of the magnet.
3. Repeat the procedure by pushing the tabletop plate to the **right** side.
Check the distance of < 8 mm on the **left** side, at the front of the magnet.
 - ⇒ The distance in both tests has to be within the range of < 8 mm.
 - ⇒ If the distance is not within the range of < 8 mm, the patient table must be adjusted. Refer to Installation > Start-up > Ramping the magnet > Check the mechanical adjustment of the patient table.

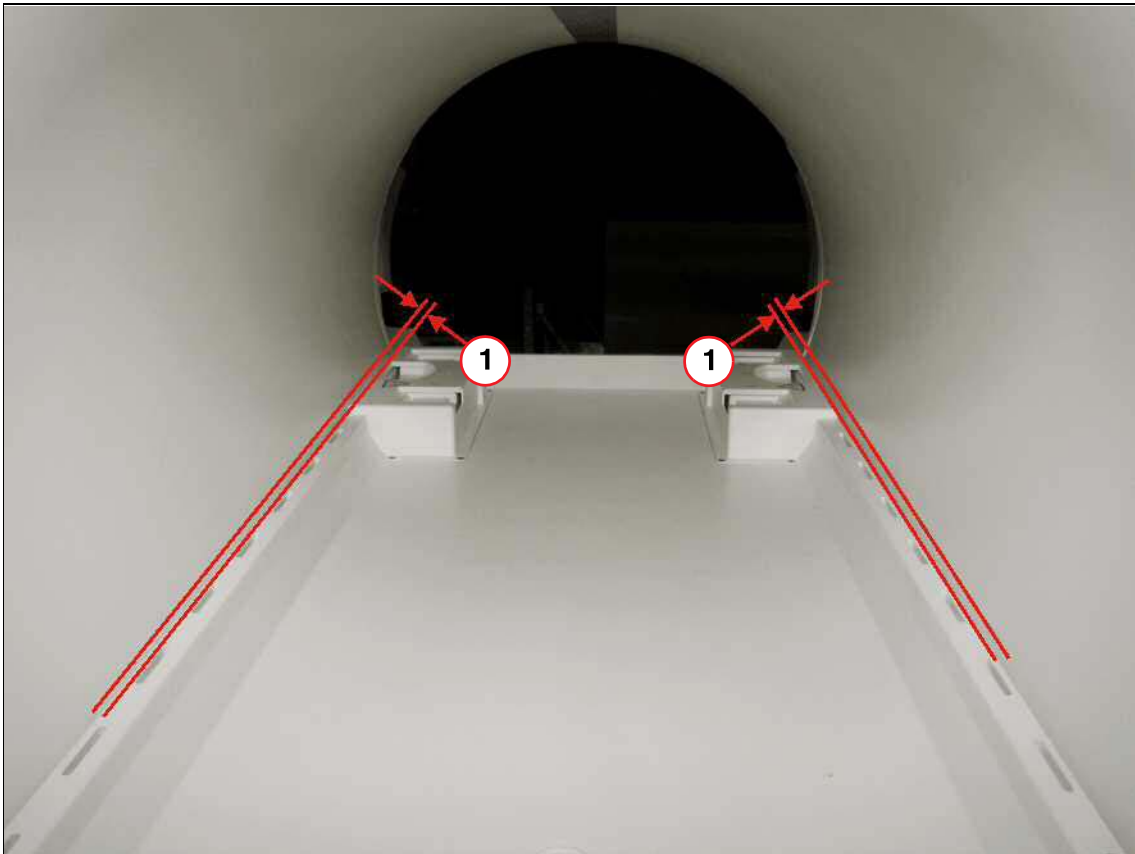


Fig. 444: Case 3: Checking the distance between patient table and Body Coil / cover (service end)
Pos. 1 Distance to be measured

Case 3:

1. Move the complete patient table into the magnet.
2. Check the distance of $< 8\text{ mm}$ between the patient table and cover / body coil on both sides along the entire length of the tabletop plate.
 - ⇒ The distance has to be within the range of $< 8\text{ mm}$.
 - ⇒ If the distance is not within the range of $< 8\text{ mm}$, the patient table must be adjusted. Refer to Installation > Start-up > Ramping the magnet > Check the mechanical adjustment of the patient table.

Checking the squeeze bulb

HP-EH102468-160206-02-008

System Type: Avanto
Espreo
TrioTim
SymphonyTim
Verio

Squeeze bulb

Required time: 5 min

Required tools: n.a.

SI Checking the squeeze bulb

Note: Perform the following test during an MR measurement.

The door to the RF-room shall be closed (direct acoustic coupling between the RF-room and the operating console is momentarily down).

- Press the squeeze bulb at the patient table.

The following must occur:

- ⇒ A continuous audible signal is heard at the intercom on the operating console (MRC).
- ⇒ The red LED above the speak button is flashing.
- ⇒ The audible signal and the red LED go off when you press the “speak” button.
- ⇒ Communication with the patient is established when you press the “listen” button on the control unit (MRC).
- ⇒ You can speak with the patient when you press the “speak” button on the operating console.
- ⇒ You change the volume of the intercom by turning the “volume1” (speaker, intercom) and “volume 2” (speaker, cabin) button as required.

Checking the patient table movement

 HP-EH102468-160206-02-009

System Type:Avanto
 Espreo
 TrioTim
 SymphonyTim
 Verio

Table movement

Required time: 5 min

Required tools:n.a.

SI Checking the patient table movement

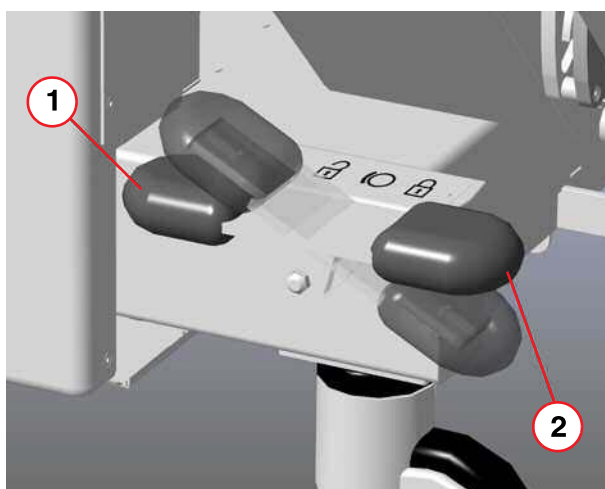
Prerequisites: The system is running.

- Move the table all the way out.
 - ⇒ The table must move without any noticeable noise.
 - ⇒ The table stops in the horizontal end position.
- Lower the table all the way.
 - ⇒ The table must move without any noticeable noise.
 - ⇒ The table stops in the vertical end position.
- Move the table all the way up again.
 - ⇒ The table must move without any noticeable noise.
 - ⇒ The table stops in the vertical end position.
- Move the table all the way in.
 - ⇒ The table must move without any noticeable noise.
 - ⇒ The table stops in the horizontal end position.
- Move the table to the Home position.

Option trolley: Checking the locking mechanism

HP-EH102468-160206-02-012

System Type: Avanto
Espreo
TrioTim
SymphonyTim
Verio

Option: Trolley**Required time:** 10 min**Required tools:** n.a.

- Press down on the brake pedal (2)



Fig. 445: Trolley brake, foot end

Pos. 1 Pedal side: Release

Pos. 2 Pedal side: Locking mechanism

NOTE

If the removable tabletop is not locked, the transfer configuration of the trolley is always in its lowest position. The height of the transfer configuration can be adjusted only after transfer of the removable tabletop to the trolley.

Condition: Trolley without removable tabletop.

- Try to push down the lifting pedal.
 - ⇒ It must not be possible to push down the lifting pedal.

1. Push the lever backward and hold the lever in this position (1/Trolley).

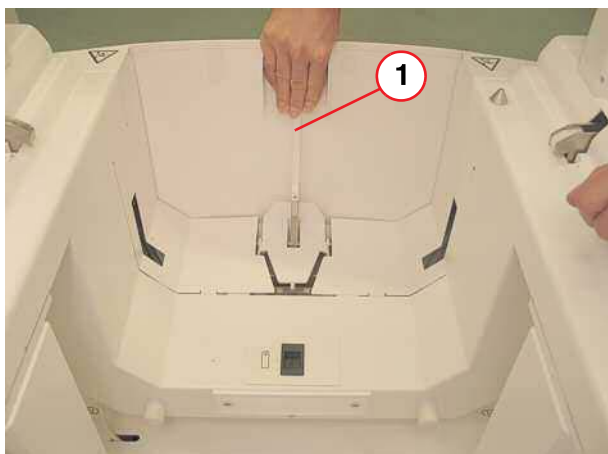


Fig. 446: Trolley
Pos. 1 Lever

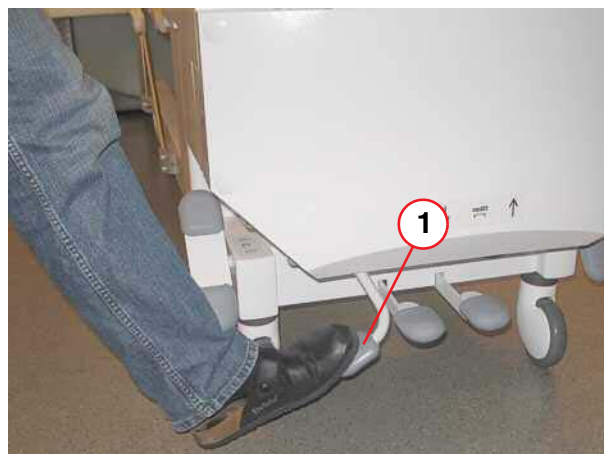


Fig. 447: Trolley locking/releasing pedal

2. Step down fully on the locking/releasing pedal (1/Trolley locking/releasing pedal).
 - ⇒ The locking/releasing pedal locks audibly into position and remains in the inclined position.
3. Check the position of the disk (2/Trolley).
 - ⇒ The slot (3/Trolley) has to be positioned aligned below the nib (1/Trolley). Slowly move the lever (1/Trolley) forward. The nib must not grind to a slot side. In case of misalignment, repair in the factory is required.

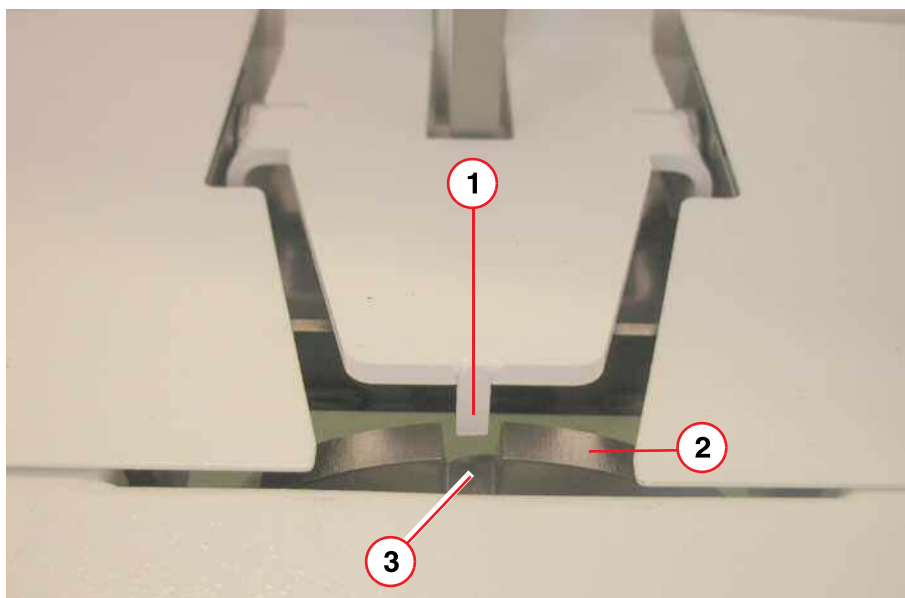


Fig. 448: Trolley
Pos. 1 Nib
Pos. 2 Disk
Pos. 3 Slot

4. Push the lever backward and hold the lever in this position (1/Trolley).
5. Step fully on the locking/releasing pedal (1/Trolley locking/releasing pedal) and let it swing back into the rest position.
6. Release the trolley brakes.



Fig. 449: Patient table, load simulation

1. Move the tabletop plate into the home position.
2. Position several coils and phantoms on the tabletop plate (load simulation).
3. Disconnect all coil plugs and close all protective covers at the coil plug-in locations.
4. Disconnect all hoses (squeeze bulb, vacuum cushion, headphones).
5. Dock the trolley to the patient table.

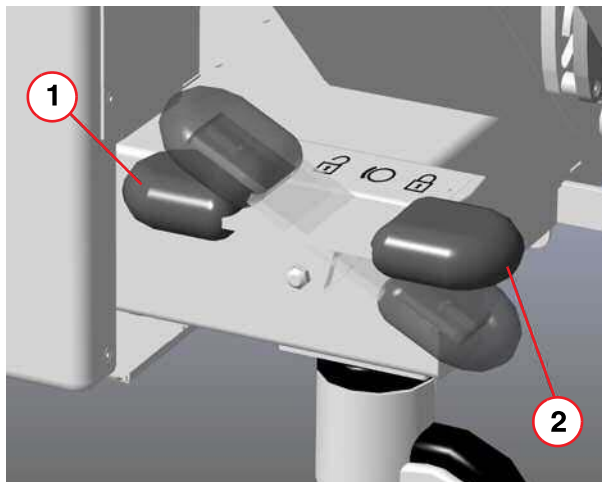


Fig. 450: Trolley brake, foot end

- Pos. 1 Pedal side: Release
Pos. 2 Pedal side: Locking mechanism

- Press down on the brake pedal (2)



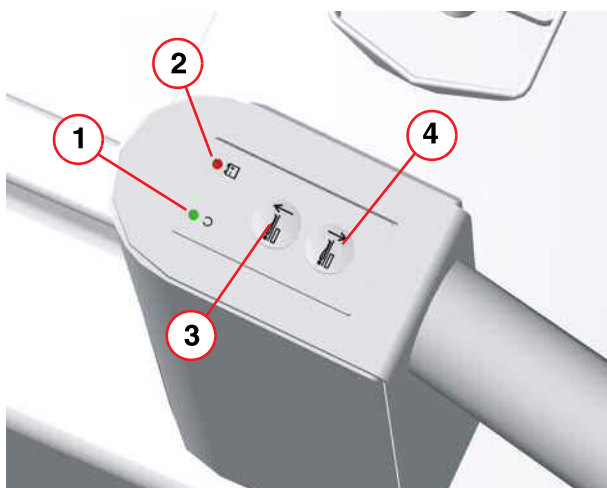


Fig. 451: Trolley remote control

- Pos. 1 Operation LED
- Pos. 2 Battery LED
- Pos. 3 Key for raising the removable tabletop
- Pos. 4 Key for lowering the removable tabletop

- Press the **Lower key** (4) on the remote control to lower the tabletop plate until it stops by itself. While doing this, check both remote control keys to the right and left of the trolley.

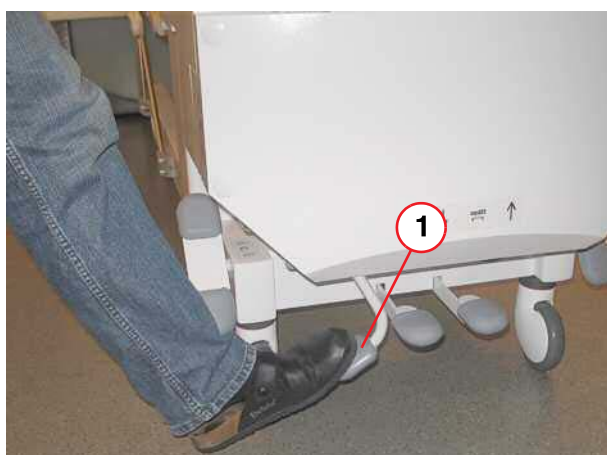


Fig. 452: Trolley locking/releasing pedal

- Step down fully on the locking/releasing pedal.
 - ⇒ The locking/releasing pedal locks audibly into position and remains in the inclined position.
 - ⇒ The removable tabletop is secured at the trolley.
 - ⇒ The table display shows the request to disconnect the connectors for the coils as well as other components (hoses for squeeze bulb etc.)

NOTICE

Damage of coils and tabletop plate

When lowering the carrier plate while the coil plugs or hoses are connected, both the removable tabletop and the coils may sustain considerable damage.

- ⇒ Ensure that all hoses and coils connectors have been disconnected and all protective covers at the coil plug-in locations are closed (**Patient table / foot end**).

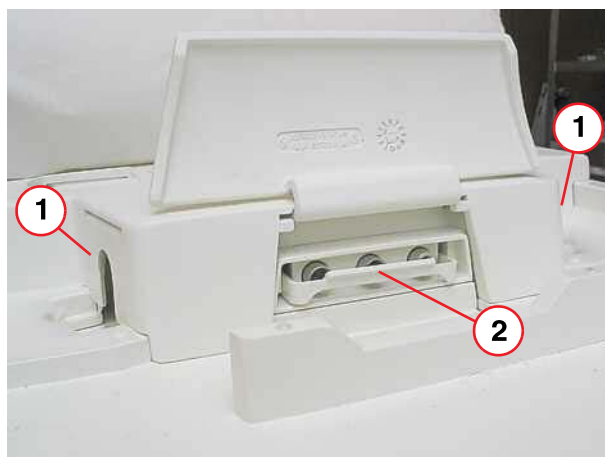


Fig. 453: Patient table / foot end

Pos. 1 Protective covers / coil plug-in

Pos. 2 Protection bracket

1. Press the **Lower key** at the remote control and keep it pressed until the carrier plate has been lowered completely.
2. Release the trolley brake (1/Trolley brake, foot end).
3. Move the trolley including the tabletop plate away from the patient table.

SI Option trolley: Checking the locking mechanism



- Activate all trolley brakes.
- Shake the head end of the removable tabletop. You are checking the locking mechanism.

Fig. 454: Trolley: Head end (magnet side)

- Release all trolley brakes and move the trolley back to the patient table (1/Trolley brake, foot end).
- Dock the trolley on the patient table.
- Activate the trolley brakes(1/Trolley brake, foot end) .
- Press the **Raise key**(3/Trolley remote control) on the remote control to lift the table carrier plate until it stops by itself. While doing this, check both remote control keys to the right and left of the trolley.
- Step fully on the locking/releasing pedal (1/Trolley brake, foot end) and let it swing back into the rest position.
- Press the **Raise key** on the remote control

- Release the trolley brakes ([1/Trolley brake, foot end](#)).
- Move the trolley away from the patient table.

ERDU Test

HP-EH102468-171012-02-002

System Type: SymphonyTim

ERDU

Required time: 30 min

Required tools:

ERDU Test Load Kit III (ETL) 03861197,
Adaptor cable B - 10129824 (if required, this adaptor should be in the ETL kit),
MSUP Display Unit - 07758522,
Flat bladed screwdriver (non-magnetic).

NOTE

The Adaptor cable B is available from CSML as a spare (part no. 10129824)

NOTE

The Mains cable supplied with the ETL is a UK, Euro and a US AC cable. When the ETL is used in another country an IEC 320 C13 mains lead for that country must be obtained.

SI ERDU Test

Prerequisites

 WARNING

Restrict Access.
Danger of Ferrous materials entering imaging suite.
Unauthorized personnel must be excluded from the imaging suite.
⇒ **Secure the room and prevent unauthorized access.**

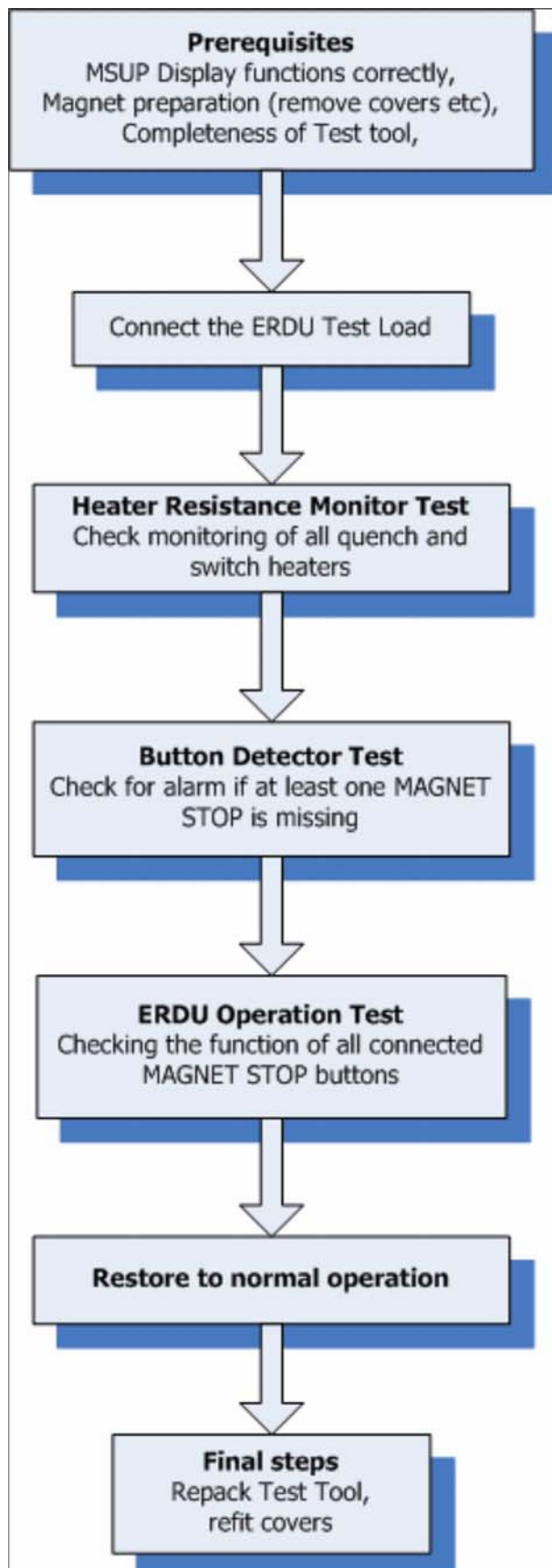


Fig. 455: ERDU testing flow chart

- Restrict access to the magnet if it is at field while the ETL is connected to the MSUP.
- Check that all alarm and fault indications show that the system is OK.
- Check that the **ERDU Test Load Kit** is complete with all the listed parts present:

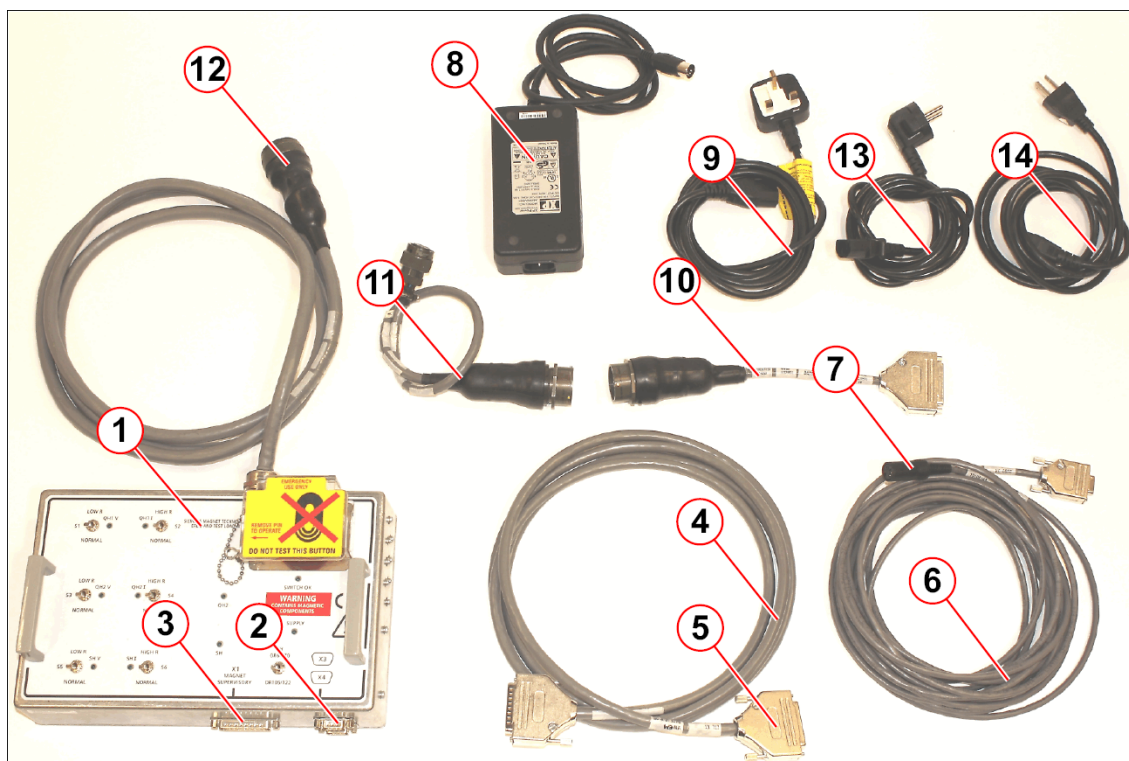


Fig. 456: ERDU Test Load Kit

- | | |
|---------|---|
| Pos. 1 | ERDU Test Load III (ETL) |
| Pos. 2 | 9-pin Connection X3 or X4 for PSU |
| Pos. 3 | 25-pin Connection X1 |
| Pos. 4 | MSUP - ETL Cable |
| Pos. 5 | 25-pin Connector to MSUP X1 |
| Pos. 6 | PSU Output Extension Lead |
| Pos. 7 | PSU Output Extension Lead Connector to PSU |
| Pos. 8 | Power Supply Unit (ETL PSU) |
| Pos. 9 | AC Supply Lead (UK) |
| Pos. 10 | Adaptor Cable B |
| Pos. 11 | Adaptor Cable for OR64 / OR70, 41-pin Connector to 10-pin Magnet Turret Connector 'B' |
| Pos. 12 | 41-pin Connector, Magnet Turret |
| Pos. 13 | AC Supply Lead (EURO) |
| Pos. 14 | AC Supply Lead (US) |

- Remove the service access cover, refer to 'Replacement of Parts'.
- Remove the turret cover, refer to 'Replacement of Parts'.

Helium remote display unit.

NOTE

The helium remote display unit is a Service Tool and must be disconnected during imaging to avoid interference to the MR Signal.

General

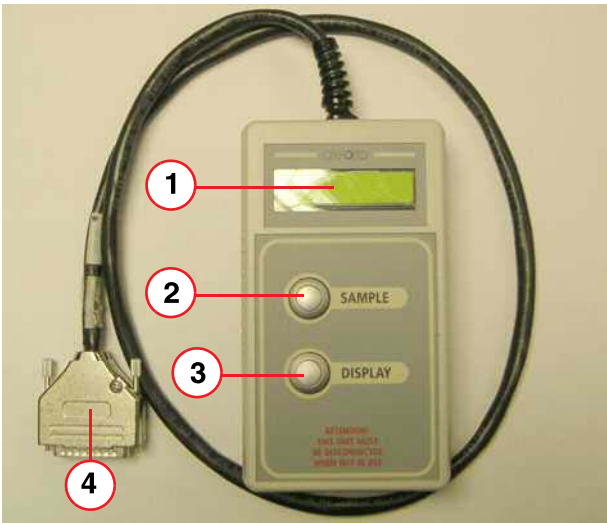


Fig. 457: MSUP- Display Unit

- Pos. 1 LCD Display
- Pos. 2 Sample button
- Pos. 3 Display scroll button
- Pos. 4 25 way 'D' connector

The helium remote display unit is a monitoring device that is connected to **X9** of the MSUP. It provides local monitoring of the magnet parameters, including alarms and warnings if values are not within specification. In addition, some functions can be toggled **ON** or **OFF** manually.

The unit will also operate in battery mode from the MSUP battery, to provide liquid helium measurement during transportation/installation.

Connecting the Helium remote display unit

- Connect the Unit to **X9** of the MSUP.
- Press both the **SAMPLE** (2) and **DISPLAY** buttons (3), at the same time, for approximately **5 seconds**.
- Press the **SAMPLE** (2) to start the test.
 - The display will show '**TEST IN PROGRESS**' whilst carrying out an internal **Self Test** and a helium level sample.

Display Magnet Parameters

- Press the **DISPLAY** button (3) repeatedly to display the different monitored sections. The sections used during the ERDU test are 7, 8 and 9.

If a parameter is outside it's specified limits, a "**W**" or "**A**" will be displayed next to the parameter reading to indicate a **WARNING** or **ALARM**. If this is the situation then trouble shoot the error.

Tab. 38 OR70/64 MSUP-Display Parameters

Item #	Parameter	Typical Value +/-20% Tolerance
7	Switch Heater (Ohms, status)	33 Ohms, Off
8	Quench heaters 1, 2 (Ohms)	22.5 Ohms, 22.5 Ohms

Connecting the ERDU Test Load (ETL)

NOTE**Note for Trio A Tim System**

Always use the magnet service platform when working on the service turret. For correct installation and usage of the service platform, refer to the 'Magnet Troubleshooting Guide > Service Platform Installation'

 WARNING

It is not possible to quench the magnet using the normal system **MAGNET STOP** buttons while the ERDU Test Load is connected to the MSUP and magnet.

⇒ The magnet may be quenched only by using the **MAGNET STOP** button on the ERDU Test Load.

⇒ Restrict access to the magnet accordingly.

- The **ETL PSU** has to stay outside the magnet room. Connect it to a convenient AC outlet using the mains lead for your country.

NOTE

Ensure that the **ETL PSU** is as far away from the magnet as possible (magnetic field < 50 Gauss).

- Connect the ETL PSU to the ETL **X3** or **X4** using the PSU output extension lead.
 - ⇒ Check that the **SUPPLY** and **SWITCH OK** indicators glow on the ETL.

 WARNING**Magnet quench!**

A magnet quench is caused by pressing the ETL Magnet Stop button.

⇒ Ensure that the **MAGNET STOP** button on the ETL is **NOT OPERATED!**

⇒ **DO NOT** operate the Magnet Stop button on the ERDU Test Load unless an emergency magnet quench is required.

- Connect the **MSUP-ETL** cable to the 25-pin connector **X1** on the ERDU Test Load. [\(3/ERDU Test Load Kit\)](#)
- Set the mode switch to **OR64/70**.
- Set all **HIGH R / LOW R** switches on the ETL to **NORMAL** position (downward). [\(ERDU Test Load \(front panel\)\)](#)
- Ensure that the **MAGNET STOP** button on the ETL is **NOT OPERATED!**
- Connect the **adaptor cable B** [\(12/ERDU Test Load Kit\)](#) to the **41-pin turret connector cable**.

- Disconnect the lead **X1** on the MSUP and connect the end of **adaptor cable B** onto the cable which was plugged into **X1**.

Check that the alarm sounds and that the **MAG STOP** supply indicator lights up on the **Alarm Box**. ([Alarm Box \(buttons and controls\)](#))

⇒ Press **ACKNOWLEDGE** to cancel the audible alarm.

Check that the **QH1**, **QH2**, **SH** and **SWITCH OK** and **SUPPLY** indicators glow on the ETL. ([ERDU Test Load \(front panel\)](#))

Connect the **ETL- MSUP lead** to **X1** on the MSUP.

⇒ Check **audible alarm** sounds in the ETL.

NOTE

The ETL will sound a continuous buzzing alarm, until the ETL is disconnected.

Silence any Alarm Box alarms by pressing the **ACKNOWLEDGE** button.

Heater resistance monitor tests

Resistance figures are nominal- +/-20% tolerance are acceptable.



Fig. 458: ERDU Test Load (front panel)



Fig. 459: Alarm Box (buttons and controls)

- Operate the **SH LOW R** switch on the ERDU Test Load.
 - ⇒ Check for the Alarm Box buzzer ON and MAG STOP LED ON.
 - ⇒ Press the **ACKNOWLEDGE** button at the alarm box, Check the alarm box buzzer turns off.
 - ⇒ Check for MSUP-Display Unit indication 7. SWITCH HEATER **12R A**.
- Reset the **SH LOW R** switch on the ERDU Test Load to normal (downwards).
 - ⇒ Check for MSUP-Display Unit indication 7. SWITCH HEATER **31R**.
- Operate the **SH HIGH R** switch on the ERDU Test Load.
 - ⇒ Check for Alarm Box buzzer ON and MAG STOP LED ON.
 - ⇒ Press the **ACKNOWLEDGE** button at the alarm box, Check the alarm box buzzer turns off.
 - ⇒ Check for MSUP Display Unit indication 7. SWITCH HEATER **70R A**.
- Reset the **SH HIGH R** switch on the ERDU Test Load to normal (downwards).
 - ⇒ Check for MSUP-Display Unit indication 7. SWITCH HEATER **31R**.

- Operate the **QH1 LOW R** switch on the ERDU Test Load.
 - ⇒ Check for Alarm Box buzzer ON and MAG STOP LED ON.
 - ⇒ Press the **ACKNOWLEDGE** button at the alarm box, Check the alarm box buzzer turns off.
 - ⇒ Check for MSUP Display Unit indication 8. QUENHTR **8R A 21R**.
- Reset the **QH1 LOW R** switch on the ERDU Test Load to normal (downwards).
 - ⇒ Check for MSUP Display Unit indication 8. QUENHTR **21R 21R**.
- Operate the **QH1 HIGH R** switch on the ERDU Test Load.
 - ⇒ Check for Alarm Box buzzer ON and MAG STOP LED ON.
 - ⇒ Press the **ACKNOWLEDGE** button at the alarm box, Check the alarm box buzzer turns off.
 - ⇒ Check for MSUP Display Unit indication 8. QUENHTR **42R A 21R**.
- Reset the **QH1 HIGH R** switch on the ERDU Test Load to normal (downwards).
 - ⇒ Check for MSUP Display Unit indication 8. QUENHTR **21R 21R**.
- Operate the **QH2 LOW R** switch on the ERDU Test Load.
 - ⇒ Check for Alarm Box buzzer ON and MAG STOP LED ON.
 - ⇒ Press the **ACKNOWLEDGE** button at the alarm box, Check the alarm box buzzer turns off.
 - ⇒ Check for MSUP Display Unit indication 8. QUENHTR **21R A 8R**.
- Reset the **QH2 LOW R** switch on the ERDU Test Load to normal (downwards).
 - ⇒ Check for MSUP Display Unit indication 8. QUENHTR **21R 21R**.
- Operate the **QH2 HIGH R** switch on the ERDU Test Load.
 - ⇒ Check for Alarm Box buzzer ON and MAG STOP LED ON.
 - ⇒ Press the **ACKNOWLEDGE** button at the alarm box, Check the alarm box buzzer turns off.
 - ⇒ Check for MSUP Display Unit indication 8. QUENHTR **21R A 42R**.
- Reset the **QH2 HIGH R** switch on the ERDU Test Load to normal (downwards).
 - ⇒ Check for MSUP Display Unit indication 8. QUENHTR **21R 21R**.

Button detector test

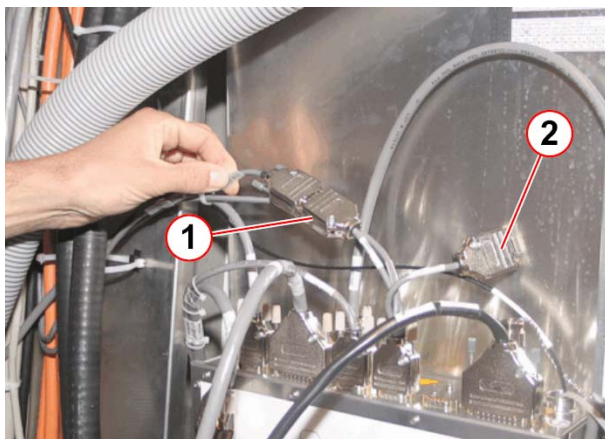


Fig. 460: Magnet - MSUP

Pos. 1 Flying connector at MSUP X6

Pos. 2 Optional: for 3rd magnet stop button at MSUP X6

- Disconnect the < ERDU button 1 > cable from the flying connector at the MSUP X6 ERDU / FIELD connector.
 - ⇒ Check for Alarm Box buzzer and **MAG STOP** indicator **ON**.
 - ⇒ Check for MSUP Display Unit “**ERDU BUTTONS**” window 9 “**TOO FEW BUTTONS**”.
 - ⇒ Reconnect the button cable to the flying connector and press the **ACKNOWLEDGE** button on the Alarm Box to clear the buzzer.
 - ⇒ Check the MSUP display, ensure it states “**ERDU BUTTONS**” windows 9 “**ERDU BUTTONS OK**”.
- If the optional (**3rd**) **MAGNET STOP** button is installed, repeat the steps above for the 2nd flying connector .

ERDU operation tests

WARNING**Magnet quench!**

A magnet quench is caused by pressing the ETL Magnet Stop button.

- ⇒ **DO NOT** operate the Magnet Stop button on the ERDU Test Load unless an emergency magnet quench is required.

NOTE

The thermal switches protecting the load resistors in the ERDU Test Load from overheating will cut out after approximately three minutes of continuous operation. Keep the operating periods in the following tests as short as possible.

Table shows the various indications expected when the Magnet Stop or a remote button is pressed. The subsequent text describes the procedure in detail. Resistance values are nominal - **+/-20% tolerance is acceptable**.

Tab. 39 Magnet Stop or remote button is pressed (OR64/OR70)

ERDU Test Load	MSUP Display Unit	Alarm Box
QH1 V LED ON QH2 V LED ON SH V LED ON QH1 I LED ON QH2 I LED's ON SH I LED ON	7. SWITCH HEATER 260R A OFF A 8. QUENHTR 52R A 52R 9. ERDU BUTTONS ERDU OPERATED	MAG STOP LED ON Buzzer ON

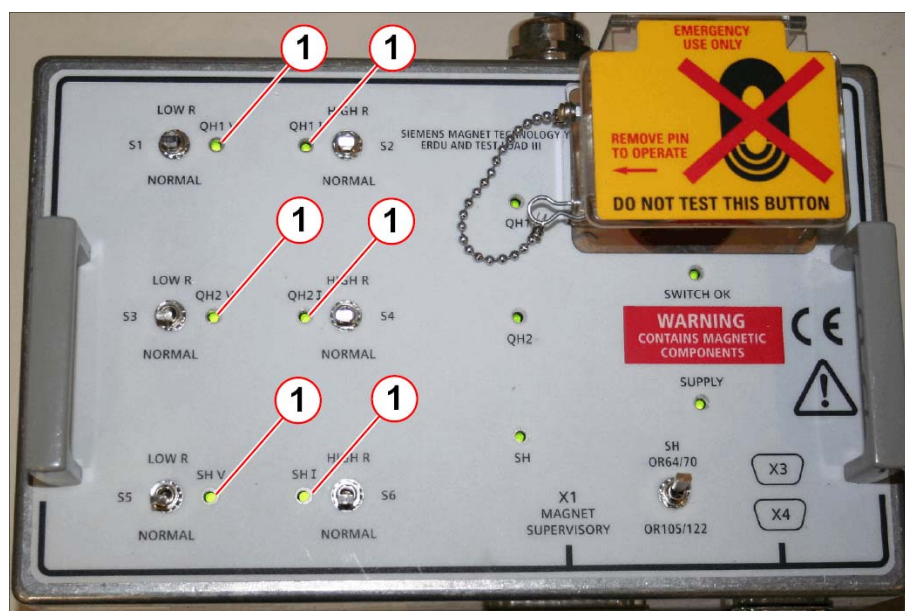


Fig. 461: ERDU Test load with ERDU pressed

Pos. 1 LEDs lit during ERDU activation

- Press the **Acknowledge** button on the **Alarm box** to clear any indicators/alarms except on **ETL**.
- Scroll the **MSUP** display to #9.
- Press the **MAGNET STOP** button on the **Alarm Box**.
 - ⇒ Check to make sure displays, indicators, and buzzers operate as in ([Magnet Stop or remote button is pressed \(OR64/OR70\)](#)).
- Release the **MAGNET STOP** button at the **Alarm Box**.
- Check the **MSUP Display** #9 “ERDU BUTTONS OK”.
- Press **Acknowledge** button on **Alarm box** to clear any indicators/alarms except on **ERDU Test Load**.
- Scroll the **MSUP** display to #9.

- Press the remote **MAGNET STOP** button.
 - ⇒ Check to make sure displays, indicators, and buzzers operate as in [\(Magnet Stop or remote button is pressed \(OR64/OR70\) \)](#).
- Release the remote **MAGNET STOP** button.
- Check the **MSUP Display # 9** “ERDU BUTTONS OK”.
- Press the **Acknowledge** button on the **Alarm Box** to clear any indicators/alarms except on **ETL**.
- If installed, press the remote **optional (3rd) MAGNET STOP** button.
 - ⇒ Check to make sure displays, indicators, and buzzers operate as in [\(Magnet Stop or remote button is pressed \(OR64/OR70\) \)](#).
- Release the remote **optional (3rd) MAGNET STOP** button.
- Press the **Acknowledge** button on the **Alarm Box** to clear any indicators/alarms except on **ETL**.

Restore normal operation

NOTE

Ensure all **MAGNET STOP** buttons are released and are **NOT ACTIVE!**

- Disconnect **X1** from the MSUP.
- Check that the MSUP Display Unit shows **(9) ERDU BUTTONS** are **OK**.
- Disconnect the **X1** turret cable connector from **adaptor cable B** connection.
- Reconnect **X1** turret cable connector to the **X1** connection on the MSUP
- Press the **ACKNOWLEDGE** button on the Alarm Box. Check that all indicators are now normal.
- The ERDU test generates some error / warning messages on the MRC. These error messages can be ignored.
- Click **< OK >** to confirm the messages.

Final steps

- Disconnect the **ERDU Test Load** from its **PSU**, and the **PSU** from the **AC** supply.
- Repack all items in the ERDU Test Load Kit to prevent loss.
- Disconnect the **Magnet Display Unit**.

NOTE

The helium remote display unit is a **Service Tool** and must be disconnected during imaging to avoid interference to the MR Signal.

- Install the **covers**

- Affix the tamper-proof seal to the alarm box.



Fig. 462: Tamper-proof seal location

Pos. 1 Tamper-proof seal

Measurement of the coil power loss

HP-EH102468-170206-02-001

System Type: Avanto
Espreo
TrioTim
SymphonyTim

Measurement of body Coil Power Loss

Required time: 5 min
Required tools: Small spherical phantom with the holder

NOTE

Use the Online Help to obtain detailed information regarding, e.g., table control, phantom positioning, or specific items and procedures. Press the 'Help' button in the header bar of the corresponding platform.

- Preliminary steps:
1. Position the small spherical phantom in the magnet isocenter (without loader).
 2. Start the Service platform **Options > Service > Local Service**.
 3. Select the Tune-up platform.

SI **Measurement of the Coil Power Loss**

1. Select the **Coil Power Losses** measurement only.
2. Start the **Coil Power Losses** measurement by clicking the **Go** button.

➡ The measurement has been completed successfully, if the **Coil Power Losses** Tune-up step indicates the status '**done**'.

Checking the gradient supervision

HP-EH102468-130711-02-006

System Type:SymphonyTim

Gradient supervision

Symphony, A Tim System

Required time: 40 min

Required tools: Smoke wick material number 10684293 or test spray material number 10684091, pocket lighter and screwdriver

Prerequisites: The display of the aspiration smoke detector shows OK ([1/Aspiration smoke detector](#)). System is on.

- Remove the following cover sections; please refer to Replacement of Parts > System > Cover
 - Service access cover
 - Left rear cover

WARNING

Accumulation of oxygen during helium filling.

The use of burning smoke wicks during helium filling could result in an open fire in the area of the magnet service turret.

⇒ **Do NOT perform a smoke test during helium filling!**

SI Checking the gradient supervision

- Take the smoke wick out of the box.



Fig. 463: Smoke wick in the box

NOTICE**On-site smoke detectors****Avoid false alarm of the on-site smoke detector!**

- ⇒ Check if there is an on-site smoke detector. The smoke or test gas must not enter the on-site smoke detector.
- ⇒ Keep the smoke or test gas away from on-site smoke detectors.
- ⇒ Keep the smoke time as short as possible to minimize smoke concentration in the air.

- Light the tip of the smoke wick and allow it to burn for 10 - 15 seconds. Then blow out the flame.



Fig. 464: Burning smoke wick

Pos. 1 Plastic handle

- ⇒ The smoke wick provides a stable and continuous trail of smoke.
- Hold the smoke wick under the intake tube for 5 seconds, that the smoke is aspirated and transported to the detector ([1/Magnet service end / gradient coil connection](#)).

NOTE

Alternatively, the test spray can be used. Spray the test gas dosed toward the intake manifold for 5 seconds. Dosed means in this context to press the spray button not completely. Distance between spray and intake manifold should be approximately 15-20 cm.



Fig. 465: Test spray

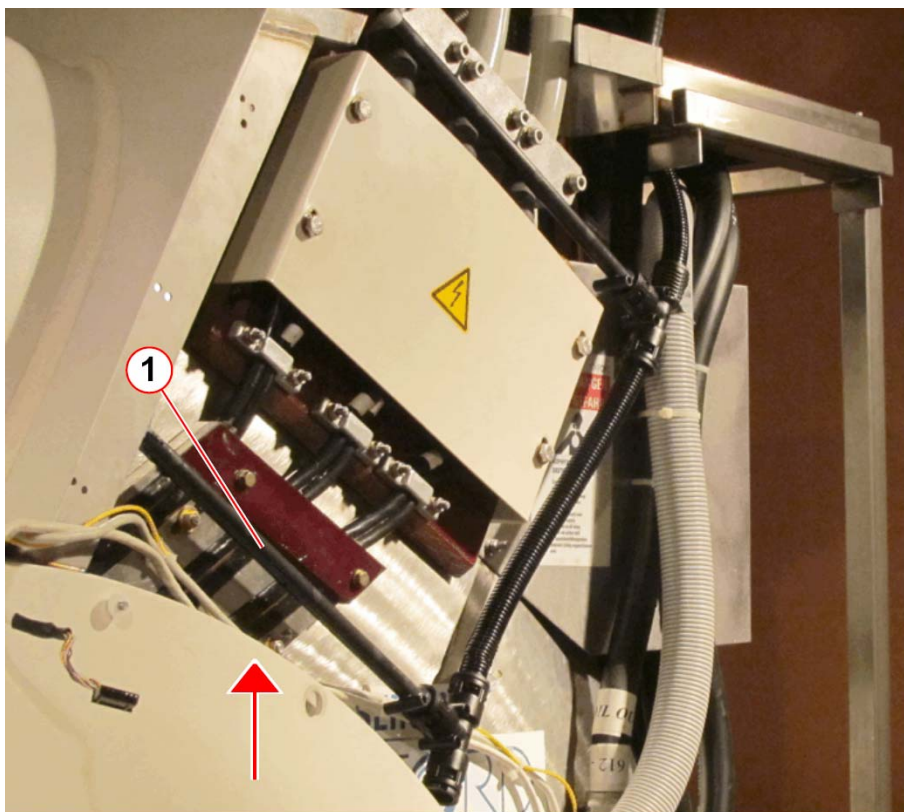


Fig. 466: Magnet service end / gradient coil connection

Pos. 1 Intake manifold (The assembly of the manifold can vary)

- After use, immediately place the smoke wick into the empty box and close the lid carefully.

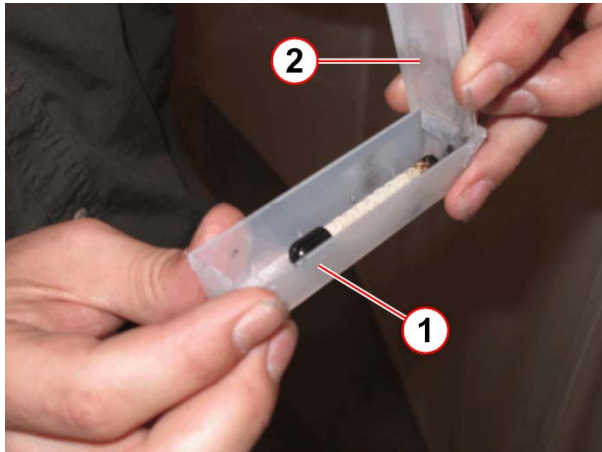


Fig. 467: Extinguishing the smoke wicks

Pos. 1 Box

Pos. 2 Lid

⇒ This will extinguish the smoke wick after approximately 15 seconds.

Expected test results:

- ⇒ The smoke moves from the gradient cable connection to the aspiration smoke detector. This will take up to 60 seconds, then:
- ⇒ The display of the aspiration smoke detector shows **FIRE**.
- ⇒ An alarm message appears at the MRC.
Scanner hardware error
Automatic shutdown! Evacuate patient immediately ...
- ⇒ The gradient modulator in the GPA / GSSU is disabled.
- ⇒ Within the next 60 seconds, the system will be switched to stand-by mode automatically.

Reset the system:

- Press a screwdriver into the security tab ([1/Aspiration smoke detector ASD](#)) to open the service access cover with the finger clip ([3/Aspiration smoke detector ASD](#)).

NOTE

If the reset fails, the concentration of smoke inside the detector chamber is too high. Retry the reset after 1-2 minutes.

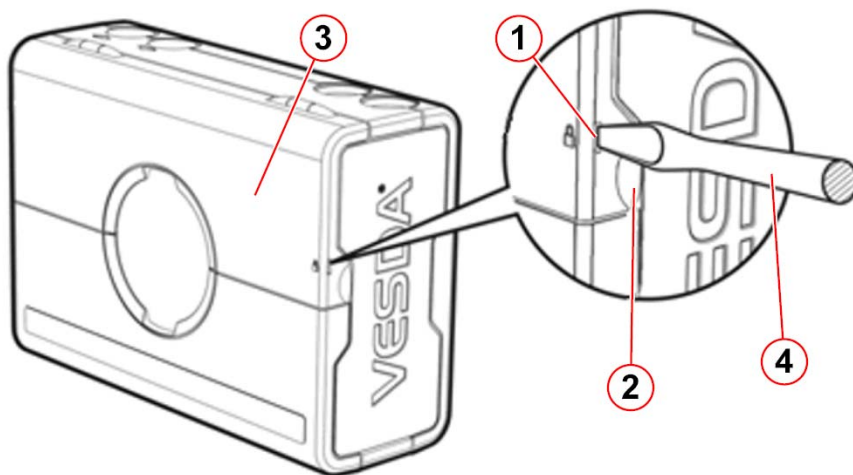


Fig. 468: Aspiration smoke detector ASD

- Pos. 1 Security tab
- Pos. 2 Finger clip
- Pos. 3 Service access cover
- Pos. 4 Screwdriver

- Reset the alarm by pressing the **reset button** (2/Aspiration smoke detector) for 2 seconds.

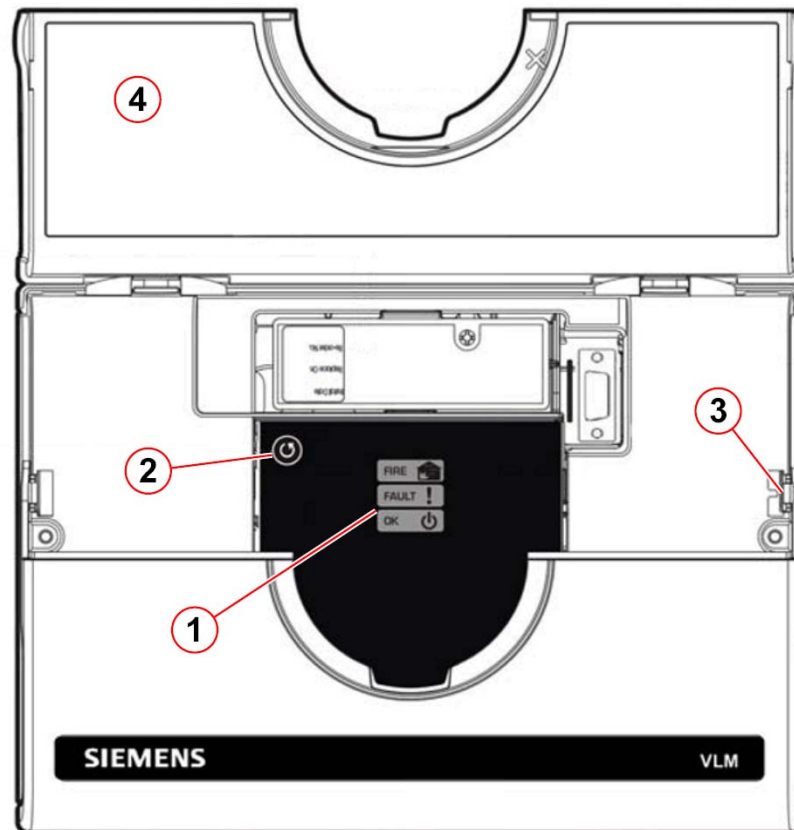


Fig. 469: Aspiration smoke detector

- Pos. 1 Status display
 Pos. 2 Reset button
 Pos. 3 Security tab
 Pos. 4 Service access cover

⇒ The aspiration smoke detector shows OK (1/Aspiration smoke detector).

- Start the system by pressing the **ON button** at the alarm box.

Quality Assurance

General QA

HP-EH102468-121012-02-002

System Type:SymphonyTim

Note for using the online 'Install&Maintenance Protocol'

NOTE

The interactive 'Install & Maintenance' protocols take the status from the 'General QA'. If all general QA steps show the 'Done' status, then the 'Install & Maintenance' protocols display the OK status. If one or more steps have the 'ToDo' or 'NotOK' status, then the 'Install & Maintenance' protocols display the 'NotOK' status.

CAUTION: When the Open QA button is pressed, the QA platform opens automatically and all QA steps change to the 'ToDo' status. Please note: the "Open QA" button is only one way to open the QA platform. This notice also applies if the QA platform is opened by different means. The data will be taken over even if you do not use the "Open QA" button.

Quality Assurance (QA)

Required time: 70 - 120 min depending on installed options.

Required tools: QA Phantom Setup, Service Plug (if Multi Nucleus Spectroscopy Option is installed)

The quality measurement of the body coil is a fully automated procedure. The results are entered in the on-site database.

NOTE

Use the Online Help to obtain detailed information regarding, e.g., table control, phantom positioning, or specific items and procedures. Press the 'Help' button in the header bar of the corresponding platform.

Preliminary steps

Starting the service platform

All actions of the quality assurance procedure are performed via the service platform.

- Start the Service platform > **Options** > **Service** > **Local Service**.

SI

General Quality Assurance

Go to the Home menu and press the corresponding button to access the **Quality Assurance** sub platforms.

- Start the measurement by clicking the "Go" button.

- ⇒ The **General Quality Assurance** check is successfully completed when every individual QA step indicates the status < **done** >.

NOTE

If the results are out of tolerance, perform “Tune-Up” or troubleshooting, if necessary. Repeat “Quality Assurance”.

Test Result / Evaluation

HP-EH102468-170206-02-099

System Type:Avanto

Espre

TrioTim

SymphonyTim

Verio

After completing the test activities, fill out this page.

The tester has to evaluate the test results as defined and listed under the test certificate points.

The operator or the person assigned by the operator must sign the system evaluation form for product liability reasons.

26.2.4 Maintenance Tests

Preventive Maintenance

Cleaning the computer

HP-EH102468-130711-02-015

System Type: Avanto
 Espree
 TrioTim
 SymphonyTim
 Verio

Computer MRC (MRAWP) - MRSC (MRWP)

Required time: 10 min

Required tools: Vacuum cleaner

PM Cleaning the computer

MRC

- Check all air inlets and outlets for dust. If necessary, clean the computer with a vacuum cleaner.

MRSC if installed

- Check all air inlets and outlets for dust. If necessary, clean the computer with a vacuum cleaner.



Fig. 470: Computer



Fig. 471: Computer

Cooling system general checks

HP-EH102468-170206-02-054

System Type:SymphonyTim

Cooling**Required time:** 10 min**Required tools:**n.a.**PMP Cooling system - general checks**

Check the entire water system for leaks and condensation, especially all water connections and water hoses within the:

- ACC
- GPA
- CCS
- RCA (Helium compressor)
- Filterplate
- Magnet
- Option: Transfer Station

Service Software

- Open > Service Software > Magnet & Cooling > Cooling Status
- Check the cooling status of the system

Checking the fan assembly

HP-EH102468-200206-02-013

System Type: SymphonyTim

GPA, ACC Cabinet

Required time: 10 min

Required tools: 5-mm Allen key, flashlight

PMP Checking the fan assembly

The fan assembly is integrated into the ACC cabinet.

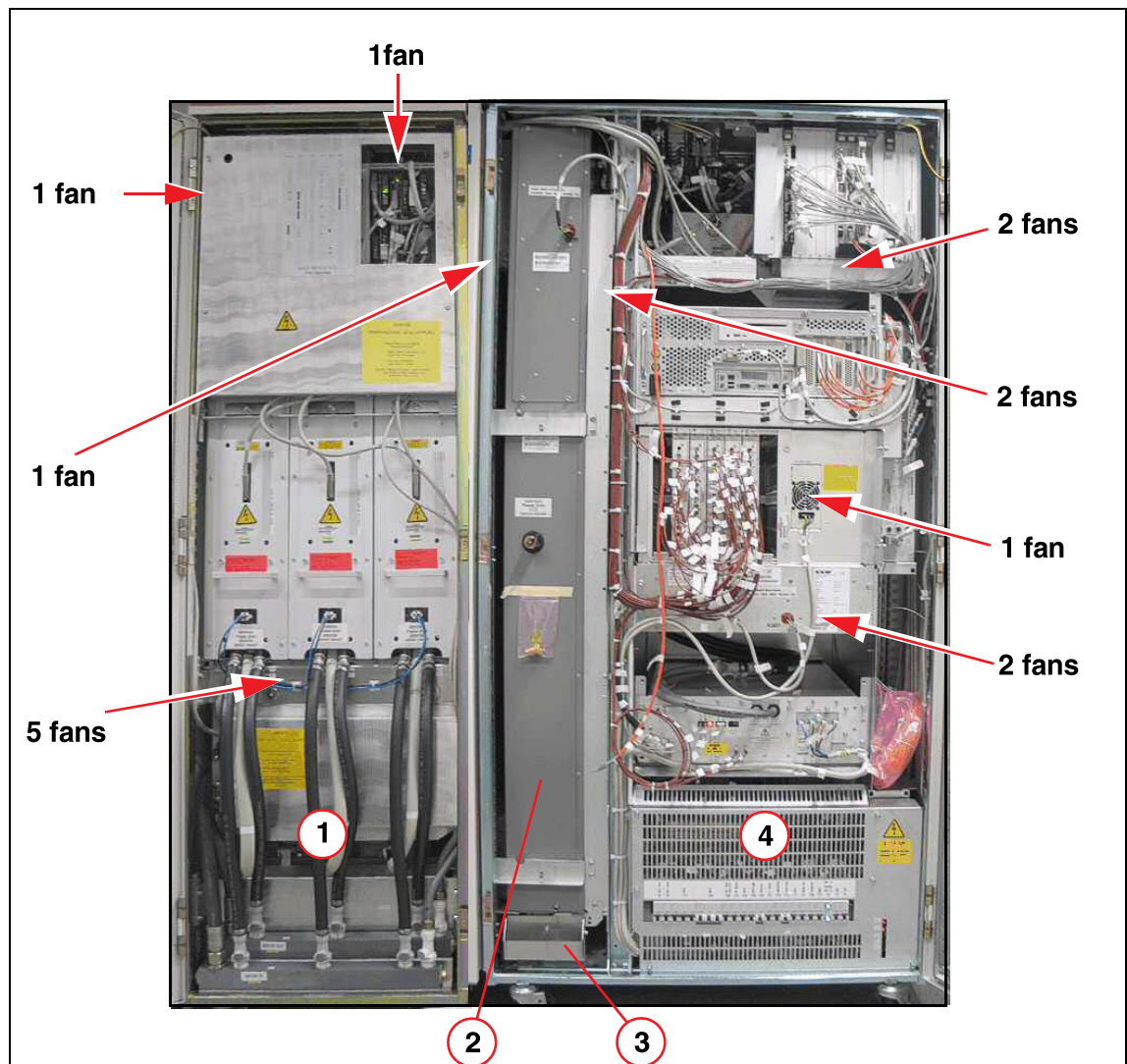


Fig. 472: GPA - CCS - ACC

- Pos. 1 GPA
Pos. 2 CCS blower assembly (3 fans)
Pos. 3 CCS condensate reservoir
Pos. 4 ACC

- Switch off the GPA and ACC cabinets.

- Carefully swivel out the electronic section of the ACC.

WARNING**High-voltage!**

Contact with live parts may result in death or serious injury.

⇒ Keep away from live parts

- Switch on the GPA and ACC cabinets.
- Visually inspect the operation of the fans in the ACC cabinet by using a flashlight. Replace the parts if necessary.

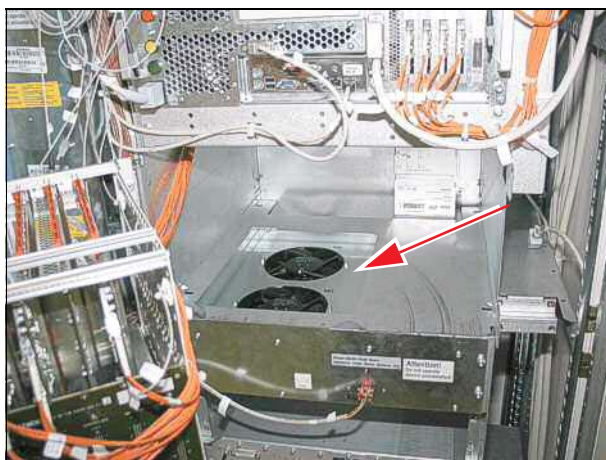


Fig. 473: Fans in ACC view

CAUTION**Fans in operation**

Sharp edges may cause injury.

⇒ Do not touch fans during operation.

- Switch off the GPA and ACC cabinets.
- Carefully swivel in the electronic section of the ACC

NOTE

The fans of the GPA Light 5938704 and GPA 059 5726752 are monitored. There is no need to check.

Checking the air filter of the patient fan

HP-EH102468-211108-02-007

System Type: SymphonyTim

Patient fan

PM Checking the air filter of the patient fan

Required time: 10 min

Required tools and parts: 5mm Allen key, air filter in case of pollution or damage.

- Switch off the **patient fan** on the control unit.
- Remove magnet turret side cover

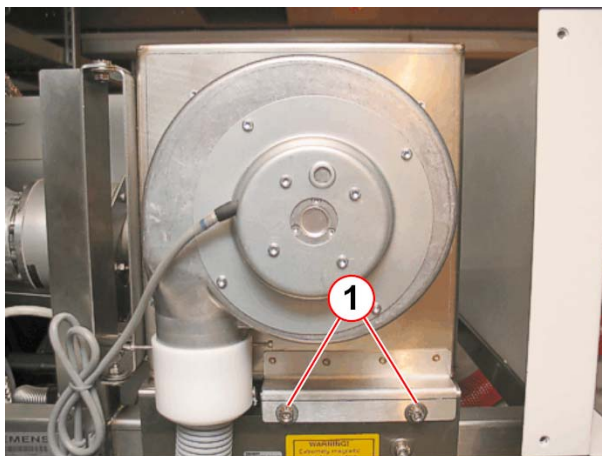


Fig. 474: Patient fan

Pos. 1 5 mm Allen screws

- Loosen the Allen screws (1).

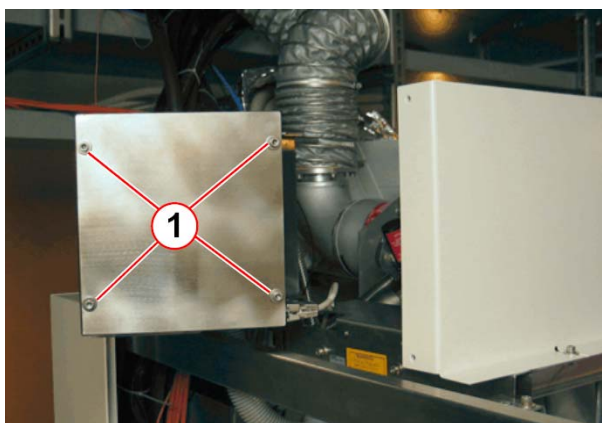


Fig. 475: Patient fan

Pos. 1 5 mm Allen screws

- Turn the patient fan outwards.
- Remove the Allen screws (1).
- Remove the cover of the patient fan.

NOTICE**Ferromagnetic parts.**

The filter grid and the wing nut are slightly ferromagnetic.

⇒ Hold the parts tightly. Handle the filter grid and wing nut with care.

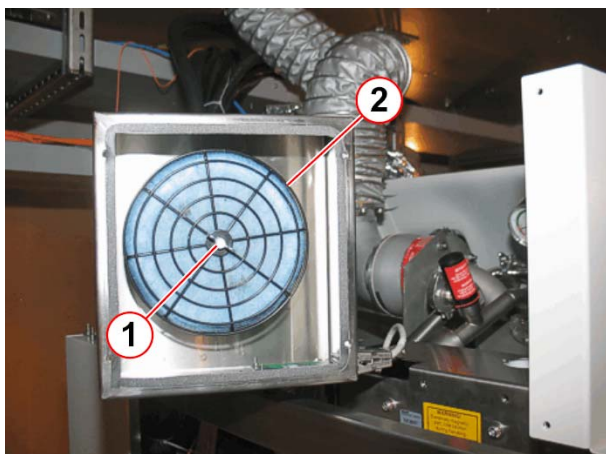


Fig. 476: Patient fan

Pos. 1 Wing nut

Pos. 2 Filter grid

- Loosen the wing nut and remove the filter grid.

The filter grid is protected with a cable tie.

- Cut the cable tie.



Fig. 477: Patient fan air filter

Pos. 1 Air filter

Pos. 2 Filter grid

- Replace the air filter (1).

- Install the new air filter with grid on the fan body in reverse order.
- Switch on the patient fan on the control unit.

Checking the gradient filter fans

HP-EH102468-241108-02-001

System Type: SymphonyTim

Gradient filters

Required time: 10 min

Required tools:n.a.

PM Checking the gradient filter fans

There are 2 different types of gradient filters.

Type A: Gradient filters without fans are used for Upgrades to Symphony, A Tim System.

Type B: Gradient filters with fans are used for Symphony, A Tim Systems ex factory

Verify what type is installed.

Type A:

No need for further checks.

Type B:The gradient filters **type B** are ventilated by 3 fans per axis.*Fig. 478: Gradient filter fans*

- ⇒ If one or 2 fans per axis are defective, replace the fan assembly.
- ⇒ If all 3 fans per axis are defective, the corresponding gradient filter has been damaged due to overheating. Replace the fan assembly **and the corresponding gradient filter**.

Cleaning the strainer of the primary water circuit

HP-EH102468-091208-02-011

System Type:SymphonyTim

RCA

NOTE

Perform the maintenance steps

- Cleaning the strainer of the primary water circuit
- Cleaning the strainer of the secondary water circuit at once.

- PMP Cleaning the strainer of the primary water circuit**
- Required time:** 20 min
- Required tools:** 37mm ring wrench; medium screwdriver
- Cleaning of the strainer**
- Switch off compressor and F1, F3 in the RCA .
 - Switch off F5 in the ACC.
 - Switch off F102 on top of ACC.
 - Remove the electronics box.

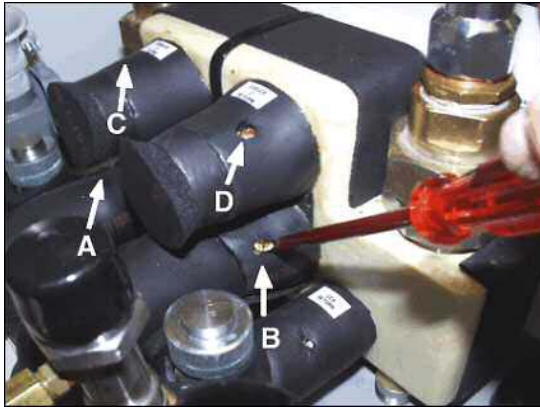


Fig. 479: Intake and outlet valves

- Shut off the intake (A) and outlet (B) valves to isolate the primary water circuit.

Valves C and D are not used (option second cooler).



Fig. 480: Drain valves

Pos. 1 Primary circuit - drain valves

- Open the primary circuit drain valves (1).

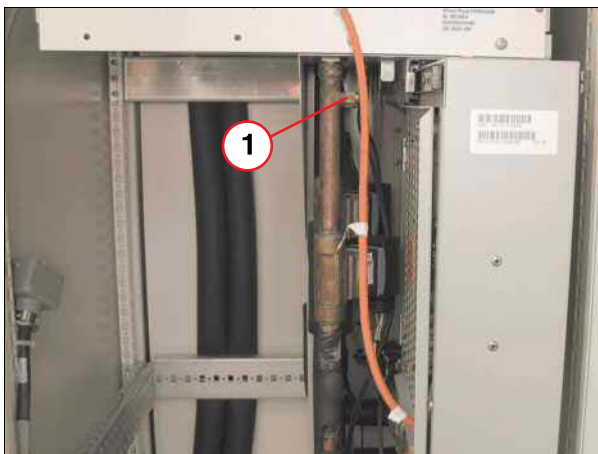


Fig. 481: RCA - bleeding valves

- Open the bleeding valves (I) or remove the filter cap to speed up the draining process .

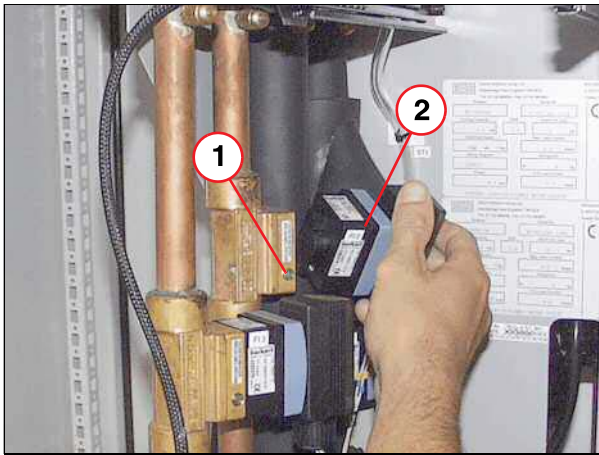


Fig. 482: Flow meter

Pos. 1 Screw

Pos. 2 Flow meter

- Loosen screw (1) and rotate the flow meter (2) CCW and pull away from the mount .



Fig. 483: Primary strainer

- Open the primary strainer using a 27mm ring wrench.
- Inspect the strainer for contamination and clean if necessary .
- Reassemble strainer.

- Close drain valves.
- Assemble the flow meter.
- Assemble the electronics box.
- Open intake (A) and outlet valves (B) .

PMP Cleaning the strainer of the secondary water circuit

- Completely de-energize the system.
- Switch off compressor and F1, F3 in the RCA .
- Switch off F5 in the ACC.
- Switch off F102 on top of ACC.

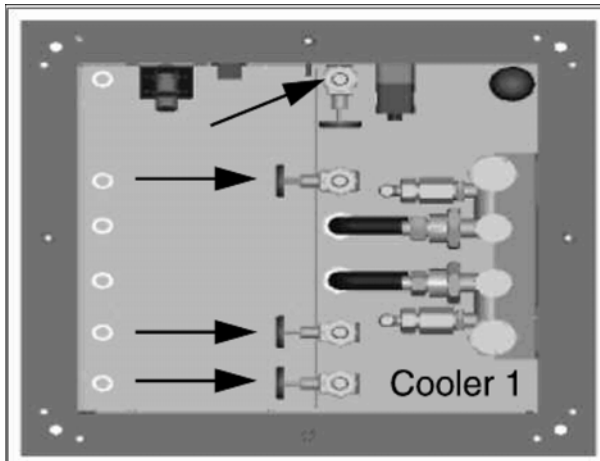


Fig. 484: Secondary valves

- Shut off the inlet and outlet valves of Cooler 1 on top of RCA to isolate the secondary water circuit.

- Remove the electronics box.



Fig. 485: Drain valves

Pos. 1 Secondary circuit - drain valves

- Open the secondary circuit drain valves (2 valves on the left).

- Open the bleeding valves or remove the filter cap to speed up the draining process.



Fig. 486: Secondary strainer

- Open the strainer on Cooler 1 using a 37mm ring wrench. Inspect the strainer for contamination and clean it by rinsing under tap water.
- Install the filter.

- Close the drain valves .
- Open the inlet and outlet valves.
- Switch on F5 in the ACC.
- Switch on F102 on top of ACC.
- Switch on F1, F3, and the compressor in the RCA .
- Check the water level. (using Service Software: 'Magnet & Cooling'). If necessary, refill the water tank with distilled water (refer to 'Start-up' chapter 'Switch on').
 - ⇒ The yellow light on the electronics box indicates the water level, it is low, if the light is on.
- Adjust the flow rates for gradient coil and ACC/GPA cabinet.
- Check the water temperature using Service Software: 'Magnet & Cooling' (refer to 'Start-up' chapter 'Switch on').
 - ⇒ **Temperature 20 ± 3 C**

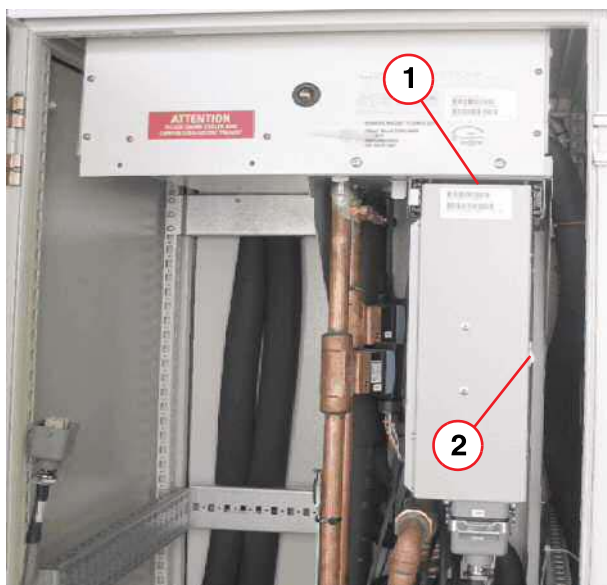


Fig. 487: RCA electronics box

Pos. 1 Water indicator

Pos. 2 Switch

Cleaning the strainer of the secondary water circuit

HP-EH102468-091208-02-012

System Type: SymphonyTim

RCA

NOTE

Perform the maintenance steps

- Cleaning the strainer of the primary water circuit
- Cleaning the strainer of the secondary water circuit at once.

- PMP Cleaning the strainer of the primary water circuit**
- Required time:** 20 min
- Required tools:** 37mm ring wrench; medium screwdriver
- Cleaning of the strainer**
- Switch off compressor and F1, F3 in the RCA .
 - Switch off F5 in the ACC.
 - Switch off F102 on top of ACC.
 - Remove the electronics box.

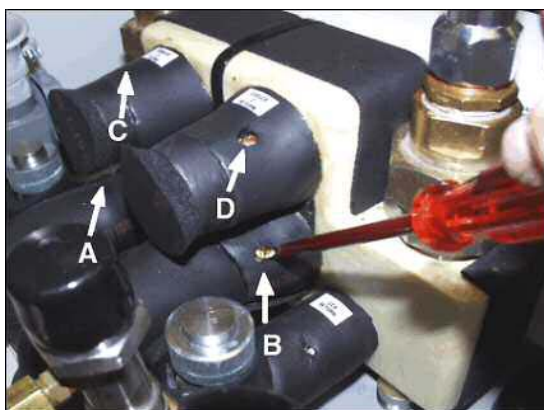


Fig. 488: Intake and outlet valves

- Shut off the intake (A) and outlet (B) valves to isolate the primary water circuit.

Valves C and D are not used (option second cooler).



Fig. 489: Drain valves

Pos. 1 Primary circuit - drain valves

- Open the primary circuit drain valves (1).

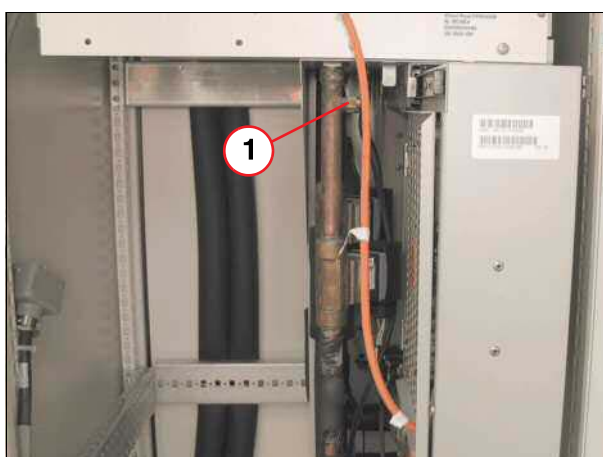


Fig. 490: RCA - bleeding valves

- Open the bleeding valves (I) or remove the filter cap to speed up the draining process .

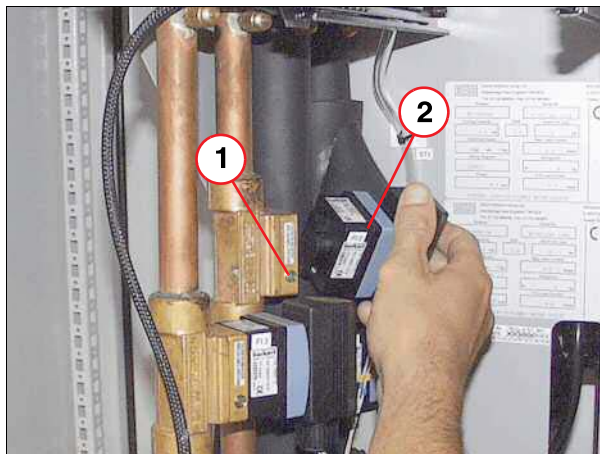


Fig. 491: Flow meter

Pos. 1 Screw

Pos. 2 Flow meter

- Loosen screw (1) and rotate the flow meter (2) CCW and pull away from the mount .



Fig. 492: Primary strainer

- Open the primary strainer using a 27mm ring wrench.
- Inspect the strainer for contamination and clean if necessary .
- Reassemble strainer.

- Close drain valves.
- Assemble the flow meter.
- Assemble the electronics box.
- Open intake (A) and outlet valves (B) .

PMP Cleaning the strainer of the secondary water circuit

- Completely de-energize the system.
- Switch off compressor and F1, F3 in the RCA .
- Switch off F5 in the ACC.
- Switch off F102 on top of ACC.

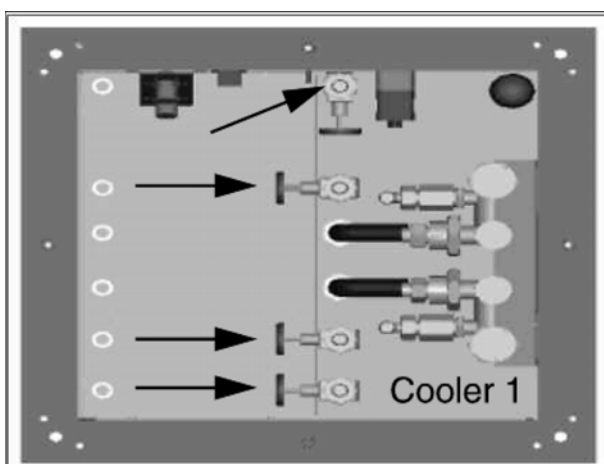


Fig. 493: Secondary valves

- Shut off the inlet and outlet valves of Cooler 1 on top of RCA to isolate the secondary water circuit.

- Remove the electronics box.



Fig. 494: Drain valves

Pos. 1 Secondary circuit - drain valves

- Open the secondary circuit drain valves (2 valves on the left).

- Open the bleeding valves or remove the filter cap to speed up the draining process.



Fig. 495: Secondary strainer

- Open the strainer on Cooler 1 using a 37mm ring wrench. Inspect the strainer for contamination and clean it by rinsing under tap water.
- Install the filter.

- Close the drain valves .
- Open the inlet and outlet valves.
- Switch on F5 in the ACC.
- Switch on F102 on top of ACC.
- Switch on F1, F3, and the compressor in the RCA .
- Check the water level. (using Service Software: 'Magnet & Cooling'). If necessary, refill the water tank with distilled water (refer to 'Start-up' chapter 'Switch on').
 - ⇒ The yellow light on the electronics box indicates the water level, it is low, if the light is on.
- Adjust the flow rates for gradient coil and ACC/GPA cabinet.
- Check the water temperature using Service Software: 'Magnet & Cooling' (refer to 'Start-up' chapter 'Switch on').
 - ⇒ **Temperature 20 ± 3 C**

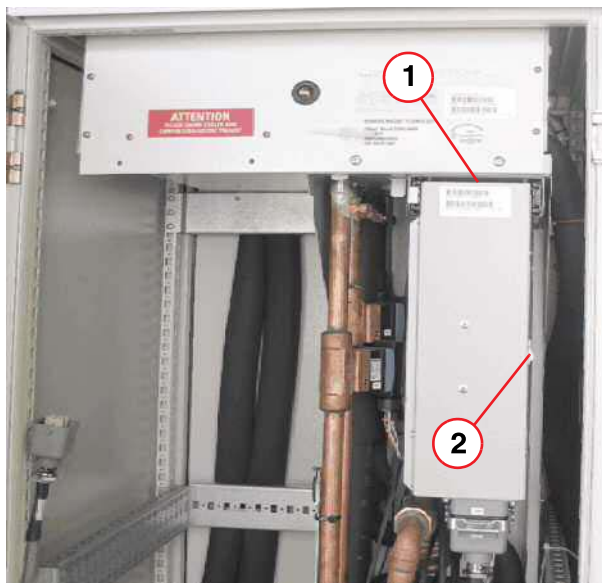


Fig. 496: RCA electronics box

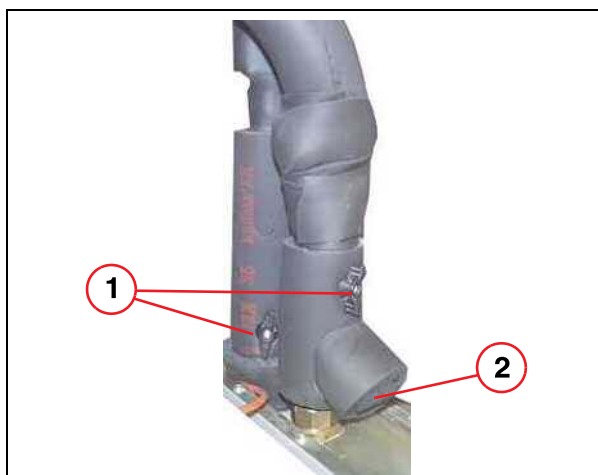
Pos. 1 Water indicator

Pos. 2 Switch

Cleaning the strainer of the water-cooled CCS

HP-EH102468-091208-02-013

System Type:SymphonyTim

CCS**Required time:** 15 min**Required tools:** n.a.**PMP Cleaning the strainer of the water-cooled CCS***Fig. 497: Top of CCS*

Pos. 1 Valves

Pos. 2 Strainer

- Shut off water flow and return on top of the CCS .
- Remove the strainer using the open-end wrench.
- The strainer can be cleaned by rinsing it under tap water.
- Install the filter and open the valves.

Measuring the center frequency

HP-EH102468-210206-02-030

System Type: SymphonyTim

Center frequency

Required time: 5 min

Required tools:n.a.

PMP Measuring the center frequency

- Register patient.
- Select protocol; and drag and drop it to the bottom left.
- Open the protocol by clicking <Open >
- Select <Options>, <Adjustments> in the main menu.
 - ⇒ The current center frequency will be displayed under Frequency (sys.).

Tolerance range for center frequency: 63.5 MHz - 63.7 MHz.

The magnetic field decay is: approx. 4.6 KHz / month

NOTE

Prior to reaching the lower limit of the tolerance range, the magnet has to be scheduled for re-ramping during the next maintenance.

For the ramping procedure, refer to the Start-up register in “Technical Documents”

Tab. 40 Measured value

Measured value:

Checking the cold head

HP-EH102468-091208-02-016

System Type:SymphonyTim

Cold head

Required time: 5 min

Required tools:n.a.

PMP Checking the cold head

- Check the cold head performance.

Use the ' **Magnet & Cooling UI > (Magnet) Status** ' in the Service Software; for details refer to the 'Troubleshooting Guide'.

Or use the MR Quality Database (**CS Intranet > For Service > Product Information > MR Systems > MR Quality Database > Reports**).

- ⇒ If the shield temperature > 75 K, the cooling performance is insufficient. The displacer on the cold head must be replaced.

Replacing the adsorber (Leybold Coolpak)

HP-EH102468-160307-02-001

System Type: SymphonyTim

Replacing the adsorber (Leybold Coolpak; every 2 years)

The compressor is a stand-alone component or mounted in the cooling cabinet.

Required time: 70 min

The adsorber (material number: 47 63 707) has to be replaced every two years.

Materials and tools required:

- Adsorber
- Medium Phillips screwdriver, large Phillips screwdriver
- Set of nuts
- Open-end wrench: 1 1/8"; 1 3/8"; included with magnet accessories
- Venting adapter from the cryo service kit
- Switch OFF the MR system
- Switch off F22 in the CCA
- Switch off F4 on top of the CCA (mains box)

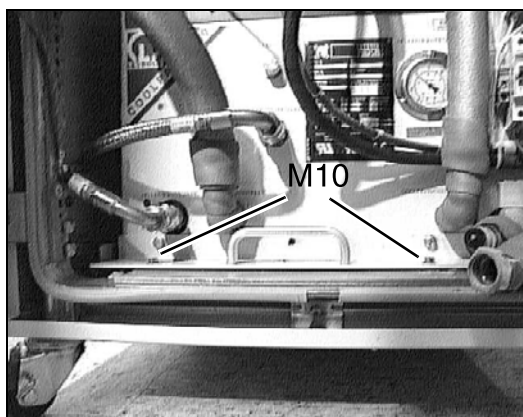


Fig. 498: Disconnection and dismantling of the cool pack

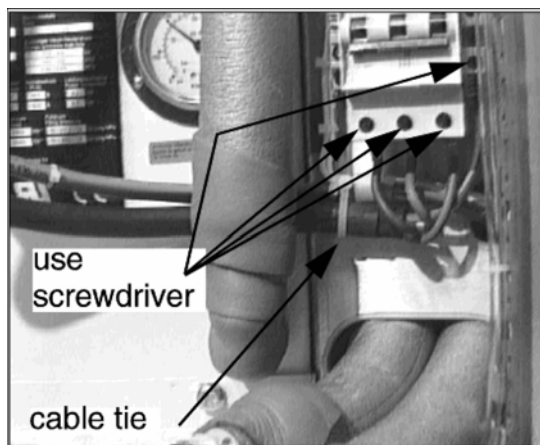


Fig. 499: Electrical disconnection of the cool pack

- Switch **off** the compressor and F3, (Cool pack)
- Remove the helium flexlines from the compressor (Cool pack). Use the tools supplied for this purpose (two open-end wrenches).
- Disconnect the two flexible water lines; (Cool pack).
- Remove the two M10 screws; (Cool pack).
- Remove the cover using an Allen key; (Cool pack)
- Cut the cable ties, (Electrical disconnection of the cool pack) .

- Remove the power cables using a screwdriver, ([Electrical disconnection of the cool pack](#)).
- Disconnect cables X7, X5; ([Cool pack](#)).
- Remove the two M10 screws; ([Disconnection and dismantling of the coolpak](#)).
- Pull out the compressor carefully from the RCA cabinet; ([Coolpak](#)) and ([Cool pack](#)). You can support the weight of the compressor using a strong piece of wood or other similar material.

NOTE

The compressor weighs approximately 95 kg.

- Disconnect the Aeroquip screw connection between the adsorber and oil separator ([Cool pack](#)).

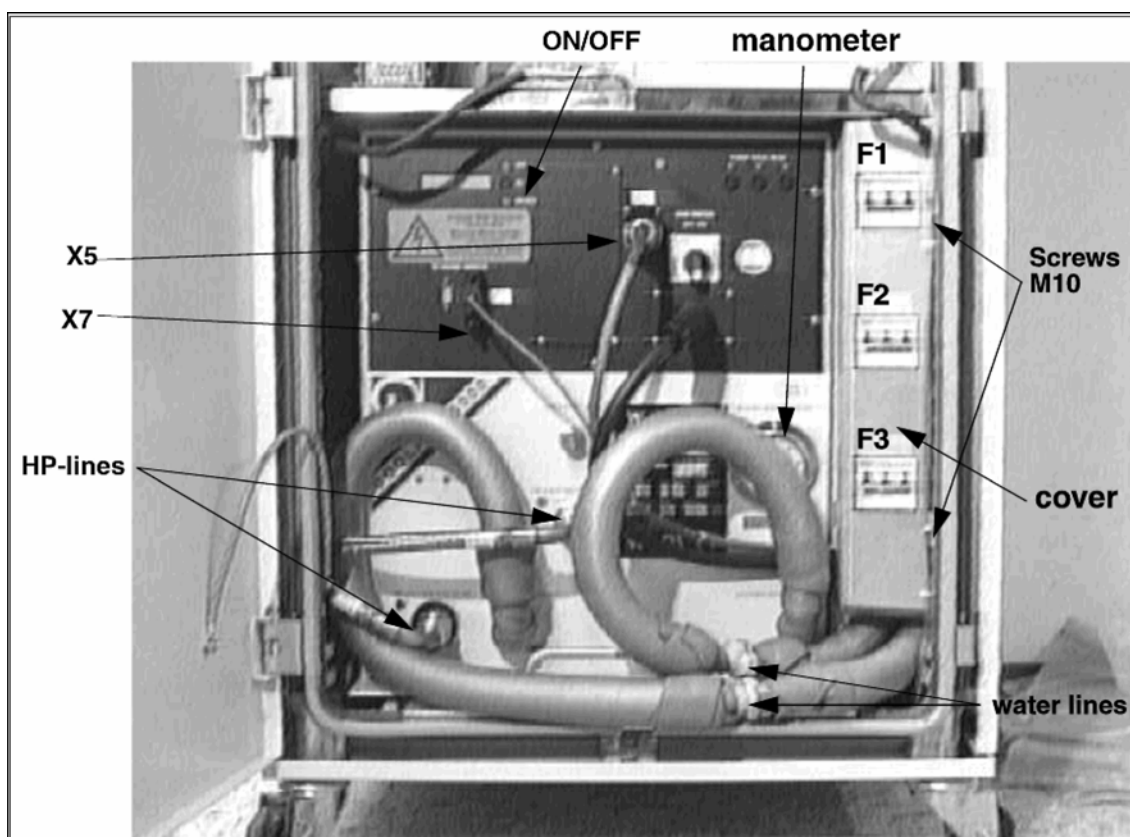


Fig. 500: Cool pack

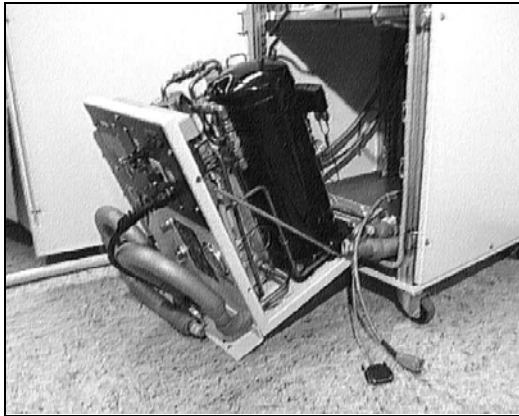


Fig. 501: Cool pack



Fig. 502: Cool pack

- Remove the two Allen screws M10 ([Adsorber](#)).

NOTE

Use only the open-end wrench from the delivery volume to install the helium pressure lines.

- Remove the adsorber.

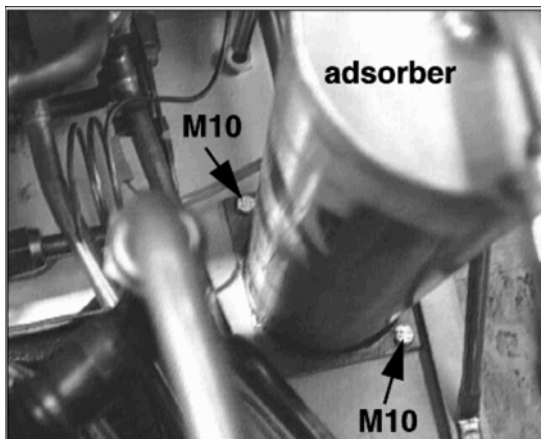


Fig. 503: Adsorber

- Remove the dummy caps from the new adsorber.

NOTE

Adsorbers are filled with 99.99 grade helium to a pressure of 14 to 16 bar. Do not vent the new adsorber.

- Install the new adsorber following the instructions above in reverse order.

NOTE

Tighten all Aeroquip screw connections to the end stop and turn them back by 1/4 turn.

CAUTION

The adsorber is charged with helium gas!

Improper handling of the used adsorber may cause injury.

- ⇒ **Follow the venting procedure for safe disposal of the used adsorber.**

Switch-on procedure

1. Switch on fuse F3.
 2. Close the MAIN POWER and CONTROL SYSTEM circuit breaker at the compressor.
 3. Switch on the compressor.
 4. Check the equalization (static) pressure on the pressure gauge.
- **Gas charging** will be required if pressure is below expected. Refer to **Troubleshooting guide** helium gas charging.

Tab. 41 Helium Gas Pressures

Cold Head	Static Pressure	Dynamic Pressure
10K - 50Hz	15 - 14 bar (218 - 203 psig)	20 - 21 bar (290- 304 psig)
10K - 60Hz	14 - 13 bar (203 - 189 psig)	19- 20 bar (276 - 290psig)

Safe disposal of the used adsorber

CAUTION

The adsorber is charged with helium gas!

Improper handling of the used adsorber may cause injury.

- ⇒ **Follow the venting procedure for safe disposal of the used adsorber .**

A venting adapter fitting is included with the new adsorber.

- Attach the adapter to one of the self-sealing couplings on the used adsorber.
- Vent the used adsorber to atmospheric pressure.
- Turn the used adsorber over to a disposal company.

Final steps

- Enter the replacement date in the Maintenance Protocol.

Replacing the adsorber (APD HC10)

HP-EH102468-160307-02-002

System Type: TrioTim
SymphonyTim

Replacing the Adsorber (APD HC 10; every two years)

The compressor is a stand-alone component or mounted in the cooling cabinet.

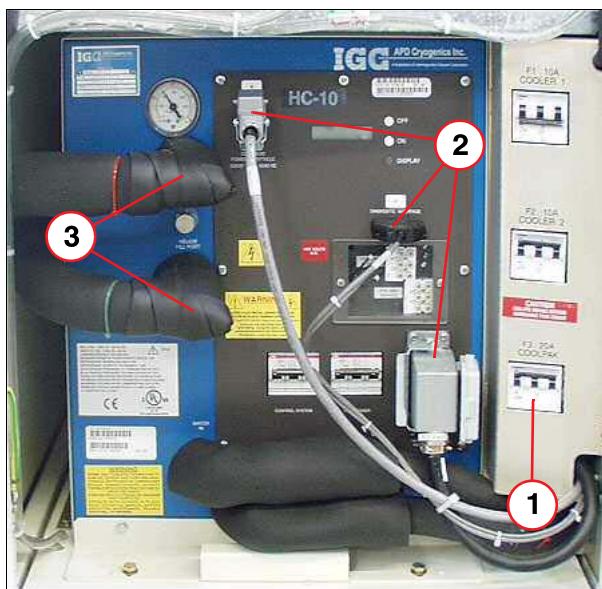
The adsorber has to be replaced every two years.

Required time: 70 min

Materials and Tools required

- Adsorber 7461127
- Medium Phillips screwdriver, large Phillips screwdriver
- Set of nuts
- open-end wrench: 13 mm.
- Open-end wrench: 1 1/8"; 1 3/8"; included with magnet accessories

Compressor Removal



- Switch off compressor.
- Switch off MAIN POWER and CONTROL SYSTEM circuit breaker
- Switch off F3.
- Disconnect cables (2).

Fig. 504: Compressor

- Pos. 1 Fuse F3
Pos. 2 Cables
Pos. 3 Helium flexlines

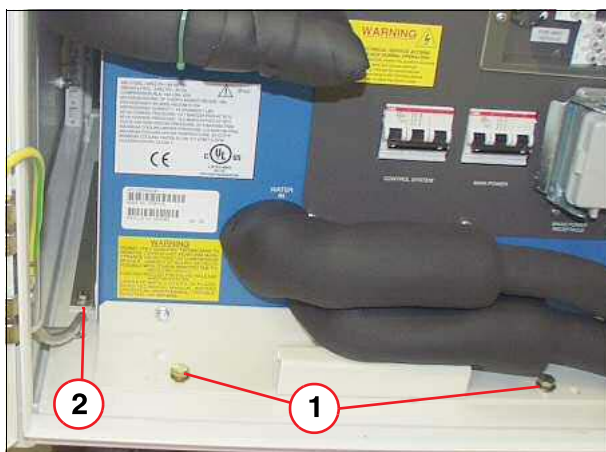


Fig. 505: Compressor attachment

- Remove screws (1).
- Removing the compressor can be facilitated by removing the guide rail screw (2) first.

Adsorber Exchange

NOTE

Always hold the stationary nut on the gas line coupling with one open-end wrench while turning the moveable coupling with the other open-end wrench.

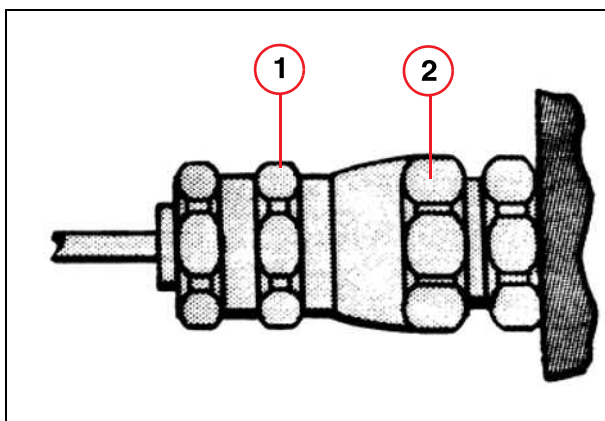


Fig. 506: Helium flexlines coupling

Pos. 1 Nut $1\frac{1}{8}$ "
Pos. 2 Nut $1\frac{3}{16}$ "

Remove helium flexlines

- Hold nut (1) using the open-end wrench (1 $\frac{1}{8}$ in.) and loosen nut (2) using the open-end wrench (1 $\frac{3}{16}$ in.) .

There is no need to remove the water lines.

NOTICE

Avoid equipment damage!

- ⇒ Do not tip the compressor by more than 10 degrees from the horizontal to avoid oil from flowing into unwanted places.

NOTE

The weight of the compressor is approximately 100 kg.



Fig. 507: Compressor out

- Carefully pull out the compressor.
The compressor should be pulled onto a sturdy support (e.g., wood) of approx. 12 cm.
- Remove housing screws. Lift housing cover at the front end, pull forward, and remove.

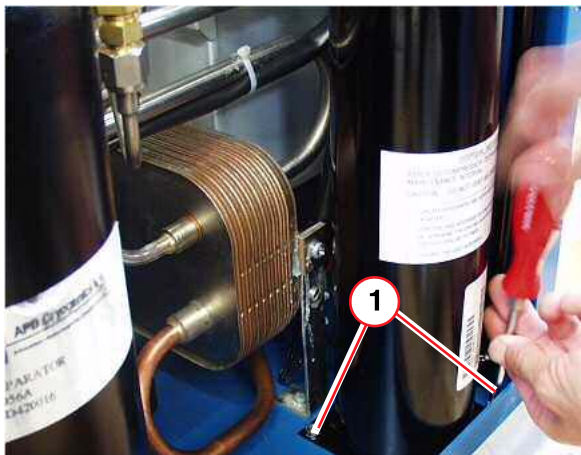


Fig. 508: Adsorber support

Remove Phillips screws and adsorber .



Fig. 509: Compressor, front panel

Remove nut.

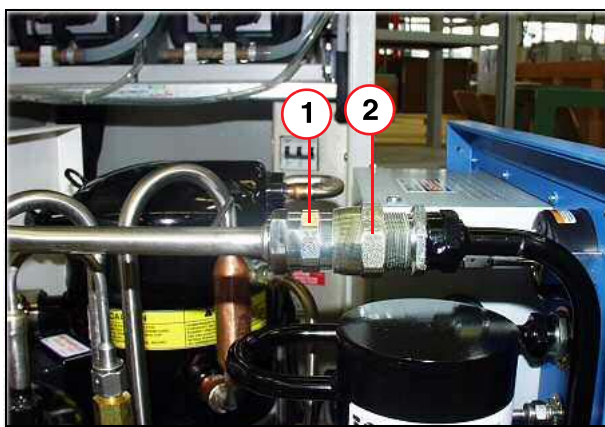


Fig. 510: Adsorber coupling

Pos. 1 Nut $1\frac{1}{8}$ "
Pos. 2 Nut $1\frac{3}{16}$ "

Remove helium gas coupling.

- Hold nut (1) using the open-end wrench ($1\frac{1}{8}$ ") and loosen nut (2) using the open-end wrench ($1\frac{3}{16}$ ").



Fig. 511: Adsorber

- Remove adsorber.
- Retain all hardware to reuse with the new adsorber.

NOTE

The adsorber is filled with 99.99 grade helium at a pressure of 14 -16 bar.

Do not vent the new adsorber

- Remove the dummy caps from the new adsorber.
- Install the lock washer onto the supply coupling of the new adsorber. Insert the supply coupling through the front panel and position the adsorber
- Insert and tighten the screws to fasten the new adsorber to the base.
- Install the nylon washer and the locknut on the supply coupling.
- Connect the adsorber's self-sealing coupling at its inlet side to the oil separator's outlet coupling.
- Assemble compressor housing.
- Mount compressor in the cabinet.
- Install helium flexlines.
 - High pressure (red)
 - Low pressure (green)

Switch-on procedure

1. Switch on fuse F3.
 2. Close the MAIN POWER and CONTROL SYSTEM circuit breaker at the compressor.
 3. Switch on the compressor.
 4. Check the equalization (static) pressure on the pressure gauge.
- **Gas charging** will be required if pressure is below expected. Refer to **Troubleshooting guide** helium gas charging.

Tab. 42 Helium Gas Pressures

Cold Head	Static Pressure	Dynamic Pressure
10K - 50Hz/ 60HZ	14,9 - 15,2 bar (216 - 220 psig)	20 - 22,8 bar (290- 330 psig)
10K - 60Hz	14,9 - 15,2 bar (216 - 220 psig)	20 - 22,8 bar (290 - 330 psig)

Safe disposal of the used adsorber

CAUTION

The adsorber is charged with helium gas!

Improper handling of the used adsorber may cause injury.

⇒ **Follow the venting procedure for safe disposal of the used adsorber .**

A venting adapter fitting is included with the new adsorber.

- Attach the adapter to one of the self-sealing couplings on the used adsorber.
- Vent the used adsorber to atmospheric pressure.
- Turn the used adsorber over to a disposal company.

Final steps

- Enter the replacement date in the Maintenance Protocol.

Replacing the adsorber (Sumitomo CSW 71)

HP-EH102468-211108-02-111

System Type:SymphonyTim
Verio
TrioTim

Not applicable for this system.

Replacing the adsorber (Sumitomo F70)

HP-EH102468-250608-02-001

System Type: SymphonyTim

Replacing the adsorber (F70; every three years)

Required time: 70 min.

Required parts and tools:

- Adsorber part number: (07461127; includes venting adaptor)
- #2 Philips screwdriver
- F70 Spanner kit 10125650 (delivered with the system)

NOTE

Adsorbers are filled with 99.999 grade helium to a pressure of 14-16 bar. Do not vent the new adsorber.

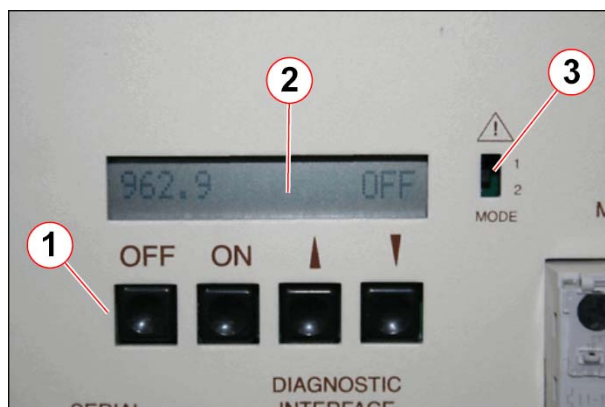


Fig. 512: F70 System status display

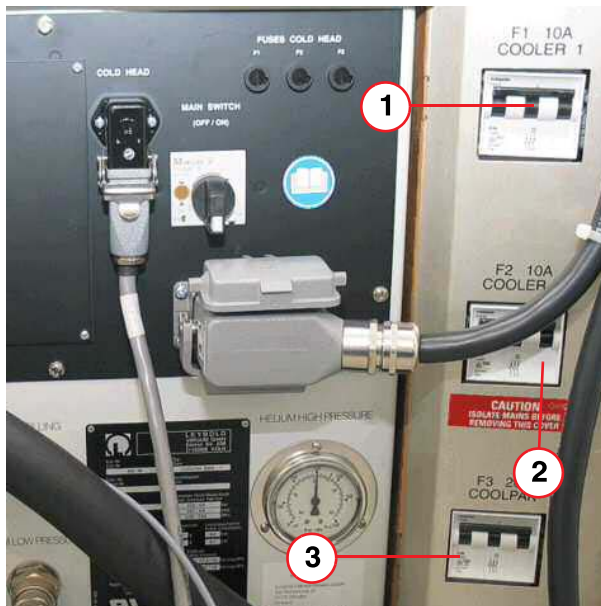
- Pos. 1 Control buttons
Pos. 2 LCD display
Pos. 3 Configuration selector switch

- If the **configuration selector switch** has been set to **configuration 2**, both **ON** and **OFF** buttons are **disabled**. (the compressor is controlled via the pressure within the magnet).
- Switch the compressor **ON** or **OFF** using the **main power switch**.

Adsorber Removal

- The compressor may be a **stand alone** unit, or fitted into a **RCA cabinet**.
- Press the **OFF** button on the Control panel.

- Turn **off** the compressor at the **main power switch**.



- **Isolate power** to the compressor:
 ➔ **RCA cabinet** - switch off circuit breaker **F3**.

Fig. 513: RCA cabinet - Circuit Breakers F1, F2, F3

- Pos. 1 F1 - isolates cooler unit 1
 Pos. 2 F2 - isolates cooler unit 2 (if fitted)
 Pos. 3 F3 - isolates the compressor

- The Alarm Box will display a red warning LED, and an audible alarm will sound
- Press the **ACKNOWLEDGE** button to cancel the audible alarm.

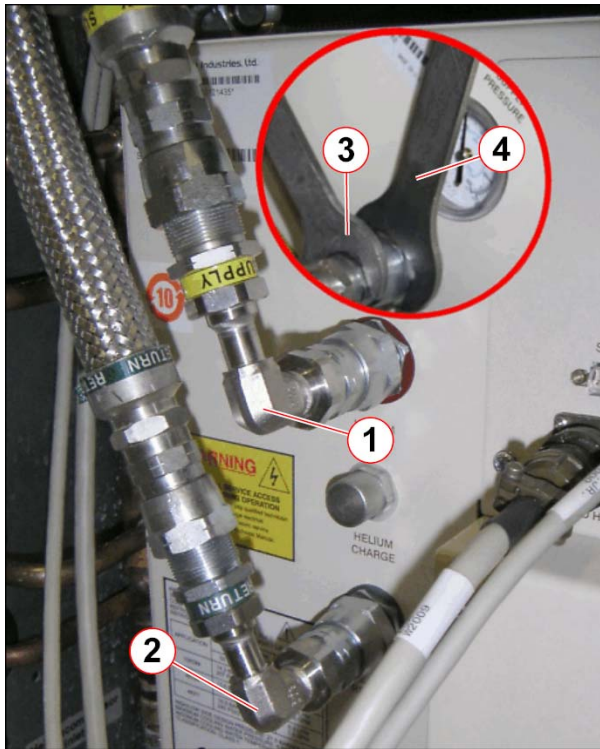


Fig. 514: F70 He compressor connections

- Pos. 1 SUPPLY (high pressure) 90-degree elbow
Pos. 2 RETURN (low pressure) 90-degree elbow
Pos. 3 Hold nut (1" spanner/wrench)
Pos. 4 Turn nut (1 3/16" open end crowfoot spanner/wrench)

RCA Cabinet option

Each **helium pressure line** is connected to the compressor via a **90 degree elbow**.

- **Remove** the self sealing helium pressure lines in the following order:-
 - **Return** (low pressure)
 - **Supply** (high pressure)
- **Hold** nut (3) and **turn** nut (4).

NOTE

90° elbow connections are not required for a stand-alone unit.



Fig. 515: F70 He compressor connections (stand-alone unit)

- Pos. 1 Turn nut (1 3/16" open-end crowfoot spanner/wrench)
- Pos. 2 Hold nut (1" spanner/wrench)

Stand-alone unit

- **Remove** the self-sealing helium pressure lines in the following order:
 - **Return** (low pressure).
 - **Supply** (high pressure).
- **Hold** nut (2) and **turn** nut (1).

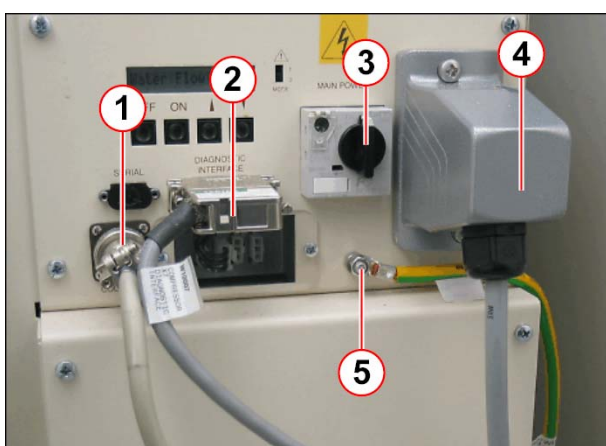


Fig. 516: F70 Electrical cable connections

- Pos. 1 Cold head cable
- Pos. 2 MSUP DB25 Diagnostic interface cable
- Pos. 3 Main power switch
- Pos. 4 Mains supply cable
- Pos. 5 Ground wire

- **Disconnect** the line power supply, cold head motor, and MSUP interface cables.
- **Disconnect** the ground wire.

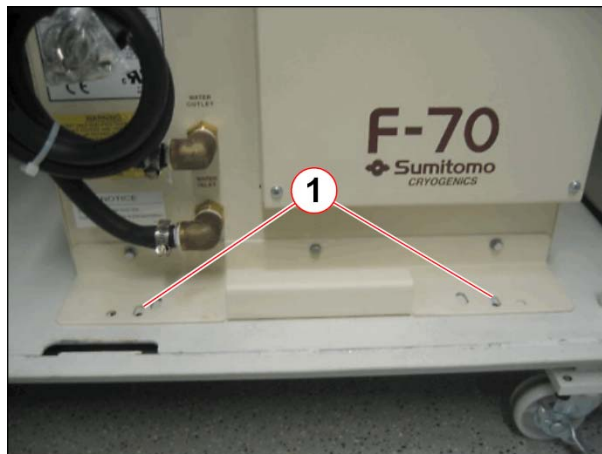


Fig. 517: F70 Compressor fixing plate

Pos. 1 M8 Hex hd screws (2 off)

RCA Cabinet option

- Remove the 2 off M8 hex hd screws securing the **fixing plate** to the cabinet.
- Use the **fixing plate handle** to carefully pull the compressor onto the wooden support 'ramp'.

The unit is fitted with **nylon slide rails** to ease movement.

WARNING

Risk of personal injury and damage to equipment.

The weight of the compressor is 100kg (220lbs).

- ⇒ When installing or removing from the cabinet, reduce the height difference by ensuring there is a sturdy support 'ramp' in position to support the unit.

NOTICE

Prevent equipment failure.

- ⇒ To prevent oil from flowing into unwanted places, do not tip the compressor by more than 10 degrees horizontally.

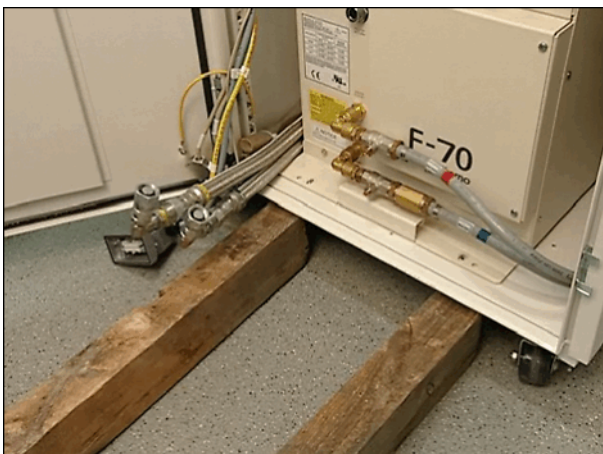
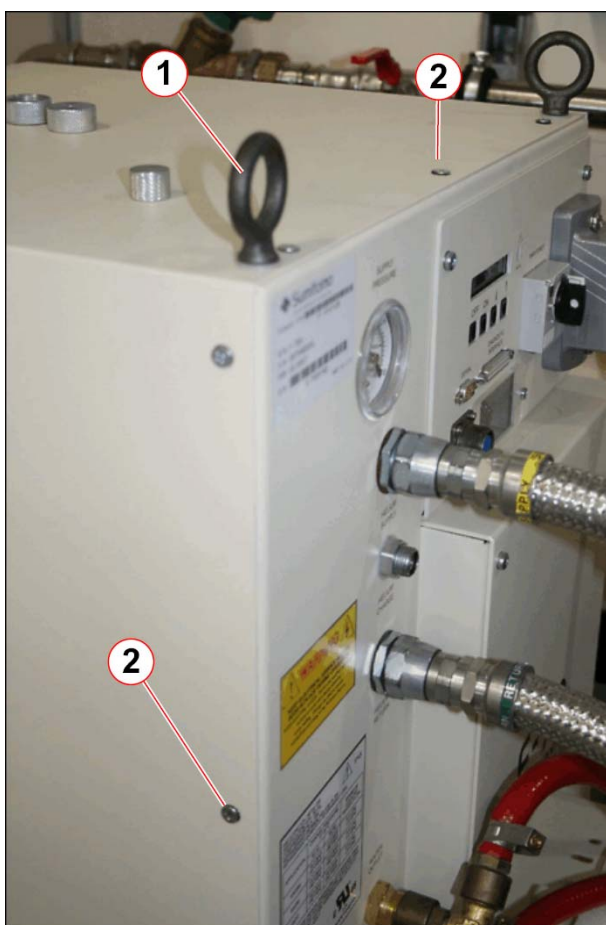


Fig. 518: Supporting ramp



Fig. 519: Move the compressor onto the supporting ramp



- **Remove the 3 lifting eyebolts and all cover panel screws.**

Fig. 520: F70 Lifting eyebolts and cover screws

Pos. 1 Lifting eyebolts (3 positions)

Pos. 2 Cover panel securing screws



- Lift off the U-shaped **cover panel**.

Fig. 521: F70 Removing/replacing the cover
Pos. 1 U-shaped cover

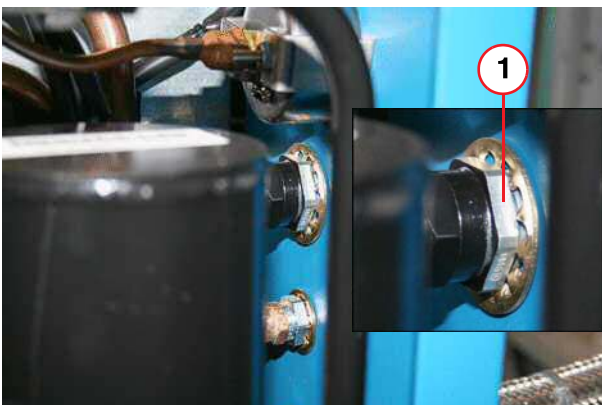


Fig. 522: F70 Adsorber - high pressure supply coupling (inlet side)
Pos. 1 HOLD stationary nut (1" open-end spanner/wrench)

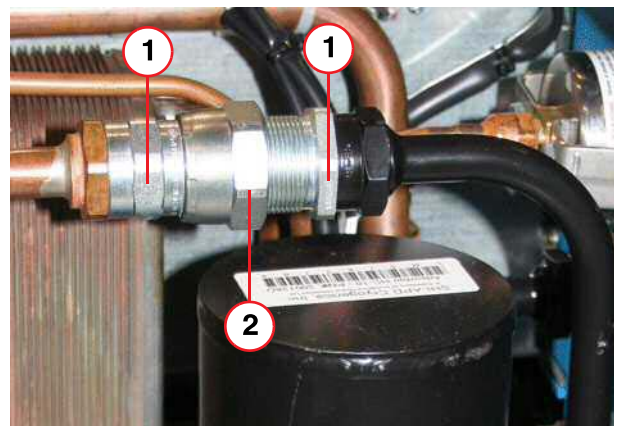


Fig. 523: F70 Adsorber - oil separator Aeroquip coupling
Pos. 1 HOLD stationary nut (1" open-endspanner/ wrench)
Pos. 2 TURN moveable coupling (1 3/16" open-end crow foot)

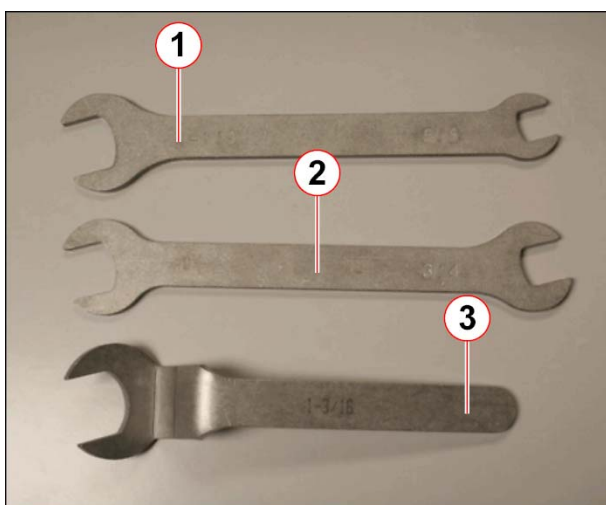


Fig. 524: F70 Spanner kit (part no. 10125650)

- Pos. 1 1 1/8" open-end and 5/8" open-end spanner/wrench
- Pos. 2 1" open-end and 3/4" open-end spanner/wrench
- Pos. 3 1 3/16" open-end crowfoot spanner/wrench

- Always use the three open-ended wrenches when connecting and disconnecting the couplings.
 - **Hold** the stationary nut(s).
 - **Turn** the moveable coupling.

- **Disconnect** the self-sealing coupling on the **inlet side** of the adsorber:- ([F70 Adsorber - disconnecting coupling \(inlet side\)](#))
 - **Hold** the stationary nut (1) on the inlet side.
 - **Turn** the **high pressure locknut** on the front panel ([1/F70 High pressure Aeroquip coupling](#)).
- **Disconnect** the adsorber at the oil separator coupling ([F70 Adsorber - oil separator Aeroquip coupling](#)).



Fig. 525: F70 Adsorber base securing screws

Pos. 1 Securing screws (2 off)

- Remove the 2 off **base securing screws**.



Fig. 526: F70 High pressure Aeroquip coupling

Pos. 1 Locknut
Pos. 2 Flat rubber gasket
Pos. 3 Nylon washer

- Remove the high pressure **locknut** and **nylon washer**.

- Carefully remove the adsorber.

- Retain all removed fixings.

NOTE

Adsorbers are filled with 99.999 grade helium to a pressure of 14-16 bar. Do not vent the new adsorber.

Adsorber replacement

- Remove the caps from the gas lines of the new adsorber.
- Install the **lock washer** to the **supply coupling** of the new adsorber.
- Insert the **supply coupling** through the **front panel** and position the adsorber.
- Secure the adsorber to the **base** using the screws ([1/F70 Adsorber base securing screws](#)).
- Install the **nylon washer** and **locknut** on the high pressure **supply** coupling ([F70 High pressure Aeroquip coupling](#)).
- **Connect** the adsorber at the **oil separator coupling** and perform a leak check. ([F70 Adsorber - oil separator Aeroquip coupling](#)).
- Re-fit the **cover panel** using the retained screws.
- Slide the compressor into the RCA cabinet.
- Re-secure the **fixing plate** using the 2 off M8 hex hd screws.

NOTE

Tighten all Aeroquip screw connections to the end stop and turn them back by 1/4 turn.

- **Re-connect** the high pressure (**supply**) and low pressure (**return**) helium gas lines.
- Reconnect the **cold head cable, diagnostic cable, and power cable** and **ground wire**([F70 Electrical cable connections](#)).

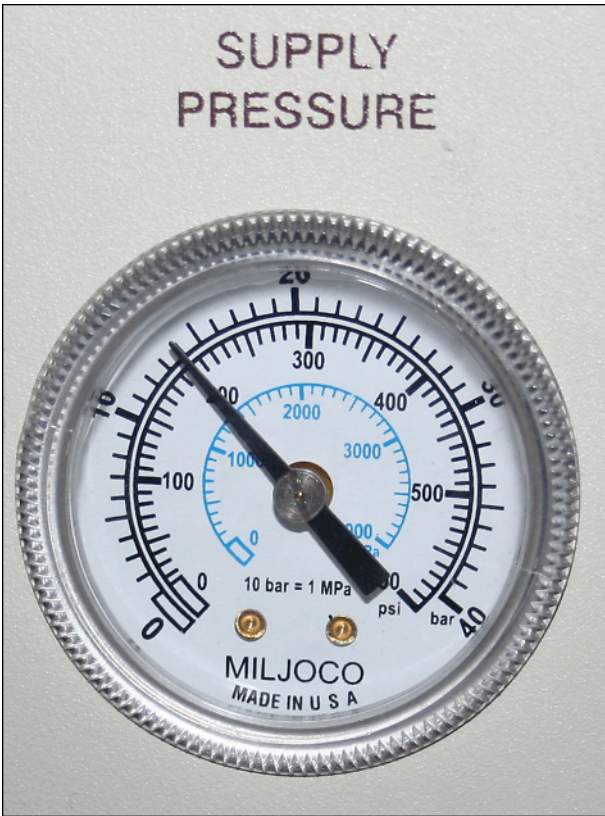


Fig. 527: F70 Static pressure reading

- Check the **equalization (static)** pressure on the pressure gauge.
 - ⇒ **Gas charging** will be required if pressure is below expected. Refer to **Troubleshooting Guide** - Helium Gas charging.

Equalization (static) pressure at 20° C (68° F) for 12 to 20m long gas lines.

Tab. 43 Helium Gas Pressures

Cold Head	Static Pressure	Dynamic Pressure
10K - 50Hz	14.3 - 14.6 bar (207 - 212 psig)	21 - 22 bar (304 - 319 psig)
10K - 60Hz	13.6 - 13.9 bar (197 - 202 psig)	21 - 22 bar (304 - 319 psig)

- Switch **on F3** (RCA cabinet)
- Switch **on** the compressor at the **main power switch**.
- **Configuration 1 only** - switch **on** the compressor using the **ON** button on the Control panel.
- Check the Alarm box status and clear any faults.
- **Leak check** the gas Aeroquip couplings.

CAUTION

The adsorber is charged with helium gas!

Improper handling of the used adsorber may cause injury.

- ⇒ Follow the venting procedure for safe depressurization and disposal of the used adsorber.

Safe disposal of the used adsorber

A **venting adapter** is included with the new adsorber.

- **Attach** the **adapter** to one of the **self-sealing couplings** on the used adsorber.
- **Vent** the used adsorber to **atmospheric pressure**.
- Give the **used adsorber** to a **disposal company**.

Checking the TFT monitor

HP-EH102468-200206-02-009

System Type: Avanto
Espreo
TrioTim
SymphonyTim
Verio

TFT monitor

Required time: 10 min

Required tools: n.a.

Calling up the test image

Windows XP

- Simultaneously press buttons **Ctrl** and **Esc**.
 - ⇒ the Windows **Start Menu** appears
- Click: **Start > Programs > Advanced User** > enter PW > press OK button
- Simultaneously press **Ctrl + Shift + Esc**
- Select the Task Manager > Applications
- Right-click on the CapGM-xxxx in the list and select **Minimize**.
- Minimize the Task Manager (WIN Desktop is now displayed in the foreground)
- Right-click on the desktop background
- Click Properties and choose the Desktop task card
- Click Browse
- Select path: C:\MedCom\utils\Imagequality\Master2AllRev20.bmp
- Click Open and OK.

Windows 7

- Simultaneously press buttons **Ctrl** and **Esc**.
 - ⇒ the Windows **Start Menu** appears
- Click: All Programs > Advanced User > enter PW > press OK button
- Simultaneously press **Ctrl + Shift + Esc**
- Select the Task Manager and choose the **Applications** task card.
- Right-click on the CapGM-xxxx in the list and select **Minimize**.
- Minimize the Task Manager (WIN Desktop is now displayed in the foreground)
- Right-click on the desktop background
- Click **Personalize** and choose **Desktop Background**
- Click Browse

- Select path: Computer > C:\MedCom\utils\Imagequality
- Select file Master2AllRev20.bmp
- Click **save changes**.
- If the syngo screen appears again, select the task manager (Alt + Tab) then right click on the CapGM-xxxx and select Minimize.

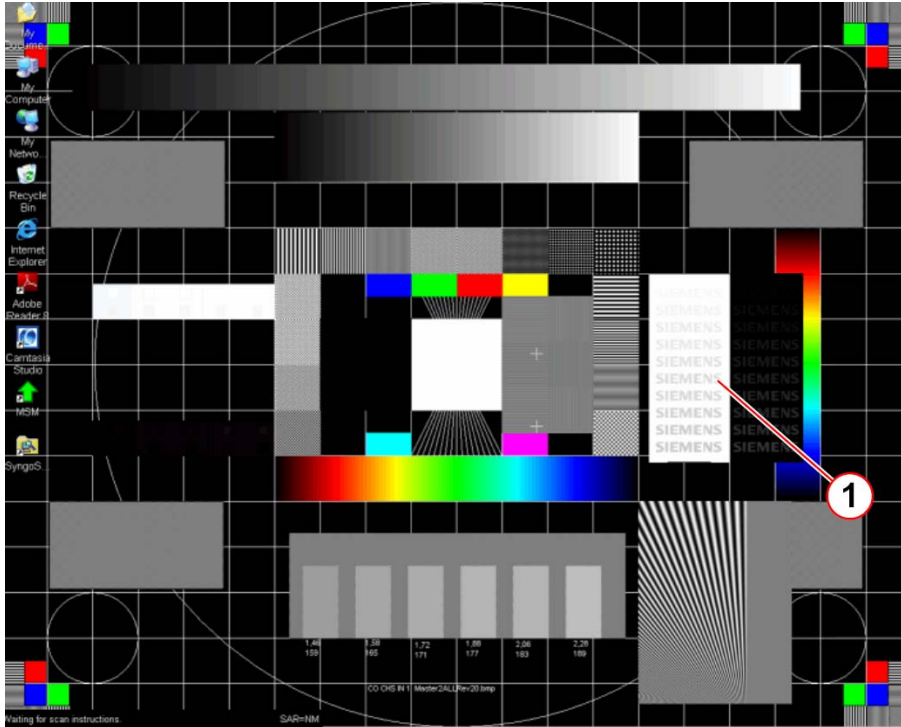


Fig. 528: Test image - master2AllRev.20

Pos. 1 Siemens test logo

Check

PMA Checking the TFT monitor

- Check the MRC TFT monitor using the test image.
 - ⇒ All colors are displayed
 - ⇒ Gray scale is correctly displayed
 - ⇒ Geometry is correct
 - ⇒ 10 SIEMENS test logos are visible in the black and white field
- If the MRSC TFT monitor (option) installed, check the MRSC TFT monitor following the same procedure.
- If the In-Room MRC TFT monitor (option) installed, check the In-Room MRC TFT monitor first following the same procedure.

If any adjustment is necessary, refer to 'Replacement of Parts REP'; 'Visually inspecting/evaluating the TFT monitor' in detail.

Set the normal mode

WindowsXP

- Right-click on the desktop background
- Click Properties and choose the Desktop task card
- Select None in the menu.
- Click OK.
- Simultaneously press **Alt + Tab** until syngo screen appears.

Windows7

- Right-click on the desktop background
- Click **Personalize** and choose **Desktop Background**
- Select **Solid Colors** in the **Picture Location** menu.
- Select color black
- Click **save changes**.
- Close the **Personalization** window and the **Task Manager**
- Simultaneously press **Alt + Tab** until syngo screen appears.

Checking the phantoms

```
HP-EH102468-200206-02-007
System Type: . . . . .Avanto
               Espree
               TrioTim
               SymphonyTim
               Verio
```

Phantoms

Required time: 10 min
Required tools: n.a.

PMP Checking the phantoms

Please read the “MR-Specific Safety Information” prior to inspecting the phantoms.

- 1. Inspect every phantom used on-site for cracks, leaks, or extensive wear and tear.
 ⇒ Replace the phantom if there is any evidence of damage that may affect normal operation.

NOTE

Defective phantoms must be disposed of. Refer to the “Disposal Instructions”.

- 2. Inspect every phantom used on-site for excessive air bubbles. Air bubbles can influence your Tune-up results and quality measurements.
- 3. Ensure that fluid is at an adequate level so that measurement performance is not adversely affected.

Water mixture phantoms with filler cap:

- Refill the phantom, if necessary.
- Phantoms with distilled water mixture - fill with distilled water only.

Water mixture phantoms without filler cap (welded):

- Refilling is not possible. Water mixture phantoms without filler cap must be replaced

Oil phantoms without filler cap (welded):

- Refilling is not possible. Oil phantoms without filler cap must be replaced

Oil phantoms with filler cap:

- Refilling is not possible. Oil phantoms with filler cap must be replaced.

Do not open the screw cap. It will lead to leakage.
Example spherical phantom blue p/n 7577666: The diameter of the air bubble should not exceed 12 cm.

Checking the RF-room door

HP-EH102468-200206-02-004

System Type: Avanto
Espreo
TrioTim
SymphonyTim
Verio

Required time: 10 min.
Required tools: 10 cm wide strip of paper,

Door

PMP Checking the RF room door

- 1. Check contact springs for mechanical damage. The contact springs should bounce back. Test the line contact at the outer surface of the contact by inserting a 10 cm wide strip of paper and pulling it out. Repair minor damage using appropriate tools, such as flat nose pliers.
- 2. The metal of the contact surfaces and contact springs has to be clean. Remove any dirt, contamination, or hardened grease.
- 3. Check the hinges for cracks or possible damage.

NOTE

In case of a defect (e.g. broken or missing contact springs, damaged hinges), inform the customer to initiate corrective action immediately.

Checking door lock of the RF-room

HP-EH102468-200206-02-003	
System Type:	Avanto
	Espre
	TrioTim
	SymphonyTim
	Verio

Required time: 5 min.
Required tools:n.a.

Door lock

PM Checking the door lock of the RF room

- 1. The locking and unlocking mechanism has to function properly.
- 2. The door handle or knob has to be securely fastened.

NOTE	In case of a defect (e.g. defective unlocking mechanism, door handle,), inform the customer to initiate corrective action immediately.
------	--

Checking the patient table drive unit

HP-EH102468-160608-02-012

System Type: SymphonyTim

Patient Table - Drive Unit

Required time: 15 min

Required tools: 4mm Allen key

PM Checking the patient table drive unit

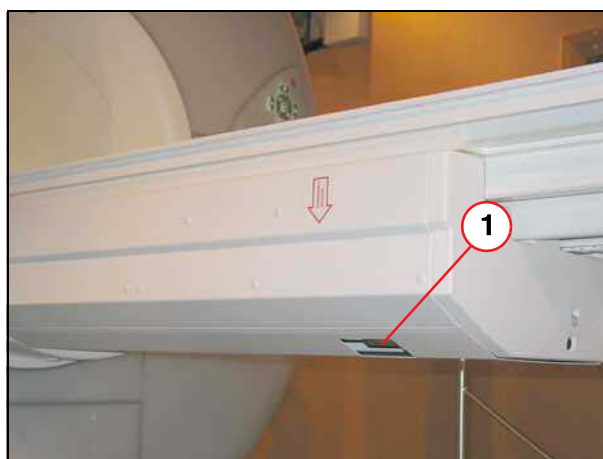


Fig. 529: Patient table

Pos. 1 Emergency release

There are different types of drives mechanisms for the patient tables (whole body and partial body). To differentiate the table type, check that there is an emergency unlocking device available ([1/Patient table](#)).

Whole body type:

- ⇒ The emergency unlocking device is available.

Partial body type:

- ⇒ The emergency unlocking device is not available.

The toothed belt tension **must be** checked for the **partial body type only**.

For patient tables with **whole body drive unit** there is **no need to check** the toothed belt tension. Mark OK in the protocol.

Check the toothed belt tension for the partial body drive unit

- Move the table top in **Home Position**

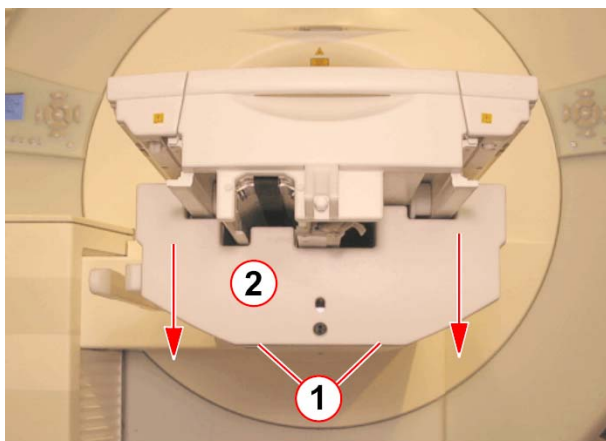


Fig. 530: Patient tabletop support frame foot end cover

Pos. 1 4 mm allen screws
Pos. 2 Foot end cover

- Remove the 2 screws on the underside of the foot-end cover(1).
- Remove the foot-end cover (2) by sliding down.

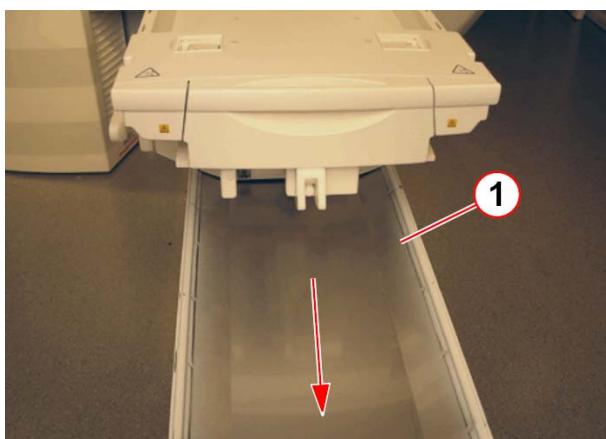


Fig. 531: Patient table support frame bottom cover

Pos. 1 Bottom cover

- Remove the bottom support frame cover (1) by sliding out.

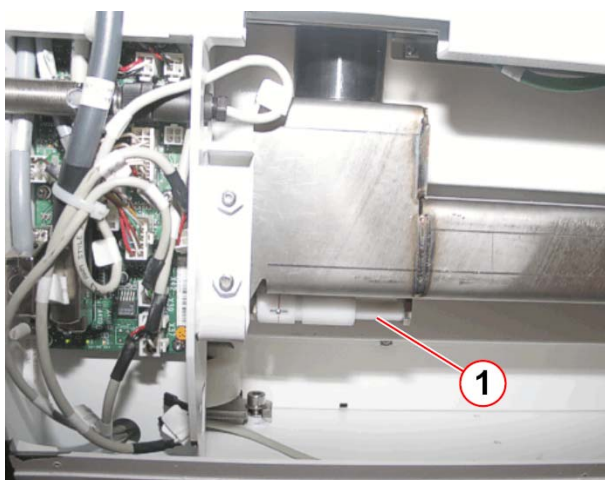


Fig. 532: Support frame; bottom side

Pos. 1 Storage position of the service tool - toothed belt tension

The toothed belt tension service tool is located under the table top in the support frame.

- Remove the toothed belt tension service tool.



Fig. 533: Support frame bottom side

Pos. 1 Service tool - toothed belt tension in use

- Insert the toothed belt tension service tool (1) in the hole on the support frame bottom side, refer to [\(Service tool - toothed belt tension\)](#).

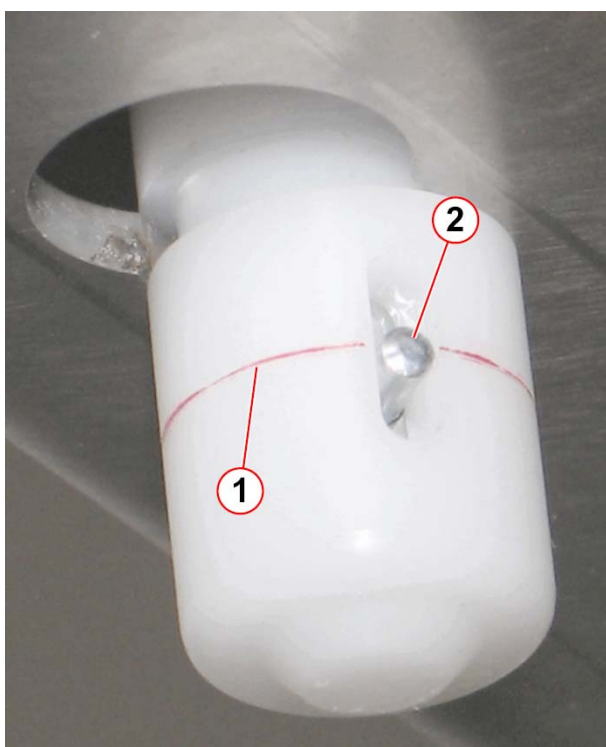


Fig. 534: Service tool - toothed belt tension

Pos. 1 Indication pin

Pos. 2 Ring marking

The indication pin shows the belt tension.

- ⇒ The indication pin must be aligned with the marking ring.
- ⇒ If the indication pin does not align with the ring marking, adjustment is necessary; refer to [\(Adjustment of the toothed belt tension\)](#).

- Remove the toothed belt tension service tool and return it to its storage position under the table top in the support frame, refer to [\(Support frame; bottom side\)](#).
- Reinstall the bottom support frame cover, refer to [\(Patient table support frame bottom cover\)](#).
- Reinstall the foot-end cover, refer to [\(Patient table support frame foot-end cover\)](#).
- Mark OK in the protocol.

Adjustment of the toothed belt tension

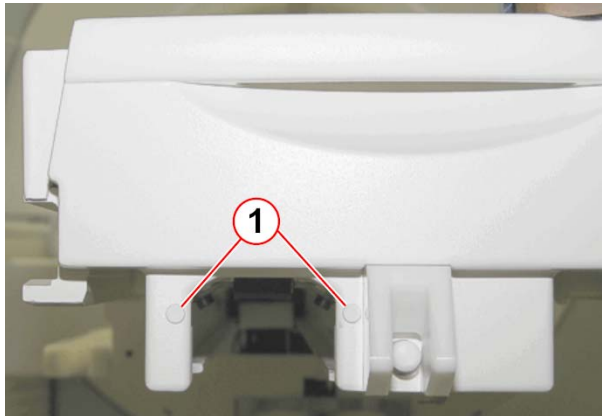


Fig. 535: Partial body patient table foot end side
Pos. 1 Blanking plug

- Remove the blanking plugs.

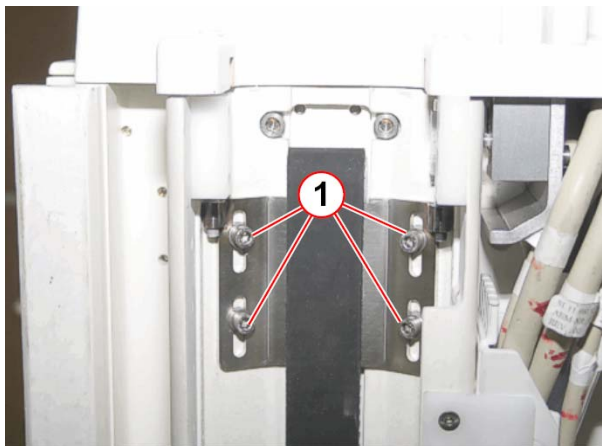


Fig. 536: Tension unit for the toothed belt
Pos. 1 Fixing screws (4 mm Allen screw)

- Loosen the fixing screws, do not remove.

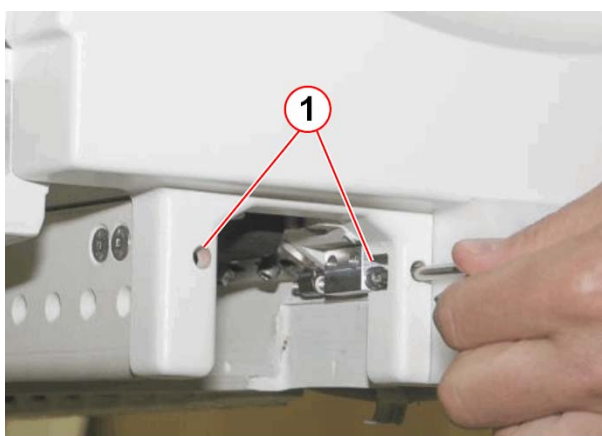


Fig. 537: Toothed belt tension adjustment
Pos. 1 Adjustment screws for the belt tension

- Adjust the toothed belt tension with a 4 mm Allen key.
- Turn both adjustment screws (cw) equally until the indication pin is aligned with the marking ring, refer to [\(Service tool - toothed belt tension\)](#).
- Tighten the fixing screws, refer to [\(Tension unit for the toothed belt\)](#).

- Check the toothed belt tension, refer to [\(Service tool - toothed belt tension\)](#).
- Remove the toothed belt tension service tool and return it to its storage position under the table top in the support frame, refer to [\(Support frame; bottom side\)](#).
- Insert the blanking plugs, refer to [\(Partial body patient table foot-end side\)](#).
- Reinstall the bottom support frame cover, refer to [\(Patient table support frame bottom cover\)](#).
- Reinstall the foot-end cover, refer to [\(Patient table support frame foot-end cover\)](#).

Replacing the patient trolley battery (option)

HP-EH102468-200206-02-002

System Type: Avanto
Espreo
TrioTim
SymphonyTim
Verio

Patient Trolley (option)

Required time: 10 min

Required tools: 2 batteries (type “AA”)

PM Replacing the patient trolley battery (option)



Fig. 538: Patient trolley

Pos. 1 Battery cover

Remove the battery cover.

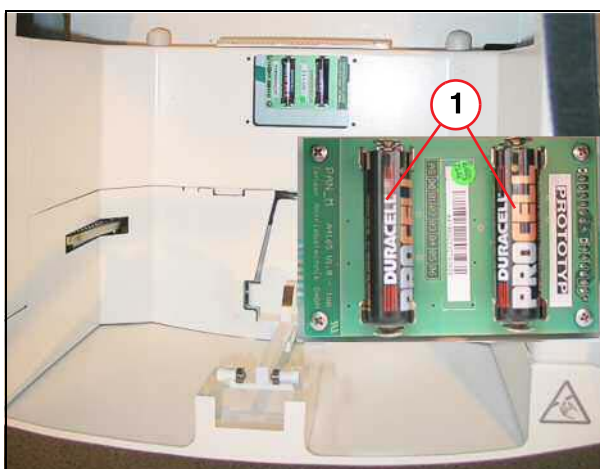


Fig. 539: Patient trolley

Pos. 1 2 mignon batteries

- Replace the batteries (battery type “AA”).
- Mount the battery cover.
- Turn the old batteries over to a disposal company in accordance with local regulations.

Replacing the vacuum pump filter

HP-EH102468-200206-02-001

System Type:Avanto
 Espree
 TrioTim
 SymphonyTim
 Verio

Vacuum pump (comfort kit option)

Required time: 5 min

Required tools: Filter

PMP Replacing the vacuum pump filter (every 2 years)

The vacuum pump is an option (comfort kit) mounted on the RF filter plate.



Fig. 540: Vacuum pump filter

Pos. 1 Connection - vacuum pump
 Pos. 2 Connection - hose to the examination room

It is no longer necessary to replace the filter inside the pump. An external filter is now available. The external filter prevents the pump and the filter inside the pump housing from dirt.

The old filter can remain inside the pump housing.

- Check the filter inside the vacuum pump if the throughput is insufficient.
 ➞ Replace the filter if required

If no external filter is installed, install the external filter in the same manner as described below.

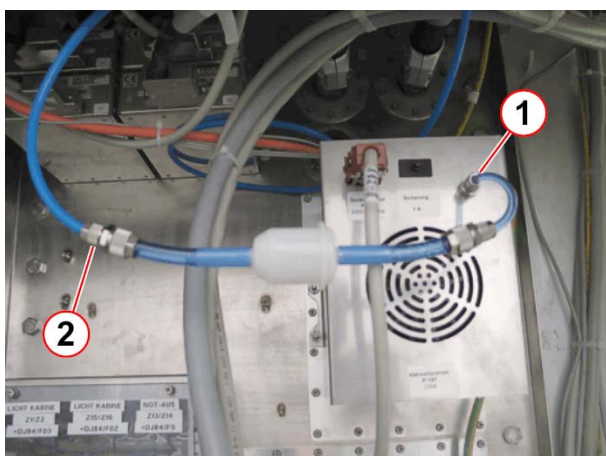


Fig. 541: External air filter of the vacuum pump

Pos. 1 Coupling nut - vacuum pump
 Pos. 2 Coupling nut - hose to the examination room

- Loosen the coupling nut (1) at the housing of the filter pump.
- Remove the hose.
- Remove the coupling nut and place it on the hose of the new filter.
- Connect the new filter to the vacuum pump and mount it with the coupling nut.
- Loosen the coupling nut (2) of the hose to the examination room.
- Remove the old filter.
- Connect the new filter to the hose of the examination room and mount it with the coupling nut.

- Enter the replacement date in the Maintenance Protocol.

Tab. 44 Replacement Date

Startup Date: . . .
Date of Last Replacement: DD MMM YYYY

Tab. 45 Maintenance performed

Maintenance of this section was performed by:

Signature: _____
Date: Name:

HP-EH102468-091208-02-020

Not applicable for this system.

Downloading the Shock-log data (mobile systems only)

HP-EH102468-091208-02-021

System Type: SymphonyTim
TrioTim
Verio

Not applicable for this system.

Software Maintenance

Saving site-specific data

HP-EH102468-170206-02-050

```
System Type: . . . . .Avanto
              Espree
              TrioTim
              SymphonyTim
              Verio
```

Site-specific data

⇒ Although site-specific data should be saved once a year, we recommend to perform it every three months.

SW Saving site-specific data

Backup of the site-specific data is possible on CD ([Backup site-specific data on CD](#)) and on the USB device ([Backing up site-specific data on external USB device](#)).

Backup site-specific data on CD

Required time: 10-60 min

Required tools / parts: CD-R or USB device

NOTE

Use only CD-R 's with medical grade, e.g. Siemens Medical CD-R 80 or TDK Medical CD-R.

Save the site-specific data each time you make changes to the site-specific configuration, tune-up, or sequences and protocols. In case of a hard disk crash, you will be able to restore a valid set of site-specific data.

- Login as **administrator**
- Start the service platform, select **Backup & Restore**.
- Insert a recordable CD-disk in the **CD/DVD-burner**.
- Select **Backup** in the **“Command”** pull-down menu.
- In the menu **Drives**, select the target drive **CD/DVD-burner** (drive letter **S:**) used to save the data.
- Select **Master-Backup** in the **“Packages”** menu.
- Click **Go** to start the “Backup”, check for errors in the log-window after the backup has been completed.

Setting that are not part of the site-specific data:

Remote access mode (No Access, Limited Access, etc. in the customer UI).

Auto-transfer flag (“Configuration“, “Service“, “Auto Transfers“)

Setting that are not part of the site-specific data:
Postscript Printer Configuration (Customer paper printer)
ImageText Settings (ImageText Settings from ImageText Dialog)
Time zone
Transfer Rules

NOTE

During backup at the MRSC, a “CorRea.bru”error may be displayed, please ignore this error.

Backing up site-specific Data on external USB device

Required time: 10-60 min

Required tools/parts: USB device

NOTE

Please check the external medium (e.g. USB-disk) first with the virus scanner of the service PC. Be sure that the external medium used is free of viruses.

- Log in as **administrator** ↴.
- Start the “Internet Explorer”.
- Connect the USB device.
- Check what drive letter was set for the USB device with the Windows File Explorer.
- Log in to the service platform and select **Backup & Restore**.
- Select Backup in the Command menu, select the drive letter of the USB device, e.g., **F** in the Drives menu.
- Select **Master-Backup** in the Package menu, start the backup by clicking **OK**.
- (Estimated time ~10-60 min) Check for errors using the log window.
- Click **Home** to leave Backup & Restore.

Deleting the MR save log files

HP-EH102468-170206-02-051

System Type:Avanto
 Espree
 TrioTim
 SymphonyTim
 Verio

Cleaning up directories

Required time: 10 min

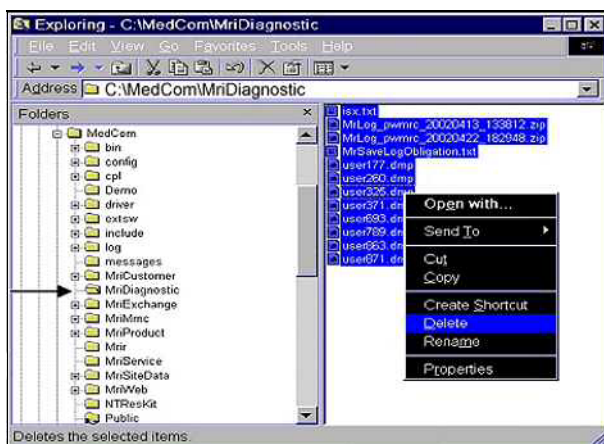
Required tools / parts: n.a.

SW **Deleting the MR save log files**

Starting the Explorer

- Press <Ctrl-Esc>
- Select <Start>, <Programs / All Programs>, <Advanced User>, enter the password (see password list).
- Press <Ctrl-Esc>
- Select <Programs / All Programs>, <Accessories>, <Windows Explorer>

Delete files



- Navigate to "C:\MedCom\MriDiagnostic"
- Click once in the right window and select all files <Ctrl-a>
- Right-click one of the files and select <Delete>.
- This opens the pop-up window "Confirm Multiple File Delete"
- Click <Yes> to delete the files
- Close the Windows Explorer

Fig. 542: Exploring C:\MedCom\MriDiagnostic

Quality Assurance

General QA

HP-EH102468-121012-02-003

System Type: SymphonyTim

Note for using the online 'Install&Maintenance Protocol'

NOTE

The interactive 'Install & Maintenance' protocols take the status from the 'General QA'. If all general QA steps show the 'Done' status, then the 'Install & Maintenance' protocols display the OK status. If one or more steps have the 'ToDo' or 'NotOK' status, then the 'Install & Maintenance' protocols display the 'NotOK' status.

CAUTION: When the Open QA button is pressed, the QA platform opens automatically and all QA steps change to the 'ToDo' status. Please note: the "Open QA" button is only one way to open the QA platform. This notice also applies if the QA platform is opened by different means. The data will be taken over even if you do not use the "Open QA" button.

Quality Assurance (QA)

Required time: 70 - 120 min depending on installed options.

Required tools: QA Phantom Setup, Service Plug (if Multi Nucleus Spectroscopy Option is installed)

The quality measurement of the body coil is a fully automated procedure. The results are entered in the on-site database.

NOTE

Use the Online Help to obtain detailed information regarding, e.g., table control, phantom positioning, or specific items and procedures. Press the 'Help' button in the header bar of the corresponding platform.

Preliminary steps

Starting the service platform

All actions of the quality assurance procedure are performed via the service platform.

- Start the Service platform > **Options** > **Service** > **Local Service**.

SI

General Quality Assurance

Go to the Home menu and press the corresponding button to access the **Quality Assurance** sub platforms.

- Start the measurement by clicking the "Go" button.

- ⇒ The **General Quality Assurance** check is successfully completed when every individual QA step indicates the status < **done** >.

NOTE

If the results are out of tolerance, perform “Tune-Up” or troubleshooting, if necessary. Repeat “Quality Assurance”.

Test Result / Evaluation

HP-EH102468-170206-02-089

System Type: Avanto

Espre

TrioTim

SymphonyTim

Verio

After completing the test activities, fill out this page.

The tester has to evaluate the test results as defined and listed under the test certificate points.

The operator or the person assigned by the operator must sign the system evaluation form for product liability reasons.

26.3 Protocols

26.3.1 Start-up Report

HP-EH102468-211009-02-010

```
System Type: . . . . .Espree
              Avanto
              TrioTim
              SymphonyTim
              Verio
```

Preview

Preview provides an overview of the protocols.

- Check to ensure that the protocols are completed in full and are correct.
 - ⇒ If values entered in the protocol are incorrect, they must be corrected at the corresponding 'Install & Maintenance' platform.
 - ⇒ Please note that only saved print views will be stored on the system.
- Enter your name in the 'signature field' and click the **Save** button to save the protocol.
 - ⇒ The protocol will be frozen and is then part of the 'Site-specific Data'.
- Perform '**Backup & Restore**' to save site-specific data.

The Install & Maintenance protocols are part of the site-specific data.

Use the '**Reports**' platform to display or to print the protocols.

Printing protocols

Preconditions: The protocols have to be available, i.e. the **Previews** have been saved.

If there is no printer installed, an external medium (e.g. USB drive) is required.

Printer is installed

Navigate to the 'Install & Maintenance **Reports**' platform.

- Click > **Reports** > SiteInfo > **Install & Maintenance** and select the required protocol.
- Click > **Show as PDF**.
- Click **Print** and the protocol will be printed.
 - ⇒ The HTML protocol will be converted to a PDF protocol.

Printer is not available

With an external media (e.g. USB drive), it is possible to save the protocol on the service laptop or print out the protocol from another computer that has a printer.

NOTE

Please check the external medium first with the virus scanner of the service PC. Be sure that the external medium used is free of viruses.

- Connect a medium to the USB port.
- Navigate to the 'Install & Maintenance **Reports**' platform.
- Click > **Reports** > SiteInfo > **Install & Maintenance** and select the required protocol.
- Click > **Show as PDF**.
 - ⇒ The HTML protocol will be converted to a PDF protocol.
- Click the **Save** icon
 - ⇒ The pop-up window opens ' Save a Copy ...'.
- Navigate to the drive where the medium is connected.
- Save the protocol on the medium.
 - ⇒ Now you can copy the protocol to your service laptop or print out the protocol from another computer that does have a printer.

26.3.2 Safety Related Protocol

 HP-EH102468-210206-02-071

System Type:Avanto
 Espreo
 TrioTim
 SymphonyTim
 Verio

Preview

Preview provides an overview of the protocols.

- Check to ensure that the protocols are completed in full and are correct.
 - ⇒ If values entered in the protocol are incorrect, they must be corrected at the corresponding 'Install & Maintenance' platform.
 - ⇒ Please note that only saved print views will be stored on the system.
- Enter your name in the 'signature field' and click the **Save** button to save the protocol.
 - ⇒ The protocol will be frozen and is then part of the 'Site-specific Data'.
- Perform '**Backup & Restore**' to save site-specific data.

The Install & Maintenance protocols are part of the site-specific data.

Use the '**Reports**' platform to display or to print the protocols.

Printing protocols

Preconditions: The protocols have to be available, i.e. the **Previews** have been saved.

If there is no printer installed, an external medium (e.g. USB drive) is required.

Printer is installed

Navigate to the 'Install & Maintenance **Reports**' platform.

- Click > **Reports** > SiteInfo > **Install & Maintenance** and select the required protocol.
- Click > **Show as PDF**.
- Click **Print** and the protocol will be printed.
 - ⇒ The HTML protocol will be converted to a PDF protocol.

Printer is not available

With an external media (e.g. USB drive), it is possible to save the protocol on the service laptop or print out the protocol from another computer that has a printer.

NOTE

Please check the external medium first with the virus scanner of the service PC. Be sure that the external medium used is free of viruses.

- Connect a medium to the USB port.
- Navigate to the 'Install & Maintenance **Reports**' platform.
- Click > **Reports** > SiteInfo > **Install & Maintenance** and select the required protocol.
- Click > **Show as PDF**.
 - ⇒ The HTML protocol will be converted to a PDF protocol.
- Click the **Save** icon
 - ⇒ The pop-up window opens 'Save a Copy ...'.
- Navigate to the drive where the medium is connected.
- Save the protocol on the medium.
 - ⇒ Now you can copy the protocol to your service laptop or print out the protocol from another computer that does have a printer.

26.3.3 Maintenance Protocol

 HP-EH102468-210206-02-072

System Type:Avanto
 Espreo
 TrioTim
 SymphonyTim
 Verio

Preview

Preview provides an overview of the protocols.

- Check to ensure that the protocols are completed in full and are correct.
 - ⇒ If values entered in the protocol are incorrect, they must be corrected at the corresponding 'Install & Maintenance' platform.
 - ⇒ Please note that only saved print views will be stored on the system.
- Enter your name in the 'signature field' and click the **Save** button to save the protocol.
 - ⇒ The protocol will be frozen and is then part of the 'Site-specific Data'.
- Perform '**Backup & Restore**' to save site-specific data.

The Install & Maintenance protocols are part of the site-specific data.

Use the '**Reports**' platform to display or to print the protocols.

Printing protocols

Preconditions: The protocols have to be available, i.e. the **Previews** have been saved.

If there is no printer installed, an external medium (e.g. USB drive) is required.

Printer is installed

Navigate to the 'Install & Maintenance **Reports**' platform.

- Click > **Reports** > SiteInfo > **Install & Maintenance** and select the required protocol.
- Click > **Show as PDF**.
- Click **Print** and the protocol will be printed.
 - ⇒ The HTML protocol will be converted to a PDF protocol.

Printer is not available

With an external media (e.g. USB drive), it is possible to save the protocol on the service laptop or print out the protocol from another computer that has a printer.

NOTE

Please check the external medium first with the virus scanner of the service PC. Be sure that the external medium used is free of viruses.

- Connect a medium to the USB port.
- Navigate to the 'Install & Maintenance **Reports**' platform.
- Click > **Reports** > SiteInfo > **Install & Maintenance** and select the required protocol.
- Click > **Show as PDF**.
 - ⇒ The HTML protocol will be converted to a PDF protocol.
- Click the **Save** icon
 - ⇒ The pop-up window opens 'Save a Copy ...'.
- Navigate to the drive where the medium is connected.
- Save the protocol on the medium.
 - ⇒ Now you can copy the protocol to your service laptop or print out the protocol from another computer that does have a printer.

27 List of Hazard IDs

HP-DUMMY00-999999-00-000

There are no Hazard IDs in this document.